



Advanced Problems in **Inorganic Chemistry**

for IIT-JEE & Other Competitive Examinations

Ramashish Paul



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Inorganic Chemistry

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&
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Ramashish Paul
B.Tech. Calcutta University



G K Publications (P) Ltd

CL MEDIA (P) LTD.

Edition : 2017

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ISBN : 978-81-8355-864-8

Administrative and Production Offices

Published by : CL Media (P) Ltd.

KH No. 1027, Ramnagar, Salempur,
Rajputana, Roorkee (U.K.)-247667

Marketed by : G.K. Publications (P) Ltd.

A-41, Lower Ground Floor,
Espire Building,
Mohan Cooperative Industrial Area,
Main Mathura Road,
New Delhi - 110044

For product information :

Visit www.gkpublications.com or email to gkp@gkpublications.com

Preface

As we know students are the backbone of our country and they are having the power to move the country ahead of time. The whole world would salute them in future for their great research work for revolutionary invention of Medical drug. Inorder to help them to achieve the target we tried to publish the book based on **“INORGANIC CHEMISTRY”** which will serve as an important tool for those who want to shine in various competitive examinations (Engineering & Medical). It helps to get an edge over other to become an “IIT”ian ; a “NIIT”ian or an AIIMS scholar. My aim in this book is to provide a comprehensive and contemporary introduction about the subject. The problems given in this book are thoroughly innovative.

Most of the students find Inorganic Chemistry as boring subject. In my opinion it's not a boring subject, it's a scoring subject. But to score it one needs to develop interest within the subject.

Along with so many objective type questions which are designed scientifically for various engineering examinations, a lot of other type questions e.g., multiple choice questions, integer types questions are given in the book which prove to be extremely useful to crack the IIT-JEE Advanced examinations. Also emphasis is given to provide “Comprehension type” & “Matrix Match type” questions which make the book unique and different from all other books available in the market. So by solving all types of problems an average student can develop confidence in IIT-JEE (Main & Advanced) examination. By this way this book can set a visionary example to connect best chunk of students with IIT-JEE examination.

I have the privilege to teach the students **“INORGANIC CHEMISTRY”** every year. But a large number of students who are not taught by me can be immensely benefitted by this book. They can connect myself with my teaching methodology through this book. My purpose will be accomplished when someone can feel **“Paul Sir's”** methods to teach Inorganic chemistry in absence of myself.

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Acknowledgement

Firstly, I would like to thank the almighty God for his blessings to complete this book successfully without my parents unconditional support, I could have not been motivated to take such a big step in my life. The continuous support of my wife & family was very much needed during this process and I am really thankful to them for it.

I would also like to thank Mr. Jitendra Gaur of GK Publications for all his editorial support to mould the content.

I am also thankful to Mr. Ashish Sharma for all his data support. Finally I would like to thank my students as my career would be incomplete without them.

With Best Regards

Ramashish Paul

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Quantitative Analysis of Questions in Each Chapter

No.	NAME OF CHAPTER	NUMBER OF QUESTIONS ON DIFFERENT TOPICS				
		Single Objective Types	Multiple Choice Questions	Comprehension Type	Matrix Match Type	Integer Type
1.	Chemical Bonding	200	150	65	35	80
2.	Coordination Compounds	150	100	75	31	50
3.	Metallurgy	100	60	20	15	40
4.	Qualitative Salt Analysis	120	65	100	25	50
5.	The s-block Elements and Their Compounds	75	65	28	17	21
6.	The p-block Elements and Their Compounds	150	80	60	30	50
7.	The d-block Elements and Some of Their Compounds	65	45	38	17	30
8.	Hydrogen and The Hydride	46	20	15	11	15
9.	Periodic Table	150	100	100	40	50
TOTAL QUESTIONS :		1056	685	456	221	386

Note : Along with this, More than 500, IIT-JEE (Main & Advanced) Questions are given with solutions. So near about total 3300 questions are given in this book which are more than sufficient for IIT-JEE preparation.

CHEMICAL BONDING

PROBLEMS BASED ON GIVEN TOPICS

- Attainment of s stable configuration
- Types of Bonds
- Transitions between the main types of bonding
 - Ionic bonds
 - Covalent bonds
 - Oxidation numbers
 - Coordinate bonds
 - Double and triple bonds
 - Metallic bonds and metallic structures
- The Covalent bond
 - The Lewis theory
 - Sidgwick-Powell theory
- Valence bond theory
 - What is the essence of hybridization?
 - Features of hybrid orbitals
 - Calculation of steric number
- Valence shell electron pair repulsion (VSEPR) theory
 - Effect of lone pair
 - Effect of double bond
 - Effect of electronegativity
 - Back bonding
- The extent of d orbital participation in molecular bonding
- Types of covalent bonds (sigma (σ) and pi (π)) bonds
 - Bridge bonding
- Molecular orbital method
- LCAO method
 - s-s combination of orbitals
 - s-p combination of orbitals
 - p-p combination of orbitals
 - d-d combination of orbitals
 - Non-bonding combination of orbitals
- Rules for linear combination of atomic orbitals
 - H_2^+ molecule ion
 - H_2 molecule
 - He_2^+ molecule ion
 - He_2 molecule
 - Li_2 molecule
 - Be_2 molecule
 - B_2 molecule
 - C_2 molecule
 - N_2 molecule
 - O_2 molecule
 - O_2^- ion
 - F_2 molecule
- Examples of molecular orbital treatment for heteronuclear diatomic molecules
 - MO molecule
 - CO molecule
- The ionic bond
 - Radius ratio rules
 - Calculation of some limiting radius ratio values
- Dipole moment
 - Application of dipole moment
 - Some special cases
- Close packing
- Ionic compound of the type AX (ZnS , NaCl , CsCl)
 - Structures of zinc sulphide
 - Sodium chloride structure
 - Caesium chloride structure
- Ionic compounds of the type AX_2 (CaF_2 , TiO_2 , SiO_2)
 - Calcium fluoride (fluorite) structure
 - Rutile structure
 - β -cristobalite (silica) structure

- Layer structures (CdI_2 , CdCl_2 , $[\text{NiAs}]$)
 - Cadmium iodide structure
 - Cadmium chloride structure
 - Nickel arsenide structure
- Lattice energy
- Stoichiometric defectes
 - Schottky defects
 - Frenkel defects
- Nonstoichiometric defects
 - Metal excess
 - Metal deficiency
- Born - Haber cycle
- Polarizing power and polarizability - Fajans' Rules
 - Properties of ionic compounds affected by polarization
- Melting point of ionic compounds
- Solubility of ionic compounds
 - Prediction of solubility order in ionic compounds
- Electrical conductivity and colour
- Acidic nature of oxides
- Thermal stability of ionic compounds
- Weak forces
 - Attractive intermolecular forces
 - Repulsive intermolecular forces
 - Lennard- Jones potential
- Interactions between ions and covalent molecules
- The metallic bond
 - Conductivity
 - Lustre
 - Malleability and cohesive force
 - Bond lengths
- Theories of bonding in metals
 - Free electron theory
 - Valence bond theory
 - Molecular orbital or band theory
- Conductors, Insulators and semiconductors

Chemical Bonding

EXERCISE # I

❑ Only one correct answer :

1. Compound having ionic bond as well as covalent bond -

(a) KI_3 (b) NaN_3
(c) LiAlH_4 (d) All

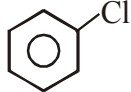
2. Which is planar ion ?

(a) $\text{CH}_3^\ominus$ (b) $\text{CMe}_3^\ominus$
(c) $\text{BF}_4^\ominus$ (d) C(CN)_3^-

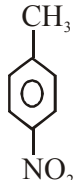
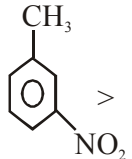
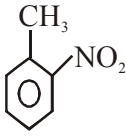
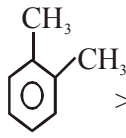
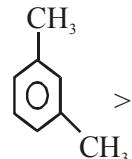
3. Which of the molecule is non polar as well as nonplanar ?

(a) CS_2 (b) BF_3
(c) CCl_4 (d) $\text{B}_3\text{N}_3\text{H}_6$

4. Which of the molecule is polar as well as planar ?

(a) $\text{H}_3\text{C}-\text{C}\equiv\text{CH}$ (b) SF_4
(c)  (d) CHCl_3

5. What is the correct order of dipole moment of the molecule ?

(a)  >  > 
(b)  >  > 

(c) $\text{HF} > \text{HCl} > \text{HBr} > \text{HI}$
(d) All

6. Which of the following molecule has highest dipole moment among methyl halide ?

(a) CH_3F (b) CH_3Cl
(c) CH_3Br (d) CH_3I

7. In which of the following compound, central atom has highest oxidation state -

(a) N_2O_5 (b) IF_7
(c) OsO_4 (d) HMnO_4

8. Number of planes containing maximum number of atoms in SeF_6 is x and number of planes containing at least three chlorine atoms in PCl_5 is y. Then the value of x - y is -

(a) 1 (b) 2
(c) 3 (d) 4

9. In trigonal bipyramidal geometry -

(a) There are 3 arrangement where 3 atoms are at an angle 120° with respect to each other.
(b) There are 6 arrangement where 2 atoms are at an angle 90° with respect to each other
(c) There is 1 arrangement where 2 atoms are at an angle 180° with respect to each other
(d) All

10. The shape of cationic part and anionic part when molten I_2Cl_6 (liquid) undergo ionisation -

(a) trigonal planar and trigonal bipyramidal respectively
(b) linear and octahedral respectively
(c) angular and square planar respectively
(d) trigonal planar and tetrahedral respectively

11. Which of the molecule does not exist ?

(a) ClF_3 (b) ICl_3
(c) BrF_3 (d) CH_3

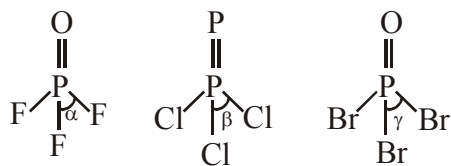
12. Which compound has N-O-N linkage?

(a) N_2O_3 (b) N_2O_4
(c) N_2O_5 (d) All

1.4 CHEMICAL BONDING

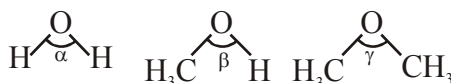
13. The maximum number of 90° angles between bond pair - bond pair electrons observed in which hybridisation -
 (a) $d\ sp^2$ (b) sp^3d
 (c) sp^3d^2 (d) sp^3d^3
14. Which of the following case C - C bond length will be highest?
 (a) $CF_3 - CF_3$ (b) $F_2CH - CHF_2$
 (c) $FCH_2 - CH_2F$ (d) $F_2CH - CF_3$
15. The hybridisation state of central atom of anionic part of product when XeF_2 and PF_5 combined is -
 (a) sp^3 (b) sp^3d
 (c) sp^3d^2 (d) sp^3d^3
16. Which of the following pair has same shape and total number of lone pair and σ bond pair on central atom -
 (a) XeF_6 , XeF_5^- (b) $XeOF_4$, XeF_5^+
 (c) XeO_2F_2 , $XeOF_4$ (d) XeF_4 , XeO_3
17. Which of the following is bent in shape with sp^2 hybridisation in central atom?
 (a) $SnCl_2$ (b) Triplet carbene (1CH_2)
 (c) Both (a) & (b) (d) None
18. Silicate in which discrete tetrahedral unit is present are also called nasosilicate. Identify example of nasosilicate :-
 (a) $BaTi(Si_3O_9)$
 (b) $Sc_2Si_2O_7$
 (c) $Ca_2Mg_5(Si_4O_{11})_2(OH)_2$
 (d) Be_2SiO_4
19. Bond order of SO_2 and O_3 are respectively :-
 (a) 2 & 1.5 (b) 1.5 & 1.5
 (c) 1.5 & 1.33 (d) 2.4 & 1.5
20. Sorosilicates contain units made of two tetrahedral that share an oxygen. The example of sorosilicate -
 (a) $BaTi(Si_3O_9)$
 (b) $Sc_2Si_2O_7$
 (c) $Ca_2Mg_5(Si_4O_{11})_2(OH)_2$
 (d) Be_2SiO_4
21. Two compounds X_1 & X_2 have general molecular formula AB_4 .
 X_1 is nonplanar all A - B bond lengths are identical
 X_2 is nonplanar but all A - B bond lengths are not identical
 X_1 and X_2 can be -
 (a) Square planar structure and tetrahedral structure
 (b) Square planar structure and sea saw structure
 (c) Sea saw structure and tetrahedral structure
 (d) tetrahedral structure and Sea saw structure
22. The minimum number of 90° angles between bond pair - bond pair of electrons is observed in -
 (a) sp^3 hybridisation
 (b) sp^3d hybridisation
 (c) sp^3d^2 hybridisation
 (d) sp^3d^3 hybridisation
23. Incorrect option regarding bond angle -
 (a) $PH_3 > PF_3$
 (b) $H_2O < F_2O < Cl_2O$
 (c) $NH_3 > NF_3$
 (d) $\widehat{HOH}$ in $H_2O < \widehat{HOC}$ in CH_3OH
24. $F - \widehat{M} - F$ bond angle is maximum for -
 (a) BeF_2 (b) MgF_2
 (c) CaF_2 (d) SrF_2
25. Number of sp^3 hybridised carbon and sp hybridised carbon present in $C(CN)_4$ molecule is -
 (a) 0, 5 (b) 0, 5
 (c) 1, 4 (d) 4, 1
26. $\widehat{OXeO}$ bond angle in XeO_4 and XeO_3F_2 are respectively :-
 (a) 120° & $109^\circ 28'$
 (b) $109^\circ 28'$ & 120°
 (c) $> 109^\circ 28'$ & $< 120^\circ$
 (d) 110° & 110°
27. $\widehat{ONX}$ ($X = \text{halogen atom}$) in NOF (A) ; $NOCl$ (B) ; $NOBr$ (C) -
 (a) $A = B = C$ (b) $A > B > C$
 (c) $C > B > A$ (d) $A > C > B$

28. Which of the following option regarding bond angle is correct ?



- (a) $\alpha < \beta < \gamma$ (b) $\alpha = \beta = \gamma$
(c) $\alpha > \beta > \gamma$ (d) $\alpha > \gamma < \beta$

29. Which of the following option regarding bond angle is correct ?



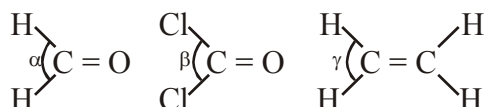
- (a) $\alpha < \beta < \gamma$ (b) $\alpha = \beta = \gamma$
(c) $\alpha > \beta > \gamma$ (d) $\alpha > \gamma < \beta$

30. Which of the following option regarding bond angle is correct ?



- (a) $\alpha < \beta < \gamma < \delta$ (b) $\alpha = \beta = \gamma = \delta$
(c) $\alpha > \beta > \gamma > \delta$ (d) $\alpha > \gamma > \beta > \delta$

31. Which of the following option regarding bond angle is correct ?



- (a) $\alpha < \beta < \gamma$ (b) $\alpha = \beta = \gamma$
(c) $\alpha > \beta > \gamma$ (d) None

32. Which of the following statements is correct ?

- (a) In trigonal bipyramidal electron geometry, if two lone pair is present then shape of the molecular is linear
(b) In pentagonal bipyramidal electron geometry, if two lone pair is present then shape of the molecule is distorted octahedron.
(c) In octahedral electron geometry, if two lone pair is present then shape of the molecule is square planar.
(d) In tetrahedral electron geometry, if two lone pair is present the shape of the molecule is trigonal planar.

33. F-P-F bond angle in PF_3Cl_2 can be approximately:-

- (a) only 120° (b) only 180°
(c) 90° & 180° (d) 90° & 120°

34. Which one is correct order regarding property of molecule ?

- (a) % p character in hybrid orbital $sp > sp^2 > sp^3$
(b) bond angle $\text{NO}_2^- > \text{NO}_3^- > \text{NO}_2^+$
(c) C-H bond length $\text{CH}_3-\text{F} > \text{CH}_2\text{F}_2 > \text{CHF}_3$
(d) $\widehat{\text{HMH}}$ $\text{CH}_4 > \text{H}_2\text{O} > \text{NH}_3$

35. What is the difference in between bond angles in cationic species of PCl_5 and PBr_5 in solid state ?

- (a) 60° (b) $109^\circ 28'$
(c) 0° (d) 90°

36. Select the incorrect statement for molecule $\text{Cl}_3\text{P}=\text{CH}_2$

- (a) P is sp^3 hybridised and Sea saw strcuture
(b) molecule has $p\pi-d\pi$ bond
(c) C atom is sp^2 hybridised
(d) Planar molecule

37. Number of exactly 90° angle in SF_4 is -

- (a) 4 (b) 3
(c) 1 (d) zero

38. The maximum % of s character on central atom present in -

- (a) N_2H_2 (b) NH_3
(c) NH_4^+ (d) NH_2OH

39. Product of which reaction can form dimer ?

- (a) $\text{Cl}_2 + \text{F}_2 \rightarrow \text{Product}$ (excess)
(b) $\text{I}_2 + \text{Cl}_2 \rightarrow \text{Product}$ (equimolar)
(c) $\text{I}_2 + \text{Cl}_2 \rightarrow \text{Product}$ (excess)
(d) $\text{Br}_2 + \text{F}_2 \rightarrow \text{Product}$ (excess)

40. XeF_4 when combines with SbF_5 the product formed is ionic. The hybridisation of cationic part and anionic part is respectively.

- (a) sp^3d^2 , sp^3d^2 (b) sp^3d , sp^3d
(c) sp^3d^2 , sp^3d (d) sp^3d , sp^3d^2

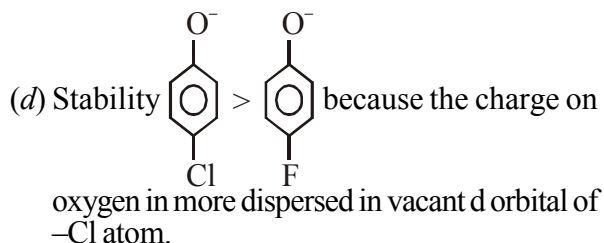
1.6 CHEMICAL BONDING

41. Which of the following geometry is not formed from sp^3d hybridisation of the central atom ?
 (a) Linear (b) Tetrahedral
 (c) T-shaped (d) Sea-saw
42. The orbital not participated in sp^3d^2 hybridisation-
 (a) p_x (b) d_{z^2}
 (c) p_z (d) d_{xz}
43. $SnCl_2$; $SnCl_3^+$; I_3^+ ; I_3^- ; H_3O^+
 Which of the following shape does not describe any of the above species ?
 (a) Angular (b) Linear
 (c) Pyramidal (d) Tetrahedral
44. All possible bond angles in octahedral geometry like PCl_6^- is -
 (a) $109^\circ 28'$ (b) 90° , 180°
 (c) 90° , 120° (d) $109^\circ 28'$, 180°
45. % s character in hybrid orbital -
 (a) $sp^3d < sp^3 < sp^2 < sp$
 (b) $sp < sp^2 < sp^3 < sp^3d$
 (c) $sp^3d < sp < sp^2 < sp^3$
 (d) $sp^2 < sp^3 < sp < sp^3d$
46. Which of the following pair of species have identical shape ?
 (a) I_3^+ ; I_3^- (b) NO_2^+ ; NO_2^-
 (c) TeF_5^- ; XeF_5^+ (d) SF_4 , XeO_4
47. In which case maximum number of atoms are present in same plane -
 (a) $B_3N_3H_6$ (b) $C(CN)_4$
 (c) I_2Cl_6 (d) IF_7
48. Cl – O bond order -
 (a) $ClO_4^- > ClO_3^- > ClO_2^- > ClO^-$
 (b) $ClO^- > ClO_2^- > ClO_3^- > ClO_4^-$
 (c) $ClO_3^- > ClO_3^- > ClO^- > ClO_4^-$
 (d) $ClO_2^- > ClO^- > ClO_3^- > ClO_4^-$
49. Find the overlapping which do not result in σ bond formation, if z axis is considered to be intermolecular axis -
 (a) $s + p_x$ (b) $p_z + d_{z^2}$
 (c) $s + p_z$ (d) $d_{x^2-y^2} + d_{z^2}$
50. Molecule AH_n does not exist because d orbital contraction is not possible. Select the correct statement -
 (a) A belongs to 2nd period
 (b) Hybridisation of A may be sp^2 or sp^3
 (c) A must have more than 4 valence electrons
 (d) Minimum value of n must be 5
51. $BF_3 + O(SiH_3)_3 \rightarrow$ No reaction.
 Which following factor has no significance ?
 (a) lone pair of oxygen are involved in back bonding with Si
 (b) F can easily donate its lone pair to boron in comparison of oxygen
 (c) $O(SiH_3)_2$ is very stable due to back bonding
 (d) $O(SiH_3)_2$ can't act as Lewis base
52. Choose the correct order of bond angles for given compound -
 (a) $NO_2 > O_3$ (b) $O_3 > NO_2$
 (c) $NO_2^- = O_3$ (d) Can't be predicted
53. The molecule having only one lone pair is / are -
 (a) ClF_3 (b) NH_3
 (c) H_2O (d) PCl_3
54. Which of the following species is non polar ?
 (a) ClF_3 (b) ICl_4^-
 (c) I_3^+ (d) OCN^-
55. The molecule which does not consist of any F–X–F bond angle which is less than 90° -
 (X = central atom) -
 (a) ClF_3 (b) IF_7
 (c) PF_5 (d) SF_4
56. Highest boiling point is observed in -
 (a) CH_4 (b) BF_3
 (c) Silica (d) CO_2
57. In a compound AB_3 , A is a central atom and B is surrounding atom. Then which of the following combination of A and B gives minimum $\hat{B}AB$ bond angle :-
 (a) A = N ; B = Cl (b) A = P ; B = Cl
 (c) A = N ; B = H (d) A = P ; B = H

58. % of s character in the orbital occupied by lone pair in H_2O .
(Given : $\cos 104.5^\circ = -0.25$, $\widehat{\text{HOH}} = 104.5^\circ$)
(a) 25% (b) 20%
(c) 80% (d) 30%
59. % of s character in the orbital occupied by lone pair in NH_3 molecule.
(Given : $\cos 107^\circ = -0.292$)
(a) 68% (b) 32%
(c) 38% (d) 25%
60. Which of the following orbital is commonly involved in hybridisation of only in two of following species PCl_5 ; PCl_4^+ ; PCl_6^- ; PCl_3 ?
(a) d_{z^2} (b) $d_{x^2-y^2}$
(c) s (d) p_x, p_y, p_z
61. Which of the following order is correct for increasing p-character in orbitals used for bonding by central atom?
(a) $\text{SiH}_4 > \text{CH}_4$ (b) $\text{PH}_4^+ > \text{PH}_3$
(c) $\text{H}_2\text{S} > \text{H}_2\text{O}$ (d) $\text{NH}_3 > \text{PH}_3$
62. Which of the following molecule does exist as oxy acid of fluorine?
(a) HFO_4 (b) HFO_3
(c) HFO_2 (d) HOF
63. Which of the following fluoride does exist?
(a) OF_6 (b) OF_4
(c) IF_7 (d) ClF_7
64. Which of the following is thermally most stable?
(a) H_2O (b) H_2S
(c) H_2Se (d) H_2Te
65. Shape of the molecule is decided by -
(a) σ bond (b) π bond
(c) both (d) None
66. Which of the following bond has highest bond dissociation energy?
(a) σ bond (b) π bond
(c) H-bond (d) None
67. Which of the following is example of strongest π bond if the molecular axis is x axis?
(a) $2p_x - 4p_x$ (b) $2p_y + 2p_y$
(c) $2p_y + 3d_{xy}$ (d) $2p_z + 4p_z$
68. Which of the following set of overlap can not provide π bond formation?
(a) $3d - 2p$ (b) $2p - 3p$
(c) $2p - 2p$ (d) $3p - 1s$
69. The number of σ bond and π bond ratio in N_2 and P_2 molecule are -
(a) $\frac{1}{2}, \frac{1}{2}$ (b) 1, 1
(c) $\frac{1}{2}, 1$ (d) $1, \frac{1}{2}$
70. Which orbitals of two atoms can produce σ bond?
(a) $d_{z^2} \longrightarrow$ overlap on z axis $\leftarrow d_{z^2}$
(b) $d_{xy} \longrightarrow$ overlap on x axis $\leftarrow d_{xy}$
(c) $d_{x^2-y^2} \longrightarrow$ overlap on y axis $\leftarrow d_{x^2-y^2}$
(d) $d_{xz} \longrightarrow$ overlap on y axis $\leftarrow d_{xz}$
71. Consider the following molecules,
I $(\text{CH}_3)_3\text{N}$; II $\text{CH}_3 - \underline{\text{N}} = \underline{\text{C}} = \text{O}$; III $(\text{CH}_3)_3\text{P}$
In which molecule, geometry around underlined atom is not changed when all CH_3 groups are replaced by SiH_3 group.
(a) II, III
(b) III only
(c) II only
(d) I & II
72. $p\pi - d\pi$ bonding is most effective in -
(a) PF_3 (b) PCl_5
(c) PBr_3 (d) PI_3
73. In which case back bonding takes place -
 $\text{CCl}_3 - \underset{\text{O}}{\underset{\parallel}{\text{C}}} - \text{O}^-$ (A) ; CCl_3 (B) ;
 CCl_2 (C) ; $\text{Cl}_3\text{C} - \text{O}^-$ (D)
(a) only B
(b) B, C & D
(c) B & C
(d) A, B, C, D
74. p-fluorophenol is less acidic than p-chlorophenol because -
(a) -I effect $-\text{F} > -\text{Cl}$
(b) -I effect of $-\text{Cl} > -\text{F}$

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- (c) Conjugate base, of p-fluoro phenol is more stable than conjugate base of p-chloro phenol due to vacant d-orbital present in F.



75. The correct order of viscosity of ethanol, ethylene glycol and glycerol is :-

- (a) ethanol > ethylene glycol > glycerol
(b) ethanol > glycerol > ethylene glycol
(c) ethylene glycol > ethanol > glycerol
(d) glycerol > ethylene glycol > ethanol

76. Which of the following pair of molecular orbital is the pair of degenerate orbital ?

- (a) $\pi 2p_y$, $\pi^* 2p_y$ (b) $\sigma 2p_x$, $\pi 2p_y$
(c) $\pi^* 2p_y$, $\pi^* 2p_{yz}$ (d) $\sigma 1s$, $\sigma 2s$

77. The O – O bond length in O_2 ; $O_2[AsF_4]$; KO_2 is :-

- (a) $O_2[AsF_4] < O_2 < KO_2$
(b) $O_2 < O_2[AsF_4] < KO_2$
(c) $KO_2 < O_2[AsF_4] < O_2$
(d) $O_2[AsF_4] < KO_2 < O_2$

78. In $B(OCH_3)_3$ (methyl borate), due to back bonding $\widehat{COB}$ angle.

- (a) becomes equal to $109^\circ 28'$
(b) less than $109^\circ 28'$
(c) more than $109^\circ 28'$
(d) can't be predicted

79. For succinic acid which of the following option is correct ?

(K_1 & K_2 are 1st dissociation constant and 2nd dissociation constant respectively)

- (a) $\frac{K_1}{K_2} > 1$ (b) $\frac{K_1}{K_2} < 1$
(c) $\frac{K_1}{K_2} = 1$ (d) All

80. If back bonding does not take place in $N(SiH_3)_3$ then what will be the hybridisation of N atom and Si atom -

- (a) sp^2 , sp^3 (b) sp^2 , sp^2
(c) sp^3 , sp^2 (d) sp^3 , sp^3

81. In which of the following, correct shape and hybridisation of central atom is given ?

Ion	Shape	Hybridisation of central atom
-----	-------	-------------------------------

- | | | |
|-----------------------|---------------|-----------|
| (a) I. MnO_4^{-1} | Tetrahedral | sp^3 |
| (b) II. CrO_4^{-2} | Square planar | d^3s |
| (c) III. XeF_4 | Tetrahedral | sp^3d^2 |
| (d) IV. $CoCl_4^{-2}$ | Tetrahedral | sp^3 |

82. The hybridisation of N in $N(SiH_3)_3$; N in $H_3Si-N=C=S$; O in $O(SiH_3)_3$ are respectively :-

- (a) sp^3 , sp^2 , sp^3 (b) sp^2 , sp^2 , sp^2
(c) sp^2 , sp , sp^2 (d) sp^3 , sp , sp^2

83. Back bonding does not take place for which molecule :-

- (a) H_2N-BH_2 (b) SCl_2
(c) H_2N-PH_2 (d) ${}^{\ominus}CF_3$

84. In the dimer of BeH_2 , the kind of overlap present in bridge bond :-

- (a) sp^3-s-sp^3 (b) sp^3-s
(c) sp^2-s-sp^2 (d) $sp^2-sp^2-sp^2$

85. In pentagonal bipyramidal geometry -

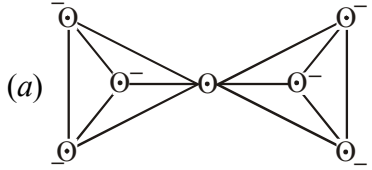
- (a) Groups / atoms which are occupied at equatorial position will be at 72°
(b) Groups / atoms which are occupied at axial position will be at 180°
(c) Equatorial groups & axial groups are at 90° w.r.t. each other
(d) All

86. In which of the following molecule 3C–2e bonding is present ?

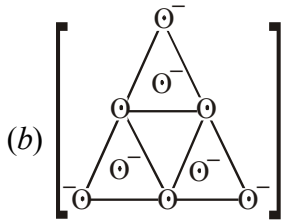
- (a) Monomer of diborane
(b) Monomer of $(BeH_2)_n$
(c) $(AlH_3)_n$
(d) All of these

87. Find the number of molecule having $3c-2e$ bond:-
 (i) O_2Cl_2 (ii) Al_2Cl_6
 (iii) $Al(CH_3)_3$ (iv) $[Be(CH_3)_2]$
 (a) 0 (b) 1
 (c) 2 (d) 3
88. Which of the following is correct regarding I_2Cl_6 ?
 (a) It is planar like Al_2Cl_6
 (b) It is non-planar like Al_2Cl_6
 (c) During ionisation of I_2Cl_6 , cationic part and anionic part formed are bent and square planar respectively.
 (d) It has $2c-2e$ as well as $3c-2e$ bond
89. Find the molecule in which different bond angle is observed but dipole moment of the molecule is zero :-
 (a) PCl_4^+ (b) BF_3
 (c) PCl_3F_2 (d) PCl_2F_3
90. Which of the following molecule is planar as well as polar?
 (a) $B_3N_3H_6$ (b) ClO_2
 (c) NH_3 (d) SOF_4
91. Which of the following molecule contains atleast one lone pair of electron on central atom and has non-zero dipole moment?
 (a) XeF_2 (b) $CClF_3$
 (c) XeF_4 (d) NH_3
92. During the hydrolysis of $SiCl_4$ transition state is formed in 1st step. What is the hybridisation of central atom in that transition state?
 (a) sp^3d (b) sp^3d^2
 (c) sp^3 (d) sp^2
93. Which of the following oxide when reacts with water, mixture of acids is formed?
 (a) CO_2 (b) N_2O_5
 (c) Mn_2O_7 (d) NO_2
94. Which of the following compounds gives only oxyacid as a final product by the hydrolysis in cold condition?
 (a) NF_3 (b) $SbCl_3$
 (c) $H_2S_2O_7$ (d) SO_2Cl_2
95. $X(1\text{ mole}) + H_2O \longrightarrow H_3PO_3(1\text{ mole}) + H_3PO_4(1\text{ mole})$;
 X is :-
 (a) $H_4P_2O_8$ (b) $H_4P_2O_6$
 (c) $H_4P_2O_7$ (d) $H_4P_2O_5$
96. $AsCl_3 + H_2O \longrightarrow X_1 + HCl$
 $SbCl_3 + H_2O \longrightarrow X_2 + HCl$
 $BiCl_3 + H_2O \longrightarrow X_3 + HCl$
 $X_1 ; X_2 ; X_3$ are respectively :-
 (a) $AsOCl$, $SbOCl$, $BiOCl$
 (b) $As(OH)_3$, $Sb(OH)_3$, $Bi(OH)_3$
 (c) $As(OH)_3$, $SbOCl$, $BiOCl$
 (d) $AsOCl$, $Sb(OH)_3$, $Bi(OH)_3$
97. Which of the following compound when undergoes hydrolysis oxyacid of halogen is formed as one of the product?
 (a) SF_4 (b) XeF_4
 (c) NCl_3 (d) SeF_6
98. $SiCl_4$ undergoes hydrolysis by :-
 (a) S_N1 mechanism (b) S_N2 mechanism
 (c) S_Ni mechanism (d) E_2 mechanism
99. Which of the molecule do not undergo hydrolysis at room temperature :-
 (I) NF_3 (II) CCl_4
 (III) SF_4 (IV) SF_6
 (a) II, III (b) I, II, III
 (c) only II (d) I, II, IV
100. PCl_3 and PCl_5 when undergo hydrolysis, products formed are respectively :-
 (a) Tribasic acid (H_3PO_3) & Tribasic acid (H_3PO_4)
 (b) Dibasic acid (H_3PO_3) & Dibasic acid (H_3PO_4)
 (c) Dibasic acid H_3PO_3 & Tribasic acid H_3PO_4
 (d) Tribasic acid H_3PO_3 & Dibasic acid H_3PO_4
101. $TeF_6 + H_2O \longrightarrow \text{Oxyacid } P_1$
 $SeF_6 + H_2O \longrightarrow \text{Oxyacid } P_2$
 P_1 & P_2 are respectively :-
 (a) H_2TeO_4 , H_2SeO_4 (b) H_6TeO_6 , H_6SeO_6
 (c) H_2TeO_4 , H_6SeO_6 (d) H_6TeO_6 , H_2SeO_4

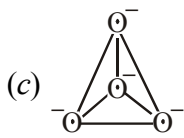
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- 102.** NF_3 at drastic condition undergoes hydrolysis through :-
 (a) $\text{S}_{\text{N}}1$ mechanism
 (b) $\text{S}_{\text{N}}2$ mechanism
 (c) E_2 mechanism
 (d) E_1 mechanism
- 103.** CCl_4 when undergoes reaction with superheated steam the product formed is :-
 (a) CO_2 (b) CHCl_3
 (c) COCl_2 (d) $\text{CCl}_3(\text{OH})$
- 104.** Which xenon fluoride when undergoes hydrolysis, redox reaction does not take place :-
 (a) XeF_2 (b) XeF_4
 (c) XeF_6 (d) All
- 105.** In which of the molecule, hydrolysis can not generate oxyacid of central atom :-
 (a) NCl_3 (b) PCl_3
 (c) SF_4 (d) P_4O_{10}
- 106.** Which of the following compounds are obtained as common product during hydrolysis of XeF_4 and XeF_6 ?
 (a) Xe
 (b) HF
 (c) XeO_3
 (d) both (b) and (c)
- 107.** In methyl radical, % of p character of C–H bond is :-
 (a) greater than % p character of C–F bond in trifluoro methyl radical
 (b) less than % p character of C–F bond in trifluoro methyl radical
 (c) is same as % p character of C–F bond is trifluoro methyl radical
 (d) none
- 108.** NeF_2 , KrCl_2 ; ArH_2 do not exist because :-
 (a) Nonavailability of d orbital for Ne, Kr, Ar
 (b) d orbital contraction is not possible for NeF_2 , KrCl_2 , ArH_2
 (c) Inertness of noble gases to form compounds
 (d) Ne^{2+} , Kr^{2+} , Ar^{2+} have noble gas configuration
- 109.** Which of the following statement is incorrect ?
 (a) In $\dot{\text{C}}\text{ClO}_3$, odd electron resides in one of the sp^3 hybridised orbital
 (b) $\dot{\text{C}}\text{HF}_2$ is pyramidal in nature
 (c) $\dot{\text{C}}\text{H}_3 + \dot{\text{C}}\text{H}_3 \longrightarrow \text{H}_3\text{C} - \text{CH}_3$. ΔG° of this reaction in +ve.
 (d) NO_2 is called mixed anhydride
- 110.** Correct order of boiling point of noble gases :-
 (a) $\text{He} > \text{Ne} > \text{Ar} > \text{Kr} > \text{Xe}$
 (b) $\text{He} > \text{Xe} > \text{Kr} > \text{Ar} > \text{Ne}$
 (c) $\text{Xe} > \text{Kr} > \text{Ar} > \text{Ne} > \text{He}$
 (d) $\text{Kr} > \text{Ar} > \text{He} > \text{Xe} > \text{Ne}$
- 111.** Correct order of boiling point :-
 (a) $\text{HF} > \text{HI} > \text{HBr} > \text{HCl}$
 (b) $\text{H}_2\text{O} > \text{H}_2\text{Te} > \text{H}_2\text{Se} > \text{H}_2\text{S}$
 (c) $\text{PbH}_4 > \text{SnH}_4 > \text{SiH}_4 > \text{CH}_4$
 (d) All
- 112.** Which diagram correctly represents Wallastonite $\text{Ca}_3\text{Si}_3\text{O}_9$.
- 

(a)

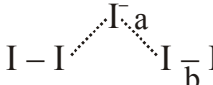


(b)



(c)

(d) None
- 113.** In amphibole which are essentially double chain silicate, which of the following statement is correct :-
 (a) No. of oxygen shared in every tetrahedral unit is 2
 (b) No. of oxygen shared in every tetrahedral unit is 3
 (c) 2 & 3 oxygen atoms shared in tetrahedral unit alternatively
 (d) None

114. In diopside $\text{CaMg}(\text{SiO}_3)_2$, the type of silicate and number of oxygens shared in tetrahedral unit are respectively :-
 (a) Double chain silicate ; 3
 (b) Single chain silicate ; 2
 (c) Single chain silicate ; 3
 (d) Double chain silicate ; 2
115. Which of the compound has odd e^- molecule and central atom is sp^3 hybridised :-
 (a) NO_2 (b) ClO_2
 (c) ClO_3 (d) NO
116. Which can form dimer ?
 (a) BCl_3 (b) $\text{B}(\text{OCH}_3)_3$
 (c) $\text{Al}(\text{CH}_3)_3$ (d) BBr_3
117. In boric acid hybridisation state of B and O are respectively :-
 (a) sp^2 , sp^2
 (b) sp^2 , sp^3
 (c) sp^3 , sp^2
 (d) sp^3 , sp^3
118. Correct statement :-
 (a) 8-hydroxy quinoline can be separated from 4-hydroxy quinoline by fractional distillation
 (b) As branching of isomeric alkane increases boiling point decreases
 (c) Salicylic acid has less boiling point than p hydroxy benzoic acid because former compound can form intramolecular H bonding but latter compound form intermolecular H bonding.
 (d) All
119. Which of the following statement is correct ?
 (a) 12 crown-4-ether dissolves Li^+ most effectively because cavity size is 75 pm
 (b) 15 crown-5-ether dissolves Na^+ most effectively because cavity size is 110 pm
 (c) 18 crown-6-ether dissolves Rb^+ most effectively because cavity size is 160 pm
 (d) All
120. Dipole-dipole interaction exist between which pair :-
 (a) KCl , H_2O (b) , EtOH
 (c) , CCl_4 (d) Acetone and acetonitrile
121.  In I_3^- :-
 (a) $a = b$ (b) $a > b$
 (c) $b > a$ (d) $a = b = 2 \text{ pm}$
122. The internuclear interaction that is independent on the inverse cube of distance between the molecule is :-
 (a) London force
 (b) Ion-Ion interaction
 (c) Ion dipole interaction
 (d) H-bonding
123. Which of the compound does not have H-bonding?
 (a) K_2HPO_4 (b) KHCO_3
 (c) K_2HPO_3 (d) KHF_2
124. Select incorrect statement :-
 (a) Ammonium is more soluble than Na^+
 (b) He/Ne can not form clathrate
 (c) ICl is more reactive than I_2
 (d) H_3O_4^+ , H-bonding is absent
125. Which of the following has highest pK_a in aqueous solution ?
 (a) CH_3NH_2 (b) $(\text{CH}_3)_2\text{NH}$
 (c) $(\text{CH}_3)_3\text{N}$ (d) NH_3
126. Which of the following compound on heating gives N_2 gas ?
 (a) NH_4NO_3 (b) NH_4NO_2
 (c) NaNO_3 (d) Li_3N
127. KCl and KI will be soluble (highly) in :-
 (a) H_2O , H_2O respectively
 (b) H_2O & acetone respectively
 (c) Acetone, H_2O respectively
 (d) Acetone, Acetone respectively

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128. Highest polarising power is observed in :-

- (a) Na^+ (b) K^+
(c) Cu^+ (d) Zn^{2+}

129. Highest polarisability is observed in :-

- (a) F^- (b) Cl^-
(c) Br^- (d) I^-

130. Most covalent halide :-

- (a) CaCl_2 (b) PbCl_2
(c) AlCl_3 (d) FeCl_2

131. Which of the following is thermally most stable ?

- (a) Na_2CO_3 (b) BeCO_3
(c) MgCO_3 (d) CaCO_3

132. Which of the following compound has highest lattice energy ?

- (a) NaCl (b) MgO
(c) ScN (d) TiC

133. Which compound has highest covalent character?

- (a) NaCl (b) MgCl_2
(c) AlCl_3 (d) SiCl_4

134. Which of the following order is correct regarding solubility of group IIA halide ?

- (a) $\text{BeX}_2 < \text{MgX}_2 < \text{CaX}_2 < \text{SrX}_2 < \text{BaX}_2$
(b) $\text{LiX} > \text{NaX} > \text{KX} > \text{RbX} < \text{CsX}$
(c) $\text{BeX}_2 > \text{MgX}_2 > \text{CaX}_2 > \text{SrX}_2 > \text{BaX}_2$
(d) $\text{LiX} < \text{NaX} < \text{KX} < \text{RbX} < \text{CsX}$

135. Which compound is least thermally stable ?

- (a) LiNO_3 (b) NaNO_3
(c) KNO_3 (d) RbNO_3

136. Out of BeCl_2 , CaCl_2 , CsCl , KCl ; BeCl_2 has :-

- (a) highest ionic character
(b) highest solubility in H_2O
(c) highest lattice energy
(d) lowest charge : size ratio in cation

137. % s-character of the hybrid orbitals used for C–H bond formation in CH_2F_2 molecule :-

- $(\widehat{\text{HCH}})_{\text{CH}_2\text{F}_2} = 111.9^\circ$, $\cos 111.9^\circ = -0.3729$
(a) 27.1% (b) 37.3%
(c) 29% (d) 30%

138. In PTFE (polytetrafluoro ethene) the hybridisation of carbon atom is :-

- (a) sp (b) sp^2
(c) sp^3 (d) sp^3d

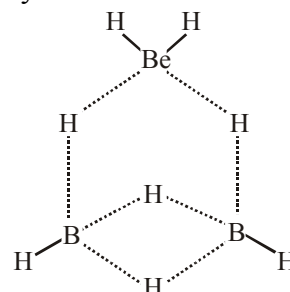
139. In vinyl acetylene the hybridisation of carbon atom :-

- (a) only sp^2 (b) only sp^3
(c) both sp^2 , sp^3 (d) both sp, sp^2

140. The correct order of B–F bond lengths :-

- (a) $\text{BF}_3 < \text{BF}_2-\text{NH}_2 < \text{BF}_2-\text{OH} < \text{BF}_4^-$
(b) $\text{BF}_2\text{NH}_2 < \text{BF}_2-\text{OH} < \text{BF}_3 < \text{BF}_4^-$
(c) $\text{BF}_3 < \text{BF}_4^- < \text{BF}_2-\text{OH} < \text{BF}_2-\text{NH}_2$
(d) $\text{BF}_3 < \text{BF}_2-\text{OH} < \text{BF}_2-\text{NH}_2 < \text{BF}_4^-$

141. Total 2c–2e and 3c–2e bonds in $\text{Be}(\text{BH}_4)_2$ are respectively :-



- (a) 6,4 (b) 4,6
(c) 4,4 (d) 4,8

142. What will be the oxidation state of P in $\text{H}_4\text{P}_2\text{O}_8$ & HPO_3 ?

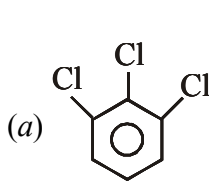
- (a) +5, +5 (b) +5, +3
(c) +5, 0 (d) +5, +7

143. The mineral $\text{Na}_2\text{Fe}_3^{II}\text{Fe}_2^{III}[\text{Si}_8\text{O}_{22}](\text{OH})_2$ (chrocidolite) is a :-

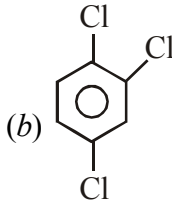
- (a) pyroxene chain silicate
(b) Sheet silicate
(c) amphiboles chain silicate
(d) 3D-silicate

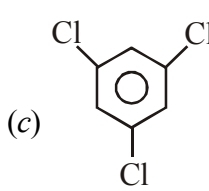
144. The silicate anion in the mineral kinoite is a chain of three SiO_4 tetrahedra that share corners with adjacent tetrahedra. The mineral also contain Ca^{2+} ions, Cu^{2+} ions, and water molecules in a 1 : 1 : 1 ratio. Mineral is represented as :-

- (a) $\text{CaCuSi}_3\text{O}_{10} \cdot \text{H}_2\text{O}$
(b) $\text{CaCuSi}_3\text{O}_{10} \cdot 2\text{H}_2\text{O}$
(c) $\text{Ca}_2\text{Cu}_2\text{Si}_3\text{O}_{10} \cdot 2\text{H}_2\text{O}$
(d) None of these

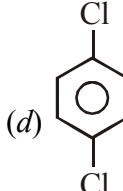
145. An atom A has atomic number less than 21. What will be the hybridisation of AlCl_3 , if AlCl_3 has zero dipole moment ?
 (a) pure p (b) sp
 (c) sp^2 (d) sp^3
146. Which of the following halide of silver is water soluble ?
 (a) AgF (b) AgCl
 (c) AgBr (d) AgI
147. Acetylene gets dissolved in acetone, it is because of :-
 (a) intramolecular hydrogen bonding
 (b) intermolecular hydrogen bonding
 (c) London forces
 (d) ionic interaction
148. $\text{Ca}_3(\text{PO}_4)_2 + \text{SiO}_2 \rightarrow \text{Calcium silicate} + \text{P}_1$
 $\text{P}_1 + \text{C} \rightarrow \text{P}_2 + \text{CO}$
 P_1, P_2 are respectively :-
 (a) $\text{P}_4\text{O}_6, \text{P}_4$ (b) $\text{P}_4\text{O}_{10}, \text{P}_4$
 (c) $\text{P}_4\text{O}_6, \text{PH}_3$ (d) $\text{P}_4\text{O}_{10}, \text{PH}_3$
149. In methylene sulphur tetrafluoride ($\text{CH}_2 = \text{SF}_4$); the hydrogen atoms are in the same plane with the :-
 (a) equatorial fluorine atoms
 (b) axial fluorine atoms
 (c) axial as well as equatorial fluorine atoms
 (d) None
150. In which of the following compounds phosphorous atoms are at the corner of tetrahedral unit :-
 (a) P_4 (b) $\text{P}_2\text{O}_7^{2-}$
 (c) PO_4^{3-} (d) KH_2PO_3
151. Which of the following equilibria would have the least value of K_p at a common temperature?
 (a) $\text{MgCO}_3 \rightleftharpoons \text{MgO} + \text{CO}_2$
 (b) $\text{CaCO}_3 \rightleftharpoons \text{CaO} + \text{CO}_2$
 (c) $\text{SrCO}_3 \rightleftharpoons \text{SrO} + \text{CO}_2$
 (d) $\text{BaCO}_3 \rightleftharpoons \text{BaO} + \text{CO}_2$
152. The solubility of anhydrous AlCl_3 and hydrated AlCl_3 in diethyl ether are S_1 and S_2 respectively. Then :-
 (a) $S_1 = S_2$ (b) $S_1 > S_2$
 (c) $S_1 < S_2$ (d) none
153. The critical temperature of water is higher than that of O_2 because the H_2O molecule has :-
 (a) fewer electrons than O_2
 (b) two covalent bonds
 (c) V-shape
 (d) more dipole moment
154. Which of the following boiling point order is correct ?
 (a) $\text{He} > \text{T}_2 > \text{D}_2$
 (b) $\text{He} < \text{T}_2 < \text{D}_2$
 (c) $\text{T}_2 > \text{He} > \text{D}_2$
 (d) $\text{He} < \text{D}_2 < \text{T}_2$
155. Two ice cubes are pressed over each other and unite to form one cube. Which force is responsible for holding them together :-
 (a) Vander wall's forces
 (b) Covalent attraction
 (c) H-bond formation
 (d) Dipole-dipole attraction
156. Which has maximum dipole moment ?
- 

(a)



(b)
- 

(c)



(d)
157. $\text{NMe}_3 + \text{H}_2\text{O}_2 \rightarrow \text{product}$
 What is the formal charge of N in product ?
 (a) +1 (b) -1
 (c) +2 (d) 0

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158. When container containing calcium carbide and calcium phosphide are pierced and thrown in the sea, then the gases evolved burn and serve as a signal called as Holme's signal. The gases evolved are :-

- (a) C_2H_4 , P_2H_4 (b) CH_4 , P_2H_4
 (c) C_2H_4 , PH_3 (d) C_2H_2 , PH_3

159. Elements of which groups are polymorphic (exist in more than one allotropic form) :-

- (a) group 13 (b) group 14
 (c) group 15 (d) group 16

160. Boric acid and fluoro boric acid are respectively :-

- (a) Monobasic acid and monobasic acid
 (b) Monobasic acid and tribasic acid
 (c) Tribasic acid and tribasic acid
 (d) Tribasic acid and monobasic acid

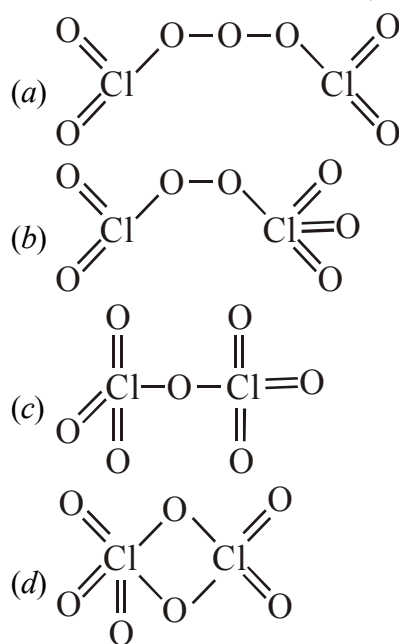
161. In CO_2 , SO_2 , SiO_2 central atom is covalently bonded with m_1 , m_2 , m_3 number of oxygen atoms respectively then :-

- (a) $m_1 = 2$, $m_2 = 2$, $m_3 = 2$
 (b) $m_1 = 4$, $m_2 = 4$, $m_3 = 4$
 (c) $m_1 = 2$, $m_2 = 4$, $m_3 = 4$
 (d) $m_1 = 2$, $m_2 = 2$, $m_3 = 4$

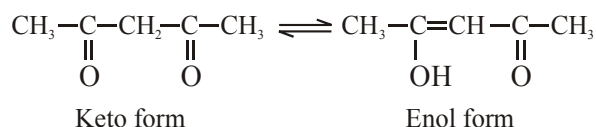
162. Which of the following reactions is spontaneous ?

- (a) $Pb^{2+} \rightarrow Pb^{4+} + 2e^-$ (b) $Sn^{2+} \rightarrow Sn^{4+} + 2e^-$
 (c) $Tl^{1+} \rightarrow Tl^{3+} + e^-$ (d) $Bi^{3+} \rightarrow Bi^{5+} + 2e^-$

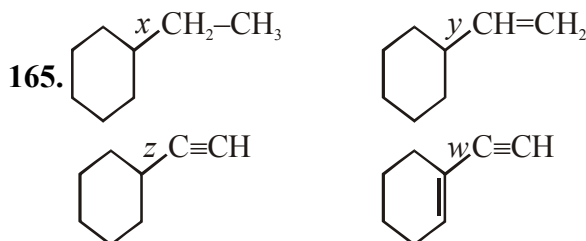
163. The correct structure of Cl_2O_7 :-



164. Which of the following option is correct :-



- (a) enol form is less stable than keto form due to hydrogen bonding
 (b) keto and enol form both are of same stability
 (c) enol form is more stable than keto form due to hydrogen bonding
 (d) none



x, y, z, w are bond length between C-C bond. The correct order is :-

- (a) $x > y > z > w$ (b) $x > z > w > y$
 (c) $x > y > w > z$ (d) none

166. In energy level diagram of O_2^{2+} , 13th electron is present in :-

- (a) σ_{2p_x} orbital
 (b) π_{2p_y} or π_{2p_z} orbital
 (c) $\pi_{2p_y}^*$ or $\pi_{2p_z}^*$ orbital
 (d) $\sigma_{2p_x}^*$ orbital

167. There is largest energy difference between M.O. of dioxygen :-

- (a) σ_{2p_x} , π_{2p_y} (b) π_{2p_y} , $\pi_{2p_y}^*$
 (c) σ_{2p_x} , σ_{2s}^* (d) $\sigma_{2p_x}^*$, σ_{2s}

168. Which of the energy order is same for both M.O. diagram of N_2 and O_2 ?

- (a) $\sigma_{2s}^* < \sigma_{2p_x}$ (b) $\sigma_{2s}^* < \sigma_{2s}$
 (c) $\sigma_{2p_x} < \pi_{2p_y}$ (d) $\pi_{2p_y}^* < \pi_{2p_y}$

169. Assuming 2s-2p mixing is not operative the paramagnetic species among the following :-

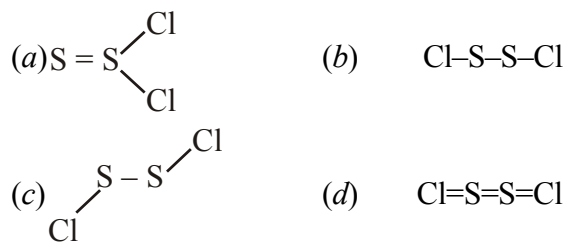
- (a) Be_2 (b) B_2
 (c) C_2 (d) N_2

170. If the filling of electrons in M.O. of O_2 does not obey Hund's rule, then which of the following property in O_2 gets changed :-
 (a) bond order
 (b) magnetic behaviour
 (c) number of electrons in bonding M.O.
 (d) number of electrons in antibonding M.O.
171. Which allotrope of phosphorous is thermodynamically most stable ?
 (a) white phosphorous
 (b) black phosphorous
 (c) purple phosphorous
 (d) red phosphorous
172. In which case, both of the following species are paramagnetic ?
 (a) O_2 , Na_2O_2 (b) BaO_2 , KO_3
 (c) O_2 , KO_3 (d) N_2 , KN_3
173. Which of the following species is expected to be coloured ?
 (a) KO_3 (b) Li_2O
 (c) Na_2O (d) K_2O_2
174. If inter nuclear axis is z axis then HOMO orbital(s) of CO is :-
 (a) $\sigma^*_{2p_z}$ (b) N.B.M.O
 (c) $\pi^*_{2p_y} = \pi^*_{2p_z}$ (d) None of these
175. Find the number of nodal plane in δ^* (ABMO) :-
 (a) 1 (b) 2
 (c) 3 (d) 4
176. If internuclear axis is assumed to be z axis then which of the following pair(s) of orbitals are HOMO of O_2 molecule :-
 (a) π_{2p_x} , π_{2p_y} (b) $\pi^*_{2p_y}$, $\pi^*_{2p_z}$
 (c) $\pi^*_{2p_x}$, $\pi^*_{2p_y}$ (d) $\sigma^*_{2p_z}$, σ_{2p_z}
177. Which of the following molecule or molecular ion has highest number of total electron in antibonding molecular orbital ?
 (a) O_2 (b) N_2
 (c) N_2^+ (d) O_2^{2-}
178. If we consider no mixing of 2s and 2p orbitals, then the bond order and magnetic moment of the diatomic molecule C_2 is :-
 (a) 3 & diamagnetic (b) 2 & diamagnetic
 (c) 2 & paramagnetic (d) 2.5 & diamagnetic
179. If C-C bond in C_2H_6 undergoes heterolytic fission, the hybridisation of two resulting carbon atoms is / are :-
 (a) sp^2 both (b) sp^3 both
 (c) sp^2 , sp^3 (d) sp^3 , sp^3
180. Which of the following halide when undergo hydrolysis the intermediate form during the reaction have sp^3 , sp^3d , sp^3d^2 hybridisation ?
 (a) SF_6 (b) SF_4
 (c) PCl_5 (d) PCl_3
181. Which halogen does not react with water ?
 (a) F_2 (b) Cl_2
 (c) Br_2 (d) I_2
182. Which halogen when reacts with water, it oxidises water to form O_2 gas at a fastest rate ?
 (a) F_2 (b) Cl_2
 (c) Br_2 (d) I_2
183. $BCl_3 + H_2O \rightarrow P_1 + P_2$
 Correct statement regarding P_1 and P_2 :-
 (a) P_1 , P_2 both are proton donor acid
 (b) P_1 , P_2 no one is proton donor acid
 (c) In between P_1 , P_2 one is proton donor acid & another is not a proton donor acid
 (d) P_1 , P_2 both are base
184. $BF_3 + H_2O \rightarrow B(OH)_3 + X_1$ $X_1 + BF_3 \rightarrow X_3$
 $SiF_4 + H_2O \rightarrow Si(OH)_4 + X_2$ $X_2 + SiF_4 \rightarrow X_4$
 X_3 & X_4 are respectively :-
 (a) HF, HF (b) $HSiF_4$, $HSiF_5$
 (c) H_2BF_5 , H_2SiF_6 (d) $HSiF_4$, H_2SiF_6
185. In the cyclo- S_8 molecule of rhombic sulphur, all the S-S bond length and all the S-S-S bond angles are respectively
 (a) 204 pm and 105° (b) 102 pm and 120°
 (c) 204 pm and 180° (d) 102 pm and 60°

1.16 CHEMICAL BONDING

- 186.** Which is correct physical state of boron halide ?
 (a) BF_3 is gas ; BCl_3 , BBr_3 both are liquids ; BI_3 is solid
 (b) BF_3 is solid ; BCl_3 , BBr_3 both are liquids ; BI_3 is gas
 (c) BF_3 , BCl_3 both are gases ; BBr_3 is liquid ; BI_3 is solid
 (d) None
- 187.** Which of the following is correct statement ?
 (a) HF forms more effective hydrogen bonding than H_2O but ΔH_{vap} of HF is less than ΔH_{vap} of water
 (b) HF forms less effective hydrogen bonding than H_2O but ΔH_{vap} of HF is more than ΔH_{vap} of water
 (c) HF forms less effective hydrogen bonding than H_2O and ΔH_{vap} of HF is less than ΔH_{vap} of water
 (d) HF forms more effective hydrogen bonding than H_2O and ΔH_{vap} of HF is more than $(\Delta H)_{\text{vap}}$ of water
- 188.** SnCl_2 , HgCl_2 can not coexist together because :-
 (a) Sn^{2+} oxidises Hg^{2+}
 (b) Sn^{2+} reduces Hg^{2+} to Hg^{+1} & finally Hg^{+1} converted into Hg metal
 (c) Sn^{2+} oxidises Hg to Hg^{2+}
 (d) Sn^{2+} reduces Hg^{4+} to Hg^{2+}
- 189.** In aqueous solution, the hydronium ion is further hydrated to give species like :-
 (a) H_5O_2^+ (b) H_7O_3^+
 (c) H_9O_4^+ (d) All
- 190.** In aqueous solution, the hydroxyl ion (OH^-) is hydrated to give species like :-
 (a) H_3O_2^- (b) H_5O_3^-
 (c) H_7O_4^- (d) All
- 191.** The number of P—O—P linkage in cyclotetrametaphosphoric acid ($\text{H}_4\text{P}_4\text{O}_{12}$) which is formed during stepwise hydrolysis of P_4O_{10} ?
 (a) zero (b) two
 (c) three (d) Four
- 192.** Which type of π bonds present in solid Cl_2O_6 ?
 (a) $2p\pi-3d\pi$
 (b) $2p\pi-3p\pi$
 (c) $3p\pi-3d\pi$
 (d) both (a) & (b)
- 193.** Which of the following molecule is diamagnetic and has last electron in sigma (σ) B.M.O. :-
 (a) O_2 (b) N_2
 (c) B_2 (d) C_2
- 194.** Which of the following has dipole-dipole interaction between the species possessing permanent dipole ?
 (a) liquid- NH_3 (b) liquid-He
 (c) solid- I_2 (d) liquid- Br_2
- 195.** Select the most ionic and most covalent compound respectively from the following :-
 CrO_5 , Mn_2O_7 , PbO , P_4O_{10} , SnO_2
 (a) CrO_5 , Mn_2O_7 (b) PbO , Mn_2O_7
 (c) CrO_5 , SnO_2 (d) CrO_5 , P_4O_{10}
- 196.** Which of the following statements is true for IO_2F_2^- according to VSEPR theory ?
 (a) the lone pair and two I—O double bonds occupy the equatorial positions of trigonal bipyramid
 (b) it has sp^3d hybridisation and T-shaped
 (c) its structure is analogous of SF_4
 (d) (a) and (c) both
- 197.** What is the hybridisation of boron atoms in compound $\text{Mg}[\text{B}_2\text{O}(\text{OH})_6]$?
 (a) both sp^3
 (b) one sp^2 and other sp^3
 (c) one sp^3 and other sp^3d
 (d) both sp^2
- 198.** A diatomic molecule has a dipole moment of 1.2 D. If its bond length is equal to 10^{-10} m. Then the fractions of an electronic charge on each atom will be :-
 (a) 42% (b) 52%
 (c) 37% (d) 25%

199. Which of the following is correct structure of S_2Cl_2 ?



200. Which of the following atomic orbital does not participate in trigonal bipyramidal electron geometry i.e., sp^3d hybridisation :-

- (a) d_{z^2} (b) $d_{x^2-y^2}$
(c) p_y (d) p_x

EXERCISE # II

□ One or More Than One Correct Answer :

1. In which of the following compounds, central atom has one lone pair of e^- ?

- (a) BrF_5 (b) $XeOF_4$
(c) SF_2Cl_2 (d) $SOCl_2$

2. For which of the following compounds, underlined atom has +1 oxidation state?

- (a) $\underline{Tl}I_3$ (b) $\begin{array}{c} H_3C \\ \diagdown \\ S \\ \diagup \\ H_3C \end{array} = O$
(c) $\underline{Cl}-F$ (d) $\underline{K}I_3$

3. In which of the following compounds, total six lone pair is present?

- (a) Melamine $[C_3N_3(NH_2)_3]$
(b) $HClO_2$
(c) Cl_2O_7
(d) SO_3

4. Which of the following are linear in shape?

- (a) N_3^- (b) I_3^+
(c) I_3^- (d) XeF_2

5. How many ions are non planar ions?

- (a) XeF_5^- (b) SO_3^{2-}
(c) SO_4^{2-} (d) ClO_4^-

6. Which of the following molecule have sp^3d^2 hybridised atom with square planar shape?

- (a) XeF_4 (b) ICl_4^-
(c) BrF_4^- (d) I_2Cl_6

7. Which of the following molecule have sp^3d^2 hybridised atom with square pyramidal shape?

- (a) BrF_5 (b) XeF_5^+
(c) $XeOF_4$ (d) TeF_5^-

8. Choose the correct option :-

- (a) In d^3s hybridisation, the d-orbital involved is $d_{x^2-y^2}$, d_{z^2} , d_{xy}
(b) In sp^3d hybridisation, the d-orbital involved is d_{z^2}
(c) In sp^3d^2 hybridisation, the d-orbital involved is $d_{x^2-y^2}$, d_{z^2} respectively
(d) In sp^3d^2 hybridisation, the d-orbital involved is d_{xy} , d_{zx} respectively

9. Which of the following are polar and planar?

- (a) CO
(b) SF_6
(c) CH_2Cl_2
(d) $\begin{array}{c} HO_2C \\ \diagdown \\ CH=CH \\ \diagup \\ CO_2H \end{array}$ (fumaric acid)

10. Which of the following are nonpolar and nonplanar?

- (a) PCl_5
(b) SF_6
(c) $H_3C-C \equiv C-CH_3$
(d) XeF_4

11. Which of the following molecule do exist?

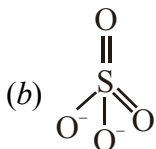
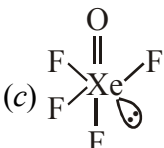
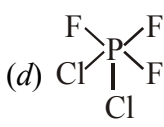
- (a) NeF_2 (b) NF_4^+
(c) IF_3^{2-} (d) $GeCl_6^{2-}$

12. Choose the correct statements -

- (a) the correct structure of NOF_3 is $\begin{array}{c} O^- \\ | \\ N^+ \\ / \quad \backslash \\ F \quad F \end{array}$
(b) maximum number of atoms lying in one plane of $C(CN)_4$ molecule is 5
(c) among halogens Cl_2 only exist as solid state
(d) only oxyacid of fluorine which can be isolated is HOF where fluorine has +1 oxidation state

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13. In which of the following compound, d_{z^2} orbital is involved in hybridisation of their central atom-
- (a) BrF_3 (b) H_3O^+
(c) CH_3^+ (d) SF_6
14. Which of the following are pyramidal in shape ?
- (a) Amide ion (NH_2^-)
(b) XeO_3
(c) ClO_3^- (Chlorate ion)
(d) Sulphite ion
15. Which of the following are angular in shape with sp^3 hybridisation on central atom ?
- (a) I_3^+
(b) ClO_2^- (chlorite ion)
(c) HOCl (Hypochlorous acid)
(d) Water
16. Which process are non spontaneous ?
- (a) $\text{NO}_2 + \text{NO}_2 \rightarrow \begin{array}{c} \text{O}=\text{N}^+-\text{N}^+=\text{O} \\ | \quad | \\ \text{O}^- \quad \text{O}^- \end{array}$
- (b) $\text{ClO}_2 + \text{ClO}_2 \rightarrow \text{O}=\text{Cl}-\text{O}-\text{O}-\text{Cl}=\text{O}$
- (c) $\text{CMe}_3-\text{C}_6\text{H}_2(\text{CMe}_3)_3-\text{O}^\bullet + \bullet\text{O}-\text{C}_6\text{H}_2(\text{CMe}_3)_3-\text{CMe}_3 \rightarrow$
 $\text{Me}_3\text{C}-\text{C}_6\text{H}_2(\text{CMe}_3)_3-\text{O}-\text{O}-\text{C}_6\text{H}_2(\text{CMe}_3)_3-\text{CMe}_3$
- (d) $\bullet\text{CH}_3 + \bullet\text{CH}_3 \rightarrow \text{CH}_3-\text{CH}_3$
17. Colourless gas (X) + $\text{O}_2 \rightarrow$ Brown colour gas NO_2 (Y)
Correct statements are :
- (a) X, O_2 , Y all are example of odd electron molecule
(b) Y maintains planarity on dimerisation
(c) X, Y are examples of odd electron molecule but O_2 is not
(d) Y during dimerisation forms compound which is diamagnetic in nature
18. For which of the following compounds, d_{z^2} orbital is involved in hybridisation ?
- (a) BrF_3 (b) SF_6
(c) H_3O^+ (d) SF_4
19. For the molecule, OF_2 (I) ; HOF (II) ; H_2O (III), which statements are correct -
- (a) $d_{\text{O-F}}$ bond length I > II
(b) $d_{\text{O-F}}$ bond length II > I
(c) $d_{\text{O-H}}$ bond length II > III
(d) $d_{\text{O-H}}$ bond length III > II
20. Which of the following statements is correct ?
- (a) $\bullet\text{CH}_3 \Rightarrow$ Hybridisation of carbon is sp^2
(b) $\bullet\text{CF}_3 \Rightarrow$ Hybridisation of carbon is sp^2
(c) $\text{C}_2\text{F}_4 \Rightarrow$ Hybridisation of carbon is sp^3
(d) $\text{CH}_2^+ \Rightarrow$ Hybridisation of carbon having positive charge is sp^2
21. Incorrect order -
- (a) C – C bond length $\text{FCH}_2\text{CH}_2\text{F} > \text{CHF}_2\text{CHF}_2 > \text{F}_3\text{C}-\text{CF}_3$
(b) Oxidation state of P $\text{H}_3\text{PO}_4 > \text{PCl}_3 > \text{NaH}_2\text{PO}_2 > \text{P}_4$
(c) Number of 90° repulsion between bond pair $\text{XeF}_4 > \text{PF}_5 > \text{PF}_6^-$
(d) Number of lone pair present on central atom $\text{XeOF}_2 > \text{Cl}^+\text{OF}_4^- > \text{ClOF}_2 > \text{BH}_4^-$
22. Which molecular geometry are most likely to result from an octahedral electron geometry ?
- (a) square planar (b) square pyramidal
(c) linear (d) v-shaped
23. Bond energy order -
- (a) $\text{sp}^3\text{C-H} > \text{sp}^2\text{C-H} > \text{spC-H}$
(b) $\text{spC-H} > \text{sp}^2\text{C-H} > \text{sp}^3\text{C-H}$
(c) $\text{F-F} > \text{Cl-Cl} > \text{Br-Br} > \text{I-I}$
(d) $\text{C-F} > \text{C-Cl} > \text{C-Br}$
24. Which of the following statements are correct ?
- (a) ClF_2^- is linear but ClF_2^+ is bent
(b) In pentagonal bipyramidal geometry, axial orbital length is greater than equatorial bond length

- (c) In trigonal bipyramidal geometry, axial orbital length is greater than equatorial bond length
 (d) Dimer of ICl_3 i.e., I_2Cl_6 has planar structure with sp^3d^2 hybridisation
- 25.** Which of the following molecule itself exist but has no definite hybridisation ?
 (a) XeF_3^- (b) PH_3
 (c) H_2S (d) None
- 26.** Which of the following pair of species have different hybridisation but same shape ?
 (a) ICl_2^- and BeCl_2 (gas phase)
 (b) PCl_5 (liq.) and XeOF_4
 (c) ICl_4^- , $\text{Pt}(\text{NH}_3)_2\text{Cl}_2$
 (d) XeF_2 and $\text{HC} \equiv \text{CH}$
- 27.** Which of the following options are correct regarding XeO_3F_2 and XeOF_4 molecule ?
 (a) $d_{\text{Xe-O}}(\text{XeO}_3\text{F}_2) < d_{\text{Xe-O}}(\text{XeOF}_4)$
 (b) $d_{\text{Xe-F}}(\text{XeO}_3\text{F}_2) < d_{\text{Xe-F}}(\text{XeOF}_4)$
 (c) shapes are trigonal bipyramidal and square pyramidal respectively
 (d) the nature of π bond is $5\text{d}\pi - 2\text{p}\pi$ type
- 28.** Considering internuclear axis is x axis, then which of the following overlapping are possible -
 (a) $s + p_x \rightarrow \sigma$
 (b) $p_x + p_x \Rightarrow \pi$
 (c) $p_x + d_{xy} \Rightarrow \pi$
 (d) $d_{xy} + p_y \Rightarrow \pi$
- 29.** Which of the following molecule has dative bond ?
 (a) NH_4^+ (b) NOF_3
 (c) PCl_3 (d) SF_6
- 30.** Which of the following process involves absorption of energy ?
 (a) $\text{S}(\text{g}) + \text{e}^- \rightarrow \text{S}^-(\text{g})$
 (b) $\text{Xe}(\text{g}) + \text{e}^- \rightarrow \text{Xe}^-(\text{g})$
 (c) $\text{O}^-(\text{g}) + \text{e}^- \rightarrow \text{O}^{2-}(\text{g})$
 (d) $\text{Cl}^-(\text{g}) \rightarrow \text{Cl}(\text{g}) + \text{e}^-$
- 31.** Which of the following representation of molecules are not correct -
 (a) $\text{N} = \overline{\text{N}} = \text{N}$ (b) 
 (c)  (d) 
- 32.** Which of the following molecule have only $\text{p}\pi - \text{d}\pi$ type of π bonds ?
 (a) SO_3 (b) XeO_4
 (c) XeO_2F_2 (d) SO_2
- 33.** The correct order of single bond energy ?
 (a) $\text{P-P} > \text{N-N}$ (b) $\text{S-S} > \text{O-O}$
 (c) $\text{C-C} > \text{Si-Si}$ (d) $\text{C-H} > \text{C-D}$
- 34.** Which of the following compound are planar ?
 (a) ClF_3 (b) ICl_4^-
 (c) I_3^+ (d) OCN^-
- 35.** Choose the correct statements regarding SF_4 molecule -
 (a) Hybridisation of sulphur occurs in 1st excited state.
 (b) Total number of l.p - b.p. repulsion at 90° is 2
 (c) There are six possible F-S-F bond angles which are less than 180°
 (d) It is hypervalent molecule
- 36.** In which of the following compounds, sigma coordinate bond is absent -
 (a) NO_3^- (b) CO
 (c) NH_4^+ (d) HNC
- 37.** Which of the following molecule is aromatic ?
 (a) $\text{B}_3\text{N}_3\text{H}_6$ (b) $\text{B}_3\text{H}_3\text{O}_3$
 (c) $\text{C}_3\text{N}_3\text{Cl}_3$ (d) C_{12}O_9
- 38.** Correct statements about C_{12}O_9 is :-
 (a) It is example of cyclic trianhydride
 (b) It is formed when mellitic acid undergoes heating
 (c) It is example of cyclic ether
 (d) It is non aromatic

1.20 CHEMICAL BONDING

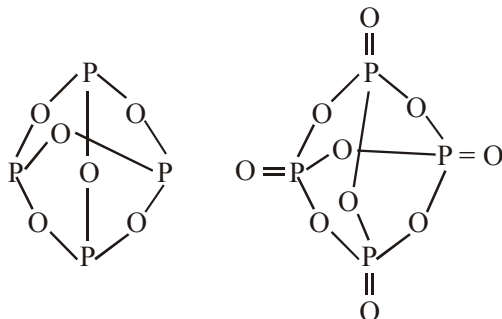
39. Correct order regarding dipole moment -

- (a) $\text{HF} < \text{HCl} < \text{HBr} < \text{HI}$
 (b) $\text{CH}_3\text{F} < \text{CH}_3\text{Cl} < \text{CH}_3\text{Br} < \text{CH}_3\text{I}$
 (c) $\text{CH}_3\text{Cl} > \text{CH}_3\text{F} > \text{CH}_3\text{Br} > \text{CH}_3\text{I}$
 (d) $\text{CH}_3\text{Cl} > \text{CH}_2\text{Cl}_2 > \text{CHCl}_3 > \text{CCl}_4$

40. What is the correct order of dipole moment ?

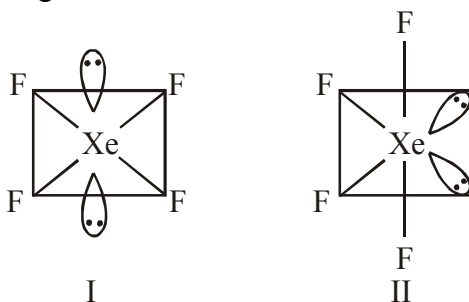
- (a) $\text{NH}_3 > \text{NF}_3$ (b) $\text{H}_2\text{O} > \text{F}_2\text{O}$
 (c) $\text{XeO}_4 > \text{XeO}_3$ (d) $\text{XeO}_3\text{F}_2 > \text{XeO}_2\text{F}_2$

41. Correct statement regarding P_4O_6 & P_4O_{10} -



- (a) Both have same number of P-O-P linkage
 (b) Both have same number of P=O linkage
 (c) Both have same valency of phosphorus
 (d) Both have same number of P-P linkage

45. In XeF_4 , 2 possible structures are given as follows -



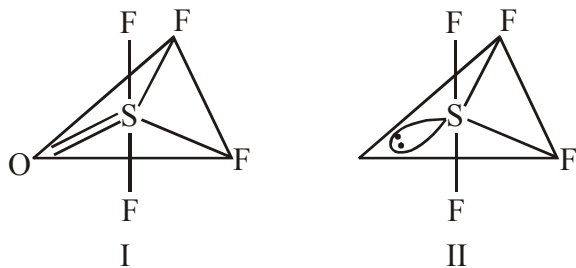
I.p.l.p repulsion at 90°

I.p.-b.p repulsion at 90°

b.p-b.p repulsion at 90°

(a) Str.-I	0	7	5
(b) Str.-I	0	8	4
(c) Str.-II	0	7	4
(d) Str.-II	1	6	4

46.



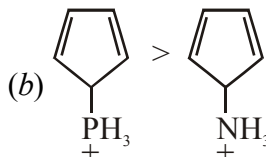
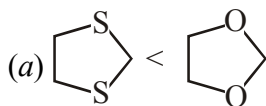
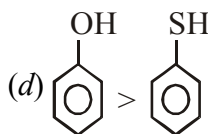
Correct option regarding I & II :-

- (a) $[\text{F}_{\text{ax}} \text{S} \text{F}_{\text{ax}}]_{\text{I}} = 180^\circ$, $[\text{F}_{\text{ax}} \text{S} \text{F}_{\text{ax}}]_{\text{II}} = 180^\circ$
 (b) $[\text{F}_{\text{ax}} \text{S} \text{F}_{\text{ax}}]_{\text{I}} < 180^\circ$, $[\text{F}_{\text{ax}} \text{S} \text{F}_{\text{ax}}]_{\text{II}} < 180^\circ$
 (c) $[\text{F}_{\text{eq}} \text{S} \text{F}_{\text{eq}}]_{\text{I}} = 120^\circ$, $[\text{F}_{\text{eq}} \text{S} \text{F}_{\text{eq}}]_{\text{II}} = 120^\circ$
 (d) $[\text{F}_{\text{eq}} \text{S} \text{F}_{\text{eq}}]_{\text{I}} < 120^\circ$, $[\text{F}_{\text{eq}} \text{S} \text{F}_{\text{eq}}]_{\text{II}} < 120^\circ$

47. Correct order of bond angle -
 (a) $\text{PF}_3 < \text{PCl}_3 < \text{PBr}_3 < \text{PI}_3$
 (b) $\text{F}_2\text{O} < \text{H}_2\text{O} < \text{Cl}_2\text{O}$
 (c) $\text{PH}_3 < \text{PF}_3$
 (d) $\text{NH}_3 < \text{NF}_3$
48. Correct order of bond angle -
 (a) $\text{H}_2\text{O} > \text{F}_2\text{O}$
 (b) $\text{H}_2\text{O} > \text{H}_2\text{S} > \text{H}_2\text{Se} > \text{H}_2\text{Te}$
 (c) $\text{SbH}_3 > \text{AsH}_3 > \text{PH}_3 > \text{NH}_3$
 (d) $\text{N}^+\text{O}_2 > \text{NO}_2 > \text{NO}_2^- > \text{NO}_3^-$
49. The correct statements are -
 (a) the reduction of bond angle in H_2O is greater than reduction of bond angle in NH_3 from $109^\circ 28'$.
 (b) in both compounds hybrid orbitals containing lone pair have p character more than 75%.
 (c) % s character of lone pair at nitrogen atom in NH_3 is less than % s character of each lone pair at oxygen atom in H_2O .
 (d) in NH_3 & BF_3 , all bonds in each molecule is identical.
50. Which of the following molecules have cation and anion pair?
 (a) $\text{PBr}_5(\text{g})$ (b) $\text{PBr}_5(\text{s})$
 (c) $\text{N}_2\text{O}_5(\text{s})$ (d) dry ice
51. Which of the following molecules do not exist?
 (a) CF_6^{2-} (b) NeF_2
 (c) XeCl_8^{2-} (d) XeH_6
52. Which of the following molecules do not exist?
 (a) PH_5 (b) PH_3
 (c) NH_3 (d) SH_6
53. If y axis is the approaching axis then which set of orbitals can not form the π bond between two atom in general -
 (a) $p_z - p_z$ (b) $p_x - p_x$
 (c) $p_y - p_y$ (d) $p_y - p_x$
54. Which of the following overlapping is involved in formation of only σ bond?
 (a) s-p overlapping (b) s-s overlapping
 (c) p-d overlapping (d) p-p overlapping
55. As per VBT, which of the following overlapping are possible?
 (a) $d_{xy} + d_{xy} \xrightarrow{\text{x axis}}$
 (b) $p_y + p_x \xrightarrow{\text{x axis}}$
 (c) $d_{xy} + p_{zy} \longrightarrow$
 (d) $sp^2 - sp^2 \longrightarrow$
56. Find the number of molecule in which direction of back bonding is from central atom to surrounding atom -
 (a) BBr_3 (b) OF_2
 (c) OCl_2 (d) $\text{CCl}_3^\ominus$
57. Find out the correct statements -
 (a) $\widehat{\text{SiNSi}}$ bond angle in $\text{N}(\text{SiH}_3)_3$ is greater than $\widehat{\text{SiPSi}}$ bond angle in $\text{P}(\text{SiH}_3)_3$
 (b) B-F bond dissociation energy in $\text{BF}(\text{CH}_3)_2$ is greater than B-F bond dissociation energy in BF_3
 (c) In BF_3 back bonding is possible due to this B-F bond has partial double bond character whereas in BF_4^- , back bonding is not possible so B-F bond has 100% single bond character.
 (d) All hybrid orbitals of O-atom in H_2O lie in the same plane.
58. Which of the following statements are correct?
 (a) B-F bond length in BF_4^- is more w.r.t. B-F bond length in BF_3
 (b) Correct order of lewis acidic strength $\text{SiF}_4 > \text{SiCl}_4 > \text{SiBr}_4 > \text{SiI}_4$
 (c) Correct order of lewis acidic strength $\text{BF}_3 > \text{BCl}_3 > \text{BBr}_3 > \text{BI}_3$
 (d) AlF_3 does not show back bonding
59. Which of the following statements are correct?
 (a) there is no scope of axial or equatorial position in octahedral and tetrahedral geometry
 (b) $\text{CF}_2 = \text{C} = \text{CF}_2$ molecule has all atoms in one plane
 (c) the bond order in NO is 2.5 and while that in NO^+ is 3
 (d) X^+ ion is smaller than X^- ion

1.22 CHEMICAL BONDING

60. Correct acidic strength order :

(c) SiH_3OH (Silanol) $>$ CH_3OH (methanol)61. Silyl isocyanate is a compound having molecular formula H_3SiNCO . Which of the following bond has significant π bond character -

- (a) C–O linkage (b) N–C linkage
(c) Si–N linkage (d) Si–H linkage

62. Which of the following statements are correct ?

- (a) In back bonding of CCl_3^- , e^- movement is from surrounding atom (Cl) to central atom carbon.
(b) In back bonding of $:\text{CCl}_2$, e^- movement is from surrounding atom (Cl) to central atom carbon.
(c) When boric acid is dissolved in water, back bonding is going to be lost in product side.
(d) $\text{CH}_2^+ - \text{CH}_3$ (ethyl carbocation) is more stable than $\text{CH}_2^+ - \text{OCH}_3$ (methoxy methyl carbocation)

63. I. $\text{O}(\text{CH}_3)_2 + \text{BF}_3 \rightarrow$ II. $\text{O}(\text{SiH}_3)_2 + \text{BF}_3 \rightarrow$ III. $\text{NMe}_3 + \text{HCl} \rightarrow$ IV. $\text{N}(\text{SiH}_3)_3 + \text{HCl} \rightarrow$

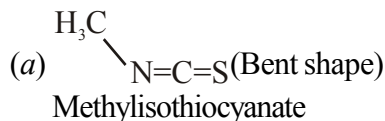
Correct statement :-

- (a) Reaction I is faster than reaction II
(b) Reaction II is faster than reaction I
(c) Reaction III is faster than reaction IV
(d) Reaction IV is faster than reaction III

64. In which of the following compound, bond angle increases due to back bonding ?

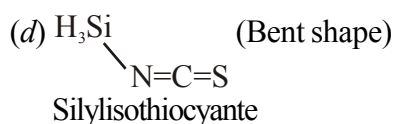
- (a) BI_3 (b) $\text{N}(\text{CH}_3)_3$
(c) $\text{N}(\text{SiH}_3)_3$ (d) $\text{H}_3\text{Si}-\text{N}=\text{C}=\text{S}$

65. Which structure shown below are correct shape ?



(b) $\text{H}_3\text{C}-\text{N}=\text{C}=\text{S}$ (Linear shape)
Methylisothiocyanate

(c) $\text{H}_3\text{Si}-\text{N}=\text{C}=\text{S}$ (Linear shape)
Silylisothiocyanate



66. In which of the following molecules during monomer formation, central atom has incomplete octet (less than 8 electrons) :-

- (a) Al_2Cl_6 (b) N_2O_4
(c) $(\text{BeCl}_2)_n$ (d) I_2Cl_6

67. Which molecule under following condition does not have bridging bond :-

- (a) SO_3 during trimer formation
(b) P_2O_5 during dimer formation
(c) BeCl_2 during polymer formation
(d) $\text{Cl}_2\text{C}=\text{CCl}_2$ during polymer formation

68. Which molecule has $3c-4e$ bond ?

- (a) Al_2Br_6 (b) $(\text{BeH}_2)_n$
(c) I_2Cl_6 (d) Fe_2Cl_6

69. Reaction of diborane with NMe_3 in product formation :-

- (a) Hybridisation of B does not change
(b) Hybridisation of N does not change
(c) Hybridisation of B changes
(d) Hybridisation of N changes

70. Which molecules are electron deficient ?

- (a) SO_3 (b) SOCl_2
(c) B_2H_6 (d) NH_3

71. Which of the following reactions products formed are correctly written :-

- (a) $\text{BCl}_3 + \text{H}_2\text{O} \rightarrow \text{B}^{3+}(\text{aq}) + 3\text{Cl}^-(\text{aq})$
(b) $\text{POCl}_3 + 3\text{H}_2\text{O} \rightarrow \text{H}_3\text{PO}_4 + 3\text{HCl}$
(c) $\text{SF}_4 + \text{H}_2\text{O} \rightarrow \text{H}_2\text{SO}_3 + \text{HF}$
(d) $\text{NaCl} + \text{H}_2\text{O} \rightarrow \text{Na}^+(\text{aq}) + \text{Cl}^-(\text{aq})$

1.23

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1.24 CHEMICAL BONDING

83. Select correct statement :-

- (a) If molecule has any polar bond then it is always polar
 (b) Solubility of noble gas increases in water down the group
 (c) London dispersion forces also contribute in net interaction between nonpolar molecule
 (d) Molecular interaction between CF_4 molecule is higher as compared to molecular interaction between NF_3 molecule

84. Which compounds undergo hydrolysis at very high temperature through $\text{S}_\text{N}1$?

- (a) SF_6 (b) NF_3
 (c) CCl_4 (d) SF_4

85. Which reactions product are correctly written ?

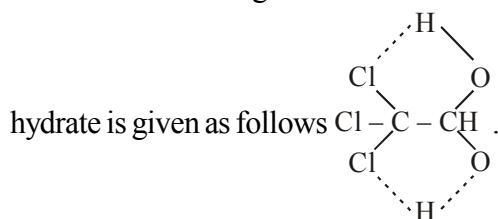
- (a) $\text{B}_2\text{H}_6 + \text{cyclopentoxide} \longrightarrow \text{H}_3\text{B}^\ominus - \text{O}^\oplus(\text{cyclopentyl})$
 (b) $\text{B}_2\text{H}_6 + \text{R}_3\text{N} \longrightarrow \text{H}_3\text{B}^\ominus - \text{NH}_3^+$
 (c) $\text{B}_2\text{H}_6 + \text{NH}_3 \longrightarrow \text{H}_3\text{B}^\ominus - \text{NH}_3^+$
 (d) All

86. Which of the following compound on heating give NO_2 gas ?

- (a) AgNO_3 (b) LiNO_3
 (c) $\text{Be}(\text{NO}_3)_2$ (d) KNO_3

87. Which of the following statements are correct ?

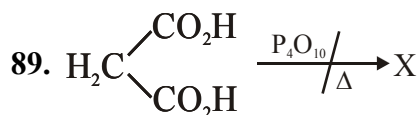
- (a) OF during dimer formation planarity is lost
 (b) Pentasulphide S_5^{2-} is bent in shape
 (c) Chloral $\text{CCl}_3\text{CH}=\text{O}$ forms stable hydrate because of H bonding. The structure of chloral



- (d) In S_2F_{10} , all S-F bond lengths are identical.

88. Which are correct representation of boranes ?

- (a) B_nH_{n+4} (b) B_nH_{n+6}
 (c) $\text{B}_n\text{H}_{2n+2}$ (d) None



Correct statements regarding X :-

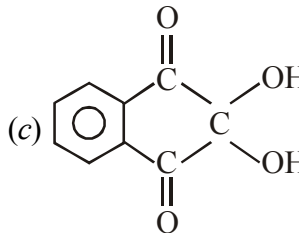
- (a) X is linear in shape
 (b) X has 3σ and 3π bond
 (c) all carbons in X have same hybridisation state
 (d) X is $\text{CH}_3\text{CO}_2\text{H}$

90. Which of the compounds do have H-bonding ?

- (a) NH_3 (b) CH_3OCH_3
 (c) $\text{CH}_3\text{CH}_2\text{OH}$ (d) CH_3NHCH_3

91. For which compounds, intramolecular hydrogen bonding is possible :-

- (a) o-nitrophenol
 (b) p-nitrophenol



- (d) HNO_3

92. H-bonding is responsible for which of the following phenomenon :-

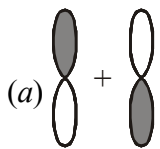
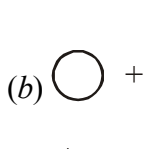
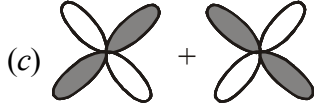
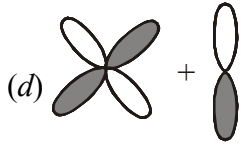
- (a) Acetylene is dissolved in acetone
 (b) DDT is soluble in water
 (c) Formic acid is more acidic than acetic acid
 (d) H_3PO_4 is more viscous w.r.t. Me_3PO_4

93. Which of the following are correct statements ?

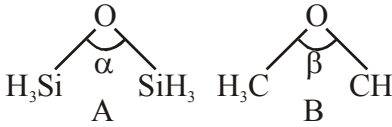
- (a) the direction of dipole moment in CO is from O to C
 (b) white phosphorus is less stable than red phosphorus
 (c) H_2 molecule is more stable than He-H molecule
 (d) π_b & π^* orbitals obtained from 2p orbital are lying in the same plane

94. Which of the following order are correct regarding electronegativity of elements ?

- (a) $\text{P} > \text{S}$ (b) $\text{P} > \text{N}$
 (c) $\text{Cl} > \text{S}$ (d) $\text{Cl} > \text{Br}$

95. Correct order of stability :-
 (a) $\text{PbCl}_2 > \text{PbCl}_4$ (b) $\text{CCl}_2 > \text{PbCl}_2$
 (c) $\text{SnCl}_2 > \text{SnCl}_4$ (d) $\text{PbCl}_4 > \text{PbI}_4$
96. All the given molecule have triangular faces. Find molecules in which all the triangular faces are equilateral triangular faces :-
 (a) CHCl_3 (b) PCl_5
 (c) CH_4 (d) SF_6
97. During linear combination of atomic orbitals to form molecular orbitals, which of the following conditions must be satisfied :-
 (a) the combining atomic orbitals must have the same or nearly the same energy
 (b) the combining atomic orbitals must have the same symmetry about the molecular axis
 (c) the combining atomic orbitals must overlap to the maximum extent
 (d) the combining atomic orbitals must overlap to the minimum extent
98. For which of the following ion, correct resonating structures are given :-
 (a) $\text{:}\ddot{\text{S}}^- - \text{C} \equiv \text{N} \longleftrightarrow \text{:}\ddot{\text{S}} = \text{C} = \ddot{\text{N}}^+$
 (b) $\text{:}\ddot{\text{N}}^+ = \text{C} = \ddot{\text{N}}^- \longleftrightarrow \text{N} \equiv \text{C} - \text{N}^{2-}$
 (c) $\begin{array}{c} \text{O} \\ \parallel \\ \text{O}=\text{S}-\text{O}^- \\ \parallel \\ \text{O} \end{array} \longleftrightarrow \begin{array}{c} \text{O}^+ \\ \parallel \\ \text{O}=\text{S}-\text{O}^- \\ \parallel \\ \text{O} \end{array}$
 (d) $\begin{array}{c} \text{O}^+ \\ \parallel \\ \text{O}-\text{O} \end{array} \longleftrightarrow \begin{array}{c} \text{O} \\ \parallel \\ \text{O}=\text{O} \end{array}$
99. Select incorrect statement :-
 (a) In general no S-H bond is present in oxy acid of sulphur
 (b) Oxyacids of P containing P-H bonds are reducing in nature
 (c) Both O_2 and S_2 are diamagnetic in nature
 (d) General reaction of O_2 with many metals and non metals are endothermic in nature because energy is required to break double bond present in O_2 .
100. B_2 molecule will be diamagnetic when :-
 (a) s-p mixing is not operative
 (b) s-p mixing is operative and Hund's rule is violated
 (c) s-p mixing is operative and Hund's rule is followed
 (d) None
101. Which of the following molecule / ions do not contain unpaired electron ?
 (a) N_2^+ (b) O_2
 (c) O_2^{2-} (d) Br_2
102. Which of the following overlapping(s) result antibonding molecular orbital with two nodal plane?
 (a) 
 (b) 
 (c) 
 (d) 
103. Which of the following process is not correct against the mentioned property ?
 (a) $\text{N}_2^+ \rightarrow \text{N}_2$ (magnetic moment increases)
 (b) $\text{O}_2 \rightarrow \text{O}_2^+$ (bond order decreases)
 (c) $\text{Li}_2^+ \rightarrow \text{Li}_2^-$ (magnetic property changes)
 (d) $\text{C}_2 \rightarrow \text{C}_2^+$ (electron is removed from bonding orbital)
104. Which of the following options are incorrect ?
 (a) dimer of NO_2 contains N-O-N linkage
 (b) all diatomic halogens are coloured due to HOMO-LUMO electron transition
 (c) all diatomic halogens are coloured due to LUMO-HOMO electron transition
 (d) dimer of NO_2 does not contain N-O-N linkage
105. The correct statements are :-
 (a) B_2 solid does not exist
 (b) B_2 have basic building B_{12} icosahedral
 (c) B_{12} icosahedral is made up of polyhedron having 20 faces and 12 corners
 (d) B_{12} icosahedral is made up of polyhedron having 10 faces and 5 corners

1.26 CHEMICAL BONDING

- 106.** Which of the following compounds when undergo reaction with water give methane gas ?
 (a) Be_2C (b) CaC_2
 (c) Mg_2C_3 (d) Al_4C_3
- 107.** Which of the following compounds when undergo reaction with water give terminal alkyne ?
 (a) Li_2C_2 (b) CaC_2
 (c) Al_4C_3 (d) Mg_2C_3
- 108.** Which compounds when react with water, it gives ammonia as one of the product ?
 (a) CaCN_2 (b) AlN
 (c) Ca_3N_2 (d) Li_3N
- 109.** Which compounds consume six moles of water per one mole of compound ?
 (a) Ca_3N_2 (b) P_4O_6
 (c) BrF_5 (d) SiF_4
- 110.** Which of the following statements are incorrect ?
 (a) maximum covalency of nitrogen is 3
 (b) in glyoxal, carbon atom is sp^2 hybridised
 (c) size of d orbital decreases : $\text{Si} > \text{P} > \text{S} > \text{Cl}$
 (d) maximum covalency of Sulphur is 4
- 111.** How many statements are correct :-
 (a) In PCl_5 ; axial P–Cl bond length > equatorial P–Cl bond length
 (b) In PF_4Cl ; axial P–F bond length < equatorial P–Cl bond length
 (c) PCl_4F has more dipole moment than PCl_3F_2
 (d) PF_3Cl_2 has more dipole moment than PCl_5
- 112.** Correct statement :-
 (a) Solid state PCl_5 exists as PCl_4^+ and PCl_6^-
 (b) Solid state PBr_5 exists as PBr_4^+ and Br^-
 (c) Solid state N_2O_4 exists as NO^+ and NO_3^-
 (d) Solid state N_2O_5 exists as NO_2^+ and NO_3^-
- 113.** Which of the following statement are incorrect for PO_4^{3-} ?
 (a) Number of identical resonating structures are 3
 (b) Bond order of P–O bond is 1.25
 (c) No. of identical P–O bonds are 3
 (d) Maximum no. of atoms are in one plane is 4
- 114.** Which of the following order are correct ?
 (a) $\text{CHF} > \text{CHCl} > \text{CHBr}$ (stability)
 (b) $\text{CHF} > \text{CHCl} > \text{CHBr}$ (extent of back bonding)
 (c) $\text{CF}_2 > \text{CCl}_2 > \text{CBr}_2$ (extent of back bonding)
 (d) $\text{CF}_2 > \text{CCl}_2 > \text{CBr}_2$ (Lewis acidity)
- 115.** 
 Correct option is :-
 (a) $\alpha > \beta$
 (b) A is more Lewis basic than B
 (c) $\beta > \alpha$
 (d) B is more Lewis basic than A
- 116.** Which of the following compounds on heating give O_2 gas ?
 (a) BeC_2O_4 (b) MgCO_3
 (c) NaNO_3 (d) H_2O_2
- 117.** Out of BeCl_2 , CaCl_2 , CsCl , KCl :-
 (a) highest covalent character is observed in BeCl_2
 (b) highest covalent character is observed in CaCl_2
 (c) Least covalent character is observed in KCl
 (d) Least covalent character is observed in CsCl
- 118.** In $\text{Cr}_2\text{O}_7^{2-}$ (dichromate ion) :-
 (a) 8 Cr–O bond lengths are identical
 (b) 6 Cr–O bond lengths are identical
 (c) Cr is in +6 oxidation state
 (d) Cr is sp^3 hybridised
- 119.** In which all hydrogens are not of same acidic character :-
 (a) H_3PO_5 (b) $\text{H}_2\text{S}_2\text{O}_5$
 (c) $\text{H}_2\text{S}_2\text{O}_8$ (d) H_3PO_4
- 120.** The correct order of boiling point :-
 (a) $\text{C}_2\text{H}_6 > \text{C}_2\text{F}_6$ (b) $\text{NH}_3 > \text{NF}_3$
 (c) $\text{C}_{10}\text{H}_{22} > \text{C}_{10}\text{F}_{22}$ (d) $\text{CCl}_4 > \text{SiCl}_4$
- 121.** Which of the following order are correct regarding solubility in water for following compounds ?
 (a) $\text{KCl} > \text{AgCl}$ (b) $\text{CaCl}_2 > \text{ZnCl}_2$
 (c) $\text{BeSO}_4 > \text{BaSO}_4$ (d) $\text{LiI} > \text{RbCl}$

- $$\text{XeFOSO}_2\text{F} ? \left\{ \begin{array}{c} \text{F}-\text{Xe}-\text{O}-\text{S}-\text{F} \\ \quad \quad \quad \diagdown \quad \diagup \\ \quad \quad \quad \text{O} \quad \text{O} \end{array} \right.$$

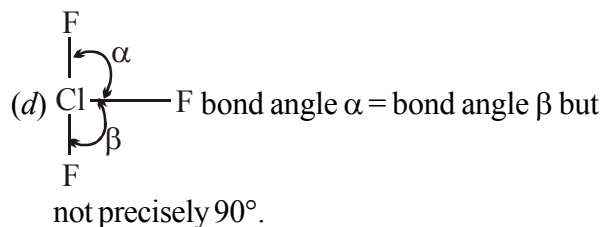
1.28 CHEMICAL BONDING

135. Which of the following are incorrect order against indicated properties ?

- (a) $\text{AgF} > \text{AgCl} > \text{AgI}$: covalent character
- (b) $\text{NaHCO}_3 > \text{KHCO}_3 > \text{RbHCO}_3$: solubility in water
- (c) $\text{NaF} < \text{MgF}_2 < \text{AlF}_3$: melting point
- (d) $\text{MgC}_2\text{O}_4 > \text{CaC}_2\text{O}_4 > \text{BaC}_2\text{O}_4$: solubility in water

136. Which of the following statements are correct ?

- (a) compounds of Hg^{2+} ions having an ionic radius of 116 pm are more covalent in character than those of Ca^{2+} ion with almost identical size (114 pm) and same charge
- (b) ethers behave as bases in the presence of mineral acids
- (c) carbon has unique ability to form $p\pi$ - $p\pi$ multiple bonds with itself and with other atoms of small size and high electronegativity.



137. From the following options which are correct for double chain silicate :-

- (a) average charge on each tetrahedral = -1.5
- (b) all the tetrahedron have one unshared oxygen / corner
- (c) all the silicon atoms have +4 oxidation state
- (d) average shared oxygen / corner per tetrahedron = 2.5

138. Which of the following options are correct ?

- (a) in alkaline medium, the hydrolysis of BeCl_2 produces clear solution consisting of $[\text{Be}(\text{OH})_4]^{2-}$ and HCl .
- (b) in the clear solution of BiCl_3 when large quantity of water is added, the white turbidity of BiOCl is obtained
- (c) SiF_4 undergoes partial hydrolysis
- (d) the final hydrolysis products of PCl_3 and POCl_3 are identical

139. If polarising power is in the order of $M_a^+ > M_b^+ > M_c^+$ and polarisability is in the order of $X^- > Y^- > Z^-$ then select which have more covalent character as compared to $M_b^+Y^-$:-

- (a) $M_a^+Y^-$ (b) $M_b^+X^-$
- (c) $M_a^+X^-$ (d) $M_b^+Y^-$

140. Which of the following statements are incorrect ?

- (a) P-F bond length is longer than that of P-Cl bond length in PF_2Cl_3
- (b) F atom occupy in equatorial position of T.b.p structure of PF_2Cl_3
- (c) In PF_3Cl_2 all axial and all equatorial bond length are identical
- (d) Fluorine atoms prefers to attach at the axial position than that of equatorial position of PF_2Cl_3

141. Which of the following options are incorrect ?

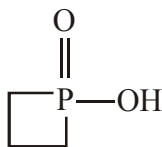
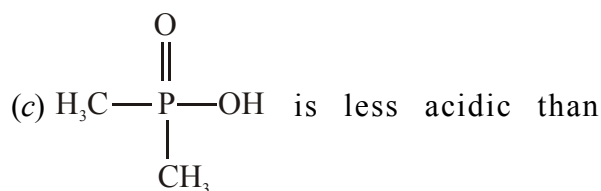
- (a) Salicylaldehyde has a higher melting point as compared to its positional isomer because of intra molecular hydrogen bonding
- (b) Ionic forces are directional in nature
- (c) In a crystals of I_2 , the forces of attraction will be covalent
- (d) crystalline boric acid forms two dimensional sheet with almost hexagonal symmetry

142. Which of the following statement is incorrect ?

- (a) the number of bonding electrons in N_2 and N_2^- is same
- (b) the bond order and bond length of N_2^+ and N_2^- is same
- (c) among O_2^- , O_2^{2-} , O_2^{2+} ; O_2^{2+} has longest bond length
- (d) bond order $(\text{O}-\text{O})$ $\text{O}_2^{2+} > \text{O}_2^{+1} > \text{O}_2 > \text{O}_2^{-1} > \text{O}_2^{2-}$

143. Which of the following statement are correct :-

- (a) the C-Cl distance in CH_3Cl and CF_3Cl are 1.78 Å and 1.75 Å respectively
- (b) the C-C single bond distance in methyl acetylene $\text{H}_3\text{C}-\text{C}\equiv\text{CH}$ is only 1.46 Å compared to 1.54 Å is CH_3-CH_3



(d) SF_6 sublimes at -64°C where graphite sublimes at 3700°C

144. Incorrect statement -

- (a) POF_3 and NOF_3 both have coordinate bond
- (b) POF_3 and NOF_3 does not have coordinate bond
- (c) POF_3 has coordinate bond but NOF_3 does not have coordinate bond
- (d) POF_3 does not have coordinate bond but NOF_3 has coordinate bond

145. Select the correct statements for sulphuric acid :-

- (a) it has high boiling point and viscosity
- (b) there are two types of bond lengths in its bivalent anion
- (c) $p\pi-d\pi$ bonding between sulphur and oxygen is observed
- (d) sulphur has the same hybridisation as that of boron in diborane

146. Which of the following statement are correct ?

- (a) the percentage of s character in the orbital forming S-S bond and P-P bonds in S_8 and P_4 molecules respectively are same
- (b) aqueous H_3PO_4 is syrupy
- (c) SiO_2 crystal may be considered as giant molecule in which eight - membered rings are formed with alternate silicon and oxygen atoms.
- (d) In SF_4 the bond angles instead of being 90° and 180° are 89° and 177° respectively due to the repulsion between lone pair and bond pairs of electrons

147. Which of the following molecules are nonplanar ?

- (a) NH_3
- (b) $\text{N}(\text{CH}_3)_3$ [trimethyl amine]
- (c) $\text{N}(\text{SiH}_3)_3$ (trisilylamine)
- (d) $\text{P}(\text{SiH}_3)_3$

148. Select the correct statement(s) :-

- (a) in ClO_4^- , all Cl-O bonds are identical and there is strong $p\pi-d\pi$ bonding between chlorine and oxygen atom
- (b) in P_4S_3 molecules there are a six P-S bonds, three P-P bonds and ten lone pairs of electrons
- (c) N_2H_4 is pyramidal about each N-atom
- (d) Br_2 and ICl have same reactivity and same boiling point

149. Which of the following are correct ?

- (a) thio-ether have less boiling point than ether
- (b) thiol have more boiling point than alcohol
- (c) the maximum possible number of hydrogen bonds a water molecule can form is 4
- (d) Al_2S_3 when undergoes reaction with water the products formed are $\text{Al}(\text{OH})_3$ and H_2S

150. Which are correct statements ?

- (a) Borazine has less intermolecular force of attraction as compared to benzene
- (b) D_2O has less boiling point than H_2O
- (c) CuI_2 is unstable even at ordinary temperature
- (d) NaClO_4 is about 1000 times as soluble as KClO_4 in water

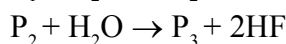
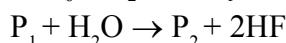
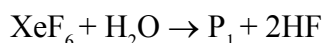
EXERCISE # III

□ Linked Comprehension Type :

Passage Type:

Passage for Q.1 to Q.3

XeF_6 undergo hydrolysis in 3 stages giving different products in each stage. Here H_2O acts as nucleophile and XeF_6 acts as electrophile. The reaction is as follows :-



1.30 CHEMICAL BONDING

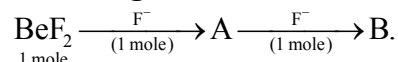
- The hybridisation of Xe atom in P_1 :-
 (a) sp^3 (b) sp^3d
 (c) sp^3d^2 (d) sp^3d^3
- The hybridisation of Xe atom in P_2 :-
 (a) sp^2 (b) sp^3
 (c) sp^3d (d) sp^3d^2
- The hybridisation of Xe atom in P_3 :-
 (a) sp^2 (b) sp^3
 (c) sp^3d (d) sp^3d^2

Passage for Q.4 to Q.6

Interhalogen compound of the type A_mB_n are possible. It can have formula AB ; AB_3 ; AB_5 ; AB_7 where A is halogen of larger size and B is halogen of smaller size and B is more electronegative than A. As the ratio between radii of A,B increases, the number of atoms per molecule increases-

- Correct geometry and polar character of AB :-
 (a) AB : linear ; polar
 (b) AB : linear ; nonpolar
 (c) AB : bent ; polar
 (d) AB : bent ; nonpolar

- Correct geometry and polar character of AB_7 :-
 (a) AB_7 : octahedral ; polar
 (b) AB_7 : octahedral ; nonpolar
 (c) AB_7 : pentagonal bipyramidal ; polar
 (d) AB_7 : pentagonal bipyramidal ; non polar
- Correct shape geometry and polar character of AB_3 -
 (a) AB_3 : bent ; nonpolar
 (b) AB_3 : Trigonal planar ; nonpolar
 (c) AB_3 : T shaped ; nonpolar
 (d) AB_3 : T shaped ; polar

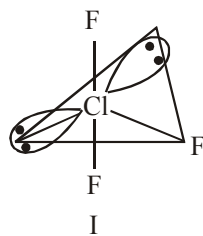
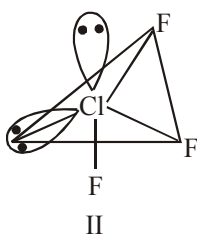
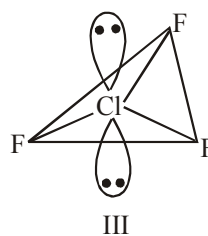
Passage for Q.7 to Q.8

- In above reaction, hybridisation of central atom in species A is -
 (a) sp (b) sp^2
 (c) sp^3 (d) sp^3d
- In above reaction, hybridisation of central atom in species B is -
 (a) sp (b) sp^2
 (c) sp^3 (d) sp^3d

Passage for Q.9 to Q.11

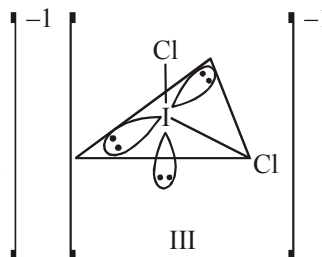
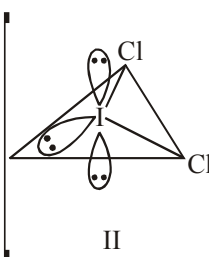
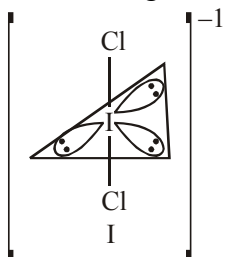
When the central atom bears both the bond pairs & lone pairs the structure are deviated from regular geometries produced from rule-1 of VSEPR. (Valence shell electron pair repulsion) theory -

- In ClF_3 , 3-possible structures given as follows -

l.p.-l.p repulsion at 90° l.p.-b.p repulsion at 90° b.p.-b.p. repulsion at 90°

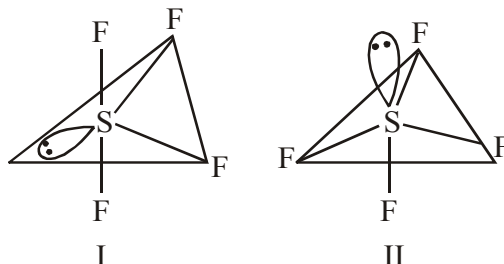
- | | | | |
|---------------------|---|---|---|
| (a) Str.-I | 0 | 4 | 2 |
| (b) Str.-II | 1 | 3 | 2 |
| (c) Str.-III | 0 | 6 | 0 |
| (d) All are correct | | | |

- In ICl_2^- , 3 possible structures given as follows -



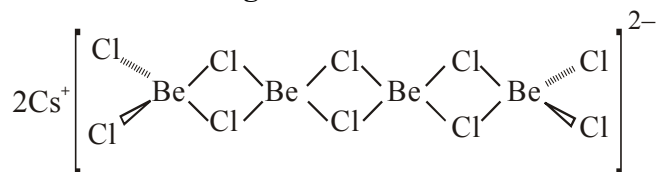
	l.p.-l.p repulsion at 90°	l.p.-b.p repulsion at 90°	b.p.-b.p.repulsion at 90°
(a) Str.-I	0	6	0
(b) Str.-II	2	4	0
(c) Str.-III	2	3	1
(d) All are correct			

11. In SF_4 , 2 possible structures given as follows -



	l.p.-l.p repulsion at 90°	l.p.-b.p repulsion at 90°	b.p.-b.p.repulsion at 90°
(a) Str.-I	0	3	3
(b) Str.-I	0	2	4
(c) Str.-II	0	2	4
(d) Str.-II	0	3	3

Passage for Q.12 to Q.14



A segment of polymeric BeCl_2 can be given as follows.

12. What is maximum number of atoms present in same plane ?

- (a) 6 (b) 8
(c) 10 (d) 12

13. What is the number of $2c-2e$ bonds present ?

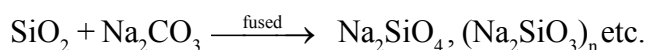
- (a) 12 (b) 4
(c) 8 (d) 6

14. What is the number of $3c-2e$ bonds present ?

- (a) 12 (b) 4
(c) 8 (d) 6

Passage for Q.15 to Q.17

Silicate may be regarded as the metal derivative of silicic acid H_4SiO_4 . Silicate can be prepared by fuming metal oxide or metal carbonate with sand.



Silicate are classified according to the nature of linking between the tetrahedral SiO_4^{4-} anions.

15. Hemimorphite having formula $\text{Zn}_4(\text{OH})_2 \cdot \text{Si}_2\text{O}_7 \cdot \text{H}_2\text{O}$. It is example of :-

- (a) Orthosilicate
(b) Pyrosilicate
(c) single chain silicate
(d) double chain silicate

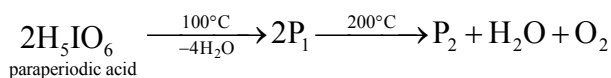
16. In Beryl $\text{Be}_2\text{Al}_2(\text{Si}_6\text{O}_{18})$, the number of oxygen shared in every tetrahedral :-

- (a) 2 (b) 3
(c) 6 (d) 4

17. Kaolin, an important clay mineral having molecular formula $\text{Al}_2(\text{OH})_4(\text{Si}_2\text{O}_5)_n$ is example of :-

- (a) Single chain silicate
(b) 3-D-silicate
(c) 2D/Sheet silicate
(d) Double chain silicate

Passage for Q.18 to Q.20



18. The structure of P_1 is :-

- (a) trigonal (b) tetrahedral
(c) octahedral (d) bent

1.32 CHEMICAL BONDING

19. Hybridisation of iodine atom in P_2 :-
 (a) sp^2 (b) sp^3
 (c) sp (d) sp^3d
20. The shape of paraperiodic acid is :-
 (a) tetrahedral (b) pentagonal bipyramidal
 (c) octahedral (d) angular

Passage for Q.21 to Q.23

Powdered borax $Na_2B_4O_7 \cdot 10H_2O$ is used in borax – bead test to identify different metal ions. For this, it is taken in a Pt wire, heated, the salt swells up and shrinks upon the loop forming a colourless transparent glass – like bead.

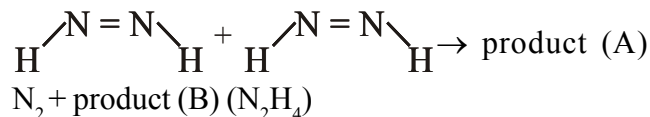
21. Correct statement about borax :-
 (a) each boron atom has one –OH group
 (b) each boron atom has two –OH group
 (c) each boron atom has three –OH group
 (d) each boron atom has four –OH group
22. Borax is actually made of :-
 (a) two tetrahedra and one triangular unit
 (b) one tetrahedra and one triangular unit
 (c) two tetrahedra and two triangular unit
 (d) None
23. Correct statement regarding borax :-
 (a) each boron atom has four B–O bonds
 (b) each boron atom has three B–O bonds
 (c) two boron atoms has four B–O bonds while other two have three B–O bonds
 (d) none

Passage for Q.24 to Q.26

The hydrogen bond is generally represented as $A-H \cdots B$. Where A is sufficiently electronegative element to cause acidic nature on the hydrogen and B has sufficient electron density to act as a base to this hydrogen.

24. Which hydrogen bonding is strongest :-
 (a) $F-H \cdots N$ (b) $N-H \cdots O$
 (c) $F-H \cdots O$ (d) $F-H \cdots F$
25. Hydrogen bonding takes place in :-
 (a) KHF_2 (b) KH_2PO_4
 (c) KH_2PO_2 (d) both (a) and (b)

26. Which of the following compound can form hydrogen bonding with each other :-
 (a) NH_4Cl + Water
 (b) KF + HF
 (c) CH_3OCH_3 + Water
 (d) All

Passage for Q.27 to Q.29

27. What kind of hybridisation change is observed for N atom from reactant to product (A) :-
 (a) $sp^2 \rightarrow sp^3$ (b) $sp^3 \rightarrow sp$
 (c) $sp \rightarrow sp^2$ (d) $sp^2 \rightarrow sp$
28. What kind of hybridisation change is observed for N atom from reactant to product (B) :-
 (a) $sp^2 \rightarrow sp^3$ (b) $sp^3 \rightarrow sp$
 (c) $sp \rightarrow sp^2$ (d) $sp^2 \rightarrow sp$
29. This reaction is example of :-
 (a) non-redox reaction
 (b) disproportionation reaction
 (c) comproportionation reaction
 (d) acid base reaction

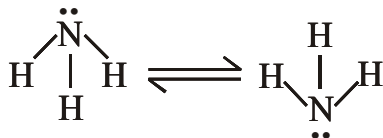
Passage for Q.30 to Q.31

Polystyrene is a polymer which is used as insulator, manufacture of toys, radio, television cabinet. Its monomeric unit is styrene ($CH_2 = CHPh$). Polyacrylonitrile is another polymer which is used as substitute for wool in making commercial fibres. Its monomeric unit is acrylonitrile. ($CH_2 = CH-CN$)

30. The hybridisation of carbon in polystyrene :-
 (a) sp^2 and sp
 (b) sp and sp^3
 (c) sp^3 and sp^2
 (d) only sp^2
31. The hybridisation of carbon in Polyacrylonitrile :-
 (a) sp^2 and sp
 (b) sp and sp^3
 (c) sp^3 and sp^2
 (d) only sp^2

Passage for Q.32 to Q.33

Ammonia molecule undergoes a type of motion known as inversion; The N atom oscillate through the plane of the three H atoms much as an umbrella turns inside out. This is called umbrella inversion.



32. During umbrella inversion the hybridisation of N in intermediate stage :-

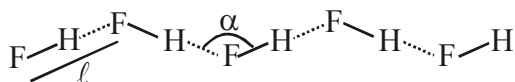
- (a) sp (b) sp^2
(c) sp^3 (d) sp^3d

33. The potential energy barrier for inversion is :-

- (a) 6 KJ mole⁻¹
(b) 12 KJ mole⁻¹
(c) 18 KJ mole⁻¹
(d) 24 KJ mole⁻¹

Passage for Q.34 to Q.35

Because of very high electronegativity of F(4.0), it has the tendency to form hydrogen bonds. The liquid acid and its vapour at low temperature consists of polymeric chain $(HF)_x$. The association of HF molecule is attributed to hydrogen bonding.



34. l and α are respectively :-

- (a) 1.22 Å ; 120°
(b) 0.77 Å ; 70°
(c) 2.55 Å ; 140°
(d) 3.66 Å ; 210°

35. Which of the following option is correct regarding HF ?

- (a) HF has highest boiling point among all HX
(b) HF is most polar among all HX
(c) HF is least acidic among all HX
(d) All

Passage for Q.36 to Q.38

According to wave mechanics, the atomic orbitals can be expressed by wave functions (ψ 's) which represent

the amplitude of the electron waves. These are obtained from the solution of Schrodinger wave equation. Molecular orbitals which are one electron wave functions for molecules are difficult to obtain directly from the solution of Schrodinger wave equation. To overcome this problem, an approximate method known as linear combination of atomic orbitals (LCAO) has been adopted.

36. If the atomic orbitals of these atoms are represented by the wave functions Ψ_A and Ψ_B , then which of the following options is correct regarding bonding molecular orbital :-

- (a) bonding molecular orbitals are formed as $\sigma = \Psi_A + \Psi_B$
(b) antibonding molecular orbitals are formed as $\sigma = \Psi_A - \Psi_B$
(c) both (a) and (b)
(d) None

37. Which of the following option is correct regarding bonding molecular orbital ?

- (a) In a bonding molecular orbital, electron density is located between the nuclei of the bonded atoms
(b) A bonding molecular orbital always possesses lower energy than either of the atomic orbital
(c) In the formation of the bonding molecular orbital, the two electron waves of bonding atoms reinforce each other due to constructive interference.
(d) All

38. Which of the following option is correct regarding antibonding molecular orbital ?

- (a) In a antibonding molecular orbital, most of the electron density is located away from the space between the nuclei.
(b) An antibonding molecular orbital always possesses higher energy than either of the atomic orbital
(c) In the formation of the antibonding molecular orbital, the two electron waves cancel each other due to destructive interference.
(d) All

1.34 CHEMICAL BONDING

Passage for Q.39 to Q.41

Type of molecule	Example	Dipole moment $\mu(\text{D})$
AB_2	H_2O	x_1
	H_2S	y_1
	CO_2	z_1
AB_3	NH_3	x_2
	NF_3	y_2
	BF_3	z_2
AB_4	CH_4	x_3
	CHCl_3	y_3
	CCl_4	z_3

39. Correct option is :-

- (a) $x_1 > y_1 > z_1$ (b) $x_1 < y_1 < z_1$
 (c) $x_1 = y_1 = z_1$ (d) $x_1 > z_1 > y_1$

40. Correct option is :-

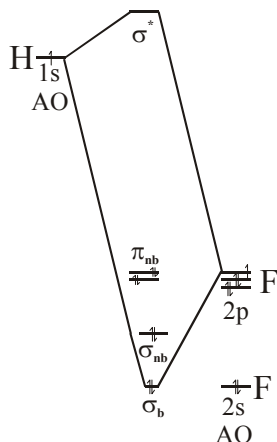
- (a) $x_2 > y_2 > z_2$ (b) $y_3 > x_3 = z_3$
 (c) $x_2 = y_2 = z_2$ (d) both (a) & (b)

41. Which one has zero dipole moment ?

- (a) CO_2 (b) BF_3
 (c) $\text{CCl}_4, \text{CH}_4$ (d) All

Passage for Q.42 to Q.44

M.O. diagram of HF molecule is given below :



42. HOMO of HF is :- (HOMO = highest occupied molecular orbital)

- (a) σ^* (b) σ_b
 (c) π_{nb} (d) σ_{nb}

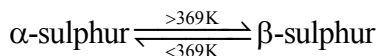
43. LUMO of HF is :- (LUMO = lowest unoccupied molecular orbital)

- (a) σ^* (b) σ_b
 (c) π_{nb} (d) σ_{nb}

44. HF is :-

- (a) paramagnetic
 (b) diamagnetic
 (c) can not be predicted
 (d) base

Passage for Q.45 to Q.47

45. α -sulphur is :-

- (a) rhombic sulphur (b) monoclinic sulphur
 (c) plastic sulphur (d) colloidal sulphur

46. β -sulphur is :-

- (a) monoclinic sulphur
 (b) rhombic sulphur
 (c) plastic sulphur
 (d) colloidal sulphur

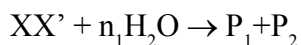
47. Correct statements are :-

- (a) both α and β variety are soluble in CS_2
 (b) both α and β variety are insoluble in water
 (c) α variety is lemon yellow and β variety is needle like transparent crystal
 (d) all

Passage for Q.48 to Q.50

Interhalogen compounds are represented as XX' , XX'_3 , XX'_5 , XX'_7 when X' is more electronegative element X is less electronegative element

when they undergo hydrolysis a mixture of hydraacid and oxyacid is formed.

48. n_1, n_2, n_3, n_4 are respectively

- (a) 2,3,4,5 (b) 0,1,2,3
 (c) 1,2,3,4 (d) 0,2,4,6

49. The common product P_1 is :-

- (a) HOX' (b) HX'
 (c) HOX (d) HX

50. P_2, P_3, P_4, P_5 are respectively :-
 (a) $XO_0(OH); XO_1(OH); XO_2(OH); XO_3(OH)$
 (b) $X'O_0(OH); X'O_1(OH); X'O_2(OH); X'O_3(OH)$
 (c) $HOX; HOX_2; HOX_3; HOX_4$
 (d) $HOX'; HOX'_2; HOX'_3; HOX'_4$

Passage for Q.51 to Q.53

Marshall's acid + $H_2O \rightarrow X(\text{acid}) + Y(\text{acid})$

$Y(\text{acid}) + H_2O \rightarrow X(\text{acid}) + Z(\text{unstable compound})$

51. The product X, Y are respectively :-
 (a) $H_2SO_3; H_2SO_4$ (b) $H_2SO_4; H_2SO_5$
 (c) $H_2SO_3; H_2SO_5$ (d) $H_2SO_5; H_2SO_3$
52. Unstable compound Z is :-
 (a) H_2O (b) H_2
 (c) O_2 (d) H_2O_2
53. Which compound has peroxide linkage ?
 (a) X, Y, Z all (b) X, Z both
 (c) X, Y both (d) Y, Z both

Passage for Q.54 to Q.56

Carbon exists in two allotropic forms : (i) crystalline form, (ii) amorphous form. The crystalline form of the allotropes of carbon includes (a) Diamond, (b) Graphite. The amorphous form of allotropes of carbon include (a) Charcoal (b) Lampblack (c) Coke (d) Gas carbon.

54. Which of the following options are correct regarding Diamond ?
 (a) Diamond is chemically inert, it is not attacked by acids and alkalies.
 (b) It is the hardest substance, specific gravity is 3.5
 (c) It has carbon with sp^3 hybridisation
 (d) All
55. Which of the following options are correct regarding Graphite ?
 (a) Graphite is used as dry lubricant in machine in place of oil
 (b) It is a good conductor of heat and electricity
 (c) It is soft and slippery, specific gravity is 2.2
 (d) All

56. Correct order regarding property between diamond and graphite :-
 (a) thermal conductivity : diamond > graphite
 (b) electrical conductivity : diamond < graphite
 (c) C-C bond length : diamond > graphite
 (d) All

Passage for Q.57 to Q.59

Nitrogen forms $p\pi-p\pi$ multiple bonds, nitrogen exists as triply bonded diatomic gaseous molecule. Bond strength is very high which is responsible for inertness at ordinary conditions, while other members of nitrogen family form $d\pi-p\pi$ bonding :-

57. Thermal and electrical conductivity is highest
 (a) N (b) P
 (c) As (d) Bi
58. Which is most acidic in nature ?
 (a) Bi_2O_3 (b) Bi_2O_4
 (c) Sb_2O_3 (d) Bi_2O_5
59. Maximum covalency of Sb will be :-
 (a) 0 (b) 2
 (c) 4 (d) 6

Passage for Q.60 to Q.62

Chemical bonding between two atoms is necessarily associated with an electrical moment arising out of the difference in electronegativity of two atoms. This means that every bond carries with it an electrical moment called to bond moment'. To compute the dipole moment it is necessary to find out the values of various bond moment. in the following table dipole moment of different bonds are as given.

Bond	$\overline{H-C}$	$\overline{C-Cl}$	$\overline{C=O}$
Bond moments	0.4D	1.5D	2.5D

The group moments of few group as given :

Group	NO_2	OH	CN	CH_3
Direction of dipole	toward N	toward O	toward N	away from CH_3
Dipole moment	4D	1.6D	3.8D	0.4D

1.36 CHEMICAL BONDING

60. The bond angle in H_2S is 97° and its dipole moment is 1.5D. The S–H bond distance is 0.15 nm. Therefore approximate percentage ionic character of S–H bond is (neglect the effect of dipole moment of lone pair on sulphur atom in H_2S). (Given $\cos 97^\circ = -0.12$ and $\sqrt{0.88} = 0.94$)

- (a) 32% (b) 16%
(c) 84% (d) 10%

61. In CH_3CCl_3 (I), CHCl_3 (II) and CH_3Cl (III) the normal tetrahedral bond angle is maintained. Also given $\cos 70.5^\circ = \frac{1}{3}$. Therefore dipole moments of the given compounds are :-

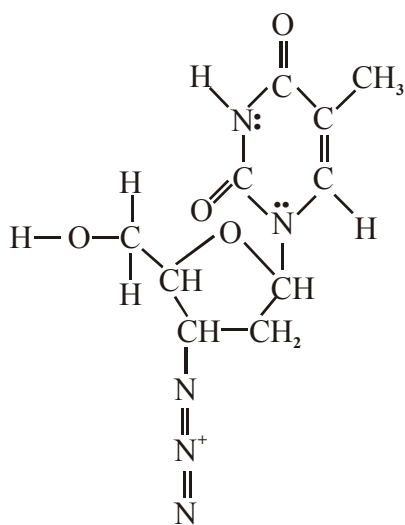
- (a) I = 1.9 D, II = 1.9 D, III = 1.7 D
(b) I = 1.9 D, II = 1.7 D, III = 1.9 D
(c) I = 1.9 D, II = 1.7 D, III = 1.0 D
(d) I = 1.9 D, II = 1.1 D, III = 1.9 D

62. In the acetone molecule considering the normal planer structure, the observed dipole moment of acetone molecule is :-

- (a) 2.9D (b) 2.75D
(c) 3D (d) none of these

Passage for Q.63 to Q.65

The structure of AZT drug (azidothymidine) also called zidovudone, is given below.



It is used to delay development of AIDS in patients infected with HIV (Human immuno deficiency virus)

63. Number of carbon atoms with sp^3 hybridisation

- (a) 2 (b) 4
(c) 6 (d) 8

64. Number of carbon atoms with sp^2 hybridisation

- (a) 2 (b) 4
(c) 6 (d) 8

65. Number of nitrogen atoms with sp hybridisation

- (a) 2 (b) 4
(c) 6 (d) 1

MATRIX MATCH TYPE

66. Match list-I with list-II and select the correct answer :-

Species	O–N–O angle
(A) NO_2^+	(P) 180°
(B) NO_2	(Q) 134°
(C) NO_2^-	(R) 120°
(D) NO_3^-	(S) 115°

67. Match list-I with list-II and select the correct answer :-

Species	HMH/FMF bond angle
(A) NF_3	(P) 102°
(B) PF_3	(Q) 97.8°
(C) NH_3	(R) 107.3°
(D) PH_3	(S) 93.8°

68. Match list-I with list-II and select the correct answer :-

Molecule/Species	Unpaired electron resides in
(A) NO_2	(P) vacant d-orbital
(B) ClO_2	(Q) sp^2 -orbital
(C) ClO_3	(R) sp^3 -orbital
(D) CH_3	(S) vacant p-orbital

69. Match the column :-

Column - I Compound	Column - II Characteristic feature
(A) POCl_3	(P) Covalent bond present
(B) XeO_2F_2	(Q) Molecule violates octet rule
(C) $\text{CH}_3\text{—NC}$	(R) One lone pair on any atom
(D) CO	(S) π -bond present

70. Match the column :-

Column - I	Column - II
Compound	Characteristic feature
(A) XeO_4	(P) $p_\pi-p_\pi$ bond present
(B) $\text{CH}_2=\text{SF}_4$	(Q) $p_\pi-d_\pi$ bond present
(C) $\text{O}=\text{C} \begin{array}{l} \diagup \text{Cl} \\ \diagdown \text{Cl} \end{array}$	(R) Number of total lone pair present is even
(D) XeO_3F_2	(S) Non polar molecule

71. Match the column :-

Column - I	Column - II
Compound	Hybridisation
(A) XeO_6^{4-}	(P) sp
(B) IF_7	(Q) sp^2
(C) CO_3^{2-}	(R) sp^3d^3
(D) $\text{Ag}(\text{CN})_2^-$	(S) sp^3d^2

72. Match the column :-

Column - I	Column - II
Atom in a compound	Hybridisation
(A) Carbon atom in cyclopentadienyl anion	(P) sp^2 hybridisation
(B) Oxygen atom in diethyl ether	(Q) sp^3 hybridisation
(C) Carbon atom in singlet carbene	(R) one lone pair of e^- present (localised)
(D) Nitrogen atom in triethyl amine	(S) two lone pair of e^- present (localised)

73. Match the column :-

Column - I	Column - II
Molecule/ Intermediate	Shape/Hybridisation of central atom
(A) NO_2	(P) sp^2
(B) Trifluoro methyl radical	(Q) sp^3
(C) Methyl radical	(R) Trigonal planar
(D) ClO_3	(S) Pyramidal

74. Match the column :-

Column - I	Column - II
Compound	Features
(A) XeO_4	(P) $p\pi-p\pi$ bond present
(B) SO_3	(Q) $p\pi-d\pi$ bond present
(C) COCl_2	(R) Number of total lone pair is even
(D) $\text{H}_2\text{S}_2\text{O}_7$	(S) sp^2 hybridisation
	(T) sp^3 hybridisation

75. Match the column :-

Column - I	Column - II
Compound	Features
(A) Be_2Cl_4	(P) $3c-4e$ bond is present
(B) C_2Cl_6	(Q) $3c-2e$ bond is present
(C) B_2H_6	(R) $2c-2e$ bond is present
(D) $\text{Al}_2(\text{C}_6\text{H}_5)_6$	(S) Molecule is planar

76. Match the column :-

Column - I	Column - II
Compound	Features
(A) BF_3	(P) electron deficient molecule either in monomer or in dimer form
(B) SiCl_4	(Q) non-polar in dimeric form
(C) ICl_3	(R) bridging atoms are present in molecular plane in its dimeric form
(D) BeCl_2	(S) dimer form does not exist

77. Match the column :-

Column - I	Column - II
Species	Features
(A) XeF_5^+	(P) Axial bond length is longer than equatorial bond length
(B) IF_4^-	(Q) sp^3d hybridisation of central atom
(C) PCl_5	(R) Non-planar species
(D) ClF_3	(S) Central atom has 2 lone pair of electrons on it
	(T) d_z^2 orbital is involved in hybridisation

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78. Match the column :-

Column - I**Process**

- (A) Solubility of noble gas is in water
 (B) The interaction between chlorobenzene molecule
 (C) Interaction between alkane
 (D) Dissolution of ionic compound in water

Column - II**Operating interaction involved**

- (P) Debye forces
 (Q) Keesom forces
 (R) London forces
 (S) Weak forces

79. Match the column :-

Column - I**Process**

- (A) Clathrate compound of Xe in ice
 (B) Liquefaction of Xe gas
 (C) Liquefaction of HCl gas
 (D) Hydration of Na^+

Column - II**Operating interaction involved**

- (P) Ion-dipole
 (Q) Dipole-dipole
 (R) Dipole-Induced dipole
 (S) London forces

80. Match the column :-

Column - I**Compound**

- (A) $\text{P}(\text{SiH}_3)_3$
 (B) $\text{N}(\text{CH}_3)_3$
 (C) $\text{N}(\text{SiH}_3)_3$
 (D) SiF_4

Column - II**Features**

- (P) pyramidal in shape
 (Q) no back bonding possible
 (R) trigonal planar in shape
 (S) $2p_\pi - 3d_\pi$ back bonding possible
 (T) tetrahedral in shape

81. Match the column :-

Column - I**Compound**

- (A) Dimer of acetic acid
 (B) Dimer of borane
 (C) Dimer of AlCl_3
 (D) Dimer of NO_2

Column - II**Features**

- (P) $2c-2e$ bond present
 (Q) $3c-2e$ bond present
 (R) $3c-4e$ bond present
 (S) H bond present
 (T) Molecule existing

82. Match the column :-

Column - I**Compound**

- (A) CN_2^{2-} , CO_2
 (B) Inorganic benzene, borazene
 (C) PCl_3F_2 , PF_3Cl_2
 (D) BrF_5 , XeF_5^+

Column - II**Features**

- (P) same number of total electrons
 (Q) pair of isostructural species
 (R) same number of valence electrons
 (S) both structures are planar
 (T) both cases, central atom has same hybridisation

83. Match the column :-

Column - I**Species**

- (A) C_{12}O_9
 (B) $\text{Si}_2\text{O}_7^{6-}$
 (C) $\gamma - \text{SO}_3$
 (trimeric form)
 (D) Cl_2O_7

Column - II**Features**

- (P) M–O–M linkage is present
 (Q) open chain silicate
 (R) cyclic structure
 (S) anhydride of HClO_4

84. Match the column :-

Column - I**Species**

- (A) PtCl_4^{2-}
 (B) CrO_4^{2-}
 (C) XeF_4
 (D) BF_4^-

Column - II**Hybridisation of central atom**

- (P) sp^3
 (Q) sp^3d^2
 (R) d^3s
 (S) dsp^2

85. Match the column :-

Column - I	Column - II
Species	Features
(A) AlF_3	(P) dipole moment is zero
(B) $\text{I}(\text{CN})_2^-$	(Q) overall molecule is linear
(C) Ba_2XeO_6	(R) ionic compound exist as lattice
(D) $\text{H}_3\text{Ge} - \text{N} = \text{C} = \text{S}$	(S) the central atom has d_z^2 orbital for hybridisation
	(T) at least one of the atom is sp hybridised

86. Match the column :-

Column - I	Column - II
Species	Features
(A) H_3BO_3 ; H_3PO_4	(P) Ist cpd is planar ; IInd cpd is non planar
(B) $\text{CH}_2=\text{C}=\text{O}$; $\text{CH}_2=\text{C}=\text{CH}_2$	(Q) Ist cpd is not proton donor but IInd cpd is proton donor
(C) LiI , LiF	(R) Ist cpd is covalent but IInd cpd is ionic in nature
(D) H_2O , H_2S	(S) For Ist cpd hybridisation of central atom is sp^3 whereas for IInd cpd, hybridisation of central atom in molecule is not defined

87. Match the column :-

Column - I	Column - II
Oxyacids of N	Features
(A) HNO_4	(P) Monobasic acid
(B) HNO_3	(Q) C.B. has 2 identical R.S.
(C) HNO_2	(R) Compound showing G.I.
(D) $\text{H}_2\text{N}_2\text{O}_2$	(S) Highest oxidation state of central atom
	(T) Peroxy linkage present

88. Match the column :-

Column - I	Column - II
Type of π bond	Molecules
(A) $2p\pi-2p\pi$	(P) $\text{O}=\text{C}=\text{O}$
(B) $2p\pi-3p\pi$	(Q) $\text{S}=\text{C}=\text{S}$
(C) $2p\pi-3d\pi$	(R) POCl_3
(D) $3p\pi-3d\pi$	(S) PSCl_3
	(T) SOCl_2

89. Match the column :-

Column - I	Column - II
Oxyacid of chlorine	Features
(A) HClO_4	(P) weakest acid
(B) HClO_3	(Q) strongest acid
(C) HClO_2	(R) C.B. has 3 identical R.S.
(D) HOCl	(S) sp^3 hybridisation of central atom
	(T) Formula of anhydride is Cl_2O_7

90. Match the column :-

Column - I	Column - II
Miscellaneous oxy acid	Features
(A) HMnO_4	(P) Monobasic acid
(B) H_2CrO_4	(Q) Pink / purple colour
(C) H_2CO_3	(R) Yellow colour
(D) $\text{CH}_3\text{CO}_2\text{H}$	(S) C-O bond lengths are not identical
	(T) Thermally unstable and gives CO_2 gas on heating

91. Match the column :-

Column - I	Column - II
Compound/molecule	Structures
(A) S_8	(P) Zig-zag structure
(B) H_2O_2	(Q) open book like structure
(C) $(\text{HF})_n$	(R) crown shape
(D) CrO_5	(S) Butterfly structure

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92. Match the column :-

Column - I**Compound**

- (A) NH_4ClO_4
 (B) MgC_2O_4
 (C) HCOONa (400°C)
 (D) $(\text{NH}_4)_2\text{CO}_3$

Column - II**Products on heating**

- (P) CO_2 gas is evolved
 (Q) H_2 gas is evolved
 (R) N_2 gas is evolved
 (S) Same gas is evolved which is obtained by heating $(\text{NH}_4)_2\text{SO}_4$
 (T) Intramolecular redox reaction

93. Match the column :-

Column - I**Compounds**

- (A) AlF_3 , SF_4 , PF_5
 (B) $\text{O}(\text{CH}_3)_2$, $\text{S}(\text{CH}_3)_2$, $\text{Se}(\text{CH}_3)_2$
 (C) HCl , HF , HBr
 (D) CH_4 , CH_3Cl , CH_3OH , $\text{CH}_3\text{CO}_2\text{H}$

Column - II**Boiling point**

- (P) 1st compound has highest boiling point
 (Q) 3rd compound has highest boiling point
 (R) 2nd compound has highest boiling point
 (S) 4th compound has highest boiling point

94. Match the column :-

Column - I**Oxyacids of P**

- (i) H_3PO_4
 (ii) H_3PO_3
 (iii) H_3PO_2
 (iv) $\text{H}_4\text{P}_2\text{O}_7$
 (v) $\text{H}_5\text{P}_3\text{O}_{10}$
 (vi) $\text{H}_3\text{P}_3\text{O}_9$
 (vii) $\text{H}_4\text{P}_2\text{O}_6$
 (viii) H_3PO_5

Column - II**Features**

- (P) All Hs' are not ionisable
 (Q) Dibasic reducing agent
 (R) Monobasic reducing agent
 (S) P-H linkage present
 (T) P-O-P linkage present
 (U) P-P linkage present
 (V) Cyclic structure
 (W) Peroxy linkage present
 (X) Highest oxidation state of P
 (Y) Tetrabasic acid
 (Z) Tribasic acid

95. Match the column :-

Column - I**Oxyacids of S**

- (i) H_2SO_2 (sulphylic acid)
 (ii) H_2SO_3
 (iii) H_2SO_4
 (iv) H_2SO_5 (Caro's acid)
 (v) $\text{H}_2\text{S}_2\text{O}_7$ (Oleum)
 (vi) $\text{H}_2\text{S}_2\text{O}_8$ (Marshall acid)
 (vii) $\text{H}_2\text{S}_2\text{O}_3$ (Thiosulfuric acid)
 (viii) $\text{H}_2\text{S}_2\text{O}_2$ (Thiosulfurous acid)
 (ix) $\text{H}_2\text{S}_n\text{O}_6$ (Polythionic acid)
 (x) $\text{H}_2\text{S}_n\text{O}_4$ (Polythionous acid)

Column - II**Features**

- (P) S-O-S linkage present
 (Q) sp^3 hybridisation of S
 (R) Peroxy linkage present
 (S) S=S linkage present
 (T) S-S linkage present
 (U) At least one sulphur atom having oxidation state of (+6)
 (V) Oxidation state of S is zero in S-S linkage
 (W) Dibasic acid
 (X) Sulphur can have -ve oxidation state
 (Y) Oxyanion undergoes resonance

96. Match the column :-

Column - I**Compounds**

- (A) Blue vitriol
 (B) Gypsum
 (C) Pure orthophosphoric acid
 (D) Chloral hydrate

Column - II**Type of bond/ interaction**

- (P) Ionic bond
 (Q) Covalent bond
 (R) Hydrogen bond
 (S) Resonance stabilisation

97. Match the column :-

Column - I**Different system**

- (A) Liquid bromine
 (B) Noble gas clathrate
 (C) Solution of sodium fluoride in water
 (D) Liquid methylamine

Column - II**Type of bond/interaction**

- (P) hydrogen bond
 (Q) ion-dipole force
 (R) london dispersion force
 (S) dipole induced dipole interaction

98. Match the column :-

Column - I**Molecules**

- (A) F_2 , N_2H_4 , H_2O_2 , NF_3
 (B) XeF_4 , SO_2 , PCl_3F_2 , $PCl_{5(g)}$
 (C) H_3BO_3 , Graphite, NH_4Cl , Na_2CO_3 , SO_3
 (D) Inorganic benzene, Benzene,

Column - II**Features**

- (P) all except one are examples of non polar compounds
 (Q) all except one are planar molecules
 (R) all except one have at least one atom which is sp^2 hybridised
 (S) all except one have less bond energy because ICl_4^- , SO_4^{2-} , NO_3^- of lone pair - lone pair repulsion

99. Match the column :-

Column - I**Compounds**

- (A) H_2O
 (B) H_2O_2
 (C) D_2O
 (D) CH_3OH

Column - II**Feature/Use**

- (P) polar solvent
 (Q) non linear molecule
 (R) more acidic than C_2H_5OH
 (S) used as moderator in nuclear reactions

100. Match the column :-

Column - I**Observed order**

- (A) $H_2 < CO_2 < H_2O$
 (B) $PH_3 < AsH_3 < NH_3 < SbH_3$
 (C) $D_2O > H_2O$
 (D) $(CH_3)_3N < CH_3NH_2 < (CH_3)_2NH$

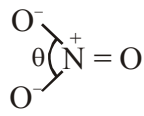
Column - II**Property**

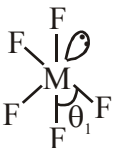
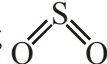
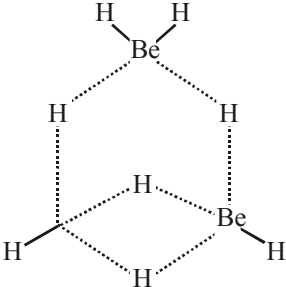
- (P) order of boiling point
 (Q) order of density
 (R) order of intermolecular forces of attraction
 (S) order of basic strength in aqueous medium

EXERCISE # IV**Integer Type :**

- Molecules having total eight lone pair present :-
 XeO_4 ; $HClO_4$; H_2SO_4 ; N_2O_3 ; $C_3N_3Cl_3$; $COCl_2$; H_2SO_3
- How many correct bond energy order is given :-
 (a) $H-H > C-H > C-C$
 (b) $C-F > C-Cl > C-Br > C-I$
 (c) $Cl-Cl > Br-Br > F-F > I-I$
 (d) $C-C > Si-Si > Ge-Ge$
 (e) $N-N > P-P > As-As > Sb-Sb > Bi-Bi$
 (f) $C \equiv N > C=N > C-N$
- How many molecules /ions are result of dative σ -bond ?
 (a) NH_4^+ (b) $H_3N^+ - BF_3^-$
 (c) I_3^- (d) BH_4^-
 (e) SbF_6^- (f) H_3O^+
- How many molecule having negative fractional oxidation state for nitrogen -
 N_2O_3 ; N_2O_5 ; Hydrazine; Hydrazoic acid; NF_3 ; N_2O ; NH_3
- How many molecules do not exist ?
 CCl_6^{2-} ; CBr_6^{2-} ; SiF_6^{2-} ; AsF_5 ; AsH_5 ; XeF_2 ; $KrCl_2$

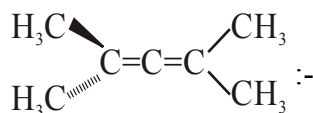
1.42 CHEMICAL BONDING

6. How many of the following has M–O–M linkage?
 (a) trimer of SO_3 which is represented as S_3O_9
 (b) $\text{H}_2\text{S}_2\text{O}_8$
 (c) Cl_2O_7
 (d) P_4O_{10}
 (e) P_4O_6
 (f) H_2SO_5
 (g) $\text{H}_2\text{S}_2\text{O}_7$
7. For how many molecule where all following points are satisfied :
 (i) central atom is sp^3 hybridised
 (ii) tetrahedral in shape
 (iii) π bond is present along with σ bond
 CH_4 , SiCl_4 , NH_4^+ , BF_4^- , XeO_4 , ClO_4^- , SO_4^{2-} , PO_4^{3-}
8. How many molecules / species are example of 16 e^- system with linear in shape -
 (a) CO_2 (b) Nitronium ion
 (c) Azide ion (d) Nitrous oxide
 (e) Cyanate ion (f) Isocyanate ion
 (g) Cyanamide (CN_2^{2-}) ion
9. Total number of odd electron system :
 NO_2 ; NO ; ClO_2 ; OF ; O_3^- ; O_2 ; ClO_3
10. Find the number of sp^3 hybrid orbital in γSO_3 :-
11. How many molecules having molecular formula PCl_nF_m will be polar -
 (a) $m = 0$; $n = 5$ (b) $n = 0$; $m = 5$
 (c) $n = 3$; $m = 2$ (d) $m = 3$; $n = 2$
 (e) $n = 4$; $m = 1$
12. Identify number of molecules in which only two types of bond angles are present
 CH_4 ; CH_3Cl ; CH_2Cl_2 ; CHCl_3 (Chloroform); CCl_4 (Carbon tetrachloride); COCl_2
13. If number of planes present in CH_4 molecule which contain maximum number of atoms of corresponding molecule is x then find out $x - 1$.
14. If number of planes present in IF_7 molecule which contain maximum number of atoms of corresponding molecule is y then find out y.
15. Find the number of planes containing maximum number of fluorine atoms (only) in SbF_5 .
16. How many molecule / species central atom is sp^3 hybridised
 BeF_2 ; BF_4^- ; SO_3 ; AlCl_4^- ; NH_4^+ ; ClO_4^- ; N_3^- ; XeO_4
17. Find the number of molecule in which p – p sideways overlapping is present -
 N_2 ; HCl ; Cl_2 ; HF ; S_8 ; F_2 ; HCP
18. Total number of bond angle of 90° in SiF_6^{2-} is
19. Find the number of atomic orbital which forms only σ bond.
 s , p_x , p_y , p_z , d_{xy} , $d_{x^2-y^2}$
20. The number of molecule having perfectly tetrahedral shape among the following are -
 NH_3 , CCl_4 , PO_4^{3-} , CHCl_3 , BF_4^-
21. Find the ratio of number of lone pairs to number of π bonds in a particular resonating structure of SO_4^{2-} -
22. In which of the following compounds all atoms are present in same plane -
 (a) I_3^- (b) ICl_4^-
 (c) XeF_4 (d) XeF_5^+
 (e) XeF_5^- (f) $\text{C}(\text{CN})_4$
23. In which of the following compound octet of central atom is incomplete or molecule is hypovalent-
 (a) BF_3 (b) ICl
 (c) ClF_3 (d) BeH_2
 (e) PCl_3 (f) AlCl_3
24. Identify the total number of molecules (amongst the following) which have bond angle lesser than θ -

 CO_2 ; CH_4 ; BF_3 ; $\text{CH}_3\text{--O--CH}_3$; SOCl_2 ; NH_3 ; H_2O
25. Find the number of chemical species in which bond angle is less than the bond angle in CCl_4 -
 PH_3 ; H_2S ; OCl_2 ; $\text{O}(\text{CH}_3)_2$; ClO_2 ; NCl_3

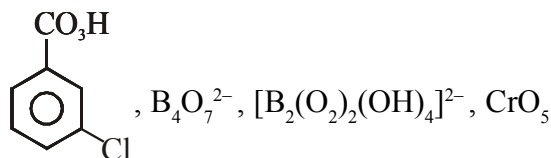
26. Find the molecule in which bond angle is equal to the expected bond angle according to hybridisation of underlined atom -
 (a) $\underline{\text{P}}\text{F}_3$ (b) $\underline{\text{B}}\text{F}_3$
 (c) $\underline{\text{P}}\text{Cl}_3$ (d) $\underline{\text{C}}\text{OCl}_2$
 (e) $\underline{\text{N}}\text{H}_4^+$ (f) $\underline{\text{N}}\text{Me}_3$
27. Find out no. of total lone pair present in trimer of isocyanic acid $(\text{HNCO})_3$.
28. If number of π bonds present in melamine = x and if no. of lone pair present in melamine = y then find out y/x.
29. If number of σ bond present in $\text{C}_3\text{N}_3(\text{N}_3)_3$ cyanauric triazide = x and if number of lone pair present in $\text{C}_3\text{N}_3(\text{N}_3)_3$ is = y & if no. of sp hybridised N present = z then $\frac{y+z}{x}$ is :-
30. Drago rule is applicable for which of the following molecule -
 SiH_4 ; PH_3 ; PF_3 ; H_2S ; H_2Te ; AsH_3 ; PH_4^+ ; SbH_3
31. How many structures are cyclic in nature?
 (a) S_3O_9 (b) $\text{P}_3\text{O}_9^{3-}$
 (c) $\text{P}_4\text{O}_{12}^{4-}$ (d) $(\text{PNCl}_2)_3$
 (e) $\text{H}_2\text{S}_2\text{O}_7$ (f) $\text{H}_2\text{S}_2\text{O}_8$
32. Number of S–O–S linkage present in cyclic trimer of SO_3 .
33. Number of bond angle which are at 90° w.r.t. each other present in SF_4 .
34. Number of bond angle which are at 90° w.r.t. each other present in BrF_5 .
35. In which case compound has bond angle which is not exactly equal according to hybridisation?
36. 
 If $\widehat{\text{FMF}}$ bond angle in MF_5 is $\theta = 87^\circ$.
 & if $\widehat{\text{FMF}}$ bond angle in MF_4 is θ_2
 Then find out $\theta_2 - \theta_1$
37. $\text{XeF}_4 + \text{PF}_5 \longrightarrow \text{XeF}_3^+ + \text{PF}_6^-$
 If total numbers of 90° angle in reactant = m then find out and if total numbers of 9° angles in product = n.
38. In how many of the following species, % s character of all bonds is not identical.
 IF_7 ; CH_4 ; SF_4 ; XeF_3^+ ; OCl_2 ; ICl_2^- ; BF_3 ; 
39. $\cos 92^\circ = -0.03$, What is the % of s-character in hybrid orbital containing lone pair of e^- s :-
40. Find out number of molecule which do not exist -
 XeF_3^- ; XeI_3^- ; IF_3^{2-} ; FeI_3 ; Al_3 ; ClI_3^- ; PI_5 in T.b.p state.
41. Find the number of molecules which has $3p_\pi - 2p_\pi$ back bonding :
 OCl_2 ; $:\text{CCl}_2$; BCl_3 ; CCl_3
42. Find the number of molecules in which $2p_\pi - 3d_\pi$ back bonding is present.
 OCl_2 ; BF_3 ; $\text{N}(\text{SiH}_3)_3$; $\text{N}(\text{CH}_3)_3$;
 $\overset{\ominus}{\text{C}}\text{Cl}_3$; $\text{O}(\text{SiH}_3)_2$; $:\text{CCl}_2$
43. Correct statement regarding B_2H_6 :-
 (a) Terminal B–H bond length 119 pm and it is $2c-2e$ bond
 (b) Bridging B–H bond length 113 pm and it is $3c-2e$ bond
 (c) It is non-polar molecule
 (d) There are only twelve bonding electrons available
 (e) B atom in B_2H_6 is approximately sp^3 hybridised
 (f) It has momentarily complete octet
 (g) The plane containing the two boron atoms and four terminal hydrogen atom is perpendicular to that containing the two bridging hydrogen atoms.
 (h) The kind of overlap present in bridge bond in B_2H_6 sp^3-s-sp^3
44. In this following molecule the number of bridging bond :-


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45. Find the maximum number of atoms may lie in a single plane for the following molecule



46. Maximum number of $2c-2e$ bonds which are formed by sp^3-sp^3 overlapping in $Al_2(CH_3)_6$:-
47. Maximum number of atoms present in one plane in $Al_2(CH_3)_6$:-
48. How many compounds on reaction with H_2O give product with $pH < 7$ in aqueous solution ?
 MgO ; Na_2O ; SO_3 ; Cl_2O_7 ; P_4O_{10} ; BaO ; N_2O_3 ; SO_2
49. Find out number of molecules which on complete hydrolysis produce sulphuric acid as one of the product :-
 SOF_4 ; SF_4 ; $H_2S_2O_8$; $H_2S_2O_7$; SO_2Cl_2 ; S_3O_9
50. Find the number of water molecules required for complete hydrolysis of one molecule of XeF_6 :-
51. Find the number of water molecules which are used for complete hydrolysis of one molecular P_4O_{10} :-
52. How many water molecules are associated with hydrogen bond but not coordinated with metal ion in $CuSO_4 \cdot 5H_2O$:-
53. Among the following find out the total number of molecules have more % of p-character in E-H bond than that of PH_3 :-
 AsH_3 ; SbH_3 ; NH_3 ; H_2O ; B_2H_6
54. Find out the number of molecules which are non-planar :-
 $C_2(CN)_4$; P_4 ; S_8 ; H_2O_2 ; F_2O_2 ; CF_4 ; S_2F_{10}
55. For how many molecule C-C bond length is shorter than C-C bond length in graphite :-
 (i) acetylene (ii) ethene (iii) benzene (iv) ethane
56. How many molecules have two peroxy linkage :-
57. In IF_7 ; what is the total number of bonds which are at 72° ?
58. For C_{70} fullerene find the number of hexagonal face :-
59. For which compound all M-O bonds have bond order 1.5 :-
 SO_2 ; CH_3CO_2H ; $CH_3CO_2^-$; NO_2^- ; NO_3^-
60. For $[ABSi_3O_x]$, where A & B have +2 & +4 oxidation state respectively. What is the value of X?
61. Number of S-S linkage in $H_2S_6O_6$:-
62. Find the number of S atom which are at zero oxidation state in $H_2S_6O_6$:-
63. Find the number of species having X-O-X linkage :-
 Pyroxene / chain silicate, Perdisulphate ion, Anion of Caro's acid, Pyrosulphate ion, Tetrathionate ion, Thiosulphate ion.
64. How many orbitals are having higher energy compared to σ_{2s} for Li_2 molecule from M.O. given below :-
 σ_{1s}^* ; σ_{2s}^* ; σ_{1s} ; π_{2py} ; π_{2py}^* ; σ_{2pz}^* ; π_{2pz}^* ; π_{2pz}
65. How many of the following are paramagnetic :-
 He_2^+ ; H_2^+ ; H_2^- ; NO_2 ; NO ; OF
66. How many molecules / ion have bond order less than or equal to two
 B_2 ; C_2 ; O_2^{2-} ; O_2^{+1} ; H_2^{+1}
67. Considering MOT find out number of diatomic species which do not have σ bond :-
 F_2 ; C_2 ; N_2 ; H_2 ; C_2^{2-}
68. Find the ratio of π electrons in the C_2 molecule with that of B_2 molecule according to MOT :-
69. Find the number of chemical species which are paramagnetic in nature :-
 O_2^{2-} ; NO ; ClO_2 ; OF ; B_2
70. Find out the number of species having paramagnetic in nature as well as fractional bond order :-
 B_2 ; NO ; OF ; N_2 ; KO_2 ; K_2O_2



71. If sp mixing is not operative then total number of electron in B.M.O. of N_2^+ will be :-
72. Find out total number of species containing N–N/P–P linkage :-
 N_2O ; N_2O_3 (symmetrical) ; N_2O_5 ; $H_2N_2O_2$;
 $H_4P_2O_6$; P_4O_6 ; P_4O_{10}
73. Find the number of chemical species which are isoelectronic and have same bond order as of CO^- :-
 CN^- ; NO^+ ; N_2 ; C_2^{2-} ; O_2^{2+}
74. Find the number of compounds which can form intermolecular hydrogen bonding but cannot form intramolecular hydrogen bonding :-
 Dimer of acetic acid ; Quinol ; Catechol ;
 $(KHCO_3)_2$; $(NaHCO_3)_n$
75. Number of bond pairs at 90° angle in XeF_5^- is x
 Number of bond pairs at 90° angle in XeF_4 is y
 The value of $y - x$:-
76. Select the no. of orbitals which can produce π bond while overlapping with an s-orbital -
 (a) p_x (b) p_y
 (c) p_z (d) d_{xy}
 (e) d_{yz} (f) s
77. Find out total number of σ bond in $P_6O_{18}^{6-}$:-
78. How many statements are correct :-
 (a) Li_2O has highest melting point among all alkali metal oxide
 (b) BeO has highest melting point among all alkaline earth metal fluoride
 (c) LiF has highest melting point among all alkali metal fluoride
 (d) BeF_2 has highest melting point among all alkaline earth metal fluoride
 (e) Li_3N has highest melting point among all alkali metal nitride
 (f) Be_3N_2 has highest melting point among all alkaline earth metal nitride
79. The bond order of the underlined species $NOHSO_4$ is :-

80. The difference in the number of σ and π bonds in trimer of SO_3 is (considering only covalent bond is present) :-

EXERCISE # V(A) JEE-MAIN

1. Which of the following statements is true? [AIEEE–2002]
 (1) HF is less polar than HBr
 (2) Water does not contain any ions
 (3) Chemical bond formation takes place when forces of attraction overcome the forces of repulsion
 (4) In covalent bond, transfer of electrons takes place
2. The reason for double helical structure of DNA is operation of: [JEE2003]
 (1) dipole–dipole interaction
 (2) hydrogen bonding
 (3) electrostatic attractions
 (4) vander Walls forces
3. Which of the following pair of molecules will have permanent dipole moments for both members [AIEEE–2003]
 (1) NO_2 and CO_2 (2) NO_2 and O_3
 (3) SiF_4 and CO_2 (4) SiF_4 and NO_2
4. The pair of species having identical shapes for molecules of both species is [AIEEE–2003]
 (1) XeF_2 , CO_2 (2) BF_3 , PCl_3
 (3) PF_5 , IF_5 (4) CF_4 , SF_4
5. The correct order of bond angles (smallest first) in H_2S , NH_3 , BF_3 and SiH_4 is :- [AIEEE–2004]
 (1) $H_2S < NH_3 < SiH_4 < BF_3$
 (2) $NH_3 < H_2S < SiH_4 < BF_3$
 (3) $H_2S < SiH_4 < NH_3 < BF_3$
 (4) $H_2S < NH_3 < BF_3 < SiH_4$
6. The bond order in NO is 2.5 while that in NO^+ is 3. Which of the following statements is true for these two species? [AIEEE–2004]
 (1) Bond length in NO^+ is equal to that NO
 (2) Bond length in NO is greater than NO^+
 (3) Bond length in NO^+ is greater than NO
 (4) Bond length is unpredictable

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7. The states of hybridization of boron and oxygen atoms in boric acid (H_3BO_3) are respectively [AIEEE–2004]
 (1) sp^3 and sp^2 (2) sp^2 and sp^3
 (3) sp^2 and sp^2 (4) sp^3 and sp^3
8. Which one of the following has the regular tetrahedral structure? [AIEEE–2004]
 (1) BF_4^- (2) SF_4
 (3) XeF_4 (4) $[\text{Ni}(\text{CN})_4]^{2-}$
 (Atomic nos.: B = 5, S = 16, Ni = 28, Xe = 54)
9. The maximum number of 90° angles between bond pair–bond pair of electrons is observed – [AIEEE–2004]
 (1) dp^2 hybridization (2) sp^3d hybridization
 (3) dsp^3 hybridization (4) sp^3d^2 hybridization
10. Beryllium and aluminium exhibit many properties which are similar. But, the two elements differ in :– [AIEEE 2004]
 (1) Forming covalent halides
 (2) Forming polymeric halides
 (3) Forming maximum covalency in compounds
 (4) Exhibiting amphoteric nature in their oxides
11. Which one of the following species is diamagnetic in nature? [AIEEE2005]
 (1) He_2^+ (2) H_2
 (3) H_2^+ (4) H_2^-
12. Lattice energy of an ionic compound depends upon [AIEEE2005]
 (1) charge on the ion only
 (2) size of the ion only
 (3) packing of the ion only
 (4) charge and size of the ion
13. The molecular shapes of SF_4 , CF_4 and XeF_4 are– [AIEEE2005]
 (1) the same with 2, 0 and 1 lone pair of electrons on the central atom, respectively
 (2) the same with 1, 1 and 1 lone pair of electrons on the central atom, respectively
 (3) the same with 1, 1 and 2 lone pair of electrons on the central atom, respectively
 (4) the same with 1, 0 and 2 lone pair of electrons on the central atom, respectively
14. Of the following sets which one does not contain isoelectronic species. [AIEEE 2005]
 (1) PO_4^{3-} , SO_4^{2-} , ClO_4^-
 (2) CN^- , N_2 , C_2^{2-}
 (3) SO_3^{2-} , CO_3^{2-} , N_3^-
 (4) BO_3^{3-} , CO_3^{2-} , NO_3^-
15. The number and type of bond between two carbon atom in calcium carbide are: [AIEEE–2005]
 (1) one sigma, one pi (2) one sigma, two pi
 (3) two sigma, one pi (4) two sigma, two pi
16. Which of the following molecules / ions does not contain unpaired electron? [AIEEE–2005]
 (1) N_2^+ (2) O_2
 (3) O_2^{2-} (4) B_2
17. Among the following mixtures, dipole–dipole as the major interaction, is present in : [AIEEE–2006]
 (1) KCl and water
 (2) benzene and carbon tetrachloride
 (3) benzene and ethanol
 (4) acetonitrile and acetone
18. A metal, M forms chlorides in its +2 and +4 oxidation states. Which of the following statement about these chlorides is correct? [AIEEE–2006]
 (1) MCl_2 is more ionic than MCl_4
 (2) MCl_2 is more easily hydrolysed than MCl_4
 (3) MCl_2 is more volatile than MCl_4
 (4) MCl_2 is more soluble in anhydrous ethanol than MCl_4
19. In which of the following molecules / ions are all the bonds not equal : [AIEEE 2006]
 (1) XeF_4 (2) BF_4^-
 (3) SF_4 (4) SiF_4
20. The decreasing values of bond angles from NH_3 (106°) to SbH_3 (91°) down group–15 of the periodic table is due to – [AIEEE 2006]
 (1) decreasing lp–bp repulsion
 (2) increasing electronegativity
 (3) increasing bp–bp repulsion
 (4) increasing p–orbital character in sp^3

21. In which of the following ionization process, the bond order has increased and the magnetic behaviour has changed [AIEEE 2007]
 (1) $\text{NO} \rightarrow \text{NO}^+$ (2) $\text{O}_2 \rightarrow \text{O}_2^+$
 (3) $\text{N}_2 \rightarrow \text{N}_2^+$ (4) $\text{C}_2 \rightarrow \text{C}_2^+$
22. Which of the following hydrogen bond is the strongest? [AIEEE 2007]
 (1) $\text{F}-\text{H} \cdots \text{F}$ (2) $\text{O}-\text{H} \cdots \text{O}$
 (3) $\text{O}-\text{H} \cdots \text{F}$ (4) $\text{O}-\text{H} \cdots \text{N}$
23. Which of the following species exhibits the diamagnetic behaviour? [AIEEE 2007]
 (1) O_2^+ (2) O_2
 (3) NO (4) O_2^{2-}
24. Which one of the following pairs of species have the same bond order [AIEEE 2008]
 (1) CN^- and NO^+ (2) CN^- and CN^+
 (3) O_2^- and CN^- (4) NO^+ and CN^+
25. The bond dissociation energy of $\text{B}-\text{F}$ in BF_3 is 646 kJ mol^{-1} whereas that of $\text{C}-\text{F}$ in CF_4 is 515 kJ mol^{-1} . The correct reason for higher $\text{B}-\text{F}$ bond dissociation energy as compared to that of $\text{C}-\text{F}$ is :- [AIEEE-2009]
 (1) Significant $\text{p}\pi-\text{p}\pi$ interaction between B and F in BF_3 whereas there is not possibility of such interaction between C and F in CF_4 .
 (2) Lower degree of $\text{p}\pi-\text{p}\pi$ interaction between B and F in BF_3 than that between C and F in CF_4
 (3) Smaller size of B-atom as compared to that of C-atom
 (4) Stronger σ bond between B and F in BF_3 as compared to that between C and F in CF_4
26. Using MO theory predict which of the following species has the shortest bond length? [AIEEE-2009]
 (1) O_2^- (2) O_2^{2-}
 (3) O_2^{2+} (4) O_2^+
27. The hybridisation of orbitals of N atom in NO_3^- , NO_2^+ and NH_4^+ are respectively:- [AIEEE-2011]
 (1) sp , sp^3 , sp^2 (2) sp^2 , sp^3 , sp
 (3) sp , sp^2 , sp^3 (4) sp^2 , sp , sp^3
28. The structure of IF_7 is :- [AIEEE-2011]
 (1) octahedral (2) pentagonal bipyramid
 (3) square pyramid (4) trigonal bipyramid
29. Among the following the maximum covalent character is shown by the compound: [AIEEE-2011]
 (1) AlCl_3 (2) MgCl_2
 (3) FeCl_2 (4) SnCl_2
30. Which of the following has maximum number of lone pairs associated with Xe
 (1) XeO_3 (2) XeF_4
 (3) XeF_6 (4) XeF_2
31. The number of types of bonds between two carbon atoms in calcium carbide is :- [AIEEE-2012]
 (1) One sigma, two pi
 (2) One sigma, one pi
 (3) Two sigma, one pi
 (4) Two sigma, two pi
32. The molecule having smallest bond angle is - [AIEEE-2012]
 (1) PCl_3 (2) NCl_3
 (3) AsCl_3 (4) SbCl_3
33. In which of the following pairs the two species are not isostructural? [AIEEE-2012]
 (1) AlF_6^{3-} and SF_6
 (2) CO_3^{2-} and NO_3^-
 (3) PCl_4^+ and SiCl_4
 (4) PF_5 and BrF_5
34. The number of $\text{S}-\text{S}$ bonds in SO_3 , $\text{S}_2\text{O}_3^{2-}$, $\text{S}_2\text{O}_6^{2-}$ and $\text{S}_2\text{O}_8^{2-}$ respectively are :- [JEE-MAINS-2012] (On-line)
 (1) 1,0,1,0 (2) 0,1,1,0
 (3) 1,0,0,1 (4) 0,1,0,1
35. Dipole moment is shown by :- [JEE-MAINS-2012] (ON-line)
 (1) trans-2,3-dichloro-2-butene
 (2) 1,2-dichlorobenzene
 (3) 1,4-dichlorobenzene
 (4) trans-1,2-dinitroethene

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36. Among the following species which two have trigonal bipyramidal shape :-
[JEE-MAINS-2012] (On-line)
- (I) NI_3 (II) I_3^-
(III) SO_3^{2-} (IV) NO_3^-
(1) II and III (2) III and IV
(3) I and IV (4) I and III
37. Among the following, the species having the smallest bond is -
[JEE-MAINS-2012] (On-line)
- (1) NO (2) NO^+
(3) O_2 (4) NO^-
38. Based on lattice energy and other considerations, which one of the following alkali metal chloride is expected to have the highest melting point :-
[JEE-MAINS-2012] (On-line)
- (1) RbCl (2) LiCl
(3) KCl (4) NaCl
39. Which of the following has the square planar structure :- [JEE-MAINS-2012] (On-line)
- (1) NH_4^+ (2) CCl_4
(3) XeF_4 (4) BF_4^-
40. The compound of Xenon with zero dipole moment is :- [JEE-MAINS-2012] (On-line)
- (1) XeO_3 (2) XeO_2
(3) XeF_4 (4) XeOF_4
41. Among the following molecule with the lowest dipole moment is: [JEE-MAINS-2012] (On-line)
- (1) CHCl_3 (2) CH_2Cl_2
(3) CCl_4 (4) CH_3Cl
42. The formation of molecular complex BF_3-NH_3 results in a change in hybridisation of boron :- [JEE-MAINS-2012]
- (1) from sp^3 to sp^3d (2) from sp^2 to dsp^2
(3) from sp^3 to sp^2 (4) from sp^2 to sp^3
(on-line)
43. Which one of the following molecules is expected to exhibit diamagnetic behaviour? [AIEEE-2013]
- (1) C_2 (2) N_2
(3) O_2 (4) S_2
44. Which of the following is the wrong statement? [JEE-M-2013]
- (1) ONCl and ONO are not isoelectronic
(2) O_3 molecule is bent
(3) Ozone is violet-black in solid state
(4) Ozone is diamagnetic gas
45. In which of the following pairs of molecules/ions, both the species are not likely to exist? [JEE-M-2013]
- (1) $\text{H}_2^+, \text{He}_2^{2-}$ (2) $\text{H}_2^-, \text{He}_2^{2-}$
(3) $\text{H}_2^{2+}, \text{He}_2$ (4) $\text{H}_2^-, \text{He}_2^{2+}$
46. Which of the following exists as covalent crystals in the solid state? [JEE-M-2013]
- (1) Iodine (2) Silicon
(3) Sulphur (4) Phosphorus
47. Stability of the species $\text{Li}_2, \text{Li}_2^-$ and Li_2^+ increases in the order of [JEE-M-2013]
- (1) $\text{Li}_2 < \text{Li}_2^+ < \text{Li}_2^-$ (2) $\text{Li}_2^- < \text{Li}_2^+ < \text{Li}_2$
(3) $\text{Li}_2 < \text{Li}_2^- < \text{Li}_2^+$ (4) $\text{Li}_2^- < \text{Li}_2 < \text{Li}_2^+$
48. Trigonal bipyramidal geometry is shown by. [JEE-Mains-2013, Online]
- (1) XeO_3F_2 (2) XeOF_2
(3) XeO_3 (4) FXeOSO_3F
49. In which of the following ionization processes the bond energy has increased and also the magnetic behaviour has changed from paramagnetic to diamagnetic? [JEE-Main 2013]
- (1) $\text{NO} \rightarrow \text{NO}^+$ (2) $\text{O}_2 \rightarrow \text{O}_2^+$
(3) $\text{N}_2 \rightarrow \text{N}_2^+$ (4) $\text{C}_2 \rightarrow \text{C}_2^+$
50. Which one of the following molecule is polar? [JEE.MAINS.2013]
- (1) CF_4 (2) SbF_5
(3) IF_5 (4) XeF_4
51. Oxidation state of sulphur in anions $\text{SO}_3^{2-}, \text{S}_2\text{O}_4^{2-}$ and $\text{S}_2\text{O}_6^{2-}$ increases in the order- [JEE.MAINS.2013, Online]
- (1) $\text{S}_2\text{O}_6^{2-}, \text{S}_2\text{O}_4^{2-} < \text{SO}_3^{2-}$
(2) $\text{SO}_3^{2-} < \text{S}_2\text{O}_4^{2-} < \text{S}_2\text{O}_6^{2-}$
(3) $\text{S}_2\text{O}_4^{2-} < \text{SO}_3^{2-} < \text{S}_2\text{O}_6^{2-}$
(4) $\text{S}_2\text{O}_4^{2-} < \text{S}_2\text{O}_6^{2-} < \text{SO}_3^{2-}$

52. Bond order normally gives idea of stability of a molecular species. All the molecules viz. H_2 , Li_2 and B_2 have the same bond order yet they are not equally stable. Their stability order is -

[JEE-MAINS-2013] (On-line)

- (1) $Li_2 > H_2 > B_2$ (2) $H_2 > B_2 > Li_2$
 (3) $B_2 > H_2 > Li_2$ (4) $Li_2 > B_2 > H_2$

53. The solubility order for alkali metal fluoride in water is-

[JEE-MAINS-2013] (On-line)

- (1) $LiF < NaF < KF < RbF$
 (2) $LiF > NaF > KF > RbF$
 (3) $RbF < KF < NaF < LiF$
 (4) $LiF < RbF < KF < NaF$

54. XeO_4 molecule is tetrahedral having :

[JEE-MAINS-2013] (On-line)

- (1) Two $p\pi-d\pi$ bonds
 (2) Four $p\pi-d\pi$ bonds
 (3) One $p\pi-d\pi$ bond
 (4) Three $p\pi-d\pi$ bonds

55. Bond distance in HF is 9.17×10^{-11} m. Dipole moment of HF is 6.104×10^{-30} Cm. The percent ionic character in HF will be :

(electron charge = 1.60×10^{-19} C)

[JEE-MAINS-2013]

- (1) 61.0% (2) 38.0%
 (3) 35.5% (4) 41.5%

56. The shape of IF_6^- is : [JEE-MAINS-2013]

- (1) Trigonally distorted octahedron
 (2) Pyramidal
 (3) Octahedral
 (4) Square antiprism

57. Which has trigonal bipyramidal shape ?

[JEE-MAINS-2013], Online

- (1) $XeOF_4$ (2) XeO_3
 (3) XeO_3F_2 (4) $XeOF_2$

58. The catenation tendency of C, Si and Ge is in the order $Ge < Si < C$. The bond energies (in $kJ\ mol^{-1}$) of C-C, Si-Si and Ge-Ge bonds are respectively :

[JEE-MAINS-2013]-(On-line)

- (1) 348, 260, 297 (2) 348, 297, 260
 (3) 297, 348, 260 (4) 260, 297, 348

59. In which of the following sets, all the given species are isostructural ? [JEE-MAINS-2013]

- (1) BF_3 , NF_3 , PF_3 , AlF_3
 (2) PCl_3 , $AlCl_3$, BCl_3 , $SbCl_3$ (On-line)
 (3) BF_4^- , CCl_4 , NH_4^+ , PCl_4^+
 (4) CO_2 , NO_2 , ClO_2 , SiO_2

60. The inter molecular distances in O-O bonds for O_2^+ , O_2 , O_2^- and O_2^{2-} respectively are :

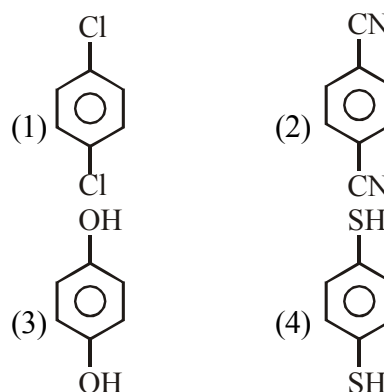
[JEE-MAINS-2013], (On-line)

- (1) 1.49 Å, 1.21 Å, 1.12 Å, 1.30 Å
 (2) 1.30 Å, 1.49 Å, 1.12 Å, 1.21 Å
 (3) 1.12 Å, 1.21 Å, 1.30 Å, 1.49 Å
 (4) 1.21 Å, 1.12 Å, 1.49 Å, 1.30 Å

61. Which one of the following properties is not shown by NO ? [JEE-Main-2014]

- (1) It combines with oxygen to form nitrogen dioxide
 (2) It's bond order is 2.5
 (3) It is diamagnetic in gaseous state
 (4) It is a neutral oxide

62. For which of the following molecule significant $\mu \neq 0$:- [JEE-Main-2014]



- (1) Only (3) (2) (3) and (4)
 (3) Only (1) (4) (1) Only (2)

63. Among the following oxoacid, the correct decreasing order of acid strength is- [JEE-M014]

- (1) $HClO_4 > HClO_3 > HClO_2 > HOCl$
 (2) $HClO_2 > HClO_4 > HClO_3 > HOCl$
 (3) $HOCl > HClO_2 > HClO_3 > HClO_4$
 (4) $HClO_4 > HOCl > HClO_2 > HClO_3$

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64. The number and type of bonds in C_2^{2-} ion in CaC_2 are— [JEE–MAINS–2014–(On–line)]
 (1) Two σ bonds and two π – bonds
 (2) Two σ bonds and two π – bonds
 (3) One σ bonds and two π – bonds
 (4) One σ bond and one π – bond
65. For the compounds CH_3Cl , CH_3Br , CH_3I and CH_3F , the correct order of increasing C–halogen bond length is:
 (1) $CH_3F < CH_3Br < CH_3Cl < CH_3I$
 (2) $CH_3F < CH_3Cl < CH_3Br < CH_3I$
 (3) $CH_3Cl < CH_3Br < CH_3F < CH_3I$
 (4) $CH_3F < CH_3I < CH_3Br < CH_3Cl$
66. Which of the following has unpaired electron(s) ? [JEE–MAINS–2014]
 (1) O_2^- (2) N_2^{2+}
 (3) O_2^{2-} (4) N_2
67. In allene (C_3H_4), the type(s) of hybridization of the carbon atoms is (are): [JEE–MAINS–2014]
 (1) only sp^2 (2) sp^2 and sp
 (3) sp and sp^3 (4) sp^2 and sp^3
68. Shapes of certain interhalogen compounds are stated below. Which one of them is not correctly stated? [JEE–MAINS–2014]
 (1) IF_7 : Pentagonal bipyramid
 (2) BrF_5 : Trigonal bipyramid
 (3) ICl_3 : Planar dimeric
 (4) BrF_3 : Planar T-shaped
69. The correct order of bond dissociation energy among N_2 , O_2 , O_2^- is shown in which of the following arrangements - [JEE–MAINS–2014 (On–line)]
 (1) $N_2 > O_2 > O_2^-$ (2) O_2 , O_2^- , N_2
 (3) N_2 , O_2^- , O_2 (4) O_2^- , O_2 , N_2
70. Which of the following molecules has two sigma (σ) and two pi (π) bonds :— [JEE–MAINS–2014, (On–line)]
 (1) HCN (2) $C_2H_2Cl_2$
 (3) N_2F_2 (4) C_2H_4
71. Which of the following xenon–oxo compounds may not be obtained by hydrolysis of xenon fluoride XeF_6 :— [JEE–MAINS–2014]
 (1) XeO_2F_2 (2) XeO_3
 (3) XeO_4 (4) $XeOF_4$
72. Which one of the following molecules is paramagnetic - [JEE–MAINS–2014]
 (1) NO (2) O_3
 (3) N_2 (4) CO
73. Amongst $LiCl$, $RbCl$, $BeCl_2$ and $MgCl_2$ the compounds with the greatest and the least ionic character, respectively are: [JEE–MAINS–2014]
 (1) $RbCl$ and $MgCl_2$ (2) $LiCl$ and $RbCl$
 (3) $MgCl_2$ and $BeCl_2$ (4) $RbCl$ and $BeCl_2$
74. Example of a three - dimensional silicate is : [JEE–MAINS–2014]
 (1) Beryls (2) Zeolites
 (3) Feldspars (4) Ultramarines
75. Which one of the following alkaline earth metal sulphates has its hydration enthalpy greater than its lattice enthalpy - [JEE–MAINS–2014]
 (1) $BeSO_4$ (2) $SrSO_4$
 (3) $CaSO_4$ (4) $BeSO_4$
76. The intermolecular interaction that is dependent on the inverse cube of distance between the molecules is " [JEE–MAINS–2015]
 (1) London force
 (2) Hydrogen bond
 (3) ion–ion interaction
 (4) ion–dipole interaction
77. Which one has the highest boiling point ? [JEE–MAINS–2015]
 (1) Kr (2) Xe
 (3) He (4) Ne
78. Which one of the following statements about water is FALSE ? [JEE–MAINS–2016]
 (1) Ice formed by heavy water sinks in normal water.
 (2) Water is oxidized to oxygen during photosynthesis.
 (3) Water can act both as an acid and as a base.
 (4) There is extensive intramolecular hydrogen bonding in the condensed phase.

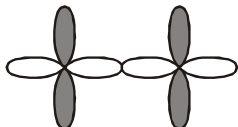
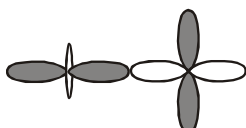
79. The species in which the N atom is in a state of sp hybridization is :- [JEE-MAINS-2016]
 (1) NO_2 (2) NO_2^+
 (3) NO_2^- (4) NO_3^-
80. The pair in which phosphorous atoms have a formal oxidation state of + 3 is: [JEE-MAINS-2016]
 (1) Pyrophosphorous and pyrophosphoric acids
 (2) Orthophosphorous and pyrophosphorous acids
 (3) Pyrophosphorous and hypophosphoric acids
 (4) Orthophosphorous and hypophosphoric acids
81. Which intermolecular force is most responsible in allowing xenon gas to liquefy? [JEE-MAINS-2016 (On-line)]
 (1) ionic
 (2) Instantaneous dipole-induced dipole
 (3) Dipole-dipole
 (4) Ion-dipole
83. The group of molecules having identical shape is: [JEE-MAINS-2016]
 (1) SF_4 , XeF_4 , CCl_4
 (2) ClF_3 , XeOF_2 , XeF_3^+
 (3) PCl_5 , IF_5 , XeO_2F_2
 (4) BF_3 , PCl_3 , XeO_3
84. The bond angle H-X-H is the greatest in the compound: [JEE-MAINS-2016 (On-line)]
 (1) NH_3 (2) H_2O
 (3) PH_3 (4) CH_4

EXERCISE # V(B) (JEE-ADVANCED)

1. The hybridisation of atomic orbitals of nitrogen in NO_2^+ , NO_3^- and NH_4^+ are: [IIT-2000]
 (a) sp^2 , sp^3 and sp^2 respectively
 (b) sp , sp^2 and sp^3 respectively
 (c) sp^2 , sp and sp^3 respectively
 (d) sp^2 , sp^3 and sp respectively
2. The correct order of hybridization of the central atom in the following species NH_3 , $[\text{PtCl}_4]^{2-}$, PCl_5 and BCl_3 is - [IIT-2001]
 (a) dsp^2 , sp^3d , sp^2 and sp^3
 (b) sp^3 , dsp^2 , sp^3d , sp^2
 (c) dsp^2 , sp^2 , sp^3 , sp^3d
 (d) dsp^2 , sp^3 , sp^2 , sp^3d
3. The common features among the species CN^- , CO and NO^+ are - [IIT-2001]
 (a) bond order three and isoelectronic
 (b) bond order three and weak field ligands
 (c) bond order two and π -acceptors
 (d) isoelectronic and weak field ligands
4. Specify the coordination geometry around and hybridisation of N and B atoms in a 1:1 complex of BF_3 and NH_3 - [IIT 2001]
 (a) N: tetrahedral sp^3 ; B: tetrahedral sp^3
 (b) N: pyramidal sp^3 ; B: pyramidal sp^3
 (c) N: pyramidal sp^3 ; B: planar sp^2
 (d) N: pyramidal sp^3 ; B: tetrahedral sp^3d
5. Specify hybridization of N and B atoms in a 1:1 complex of BF_3 and NH_3 - [JEE 2002]
 (a) N: tetrahedral sp^3 ; B: tetrahedral sp^3
 (b) N: pyramidal sp^3 ; B: pyramidal sp^3
 (c) N: pyramidal sp^3 ; B: planar sp^2
 (d) N: pyramidal sp^3 ; B: tetrahedral sp^3d
6. The nodal plane in the π -bond of ethene is located in [JEE 2002]
 (a) the molecular plane
 (b) a plane parallel to the molecular plane
 (c) a plane perpendicular to the molecular plane which bisects, the carbon-carbon bond at right angle
 (d) a plane perpendicular to the molecular plane which contains the carbon-carbon bond.
7. Which of the following molecular species has unpaired electron(s)? [JEE 2003]
 (a) N_2 (b) F_2
 (c) O_2^- (d) O_2^{2-}
8. Which of the following are isoelectronic and isostructural?
 NO_3^- , CO_3^{2-} , ClO_3^- , SO_3
 (a) NO_3^- , CO_3^{2-}
 (b) SO_3 , NO_3^-
 (c) ClO_3^- , CO_3^{2-}
 (d) CO_3^{2-} , SO_3

1.52 CHEMICAL BONDING

9. According to molecular orbital theory which of the following statement about the magnetic character and bond order is correct regarding O_2^+ : [JEE 2004]
 (a) Paramagnetic and Bond order $< O_2$
 (b) Paramagnetic and Bond order $> O_2$
 (c) Diamagnetic and Bond order $< O_2$
 (d) Diamagnetic and Bond order $> O_2$
10. Which species has the maximum number of lone pair of electrons on the central atom?
 (a) ClO_3^- (b) XeF_4
 (c) SF_4 (d) I_3^- [JEE 2005]
11. The percentage of p-character in the orbitals forming P-P bonds in P_4 is [JEE 2006]
 (a) 25 (b) 33
 (c) 50 (d) 75
12. Among the following, the paramagnetic compound is [JEE 2007]
 (a) Na_2O_2 (b) O_3
 (c) N_2O (d) KO_2
13. The species having bond order different from that in CO is [JEE 2007]
 (a) NO^- (b) NO^+
 (c) CN^- (d) N_2
14. The structure of XeO_3 is [JEE 2007]
 (a) linear (b) planar
 (c) pyramidal (d) T-shaped
15. Statement-1 : p-Hydroxybenzoic acid has a lower boiling point than o-hydroxybenzoic acid.
 Statement-2 : o-Hydroxybenzoic acid has intramolecular hydrogen bonding. [JEE 2007]
 (a) Statement-1 is True, Statement-2 is True; Statement-2 is a correct explanation for Statement-1.
 (b) Statement-1 is True, Statement-2 is True; Statement-2 is NOT a correct explanation for Statement-1.
 (c) Statement-1 is True, Statement-2 is False.
 (d) Statement-1 is False, Statement-2 is True.
16. Statement-1 : In water, orthoboric acid behaves as a weak monobasic acid. [JEE 2007]
 Statement-2 : In water, orthoboric acid acts as a proton donor.
 (a) Statement-1 is True, Statement-2 is True; Statement-2 is a correct explanation for Statement-1.
 (b) Statement-1 is True, Statement-2 is True; Statement-2 is NOT a correct explanation for Statement-1.
 (c) Statement-1 is True, Statement-2 is False.
 (d) Statement-1 is False, Statement-2 is True.
17. Statement-1 : Pb^{+4} compounds are stronger oxidizing agents than Sn^{+4} compounds
 Statement-2 : The higher oxidation states for the group 14 elements are more stable for the heavier members of the group due to 'inert pair effect'. [JEE 2008]
 (a) Statement-1 is True, Statement-2 is True; Statement-2 is a correct explanation for Statement-1.
 (b) Statement-1 is True, Statement-2 is True; Statement-2 is NOT a correct explanation for Statement-1.
 (c) Statement-1 is True, Statement-2 is False.
 (d) Statement-1 is False, Statement-2 is True.
18. Match each of the diatomic molecules in column I with its property / properties in column - II
- | Column-I | Column-II |
|-------------|------------------------------------|
| (a) B_2 | (P) paramagnetic |
| (b) N_2 | (Q) undergoes oxidation |
| (c) O_2^- | (R) undergoes reduction |
| (d) O_2 | (S) Bond order ≥ 2 |
| | (T) Mixing of 's' and 'p' orbitals |
19. The nitrogen oxides that contains N-N bonds is are - [JEE 2009]
 (a) N_2O (b) N_2O_3
 (c) N_2O_4 (d) N_2O_5

20. In the reaction $2X + B_2H_6 \rightarrow [BX_2(X_2)]^+ [BH_4]^-$ the amines X is/are - [JEE 2009]
 (a) NH_3 (b) CH_3NH_2
 (c) $(CH_3)_2NH$ (d) $(CH_3)_3N$
21. The species having pyramidal shape is - [JEE 2010]
 (a) SO_3 (b) BrF_3
 (c) SiO_3^{2-} (d) OSF_2
22. Assuming that Hund's rule is violated, the bond order and magnetic nature of the diatomic molecule B_2 is [JEE 2009]
 (a) 1 and diamagnetic
 (b) 0 and diamagnetic
 (c) 1 and paramagnetic
 (d) 0 and paramagnetic
23. In allene (C_3H_4) the type(s) of hybridisation of the carbon atom(s) is/are - [JEE 2012]
 (a) sp and sp^3 (b) sp and sp^2
 (c) only sp^2 (d) sp^2 and sp^3
24. Which ordering of compounds is according to the decreasing order of the oxidation state of nitrogen :- [JEE 2012]
 (a) HNO_3 , NO , NH_4Cl , N_2
 (b) HNO_3 , NO , N_2 , NH_4Cl
 (c) HNO_3 , NH_4Cl , NO , N_2
 (d) NO , HNO_3 , NH_4Cl , N_2
25. The shape of XeO_2F_2 molecule is : [JEE 2012]
 (a) Trigonal bipyramidal
 (b) Square planar
 (c) tetrahedral
 (d) see-saw
- Subjective :**
26. The number of water molecules directly bonded to the metal centre in $CuSO_4 \cdot 5H_2O$ is :- [JEE 2012]
27. Based on VSEPR theory, the number of 90 degree F-Br-F angles in BrF_5 is.
28. The value of n in the molecular formula $Be_nAl_2Si_6O_{18}$ is ? [JEE 2010]
29. The total number of diprotic acids among the following is?
 H_3PO_4 H_2SO_4 H_3PO_3 H_2CO_3 $H_2S_2O_7$
 H_3BO_3 H_3PO_2 H_2CrO_4 H_2SO_3
30. Among the following, the number of elements showing only one non-zero oxidation state -
 O, Cl, F, N, P, Sn, Tl, Na, Ti [JEE 2010]
31. The difference in the oxidation numbers of the two types of sulphur atoms in $Na_2S_4O_6$ - [JEE 2011]
32. The total number of lone-pairs of electrons in melanin is. [JEE Adv. 2013]
33. Assuming 2s-2p mixing is NOT operative, the paramagnetic species among the following:-
 (a) Be_2 (b) B_2
 (c) C_2 (d) N_2
34. Match the orbital overlap figures shown in List - I with the description given in List - II and select the correct answer using the code given below the lists: [JEE-Adv.2014]
- | List - I | List-II |
|--|--------------------------------|
| (P)  | (1) p - d π antibonding |
| (Q)  | (2) d - d σ bonding |
| (R)  | (3) p - d π bonding |
| (S)  | (4) d - d σ antibonding |
- | P | Q | R | S |
|-------|---|---|---|
| (a) 2 | 1 | 3 | 4 |
| (b) 4 | 3 | 1 | 2 |
| (c) 2 | 3 | 1 | 4 |
| (d) 4 | 1 | 3 | 2 |

1.54 CHEMICAL BONDING

35. Among the triatomic molecules / ions, BeCl_2 , N_3^- , N_2O , NO_2^+ , O_3 , SCl_2 , ICl_2^- , I_3^- and XeF_2 the total number of linear molecules(s) / ion(s) where the hybridization of the central atoms does not have contribution from the d-orbital(s) is :

[JEE Adv. 2015]

(Atomic number : S = 16, Cl = 17, I = 53 and Xe = 54)

36. The total number of lone pairs of electrons in N_2O_3 is :
37. The correct statement(s) regarding,
- (i) HClO , (ii) HClO_2 ,
(iii) HClO_3 and (iv) HClO_4 , is (are)

- (a) The number of Cl=O bonds in (ii) and (iii) together is two
(b) The number of lone pairs of electrons on Cl in (ii) and (iii) together is three
(c) The hybridization of Cl in (iv) is sp^3
(d) Amongst (i) to (iv), the strongest acid is (i)

38. When O_2 is adsorbed on a metallic surface, electron transfer occurs from the metal to O_2 . The TRUE, statement (s) regarding this adsorption is (are)

[JEE Adv. 2015]

- (a) O_2 is physisorbed
(b) heat is released
(c) occupancy of π_{2p}^* of O_2 is increased
(d) bond length of O_2 is increased

■ ■

COORDINATION COMPOUNDS

PROBLEMS BASED ON GIVEN TOPICS

- Double salt and coordination compounds
- Werner's work
- More recent methods of studying complexes
- Classification of ligands
- Effective atomic number (EAN)
 - Sidgwick EAN rule
- Shape of d orbitals
- Bonding in transition metal complexes
 - Valence bond theory
 - Crystal field theory
 - Molecular orbitals theory
- Valence bond theory
- Crystal field theory
 - Octahedral complexes
- Effects of crystal field splitting
 - Tetragonal distortion of octahedral complexes (Jahn-Teller Distortion)
 - Square planar arrangements
 - Tetrahedral complex
 - Magnetism
 - Extension of the crystal field theory to allow for some covalency
 - Nomenclature of coordination compounds
 - Isomerism
 - Polymerisation isomerism
 - Ionisation isomerism
 - Hydrate isomerism
 - Linkage isomerism
 - Coordination isomerism
 - Coordination position isomerism
 - Geometrical isomerism and optical isomerism

Coordination Compounds

EXERCISE # I

❑ Only one correct answer :

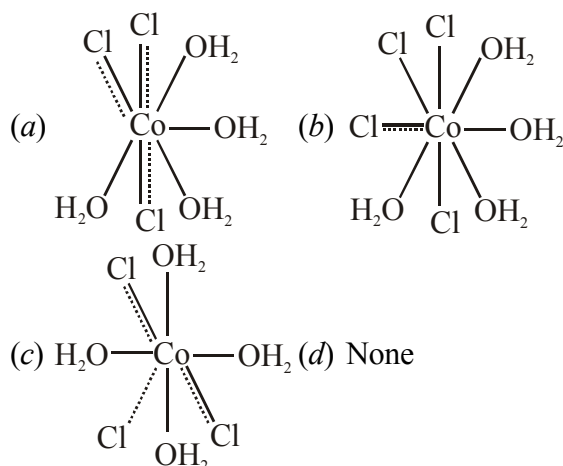
1. The formula of sodium nitropruside is $\text{Na}_2[\text{Fe}(\text{CN})_5\text{NO}]$. In sodium nitropruside, the oxidation state, coordination number and EAN of iron are respectively :-

(a) +3, 6, 36 (b) +2, 6, 36
(c) +3, 3, 36 (d) 6, +3, 35

2. What is the oxidation state of Th in organometallic compound $[\text{ThHOR}(\text{C}_5\text{Me}_5)_2]$?

(a) +1
(b) +2
(c) +4
(d) +3

3. $\text{CoCl}_3 \cdot 4\text{H}_2\text{O}$ is an anhydrous binary soluble compound. Hence its Werner's representation is:-



4. Concentrated H_2SO_4 will not dehydrate which of the following complex

(a) $[\text{Cr}(\text{H}_2\text{O})_5\text{Cl}] \text{Cl}_2 \cdot \text{H}_2\text{O}$
(b) $[\text{Cr}(\text{H}_2\text{O})_6]\text{Cl}_3$
(c) $[\text{Cr}(\text{H}_2\text{O})_4\text{Cl}_2] \text{Cl}_2 \cdot 2\text{H}_2\text{O}$
(d) All of these

5. On adding AgNO_3 solution to a solution of $\text{Pt}(\text{NH}_3)_3\text{Cl}_4$, the percentage of total chloride ion precipitated is :-

(a) 100 (b) 75
(c) 50 (d) 25

6. The effective atomic number of iron in complex $[(\eta^5\text{C}_5\text{H}_5)\text{Fe}(\text{CO})_2\text{Cl}]$:-

(a) 34 (b) 36
(c) 38 (d) 32

7. If EAN of metal is 36 in $[\text{M}(\text{CO})_2(\sigma\text{C}_5\text{H}_5)(\pi\text{C}_5\text{H}_5)]$, then what is the atomic number of metal M ?

(a) 23 (b) 24
(c) 25 (d) 26

8. What is the coordination number of central atom of the complex $[\text{M}(\text{NH}_3)_4\text{SO}_4]\text{Br}$ which has one ring?

(a) 4 (b) 5
(c) 6 (d) None

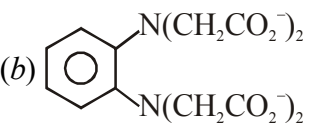
9. Oxidation state of Co in $\text{Ag}[\text{Co}(\text{CO})_4]$ and Mo in oxo complex species $[\text{Mo}_2\text{O}_4(\text{C}_2\text{H}_4)(\text{H}_2\text{O})_2]^{2-}$ are respectively :-

(a) +1, +3 (b) -1, -3
(c) +1, -3 (d) -1, +3

10. In Tollen's reagent, the oxidation number, coordination number and effective atomic number of central metal ion are respectively. (At. no. of Ag = 47) :-

(a) +1, +2, 50
(b) +1, +2, 51
(c) +2, +1, 50
(d) +1, +1, 50

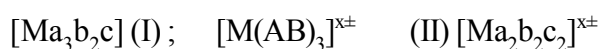
2.4 COORDINATION COMPOUNDS

11. Which will not give test of all the anions present in it ?
 (a) $K_2Fe_2(SO_4)_4 \cdot 24H_2O$
 (b) $(NH_4)_2Fe(SO_4)_2 \cdot 6H_2O$
 (c) $K_3[Fe(CN)_6]$
 (d) All
12. In the compound $CoCl_3 \cdot 4NH_3$:-
 (a) All the Cl shows primary valency only
 (b) Two Cl show primary valency and two Cl shows secondary valency
 (c) Two Cl show primary valency and one Cl shows primary valency as well as secondary valency
 (d) 3 Cl shows primary valency and two Cl shows primary valency as well as secondary valency
13. For which of the following complex, metal carbon linkage is present :-
 (a) $Cr(acac)_3$ (b) $Ni(DMG)_2$
 (c) $Fe(CO)_5$ (d) $[Co(C_2H_5O)_6]^{3-}$
14. Which of the following ligand can act as chelating ligand with more than one chiral centre ?
 (a) 1, 2 propanediamine (pn)
 (b) 2,3 butane diamine (bn)
 (c) Nitrilotriacetato(nta^{3-})
 (d) glycinate
15. Which one of the following is example of 1 : 3 electrolyte :-
 (a) $PtCl_4 \cdot 6NH_3$ (b) $CrCl_3 \cdot 6NH_3$
 (c) $CoBr_3 \cdot 5NH_3$ (d) $PtCl_4 \cdot 3NH_3$
16. Which of the following are examples of hexadentate chelating ligand where four oxygen atoms and two nitrogen atoms act as donor atoms ?
 (a) $[EDTA]^{4-}$
 (b) 
 (c) both (a) and (b)
 (d) None
17. Which of the following are examples of tetradentate ligand where all donor atoms are same ?
 (a) $N(CH_2CH_2NH_2)_3$
 (b) $N(CH_2CO_2^-)_3$
 (c) $NH_2(CH_2)_2NH(CH_2)_2NH(CH_2)_2NH_2$
 (d) Both (a) & (c)
18. In which case Pt has +2 oxidation state :-
 (a) $[Pt(NH_3)_4][PtCl_4]$ (b) $[Pt(NH_3)_2Cl_2]$
 (c) $[Pt(C_2H_4)Cl_3]^{-1}$ (d) All
19. $KCN(\text{excess}) + CuSO_4 \longrightarrow \text{Product (Complex)}$.
 Complex is :-
 (a) $K_2[Cu(CN)_4]$ (b) $K[Cu(CN)_2]$
 (c) $K_3[Cu(CN)_4]$ (d) None
20. Mohr salt in aqueous solution gives :-
 (a) 3 different ions (b) 2 different ions
 (c) 4 different ions (d) 5 different ions
21. Which of the following statement is correct ?
 (a) NH_2^- will be named as amido
 (b) SO_4^{2-} will be named as sulphito
 (c) CH_3^- will be named as methylo
 (d) $SnCl_3^-$ will be named as trichloro stannito
22. Which of the following ligand forms only two ring towards a central atom ?
 (a) en (b) bn
 (c) dien (d) trien
23. For which of the following complex, chelate is not formed ?
 (a) Bis(dimethyl glyoximato)nickel(II)
 (b) Potassium ethylene diamine tetrathiocyanato chromate(III)
 (c) Pentaammine carbonato cobalt(III) nitrate
 (d) Trans-diglycinatoplatinum(II)
24. Which of the following pair of molecule have identical shape ?
 (a) $[NiCl_4]^{2-}$ & XeF_4 (b) $[Ag(NH_3)_2]^+$, H_2S
 (c) $[Zn(H_2O)_4]^{2+}$, $SiCl_4$ (d) $Fe(CO)_5$, $XeOF_4$
25. A complex of Pt, ammonia & chlorine produces four ions per molecule in the solution. The solution consistent with the observation is :-
 (a) $[Pt(NH_3)_5Cl]Cl_3$ (b) $[Pt(NH_3)_4Cl_2]Cl_2$
 (c) $[Pt(NH_3)_3Cl_3]Cl$ (d) All of these

26. $[M(H_2O)_6]^{2+}$ complex typically absorbs light at around 600 nm. It is allowed to react with ammonia to form a new complex $[M(NH_3)_6]^{2+}$. It will absorb light at around :-
 (a) 800 nm (b) 620 nm
 (c) 580 nm (d) 320 nm
27. Arrange the following compound according to decreased C–C bond length :-
 (I) C_2F_4
 (II) C_2H_4
 (III) $[PtCl_3(C_2H_4)]^{-1}$
 (a) I > II > III (b) III > II > I
 (c) II > I > III (d) II > III > I
28. 1 mole of which complex produce minimum number of K^+ ions on dissociation :-
 (a) Potassium tris(oxalato)cobaltate(III)
 (b) Potassium tetrahydroxido zincate(II)
 (c) Potassium hexacyanidoferrate(II)
 (d) Potassium ammine penta chlorido platinate(IV)
29. Facial and Meridional isomers are associated with which of the general formula of complex -
 A, B, C, D are example of monodentate ligand.
 & AA is example of bidentate ligand.
 (a) $[M(AA)_2]$ (b) $[M(AA)_3]$
 (c) $[MABCD]$ (d) $[MA_3B_3]$
30. How many geometrical isomers possible for
 $[Pd(NH_2 - \underset{\substack{| \\ Cl}}{CH} - CO_2^-)_2]$
 (a) 2 (b) 3
 (c) 4 (d) 6
31. In $[Fe(EDTA)]^{-1}$, which of the following statements are correct ?
 (a) It shows optical isomerism
 (b) C.N. of Fe^{+3} is 6
 (c) 5 chelates are formed in the complex
 (d) All
32. A coordination complex of type Mx_2y_2 (M metal ion ; x, y monodentate ligands). can have either a tetrahedral or a square planar geometry. The number of structures in these two cases are respectively :-
 (a) 1 & 2 (b) 2 & 1
 (c) 1 & 3 (d) 3 & 2
33. The ratio of trans and cis isomers of the complex $[Ma_2bcde]$?
 (a) 2 : 3 (b) 2 : 1
 (c) 1 : 2 (d) 3 : 5
34. Which of the following can show geometrical isomerism -
 (a) $[Co(en)_2(NH_3)Cl]$
 (b) $(Ph_3P)ClPdCl_2PdCl(PPh_3)$
 (c) Both (a) and (b)
 (d) $K_4[Fe(CN)_6]$
35. Which will give a pair enantiomers (which are optically active) ?
 (a) $[Co(NH_3)_4Cl_2]NO_2$
 (b) $[Cr(NH_3)_6][Co(CN)_6]$
 (c) $[Co(en)_2Cl_2]Cl$
 (d) $[Pt(NH_3)_4][PtCl_6]$
36. For the complex, $K[CoBr_2(CN)(NO_2)(NH_3)(H_2O)]$
 X = Number of possible stereoisomer where both Br are at cis position
 Y = Number of possible stereoisomer where both Br are at trans position
 Find out X & Y :-
 (a) 6, 3 (b) 3, 6
 (c) 12, 3 (d) 3, 12
37. A complex with the composition $[MA_3B]^{x\pm}$ is found to have no geometrical isomers. The possible structures of the complex is (where A and B are monodentate ligands) :-
 (a) Tetrahedral (b) Square planar
 (c) Octahedral (d) (a) & (b) both
38. Select incorrect option code about complex $[Cr(NO_2)(NH_3)_5][ZnCl_4]$:-
 (a) IUPAC name of the compound is pentaammine nitrito-N-chromium(III) tetrachlorido zincate(II)
 (b) It shows geometrical isomers
 (c) It shows linkage isomerism
 (d) It shows coordination isomerism

2.6 COORDINATION COMPOUNDS

39. Select the correct code regarding total number of stereoisomers for the following compounds :-



I II III

- | | | |
|-------|---|---|
| (a) 4 | 4 | 6 |
| (b) 4 | 3 | 5 |
| (c) 3 | 3 | 5 |
| (d) 3 | 4 | 6 |

40. For the complex dichlorido bis(ethylenediamine) cobalt(III) ion,

Select the correct statement :-

- (a) It has three isomers, two of them are optically active and one is optically inactive.
 (b) It has three isomers, all of them are optically active
 (c) It has three isomers, all of them are optically inactive
 (d) It has one optically active isomer & two geometrical isomer.

41. When $[\text{Co}(\text{CO}_3)(\text{NH}_3)_4]^+$ is reacted with concentrated HCl, the violet coloured $[\text{CoCl}_2(\text{NH}_3)_4]\text{Cl}$ compound is formed. Predict the geometry of the product formed.

- (a) Only cis isomer
 (b) Only trans isomer
 (c) Cis and trans isomer both
 (d) Can't be predicted

42. Which of the following complex only show linkage isomerism ?

- (a) $[\text{Co}(\text{NH}_3)_4(\text{H}_2\text{O})\text{Cl}]\text{CN}$
 (b) $[\text{Co}(\text{NH}_3)_5(\text{SCN})]^{2+}$
 (c) $[\text{Co}(\text{H}_2\text{O})_6]\text{Cl}_3$
 (d) $[\text{Co}(\text{NH}_3)_5\text{SO}_4]\text{Br}$

43. Which of the following complex only show ionisation isomerism ? (among all type of structural isomerism)

- (a) $[\text{Co}(\text{NH}_3)_4(\text{H}_2\text{O})\text{Cl}]\text{Br}$
 (b) $[\text{Co}(\text{NH}_3)_5(\text{SCN})]^{2+}$
 (c) $[\text{Co}(\text{H}_2\text{O})_6]\text{Cl}_3$
 (d) $[\text{Co}(\text{NH}_3)_5\text{Br}]\text{Br}_2$

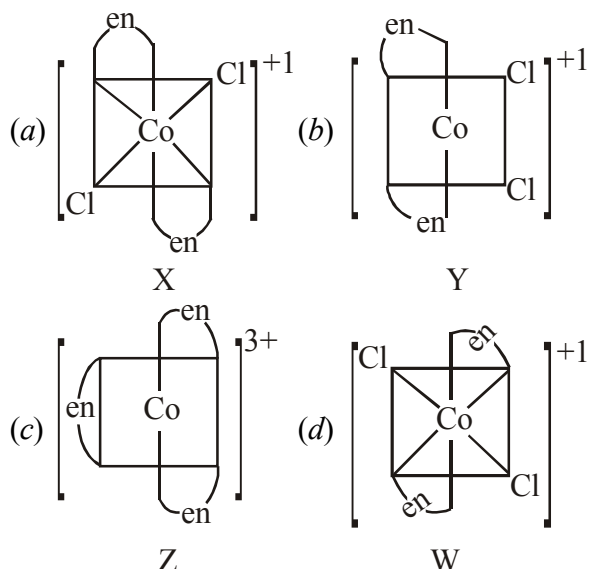
44. Total possible geometrical isomers and stereoisomers are existing respectively for $[\text{M}(\text{gly})_2(\text{en})]$ type octahedral complex :-

- (a) 3, 6 (b) 3, 4
 (c) 4, 8 (d) 3, 4

45. How many coordination isomers of $[\text{Pt}(\text{NH}_3)_4][\text{PtCl}_4]$ showing geometrical isomers :-

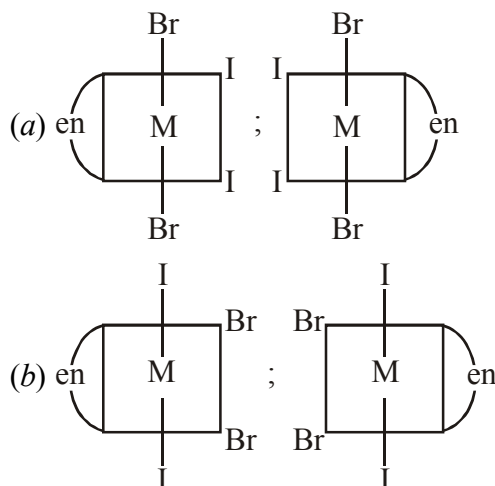
- (a) Four (b) One
 (c) Two (d) Zero

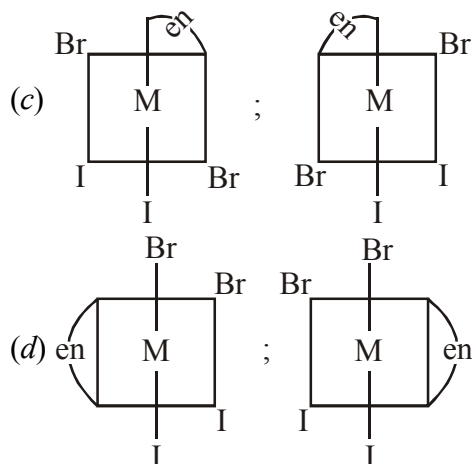
46. Which of the following ions are optically active (exists as pair of enantiomers) ?



- (a) X only (b) Y only
 (c) Y & Z both (d) W only

47. The complex has two optical isomers which are related as enantiomers. Their correct structures are-





48. Find complex which has maximum number of geometrical isomers ?

- (a) Ma_3b_3 (b) $\text{Ma}_3\text{b}_2\text{c}$
(c) $\text{Ma}_2\text{b}_2\text{c}_2$ (d) $\text{M}(\text{AA})_2\text{a}_2\text{b}_2$

49. The formula for the compound tris (ethane-1,2-diamine)cobalt(III) sulphate is -

- (a) $[\text{Co}(\text{en})_3]\text{SO}_4$ (b) $[\text{Co}(\text{SO}_4)_4(\text{en})_3]$
(c) $[\text{Co}(\text{en})_3(\text{SO}_4)_2]$ (d) $[\text{Co}(\text{en})_3]_2(\text{SO}_4)_3$

50. For which of the following d^n configuration, octahedral complexes can not exist in both high and low spin forms -

- I. d^3 II. d^5 III. d^6 IV. d^8

- (a) I, II (b) I, IV
(c) III, IV (d) II, III

51. Correct option regarding complex $[\text{Co}(\text{en})_3]_2(\text{SO}_4)_3$:-

- (a) low spin inner orbital octahedral complex with d^2sp^3 hybridisation
(b) optically active
(c) C_2 axis of symmetry is present
(d) All

52. In which ligand, only two nitrogen acts as donor site ?

- (a) nta^{3-} (b) NH_3
(c) en (d) $\text{H}_2\text{N}-\text{CH}_2-\text{CO}_2^-$

53. $[\text{Ru}(\text{NH}_3)_5(\text{NO}_2)]\text{Cl}$ vs $[\text{Ru}(\text{NH}_3)_5\text{Cl}]\text{NO}_2$ & $[\text{Ru}(\text{NH}_3)_5(\text{NO}_2)]\text{Cl}$ vs $[\text{Ru}(\text{NH}_3)_5(\text{ONO})]\text{Cl}$ are related as -

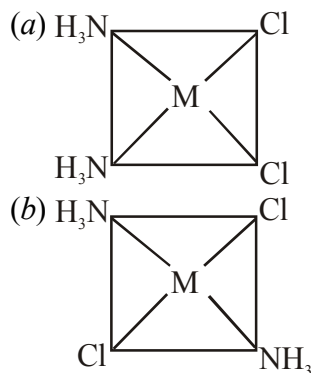
- (a) Ionisation isomer and linkage isomer respectively

(b) Linkage isomer and ionisation isomer respectively

(c) Ionisation and coordination isomer respectively

(d) Coordination isomer and coordination position isomer respectively

54. Which of the following isomers of $[\text{M}(\text{NH}_3)_2\text{Cl}_2]$ would react with silver oxalate $\text{Ag}_2\text{C}_2\text{O}_4$?



(c) Both

(d) None

55. In Zieses salt C-C bond length is -

- (a) 1.37°A (b) 1.19°A
(c) 1.87°A (d) 1.34°A

56. Identify incorrect statement for the behaviour of ethane 1,2-diamine as a ligand.

- (a) It is a neutral ligand
(b) It is a didentate ligand
(c) It is a chelating ligand
(d) It is a unidentate ligand

57. The stabilisation of coordination compounds due to chelation is called the chelate effect. Which of the following is most stable complex species ?

- (a) $\text{Fe}(\text{CO})_5$ (b) $[\text{Fe}(\text{CN})_6]^{3-}$
(c) $[\text{Fe}(\text{C}_2\text{O}_4)_3]^{3-}$ (d) $[\text{Fe}(\text{H}_2\text{O})_6]^{3+}$

58. Which of the following complex formed by Cu^{2+} ion is most stable ?

- (a) $\text{Cu}^{2+} + 4\text{NH}_3 \longrightarrow [\text{Cu}(\text{NH}_3)_4]^{2+}$ $\log k = 11.6$
(b) $\text{Cu}^{2+} + 4\text{CN}^- \longrightarrow [\text{Cu}(\text{CN})_4]^{3-}$ $\log k = 27.3$
(c) $\text{Cu}^{2+} + 2\text{en} \longrightarrow [\text{Cu}(\text{en})_2]^{2+}$ $\log k = 15.4$
(d) $\text{Cu}^{2+} + 4\text{H}_2\text{O} \longrightarrow [\text{Cu}(\text{H}_2\text{O})_4]^{2+}$ $\log k = 8.9$

59. The correct IUPAC name of $[\text{Pt}(\text{NH}_3)_2\text{Cl}_2]$ is -

- (a) Diamminedichloridoplatinum(II)
(b) Diamminedichloridoplatinum(IV)
(c) Dichloridoammineplatinum(II)
(d) Diamminedichloridoplatinum(0)

2.8 COORDINATION COMPOUNDS

60. The colour of the coordination compound depend on the crystal field splitting. What will be the correct order of absorption of wavelength of light in the visible region for complex ?
 $[\text{Co}(\text{NH}_3)_6]^{3+}$; $[\text{Co}(\text{CN})_6]^{3-}$; $[\text{Co}(\text{H}_2\text{O})_6]^{3+}$
 (a) $[\text{Co}(\text{CN})_6]^{3-} > [\text{Co}(\text{NH}_3)_6]^{3+} > [\text{Co}(\text{H}_2\text{O})_6]^{3+}$
 (b) $[\text{Co}(\text{NH}_3)_6]^{3+} > [\text{Co}(\text{H}_2\text{O})_6]^{3+} > [\text{Co}(\text{CN})_6]^{3-}$
 (c) $[\text{Co}(\text{H}_2\text{O})_6]^{3+} > [\text{Co}(\text{NH}_3)_6]^{3+} > [\text{Co}(\text{CN})_6]^{3-}$
 (d) None
61. The type of isomerism arises due to presence of ambidentate ligands attached to central metal ion is
 (a) Linkage isomerism (b) Ligand isomerism
 (c) Hydrate isomerism (d) Ionisation isomerism
62. The type of isomerism arises due to presence of isomeric ligands is
 (a) Ionisation isomerism
 (b) Coordination position isomerism
 (c) Ligand isomerism
 (d) Linkage isomerism
63. $[\text{Fe}(\text{CN})_6]^{3-}$ ion shows magnetic moment corresponding to -
 (a) 2 unpaired electrons
 (b) 3 unpaired electrons
 (c) 1 unpaired electron
 (d) None
64. Toxic metal ions are removed by chelating ligand. The chelating ligand used can be -
 (a) en (b) pn
 (c) $\text{H}_2\text{N}-\text{CH}_2-\text{CO}_2^-$ (d) EDTA
65. Complex $[\text{Fe}(\text{H}_2\text{O})_6]^{3+}$ ion has magnetic moment-
 (a) 2.83 B.M. (b) 3.87 B.M.
 (c) 4.92 B.M. (d) 5.92 B.M.
66. A chelating ligand has two or more than two donor atoms to bind to a single metal ion. Which of the following is not a chelating ligand ?
 (a)  (b) Oxalato
 (c) Glycinato (d) Ethan-1, 2 diamine
67. What kind of isomerism exist between violet colour $[\text{Cr}(\text{H}_2\text{O})_6]\text{Cl}_3$ & green colour $[\text{Cr}(\text{H}_2\text{O})_5\text{Cl}]\text{Cl}_2 \cdot \text{H}_2\text{O}$.
 (a) Linkage isomerism
 (b) Solvent isomerism
 (c) Ionisation isomerism
 (d) Coordination isomerism
68. The CFSE for octahedral $[\text{CoCl}_6]^{4-}$ is 18000 cm^{-1} . The CFSE for tetrahedral $[\text{CoCl}_4]^{2-}$ will be -
 (a) 18000 cm^{-1} (b) 16000 cm^{-1}
 (c) 8000 cm^{-1} (d) 20000 cm^{-1}
69. $[\text{Co}(\text{NH}_3)_5(\text{CO}_3)]\text{ClO}_4$, for this complex the coordination number, oxidation number, number of d-electrons and number of unpaired d electrons are respectively -
 (a) 4, 2, 6, 0 (b) 6, 3, 6, 0
 (c) 4, 2, 6, 3 (d) 6, 3, 6, 3
70. Select correct statement for $[\text{Cu}(\text{CN})_4]^{3-}$, $[\text{Cd}(\text{CN})_4]^{2-}$, $[\text{Cu}(\text{NH}_3)_4]^{2+}$:-
 (a) Both $[\text{Cd}(\text{CN})_4]^{2-}$ & $[\text{Cu}(\text{NH}_3)_4]^{2+}$ have square planar geometry
 (b) $[\text{Cu}(\text{CN})_4]^{3-}$ & $[\text{Cu}(\text{NH}_3)_4]^{2+}$ have equal number of unpaired electron
 (c) $[\text{Cu}(\text{CN})_4]^{3-}$ & $[\text{Cd}(\text{CN})_4]^{2-}$ can be distinguished by passing H_2S gas.
 (d) All complexes have magnetic moment equal to zero.
71. Which of the following cation is colourless in aqueous solution ?
 (a) Ni^{2+} (b) Co^{3+}
 (c) Cr^{3+} (d) Ti^{4+}
72. What is the magnetic moment of high spin octahedral complex that has 6 electron in 3d orbitals?
 (a) 0 B.M. (b) $\sqrt{24}$ B.M.
 (c) $\sqrt{8}$ B.M. (d) $\sqrt{15}$ B.M.
73. For the complex, $[\text{Fe}^{\text{II}}(\text{en})_3][\text{Co}^{\text{III}}(\text{NH}_3)_2(\text{NO}_2)_4]_2$ the value of n is -
 (a) 1 (b) 2
 (c) 3 (d) 4

74. In which of the following pair, hybridisation, magnetic moment and geometry are same :-
 (a) $[\text{IrF}_6]^{3-}$, $[\text{Co}(\text{NH}_3)_6]^{3+}$
 (b) $[\text{CrF}_6]^{3-}$, $[\text{Co}(\text{H}_2\text{O})_6]^{2+}$
 (c) $[\text{Cu}(\text{NH}_3)_4]^{2+}$, $[\text{Ni}(\text{NH}_3)_4]^{2+}$
 (d) $[\text{Co}(\text{H}_2\text{O})_6]^{3+}$, $[\text{Co}(\text{H}_2\text{O})_6]^{4+}$
75. What is the oxidation state, magnetic moment of central metal cation in the following oxo complex $[\text{Os}(\text{ONO})(\text{O})_2(\text{O}_2)(\text{SCN})(\text{H}_2\text{O})]\text{OH}$?
 (a) +8, $\sqrt{15}$ B.M. (b) +6 ; 0
 (c) +6, $\sqrt{35}$ B.M. (d) +8, 0 BM
76. If $10Dq_0 < P$ then crystal field splitting energy for d^7 ion in octahedral field will be :-
 (a) $20 Dq - 2P$ (b) $4 Dq$
 (c) $8 Dq$ (d) $24 Dq - 2p$
77. Which of the following complex is low spin as well as diamagnetic ?
 (a) $\text{K}_4[\text{Fe}(\text{CN})_6]$ (b) $[\text{Co}(\text{C}_2\text{O}_4)_3]^{3-}$
 (c) $[\text{Ni}(\text{H}_2\text{O})_6]^{2+}$ (d) both (a) & (b)
78. Correct statements about complex consisting Ni, NH_3 , H_2O and NO_3^- ion -
 (a) If magnetic moment is zero, then the formula of the complex will be $[\text{Ni}(\text{NH}_3)_4](\text{NO}_3)_2 \cdot 2\text{H}_2\text{O}$
 (b) If magnetic moment is $2\sqrt{2}$ & conducts electricity, then the formula of the complex is $[\text{Ni}(\text{NH}_3)_4(\text{H}_2\text{O})_2](\text{NO}_3)_2 \text{BM}$
 (c) The higher and lower value of magnetic moment of the given complex corresponds to octahedral and square planar complex.
 (d) All
79. Which of the following complex is paramagnetic as well as square planar ?
 (a) NiCl_4^{2-} (b) $[\text{Cu}(\text{CN})_4]^{3-}$
 (c) $[\text{Cu}(\text{DMG})_2]$ (d) $[\text{PtBr}_4]^{2-}$
80. Which of the following complex has lowest number of unpaired electron ?
 (a) $[\text{NiF}_6]^{2-}$ (b) $[\text{Cu}(\text{NH}_3)_4]^{2+}$
 (c) $[\text{CoF}_6]^{3-}$ (d) $[\text{Fe}(\text{EDTA})]^{-1}$
81. The spin magnetic moment of $\text{K}_3[\text{Fe}(\text{C}_2\text{O}_4)_3]$ is :-
 (a) $\sqrt{15}$ B.M. (b) $\sqrt{35}$ B.M.
 (c) $\sqrt{24}$ B.M. (d) $\sqrt{2}$ B.M.
82. What is the magnetic nature of $\text{Co}(\text{II})$ in tetrahedral and octahedral complexes respectively ?
 (a) Paramagnetic, Paramagnetic
 (b) Paramagnetic, Diamagnetic
 (c) Diamagnetic, Diamagnetic
 (d) Diamagnetic, Paramagnetic
83. For octahedral complex the value of the spin only magnetic moment for one of the following configuration is 2.84 B.M. The correct one is :-
 (a) d^4 (strong field ligand)
 (b) d^4 (weak field ligand)
 (c) d^3 (in weak as well as strong field ligand)
 (d) d^5 (strong field ligand)
84. $\left[\text{CrCl} \cdot 3 \text{CH}_2 \begin{array}{l} \nearrow \text{CH}_2-\text{CH}_2-\text{NH}_2 \\ \searrow \text{NH}_2 \end{array} \right]$ shows which type of isomerism :-
 (a) Ligand isomerism
 (b) Polymerisation isomerism
 (c) Linkage isomerism
 (d) Ionisation isomerism
85. Which of the following property of $[\text{Ni}(\text{H}_2\text{O})_6]\text{SO}_4$ can not be explained by valence bond theory ?
 (a) It has 6 M-L coordinate bond
 (b) Geometry of Ni^{2+} is octahedral
 (c) Total six Ni-O linkage is present
 (d) It is green solution
86. The correct order of magnetic moment (spin only magnetic moment in BM) among is :-
 (a) $[\text{Fe}(\text{CN})_6]^{4-} > \text{MnCl}_4^{2-} > \text{CoCl}_4^{2-}$
 (b) $\text{MnCl}_4^{2-} > \text{CoCl}_4^{2-} > [\text{Fe}(\text{CN})_6]^{4-}$
 (c) $\text{MnCl}_4^{2-} > [\text{Fe}(\text{CN})_6]^{4-} > \text{CoCl}_4^{2-}$
 (d) $[\text{Fe}(\text{CN})_6]^{4-} > \text{CoCl}_4^{2-} > \text{MnCl}_4^{2-}$
87. Which of the following is paramagnetic ?
 (a) $\text{K}_2[\text{NiF}_6]$
 (b) $[\text{Co}(\text{H}_2\text{O})_6]^{3+}$
 (c) $\text{K}_4[\text{Fe}(\text{CN})_5\text{O}_2]$ [$\text{O}_2^{2-} = \text{Peroxido}$]
 (d) None of these

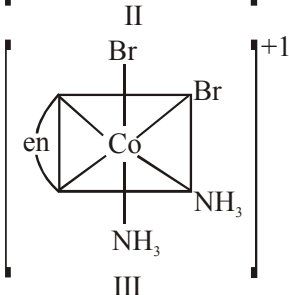
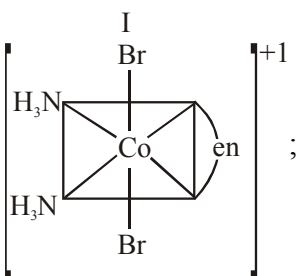
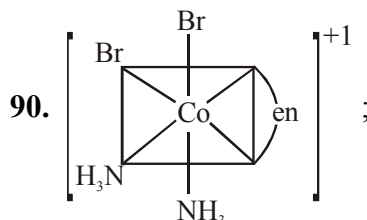
2.10 COORDINATION COMPOUNDS

88. For which complex compound, optical activity does not depend upon the orientation of the ligand around metal cation :-

- (a) $[\text{CoCl}_3(\text{NH}_3)_3]$
 (b) $[\text{Co}(\text{en})_3]\text{Cl}_3$
 (c) $[\text{Co}(\text{C}_2\text{O}_4)_2(\text{NH}_3)_2]^-$
 (d) $[\text{CrCl}_2(\text{NH}_3)_2\text{en}]$

89. Which of the following complex is paramagnetic as well as low spin complex ?

- (a) $[\text{Cr}(\text{CN})_6]^{4-}$ (b) $[\text{Ni}(\text{NH}_3)_6]^{2+}$
 (c) $[\text{Fe}(\text{CN})_6]^{4-}$ (d) $[\text{Co}(\text{H}_2\text{O})_6]^{3+}$



Incorrect statement are :-

- (a) I, II are G.I.
 (b) II, III are optical isomers
 (c) I, III are optical isomers
 (d) II & III are Geometrical isomers

91. Which of the following option has maximum number of unpaired electrons?

- (a) A tetrahedral d^6 ion
 (b) $[\text{Co}(\text{H}_2\text{O})_6]^{3+}$
 (c) A square planar complex with d^7 ion
 (d) A coordination compound with magnetic moment of 5.92 B.M.

92. In $[\text{NiCl}_4]^{2-}$, the magnetic moment is :-

- (a) 2.82 B.M. (b) 1.41 B.M.
 (c) 1.82 B.M. (d) 5.46 B.M.

93. Geometrical isomerism is found in coordination compounds having coordination number :-

- (a) 2 (b) 3
 (c) 4 (tetrahedral) (d) 6

94. Cis trans isomerism is found in square planar complexes of molecular formula (a & b are monodentate ligand) :-

- (a) Ma_4 (b) Ma_3b
 (c) Ma_2b_2 (d) Mab_3

95. The complexes $[\text{Pt}(\text{NH}_3)_4]$ $[\text{PtCl}_6]$ & $[\text{Pt}(\text{NH}_3)_4\text{Cl}_2]$ $[\text{PtCl}_4]$ are :-

- (a) Linkage isomers
 (b) Optical isomers
 (c) Coordination isomers
 (d) Ionisation isomers

96. $[\text{Sc}(\text{H}_2\text{O})_6]^{3+}$ ion is :-

- (a) Coloured and paramagnetic
 (b) Colourless and paramagnetic
 (c) Colourless and diamagnetic
 (d) Coloured and octahedral

97. Maximum paramagnetism in 3d series is shown by:-

- (a) Mn (b) Co
 (c) Ni (d) Fe

98. Which of the following complexes has geometry different from others :-

- (a) $[\text{NiCl}_4]^{2-}$ (b) $[\text{Ni}(\text{CO})_4]$
 (c) $[\text{Ni}(\text{CN})_4]^{2-}$ (d) $[\text{Zn}(\text{NH}_3)_4]^{2+}$

99. Which of the following is a high spin complex ?

- (a) $[\text{Co}(\text{NH}_3)_6]^{3+}$ (b) $[\text{Fe}(\text{CN})_6]^{4-}$
 (c) $[\text{Ni}(\text{CN})_4]^{2-}$ (d) $[\text{FeF}_6]^{3-}$

100. In the complex $\text{Co}(\text{NH}_3)_3\text{Cl}_3$ chloride ion satisfy :-

- (a) only primary valency
 (b) only secondary valency
 (c) both primary and secondary valencies
 (d) None

- 101.** A complex cation is formed by Pt^{+4} ion with ligands. Which of the following can be its correct IUPAC name :-
 (a) Diammineethylenediaminedithiocyanato-s platinum(II)ion
 (b) Diamineethylenediaminedithiocyanato-s platinum(IV)ion
 (c) Diamminebis(ethylenediamine)dithiocyanato-s platinum(IV)ion
 (d) Diammineethylenediaminedithiocyanato-s platinum(IV)ion
- 102.** The number of geometrical isomers possible for complex $[\text{M}(\text{gly})_2\text{Br}_2]$ is -
 (a) 3 (b) 5
 (c) 7 (d) 9
- 103.** If NO reacts with $[\text{Cr}(\text{CO})_6]$, how many CO groups can be replaced by NO ?
 (a) All 6 CO groups are replaced by 6 NO groups.
 (b) 4 CO groups are replaced by 6 NO groups.
 (c) 2 CO groups are replaced by 3 NO groups.
 (d) All the 6 CO groups are replaced by 4 NO groups.
- 104.** During developing of a photographic plates, one of the step involves washing the plate with hypo solution. The purpose of this is to :-
 (a) Reduce reacted AgBr to Ag
 (b) Convert unreacted AgBr to silver thiosulphate complex $[\text{Ag}(\text{S}_2\text{O}_3)_2]^{3-}$
 (c) Convert unreacted AgBr to Ag_2SO_4
 (d) Convert reacted AgBr to $\text{Ag}_2\text{S}_2\text{O}_3$
- 105.**
$$\left[(\text{NH}_3)_5 \text{Co} - \text{O} - \text{O} - \text{Co}(\text{NH}_3)_5 \right]^{4+} \xrightarrow[\text{Oxidised}]{\text{S}_2\text{O}_8^{2-}}$$

 Green colour complex

$$\left[(\text{NH}_3)_5 \text{Co} - \text{O} - \text{O} - \text{Co}(\text{NH}_3)_5 \right]^{5+}$$

 Brown colour complex
 The magnetic moment of green complex is 1.73 B.M. and for brown complex magnetic moment is zero. (O—O) is same in all replaced in both the complex. The oxidation state of Co in brown complex and green complex respectively are :-
 (a) III, II ; IV, III (b) III, II ; III, III
 Brown Green Brown Green
 (c) III, III ; III, II (d) III, IV ; III, III
 Brown Green Brown Green
- 106.** A complex $\text{K}_n[\text{MnF}_6]$ has magnetic moment 4.9 B.M. What is the value of n -
 (a) 2 (b) 3
 (c) 4 (d) 5
- 107.** Which of the following is copper (I) compound :-
 (a) $[\text{Cu}(\text{H}_2\text{O})_4]^{2+}$ (b) $[\text{Cu}(\text{CN})_4]^{3-}$
 (c) $[\text{Cu}(\text{NH}_3)_4]^{2+}$ (d) All of these
- 108.** $\text{Cu}^{2+}(\text{aq}) + \text{KCN}(\text{aq}) (\text{excess}) \longrightarrow \text{Complex } \text{X}_1$
 $\text{Co}^{2+}(\text{aq}) + \text{KNO}_2(\text{s}) + \text{H}^+(\text{aq}) \longrightarrow \text{Complex } \text{X}_2$
 $\text{Zn}^{2+}(\text{aq}) + \text{NaOH}(\text{aq}) (\text{excess}) \longrightarrow \text{Complex } \text{X}_3$
 $\text{X}_1, \text{X}_2, \text{X}_3$ are respectively :-
 (a) Tetrahedral, Octahedral and Square planar
 (b) Tetrahedral, Octahedral and Tetrahedral
 (c) Square planar, Octahedral and Tetrahedral
 (d) Octahedral, Tetrahedral and Square planar
- 109.** Which of the following is not correctly matched ?
 (a) Sodium (ethylenediaminetetraacetato) chromate(II) : $\text{Na}_2[\text{Cr}(\text{CH}_3\text{COO})_4(\text{en})]$
 (b) Dichloridobis(ethane-1,2-diamine)cobalt(III) ion : $[\text{Co}(\text{en})_2\text{Cl}_2]^+$
 (c) Tris(bipyridine) iron(II)ion : $[\text{Fe}(\text{NH}_4\text{C}_5\text{H}_4\text{N})_3]^{2+}$
 (d) Ammineaquadibromidocopper(II) : $[\text{Cu}(\text{H}_2\text{O})(\text{NH}_3)\text{Br}_2]$
- 110.** Which of the following has total 18 valence electron:-
 (a) $[\text{Fe}(\text{CO})_3(\eta^4\text{-C}_8\text{H}_8)]$
 (b) $[\text{Cr}(\eta^6\text{-C}_6\text{H}_6)_2]$
 (c) $[\text{Mo}(\text{CO})_3(\eta^6\text{-C}_7\text{H}_7)]^{+1}$
 (d) All
- 111.** Jahn-Teller distortion in octahedral field will occur for :-
 (a) d^1 ion (b) d^4 ion
 (c) d^9 ion (d) All
- 112.** Jahn-Teller distortion in octahedral field will not occur for :-
 (a) d^3 ion (b) d^8 ion
 (c) d^5 ion (d) (a) & (b) both

2.12 COORDINATION COMPOUNDS

113. Which of the following is incorrect about :-

Tetraamminedithiocyanato-Scobalt(III)
tris(oxalato)cobaltate(III)

- (a) Formula of the complex is $[\text{Co}(\text{SCN})_2(\text{NH}_3)_4][\text{Co}(\text{Ox})_3]$
(b) It is a chelating complex and show linkage isomerism
(c) It shows optical isomerism
(d) It shows geometrical isomerism

114. Irrespective of the nature of ligand, inner orbital octahedral complex is formed by :-

- (a) d^1 ion (b) d^2 ion
(c) d^3 ion (d) All

115. CoCl_4^{2-} is observed to be tetrahedral whereas CoCl_6^{4-} is observed to be high spin octahedral complex. The configuration in tetrahedral field and octahedral field are respectively :-

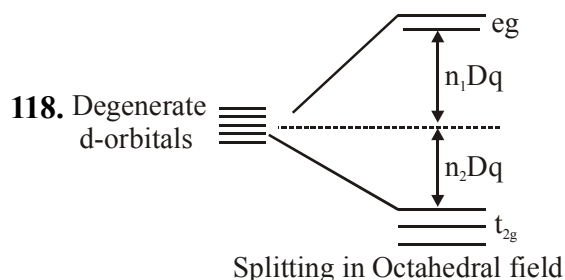
- (a) $e^4 t_2^3; t_{2g}^5 e_g^2$ (b) $e^4 t_2^3; t_{2g}^6 e_g^1$
(c) $t_2^4 e^3; t_{2g}^6 e_g^1$ (d) $t_2^4 e^3; t_{2g}^5 e_g^2$

116. Incorrect statement regarding dichromate ion :-

- (a) Six Cr–O bond lengths are identical
(b) Cr in dichromate ion is sp^3 hybridisation
(c) the colour is due to d–d transition
(d) the colour is due to $L \rightarrow M$ charge transfer (where M - Metal ion, L - Ligand)

117. In crystal field theory, each of p orbitals in octahedral field :-

- (a) equally face two ligands along the coordinate axis
(b) unequally face two ligands along the coordinate axis
(c) equally face two ligands in between the coordinate axis
(d) unequally face two ligands in between the coordinate axis



n_1 and n_2 are :-

- (a) 4 & 6 (b) 7 & 3
(c) 3 & 7 (d) 6 & 4

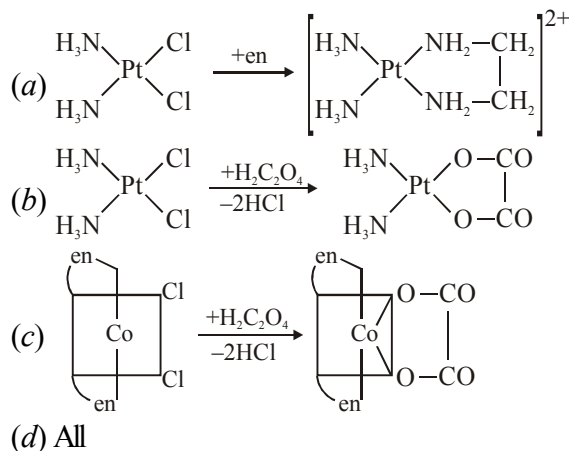
119. The order of Δ_0 :-

- (a) $\text{Co}^{3+} > \text{Rh}^{3+} > \text{Ir}^{3+}$ (for same ligand)
(b) $\text{F}^- > \text{Br}^- > \text{Cl}^-$ (for same metal ion)
(c) $\text{CN}^- > \text{NH}_3 > \text{S}^{2-}$ (for same metal ion)
(d) $\text{Fe}^{2+} > \text{Fe}^{3+}$ (for same ligand)

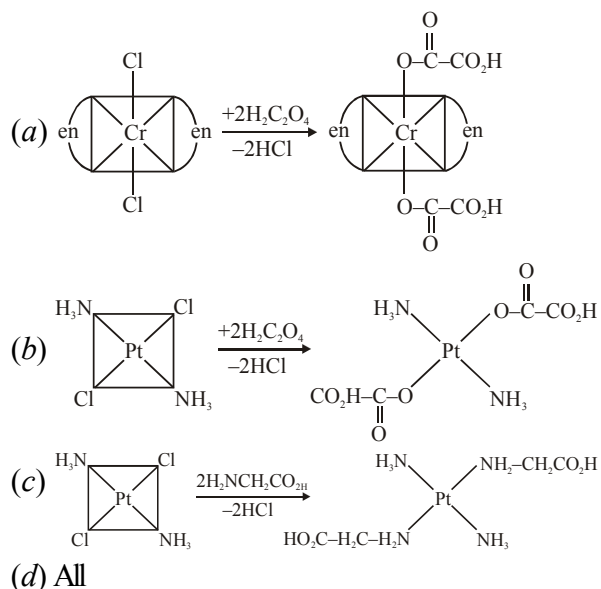
120. If the splitting energy of octahedral field is Δ_0 and if the splitting energy of tetrahedral field is Δ_t then :-

- (a) $\Delta_0 = \Delta_t$ (b) $\Delta_0 < \Delta_t$
(c) $\Delta_0 = \frac{2}{3} \Delta_t$ (d) $\Delta_t = \frac{4}{9} \Delta_0$

121. In which of the following reactions, correct product is given ?



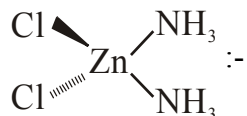
122. In which of the following reactions, correct product is given :-



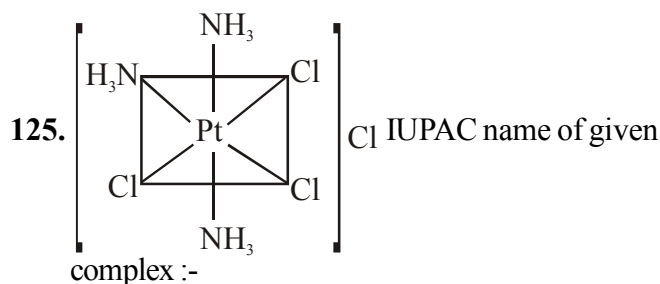
123. For $\text{Xe}[\text{PtF}_6]$; oxidation state of Xe and Pt are respectively :-

- (a) +4, +2 (b) +2, +4
(c) +1, +5 (d) +6, 0

124. Name the following coordination species



- (a) Cis-diamminedichloridozinc(II)
(b) Trans-diamminedichloridozinc(II)
(c) Diamminedichloridozinc(II)
(d) Dichloridodiamminezinc(II)



- (a) Cis-triamminetrichloridoplatinum(IV)chloride
(b) Trans-triamminetrichloridoplatinum(IV)chloride
(c) Fac-triamminetrichloridoplatinum(IV)chloride
(d) Mer-triamminetrichloridoplatinum(IV)chloride

126. IUPAC name of $\text{Xe}[\text{PtF}_6]$ is :-

- (a) Xenoniumhexafluoridoplatinum(V)
(b) Xenonhexafluoridoplatinate(V)
(c) Xenonhexafluoridoplatinate(VI)
(d) Hexafluoroxenoniumplatinum(VI)

127. Type of isomerism which may be possible for six coordinated complex of metal M having formula $\text{Cr}(\text{NO}_2)_2 \cdot 6\text{H}_2\text{O}$:-

- (a) Hydrate isomerism
(b) Linkage isomerism
(c) Geometrical isomerism
(d) All of them

128. $[\text{NiCl}_2(\text{PMe}_3)_2]$ is a paramagnetic complex of Ni(II). How many isomers are possible of analogous diamagnetic complex of Pd^{2+} :-

- (a) Zero (b) Five
(c) Two (d) Three

129. Which of the following complex can have optical isomer ?

- (a) $[\text{Pt}(\text{bn})_2]^{2+}$ (b) $[\text{PtCl}_4]^{2-}$
(c) $[\text{Pt}(\text{en})_2]^{2+}$ (d) $[\text{Pt}(\text{NH}_3)_3\text{Cl}]^{1+}$

130. Geometrical isomerism can be shown by :-

- (a) $[\text{Ag}(\text{CN})(\text{NH}_3)]$ (b) $\text{Na}_2[\text{Cd}(\text{NO}_3)_4]$
(c) $[\text{PtCl}_4\text{I}_2]^{2-}$ (d) $[\text{PtCl}(\text{NH}_3)_3]$

131. Which of the following options is correct :-

- (a) Pi-acid ligands favour spin pairing
(b) In low spin complex, the number of unpaired electrons is equal to that present in the free ion
(c) In high spin complex, the number of unpaired electron is minimum
(d) $\text{Ti}(\text{H}_2\text{O})_6^{3+}$; $\text{V}(\text{H}_2\text{O})_6^{3+}$; $\text{Cr}(\text{H}_2\text{O})_6^{3+}$ all are sp^3d^2 hybridised

132. $[\text{Cr}(\text{H}_2\text{O})_6]^{3+}$; $[\text{Fe}(\text{H}_2\text{O})_6]^{3+}$; $[\text{Co}(\text{H}_2\text{O})_6]^{3+}$ are:-

- (a) sp^3d^2 ; sp^3d^2 ; sp^3d^2 hybridised
(b) d^2sp^3 ; d^2sp^3 ; d^2sp^3 hybridised
(c) d^2sp^3 ; sp^3d^2 ; d^2sp^3 hybridised
(d) sp^3d^2 ; d^2sp^3 ; sp^3d^2 hybridised

133. In $[\text{Cu}(\text{NH}_3)_4]^{2+}$, dsp^2 hybridisation takes place for Cu^{2+} ion. The d orbital involved is :-

- (a) $(n-1)\text{d}_{z^2}$ (b) $(n-1)\text{d}_{x^2-y^2}$
(c) $(n-1)\text{d}_{xy}$ (d) $(n-1)\text{d}_{xz}$

134. $[\text{BeCl}_4]^{2-}$; $[\text{MnCl}_4]^{2-}$; $[\text{CoCl}_4]^{2-}$ all are sp^3 hybridised with :-

- (a) 2,5,3 unpaired electron respectively
(b) 0,5,2 unpaired electron respectively
(c) 1,5,1 unpaired electron respectively
(d) 0,3,3 unpaired electron respectively

135. If $\text{pn} = \begin{array}{c} \text{CH}_3\text{CHNH}_2 \\ | \\ \text{CH}_2\text{NH}_2 \end{array}$ and $\text{tn} = \begin{array}{c} \text{CH}_2\text{CH}_2\text{NH}_2 \\ | \\ \text{CH}_2\text{NH}_2 \end{array}$ then

$\text{CoCl}_2(\text{pn})_2$ and $\text{CoCl}_2(\text{tn})_2$ are example of :-

- (a) ligand isomerism
(b) linkage isomerism
(c) coordinate position isomerism
(d) ionisation isomerism

136. Which of the following is optically active :-

- (a) $[\text{Zn}(\text{gly})_2]^{-1}$ (b) $[\text{Ca}(\text{EDTA})]^{-2}$
(c) $[\text{Co}(\text{Ox})_3]^{-3}$ (d) All of these

2.14 COORDINATION COMPOUNDS

137. For the complex ion, dichloridobis(ethylenediamine)cobalt(III), which of the following is incorrect:-
 (a) it has two optically active isomers
 (b) it has two optically inactive isomers
 (c) it is example of cationic complex
 (d) it is example of octahedral complex
138. In $[\text{Cu}(\text{NH}_3)_4]^{2+}$ outermost electron is present in :-
 (a) 3d orbital (b) 4s orbital
 (c) 4p orbital (d) 4d orbital
139. The type of hybridisation of $\text{Fe}(\text{CO})_5$ is :-
 (a) $d_{x^2-y^2} sp^3$ (b) $sp^3 d_{x^2-y^2}$
 (c) $d_{z^2} sp^3$ (d) $sp^3 d_{z^2}$
140. The hybridisation of metal ion and shape of $[\text{CuCl}_5]^{3-}$ are respectively :-
 (a) $d_{x^2-y^2} sp^3$, square pyramidal
 (b) $sp^3 d_{x^2-y^2}$, trigonal bipyramidal
 (c) $d_{z^2} sp^3$, trigonal bipyramidal
 (d) $sp^3 d_{z^2}$, tetrahedral
141. VO_4^{3-} :-
 (a) square planar geometry with dsp^2 hybridisation
 (b) tetrahedral geometry with sp^3 hybridisation
 (c) tetrahedral geometry with d^3s hybridisation
 (d) None
142. In crystal field splitting diagram for square planar complex the d-orbital having highest energy is :-
 (a) $d_{x^2-y^2}$ (b) d_{xy}
 (c) d_{z^2} (d) d_{xz} & d_{yz}
143. In crystal field splitting diagram for square planar complex the d-orbital having least energy is :-
 (a) $d_{x^2-y^2}$ (b) d_{xy}
 (c) d_{z^2} (d) d_{xz} & d_{yz}
144. Among the following the compound that is both paramagnetic and coloured is :-
 (a) $\text{K}_2\text{Cr}_2\text{O}_7$ (b) $(\text{NH}_4)_2[\text{TiCl}_6]$
 (c) VOSO_4 (d) $\text{K}_3[\text{Cu}(\text{CN})_4]$
145. Which of the following is correct electronic configuration of 3d orbital in excited state of central metal ion, when $[\text{Ti}(\text{H}_2\text{O})_6]^{3+}$ absorbed yellow-green light :-
 (a) $3d^1$ (b) $t_{2g}^1 e_g^1$
 (c) $t_{2g}^1 e_g^0$ (d) $t_{2g}^0 e_g^1$
146. $[\text{Fe}(\text{H}_2\text{O})_6]^{2+}$ has crystal field splitting energy value $10,400 \text{ cm}^{-1}$ and pairing energy value $17,600 \text{ cm}^{-1}$ then it is :-
 (a) low spin complex
 (b) paramagnetic in nature
 (c) diamagnetic in nature
 (d) none of these
147. An ion M^{2+} forms the complexes $[\text{M}(\text{H}_2\text{O})_6]^{2+}$, $[\text{M}(\text{en})_3]^{2+}$ and $[\text{MBr}_6]^{4-}$, match the complex with the appropriate colour :-
 (a) Green, blue and red
 (b) Blue, red and green
 (c) Green, red and blue
 (d) Red, blue and green
148. Which one of the following species does not represent cationic species of vanadium formed in aqueous solution :-
 (a) VO_2^+ (b) VO^{2+}
 (c) $[\text{V}(\text{H}_2\text{O})_6]^{3+}$ (d) VO_2^{2+}
149. In which of the following C–O bond length is shorter as compared to CO.
 (a) CO^+ (b) H_2CO
 (c) $[\text{Ni}(\text{CO})_4]$ (d) $[\text{V}(\text{CO})_6]^{-1}$
150. $\text{PtCl}_4 \cdot 3\text{NH}_3$ is example of :-
 (a) 1 : 1 electrolyte (b) 1 : 2 electrolyte
 (c) 1 : 3 electrolyte (d) 1 : 4 electrolyte

EXERCISE # II**One or More Than One Correct Answer :**

- To form a coordination bond, one needs a ligand. Which of the following species can not be a ligand?
 (a) NH_4^+ (b) PH_4^+
 (c) H_3O^+ (d) NO^+
- Which of the following are examples of bidentate ligand?
 (a) $\text{C}_2\text{O}_4^{2-}$ (b) en
 (c) Glycinato (d) $\text{NH}_2\text{--NH}_2$
- Among the properties of CN^- (cyanid ion), which of the following properties are correct :-
 (a) oxidising (b) reducing
 (c) acidic (d) complexing

4. Which of the following are anionic didentate ligands where oxygen is donor atom ?
 (a) H_2O (b) Acetato
 (c) Carboxylato (d) Oxalato
5. Which of the following is example of ambidentate monodentate ligand ?
 (a) SCN^- (b) NO_2^-
 (c) $\text{S}_2\text{O}_3^{2-}$ (d) CN^-
6. Which of the following are colourless ?
 (a) $[\text{Cu}(\text{CN})_4]^{3-}$ (b) $[\text{Ti}(\text{NO}_3)_4]$
 (c) $[\text{Co}(\text{NH}_3)_6]^{3+}$ (d) $[\text{Cu}(\text{NH}_3)_4]^{2+}$
7. The complex ion which has no d electrons in the central metal atom is :-
 (a) MnO_4^{-1} (b) $[\text{Co}(\text{Ml}_3)_6]^{3+}$
 (c) $[\text{Cr}(\text{H}_2\text{O})_6]^{3+}$ (d) CrO_4^{2-}
8. Select the correct statement about metal carbonyl complex compound :-
 (a) metal carbon bond in metal carbonyls possess both σ & π character
 (b) due to synergic bonding metal carbon bond becomes weak
 (c) due to synergic bonding carbon oxygen bond strength increases
 (d) in metal carbonyls extent of synergic bonding will increase with increases in negative charge on central metal ion.
9. Paramagnetic species are :-
 (a) $\text{H}_2[\text{NiF}_6]$ (b) $\text{K}_3[\text{CoF}_6]$
 (c) O_2 (d) B_2
10. Which ligands are examples σ donor ligand ?
 (a) Propylene
 (b) Propylene diamine
 (c) NH_2^-
 (d) 1, 2,3-triaminopropane
11. Which of the following are example of bidentate chelating ligand ?
 (a) (b)
 (c) (d)
12. Which ligand are examples of π donor ligand ?
 (a) cyclobutadiene (b) benzene
 (c) SO_4^{2-} (d) CO_3^{2-}
13. Which can act as +ve charged ligand ?
 (a) NO_2^+ (b) $\text{NH}_2^+-\text{NH}_3^+$
 (c) CH_3^+ (d) CN^-
14. Which can act as reducing agent :-
 (a) $\text{Fe}(\text{CO})_4$ (b) $[\text{Co}(\text{CO})_4]$
 (c) $[\text{V}(\text{CO})_6]$ (d) $[\text{Mn}(\text{CO})_6]$
15. Which of the following statements are correct :-
 (a) In first transition series Cr can show highest value of magnetic moment in the ground state.
 (b) $[\text{Cr}(\text{NH}_3)_6]$, $[\text{Cr}(\text{NO}_2)_6]$ & $[\text{Cr}(\text{NH}_3)_4(\text{NO}_2)_2]$ can be distinguished from conductivity measurement in aqueous solution.
 (c) $[\text{Ti}(\text{H}_2\text{O})_6]^{3+}$ is coloured white $[\text{Sc}(\text{H}_2\text{O})_6]^{3+}$ is colourless
 (d) $[\text{Al}(\text{H}_2\text{O})_6]^{3+}$ can act as proton donor Bronsted acid.
16. Which of the following statements are incorrect ?
 (a) Among TiCl_3 , MnCl_2 , FeSO_4 , CuSO_4 ; MnCl_2 contains highest number of unpaired electron
 (b) In forming metal complex, metal ion functions as Lewis acid and ligand functions as Lewis base
 (c) In aq. solution, Mn^{3+} is stronger oxidising reagent as compared to Co^{3+}
 (d) All
17. For which of the following complex, correct IUPAC name is given :-
 (a) $\text{K}_2[\text{SiF}_6]$: Potassiumhexafluorido Silicate(IV)
 (b) $[\text{Cr}(\text{NH}_3)_6]\text{ClSO}_4$: Hexaamminechromium (III) chloridesulphate
 (c) $[\text{Ru}(\text{NH}_3)_5(\text{N}_2)]\text{Cl}_3$: Pentaammine(dinitrogen) ruthenium(III)chloride
 (d) $[\text{Pt}(\text{PPh}_3)_4]$: Tetrakis(triphenylphosphane) platinum(0)
18. When metal is present in anionic part of the complex, then correct naming of metal will be :-
 (a) Ag : Silverate (b) Co : Cobaltate
 (c) Fe : Ferrate (d) Zn : Zincate

2.16 COORDINATION COMPOUNDS

19. For which ligand in IUPAC nomenclature the term bis / tris / tetrakis is used up, when more than one such type of ligands is present :-

- (a) $(\text{CH}_3)_2\text{NH}$ (b) $\text{NH}_2\text{CH}_2\text{CH}_2\text{NH}_2$
(c) $\eta^5\text{C}_5\text{H}_5$ (d) NH_3

20. What are the correct statements ?

- (a) $[\text{Co}(\text{en})_3]\text{Cl}_3$ has four optically active stereoisomers
(b) Ma_3bcd has total five stereoisomers out of which two are optically active and three are optically inactive.
(c) Resolution of $\text{cis}[\text{CoCl}_2(\text{en})_2]\text{Cl}$ is possible
(d) Resolution of $\text{trans}[\text{CoCl}_2(\text{en})_2]\text{Cl}$ is not possible

21. Which of the following statements are incorrect ?

- (a) Charge on the complex of ferric ion with EDTA is -1
(b) $[\text{Ni}(\text{en})_3]\text{Cl}_2$ has lower stability than $[\text{Ni}(\text{NH}_3)_6]\text{Cl}_3$
(c) NF_3 is a weaker ligand than NMe_3
(d) The complex $[\text{Cr}(\text{H}_2\text{O})_6]\text{Cl}_3$ loses six water molecules to concentrated H_2SO_4 and does not give any precipitate with AgNO_3 .

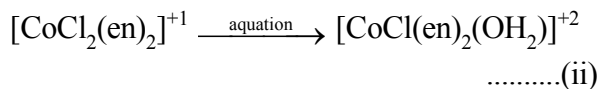
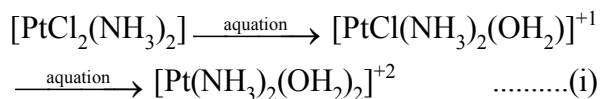
22. Which of the following statement are correct ?

- (a) Identification of cis-trans isomers of compound $\text{Ma}_2\text{b}_2\text{c}$ can be done by measuring dipole moment.
(b) Aqueous solution of copper sulphate solution gives green precipitate with aqueous potassium fluoride.
(c) The magnetic moment changed when CN^- ligands in octahedral complex are replaced by Cl^- ligand for d^6 ion.
(d) The magnetic moment remains same when CN^- ligands in octahedral complex are replaced by Cl^- for d^3 ion.

23. For which complex, facial and meridional isomers are possible ? :-

- (a) $[\text{Cr}(\text{gly})_3]$ (b) $[\text{CrCl}_3(\text{NH}_3)_3]$
(c) $[\text{RuCl}_3(\text{OH})_3]$ (d) $[\text{PtCl}_2(\text{NH}_3)_2(\text{py})_2]$

24. The reactivity of cis and trans isomeric compounds may differ widely. It may help to distinguish the cis-trans isomers :-



Correct statements :

- (a) Reactivity order in (I) : cis >>> trans
(b) Reactivity order in (I) : trans >>> cis
(c) Reactivity order in (II) : cis >>> trans
(d) Reactivity order in (II) : trans >>> cis

25. Which of the following π bonded organometallic compound follow Sidgwick EAN rule ?

- (a) Ferrocene (b) Zeises salt
(c) Cisplatin (d) $\text{Cr}(\text{C}_6\text{H}_6)_2$

26. Aqueous solution of potash alum gives :-

- (a) white precipitate with BaCl_2
(b) violet colour flame
(c) green colour flame
(d) white gelatinous ppt. with NaOH

27. Which of the following are the correct naming regarding ligand ?

- (a) $\text{NH}_3 \rightarrow$ Ammine
(b) $\text{CH}_3-\text{NH}_2 \rightarrow$ Methylamine
(c) $(\text{CH}_3)_2\text{NH} \rightarrow$ Dimethylamine
(d) $(\text{CH}_3)_3\text{N} \rightarrow$ Trimethylamine

28. Which of the following statements are correct :-

- (P) $[\text{Cr}(\text{H}_2\text{O})_4\text{Cl}_2]\text{Br} \cdot 2\text{H}_2\text{O}$
(Q) $[\text{Cr}(\text{H}_2\text{O})_4\text{Br}_2]\text{Cl} \cdot 2\text{H}_2\text{O}$
(R) $[\text{Cr}(\text{H}_2\text{O})_5\text{Cl}]\text{Br}_2 \cdot \text{H}_2\text{O}$
(S) $[\text{Cr}(\text{H}_2\text{O})_6]\text{Cl}_3$
(a) P & Q are example of ionisation isomers.
(b) Q & R are example of hydrated isomers.
(c) S is most conducting.
(d) P & R are example of ionisation isomers.

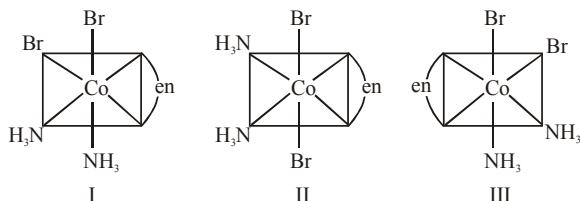
29. Which of the following have two stereoisomeric forms ?

- (a) $[\text{Cr}(\text{NO}_3)_3(\text{NH}_3)_3]$ (b) $\text{K}_3[\text{Fe}(\text{C}_2\text{O}_4)_3]$
(c) $[\text{Fe}(\text{en})\text{Br}_3\text{Cl}]^{-1}$ (d) $[\text{CoBrCl}(\text{Ox})_2]^{3-}$

30. Which of the following statements are correct for $[\text{Pt}(\text{H}_2\text{N}-\underset{\text{CH}_3}{\underset{|}{\text{CH}}}-\underset{\text{CH}_3}{\underset{|}{\text{CH}}}-\text{NH}_2)_2]^{2+}$:-

- (a) It has total 6 geometrical isomers
 (b) It has total 5 geometrical isomers
 (c) One geometrical isomers has COS and 4 geometrical isomers have POS
 (d) Two geometrical isomers has POS and 4 geometrical isomers have COS

31. Which of the following statements are correct ?

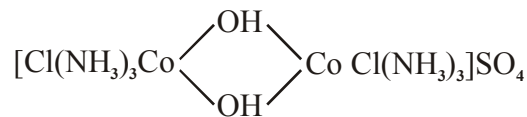
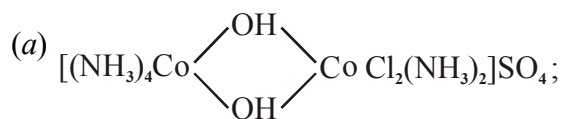


- (a) I, II are diastereoisomers
 (b) II, III are enantiomers
 (c) I, III are enantiomers
 (d) II, III are diastereoisomers
32. Which of the following statements are incorrect ?
- (a) In Mabcd, square planar complexes show both optical as well as geometrical isomerism
 (b) In Mabcd, tetrahedral complex, optical isomerism & geometrical isomerism both are observed
 (c) $[\text{Fe}(\text{EDTA})]^{-1}$ shows geometrical isomerism
 (d) tris(ethan-1,2-diamine)nickel(II)chloride is homoleptic outer orbital complex
33. Which of the following statements are correct ?
- (a) In $[\text{PtCl}_2(\text{NH}_3)_2]^{2+}$, the cis form is optically inactive while trans form is optically active
 (b) In $[\text{Cr}(\text{C}_2\text{O}_4)_3]^{3+}$, geometrical isomerism does not exist while optical isomerism exists
 (c) $[\text{Co}(\text{en})_2\text{Br}_2]^{+1}$, cis form is optically active while trans form is optically inactive
 (d) $[\text{Co}(\text{Ox})_2\text{Cl}(\text{NO}_2)]$, cis form is optically active while trans form is optically inactive (Ox - oxalato)

34. $(\text{PPh}_3)_2\text{ClPdCl}_2\text{PdCl}(\text{PPh}_3)$ shows -

- (a) geometrical isomerism
 (b) optical (space) isomerism
 (c) co-ordination position isomerism
 (d) ionisation isomerism

35. Which of the following relationship between complex are correct ?



: Coordination position isomers

- (b) $[\text{Pt}(\text{NH}_3)_4][\text{PtCl}_6]$; $[\text{PtCl}_2(\text{NH}_3)_4][\text{PtCl}_4]$: Coordination isomers

- (c) $[\text{Co}(\text{NH}_3)_3(\text{NO}_3)_3]$; $[\text{Co}(\text{NO}_2)(\text{NH}_3)_5][\text{Co}(\text{NH}_3)_2(\text{NO}_2)_4]_2$: Polymerisation isomers

- (d) $[\text{Co}(\text{NH}_3)_5\text{SCN}]^{2+}$; $[\text{Co}(\text{NH}_3)_5\text{NCS}]^{2+}$: Linkage isomers

36. Which of the following compounds show optical isomerism ?

- (a) $[\text{Pt}(\text{bn})_2]^{2+}$ (b) $[\text{CrCl}_2(\text{en})_2]^{+1}$
 (c) $[\text{Co}(\text{en})_3][\text{CoF}_6]$ (d) $[\text{Zn}(\text{gly})_2]$

37. Which of the following statements are correct about the complex ion $[\text{CrCl}(\text{NO}_2)(\text{en})_2]^{+1}$ (en = ethylenediamine) ?

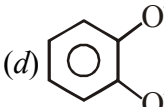
- (a) It has two geometrical isomers ; cis and trans isomer
 (b) Cis and trans isomers are diastereomers w.r.t. to each other
 (c) Only cis isomer is optically active, exists as d and l pair
 (d) It has three stereoisomers ; two of which are optically active and one is optically inactive which is meso form.

38. Which of the following statements are correct regarding the following compound $[\text{Pt}(\text{NH}_3)_4][\text{PtCl}_4]$:-

- (a) It is the polymerisation isomer of $[\text{Pt}(\text{NH}_3)_3\text{Cl}_3]$
 (b) E.A.N. of metal in cationic part and anionic part both are same
 (c) It is the coordination isomer of $[\text{Pt}(\text{NH}_3)_3\text{Cl}][\text{Pt}(\text{NH}_3)_3\text{Cl}_3]$
 (d) Synergic bonding is not involved in the complex.

2.18 COORDINATION COMPOUNDS

39. Correct option regarding Tetrakis (pyridine) dithiocyanato-N cobalt(III)nitrate.
 (a) It is chelating complex
 (b) It exist in linkage isomeric form
 (c) It exist in geometrical isomeric form
 (d) Number of ions formed in solution of the complex > 2
40. Wilkinson catalyst $[\text{RhCl}(\text{PPh}_3)_3]$ is -
 (a) square planar complex with dsp^2 hybridisation
 (b) optically inactive
 (c) diamagnetic
 (d) example of heterogeneous catalyst
41. Which of the following are dsp^2 hybridised ?
 (a) $[\text{AuCl}_4]^-$ (b) $[\text{Pt}(\text{NH}_3)_4]^{2+}$
 (c) $[\text{AuBr}_2\text{Cl}_2]^-$ (d) $[\text{Ni}(\text{CO})_4]$
42. Which of following are sp^3 hybridised for Ni-complexes ?
 (a) $[\text{NiCl}_4]^{2-}$ (b) $[\text{Ni}(\text{CN})_4]^{2-}$
 (c) $[\text{Ni}(\text{CN})_4]^{4-}$ (d) $[\text{Ni}(\text{NH}_3)_6]^{2+}$
43. $[\text{Zn}(\text{gly})_2]$ is -
 (a) sp^3 hybridised (b) optically inactive
 (c) dsp^2 hybridised (d) optically active
44. $[\text{Pt}(\text{NH}_3)_4(\text{NO}_2)_2]$ exists in 2 geometrical isomers. Which of the following options are correct ?
 (a) Cis & trans isomer both have plane of symmetry
 (b) Cis & trans isomer both have C_2 axis of symmetry
 (c) both are optically active
 (d) both have centre of symmetry
45. Brown ring - having molecular formula $[\text{Fe}(\text{H}_2\text{O})_5\text{NO}]\text{SO}_4$. Which of the following statements are correct regarding brown ring -
 (a) Outer orbital octahedral complex with sp^3d^2 hybridisation
 (b) Fe has (+2) configuration and NO is example of neutral ligand
 (c) It has 2 unpaired electron and magnetic moment 2.73 B.M.
 (d) It has 3 unpaired electron and magnetic moment 3.82 B.M.
46. Which of the following are inner orbital octahedral complex ions and diamagnetic ?
 (a) $[\text{Co}(\text{NH}_3)_6]^{3+}$ (b) $[\text{Mn}(\text{CN})_6]^{3-}$
 (c) $[\text{Fe}(\text{CN})_6]^{4-}$ (d) $[\text{Fe}(\text{CN})_6]^{3-}$
47. Which of the following are outer orbital octahedral complex with same number of unpaired electrons ?
 (a) $[\text{MnCl}_6]^{3-}$ (b) $[\text{FeF}_6]^{3-}$
 (c) $[\text{CoF}_6]^{3-}$ (d) $[\text{Ni}(\text{NH}_3)_6]^{2+}$
48. Correct statement regarding $[\text{MnCl}_4]^{2-}$:-
 (a) It has magnetic moment 5.92 B.M.
 (b) It has five unpaired electrons
 (c) It is tetrahedral complex
 (d) The hybridisation of Mn in the complex is sp^3
49. Correct order of ligand strength -
 (a) $\text{CN}^- > \text{CO}$ (b) $\text{F}^- > \text{Cl}^-$
 (c) $\text{NO}_2^- > \text{NO}_3^-$ (d) $\text{H}_2\text{O} > \text{NH}_3$
50. Which of the following complexes are homoleptic?
 (a) $[\text{Co}(\text{NH}_3)_6]^{3+}$ (b) $[\text{Co}(\text{NH}_3)_4]^{+1}$
 (c) $[\text{Ni}(\text{CN})_4]^{2-}$ (d) $[\text{Ni}(\text{NH}_3)_4\text{Cl}_2]$
51. Which of the following complexes are heteroleptic?
 (a) $[\text{Cr}(\text{NH}_3)_6]^{3+}$ (b) $[\text{Fe}(\text{NH}_3)_4\text{Cl}_2]^{+1}$
 (c) $[\text{Mn}(\text{CN})_6]^{4-}$ (d) $[\text{Co}(\text{NH}_3)_4\text{Cl}_2]$
52. Which of the following complexes are optically active?
 (a) $[\text{Co}(\text{en})_3]^{3+}$ (b) $\text{trans} [\text{Co}(\text{en})_2\text{Cl}_2]^{+1}$
 (c) $\text{Cis} [\text{Co}(\text{en})_2\text{Cl}_2]^{+1}$ (d) $[\text{Cr}(\text{NH}_3)_5\text{Cl}]^{2+}$
53. An aqueous pink solution of cobalt(III)chloride changes to deep blue on addition of excess HCl. This is because -
 (a) $[\text{Co}(\text{H}_2\text{O})_6]^{3+}$ is transformed into $[\text{CoCl}_6]^{4-}$
 (b) $[\text{Co}(\text{H}_2\text{O})_6]^{3+}$ is transformed into $[\text{CoCl}_4]^{2-}$
 (c) Tetrahedral complexes have smaller crystal field splitting than octahedral complex
 (d) Tetrahedral complexes have larger crystal field splitting than octahedral complex
54. Which of the following complex is tetrahedral but diamagnetic ?
 (a) $\text{K}_3[\text{Cu}(\text{CN})_4]$ (b) $\text{Ni}(\text{CO})_4$
 (c) $\text{K}_4[\text{Ni}(\text{CN})_4]$ (d) $[\text{Zn}(\text{H}_2\text{O})_4]^{2+}$

55. Which of the following are correct statements ?
 (a) Pd(II), Pt(II) forms mostly 4 coordination, square planar diamagnetic coordination compounds.
 (b) In $\text{Mn}_2(\text{CO})_{10}$ there are total 70 electrons present
 (c) $[\text{Ni}(\text{CN}_4)]^{4-}$ & $[\text{Ni}(\text{CN}_4)]^{2-}$ have different dipole moment
 (d) The difference of the number of unpaired electron in Zn^{2+} and $[\text{Zn}(\text{NH}_3)_4]^{2+}$ is zero
56. $\text{FeCl}_3 + \text{K}_4[\text{Fe}(\text{CN})_6] \rightarrow$ Complex product. Correct statement regarding complex product :-
 (a) IUPAC name of complex product is iron(II) hexacyanidoferrate(III)
 (b) Total number of Fe ions in the solution is 4
 (c) Average oxidation state of Fe is 2.57
 (d) EAN of central metal atom of complex is 36
57. Which complex compound having $t_2^6 e_g^0$ configuration ?
 (a) $\text{K}_3[\text{Co}(\text{NO}_2)_6]$ (b) $\text{K}_4[\text{Fe}(\text{CN})_6]$
 (c) $\text{K}_3[\text{Fe}(\text{CN})_6]$ (d) $\text{Na}_2[\text{Fe}(\text{CN})_5\text{NO}]$
58. Which complex compound having $t_2^3 e_g^2$ configuration ?
 (a) $[\text{Mn}(\text{H}_2\text{O})_6]^{2+}$ (b) $[\text{Mn}(\text{NH}_3)_6]^{2+}$
 (c) $[\text{Mn}(\text{CN})_6]^{4-}$ (d) MnO_2
59. Co-ordination entity Wave length of light absorbed (nm)
- | | |
|---|---|
| $[\text{CoCl}(\text{NH}_3)_5]^{2+}$ | P |
| $[\text{Co}(\text{NH}_3)_5(\text{H}_2\text{O})]^{3+}$ | Q |
| $[\text{Co}(\text{NH}_3)_6]^{3+}$ | R |
| $[\text{Co}(\text{CN})_6]^{3-}$ | S |
| $[\text{Cu}(\text{H}_2\text{O})_4]^{2+}$ | T |
- Select the correct orders :-
 (a) $P > R > Q$ (b) $T > P > Q > R > S$
 (c) $P > Q > R$ (d) $T > Q > P > R > S$
60. What are the basic features of crystal field theory?
 (a) ligands are considered as point like charge
 (b) the metal ligands interaction are purely electrostatic in nature
 (c) metal electrons are considered as non bonding electrons
 (d) covalent interaction takes place between ligand and metal ion
61. Which of the following are examples of bidentate chelating ligand ?
 (a) $[\text{NH}_3-\text{CH}_2-\text{CH}_2-\text{NH}_3]^{2+}$
 (b) $\text{NH}-\begin{array}{l} \text{CH}_2\text{CO}_2^- \\ \text{CH}_2\text{CO}_2^- \end{array}$
 (c) $\text{H}_2\text{N}-\text{CH}_2-\text{C}(=\text{O})-\text{O}^-$
 (d) 
62. In $[\text{Fe}_2(\text{CO})_9]$, which of following statements are correct :-
 (a) Fe has zero oxidation state
 (b) EAN of each Fe atom is 36
 (c) EAN of each Fe atom is 35
 (d) None
63. For which of the following compound secondary valency of metal is six :-
 (a) $\text{PdCl}_2 \cdot 4\text{NH}_3$ (b) $\text{NiCl}_2 \cdot 6\text{H}_2\text{O}$
 (c) $\text{PtCl}_2 \cdot 2\text{NH}_3$ (d) $\text{CoCl}_3 \cdot 4\text{NH}_3$
64. Which of the following are example of symmetrical electronic configuration :-
 (a) d^5 in weak field ligand
 (b) d^7 in strong field ligand
 (c) d^8 in weak field ligand
 (d) d^{10} in strong field ligand
65. Which of the following are example of asymmetrical electronic configuration :-
 (a) d^3 in strong or weak field ligand
 (b) d^7 in strong field ligand
 (c) d^4 in weak field ligand
 (d) d^9 in strong or weak field ligand
66. Which of the complex give same number of particles as that of LaCl_3 :-
 (a) $\text{K}_3[\text{Co}(\text{NO}_2)_6]$
 (b) $\text{Co}(\text{NO}_2)_3 \cdot \text{KNO}_2 \cdot 2\text{NH}_3$
 (c) $\text{CoCl}_3 \cdot 3\text{NH}_3$
 (d) $\text{CoCl}_3 \cdot 6\text{NH}_3$

2.20 COORDINATION COMPOUNDS

67. Correct option regarding Pentaamminethiocyanato-N cobalt(III) trioxalatochromate(III) -
 (a) It is chelating complex
 (b) It exist in linkage isomeric form
 (c) It exist in geometrical isomeric form
 (d) Number of ions formed in solution of the complex > 2
68. Which of the following IUPAC names are correct:-
 (a) $[(\text{CO})_3\text{Fe}(\text{CO})_3\text{Fe}(\text{CO})_3]$: Tri- μ -carbonyl-bis(tricarbonyliron(0))
 (b) $[(\text{NH}_3)_5\text{CoNH}_2\text{Co}(\text{NH}_3)_5](\text{NO}_3)_5$: μ -amido-bis[pentaamminecobalt(III)]nitrate
 (c) $\text{Na}_3[\text{Ag}(\text{S}_2\text{O}_3)_2]$: Sodiumbis(thiosulphato) argentate(I)
 (d) $\text{K}_2[\text{Cr}^{\text{VI}}(\text{CN})_2\text{O}_2(\text{O}_2)\text{NH}_3]$: Potassiumammine dicyanidodioxoperoxochromate(VI)
69. Which of the following statements are incorrect :-
 (a) Maximum coordination number of Ag^{+1} is 6
 (b) Maximum coordination number of Pt^{2+} is 6
 (c) Maximum coordination number of Al^{3+} is 6
 (d) Maximum coordination number of Ni^{2+} is 6
70. Which of the following have $e_g^0 t_{2g}^0$ configuration :-
 (a) FeO_4^{2-} (b) MnO_4^-
 (c) TiCl_4 (d) ZnCl_4^{2-}
71. Which of the following statements are incorrect :-
 (a) $[\text{HgI}_3]^-$ is planar and diamagnetic
 (b) In Zeise's salt central metal act as π donor, σ acceptor
 (c) Zeise's salt has planar anion
 (d) NO_3^- is example of ambidentate ligand
72. Which of the following complexes/compounds, atleast one metal ion has +3 oxidation state:-
 (a) Ferric alum (b) Cryolite
 (c) Blue vitriol (d) Potassium fericyanide
73. A complex which is meso isomer can :-
 (a) Have plane of symmetry
 (b) Have one chiral centre
 (c) Have centre of symmetry
 (d) Have nonsuperimposable mirror image
74. In spherical octahedral crystal field, average energy of the 3d orbital is :-
 (a) higher than e_g orbital
 (b) lower than e_g orbital
 (c) higher than t_{2g} orbital
 (d) lower than t_{2g} orbital
75. **Complex** **Absorption peak (cm^{-1})**
 (A) $[\text{Cr}^{\text{III}}\text{Cl}_6]^{3-}$ x
 (B) $[\text{Cr}^{\text{III}}(\text{H}_2\text{O})_6]^{3+}$ y
 (C) $[\text{Cr}^{\text{III}}(\text{NH}_3)_6]^{3+}$ z
 (D) $[\text{Cr}^{\text{III}}(\text{CN})_6]^{3-}$ w
 Correct order between x,y,z,w are :-
 (a) $x > y$ (b) $y > z$
 (c) $w > z$ (d) $w > z > y > x$
76. **Complex** **Δ_0 (cm^{-1})**
 (A) $[\text{Co}^{\text{III}}(\text{NH}_3)_6]^{3+}$ x
 (B) $[\text{Rh}^{\text{III}}(\text{NH}_3)_6]^{3+}$ y
 (C) $[\text{Ir}^{\text{III}}(\text{NH}_3)_6]^{3+}$ z
 Correct order between x,y,z are :-
 (a) $z > y$ (b) $z > x$
 (c) $y > x$ (d) $x = y = z$
77. Which of the following do not act as an oxidizing agent :-
 (a) $\text{Mn}(\text{CO})_5$ (b) $\text{Fe}(\text{CO})_5$
 (c) $\text{Mn}_2(\text{CO})_{10}$ (d) $\text{Fe}_2(\text{CO})_9$
78. Prussian blue is formed by combining between :-
 (a) Fe^{3+} (b) $[\text{Fe}(\text{CN})_6]^{3-}$
 (c) Fe^{2+} (d) $[\text{Fe}(\text{CN})_6]^{4-}$
79. For d^7 ion like Co^{2+} with very strong ligand like NO_2^- :-
 (a) Co^{2+} ion can be reduced to Co
 (b) Co^{2+} ion can be oxidised to Co^{3+}
 (c) The excited 7th electron resides in 5s orbital which is of higher energy level.
 (d) The complex formed is d^2sp^3 hybridised
80. Which of the following complexes are examples of ternary complexes :-
 (a) $[\text{CoCl}(\text{NH}_3)_5]^{2+}$ (b) $[\text{Cu}(\text{bpy})\text{en}]^{2+}$
 (c) $[\text{Cu}(\text{bpy})\text{ox}]$ (d) $[\text{Co}(\text{NH}_3)_5(\text{OH}_2)]^{3+}$
 (bpy = bipyridine ; en = ethylene diamine ; ox = oxalato)

81. Which of the following statements are incorrect ?
 (a) NO^+ has lower p accepting tendency than CO
 (b) The C-O bond length in bridging carbonyl compound is shorter than that of terminal carbonyl group
 (c) The C-O bond length in bridging carbonyl compound is longer than that of terminal carbonyl group
 (d) $\text{Ni}(\text{H}_2\text{O})_6^{2+}$ is green in colour but $[\text{Ni}(\text{CN})_4]^{2-}$ is colourless
82. Correct order of CO bond length :-
 (a) $[\text{M}(\text{CO})_3\text{PF}_3] < [\text{M}(\text{CO})_3\text{PCl}_3]$
 (b) $[\text{M}(\text{CO})_3\text{PF}_3] > [\text{M}(\text{CO})_3\text{PCl}_3]$
 (c) $[\text{M}(\text{CO})_3\text{PF}_3] < [\text{M}(\text{CO})_3\text{PMe}_3]$
 (d) $[\text{M}(\text{CO})_3\text{PCl}_3] < [\text{M}(\text{CO})_3\text{PMe}_3]$
83. The value of x in :-
 (a) $[\text{Hg}(\text{CN})_4]^x$ can be -2
 (b) $[\text{Fe}(\text{CN})_6]^x$ can be -3
 (c) $[\text{Fe}(\text{CN})_6]^x$ can be -4
 (d) $[\text{Co}(\text{NH}_3)_2\text{Cl}_4]^x$ can be -1
84. Which of the following ligands can act as flexidentate ligand ?
 (a) SO_4^{2-} ion (b) CH_3CO_2^- ion
 (c) CO_3^{2-} ion (d) HCO_2^- ion
85. Which of the following are example of perfect complex :-
 (a) $[\text{Fe}(\text{CN})_6]^{4-}$ (b) $[\text{Ag}(\text{CN})_2]^{-1}$
 (c) $[\text{Cu}(\text{CN})_4]^{3-}$ (d) $[\text{Cu}(\text{NH}_3)_4]^{2+}$
86. Which of the following are example of imperfect complex :-
 (a) $[\text{Cd}(\text{CN})_4]^{2-}$ (b) $[\text{CdI}_4]^{2-}$
 (c) $[\text{Co}(\text{NH}_3)]^{3+}$ (d) $[\text{Fe}(\text{CN})_6]^{4-}$
87. Which of the following are example of anionic complex :-
 (a) $\text{K}_2[\text{HgI}_4]$ (b) $\text{K}_3[\text{Co}(\text{NO}_2)_6]$
 (c) $\text{Ni}(\text{CO})_4$ (d) $\text{H}[\text{AuCl}_4]$
88. Which of the following are example of cationic complex :-
 (a) $[\text{Rh}(\text{PPh}_3)\text{Cl}_3]$ (b) $[\text{Ag}(\text{NH}_3)_2]\text{Cl}$
 (c) $[\text{Co}(\text{NH}_3)_6]\text{Cl}_3$ (d) $[\text{Co}(\text{NH}_3)_5\text{SO}_4]\text{Br}$
89. $\text{Cr}_2\text{O}_7^{2-} + \text{Fe}^{\text{II}} + \text{H}_2\text{SO}_4 \rightarrow$ Final solution (a)
 Correct option regarding A :-
 (a) A is $[\text{Cr}(\text{SO}_4)_3]^{3-}$ (b) A is $[\text{Cr}(\text{H}_2\text{O})_6]^{3+}$
 (c) A is violet in colour (d) A is green in colour
90. In the crystal structure of $\text{MnSO}_4 \cdot 4\text{H}_2\text{O}$, which of the following statement are correct ?
 (a) SO_4^{2-} acts as bidentate ligand where 2 oxygen atoms of SO_4^{2-} form chelate with Mn^{2+}
 (b) There are 6 water molecule associated with each Mn^{2+} ion
 (c) SO_4^{2-} acts as bridging ligand.
 (d) Mn^{2+} has coordination number six
91. Incorrect statements are :-
 (a) Ni^{4+} is more stable than Ni^{2+}
 (b) Ni^{2+} is more stable than Ni^{4+}
 (c) Pt^{2+} is more stable than Pt^{4+}
 (d) Pt^{4+} is more stable than Pt^{2+}
92. X forms pentacarbonyl of the type $\text{X}(\text{CO})_5$
 Y forms tetracarbonyl of the type $\text{Y}(\text{CO})_4$
 Z forms hexacarbonyl of the type $\text{Z}(\text{CO})_6$
 If in all the complexes metal follows EAN rule then:-
 (a) X can be Fe (b) Y can be Ni
 (c) Z can be V (d) Z can be Cr
93. When Gold is attacked by aqua regia :-
 (a) the coordination number of Au in complex formed is +2
 (b) the coordination number of Au in complex formed is +4
 (c) the complex formed can act as a bronsted acid
 (d) NO_2 gas is evolved
94. An aqueous solution of Fe(III) sulphate is paramagnetic but when an excess of KCN is added to the solution which of the following properties are changed :-
 (a) Shape of the molecule
 (b) Hybridisation of Fe(III)
 (c) Number of unpaired electron
 (d) magnetic behaviour

2.22 COORDINATION COMPOUNDS

95. Correct statement regarding $[\text{Co}(\text{en})_3]^{3+}$:-

- (a) It exists as cis and trans isomer
- (b) It is nonresolvable
- (c) It is resolvable
- (d) It is optically active

96. Correct statement regarding Ferrocene :-

- (a) It is an orange solid with melting point 174°C
- (b) It is prepared as $2\text{C}_5\text{H}_5\text{MgBr} + \text{FeCl}_2 \rightarrow \text{Fe}(\text{C}_5\text{H}_5)_2 + \text{MgCl}_2 + \text{MgBr}_2$
- (c) IUPAC name of ferrocene is bis (cyclopentadienyl) iron(II)
- (d) It is an example of sandwich compound

97. Which of the following statements are correct :-

- (a) In an octahedral ligand field the five fold degenerate 3d level will split into doubly degenerate e_g set and triply degenerate t_{2g} set
- (b) The symbol g denotes the presence of centre of inversion
- (c) In the square ligand field, the 3d orbitals will be split into four levels
- (d) In the square ligand field, the 3d orbitals will be split into two levels

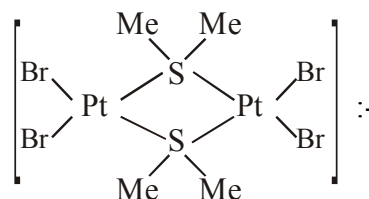
98. The substance $\text{CoBr}_3 \cdot 4\text{NH}_3 \cdot 2\text{H}_2\text{O}$ has a molar conductivity of 420 ohm^{-1} at infinite dilution. Which corresponds to $(3+, 1-)$ electrolyte like AlCl_3 . Correct option regarding that substance :-

- (a) Substance is $[\text{Co}(\text{NH}_3)_4(\text{H}_2\text{O})_2]\text{Br}_3$
- (b) IUPAC name of substance is tetraammine diaquacobalt(III)bromide
- (c) Substance is $[\text{Co}(\text{NH}_3)_3(\text{H}_2\text{O})_3]\text{Br}_3$
- (d) IUPAC name of substance is tetraammine diaquacobalt(II)bromide

99. Select the correct IUPAC name for $[(\text{NH}_3)_4\text{Co}(\text{OH})(\text{NH}_2)\text{Co}(\text{NH}_3)_4]^{4+}$:-

- (a) μ -Amido- μ -hydroxidobis {tetraamminecobalt (4+)} ion
- (b) μ -Amido- μ -hydroxidobis {tetraamminecobalt (III)} ion
- (c) μ -Amido- μ -hydroxidobis {tetraamminecobaltate (4+)} ion
- (d) μ -Amido- μ -hydroxidobis {tetraamminecobaltate (III)} ion

100. Select the correct IUPAC name for the following complex



- (a) Dibromidoplatinum (II) bis- μ -(dimethylthioether) dibromidoplatinum(II)
- (b) Bis { μ (dimethylthioether) dibromidoplatinum(II)}
- (c) Bis- μ -dimethylthioethertetrabromidoplatinum(II)
- (d) Bis- μ -dimethylthioethertetrabromidoplatinate(II)

EXERCISE # III**Linked Comprehension Type :****Paragraph for Q.1 to Q.2**

An octahedral complex of cobalt has its composition $\text{Co}(\text{NH}_3)_3(\text{H}_2\text{O})_3(\text{SO}_4)(\text{NO}_3)$ and it exists in four isomeric forms A, B, C and D. Simple heating of either A or B loses 5.6% of its original weight whereas similar treatment on C causes 11.2% loss in its original weight D did not lose any weight on simple heating. Either A or C did not result in any precipitation if treated with aqueous BaCl_2 solution, whereas B and D resulted a white precipitate when treated with aqueous BaCl_2 solution. Also C does not conduct electricity in its aqueous solution,

1. A is :-

- (a) $[\text{Co}(\text{NH}_3)_3(\text{H}_2\text{O})_3]\text{SO}_4 \cdot \text{NO}_3$
- (b) $[\text{Co}(\text{NH}_3)_3(\text{H}_2\text{O})_2\text{NO}_3]\text{SO}_4 \cdot \text{H}_2\text{O}$
- (c) $[\text{Co}(\text{NH}_3)_3(\text{H}_2\text{O})_2\text{SO}_4]\text{NO}_3 \cdot \text{H}_2\text{O}$
- (d) $[\text{Co}(\text{NH}_3)_3(\text{H}_2\text{O})\text{SO}_4\text{NO}_3]2\text{H}_2\text{O}$

2. B & C are respectively :-

- (a) $[\text{Co}(\text{NH}_3)_3(\text{H}_2\text{O})_3]\text{SO}_4 \cdot \text{NO}_3$; $[\text{Co}(\text{NH}_3)_3(\text{H}_2\text{O})\text{SO}_4 \cdot \text{NO}_3]2\text{H}_2\text{O}$
- (b) $[\text{Co}(\text{NH}_3)_3(\text{H}_2\text{O})_2\text{NO}_3]\text{SO}_4 \cdot \text{H}_2\text{O}$; $[\text{Co}(\text{NH}_3)_3(\text{H}_2\text{O})\text{SO}_4]\text{NO}_3 \cdot \text{H}_2\text{O}$
- (c) $[\text{Co}(\text{NH}_3)_3(\text{H}_2\text{O})_2\text{NO}_3]\text{SO}_4 \cdot \text{H}_2\text{O}$; $[\text{Co}(\text{NH}_3)_3(\text{H}_2\text{O})\text{SO}_4 \cdot \text{NO}_3]2 \cdot \text{H}_2\text{O}$
- (d) $[\text{Co}(\text{NH}_3)_3(\text{H}_2\text{O})_2\text{SO}_4]\text{NO}_3 \cdot \text{H}_2\text{O}$; $[\text{Co}(\text{NH}_3)_3(\text{H}_2\text{O})_2 \cdot \text{NO}_3]\text{SO}_4 \cdot \text{H}_2\text{O}$

3. D is :-

- (a) $[\text{Co}(\text{NH}_3)_3(\text{H}_2\text{O})_3]\text{SO}_4 \cdot \text{NO}_3$
 (b) $[\text{Co}(\text{NH}_3)_3(\text{H}_2\text{O})_2\text{NO}_3]\text{SO}_4 \cdot \text{H}_2\text{O}$
 (c) $[\text{Co}(\text{NH}_3)_3(\text{H}_2\text{O})_2\text{SO}_4]\text{NO}_3 \cdot \text{H}_2\text{O}$
 (d) $[\text{Co}(\text{NH}_3)_3(\text{H}_2\text{O})\text{SO}_4\text{NO}_3]2\text{H}_2\text{O}$

Paragraph for Q.4 to Q.6

A, B and C are three complexes of chromium (ii) with their formula $\text{H}_{12}\text{O}_6\text{Cl}_3\text{Cr}$. Complex A does not react with concentrated H_2SO_4 solution whereas complexes B and C loses 6.75% and 13.5% of their weight, respectively, on treatment with concentrated H_2SO_4 solution.

4. Correct option regarding A :-

- (a) A is violet with 3 chloride ions/unit formula
 (b) B is light green colour with 2 chloride ions / unit formula
 (c) C is dark green colour with 1 chloride ions / unit formula
 (d) All

5. Electrical conductivity order of A,B,C

- (a) $A > B > C$ (b) $B > A > C$
 (c) $C > A > B$ (d) $C > B > A$

6. A,B,C are related as

- (a) Ionisation isomers (b) Hydrate isomers
 (c) Optical isomers (d) Geometrical isomers

Paragraph for Q.7 to Q.9

A light green coloured substance A has 27.55% H_2O . A gives the following reaction, :-

- (i) it gives white ppt. with BaCl_2 which is soluble in conc. HNO_3
 (ii) it gives brown colour ppt when reacts with alkaline $\text{K}_2[\text{HgI}_4]$ solution
 (iii) it gives dark blue ppt when reacts with $\text{K}_3[\text{Fe}(\text{CN})_6]$

7. A is :-

- (a) $\text{KCl} \cdot \text{MgCl}_2 \cdot 6\text{H}_2\text{O}$
 (b) $\text{Fe}_2(\text{SO}_4)_3 \cdot (\text{NH}_4)_2\text{SO}_4 \cdot 6\text{H}_2\text{O}$
 (c) $\text{Cr}_2(\text{SO}_4)_3 \cdot \text{K}_2\text{SO}_4 \cdot 6\text{H}_2\text{O}$
 (d) $\text{FeSO}_4 \cdot (\text{NH}_4)_2\text{SO}_4 \cdot 6\text{H}_2\text{O}$

8. The white ppt is :-

- (a) BaSO_4 (b) BaCl_2
 (c) $\text{Ba}(\text{NO}_3)_2$ (d) BaCO_3

9. The brown and blue colour ppt are respectively :-

- (a) I HgO Hg NH_2 & ferro-ferri cyanide
 (b) I HgO Hg NH_2 & ferri-ferro cyanide
 (c) $\text{Fe}(\text{OH})_3$ & $\text{Fe}(\text{CN})_3$
 (d) $\text{Fe}(\text{OH})_3$; $[\text{Cn}(\text{NH}_3)_4]\text{SO}_4$

Paragraph for Q.10 to Q.11

In metal carbonyls, synergic bonding takes place between metal and carbonmonooxide. This leads to increase in strength of metal ligand bond and decrease in bond order of CO in carbonyl complex as compared to bond order in carbon monooxide.

10. Which among the following metal carbonyls are inner orbital complexes with diamagnetic property ?

- (X) $\text{Ni}(\text{CO})_4$; (Y) $\text{Fe}(\text{CO})_5$; (Z) $\text{V}(\text{CO})_6$; (W) $\text{Cr}(\text{CO})_6$

- (a) (X) and (Y) only (b) (Y), (Z) and (W) only
 (c) (Y) and (W) only (d) (X), (Y) and (W) only

11. Which of the following metal carbonyls involve the d^2sp^3 hybridisation for the formation of metal carbon σ bonds and is paramagnetic ?

- (a) $[\text{Cr}(\text{CO})_6]$ (b) $[\text{V}(\text{CO})_6]$
 (c) $[\text{Mo}(\text{CO})_6]$ (d) $[\text{W}(\text{CO})_6]$

Paragraph for Q.12 to Q.14

$\text{Co}^{2+}(\text{aq.}) + \text{SCN}^-(\text{aq.}) \longrightarrow \text{Complex (X)}$

$\text{Ni}^{2+}(\text{aq.}) + \text{Dimethylglyoxime} \xrightarrow{\text{NH}_4\text{OH}} \text{Complex (Y)}$

The coordination number of cobalt and nickel in complexes X and Y is four.

12. The IUPAC names of the complexes (X) and (Y) are respectively :-

- (a) tetrathiocyanato-S-cobalt(II)ion and bis(dimethylglyoximate)nickel(II)
 (b) tetrathiocyanato-S-cobaltate(II)ion and bis(dimethylglyoximate)nickel(II)
 (c) tetrathiocyanato-S-cobaltate(II)ion and bis(dimethylglyoximate)nickelate(II)
 (d) tetrathiocyanato-S-cobaltate(III)ion and bis(dimethylglyoximate)nickel(II)

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13. The geometry of complexes (X) and (Y) are respectively :-

- (a) tetrahedral and square planar
- (b) both tetrahedral
- (c) square planar and tetrahedral
- (d) both square planar

14. Select the correct statement for the complexes (X) and (Y) :-

- (a) (X) is paramagnetic with two unpaired electrons
- (b) (Y) is diamagnetic and shows intermolecular H-bonding
- (c) (X) is paramagnetic with three unpaired electrons and (Y) is diamagnetic
- (d) (X) and (Y) both are diamagnetic

Paragraph for Q.15 to Q.17

Paramagnetism is a property due to the presence of unpaired electrons. In case of transition of metals, as they contain unpaired electrons in the $(n-1)d$ orbitals, most of the transition metal ions and their compounds are paramagnetic. Paramagnetism increases with increase in number of unpaired electrons. Magnetic moment is calculated from 'Spin only formula' viz.

$$\mu = \sqrt{n(n+2)} \text{ B.M. } n = \text{number of unpaired electrons}$$

Similarly the colour of the compounds of transition metals may be attributed to the presence of incomplete $(n-1)d$ subshell. When an electron from a lower energy of d-orbital is excited to a higher energy d-orbital, the energy of excitation corresponds to the frequency of light absorbed. This frequency generally lies in the visible region. The colour observed corresponds to complementary colour of the light. The frequency of the light absorbed is determined by the nature of the ligand.

15. Which of the following pair of compounds is expected to exhibit same colour in aqueous solution?

- (a) FeCl_2 , CuCl_2 (b) VOCl_2 , CuCl_2
- (c) VOCl_2 , FeCl_2 (d) FeCl_2 , MnCl_2

16. Titanium shows magnetic moment of 1.73 B.M. in its compounds. What is the oxidation state of titanium in the compound?

- (a) +2 (b) +1
- (c) +3 (d) +4

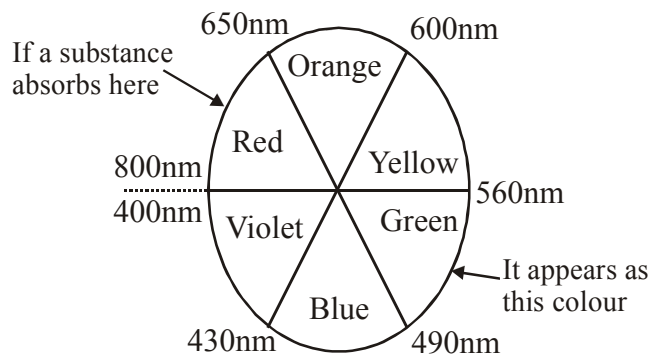
17. Identify the incorrect statement :-

- (a) Mn^{2+} has the highest paramagnetism amongst the bivalent cations of the 1st transition series
- (b) The coloured ions or compounds of transition elements are due to d-d transition, polarisation of anion and charge transfer spectrum
- (c) In 3d series the paramagnetic character first increase to maximum and then starts decreasing
- (d) None of these

Paragraph for Q.18 to Q.20

In general, the crystal field splitting energy Δ corresponds to wavelengths of light in visible region of the spectrum, and colours of the complexes can therefore be attributed to electronic transition between the lower- and higher energy sets of d-orbitals.

In general, the colour that we see is complementary to the colour absorbed.



Different metal ions have different values of Δ , which explains why their complexes with the same ligand have different colour.

Similarly the crystal field splitting also depends on the nature of ligands and as the ligand for the same metal varies from H_2O to NH_3 to ethylenediamine, Δ for complexes increases. Accordingly, the electronic transition shifts to higher energy (shorter wavelength) as the ligand varies from H_2O to NH_3 to en, thus accounting for the variation in colour.

Crystal field theory accounts for the magnetic properties of complexes in terms of the relative values of Δ and the spin pairing energy P. Small Δ values favour high spin complexes, and large Δ values favour low spin complexes.

18. The $[\text{Ti}(\text{NCS})_6]^{3-}$ ion exhibits a single absorption band at 544 nm. What will be the crystal field splitting energy (in kJ mol^{-1}) of the complex ?
 ($h = 6.626 \times 10^{-34} \text{ J.s}$; $C = 3.0 \times 10^8 \text{ m/s}$; $N_A = 6.02 \times 10^{23} \text{ ions.mjole.}$)
 (a) 240 (b) 220
 (c) 270 (d) 250
19. Which of the following statement is incorrect :-
 (a) The $\text{Ni}^{2+}(\text{aq})$ cation is coloured because Ni^{2+} ion can absorb light, which promotes electrons from the filled d-orbitals to the higher half filled d-orbitals
 (b) The $\text{Zn}^{2+}(\text{aq})$ cation is coloured because the d-orbitals are completely filled and no electrons can be promoted, so no light is absorbed
 (c) A complex which has just one absorption band at 455 nm, must be red coloured.
 (d) None
20. Which of the following complexes are diamagnetic:-
 $[\text{Pt}(\text{NH}_3)_4]^{2+}$ $[\text{Co}(\text{SCN})_4]^{2-}$ $[\text{Cu}(\text{en})_2]^{2+}$ $[\text{HgI}_4]^{2-}$
 square planar tetrahedral square planar tetrahedral
 (i) (ii) (iii) (iv)
 (a) (i) and (ii) (b) (ii) and (iii)
 (c) (i) and (iv) (d) (iii) and (iv)

Paragraph for Q.21 to Q.23

In the octahedral systems there are two possible ways to distribute the d-electron in the t_{2g} and e_g level for each d^n system. There will be two opposing factors. The crystal field splitting energy, i.e., $10Dq_0$ (or Δ_0) will try to place the electrons into the more stable t_{2g} level as many as possible while the pairing energy (P) required to pair two electrons in the same orbital will try to avoid the pairing as far as possible.

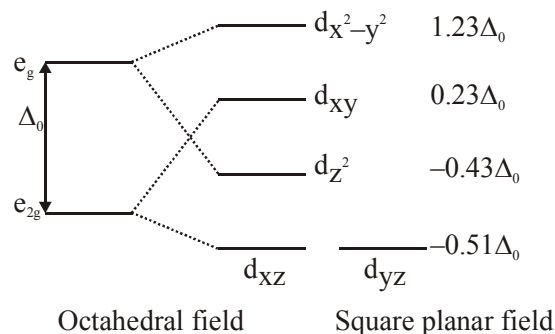
21. If $10Dq_0 > P$ the correct electronic configuration for d^4 ion is :-
 (a) $t_{2g}^3 e_g^1$ (b) $t_{2g}^4 e_g^0$
 (c) $t_{2g}^2 e_g^2$ (d) $e_g^3 t_{2g}^1$
22. If $10Dq_0 < P$ then correct electronic configuration for d^5 ion is :-
 (a) $t_{2g}^5 e_g^0$ (b) $e_g^3 t_{2g}^2$
 (c) $t_{2g}^3 e_g^2$ (d) $t_{2g}^4 e_g^1$

23. If $10Dq_0 > P$ then crystal field splitting energy for d^6 ion in octahedral field will be :-

- (a) $20 Dq - 2P$ (b) $4 Dq$
 (c) $8 Dq$ (d) $24 Dq - 2p$

Paragraph for Q.24 to Q.25

The splitting diagram for square planar complexes is more complex than for octahedral and tetrahedral complexes and is shown below with the relative energies of each orbital.



24. Crystal field stabilisation energy for a diamagnetic square planar d^8 metal complex with the help of above diagram neglecting pairing energy (P) :-
 (a) $-4.88 \Delta_0$ (b) $-2.44 \Delta_0$
 (c) $-0.68 \Delta_0$ (d) $-3.88 \Delta_0$
25. Which of the following set of d electrons have same crystal field stabilisation energy value for tetrahedral complex neglecting pairing energy (P):-
 (I) d^1, d^6 (II) d^2, d^7
 (III) d^3, d^8 (IV) d^4, d^9
 (V) d^5, d^{10}
 (a) Only V (b) II, III, V
 (c) I, III, IV (d) I, II, III, IV, V

Paragraph for Q.26 to Q.27

According to crystal field theory, the electronic configuration of ions are given below :

$$\begin{array}{lll} d^1 : & t_{2g}^1 d^8 & : & t_{2g}^6 e_g^2 \\ d^2 : & t_{2g}^2 d^9 & : & t_{2g}^6 e_g^3 \\ d^3 : & t_{2g}^3 d^{10} & : & t_{2g}^6 e_g^4 \end{array}$$

26. Irrespective of the nature of ligand outer orbital octahedral complex is formed by :-
 (a) d^8 ion (b) d^9 ion
 (c) d^{10} ion (d) All

2.26 COORDINATION COMPOUNDS

27. Irrespective of the nature of ligand inner orbital octahedral complex is formed by :-

- (a) Only d^1 ion (b) d^1, d^2 ion
(c) Only d^3 ion (d) d^1, d^2, d^3 ion

Passage for Q.28 to Q.29

AlF_3 is not soluble in analysis HF but not soluble in KF. When BrF_3 is added to the above solution, AlF_3 is precipitated.

28. What will be the product when AlF_3 is dissolved in KF ?

- (a) $K[AlF_4]$ (b) $K_2[AlF_5]$
(c) $K_3[AlF_6]$ (d) $K_4[AlF_7]$

29. What will be the product when this dissolved solution combines with BF_3 ?

- (a) $K[BF_4]$ (b) $K_2[BF_5]$
(c) $K_3[BF_6]$ (d) $K_4[BF_7]$

Passage for Q.30 to Q.32

Electrostatic theory of bonding in coordination compound consider the electrostatic attraction between the positively charged metal ion and negatively charge ligands or negative ends of polar molecule acting as the ligands. Thus complexing power of particular metal ion depends on the charge density called ionic potential. In fact ionic potential is a good parameter to measure the complexing power of metal ion.

30. Metal ion like Cs^+ , Rb^+ , K^+ , Na^+ , Li^+ , complexing power is very low. It is because these metal ions have :-

- (a) very high ionic potential (in the range of 3.7 - 5.2)
(b) moderately high ionic potential (in the range of 1.5 - 2.5)
(c) low ionic potential (in the range of 0.55 - 1.11)
(d) moderately high ionic potential (in the range of 2 - 3)

31. Ionic potential of Co^{3+} , Al^{3+} , Sn^{4+} , Be^{2+} , Pt^{4+} are very high. So these ions have :-

- (a) very poor complexing ability with ligands
(b) moderate complexing ability with ligands
(c) very high or most pronounced complexing ability with ligands
(d) do not form complex at all

32. Ionic potential of :-

- (a) Zn^{2+} is in between Fe^{3+} and Cd^{2+}
(b) Cu^{2+} is in between Cr^{3+} and Cu^+
(c) Mn^{2+} is in between Rh^{3+} and Ba^{2+}
(d) All

Passage for Q.33 to Q.35

According to Sidgwick's Effective Atomic Number rule, the ligands donate the electrons to the central metal ion through the covalent coordinate bonds and the total number of electrons on the central metal ion including those gained from ligands is the atomic number of nearest noble gas.

Given : At. No. of Cd & Pt are 48, 78 respectively.

33. In $[Cd(NH_3)_4]^{2+}$, EAN value of metal ion :-

- (a) is same as atomic number of Kr
(b) is same as atomic number of Xe
(c) is same as atomic number of Rn
(d) does not match any noble gas configuration

34. In $[Pt(NH_3)_4Cl_2]^{2+}$, EAN value of metal ion :-

- (a) is same as atomic number of Kr
(b) is same as atomic number of Xe
(c) is same as atomic number of Rn
(d) does not match any noble gas configuration

35. In $[Pt(\eta^2 C_2H_4)Cl_3]^{-1}$, EAN value of metal ion :-

- (a) is same as atomic number of Kr
(b) is same as atomic number of Xe
(c) is same as atomic number of Rn
(d) does not match any noble gas configuration

Passage for Q.36 to Q.38

The following complexes are given below :

$[Ag(NH_3)_2]^+$; X	$[Ag(CN)_2]^{-1}$; Y
$[Ag(SCN)_2]^{-1}$; Z	$[Pt(CN)_4]^{2-}$ W

36. Which of the following complex is linear in shape having all 5 atoms including metal ion are in a straight line :-

- (a) X (b) Y
(c) Z (d) W

37. Which of the following complex is square planar in shape having at least 5 atoms including metal ion are in a straight line :-

- (a) X (b) Y
(c) Z (d) W

38. Which of the following complex is linear in shape having at least 7 atoms including metal ion are in a straight line :-

- (a) X (b) Y
(c) Z (d) W

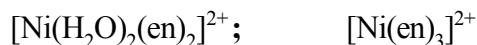
Passage for Q.39 to Q.41

The following complexes are given below :



X_1

X_2



X_3

X_4

39. Correct statement regarding X_4 :-

- (a) X_4 has 2 stereoisomers
(b) X_4 has 3 stereoisomers
(c) X_4 has 4 stereoisomers
(d) X_4 has 2 optically inactive stereoisomers

40. Correct statement regarding X_3 :-

- (a) X_3 has 2 stereoisomers
(b) X_3 has 3 stereoisomers
(c) X_3 has 4 stereoisomers
(d) X_3 has 2 optically inactive stereoisomers

41. Correct statement regarding X_1 and X_2 :-

- (a) X_1, X_2 both are optically inactive
(b) X_1, X_2 both are optically active
(c) X_1 is optically inactive but X_2 is optically active
(d) X_2 is optically inactive but X_1 is optically active

Passage for Q.42 to Q.44

A complex of cobalt, X has magnetic moment 4.89 BM and EAN value 36. Another complex of cobalt, Y has magnetic moment 3.87 BM and EAN value 37.

Third complex of cobalt, Z has magnetic moment 0 BM and EAN value 36.

42. Z can be :-

- (a) $[\text{Co}(\text{H}_2\text{O})_6]\text{SO}_4$ (b) $[\text{CoCl}_4]^{2-}$
(c) $\text{K}_3[\text{Co}(\text{NO}_2)_6]$ (d) $[\text{CoBr}_4]^{2-}$

43. X can be :-

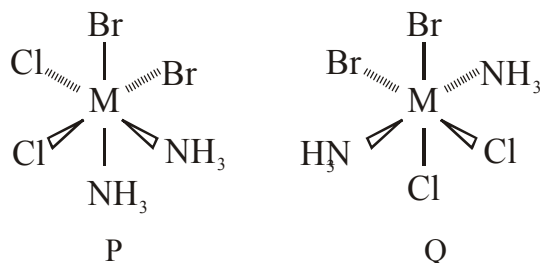
- (a) $[\text{Co}(\text{NH}_3)_6]\text{Cl}_3$ (b) $\text{K}_3[\text{Co}(\text{C}_2\text{O}_4)_3]$
(c) $[\text{Co}(\text{H}_2\text{O})_6]\text{SO}_4$ (d) $[\text{CoCl}_4]^{2-}$

44. Y can be :-

- (a) $[\text{CoBr}_6]^{4-}$ (b) $[\text{Co}(\text{CO})_4]^{-1}$
(c) $[\text{Co}_2(\text{CO})_8]$ (d) $[\text{CoF}_6]^{3-}$

Passage for Q.45 to Q.47

The following complexes P and Q are given as follows



45. P and Q are related as :-

- (a) Structural isomers (b) Geometrical isomers
(c) Optical isomers (d) None

46. P :-

- (a) can exist as enantiomeric pair.
(b) can not exist as enantiomeric pair.
(c) is resolvable
(d) both (a) and (c)

47. Q :-

- (a) can exist as enantiomeric pair.
(b) can not exist as enantiomeric pair.
(c) is resolvable
(d) both (a) and (c)

Passage for Q.48 to Q.51

A qualitative correlation between the various Lewis acids and bases is done by classifying the acids or bases into two classes hard and soft. Acid base reactions are then treated by the general principle that hard acid prefer to combine with hard bases and soft acids prefer to combine with soft bases. A firmly held electron - cloud with low polarisability makes a species "hard" while an easily polarisable electron cloud characterises the species as "soft".

48. Which of the following are example of hard acids?

- (a) Co^{3+} (b) Fe^{3+}
(c) Mg^{2+} (d) Cr^{3+}

2.28 COORDINATION COMPOUNDS

49. Which of the following are example of soft acids ?

- (a) Cu^+ (b) Ag^+
(c) Hg^{2+} (d) Pd^{2+}

50. Which of the following are example of hard bases?

- (a) NO_3^- (b) F^-
(c) NH_3 (d) ROH

51. Which of the following are example of soft base ?

- (a) RSH (b) R_3P
(c) I^- (d) CN^-

Passage for Q.52 to Q.53

Irving and Williams calculated the series of stability constants of octahedral complexes of the bivalent metal ions of first transition series with the common ligands having the donor sites like oxygen, sulphur, nitrogen and halogen. This series is called Irving and Williams stability order of octahedral complexes. This occurs due to the following contributing factors.

- ☐ decrease of ionic radii, i.e., increase of electronegativity along the series.
- ☐ increase of crystal field stabilization energy along the series.

52. The stability constant of the octahedral complexes for d block ions :-

- (a) $\text{Mn}^{2+} < \text{Fe}^{2+} < \text{Co}^{2+}$
(b) $\text{Mn}^{2+} > \text{Fe}^{2+} > \text{Co}^{2+}$
(c) $\text{Co}^{2+} < \text{Ni}^{2+} < \text{Cu}^{2+} > \text{Zn}^{2+}$
(d) $\text{Ni}^{2+} < \text{Cu}^{2+} < \text{Zn}^{2+}$

53. The stability constant of the octahedral complexes for s-block & d-block ions :-

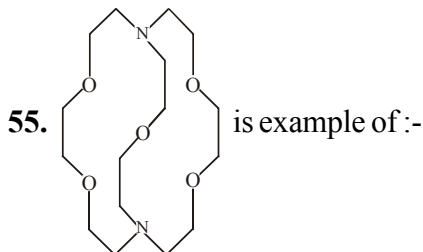
- (a) $\text{Ba}^{2+} < \text{Ca}^{2+} < \text{Mg}^{2+}$
(b) $\text{Ba}^{2+} > \text{Ca}^{2+} > \text{Mg}^{2+}$
(c) $\text{Ca}^{2+} < \text{Mg}^{2+} < \text{Mn}^{2+}$
(d) $\text{Mg}^{2+} < \text{Ba}^{2+} < \text{Ca}^{2+}$

Passage for Q.54 to Q.56

The crown ethers are monocyclic while the cryptands are bicyclic or tricyclic. The multi cyclic cryptands are basically the amino ethers where the N-sites acts as bridged atoms. The cryptands can provide the cage structure to accommodate the metal ion and other suitable guest species. Thus these multicyclic ligands provide the heterocyclic rings to encapsulate the metal ion in a cage-like structure.

54. The donor atoms of cryptands :-

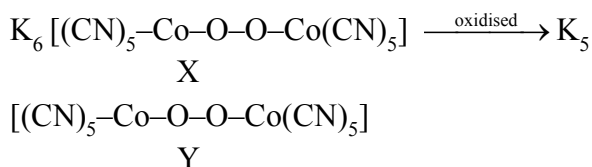
- (a) O, O (b) O, N
(c) N, N (d) None



- (a) crypt-[2,2,1] (b) crypt-[2,2,2]
(c) crypt-[2,3,2] (d) crypt-[0,2,0]

56. Which of the following statements are correct :-

- (a) Four member chelate ring produced by η^2 didentate ligands like acetato, carbonato, sulphato and nitrato are fairly stable and preferred in complexes of high coordination number 8,10,12
- (b) The five member chelate ring formed by amino acid or amino carboxylate are more stable than six member chelate ligands formed by similar ligands
- (c) Resonance or delocalisation of charge stabilises the six membered chelate rings
- (d) Generally chelate rings less than five membered or more than six membered will be less stable

Passage for Q.57 to Q.58

57. In both the complex Co have $t_{2g}^6 e_g^0$ configuration. The bond energy of (O-O) in X & Y is :-

- (a) B.E. of (O-O) in Y < B.E. of (O-O) in X
(b) B.E. of (O-O) in X < B.E. of (O-O) in Y
(c) B.E. of (O-O) in X = B.E. of (O-O) in Y
(d) None

58. Both the complex are :-

- (a) Tetrahedral and Diamagnetic
(b) Octahedral and Paramagnetic
(c) Diamagnetic
(d) Paramagnetic

Passage for Q.59 to Q.60

Complex	Λ_M ($\text{Ohm}^{-1} \text{cm}^2 \text{mol}^{-1}$)	No. of ions per molecule	No. of ionic Cl^{-1} per molecule for AgCl ppt.
$\text{CoCl}_3 \cdot 6\text{NH}_3$ (Lutecobaltic chloride)	m_1	x	p
$\text{CoCl}_3 \cdot 5\text{NH}_3$ (Purpurecobaltic chloride)	m_2	y	q
$\text{CoCl}_3 \cdot 4\text{NH}_3$ (Praseocobaltic chloride)	m_3	z	r
$\text{CoCl}_3 \cdot 5\text{NH}_3 \cdot \text{H}_2\text{O}$ (Roseocobaltic chloride)	m_4	w	s

59. Correct option is :-

- (a) $m_1 > m_2 > m_3 > m_4$ (b) $m_1 > m_4 > m_2 > m_3$
 (c) $m_1 > m_3 > m_4 > m_2$ (d) $m_1 > m_2 > m_4 > m_3$

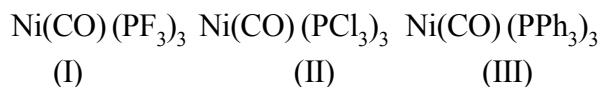
60. Correct option is :-

- (a) $w = x = 4$; $s = p = 3$
 (b) $y = 3$; $q = 2$
 (c) $z = 2$; $r = 1$
 (d) All

Passage for Q.61 to Q.62

The metal carbon bond in metal carbonyls possess both σ and π character. The M–C σ bond is formed by donation of lone pair of electrons on the metal carbonyls into a vacant orbital of the metal. The M–C π bond is formed by the donation of a pair of electrons from a filled d orbital of metal into the vacant antibonding π^* orbital of carbon monoxide. The metal to ligand bonding creates a synergic effect.

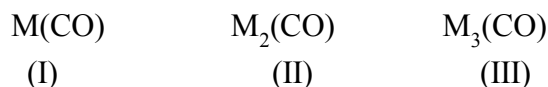
61. For the complexes -



Which of the following option is correct ?

- (a) M–C bond order : $\text{III} < \text{II} < \text{I}$
 (b) C–O bond order : $\text{III} < \text{II} < \text{I}$
 (c) C–O vibrational frequency : $\text{I} < \text{II} < \text{III}$
 (d) All

62. For the complexes -



Which of the following option is correct ?

- (a) M–C bond length : $\text{III} > \text{II} > \text{I}$

(b) M–C vibrational frequency : $\text{III} > \text{II} > \text{I}$

(c) Synergic effect order : $\text{I} > \text{II} > \text{III}$

(d) C–O vibrational frequency : $\text{III} > \text{II} > \text{I}$

Passage for Q.63 to Q.65

The stability of a complex in solution refers to the degree of association between the two species involved in the state of equilibrium. The magnitude of the equilibrium constant expresses the stability. Thus, for the reaction of the type : $\text{M} + 4\text{L} \rightleftharpoons \text{ML}_4$ if equilibrium constants K_1 , K_2 , K_3 , K_4 for the stepwise formation of the successive complexes ML , ML_2 , ML_3 , ML_4 respectively are called the stepwise stability constants.

63. If overall stability constant is β_4 for the above reaction then

- (a) $\beta_4 = K_1 + K_2 + K_3 + K_4$
 (b) $\beta_4 = K_1 - K_2 + K_3 - K_4$
 (c) $\beta_4 = K_1 \times K_2 \times K_3 \times K_4$
 (d) None

64. Thermodynamic stability of the complex -

- (a) Reflects how slowly it reacts with the ligand
 (b) Reflects how fast it reacts with the ligand
 (c) Measures the strength of metal-ligand bond in the complex
 (d) Measures how easily it dissociates

65. A complex is kinetically stable provided -

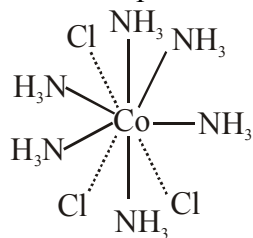
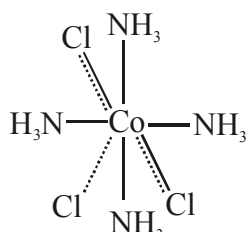
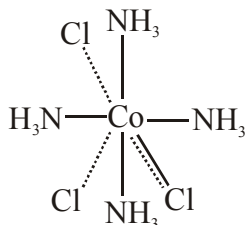
- (a) it reacts very fast with the ligand
 (b) it reacts very slow with the ligand
 (c) complex is extremely stable in nature of bonding
 (d) activation energy for the complex formation is very low

2.30 COORDINATION COMPOUNDS**Passage for Q.66 to Q.68**

According to Werner's coordination theory complex has two types of valencies.

- (i) Primary valency : It is the oxidation state of the metal ion in the complex
 (ii) Secondary valency : It is number of ligands atoms coordinated to metal ion

Three complexes are given

**(X)****(Y)****(Z)**

66. X is represented as -

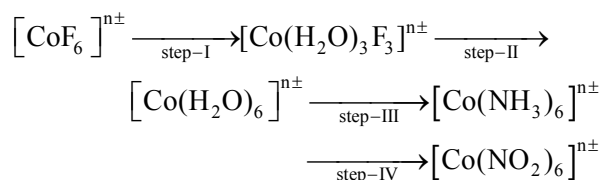
- (a) $[\text{Co}(\text{NH}_3)_6]\text{Cl}_3$ (b) $[\text{Co}(\text{NH}_3)_4\text{Cl}_2]\text{Cl}$
 (c) $[\text{Co}(\text{NH}_3)_5\text{Cl}]\text{Cl}_2$ (d) None

67. Z is represented as -

- (a) $[\text{Co}(\text{NH}_3)_6]\text{Cl}_3$ (b) $[\text{Co}(\text{NH}_3)_4\text{Cl}_2]\text{Cl}$
 (c) $[\text{Co}(\text{NH}_3)_5\text{Cl}]\text{Cl}_2$ (d) None

68. In Y which of the following option is correct ?

- (a) 1 Cl shows primary valency only
 (b) 2 Cl shows secondary valency only
 (c) 2 Cl shows primary valency as well as secondary valency
 (d) both (a) and (c)

Passage for Q.69 to Q.70

69. Magnetic behaviour is changed in -

- (a) Step-I (b) Step-II
 (c) Step-III (d) Step-IV

70. Total charge of the complex is unchanged in -

- (a) Step-I (b) Step-II
 (c) Step-III (d) Step-IV

Match the Column Type

71. Match the column :-

Column - I Complex	Column - II Features
(a) $\text{Fe}(\text{CO})_5$	(P) can act as reducing agent
(b) $\text{Mn}(\text{CO})_6$	(Q) can undergo dimerisation
(c) $\text{Mn}(\text{CO})_5$	(R) can act as oxidising reagent
(d) $[\text{Ti}(\text{CO})_6]$	(S) follows Sidgwick E.A.N. rule
	(T) does not follow Sidgwick E.A.N. rule

72. Match the column :-

Column - I Complex	Column - II Features
(a) $[\text{Co}(\text{NH}_3)_5\text{SO}_4]^{+1}$	(P) only monodentate ligand is present
(b) $[\text{Co}(\text{NH}_3)_4\text{SO}_4]^{+1}$	(Q) only bidentate ligand is present
(c) $[\text{Co}(\text{en})_3]^{+3}$	(R) monodentate & bidentate ligand both are present
(d) $[\text{Co}(\text{NH}_3)_6]^{2+}$	(S) chelating ligand present
	(T) metal has +3 oxidation state

73. Match the column :-

Column - I Complex	Column - II Features
(a) $[\text{Co}(\text{H}_2\text{O})\text{BrCl}]$	(P) show optical isomerism
(b) $[\text{Co}(\text{gly})_3]$	(Q) show geometrical isomerism
(c) $[\text{Co}(\text{NH}_3)_2\text{Cl}_2(\text{en})]\text{NO}_3$	(R) metal is in +3 oxidation state
(d) $\text{K}_3[\text{CoCl}_2(\text{C}_2\text{O}_4)_2]$	(S) aqueous solution acts as non-electrolyte
	(T) homoleptic complex

74. Match the column :-

Column - I Complex	Column - II Features
(a) $[\text{Co}(\text{H}_2\text{O})_6]^{3+}$	(P) d^2sp^3
(b) $[\text{CoF}_6]^{3-}$	(Q) sp^3d^2
(c) $[\text{Co}(\text{NO}_2)_6]^{3-}$	(R) diamagnetic
(d) $[\text{CoF}_3(\text{H}_2\text{O})_3]$	(S) paramagnetic
	(T) central metal ion with +3 oxidation state

75. Match the column :-

Column - I Complex	Column - II Features
(a) $[\text{AuF}_4]^-$	(P) dsp^2 hybridisation
(b) $[\text{Cu}(\text{CN})_4]^{3-}$	(Q) sp^3 hybridisation
(c) $[\text{Ni}(\text{NH}_3)_6]^{2+}$	(R) sp^3d^2 hybridisation
(d) $[\text{Fe}(\text{CN})_5\text{NOS}]^{4-}$	(S) d^2sp^3 hybridisation
	(T) contains two unpaired electron

76. Match the column :-

Column - I Ligand	Column - II Name
(a) N_3^-	(P) azido
(b) N^{3-}	(Q) nitrido
(c) ONO^-	(R) nitrito-O
(d) NO_3^-	(S) nitrate

77. Match the column :-

Column - I Ligand	Column - II Features
(a) $\text{NH}_2\text{-NH}_2$	(P) didentate nonbridging ligand
(b) $\text{NH}_2\text{CH}_2\text{CO}_2^-$	(Q) didentate symmetrical chelating ligand
(c) CH_3NH_2	(R) didentate unsymmetrical chelating ligand
(d) $\text{NH}_2\text{CH}_2\text{CH}_2\text{NH}_2$	(S) monodentate ligand
	(T) at least one lone pair of electron is present

78. Match the column :-

Column - I Ions	Column - II Colour in aqueous solution
(a) Cu^{2+}	(P) Purple
(b) Ni^{2+}	(Q) Green
(c) Co^{2+}	(R) Blue
(d) Ti^{3+}	(S) Pink

79. Match the column :-

Column - I Complex	Column - II Type of s, p, d orbitals participated in hybridisation
(a) $[\text{Ti}(\text{H}_2\text{O})_6]^{3+}$	(P) 4s, 4p, $4d_{x^2-y^2}$, $4d_{z^2}$
(b) $[\text{Ni}(\text{NH}_3)_6]^{2+}$	(Q) $3d_{x^2-y^2}$, $3d_{z^2}$, 4s, 4p
(c) $[\text{Zn}(\text{NH}_3)_4]^{2+}$	(R) $3d_{x^2-y^2}$, 4s, $4p_x$, $4p_y$
(d) $[\text{Cu}(\text{NH}_3)_4]^{2+}$	(S) 4s, $4p_x$, $4p_y$, $4p_z$

80. Match the column :-

Column - I (Complex)

- (a) $[\text{PtCl}(\text{NO}_2)(\text{NH}_3)_4]\text{SO}_4$
 (b) $[\text{Mn}(\text{NH}_3)_5(\sigma\text{C}_3\text{H}_5)]$
 (c) $[\text{Pt}(\text{NH}_3)_4][\text{Cr}(\text{NCS})(\text{NO}_2)_2(\text{NH}_3)_2\text{I}]$
 (d) $\left[\text{Cl}(\text{NH}_3)_3\text{Co} \begin{array}{c} \text{NH}_2 \\ \diagup \quad \diagdown \\ \text{O}_2 \end{array} \text{Co}(\text{NH}_3)_3\text{Cl} \right]^{2+}$

Column - II (Isomerism shown)

- (P) shows linkage isomerism
 (Q) shows coordination position isomerism
 (R) shows ionisation isomerism
 (S) shows coordination isomerism
 (T) shows ligand isomerism

81. Match the column :-

Column - I Ions	Column - II Colour in aqueous solution
(a) V^{4+}	(P) Blue
(b) V^{3+}	(Q) Green
(c) $\text{V}^{2+} / \text{Cr}^{3+} / \text{Mn}^{3+}$	(R) Violet
(d) Fe^{3+}	(S) Yellow

2.32 COORDINATION COMPOUNDS

82. Match the column :-

Column - I (Complex / Colour of light absorbed)

- (a) $[\text{CoCl}(\text{NH}_3)_5]^{2+}$ / Yellow
 (b) $[\text{Co}(\text{NH}_3)_5(\text{H}_2\text{O})]^{3+}$ / Blue Green
 (c) $[\text{Co}(\text{NH}_3)_6]^{3+}$ / Blue
 (d) $[\text{Cu}(\text{H}_2\text{O})_4]^{2+}$ / Red/Orange

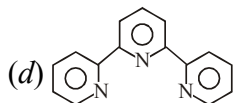
Column - II (Colour of complex)

- (P) Violet
 (Q) Red
 (R) Yellow Orange
 (S) Blue

83. Match the column :-

Column - I (Ligand)

- (a) $\text{O}_2\text{CCH}_2\text{NHCH}_2\text{CO}_2^-$
 (b) $\text{O}_2\text{CCH}_2\text{N}=\text{C}(\text{O}^-)\text{CH}_2\text{CO}_2^-$
 (c) $\text{NH}_2(\text{CH}_2)_2\text{NH}(\text{CH}_2)_2\text{NH}_2$

**Column - II (Characteristic feature)**

- (P) tridentate ligand
 (Q) chelating ligand
 (S) nitrogen and oxygen can act as donor atom
 (R) only nitrogen can act as donor atom
 (T) two oxygen atoms and one nitrogen atom act as donor atom

84. Match the column :-

Column - I**Column - II**

- (a) $[\text{Ti}(\text{CO})_6]^{2-}$ (P) Highest C–O bond vibration stretching frequency
 (b) $[\text{V}(\text{CO})_6]^{1-}$ (Q) Longest M–C bond length
 (c) $[\text{Cr}(\text{CO})_6]$ (R) Follows sidgwick rule
 (d) $[\text{Mn}(\text{CO})_6]^{+1}$ (S) Synergic effect is observed

85. Formula of complex features -

Column-I**Column-II**

- (a) $\text{Ma}_3\text{b}_2\text{c}$ (P) All stereoisomers are optically inactive
 (b) Ma_3b_3 (Q) Number of Geometrical isomers = 2

(c) Ma_3bcd (R) Number of Geometrical isomers = 4(d) Ma_4bc (S) Total 3 stereoisomers are possible

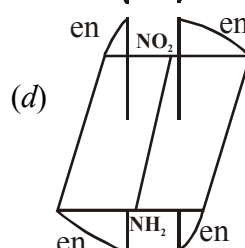
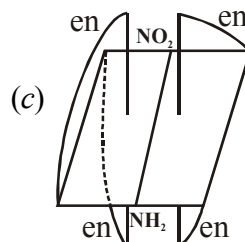
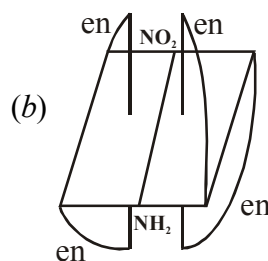
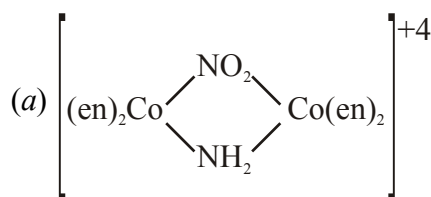
(T) Only one ENT pair is possible

86. Match the column :-

Column-I**Column-II****Complex****Type of isomerism exhibited**

- (a) $\text{Na}_2[\text{Pt}(\text{SCN})_2(\text{Ox})_2]$ (P) Ionisation
 (b) $[\text{CrCl}_2(\text{NH}_3)_4]\text{NO}_3$ (Q) Linkage
 (c) $[\text{Pt}(\text{NO}_2)(\text{gly})(\text{NH}_3)]$ (R) Geometrical
 (d) $\text{K}_3[\text{Fe}(\text{OH})_2(\text{C}_2\text{O}_4)_2]$ (S) Optical

87. Match the following :-

Column-I**Column-II**

- (P) Meso form
 (Q) Optically active form (either d or s form)
 (R) Polynuclear complex
 (S) Plane of symmetry if present

88. Match the column :-

Column - I Complex	Column - II Features
(a) $[\text{Cr}(\text{CN})_3(\text{NO}_2)_3]^{4-}$ (P)	Outer orbital octahedral complex
(b) $[\text{Co}(\text{C}_2\text{O}_4)_3]^{3-}$ (Q)	Inner orbital octahedral complex
(c) $[\text{Fe}(\text{EDTA})]^{-1}$ (R)	$\mu = 2\sqrt{2}$ B.M.
(d) $[\text{Ni}(\text{en})_3](\text{NO}_3)_2$ (S)	Shows optical activity

89. Match the column :-

Column - I Complex	Column - II Features
(a) $[\text{Co}(\text{CN})(\text{NH}_3)_5]\text{SO}_4$ (P)	Diamagnetic
(b) $\text{K}[\text{Co}(\text{CO})_4]$ (Q)	O.N. of metal ≤ 0
(c) $\text{K}_4[\text{Co}(\text{C}_2\text{O}_4)_3]$ (R)	Metal - Metal linkage
(d) $\text{Co}_2(\text{CO})_8$ (S)	Paramagnetic
	(T) Optically active

90. Match the column :-

Column - I Complex	Column - II Features
(a) Sodiumnitropruside	(P) $\mu = 0$ BM
(b) Brown ring complex	(Q) Octahedral
(c) Complex of Ag used for its extraction	(R) $\mu = \sqrt{15}$ B.M.
(d) Potassium ferrocyanide	(S) NO^+ acts as ligand

91. Match the column :-

Column - I Types of complex	Column - II Total possible geometrical isomers
(a) $[\text{M}(\text{AB})_2\text{ab}]^{n\pm} \xrightarrow{-a/+b}$	(P) 2
(b) $[\text{Ma}_2\text{b}_2\text{c}_2]^{n\pm} \xrightarrow{-b/+d}$	(Q) 6
(c) $[\text{M}(\text{AA})_2\text{a}_2]^{n\pm} \xrightarrow{-a/+c}$	(R) 3
(d) $[\text{M}(\text{AB})\text{a}_3\text{b}] \xrightarrow{-a/+b}$	(S) 5
	(T) 0

92. Match the column :-

Column - I Complex	Column - II Features
(a) $[\text{Ni}(\text{H}_2\text{O})_6]\text{Cl}_2$	(P) d^2sp^3 hybridised
(b) $[\text{Co}(\text{CN})_2(\text{NH}_3)_4]\text{OEt}$ (Q)	Ionisation isomerism
(c) $[\text{IrCl}_6]^{3-}$	(R) $\mu = 2.83$ B.M.
(d) $[\text{PtCl}_2(\text{NH}_3)_4]\text{Br}_2$	(S) 2 unpaired electron is present

93. Match the column :-

Column - I Complex	Column - II Features
(a) $[\text{NiCl}_2(\text{PPh}_3)_2]$	(P) Paramagnetic with 1 unpaired electron
(b) $\text{V}(\text{CO})_6$	(Q) Paramagnetic with 2 unpaired electron
(c) $[\text{Cr}(\text{CN})_6]^{4-}$	(R) C.N. of metal ion = 4
(d) $\text{Ni}(\text{CO})_4$	(S) Diamagnetic
	(T) C.N. of metal ion = 6

94. Match the column :-

Column - I (Compound)

- (a) Double salt
(b) Ethylene diaminetriacetato
(c) $[\text{Cr}(\text{NH}_3)_6]^{3+}$
(d) t_{2g}

Column - II (Features)

- (P) $\text{FeSO}_4 \cdot (\text{NH}_4)_2\text{SO}_4 \cdot 6\text{H}_2\text{O}$
(Q) Pentadentate ligand
(R) Paramagnetic
(S) Triply degenerate orbitals

95. Match the column :-

Column - I Geometry	Column - II Complex
(a) Trigonal bipyramidal	(P) $[\text{Cu}(\text{NH}_3)_4]^{2+}$
(b) Tetrahedral	(Q) $[\text{Cr}(\text{H}_2\text{O})_6]^{3+}$
(c) Square planar	(R) $[\text{Ni}(\text{CN})_5]^{3-}$
(d) Octahedral	(S) $[\text{NiCl}_4]^{2-}$

2.34 COORDINATION COMPOUNDS

96. Match the column :-

Column - I**(Ion)**(a) d^6 (b) d^7 (c) d^8 (d) d^9 **Column - II****(cfse in high-spin octahedral complex)**(P) $4Dq_0$ (Q) $8Dq_0$ (R) $12Dq_0$ (S) $6Dq_0$

97. Match the column :-

Column - I**(Ion)**(a) d^4 (b) d^5 (c) d^6 (d) d^7 **Column - II****(cfse in high-spin octahedral complex)**(P) $16Dq_0 - P$ (Q) $20Dq_0 - 2P$ (R) $24Dq_0 - 2P$ (S) $18Dq_0 - P$

98. Match the column :-

Column - I**Formula of complex**(a) $M(AA)(BC)de$ (P) 4 (AA = Bidentate symmetrical ligand)(b) $M(AB)(AB)cd$ (Q) 13 (CD/BC/AB = Bidentate non symmetrical ligand)(c) $Mabcdef$ (R) 11 (abcdef = Monodentate ligand)(d) $M(AB)_3$ (S) 10**Column - II****Number of stereoisomers**

99. Match the column :-

Column - I**Complex**(a) $PtCl_4 \cdot 6NH_3$ (b) $PtCl_4 \cdot 5NH_3$ (c) $PtCl_4 \cdot 4NH_3$ (d) $PtCl_4 \cdot 3NH_3$ **Column - II****Total number of ions**

(P) 2

(Q) 3

(R) 4

(S) 5

100. Match the column :-

Column - I(a) $CoCl_3 \cdot 5NH_3 \cdot H_2O$ (b) $Co(NH_3)_5BrSO_4$ (c) $CrCl_3 \cdot 6H_2O$ (d) $Cr(NO_2)_2 \cdot 6H_2O$ **Column - II**

(P) In one of the isomeric forms, the metal satisfies its secondary valences by only neutral molecules

(Q) The magnetic moment of one of the isomeric form is 3.87 B.M.

(R) Show ionisation isomerism

(S) Show hydrate isomerism

(T) Show linkage isomerism

101. Match the column :-

Column - I (Ligands)(a) CO, CN^-, C_2H_4, NO^+ (b) H^-, NH_3 (c) H_2O, NH_2^-, ROH, R_2S (d) $PF_3, PR_3, AsMe_3$ **Column - II (Nature of ligands)**(P) σ donor ligand(Q) π donor ligand(R) π acceptor ligand

(S) strong field ligand

(T) weak field ligand

EXERCISE # IV**Integer Type :**1. How many compounds are expected to yield a white precipitate with $AgNO_3$ solution ? $PtCl_4 \cdot 4NH_3$; $CoCl_3 \cdot 3NH_3$; Vinyl chloride; Allyl chloride; Carnalite; Tertiary butyl chloride2. What is the coordination number of central atom in $[Pt(trien)]^{2+}$?

3. Find out number of complexes which follow sidgwick's rule of EAN :-

 $H[AuCl_4]$; $[Pt(NH_3)_4Cl_2]^{2+}$; $[HgI_4]^{2-}$; $NiCl_4^{2-}$; $[Ti(\sigma C_5H_5)_2(\pi C_5H_5)_2]$; $K_3[Fe(CN)_6]$; $K_4[Fe(CN)_6]$; $[Ag(S_2O_3)_2]^{-1}$; $[Fe(H_2O)_5NO]^{2+}$ 4. How many statements are correct regarding $[Ni(DMG)_2]$?

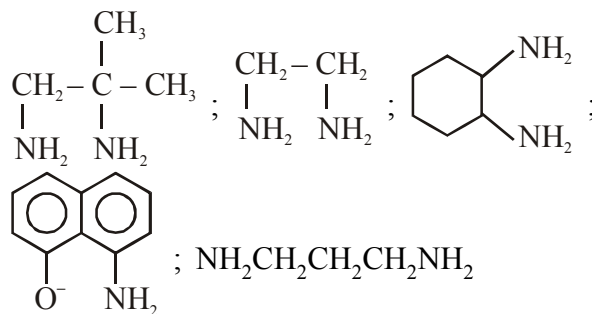
(i) the complex is diamagnetic.

(ii) the complex forming 5 membered chelate with ligand and metal ion.

(iii) the complex forming hydrogen bonding and 6 membered chelation is possible.

- (iv) the complex forming 6 membered chelate with ligand and metal ion.
 (v) it is rosy red colour.
 (vi) it is tetrahedral complex.
 (vii) it is square planar complex.
 (viii) O, O is the donor atom for ligand.
 (ix) N, N is the donor atom for ligand.
5. How many ligands are example of non classical ligand?
 C_2H_4 , CO , NO^+ , CN^- , PF_3 , PR_3 , $C_2(CN)_4$
6. In Zeise's salt, how many statements are correct :-
 (a) oxidation state of Pt is +2 & coordination number of Pt is 4
 (b) first organometallic complex discovered
 (c) it is example of cationic complex
 (d) the carbon atom exhibit some electrophilic character
 (e) HCH bond angle is 115°
 (f) carbon atom in Zeise's salt is perfectly sp^3 hybridised
 (g) ethylene moiety of Zeise's salt loses planarity
 (h) C-C bond length in $C_2H_4 < C-C$ bond length in Zeise's salt
 (i) synergic effect takes place (back donation takes place) between filled d orbital of Pt to π^* ABMO of ethylene molecule
7. How many isomers are possible for the complex $[Ir(CO)Cl(PPh_3)_2]$?
8. How many type of isomerism is exhibited by the complex $[Co(NH_3)_4(NO_2)_2]Cl$?
9. How many total stereoisomers are possible for the complex $[Rh(en)_2(NO_2)(SCN)]^+$?
10. Find the number of ions which have higher splitting energy than the Mn^{2+} in octahedral complex considering same ligands for all ions :-
 Cu^{2+} ; Ni^{2+} ; Co^{2+} ; Fe^{2+} ; Cr^{3+}
11. Find out the number of bidentate ligands where both oxygen atoms act as donor atom :-
 (a) acetylacetonato (b) biuret
 (c) dimethylglyoximate (d) glycinate
 (e) biguanido (f) oxalato
 (g) dimethylglycol

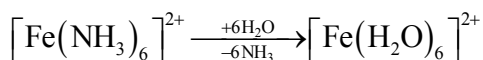
12. Find out number of bidentate ligand where both nitrogen atom act as donor atom and 5 member chelate is formed with metal ion :-



13. What is the coordination number of aluminium in cryolite?
14. Number of optically active isomers of Mabcdef in which a and e are on opposite position.
15. $[M(NH_3)_2(py)_2(NO_2)_2] \xrightarrow[+NO_2]{+NH_3} [M(NH_3)_3(py)_2(NO_2)]$
 Number of geometrical isomers reduced in product side as compared to reactants -
16. Total number of isomers for the complex $K_2[PtCl_2BrSCN]$.
17. Number of geometrical isomer in Mabcdef where a,b,c are at adjacent position with respect to each other.
18. Find out number of linkage isomers possible for $K_4[Fe(CN)_6]$:-
19. Find out number of coordination isomers for the complex $[Cr(NH_3)_6][Co(NO_2)_6]$:-
20. Find out number of coordination isomers for the complex $[Pt(NH_3)_4][CuCl_4]$
21. Find out geometrical isomers are possible for complex $[Mab(AB)_2]^{n\pm}$
22. How many geometrical isomers are possible for compound $[Co(gly)Cl_2(pn)]$.
23. How many ligands forming $d\pi-d\pi$ back bonding (synergic bonding) with metal ion,
 CO ; NO^+ ; CN^- ; PF_3 ; PPh_3 ; H_2O ; en ; AsR_3
24. In $[Co(EDTA)]^{-1}$ number of N-Co-O linkage present ?
25. Find the total number of stereo isomers present in $[Pt(bn)_2]$:-

2.36 COORDINATION COMPOUNDS

26. Find the change in magnetic moment of the complex formed in following process :-



27. Find the maximum number of atoms lying in one plane for $[\text{Cr}(\text{CN})_6]^{3-}$:-
28. How many hexagonal rings are present for C_{84} fullerene :-
29. How many of the following have square planar geometry ?
 $[\text{PtCl}_4]^{2-}$, PdCl_4^{2-} , AgF_4^{-1} , AuCl_4^{-} , ICl_4^{-} , CoCl_4^{-} , FeCl_4^{2-} , NiCl_4^{2-}
30. What are the number of all possible constitutional isomers of $\text{Rn}(\text{NH}_3)_5(\text{NO}_2)\text{Cl}$? :-
31. Maximum number of atoms which are in same plane in $\text{Ni}(\text{CO})_4$:-
32. Find the number of ligands which are stronger ligand as compound to ethylenediamine(en)
 NO_2^{-} ; H_2O ; NO_3^{-} ; F^{-} ; $\text{C}_2\text{O}_4^{2-}$; NH_3 ; Cl^{-} ; CN^{-}
33. Find the number of optically active isomers for
 $[\text{Pd}(\text{en})_2(\text{NH}_3)(\text{H}_2\text{O})]^{+}$:-
cation
34. What are the number of all possible isomeric compound for a square planar palladium(II) complex that contain two Cl^{-} and two SCN^{-} ligands ?
35. Find out the number of unpaired electron present in FeCl_4^{2-} :-
36. Find out the number of unpaired electron present in NiCl_4^{2-} :-
37. In MnCl_4^{2-} , how many unpaired electron present in e orbital according to CFT :-
38. In CoCl_4^{2-} , how many unpaired electron present in t_2 orbital according to CFT :-
39. Find out the number of stereoisomer present in $\text{M}(\text{AB})_2\text{a}_2$:-
 Where AB is bidentate nonsymmetrical ligand and a is monodentate ligand.
40. Find out the number of stereoisomer present in $\text{M}(\text{AB})_2\text{ab}$:-
 Where AB is bidentate nonsymmetrical ligand and a & b is monodentate ligand.

41. If crystal field stabilization energy of $[\text{ML}_6]^{+n}$ is $-0.8\Delta_0$, ($\Delta_0 = 10\text{dq}$)

Find minimum number of electron in t_{2g} orbital of metal ion ?

42. A coordination compound have magnetic moment 3.83 B.M.. Find out the number of unpaired electron(s) in the compound:-
43. Find number of ligands which are stronger ligand as compared to pyridine :-
 NO_2^{-} ; H_2O ; NO_3^{-} ; F^{-} ; en ; $\text{C}_2\text{O}_4^{2-}$; Cl^{-} ; CN^{-} ; CO ; o-phenanthroline ; dipyrityl.
44. In $\text{K}_2[\text{Ni}(\text{EDTA})]$ total number of 5 member ring is m and total number of G.I. possible is n. Find out $m \times n$:-
45. In $[\text{Co}(\text{gly})_3]$ how many O-Co-N linkage is present -
46. Find out number of unpaired electron for d^5 ion having d^2sp^3 hybridisation :-
47. Find out number of unpaired electron for d^7 ion having sp^3d^2 hybridisation :-
48. If number of unpaired electron for d^6 ion having sp^3d^2 hybridisation is n_1 and number of unpaired electron for d^6 ion having d^2sp^3 hybridisation is n_2 then find out $n_1 - n_2$:-
49. What is the number of valence shell electron for the following complex : $[\text{Fe}(\text{CO})_2(\text{NO})_2]$
50. Total isomeric complex salt isolated having formula $\text{CrCl}_3 \cdot 6\text{H}_2\text{O}$

EXERCISE # V(A) JEE–MAIN

1. In $[\text{Cr}(\text{C}_2\text{O}_4)_3]^{3-}$, the isomerism shown is :-
[AIEEE 2002]
 (a) Ligand (b) Optical
 (c) Geometrical (d) Ionisation
2. In the complexes $[\text{Fe}(\text{H}_2\text{O})_6]^{3+}$, $[\text{Fe}(\text{SCN})_6]^{3-}$, $[\text{Fe}(\text{C}_2\text{O}_4)_3]^{3-}$ and $[\text{FeCl}_6]^{3-}$, most stability is shown by :-
[AIEEE 2002]
 (a) $[\text{Fe}(\text{H}_2\text{O})_6]^{3+}$ (b) $[\text{Fe}(\text{SCN})_6]^{3-}$
 (c) $[\text{Fe}(\text{C}_2\text{O}_4)_3]^{3-}$ (d) $[\text{FeCl}_6]^{3-}$
3. One mole of the complex compound $\text{Co}(\text{NH}_3)_5\text{Cl}_3$, gives 3 moles of ions on dissolution in water. One mole of the same complex reacts with two moles of AgNO_3 solution to yield two moles of $\text{AgCl}(\text{s})$.

- The structure of the complex is :- [AIEEE 2003]
- (a) $[\text{Co}(\text{NH}_3)_3\text{Cl}_3] \cdot 2\text{NH}_3$
 (b) $[\text{Co}(\text{NH}_3)_4\text{Cl}_2]\text{Cl} \cdot \text{NH}_3$
 (c) $[\text{Co}(\text{NH}_3)_4\text{Cl}]\text{Cl}_2 \cdot \text{NH}_3$
 (d) $[\text{Co}(\text{NH}_3)_5\text{Cl}]\text{Cl}_2$
4. In the coordination compound $\text{K}_4[\text{Ni}(\text{CN})_4]$, the oxidation state of nickel is :- [AIEEE 2003]
 (a) 0 (b) +1
 (c) +2 (d) -1
5. The number of 3d-electrons remained in Fe^{2+} (At. no. of Fe = 26) ion is :- [AIEEE 2003]
 (a) 4 (b) 5
 (c) 6 (d) 3
6. Ammonia forms the complex ion $[\text{Cu}(\text{NH}_3)_4]^{2+}$ with copper ions in alkaline solutions but not in acidic solution. What is the reason for it :- [AIEEE 2003]
 (a) In acidic solutions hydration protects copper ions
 (b) In acidic solutions protons coordinate with ammonia molecules forming NH_4^+ ions and NH_3 molecules are not available.
 (c) In alkaline solutions insoluble $\text{Cu}(\text{OH})_2$ is precipitated which is soluble in excess of any alkali
 (d) Copper hydroxide is an amphoteric substance
7. Among the properties
 (a) reducing (b) oxidising (c) complexing,
 the set of properties shown by CN^- ion towards metal species is :- [AIEEE 2004]
 (a) c,a (b) b,c
 (c) a,b (d) a,b,c
8. The coordination number of a central metal atom in a complex is determined by :- [AIEEE 2004]
 (a) The number of ligands around a metal ion bonded by sigma and pi-bonds both
 (b) The number of ligands around a metal ion bonded by pi-bonds
 (c) The number of ligands around a metal ion bonded by sigma bonds
 (d) The number of only anionic ligands bonded to the metal
9. Which of the following complex is an outer orbital complex :- [AIEEE 2004]
 (a) $[\text{Co}(\text{NH}_3)_6]^{3+}$ (b) $[\text{Mn}(\text{CN})_6]^{4-}$
 (c) $[\text{Fe}(\text{CN})_6]^{4-}$ (d) $[\text{Ni}(\text{NH}_3)_6]^{2+}$
10. Coordination compounds have great importance in biological systems. In this context which of the following statement is incorrect ? [AIEEE 2004]
 (a) Cyanocobalamin is vitamin B_{12} and contains cobalt
 (b) Haemoglobin is the red pigment of blood and contains iron
 (c) Chlorophylls are green pigment in plants and contains calcium
 (d) Carboxypeptidase - A is an enzyme and contains zinc
11. The correct order of magnetic moments (spin only values in B.M.) among is :- [AIEEE 2004]
 (a) $[\text{Fe}(\text{CN})_6]^{4-} > [\text{MnCl}_4]^{2-} > [\text{CoCl}_4]^{2-}$
 (b) $[\text{MnCl}_4]^{2-} > [\text{Fe}(\text{CN})_6]^{4-} > [\text{CoCl}_4]^{2-}$
 (c) $[\text{MnCl}_4]^{2-} > [\text{CoCl}_4]^{2-} > [\text{Fe}(\text{CN})_6]^{4-}$
 (d) $[\text{Fe}(\text{CN})_6]^{4-} > [\text{CoCl}_4]^{2-} > [\text{MnCl}_4]^{2-}$
12. For octahedral complex, the value of the 'spin only' magnetic moment for one of the following configuration is 2.84 B.M. The correct one is :- [AIEEE 2005]
 (a) d^4 (in strong ligand field)
 (b) d^4 (in weak ligand field)
 (c) d^3 (in weak as well as strong field)
 (d) d^5 (in strong ligand field)
13. The IUPAC name for the complex $[\text{Co}(\text{NO}_2)(\text{NH}_3)_5]\text{Cl}_2$ is :- [AIEEE 2006]
 (a) pentaammine nitrito-N-cobalt(II)chloride
 (b) pentaammine nitrito-N-cobalt(III)chloride
 (c) nitrito-N-pentaamminecobalt(III)chloride
 (d) nitrito-N-pentaamminecobalt(II)chloride
14. Nickel ($Z = 28$) combines with a uninegative monodentate ligand X^- to form a paramagnetic complex $[\text{NiX}_4]^{2-}$. The number of unpaired electron in the nickel and geometry of this complex ion are, respectively :- [AIEEE 2006]
 (a) one, square planar (b) two, square planar
 (c) one, tetrahedral (d) two, tetrahedral

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15. In $\text{Fe}(\text{CO})_5$, the Fe–C bond possesses :–
 (a) ionic character [AIEEE 2006]
 (b) σ -character only
 (c) π -character only
 (d) both σ and π character
16. How many EDTA (ethylenediaminetetraacetate) molecule are required to make an octahedral complex with a Ca^{2+} ion? [AIEEE 2006]
 (a) One (b) Two
 (c) Six (d) Three
17. The “spin-only” magnetic moment [in units of Bohr magneton, (μ_B)] of Ni^{2+} in aqueous solution would be (At. no. Ni = 28) :– [AIEEE 2006]
 (a) 0 (b) 1.73
 (c) 2.84 (d) 4.90
18. Which one of the following has a square planar geometry :– [AIEEE 2007]
 (Co = 27, Ni = 28, Fe = 26, Pt = 78)
 (a) $[\text{CoCl}_4]^{2-}$ (b) $[\text{FeCl}_4]^{2-}$
 (c) $[\text{NiCl}_4]^{2-}$ (d) $[\text{PtCl}_4]^{2-}$
19. The coordination number and the oxidation state of the element ‘E’ in the complex $[\text{E}(\text{en})_2(\text{C}_2\text{O}_4)]\text{NO}_2$ (where en is ethylenediamine) are, respectively :– [AIEEE 2008]
 (a) 6 and 2 (b) 4 and 2
 (c) 4 and 3 (d) 6 and 3
20. In which of the following octahedral complex of Co (at. no. 27), will the magnitude of Δ_0 be the highest :– [AIEEE 2008]
 (a) $[\text{Co}(\text{CN})_6]^{3-}$ (b) $[\text{Co}(\text{C}_2\text{O}_4)_3]^{3-}$
 (c) $[\text{Co}(\text{H}_2\text{O})_6]^{3+}$ (d) $[\text{Co}(\text{NH}_3)_6]^{3+}$
21. Which of the following pairs represents linkage isomers :– [AIEEE 2009]
 (a) $[\text{Co}(\text{NH}_3)_5\text{NO}_3]\text{SO}_4$ and $[\text{Co}(\text{NH}_3)_5\text{SO}_4]\text{NO}_3$
 (b) $[\text{PtCl}_2(\text{NH}_3)_4]\text{Br}_2$ and $[\text{PtBr}_2(\text{NH}_3)_4]\text{Cl}_2$
 (c) $[\text{Cu}(\text{NH}_3)_4][\text{PtCl}_4]$ and $[\text{Pt}(\text{NH}_3)_4][\text{CuCl}_4]$
 (d) $[\text{Pd}(\text{PPh}_3)_2(\text{NCS})_2]$ and $[\text{Pd}(\text{PPh}_3)_2(\text{SCN})_2]$
22. Which of the following has an optical isomer :– [AIEEE 2009]
 (a) $[\text{Co}(\text{H}_2\text{O})_4(\text{en})]^{3+}$ (b) $[\text{Co}(\text{en})_2(\text{NH}_3)_2]^{3+}$
 (c) $[\text{Co}(\text{NH}_3)_3\text{Cl}]^+$ (d) $[\text{Co}(\text{en})(\text{NH}_3)_2]^{2+}$
23. Which one of the following has an optical isomer? (en = ethylene diamine) [AIEEE 2010]
 (a) $[\text{Zn}(\text{en})_2]^{2+}$ (b) $[\text{Zn}(\text{en})(\text{NH}_3)_2]^{2+}$
 (c) $[\text{Co}(\text{en})_3]^{3+}$ (d) $[\text{Co}(\text{H}_2\text{O})_4(\text{en})]^{3+}$
24. A solution containing 2.675 g of $\text{CoCl}_3 \cdot 6\text{NH}_3$ (molar mass = 267.5 g mol^{-1}) is passed through a cation exchanger. The chloride ions obtained in solution were treated with excess of AgNO_3 to give 4.78 g of AgCl (molar mass = 143.5 g mol^{-1}). The formula of the complex is :– [AIEEE 2010]
 (At. mass of Ag = 108 u)
 (a) $[\text{CoCl}(\text{NH}_3)_5]\text{Cl}_2$ (b) $[\text{Co}(\text{NH}_3)_6]\text{Cl}_3$
 (c) $[\text{CoCl}_2(\text{NH}_3)_4]\text{Cl}$ (d) $[\text{CoCl}_3(\text{NH}_3)_3]$
25. Which of the following facts about the complex $[\text{Cr}(\text{NH}_3)_6]\text{Cl}_3$ is wrong :– [AIEEE 2011]
 (a) The complex is an outer orbital complex
 (b) The complex gives white precipitate with silver nitrate solution
 (c) The complex involves d^2sp^3 hybridisation and is octahedral in shape
 (d) The complex is paramagnetic
26. The magnetic moment (spin only) of $[\text{NiCl}_4]^{2-}$ is :– [AIEEE 2011]
 (a) 2.82 B.M. (b) 1.41 B.M.
 (c) 1.82 B.M. (d) 5.46 B.M.
27. Among the ligands NH_3 , en, CN^- and CO the correct order of their increasing field strength, is :– [AIEEE 2011]
 (a) $\text{CO} < \text{NH}_3 < \text{en} < \text{CN}^-$
 (b) $\text{NH}_3 < \text{en} < \text{CN}^- < \text{CO}$
 (c) $\text{CN}^- < \text{NH}_3 < \text{CO} < \text{en}$
 (d) $\text{en} < \text{CN}^- < \text{NH}_3 < \text{CO}$
28. Which one of the following complex ions has geometrical isomers :– [AIEEE 2011]
 (a) $[\text{Co}(\text{en})_3]^{3+}$ (b) $[\text{Ni}(\text{NH}_3)_5\text{Br}]^+$
 (c) $[\text{Co}(\text{NH}_3)_2(\text{en})_2]^{3+}$ (d) $[\text{Cr}(\text{NH}_3)_4(\text{en})]^{3+}$
29. Which among the following will be named as dibromidobis (ethylene diamine) chromium (III) bromide [AIEEE 2012]
 (a) $[\text{Cr}(\text{en})\text{Br}_2]\text{Br}$ (b) $[\text{Cr}(\text{en})_3]\text{Br}_3$
 (c) $[\text{Cr}(\text{en})_2\text{Br}_2]\text{Br}$ (d) $[\text{Cr}(\text{en})\text{Br}_4]^-$

30. The complex ion $[\text{Pt}(\text{NO}_2)(\text{Py})(\text{NH}_3)(\text{NH}_2\text{OH})]^+$ will give :-
[JEE–MAIN–2012, Online]
(a) 4 isomers (Geometrical)
(b) 2 isomers (Geometrical)
(c) 3 isomers (Geometrical)
(d) 6 isomers (Geometrical)
31. Which of the following complex ions will exhibit optical isomerism :- [JEE–Main–2012, Online]
(en = 1,2–diamine ethane)
(a) $[\text{Co}(\text{en})_2\text{Cl}_2]^+$ (b) $[\text{Zn}(\text{en})_2]^{2+}$
(c) $[\text{Co}(\text{NH}_3)_4\text{Cl}_2]^+$ (d) $[\text{Cr}(\text{NH}_3)_2\text{Cl}_2]^+$
32. Which of the following complex species is not expected to exhibit optical isomerism :-
[JEE–Main–2013]
(a) $[\text{Co}(\text{en})_3]^{3+}$ (b) $[\text{Co}(\text{en})_2\text{Cl}_2]^+$
(c) $[\text{Co}(\text{NH}_3)_3\text{Cl}_3]$ (d) $[\text{Co}(\text{en})(\text{NH}_3)_2\text{Cl}_2]^+$
33. Type of isomerism which exist between $[\text{Pd}(\text{C}_6\text{H}_5)_2(\text{SCN})_2]$ and $[\text{Pd}(\text{C}_6\text{H}_5)_2(\text{NCS})_2]$ is :-
[JEE–Main–2002, Online]
(a) Solvate isomerism
(b) Ionisation isomerism
(c) Linkage isomerism
(d) Coordination isomerism
34. Which of the following is diamagnetic ?
[JEE–MAIN–2013, Online]
(a) $[\text{CoF}_6]^{3-}$ (b) $[\text{FeF}_6]^{3-}$
(c) $[\text{Fe}(\text{CN})_6]^{3-}$ (d) $[\text{Co}(\text{Ox})_3]^{3-}$
35. The magnetic moment of the complex anion $[\text{Cr}(\text{NO})(\text{NH}_3)(\text{CN})_4]^{2-}$ is :-
[JEE–Main–2013, Online]
(a) 2.82 B.M. (b) 5.91 B.M.
(c) 1.73 B.M. (d) 3.87 B.M.
36. The octahedral complex of a metal ion M^{3+} with four monodentate ligands $\text{L}_1, \text{L}_2, \text{L}_3$ and L_4 absorb wavelength in the region of red, green, yellow and blue respectively. The increasing order of ligand strength of the four ligands is : [JEE–Main–2014]
(a) $\text{L}_3 < \text{L}_2 < \text{L}_4 < \text{L}_1$ (b) $\text{L}_1 < \text{L}_2 < \text{L}_4 < \text{L}_3$
(c) $\text{L}_4 < \text{L}_3 < \text{L}_2 < \text{L}_1$ (d) $\text{L}_1 < \text{L}_3 < \text{L}_2 < \text{L}_4$
37. The equation which is balanced and represents the correct product (s) is : [JEE–MAIN–2014]
(a) $[\text{Mg}(\text{H}_2\text{O})_6]^{2+} + (\text{EDTA})^{4-} \xrightarrow{\text{excess NaOH}} [\text{Mg}(\text{EDTA})]^{2+} + 6\text{H}_2\text{O}$
(b) $\text{CuSO}_4 + 4\text{KCN} \rightarrow \text{K}_2[\text{Cu}(\text{CN})_4] + \text{K}_2\text{SO}_4$
(c) $\text{Li}_2\text{O} + 2\text{KCl} \rightarrow 2\text{LiCl} + \text{K}_2\text{O}$
(d) $[\text{CoCl}(\text{NH}_3)_5]^+ + 5\text{H}^+ \rightarrow \text{Co}^{2+} + 5\text{NH}_4^+ + \text{Cl}^-$
38. The correct statement about the magnetic properties of $[\text{Fe}(\text{CN})_6]^{3-}$ and $[\text{FeF}_6]^{3-}$ is : (Z = 26). [J–MAIN–2014, Online]
(a) $[\text{Fe}(\text{CN})_6]^{3-}$ is paramagnetic, $[\text{FeF}_6]^{3-}$ is diamagnetic.
(b) both are diamagnetic.
(c) $[\text{Fe}(\text{CN})_6]^{3-}$ is diamagnetic, $[\text{FeF}_6]^{3-}$ is paramagnetic.
(d) both are paramagnetic
39. An octahedral complex of Co^{3+} is diamagnetic. The hybridisation involved in the formation of the complex is [J–MAIN–2014]
(a) d^2sp^3 (b) sp^3d
(c) dsp^2 (d) sp^3d^2
40. Which of the following name formula combinations is not correct? [J–MAIN–2014, Online]
- | Formula | Name |
|---|--|
| (a) $\text{K}[\text{Cr}(\text{NH}_3)_2\text{Cl}_4]$ | Potassium diammine Tetrachlorochromate (III) |
| (b) $[\text{Co}(\text{NH}_3)_4(\text{H}_2\text{O})\text{I}]\text{SO}_4$ | Tetraammine aquaiodo cobalt (III) sulphate |
| (c) $[\text{Mn}(\text{CN})_5]^{2-}$ | Pentacyanomagnate (II) ion |
| (d) $\text{K}_2[\text{Pt}(\text{CN})_4]$ | Potassiumtetracyanoplatinate(II) |
41. Consider the coordination compound, $[\text{Co}(\text{NH}_3)_6]\text{Cl}_3$. In the formation of this complex, the species which acts as the Lewis acid is : [J–MAIN–2014, Online]
(a) $[\text{Co}(\text{NH}_3)_6]^{3+}$ (b) NH_3
(c) Co^{3+} (d) Cl^-

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42. Among the following species the one which causes the highest CFSE, Δ_0 as a ligand is:

[J–MAIN–2014, Online]

- (a) CN^- (b) NH_3
(c) CO (d) F^-

43. Which one of the following complexes will most likely absorb visible light ?

[J–MAIN–2014, Online]

(At. nos. Sc = 21, Ti = 22, V = 23, Zn = 30) :-

- (a) $[\text{Ti}(\text{NH}_3)_6]^{4+}$ (b) $[\text{V}(\text{NH}_3)_6]^{3+}$
(c) $[\text{Zn}(\text{NH}_3)_4]^{2+}$ (d) $[\text{Sc}(\text{H}_2\text{O})_6]^{3+}$

44. Nickel ($Z = 28$) combines with a uninegative monodentate ligand to form a diamagnetic complex $[\text{NiL}_4]^{2-}$. The hybridisation involved and the number of unpaired electrons are respectively:

[J–MAIN–2014, Online]

- (a) sp^3 , zero (b) sp^3 , two
(c) dsp^2 , one (d) dsp^2 , zero

45. The number of geometrical isomers that can exist for square planar $[\text{Pt}(\text{Cl})(\text{py})(\text{NH}_3)(\text{NH}_2\text{OH})]^+$ is (py = pyridine) :-

[J–MAIN–2015]

- (a) 4 (b) 6
(c) 2 (d) 3

46. The colour of KMnO_4 is due to :-

[J–MAIN–2015]

- (a) $L \rightarrow M$ charge transfer transition
(b) $\sigma - \sigma^*$ transition
(c) $M \rightarrow L$ charge transfer transition
(d) $d - d$ transition

47. Which of the following complex ions has electrons that are symmetrically filled in both t_{2g} and e_g orbitals ?

[J–MAIN–2015, Online]

- (a) $[\text{Co}(\text{H}_2\text{O})_6]^{2+}$ (b) $[\text{CoCl}_6]^{3-}$
(c) $[\text{CoCl}_4]^{2-}$ (d) $[\text{CoCl}_6]^{4-}$

49. The correct statement on the isomerism associated with the following complex ions,

- (a) $[\text{Ni}(\text{H}_2\text{O})_5\text{NH}_3]^{2+}$ [J–MAIN–2015, Online]
(b) $[\text{Ni}(\text{H}_2\text{O})_4(\text{NH}_3)_2]^{2+}$ and
(c) $[\text{Ni}(\text{H}_2\text{O})_3(\text{NH}_3)_3]^{2+}$ is

- (a) (a) and (b) show geometrical and optical isomerism

- (b) (b) and (c) show geometrical and optical isomerism

- (c) (a) and (b) show only geometrical isomerism

- (d) (b) and (c) show only geometrical optical isomerism

50. Which one of the following complexes shows optical isomerism :-

[J–MAIN–2016]

- (a) $[\text{Co}(\text{NH}_3)_4\text{Cl}_2]\text{Cl}$
(b) $[\text{Co}(\text{NH}_3)_3\text{Cl}_3]$
(c) $\text{cis}[\text{Co}(\text{en})_2\text{Cl}_2]\text{Cl}$
(d) $\text{trans}[\text{Co}(\text{en})_2\text{Cl}_2]\text{Cl}$
(en = ethylenediamine)

51. The pair having the same magnetic moment is :-

[J–MAIN–2016]

(At. No. : Cr = 24, Mn = 25, Fe = 26, Co = 27)

- (a) $[\text{CoCl}_4]^{2-}$ and $[\text{Fe}(\text{H}_2\text{O})_6]^{2+}$
(b) $[\text{Cr}(\text{H}_2\text{O})_6]^{2+}$ and $[\text{CoCl}_4]^{2+}$
(c) $[\text{Cr}(\text{H}_2\text{O})_6]^{2+}$ and $[\text{Fe}(\text{H}_2\text{O})_6]^{2+}$
(d) $[\text{Mn}(\text{H}_2\text{O})_6]^{2+}$ and $[\text{Cr}(\text{H}_2\text{O})_6]^{2+}$

52. Which one of the following complexes will consume highest equivalents of aqueous solution of $\text{Ag}(\text{NO}_3)?$

[J–MAIN–2016-Online]

- (a) $[\text{Cr}(\text{H}_2\text{O})_6]\text{Cl}_3$ (b) $\text{Na}_2[\text{CrCl}_5(\text{H}_2\text{O})]$
(c) $\text{Na}_3[\text{CrCl}_6]$ (d) $[\text{Cr}(\text{H}_2\text{O})_5\text{Cl}]\text{Cl}_2$

53. Identify the correct trend given below :-

[J–MAIN–2016, Online]

(At. No. : Ti = 22, Cr = 24 and Mo = 42)

- (a) Δ_0 of $[\text{Cr}(\text{H}_2\text{O})_6]^{2+} < [\text{Mo}(\text{H}_2\text{O})_6]^{2+}$ and Δ_0 of $[\text{Ti}(\text{H}_2\text{O})_6]^{3+} < [\text{Ti}(\text{H}_2\text{O})_6]^{2+}$
(b) Δ_0 of $[\text{Cr}(\text{H}_2\text{O})_6]^{2+} < [\text{Mo}(\text{H}_2\text{O})_6]^{2+}$ and Δ_0 of $[\text{Ti}(\text{H}_2\text{O})_6]^{3+} > [\text{Ti}(\text{H}_2\text{O})_6]^{2+}$
(c) Δ_0 of $[\text{Cr}(\text{H}_2\text{O})_6]^{2+} > [\text{Mo}(\text{H}_2\text{O})_6]^{2+}$ and Δ_0 of $[\text{Ti}(\text{H}_2\text{O})_6]^{3+} > [\text{Ti}(\text{H}_2\text{O})_6]^{2+}$
(d) Δ_0 of $[\text{Cr}(\text{H}_2\text{O})_6]^{2+} > [\text{Mo}(\text{H}_2\text{O})_6]^{2+}$ and Δ_0 of $[\text{Ti}(\text{H}_2\text{O})_6]^{3+} < [\text{Ti}(\text{H}_2\text{O})_6]^{2+}$

EXERCISE # V(B) JEE-ADVANCED**True/False**

1. The electron density in the xy plane in $3d_{x^2-y^2}$ orbital is zero. [JEE 1986]
2. Amongst the following the lowest degree of paramagnetism per mole of the compound at 298 K will shown by ; [JEE 1988]

(a) $\text{MnSO}_4 \cdot 4\text{H}_2\text{O}$	(b) $\text{CuSO}_4 \cdot 5\text{H}_2\text{O}$
(c) $\text{FeSO}_4 \cdot 6\text{H}_2\text{O}$	(d) $\text{NiSO}_4 \cdot 6\text{H}_2\text{O}$

True/False

3. Both potassium ferrocyanide and potassium ferricyanide are diamagnetic. [JEE 1989]
4. Mixture X = 0.02 mole of $[\text{Co}(\text{NH}_3)_5\text{SO}_4]\text{Br}$ and 0.02 mole of $[\text{Co}(\text{NH}_3)_5\text{Br}]\text{SO}_4$ was prepared in 2 L of solution. [JEE 1991]

1 L of mixture X + excess $\text{AgNO}_3 \longrightarrow \text{Y}$
 1 L of mixture X + excess $\text{BaCl}_2 \longrightarrow \text{Z}$
 Number of moles of Y and Z are

(a) 0.01, 0.01	(b) 0.02, 0.01
(c) 0.01, 0.02	(d) 0.02, 0.02
5. Amongst $[\text{Ni}(\text{CO})_4]$, $[\text{Ni}(\text{CN})_4]^{2-}$ and $[\text{NiCl}_4]^{2-}$ [JEE 1991]

(a) $[\text{Ni}(\text{CO})_4]$ and $[\text{Ni}(\text{CN})_4]^{2-}$ are diamagnetic and $[\text{NiCl}_4]^{2-}$ is paramagnetic
(b) $[\text{NiCl}_4]^{2-}$ and $[\text{Ni}(\text{CN})_4]^{2-}$ are diamagnetic and $[\text{Ni}(\text{CO})_4]$ is paramagnetic
(c) $[\text{Ni}(\text{CO})_4]$, $[\text{NiCl}_4]^{2-}$ and $[\text{Ni}(\text{CN})_4]^{2-}$ are diamagnetic.
(d) $[\text{Ni}(\text{CO})_4]$ is diamagnetic and $[\text{Ni}(\text{CN})_4]^{2-}$, $[\text{NiCl}_4]^{2-}$ are paramagnetic
6. The compound which does not show paramagnetism is [JEE 1992]

(a) $[\text{Cu}(\text{NH}_3)_4]\text{Cl}_2$	(b) $[\text{Ag}(\text{NH}_3)_2]\text{Cl}$
(c) NO	(d) NO_2
7. The number of d-electron is $[\text{Cr}(\text{H}_2\text{O})_6]^{3+}$ [At No. of Cr = 24] is [JEE 1993]

(a) 2	(b) 3
(c) 4	(d) 5
8. Amongst the following ions which one has the highest paramagnetism [JEE 1993]

(a) $[\text{Cr}(\text{H}_2\text{O})_6]^{3+}$	(b) $[\text{Fe}(\text{H}_2\text{O})_6]^{2+}$
(c) $[\text{Cu}(\text{H}_2\text{O})_6]^{2+}$	(d) $[\text{Zn}(\text{H}_2\text{O})_6]^{2+}$

Fill in the Blanks

9. The type of magnetism exhibited by $[\text{Mn}(\text{H}_2\text{O})_6]^{2+}$ ion is (paramagnetic) [JEE 1994]
10. The IUPAC name of $[\text{Co}(\text{NH}_3)_6]\text{Cl}_3$ is [JEE 1994]
11. Identify the complexes which are expected to be coloured [JEE 1994]

(a) $\text{Ti}(\text{NO}_3)_4$
(b) $[\text{Cu}(\text{NCCH}_3)_4]^+\text{BF}_4^-$
(c) $[\text{Cr}(\text{NH}_3)_6]\text{Cl}_3$
(d) $\text{K}_3[\text{VF}_6]$
12. Which of the following is formed when excess of KCN is added to aqueous solution of copper sulphate? [JEE 1996]

(a) $\text{Cu}[\text{CN}]_2$	(b) $\text{K}_2[\text{Cu}[\text{CN}]_4]$
(c) $\text{K}[\text{Cu}[\text{CN}]_2]$	(d) $\text{K}_3[\text{Cu}(\text{CN})_4]$
13. Which of the following is an organometallic compound [JEE 1997]

(a) Lithium methoxide
(b) Lithium acetate
(c) Lithium dimethylamide
(d) Methyl lithium
14. In nitroprusside ion, the iron and NO exist as Fe^{II} and NO^+ rather than Fe^{III} and NO. These forms can be differentiated by [JEE 1998]

(a) estimating the concentration of iron
(b) measuring the concentration of CN^-
(c) measuring the solid state magnetic moment
(d) thermally decomposing the compound
15. The geometry of $\text{Ni}(\text{CO})_4$ and $\text{Ni}(\text{PPh}_3)_2\text{Cl}_2$ are [JEE 1999]

(a) both square planar
(b) tetrahedral and square planar
(c) both tetrahedral
(d) square planar and tetrahedral
16. The complex ion which has no 'd' electrons in the central atom is : [JEE 2001]

(a) $[\text{MnO}_4]^-$	(b) $[\text{Co}(\text{NH}_3)_6]^{3+}$
(c) $[\text{Fe}(\text{CN})_6]^{3-}$	(d) $[\text{Cr}(\text{H}_2\text{O})_6]^{3+}$

2.42 COORDINATION COMPOUNDS

17. The correct order of hybridisation of the central atom in the following species. [JEE2001]
 NH_3 , $[\text{PtCl}_4]^{2-}$, PCl_5 and BCl_3 , is [At No. Pt = 78]
 (a) $\text{dsp}^2, \text{sp}^3\text{d}, \text{sp}^2$ and sp^3
 (b) $\text{sp}^3, \text{dsp}^2$, sp^3d , sp^2
 (c) $\text{dsp}^2, \text{sp}^2, \text{sp}^3$ and sp^3d
 (d) dsp^2 , sp^3, sp^2 and sp^3d
18. The species having tetrahedral shape is : [JEE2004]
 (a) $[\text{PdCl}_4]^{2-}$ (b) $[\text{Ni}(\text{CN})_4]^{2-}$
 (c) $[\text{Pd}(\text{CN})_4]^{2-}$ (d) $[\text{NiCl}_4]^{2-}$
19. The pair of compounds having metals in their highest oxidation state is [JEE2004]
 (a) MnO_2 , FeCl_3
 (b) $[\text{MnO}_4]^-$, CrO_2Cl_2
 (c) $[\text{Fe}(\text{CN})_6]^{3-}$, $[\text{Co}(\text{CN})_3]$
 (d) $[\text{NiCl}_4]^{2-}$, $[\text{CoCl}_4]^-$
20. Spin only magnetic moment of the compound $\text{Hg}[\text{Co}(\text{SCN})_4]$ is [JEE2004]
 (a) $\sqrt{3}$ (b) $\sqrt{15}$
 (c) $\sqrt{24}$ (d) $\sqrt{8}$
21. Which of the following pair is expected to exhibit same colour in solution? [JEE2005]
 (a) VOCl_2 ; FeCl_2 (b) CuCl_2 ; VOCl_2
 (c) MnCl_2 ; FeCl_2 (d) FeCl_2 ; CuCl_2
22. Which type of isomerism is shown by $\text{Co}(\text{NH}_3)_4\text{Br}_2$? [JEE2005]
 (a) Geometrical and Ionisation
 (b) Optical and Ionisation
 (c) Geometrical and Optical
 (d) Geometrical only
- Question No. 23 to 25 (3 questions)**
 The coordination number of Ni^{2+} is 4. [JEE2006]
 $\text{NiCl}_2 + \text{KCN (excess)} \rightarrow \text{A (cyano complex)}$
 $\text{NiCl}_2 + \text{KCl (excess)} \rightarrow \text{B (chloro complex)}$
23. The IUPAC name of A and B are
 (a) Potassium tetracyanonickelate (II), potassium tetrachloridonickelate (II)
 (b) Tetracyanidopotassiumnickelate (II), tetrachloridopotassiumnickelate (II)
 (c) Tetracyanonickel (II), tetrachloridonickel (II)
 (d) Potassium tetracyanonickel (II), potassium tetrachloridonickel (II)
24. Predict the magnetic nature of A and B.
 (a) Both are diamagnetic
 (b) A is diamagnetic and B is paramagnetic with two unpaired electrons
 (c) A is diamagnetic and B is paramagnetic with one unpaired electron
 (d) Both are paramagnetic
25. The hybridisation of A and B –
 (a) dsp^2 , sp^3 (b) sp^3 , sp^3
 (c) dsp^2 , dsp^2 (d) sp^3d^2 , d^2sp^3
26. If the bond length of CO bond in carbon monoxide is 1.128 Å, then what is the value of CO bond length in $\text{Fe}(\text{CO})_5$? [JEE 2006]
 (a) 1.15 Å (b) 1.128 Å
 (c) 1.72 Å (d) 1.118 Å
27. Among the following metal carbonyls, the C–O bond order is lowest in [JEE 2007]
 (a) $[\text{Mn}(\text{CO})_6]^+$ (b) $[\text{Fe}(\text{CO})_5]$
 (c) $[\text{Cr}(\text{CO})_6]$ (d) $[\text{V}(\text{CO})_6]^-$
28. Match the complexes in Column I with their properties listed in Column II. Indicate your answer by darkening the appropriate bubbles of the 4 x 4 matrix given in the ORS. [JEE 2007]
- Column I**
 (a) $[\text{Co}(\text{NH}_3)_4(\text{H}_2\text{O})_2]\text{Cl}_2$
 (b) $[\text{Pt}(\text{NH}_3)_2\text{Cl}_2]$
 (c) $[\text{Co}(\text{H}_2\text{O})_5\text{Cl}]\text{Cl}$
 (d) $[\text{Ni}(\text{H}_2\text{O})_6]\text{Cl}_2$
- Column II**
 (P) Geometrical isomers
 (Q) Paramagnetic
 (R) Diamagnetic
 (S) Metal ion with 2+ oxidation state
29. Among the following, the coloured compound is [JEE 2008]
 (a) CuCl (b) $\text{K}_3[\text{Cu}(\text{CN})_4]$
 (c) CuF_2 (d) $[\text{Cu}(\text{CH}_3\text{CN})_4]\text{BF}_4$

2.43

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2.44 COORDINATION COMPOUNDS

42. The volume (in mL) of 0.1 M AgNO_3 required for complete precipitation of chloride ions present in 30 mL of 0.01 M solution of $[\text{Cr}(\text{H}_2\text{O})_5\text{Cl}]\text{Cl}_2$, as silver chloride is close to. [JEE 2011]
43. As per IUPAC nomenclature, the name of the complex $[\text{Co}(\text{H}_2\text{O})_4(\text{NH}_3)_2]\text{Cl}_3$ is : [JEE 2011]
 (a) Tetraaquadiaminocobalt(III)chloride
 (b) Tetraaquadiaminocobalt(III)chloride
 (c) Diaminetetraaquacobalt(III)chloride
 (d) Diamminetetraaquacobalt(III)chloride
44. The colour of light absorbed by an aqueous solution of CuSO_4 is [JEE 2012]
 (a) orange-red (b) blue-green
 (c) yellow (d) violet
45. $\text{NiCl}_2 \cdot \{\text{P}(\text{C}_2\text{H}_5)_2(\text{C}_6\text{H}_5)\}_2$ exhibits temperature dependent magnetic behaviour (paramagnetic / diamagnetic). The coordination geometries of Ni^{2+} in the paramagnetic and diamagnetic states are respectively : [JEE 2012]
 (a) tetrahedral and tetrahedral
 (b) square planar and square planar
 (c) tetrahedral and square planar
 (d) square planar and tetrahedral
46. Consider the following complex ions P, Q and R,
 $\text{P} = [\text{FeF}_6]^{3-}$, $\text{Q} = [\text{V}(\text{H}_2\text{O})_6]^{2+}$
 and $\text{R} = [\text{Fe}(\text{H}_2\text{O})_6]^{2+}$
 The correct order of the complex ions, according to their spin-only magnetic moment values (in B.M) [JEE 2013]
 (a) $\text{R} < \text{Q} < \text{P}$
 (b) $\text{Q} < \text{R} < \text{P}$
 (c) $\text{R} < \text{P} < \text{Q}$
 (d) $\text{Q} < \text{P} < \text{R}$
48. The pair(s) of coordination complex/ion exhibiting the same kind of isomerism [JEE 2013]
 (a) $[\text{Cr}(\text{NH}_3)_5\text{Cl}]\text{Cl}_2$ and $[\text{Cr}(\text{NH}_3)_4\text{Cl}_2]\text{Cl}$
 (b) $[\text{Co}(\text{NH}_3)_4\text{Cl}_2]^+$ and $[\text{Pt}(\text{NH}_3)_2(\text{H}_2\text{O})\text{Cl}]^+$
 (c) $[\text{CoBr}_2\text{Cl}_2]^{2-}$ and $[\text{PtBr}_2\text{Cl}_2]^{2-}$
 (d) $[\text{Pt}(\text{NH}_3)_3(\text{NO}_3)]\text{Cl}$ and $[\text{Pt}(\text{NH}_3)_3\text{Cl}]\text{Br}$

49. Match each coordination compound in List-I with an appropriate pair and select the correct answer using the code given below the lists,

{en = $\text{H}_2\text{NCH}_2\text{CH}_2\text{NH}_2$ ' atomic numbers ; Ti = 22 ; Cr = 24 ; Co = 27 ; Pt = 78}

List-I

- (P) $[\text{Cr}(\text{CN})_4\text{Cl}_2]\text{Cl}$
 (Q) $[\text{Ti}(\text{H}_2\text{O})_5\text{Cl}](\text{NO}_3)_2$
 (R) $[\text{Pt}(\text{en})(\text{NH}_3)\text{Cl}]\text{NO}_3$
 (S) $[\text{Co}(\text{NH}_3)_4(\text{NO}_3)_2]\text{NO}_3$

List-II

- (1) Paramagnetic and exhibits ionisation isomerism
 (2) Diamagnetic and exhibits cis-trans isomerism
 (3) Paramagnetic and exhibits cis-trans isomerism
 (4) Diamagnetic and exhibits ionisation isomerism

Code :

	P	Q	R	S
(a)	4	2	3	1
(b)	3	1	4	2
(c)	2	1	3	4
(d)	1	3	4	2

50. A list of species having the formula XZ_4 is given below : [JEE Adv. 2014]

XeF_4 , SF_4 , SiF_4 , BF_4^- , BrF_4^- , $[\text{Cu}(\text{NH}_3)_4]^{2+}$, $[\text{FeCl}_4]^{2-}$, $[\text{CoCl}_4]^{2-}$ and $[\text{PtCl}_4]^{2-}$.

Defining shape on the basis of the location of X and Z atoms, the total number of species having a square planar shape is—

Subjective :

51. Give reasons in two or three sentences only for the following :
 "The species $[\text{CuCl}_4]^{2-}$ exists, while $[\text{CuI}_4]^{2-}$ does not exist" [JEE 1992]
52. Write the IUPAC name of the following compounds [JEE 1995]
 (i) $[\text{Co}(\text{NH}_3)_5\text{ONO}]\text{Cl}_2$
 (ii) $\text{K}_3[\text{Cr}(\text{CN})_6]$
53. Write the IUPAC name for $[\text{Cr}(\text{NH}_3)_5\text{CO}_3]\text{Cl}$ [JEE 1996]

54. Write the formulae of the following complexes :

[JEE 1997]

- (i) Pentamminechlorocobalt (III) ion
(ii) Lithium tetrahydridoaluminate (III)

55. A, B and C are three complexes of chromium (III) with the empirical formula $H_{12}O_6Cl_3Cr$. All the three complexes have water and chloride ion as ligands. Complex A does not react with concentrated H_2SO_4 , whereas complexes B and C lose 6.75% and 13.5% of their original mass, respectively, on treatment with concentrated H_2SO_4 . Identify A, B and C.

[JEE 1989]

56. Draw the structures of $[Co(NH_3)_6]^{3+}$, $[Ni(CN)_4]^{2-}$ and $[Ni(CO)_4]$. Write the hybridisation of atomic orbitals of the transition metal in each case.

[JEH 2000]

57. A metal complex having composition $Cr(NH_3)_4Cl_2Br$ has been isolated in two forms A and B. The form A reacts with $AgNO_3$ to give a white precipitate readily soluble in dilute aqueous ammonia, whereas B gives a pale yellow precipitate soluble in concentrated ammonia. Write the formula of A and B and state the hybridisation of chromium in each. Calculate their magnetic moments (spin-only value).

[JEE 2001]

58. Deduce the structures of $[NiCl_4]^{2-}$ and $[Ni(CN)_4]^{2-}$ considering the hybridisation of the metal ion. Calculate the magnetic moment (spin only) of the species.

[JEE 2002]

59. Write the IUPAC name of the compound $K_2[Cr(NO)(CN)_4(NH_3)]$. Spin magnetic moment of the complex $\mu = 1.73$ BM. Give the structure of anion.

[JEE 2003]

60. $NiCl_2$ in the presence of dimethyl glyoxime (DMG) gives a complex which precipitates in the presence of NH_4OH , giving a bright red colour.

[JEE 2004]

(a) Draw its structure and show H-bonding

(b) Give oxidation state of Ni and its hybridisation

(c) Predict whether it is paramagnetic or diamagnetic

61. For the octahedral complexes of Fe^{3+} in SCN^- (thiocyanato-S) and in CN^- ligand environments, the difference between the spin only magnetic moment in Bohr magnetons (when approximated to the nearest integer) is :

[Atomic number of Fe = 26] [JEE Adv. 2015]

62. In the complex acetyl bromidodicarbonylbis (triethylphosphine)iron(II), the number of Fe-C bond(s) is :-

[JEE Adv. 2015]

63. Among the complex ions, $[Co(NH_2-CH_2-CH_2-NH_2)_2Cl_2]^+$, $[CrCl_2(C_2O_4)_2]^{3-}$, $[Fe(H_2O)_4(OH)_2]^+$, $[Fe(NH_3)_2(CN)_4]^-$, $[Co(NH_2-CH_2-CH_2-NH_2)_2(NH_3)Cl]^{2+}$ and $[Co(NH_3)_4(H_2O)Cl]^{2+}$, the number of complex ion(s) that show(s) cis-trans isomerism is :-

[JEE Adv. 2016]

64. Among $[Ni(CO)_4]$, $[NiCl_4]^{2-}$, $[Co(NH_3)_4Cl_2]Cl$, $Na_3[CoF_6]$, Na_2O_2 and CsO_2 , the total number of paramagnetic compounds is :-

[JEE Adv. 2016]

(a) 2

(b) 3

(c) 4

(d) 5

65. The number of geometric isomers possible for complex $[CoL_2Cl_2]^-$ ($L = H_2NCH_2CH_2O^-$) is :-

[JEE Adv. 2016]

66. The geometries of the ammonia complexes of Ni^{+2} , Pt^{2+} and Zn^{2+} respectively are :-

[JEE Adv. 2016]

(a) octahedral, square planar and tetrahedral

(b) square planar, octahedral and tetrahedral

(c) tetrahedral, square planar and octahedral

(d) octahedral, tetrahedral and square planar



METALLURGY

PROBLEMS BASED ON GIVEN TOPICS

- Types of ores
- Principal steps in the recovery of a metal from its ore
- Concentration or dressing of ore
 - Gravity separation or levigation
 - Magnetic separation
 - Froth floatation of oil floatation
 - Chemical method of separation leaching
- Conversion of concentrated ore into its oxide
 - Calcination
 - Roasting
- Different reduction processes
 - Carbon reduction
 - Self reduction
 - Thermite reduction (or Goldschmidt-Thermite process)
 - Metal replacement method (Hydrometallurgy)
 - Electrolytic reduction
 - Thermal decomposition method
- Purification or refining of metal
 - Thermal refining
 - Electrorefining
- Thermodynamics of reaction process
- Alloy and amalgams
 - Classification of alloys
 - Characteristic of alloys
- Preparation of alloys
- Amalgam
- Different types of furnaces used in metallurgy
- Extraction of silver
 - Refining of Ag
- Extraction of Gold by Cyanide process
 - Refining Au
- Extraction of tin
 - Refining of Sn
- Extraction of Magnesium
 - Electrolytic reduction
 - Carbon reduction process
 - Other processes
- Extraction of Aluminium
 - Beneficiation of bauxite
 - Electrolytic reduction of pure Al_2O_3
 - Electrorefining of aluminium
- Extraction of lead
 - Carbon reduction
 - Self reduction process
 - Refining of lead
- Extraction of copper
 - Refining of blister copper
- Extraction of zinc
- Extraction of iron
 - Purification of iron or preparation of wrought iron
 - Steel making

Metallurgy

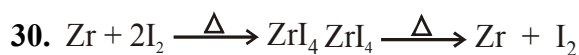
EXERCISE # I

❑ Only one correct answer :

- Which of the following is not sulfide ore ?
(a) Zinc blende (b) Galena
(c) Cinnabar (d) Baryte
- Which of the following is carbonate ore ?
(a) Siderite (b) Magnesite
(c) Dolomite (d) All
- The salt which is least abundant as mineral is ?
(a) Halide
(b) Nitrate
(c) Carbonate
(d) sulfide
- Which ore does not have water of crystallisation ?
(a) Diaspore (b) Bauxite
(c) Gibbsite (d) Pyrolusite
- Which of the following is not halide ore ?
(a) Cryolite
(b) Fluorosparg
(c) Horn silver
(d) Limonite
- Sulphides ore is common in between :
(a) Ag, Cu, Pb (b) Ag, Cu, Sn
(c) Ag, Mg, Pb (d) Al, Cu, Pb
- $\text{BO} \xrightarrow{400^\circ\text{C}} \frac{1}{2} \text{O}_2 \uparrow + \text{B}$
 $\text{ACl}_2 + \text{BCl}_2 \longrightarrow \text{ACl}_4 + \text{B}$
If ACl_2 is the reducing agent. B is the metal, then ore of A & B would be :
(a) Siderite, Cinnabar
(b) Cassiterite, Cinnabar
(c) Hornsilver, Tinstone
(d) None
- During leaching of alumina from Bauxite the leaching reagent used would be :
(a) Concentrated solution of $\text{Ba}(\text{OH})_2$
(b) Concentrated solution of NaOH or 40% KOH
(c) Concentrated HCl
(d) Concentrated H_2SO_4
- The leaching of alumina from Bauxite is done by :
(a) 473-523 K & 35-36 bar pressure
(b) 300 K & 5-6 bar pressure
(c) 1000 K & 1 bar pressure
(d) None
- In the metallurgy of silver and gold, the metal is leached with the reagent :
(a) NaOH (b) NaNH_2
(c) NaCN (d) Zn metal
- Which of the following ore is best concentrated by Froth floatation method ?
(a) Galena
(b) Cassiterite
(c) Magnetite
(d) Malachite green
- Gravity separation is primarily used for separation of impurities from which ore
(a) Cassiterite (SnO_2)
(b) Haematite (Fe_2O_3)
(c) Chromite (FeCr_2O_4)
(d) All
- The metal that can't be obtained by electrolysis of an aqueous solution of its salt is :
(a) Cu (b) Cr
(c) Ag (d) Ca

3.4 METALLURGY

14. Which metallurgical process describe best the extraction of Mn from Mn_3O_4 by the use of aluminium.
 - (a) Hydrometallurgy
 - (b) Electrometallurgy
 - (c) Pyrometallurgy
 - (d) Amalgamation
15. The main impurity present in red bauxite is :
 - (a) CuO
 - (b) ZnO
 - (c) Fe_2O_3
 - (d) SiO_2
16. The main impurities present in white bauxite is :
 - (a) Fe_2O_3
 - (b) TiO_2
 - (c) SiO_2
 - (d) FeO
17. Heating pyrites in air for oxidation of sulphur is called :
 - (a) Smelting
 - (b) Roasting
 - (c) Calcination
 - (d) Slagging
18. The process of converting hydrated ore into anhydrous ore is called :
 - (a) Roasting
 - (b) Smelting
 - (c) Dressing
 - (d) Calcination
19. Copper matte consists
 - (a) Mixture of $Cu_2S + FeS$
 - (b) Mixture of $Cu_2S + FeO$
 - (c) Metallic copper with 98-99 % purity
 - (d) Metallic copper with 1-2 % purity
20. Zincite is example of
 - (a) Oxide ore
 - (b) Halide ore
 - (c) sulphide ore
 - (d) Carbonate ore
21. Which method is based on the principle that impurities are more soluble in molten metal than in solid state.
 - (a) Vapour phase refining
 - (b) Zone refining
 - (c) Poling
 - (d) Levigation
22. Auto reduction is applicable for which of metals ?
 - (a) Al, Fe
 - (b) Cu, Sn
 - (c) Cu, Hg
 - (d) Ag, Au
23. During self reduction of Cu_2S in the Ist step when O_2 is added then.
 - (a) Oxidation state of Cu^{+1} converted into Cu^{2+}
 - (b) Oxidation state of S^{2-} converted into SO_2
 - (c) Oxidation state of Cu^{+1} & S^{2-} both undergoes change
 - (d) Oxidation state of Cu^{+1} & S^{2-} both remains unchanged
24. Slag formed by reaction between impurities and flux. If flux is basic then impurities must be
 - (a) Acidic in nature eg. SiO_2
 - (b) Acidic in nature eg. CaO
 - (c) Basic in nature eg. MgO
 - (d) Basic in nature eg. P_4O_{10}
25. Carbon reduction method is not applicable for extraction of
 - (a) Tin from SnO_2
 - (b) Iron from Fe_2O_3
 - (c) Al from Al_2O_3
 - (d) Pb from PbO
26. In which reaction aluminium acts as reducing agent:
 - (a) Baeyer's process
 - (b) Serpeck's process
 - (c) Thermite process
 - (d) Hall's process
27. Thermite reduction is not used for commercial extraction of respective metal from which of the following oxide :
 - (a) Mn_3O_4
 - (b) TiO_2
 - (c) Fe_2O_3
 - (d) Cr_2O_3
28. Spiegel (or spiegeleisen), used in the manufacture of steel by the Bessemer process, is an alloy of :
 - (a) Iron, chromium and carbon
 - (b) Iron, nickel and carbon
 - (c) Iron, tungsten and carbon
 - (d) Iron, Manganese and carbon
29. Which of the following statement is incorrect regarding thermite welding.
 - (a) Al is used as reducing agent
 - (b) Fe is obtained in the form of molten metal
 - (c) This process is used for rail rod joining
 - (d) $Fe_2O_3 + Al$ taken are in the ratio (5 : 4)



The name of this method is :

- (a) Van Arkel method (b) Zone refining
(c) Poling (d) Cupellation

31. In Mond's process during extraction of nickel, the reagent added which gives complex is X & the complex formed is Y, then X & Y are respectively:

- (a) CN^- , $\text{Ni}(\text{CN})_4^{2-}$ (b) CO , $\text{Ni}(\text{CO})_4$
(c) CN^- , $[\text{Ni}(\text{CN})_6]^{4-}$ (d) CO , $\text{Ni}(\text{CO})_5$

32. Froth floatation is based on the following principle:

- (a) Differences in wetting ability of ore particles in oil and impurity particles in water.
(b) Difference in specific gravity of ore particles and impurity particles in water.
(c) Difference in magnetic property of ore particles and impurity particles in water.
(d) All

33. Zinc blende and chalcopyrites are concentrated by :

- (a) Leaching
(b) Levigation
(c) Froth floatation
(d) Magnetic separation

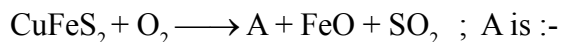
34. The oxidation state of metal in cuprite and fool's gold are respectively :

- (a) +1, +1 (b) +2, +2
(c) +1, +2 (d) +2, +1

35. Oxidation state of Fe in Haematite and Magnetite respectively :

- (a) II ; III (b) II ; (II + III)
(c) III ; III (d) III ; (II + III)

36. Observe the following partial roasting process,



- (a) Cu_2O (b) CuO
(c) Cu_2S (d) CuS

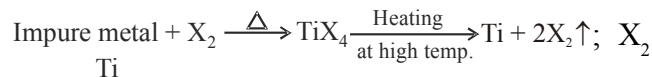
37. Steps which are involved in the extraction of Cu from its low grade ore.

- (a) Leaching by NaCN
(b) Metal displacement by Ag
(c) Leaching with acid
(d) Metal displacement by iron

38. Out of following oxides, unstable oxide at temperature 200°C is

- (a) CaO (b) Al_2O_3
(c) HgO (d) Fe_2O_3

39. Ti is prepared by the following method .



is :

- (a) Cl_2 (b) Fe
(c) Br_2 (d) I_2

40. Which alkaline earth metals are present in Dolomite.

- (a) Na, Mg (b) K, Ca
(c) Be, Ba (d) Ca, Mg

41. Zinc blende $\xrightarrow{\text{Roasting}}$ Metal oxide + Gas.
(Suffocating odour)

What is the oxidation state of S in zinc blende and gas ?

- (a) -2, +4 (b) -2, +6
(c) 0, +6 (d) 0, +4

42. Ellingham diagram normally describes

- (a) Variation of ΔG° vs temperature for formation of oxide $2\text{xM}(\text{s}) + \text{O}_2(\text{g}) \longrightarrow 2\text{M}_\text{x}\text{O}(\text{s})$
(b) Variation of ΔH° vs temperature for formation of oxide $2\text{xM}(\text{s}) + \text{O}_2(\text{g}) \longrightarrow 2\text{M}_\text{x}\text{O}(\text{s})$
(c) Variation of ΔS° vs temperature for formation of oxide $2\text{xM}(\text{s}) + \text{O}_2(\text{g}) \longrightarrow 2\text{M}_\text{x}\text{O}(\text{s})$
(d) None

43. In Ellingham diagram there is sudden increase in slope at particular point for every metals which depicts

- (a) Phase change e.g., solid $\longrightarrow$ Liq. ; Liq $\longrightarrow$ gas
(b) The metals oxide decomposition temperature
(c) Metal oxide boiling point
(d) Metal oxide fusion point

44. Which reaction is most spontaneous ?

- (a) $\text{Zn}^{2+} + 2\text{e}^- \longrightarrow \text{Zn}$
(b) $\text{Ag}^+ + \text{e}^- \longrightarrow \text{Ag}$
(c) $\text{Mg} \longrightarrow \text{Mg}^{2+} + 2\text{e}^-$
(d) $\text{Li}^{+1} + \text{e}^- \longrightarrow \text{Li}$

3.6 METALLURGY

45. The oxide of which metal having least thermal stability.
 (a) Hg (b) Fe
 (c) Al (d) Mg
46. For which reaction $\Delta S = 0$.
 (a) $C(s) + O_2(g) \rightarrow CO_2(g)$
 (b) $Ca(s) + \frac{1}{2} O_2(g) \rightarrow CaO(s)$
 (c) $2Al(s) + \frac{3}{2} O_2(g) \rightarrow Al_2O_3$
 (d) $C(s) + \frac{1}{2} O_2 \rightarrow CO(g)$
47. From Ellingham diagram in Blast furnace at 673 K, best reducing agent is :
 (a) Coke
 (b) CO
 (c) Mixture of (C+CO)
 (d) Can't be predicted
48. Sphalerite ore is used for extraction of which metal?
 (a) Fe (b) Cu
 (c) Zn (d) Al
49. Distillation is a method of refining of metals. For which metal it is applicable.
 (a) Low boiling point metals e.g., Pb, Sn
 (b) High boiling point metal e.g., Zn, Hg
 (c) Low boiling point metal e.g., Zn, Hg
 (d) High boiling point metal e.g., Pb, Sn
50. In the metallurgy of aluminium the electrolyte is :
 (a) Fused matrix of Al_2O_3 with Na_3AlF_6 and CaF_2
 (b) Al_2O_3 only
 (c) Na_3AlF_6
 (d) CaF_2
51. Impurity has more tendency to be oxidised by atmospheric O_2 than ore particles. This concept is applicable for :
 (a) Poling
 (b) Cupellation process
 (c) Zone refining
 (d) Electrolytic refining
52. In cupellation process :
 (a) Lead impurities are separated from Ag
 (b) Silver impurities are separated from Pb
 (c) Cu impurities are separated from Sn
 (d) Sn impurities are separated from Cu
53. Boiling point order :
 (a) $Cd > Zn > Fe$ (b) $Fe > Zn > Cd$
 (c) $Zn > Fe > Cd$ (d) $Zn > Cd > Fe$
54. In liquation metal can be separated from impurities. Here metal and impurities are of
 (a) Higher melting point and lower melting point respectively
 (b) Higher boiling point and lower boiling point respectively
 (c) Lower melting point and higher melting point respectively
 (d) Lower boiling point and higher boiling point respectively
55. In zone refining impurities get concentrated
 (a) In solidified zone (b) Molten zone
 (c) Freezing zone (d) None
56. The chemical process in the production of steel from haematite ore involve :-
 (a) reduction
 (b) oxidation
 (c) reduction followed by oxidation
 (d) oxidation followed by reduction
57. The temperature T_1 and T_2 for Mond process

$$Ni + 4CO \xrightarrow{T_1} Ni(CO)_4 ;$$

$$Ni(CO)_4 \xrightarrow{T_2} Ni + 4CO$$

 (a) 330 - 350 K ; 600 - 630 K
 (b) 330 - 350 K ; 450 - 470 K
 (c) 450 - 470 K ; 600 - 630 K
 (d) 600 - 630 K ; 330 - 350 K
58. In which of the following process roasting followed by self reduction takes place :
 (a) $Cu_2S \rightarrow Cu$ (b) $PbS \rightarrow PbO$
 (c) $CaCO_3 \rightarrow Ca$ (d) None

59. The pyrometallurgy requires
 (a) Electric current (b) Low temperature
 (c) High temperature (d) Atmospheric O_2
60. Which is correctly matched ?
 (a) Anthracite – Ore of iron
 (b) Fluorospir – Mineral of aluminium
 (c) Cementite – A carbide of iron in steel
 (d) Pyrolusite – An ore of Mg
61. At $1100^\circ C$ which of the following reaction is thermodynamically most feasible.
 (a) $TiO_2 + C \longrightarrow Ti + CO_2$
 (b) $CO_2 + C \longrightarrow 2CO$
 (c) $Al_2O_3 + 3C \longrightarrow 2Al + 3CO$
 (d) $Cr_2O_3 + 3C \longrightarrow 2Cr + 3CO$
62. In Parke's process, 2 layers are developed. Correct options is :-
 (a) Upper layer consists of Pb
 (b) Upper layer consists of alloy of Zn-Ag
 (c) Upper layer consists of Zn
 (d) Upper layer consists of Ag-Pb-alloy
63. Lower layer consists of
 (a) Molten zinc
 (b) Molten silver
 (c) Molten lead
 (d) Molten mixture of (Zn + Ag)
64. In Baeyer's process Fe_2O_3 does not react with NaOH, but SiO_2 does along with Al_2O_3 because.
 (a) Because Fe_2O_3 is basic oxide whereas SiO_2 is acidic oxides and Al_2O_3 is amphoteric oxide
 (b) Al_2O_3 and SiO_2 are basic oxides whereas Fe_2O_3 is acidic oxide
 (c) Fe_2O_3 is basic oxides whereas SiO_2 and Al_2O_3 are example of acidic oxide
 (d) Al_2O_3 , Fe_2O_3 are example of basic oxide whereas SiO_2 is example of acidic oxide
65. In froth floatation method sodium ethyl xanthate is used as collector. Actually it
 (a) Helps sulfide ore to collect in water medium
 (b) Helps sulfide ore to collect on air bubble
 (c) Dissolves impurities in water
 (d) Dissolves impurities in air
66. White bauxite is purified by.
 (a) Baeyer's process (b) Hoop's process
 (c) Serpeck's process (d) Hall's process
67. Red bauxite is purified by.
 (a) Baeyer's process (b) Hall's process
 (c) Both A and B (d) Hoop's process
68. Aluminium is purified by.
 (a) Baeyer's process (b) Hall's process
 (c) Serpeck's process (d) Hoop's process
69. The correct position of metal (from top to bottom) is electrochemical series.
 (a) $Li > K > Mg > Al$
 (b) $Na > Mg > K > Li$
 (c) $Li > K > Ca > Na$
 (d) $Mn > Mg > Al > K$
70. In electrolytic reduction of Al_2O_3 , the ratio of Al_2O_3 , Cryolite and fluorspar will be
 (a) 1 : 1 : 1 (b) 3 : 1 : 2
 (c) 1 : 5 : 1 (d) 1 : 3 : 1
71. In Golds mith process Fe_2O_3 and Al mixture are in the ratio
 (a) 1 : 2 (b) 5 : 1
 (c) 3 : 2 (d) 3 : 1
72. Ignition mixture which is used for thermite process consists of
 (a) $KCl + BaO$ (b) $KClO_3 + BaO$
 (c) $KClO_3 + BaO_2$ (d) $KCl + BaO_2$
73. At approximate what temperature, zinc and carbon have equal affinity for oxygen
 (a) $500^\circ C$ (b) $1000^\circ C$
 (c) $1200^\circ C$ (d) $1500^\circ C$
74. In Castner's process for extraction of sodium the electrolyte is
 (A) fused NaOH (b) $NaOH + CaCl_2$
 (c) fused NaCl (d) fused $NaCl + CaCl_2$
75. In Down's process for extraction of sodium the electrolyte is
 (a) fused NaOH
 (b) $NaOH + CaCl_2$
 (c) fused NaCl
 (d) fused $NaCl + CaCl_2$

3.8 METALLURGY

76. Which of the following options are incorrect ?
 (A) Si can't reduce MgO at all.
 (b) Al can't reduce ZrO_2 but Mg can reduce ZrO_2
 (c) Al can reduce Cr_2O_3 as well as TiO_2
 (d) H_2 does not act as strong reducing agent at high temperature
77. Sodium chloride is added during electrolysis of fused anhydrous magnesium chloride
 (a) To increase melting point of MgCl_2 from 500°C to 800°C
 (b) To reduced melting point of MgCl_2 from 700°C to 400°C
 (c) To increase melting point of MgCl_2 from 600°C to 900°C
 (d) To reduced melting point of MgCl_2 from 900°C to 600°C
78. The point at which line of $\text{M} + \text{O}_2 \rightarrow \text{MO}$ cuts the zero free energy line depicts
 (a) decomposition temperature of oxide
 (b) decomposition temperature of metal
 (c) fusion point of metal
 (d) fusion point of slag
79. The elements which are at lowest line in Ellingham diagram has
 (a) very low tendency to combine with O_2
 (b) very high tendency to combine with O_2
 (c) Oxide which are very unstable
 (d) None
80. In Polling, basic impurities associated with metal
 (a) gets reduced to form volatile elemental form
 (b) gets oxidised to form slag which is immiscible with molten metal
 (c) gets oxidised to form volatile oxide which is removed as scum
 (d) gets reduced to form solidified element which is crystallised out of molten metal
81. In Distillation process, zinc is separated from Cd at temperature T_1 and zinc is separated from Pb and Fe at temperature T_2 . T_1 and T_2 are respectively
 (a) 767°C and 920°C (b) 667°C and 940°C
 (c) 657°C and 800°C (d) 957°C and 750°C
82. Which of the following option are correct?
 (a) $\text{Ca} + \frac{1}{2} \text{O}_2 \rightarrow \text{CaO}$ line sharply increases at temperature at 1440°C
 (b) $\text{Mg} + \frac{1}{2} \text{O}_2 \rightarrow \text{MgO}$ line sharply increases at temperature at 1440°C
 (c) $\text{Hg} + \frac{1}{2} \text{O}_2 \rightarrow \text{HgO}$ line sharply increases at temperature at 1440°C
 (D) All
83. Carnallite on electrolysis gives
 (a) Ca and Cl_2 (b) Al and Cl_2
 (c) Na & CO_2 (d) Mg and Cl_2
84. In which of the following extraction, no reducing agent is used
 (a) Iron from Haematite
 (b) Tin from Cassiterite
 (c) Mercury from Cinnabar
 (d) Zinc from zincblende
85. Railway wagon axles are made by heating iron rods embedded in charcoal powder. This process is known as -
 (a) Sherardising (b) Annealing
 (c) Tempering (d) Case hardening
86. Which of the following options is incorrect -
 (a) Reduction potential of alkali metal cation is much lower than that of H^+
 (b) Magnesite and quick lime are used as acidic flux
 (c) TiI_4 is volatile stable compound
 (d) In Park's process, silver impurity is removed from lead metal
87. At 1100°C which of the following reactions is thermodynamically most favoured
 (a) $\text{TiO}_2 + \text{C} \rightarrow \text{Ti} + \text{CO}_2$
 (b) $\text{Al}_2\text{O}_3 + 3\text{C} \rightarrow 2\text{Al} + 3\text{CO}$
 (c) $\text{Cr}_2\text{O}_3 + 3\text{C} \rightarrow 2\text{Cr} + 3\text{CO}$
 (d) $\text{CO}_2 + \text{C} \rightarrow 2\text{CO}$

88. In the extraction of copper from its sulphide ore, the metal is formed by reduction of Cu_2O with
(a) FeS (b) CO
(c) Cu_2S (d) SO_2
89. Refractory materials are generally used in furnace because -
(a) they are chemically inert
(b) they can withstand high temperature
(c) they do not contain impurities
(d) they decrease melting point of ore
90. Addition of high proportions of manganese makes still useful in making rails of rail roads, because manganese -
(a) gives hardness to steel
(b) help the formation of oxide of iron
(c) can remove oxygen and sulphur
(d) can show highest oxidation state of +7
91. Silica is added to roasted copper ores during extraction copper to remove -
(a) Cuprous sulphide (b) Cuprous oxide
(c) Ferrous oxide (d) Ferrous sulphide
92. Blister copper is refined by stirring molten impure metal with green logs to wood because such wood liberates hydrocarbon gases. This process is called and the metal contains impurities of -
(a) Cupellation, CuO (b) Polling, CuO
(c) Cupellation, Cu_2O (d) Polling, Cu_2O
93. The process which does not use a catalyst is -
(a) Contact process (b) Thermite process
(c) Ostwald's process (d) Haber's process
94. Spelter may be purified by -
(a) Polling
(b) Heating with iodine
(c) Electrolysis process
(d) Fractional distillation
95. ΔG° vs T plot in Ellingham diagram slopes downward for which of the following reactions
(a) $2\text{Mg} + \text{O}_2 \rightarrow 2\text{MgO}$
(b) $2\text{CO} + \text{O}_2 \rightarrow 2\text{CO}_2$
(c) $2\text{Zn} + \text{O}_2 \rightarrow 2\text{ZnO}$
(d) $2\text{C} + \text{O}_2 \rightarrow 2\text{CO}$
96. Iron obtained from blast furnace is
(a) wrought iron (b) cast iron
(c) pig iron (d) steel
97. High temperature ($>1000^\circ\text{C}$) electrolytic reduction is necessary for extraction of
(a) Mg (b) Cu
(c) Pb (d) Al
98. From the Ellingham graph on carbon, which of the following statement is false?
(a) CO_2 is more stable than CO at less than 983 K
(b) CO reduces Fe_2O_3 to Fe at less than 983 K
(c) CO is less stable than CO_2 at more than 983 K
(d) CO reduces Fe_2O_3 to Fe in the reduction zone of blast furnace
99. Coke powder is spreaded over the molten electrolyte in electrolytic reduction of Al_2O_3 due to :
(a) prevent the heat radiation from the surface
(b) prevent the corrosion of graphite anode
(c) prevent oxidation of molten aluminium by air
(d) both (a) & (b)
100. Which of the following reaction cannot occur in blast furnace during extraction of iron.
(a) $\text{CaO} + \text{SiO}_2 \rightarrow \text{CaSiO}_3$
(b) $\text{Fe}_2\text{O}_3 + 3\text{CO} \rightarrow 2\text{Fe} + 3\text{CO}_2$
(c) $\text{FeO} \rightarrow \text{Fe} + \frac{1}{2} \text{O}_2$
(d) $\text{P}_2\text{O}_5 + 5\text{C} \rightarrow \frac{1}{2} \text{P}_4 + 5\text{CO}$

EXERCISE # II**□ One or More Than One Correct Answer :**

1. In which ore iron (Fe) can have (+2) oxidation state :
(a) Haematite (Fe_2O_3)
(b) Iron pyrite (Fool's gold) FeS_2
(c) Siderite (FeCO_3)
(d) Copper pyrite (CuFeS_2)

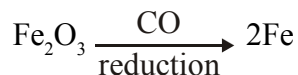
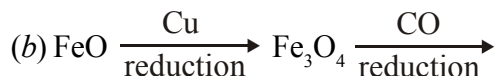
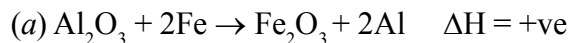
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2. Which of the following ores having potassium ?
 (a) Carnalite (b) Indian salt Petre
 (c) Gypsum (d) Cryolite
3. Which of the following ores have copper ?
 (a) Chalcopyrite (b) Azurite
 (c) Malachite green (d) Limonite
4. Which of the following changes is observed when ore is heated for calcination ?
 (a) $\text{Fe}_2\text{O}_3 \cdot 3\text{H}_2\text{O} \rightarrow \text{Fe}_2\text{O}_3 + 3\text{H}_2\text{O}$
 (b) $\text{Al}_2\text{O}_3 \cdot 2\text{H}_2\text{O} \rightarrow \text{Al}_2\text{O}_3 + 2\text{H}_2\text{O}$
 (c) $\text{PbCO}_3 \rightarrow \text{PbO} + \text{CO}_2$
 (d) $\text{ZnCO}_3 \rightarrow \text{ZnO} + \text{CO}_2$
5. Magnetic substance which are present in cassiterite for Sn extraction.
 (a) FeWO_4 (b) SnO_2
 (c) MnWO_4 (d) Fe_3O_4
6. Which of the two metal form cyano complex during its extraction?
 (a) Zn (b) Ag
 (c) Au (d) Cu
7. Electrolytic reduction is applicable for
 (a) Al from Al_2O_3 (Molten)
 (b) Ag from Ag_2S
 (c) Na from NaOH
 (d) Cu from CuFeS_2
8. In zone refining method some elements are formed in ultrapure form. The elements can be :
 (a) Si (b) Ge
 (c) Ga (d) Fe
9. Which of the following options are correct ?
 (a) Carbon cannot be used to produce magnesium by chemical reduction of MgO because magnesium reacts with carbon to form carbide
 (b) In froth floatation method activator can enhance the frothing tendency of sulphide ore
 (c) CO is preferential reducing agent in iron extraction from haematite ore
 (d) Na, K, Li are situated at the top of electrochemical series.
10. For aluminium extraction when electrolysis of molten mass Al_2O_3 is carried out then which statement are correct?
 (a) Lining of carbon acts as cathode and graphite anode is used
 (b) During electrolysis, the oxygen liberated reacts with carbon of anode to give CO and CO_2
 (c) This process is called Hall-Heroult process
 (d) CaF_2 reduces the melting point of the mixture and increases electrical conductivity
11. Mercury can't form amalgam with :
 (a) Fe (b) Ag
 (c) Au (d) Pt
12. The method used in metallurgy to refine the impure metal is :
 (a) Chromatographic separation
 (b) Vapour phase refining
 (c) Zone refining
 (d) Distillation
13. Tin can be refined by
 (a) Poling
 (b) Liquation
 (c) Distillation
 (d) Zone refining
14. $4\text{Au} + 8\text{CN}^- + 2\text{H}_2\text{O} + \text{O}_2 (\text{g}) \rightarrow 4[\text{Au}(\text{CN})_2]^- (\text{aq}) + 4\text{OH}^- (\text{aq})$
 $2[\text{Au}(\text{CN})_2]^- (\text{aq}) + \text{Zn} \rightarrow 2\text{Au}(\text{s}) + [\text{Zn}(\text{CN})_4]^{2-} (\text{aq})$
 Which of the following options are correct ?
 (a) Both the reactions are the example of nonredox reaction
 (b) Both are examples of redox reactions
 (c) In Ist reaction O_2 acts as oxidising agent
 (d) In IInd reaction Zn acts as reducing agent
15. CO can't be used as reducing agent
 (a) To prepare/extract Zn metal from ZnO
 (b) To Prepare Al metal from Al_2O_3
 (c) To prepare Fe metal from Fe_2O_3
 (d) To prepare Pb metal from PbS

16. In Ellingham diagram
- The elements in the higher line can reduce oxide of elements in the lower line
 - The elements in the lower line can reduce oxide of elements in higher line
 - Reduction of metals oxide is easier if the metals formed is in liquid state at the temperature of reduction
 - Reduction of metals oxide can take place of room temperature.
17. In Ellingham diagram Fe_2O_3 can be reduced by which of following elements.
- Ca
 - Mg
 - Hg
 - Al
18. Pick the correct statement(s) for aluminothermite process :-
- Reaction is endothermic
 - It is used for welding of rail tracks
 - To start the reaction a Mg ribbon is ignited
 - Al is used as reductant
19. Which of the following statement is / are correct :-
- Combination Lime stone and wolframite is non magnetic in nature
 - No external reducing agent is required for extraction of Hg from HgO .
 - Poling method is mainly used when impure metal oxide having oxide as impurity
 - For Cu extraction in bessemer converter process, oxidation and reduction reaction takes place.
20. Which of the following statements are not correct?
- Copper is extracted by self reduction method
 - Cast iron is the purest form of iron
 - The composition of malachite ore is $\text{Ca}(\text{OH})_2 \cdot \text{CaCO}_3$
 - Cupellation process is used for refining of Ag and Au
21. Select the statement which is correct.
- Ellingham diagram M $\rightarrow$ MO curves have negative slope.
 - MgO can be reduced by carbon at very high temperature ($\sim 800^\circ\text{C}$)
 - In blast furnace carbon does not reduce iron oxides
 - Iron scraps will be advisable and advantageous w.r.t Zn scraps for reducing leached copper ore
22. Find the number of ores which can be concentrated by magnetic separation method.
- Galena
 - Copper pyrites
 - Haematite
 - Siderite
23. Which of the following are the steps / reactions involved in extraction of iron :-
- $3\text{FeO}_3 + \text{CO} \rightarrow 2\text{FeO}_3 + \text{CO}_2$
 - $\text{Fe}_3\text{O}_4 + \text{CO} \rightarrow 3\text{FeO}_3 + \text{CO}_2$
 - $\text{FeO} + \text{CO} \rightarrow \text{Fe} + \text{CO}_2$
 - Calcium silicate is produced as slag
24. Which of the following statements are correct :-
- Anhydrous MgCl_2 can not be prepared by heating $\text{MgCl}_2 \cdot 6\text{H}_2\text{O}$
 - Anhydrous CaCl_2 is used in drying gases and organic compounds but not NH_3 or ethyl alcohol due to formation of $\text{CaCl}_2 \cdot 8\text{NH}_3$ and $\text{CaCl}_2 \cdot 4\text{C}_2\text{H}_5\text{OH}$
 - Leblanc process is used for preparation of both Na_2CO_3 and H_2CO_3 but not solvay process.
 - KOH is preferably used compared to NaOH for absorption of CO_2 because KHCO_3 formed is soluble whereas NaHCO_3 is sparingly soluble and therefore choke the tubes of apparatus used.
25. Which of the following reactions are example of thermite reduction ?
- A thermite reaction may start, if a ship (having aluminium parts) is hit by a missile
 - $\text{Fe}_2\text{O}_3 + 2\text{Al} \rightarrow 2\text{Fe} + \text{Al}_2\text{O}_3$
 - $\text{Al}_2\text{O}_3 + 3\text{Mg} \xrightarrow{\Delta} 3\text{MgO} + 2\text{Al}$
 - $\text{Cr}_2\text{O}_3 + 2\text{Al} \xrightarrow{\Delta} 2\text{Cr} + \text{Al}_2\text{O}_3$
26. Blister copper is :-
- Impure copper
 - Obtained by self reduction process in Bessemer converter
 - pure 100% Cu
 - 50% Copper

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27. Select correct option :-



(c) Out of two metal oxide FeO and CaO ; CaO more easily combine with SiO_2

(d) Impurities present in molten cast iron are possibly oxidised to Fe_2O_3

28. Which of the following statements are incorrect ?

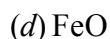
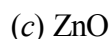
(a) During metallurgy of zinc, spelter is obtained by reduction process of ZnO

(b) Pig iron is obtained from blast furnace.

(c) Liquation is only valid for purification of Bi not for Sn and Pb.

(d) Pitch blende is ore of U

29. Which of the following is an example of amphoteric oxide ?



30. The compound with different metals as

(a) Dolomite

(b) Carnalite

(c) Copper pyrite

(d) Azurite

31. Choose incorrect statement.

(a) Mg is liquid at less than 1500°C and gas at above 1500°C

(b) Si can't reduce MgO at all

(c) The equation of line in Ellingham diagram is given by $\Delta G = \Delta H - T\Delta S$

(d) The preferable temperature for Mg to be used as a reducing agent for SiO_2 is 2200°C

32. When copper is purified by electrolytic refining of the blister copper, the correct statement :

(a) Pure Cu is deposited at cathode

(b) Impurities settle as anode mud.

(c) Acidified CuSO_4 is used as electrolyte

(d) Pure Cu is deposited at anode

33. Which of the following metals can be extracted by electrolytic reuction

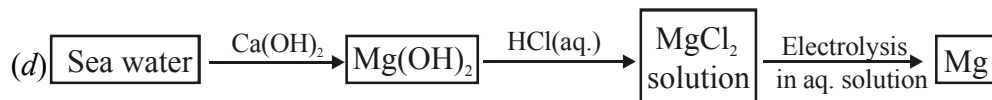
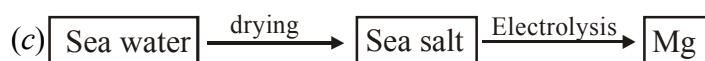
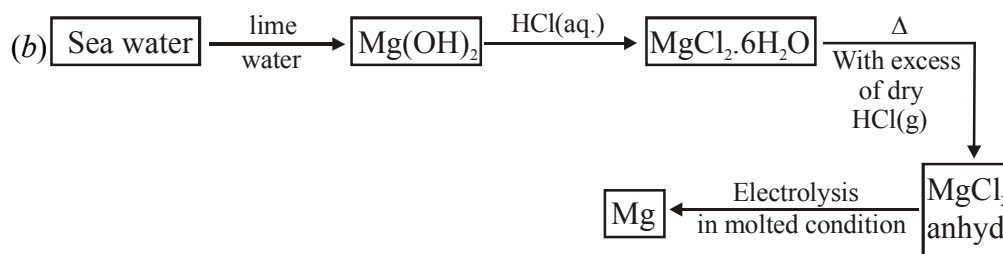
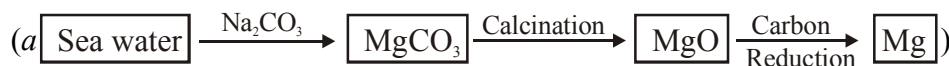
(a) Na

(b) Mg

(c) Ca

(d) Ag

34. Which of the following diagram are not related to the extraction of Mg from sea water ?



35. "In electrolysis $\text{M}^{n+}(\text{aq.})$, metal ions are discharged at negative electrodes (cathodes) and deposited there" above method of extraction is not possible for :

(a) Sodium

(b) Aluminium

(c) Silver

(d) Carbon

36. The reduction processes employed to prepare lead metal can be
(a) Carbon reduction
(b) self reduction
(c) Aluminothermic reduction
(d) Electrolytic reduction
37. The reduction process employed to prepare iron metal can be
(a) Carbon reduction
(b) self reduction
(c) Aluminothermic reduction
(d) Electrolytic reduction
38. The metallurgy/reduction process involved for extraction for Cu can be
(a) Electrometallurgy
(b) Pyrometallurgy with carbon reduction
(c) Pyrometallurgy with Auto reduction
(d) Hydrometallurgy
39. In Ellingham Diagram, Fe_2O_3 can be reduced by which of the following elements
(a) Ca (b) Mg
(c) Ag (d) Al
40. The position of element from (top to bottom) in electrochemical series
(a) $\text{Zn} > \text{Cr} > \text{Fe} > \text{Sn} > \text{Pb}$
(b) $\text{H} > \text{Cu} > \text{Hg}$
(c) $\text{Hg} > \text{Ag} > \text{Au}$
(d) $\text{Al} > \text{Mn} > \text{Zn} > \text{Cr}$
41. In Cupellation
(a) Ag impurity is removed from Pb metal
(b) Pb impurity is removed from Ag metal
(c) Zn impurity is removed from Ag metal
(d) Zn impurity is removed from Pb metal
42. Which of the following options are correct
(a) In electrolytic refining of Au, AuCl_3 solution is used as electrolyte
(b) In electrolytic refining of Ni, $\text{Ni}(\text{NO}_3)_2$ solution is used as electrolyte
(c) In electrolytic refining of Ag, $\text{AgNO}_3 + \text{HNO}_3$ solution is used as electrolyte
(d) In electrolytic refining of Zn, $\text{ZnSO}_4 + \text{H}_2\text{SO}_4$ solution is used as electrolyte
43. Which of the following statements are correct :-
(a) 24 carat Gold consists of 100 % Au
(b) 14 carat Gold consists of 54 % Au
(c) Stainless steel has Cr : 12-14%
(d) Amalgamation is carried out in Iron container
44. Which of the following statements are correct :-
(a) The metallic impurities having higher oxidation potential than that of metal to be refined are separated in the form of anode mud.
(b) Cathode is made of large slab of impure metal in electrolytic refining
(c) Anode is made of thin strip of pure metal in electrolytic refining
(d) Spinel is a mineral of Al with molecular formula $\text{MgO} \cdot \text{Al}_2\text{O}_3$
44. Which of the following can not be obtained by electrolytic reduction of their compounds in aqueous solution.
(a) Barium (b) Cadmium
(c) Potassium (d) Nickel
45. Which of the following are true or false electrolytic extraction of aluminium -
(a) anode material contains graphite
(b) cathode made of carbon
(c) cathode reacts with O_2 released to form CO_2
(d) anode reacts with O_2 released to form CO_2
46. Which of the following employ downward movement of ore -
(a) Gravity separation (b) Froth floatation
(c) Bessemer converter (d) Blast furnace
47. Metal which can be extracted by smelting process
(a) Pb (b) Fe
(c) Zn (d) Mg
48. Which of the following reactions are actually employed for commercial extraction of metal -
(a) $\text{Fe}_2\text{O}_3 + 2\text{Al} \rightarrow \text{Al}_2\text{O}_3 + 2\text{Fe}$
(b) $2\text{Na}[\text{Au}(\text{CN})_2] + \text{Zn} \rightarrow \text{Na}_2[\text{Zn}(\text{CN})_4] + 2\text{Au}$
(c) $\text{Cr}_2\text{O}_3 + 2\text{Al} \rightarrow \text{Al}_2\text{O}_3 + 2\text{Cr}$
(d) $\text{Cu}_2\text{S} + \text{Pb} \rightarrow \text{Cu} + \text{PbS}$

3.14 METALLURGY

49. In the manufacturing of metallic sodium by fused salt electrolysis method (Down's process), small amount of CaCl_2 is added. The purpose is -
 (a) To improve electrical conductance of electrolyte
 (b) To stabilise metallic sodium
 (c) To increase the temperature of electrolysis
 (d) To decrease the melting point of NaCl
50. Collectors are the substances which help in attachment of an ore particle to air bubble in froth. The example of collectors are
 (a) Sodium pyrophosphate
 (b) Sodium nitropruside
 (c) Sodium ethyl xanthate
 (d) Sodium lauryl sulphate
51. B_4C is used -
 (a) to extract boron
 (b) for making bullet-proof clothing
 (c) as an abrassive for polishing
 (d) as lubricants
52. Elingham diagram can be drawn for which of the following
 (a) Sulphide (b) Oxide
 (c) Halides (d) Nitrates
53. Choose correct options -
 (a) Reduction of metal oxide will be easier if the metal formed is in the liquid state at the temperature of reduction
 (b) Reduction of metal oxide will be easier if the metal formed is in the solid state at the temperature of reduction
 (c) Aluminium can reduce MgO above 1350°C
 (d) Aluminium can not reduce MgO above 1500°C
54. Choose correct options -
 (a) Reduction of metal oxide take place at room temperature only
 (b) A particular metal oxide can be reduced by many reducing agent, but temperature for reduction is same for all reducing agent
 (c) On increasing temperature metal oxide stability decreases
 (d) A metal oxide can be reduced by many reducing agent but temperature for reduction is different for all reductants
55. If at a temperature there is sudden rise in slope in Elingham diagram, that temperature can indicate-
 (a) Melting point of metal oxide
 (b) Boiling point of metal oxide
 (c) Melting point of metal
 (d) Boiling point of metal
56. Which alloy has Cu and Zn as constituent metals -
 (a) Brass (b) Dutch metal
 (c) Bronze (d) Bell metal
57. Correct statements are -
 (a) Black zack is ZnS
 (b) Ruby copper is Cu_2O
 (c) Parke's process is based on distribution principle
 (d) Stellite is an alloy having highest % of cobalt metal among all metals present in it
58. The crude metal zinc which is called spelter contains impurities like -
 (a) Ag (b) Fe
 (c) Cd (d) Pb
59. Which alloy has Cu and Sn as constituent metals -
 (a) Brass (b) Dutch metal
 (c) Bronzene (d) Bell metal
60. Pitch blende is not the source of
 (a) Ba (b) U
 (c) Ge (d) Tl

EXERCISE # III

□ Linked Comprehension Type :

Paragraph for Q.1 to Q.3

According to Lux concept of acids and bases, an acid is an acceptor of oxide ions and a base is a donor of oxide ion. This concept is very much applicable in the fields of ceramic and metallurgy.

1. CaO acts as flux for impurities like SiO_2 to form slag CaSiO_3 . Here :-
 (a) CaO acts as base because it is donor of oxide ion.
 (b) CaO acts as base because it acceptor of oxide ion.
 (c) SiO_2 acts as acid because it is donor of oxide ion.
 (d) SiO_2 acts as acid because it is acceptor of oxide ion.

2. $6\text{SiO}_2 + 2\text{Ca}_3(\text{PO}_4)_2 \rightarrow 6\text{CaSiO}_3 + \text{P}_4\text{O}_{10}$. Here
- SiO_2 and P_4O_{10} acts as acid
 - SiO_2 and P_4O_{10} acts as base
 - $\text{Ca}_3(\text{PO}_4)_2$ and CaSiO_3 acts as acid
 - $\text{Ca}_3(\text{PO}_4)_2$ and CaSiO_3 acts as extremely strong base
3. $\text{CaO} + \text{SiO}_2 \rightarrow \text{CaSiO}_3$
 $\text{FeO} + \text{SiO}_2 \rightarrow \text{FeSiO}_3$
 $\text{MgO} + \text{SiO}_2 \rightarrow \text{MgSiO}_3$
 $\text{CaSiO}_3, \text{FeSiO}_3, \text{MgSiO}_3$ all are :-
- Flux
 - Gangue particle
 - Slag
 - Lewis acid

Paragraph for Q.4 to Q.6

When mixture of metal in the liquid state solidify three types of alloys are formed :

- Alloys in which the metals crystallise separately.
 - Alloys in which the metals form solid solutions.
 - Alloys in which the metals form the intermetallic compound.
4. Alnico ; Duralumin and Magnelium are alloy of :-
- Zn metal
 - Cu metal
 - Al
 - As
5. Copper based alloy is :-
- Brass
 - Bronze
 - Bell metal
 - All
6. Which of the following alloy having copper and zinc metal along with other metal :-
- German silver
 - Gun metal
 - (a) and (b)
 - None

Paragraph for Q.7 to Q.9

Iron is extracted from its oxide and carbonate ores by reduction with coke in blast furnace. The process is carried out in two steps.

- Preliminary roasting or calcination : The ore is calcine with a little coal in heaps regulating the

temperature and air supply in order to burn the organic metal and drive off most of the moisture, carbon dioxide, sulphur, arsenic. Ferrous oxide is converted into ferric oxide. The roasted mass contains ferric oxide Fe_2O_3 .

- Smelting and reduction in blast furnace : The roasted ore is mixed with coke and limestone (flux) and charged into the blast furnace when the ferric oxide is reduced to metallic iron.

7. Top most zone in blast furnace, at 1070 K / 800°C which reaction can not take place :-

- $3\text{Fe}_2\text{O}_3 + \text{CO} \rightarrow 2\text{Fe}_3\text{O}_4 + \text{CO}_2$
- $\text{Fe}_3\text{O}_4 + \text{CO} \rightarrow 3\text{FeO} + \text{CO}_2$
- $\text{CaCO}_3 \rightarrow \text{CaO} + \text{CO}_2$
- $\text{FeO} + \text{C} \rightarrow \text{Fe} + \text{CO}$

8. The slag formed in blast furnace is :-

- SiO_2
- CaSiO_3
- CaCO_3
- P_4O_{10}

9. At temperature zone 1300°C to 1900°C which reaction can not take place :-

- $\text{FeO} + \text{C} \rightarrow \text{Fe} + \text{CO}$
- $\text{C} + \text{O}_2 \rightarrow \text{CO}_2$
- $\text{C} + \text{CO}_2 \rightarrow 2\text{CO}$
- $3\text{Fe}_2\text{O}_3 + \text{CO} \rightarrow 2\text{Fe}_3\text{O}_4 + \text{CO}_2$

Paragraph for Q.10 to Q.13

Steel is produced from pig iron by first removing mechanically all the impurities in the pig iron. e.g. e, Si, Mn, S, P by oxidation and then adding the current amount of carbon. The operation is carried out in Bessemer converter or in a Siemens-Martin open hearth furnace.

10. The two types of bessemer process are :

- The oxidation bessemer process and reduction bessemer process
- The redox bessemer process and non-redox bessemer process
- The acidic bessemer process and basic bessemer process
- The neutral bessemer process and amphoteric bessemer process

3.16 METALLURGY

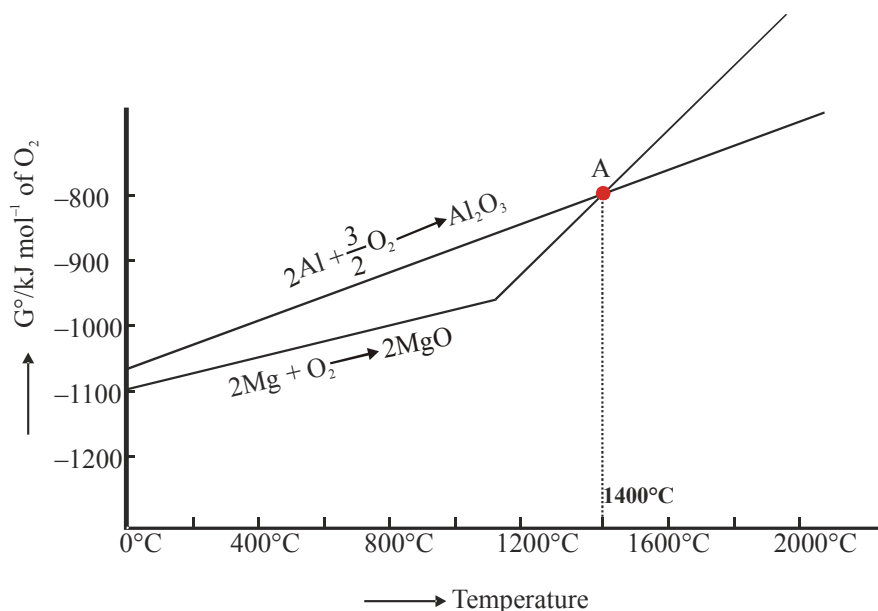
11. In acid bessemer process the slag obtained can be:-
 (a) MnSiO_3
 (b) FeSiO_3
 (c) $\text{Ca}_3(\text{PO}_4)_2$
 (d) both (a) and (b)
12. In basic bessemer process the slag obtained is called Thomas slag which is used as fertiliser. The Thomas slag is actually.
 (a) MnSiO_3
 (b) FeSiO_3
 (c) $\text{Ca}_3(\text{PO}_4)_2$
 (d) CaSiO_3
13. Which of the following statements are correct :
 (a) Silica bricks are used as lining in acidic bessemer converter :-
 (b) Calcined dolomite or magnesite is used as lining in basic bessemer converter
 (c) In acidic bessemer , the interaction between the slag which is iron phosphate and the carbon containing iron continuously regenerates iron phosphide according to the reaction.

$$2\text{P}_2\text{O}_5 + 6\text{Fe} + 3\text{O}_2 \rightarrow 2\text{Fe}_3(\text{PO}_4)_2$$

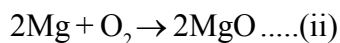
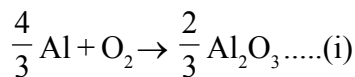
$$\text{Fe}_3(\text{PO}_4)_2 + 2\text{Fe}_3\text{C} + 3\text{Fe} \rightleftharpoons 2\text{Fe}_3\text{P} + 6\text{FeO} + 2\text{CO}$$

 (d) All

Paragraph for Q.14 to Q.16



14. The two equations are



A is the point of intersection of the Al_2O_3 and MgO curves. Which of the following option is correct.

- (a) Below temperature 1400°C , $\frac{2}{3}\text{Al}_2\text{O}_3 + 2\text{Mg} \rightarrow 2\text{MgO} + \frac{4}{3}\text{Al}$ reaction is thermodynamically feasible.
 (b) Above temperature 1400°C , $\frac{4}{3}\text{Al} + 2\text{MgO} \rightarrow \frac{2}{3}\text{Al}_2\text{O}_3 + 2\text{Mg}$ reaction is thermodynamically feasible.
 (c) At temperature 1400°C , $\Delta G^\circ = 0$ for the reaction $\frac{2}{3}\text{Al}_2\text{O}_3 + 2\text{Mg} \rightarrow 2\text{MgO} + \frac{4}{3}\text{Al}$
 (d) All

15. Al metal can't be used for reduction of MgO because :-
 (a) This reaction is nonspontaneous
 (b) This reaction is economical
 (c) This reaction is noneconomical
 (d) None

16. Al_2O_3 , MgO are examples of
 (a) amphoteric, amphoteric oxide
 (b) amphoteric, basic oxide
 (c) basic, amphoteric oxide
 (d) basic, basic oxide

Paragraph for Q.17 & Q.18

Zinc ore (X) $\xrightarrow{\text{calcination}}$ Solid(S) + Gas - 1

Zinc ore (Y) $\xrightarrow{\text{roasting}}$ Solid(S) + Gas - 2

Solid (S) $\xrightarrow[\text{At } 1673 \text{ K}]{\Delta \text{ with C}}$ Metal (M) + Gas - 3

Gas - 3 $\xrightarrow[\text{with } \text{O}_2]{\text{combustion}}$ Gas - 1

17. Which substance produces colourless, odourless gas with dil. H_2SO_4 -
 (I) Ore(X) (II) Ore (Y)
 (III) Metal(M) (IV) Solid(S)
 (a) I, III (b) I, II, III
 (c) II, III (d) I, IV

Matrix Match Type :

21. Match the column-I with column-II :-

Column - I : (Property)

- (a) Explosive
 (b) Self-reduction
 (c) Ferrimagnetic material
 (d) Verdigris

22. **Column - I : (Process)**

- (a) Mac-Arther forest cyanide process
 (b) Mond process
 (c) Van-Arkel method
 (d) Blast Furnace

23. **Column - I : (Process involved)**

- (a) Bessemerisation
 (b) Electrolytic refining using aqueous salt as an electrolyte
 (c) Extracted by the formation of soluble complex
 (d) Reaction which involves Mg ribbon and $(\text{BaO}_2 + \text{KClO}_3)$ mixture to generate ignition temperature

18. Which step is not involved in the extraction of pure metal from ore-(Y)
 (a) Froath flotation
 (b) Smelting
 (c) Self reduction
 (d) Distillation

Paragraph for Q.19 & Q.20

(I) $\text{FeCr}_2\text{O}_4 + \text{NaOH} + \text{air} \rightarrow \text{a} + \text{Fe}_2\text{O}_3$

(II) $(\text{a}) + (\text{b}) \rightarrow \text{Na}_2\text{Cr}_2\text{O}_7$

(III) $\text{Na}_2\text{Cr}_2\text{O}_7 + \text{X} \xrightarrow{\Delta} \text{Cr}_2\text{O}_3$

(IV) $\text{Cr}_2\text{O}_3 + \text{Y} \xrightarrow{\Delta} \text{Cr}$

19. Compound A and B are -

- (a) Na_2CrO_4 , H_2SO_4
 (b) $\text{Na}_2\text{Cr}_2\text{O}_7$, HCl
 (c) Na_2CrO_5 , H_2SO_4
 (d) None

20. X and Y are

- (a) C and Al
 (b) Al and C
 (c) C in both
 (d) Al in both

Column - II : (Element/ Compound)

- (P) Cu
 (Q) Fe_3O_4
 (R) $\text{Cu}(\text{CH}_3\text{COO})_2 \cdot \text{Cu}(\text{OH})_2$
 (S) $\text{Pb}(\text{NO}_3)_2$

Column - II : (Compounds formed during reaction)

- (P) $[\text{Ag}(\text{CN})_2]^{-1}$
 (Q) TiI_4
 (R) $\text{Ni}(\text{CO})_4$
 (S) CaSiO_3 (slag)

Column - II : (Pure metal extracted)

- (P) Cr
 (Q) Au
 (R) Ag
 (S) Fe
 (T) Cu

3.18 METALLURGY

24. Column - I

(Elements)

(a) Mg

(b) Al

(c) Fe

(d) Hg

Column - II

(Characteristic feature of elements and their oxide)

(P) Highest value of $|\Delta G|$ range for oxide formation for these elements

(Q) Lowest value of $|\Delta G|$ range for oxide formation for these elements

(R) Can reduce at least one of metal oxides of given elements

(S) Strongest reducing agent among these

(T) Its metal oxides has lowest thermal stability among oxide of these metal

25. Column - I

Chemical changing metallurgy

(a) $M_2O_3 + C + \text{Electrical energy} \longrightarrow M + CO / CO_2$

(b) $(M + CO + Fe) + CO \xrightarrow[50^\circ C]{\Delta} Fe(s)$

+ CO(s) + [vapours of carbonyl complex of M]

(c) $(Fe_2O_3, MgO)_g + OH^-(aq.) \longrightarrow Fe_2O_3 + [M(OH)_4]^{-1} \text{ sol.}$

(d) Fe.M.Ag.Au + Electrical oxidation

in aq. solution $Fe^{2+}(aq.) + Ag(s)$
 $\xrightarrow{\text{of metal (M)}} + M^{2+}(aq.) + Au(s)$

Column - II

Purpose of metallurgy of specified metal

(P) Purification of copper

(Q) Benefication of bauxite

(R) Extraction of metal from alumina

(S) Separation of nickel from impurities

26. Match the column :-

Column - I

(Process involved)

(a) Poling

(b) Cupellation

(c) Zone refining

(d) Parke's process

(T) Pb

Column - II

(Pure metal extracted)

(P) Sn

(Q) Ag

(R) Cu

(S) Ge

27. Match the column :-

Column - I

(Elements)

(a) Ni

(b) Pb

(c) Cu

(d) Hg

Column - II

(Correct information regarding extraction process)

(P) A solution of lead silicofluoride $PbSiF_6$ containing 8-10% of H_2SiF_6 is used as electrolyte for

(Q) Complex formation reaction is involved in extraction of

(R) Self reduction

(S) Carbon monoxide is used for extraction of metal

(T) d-block element

28. Column - I

(Metal extracted from its ore)

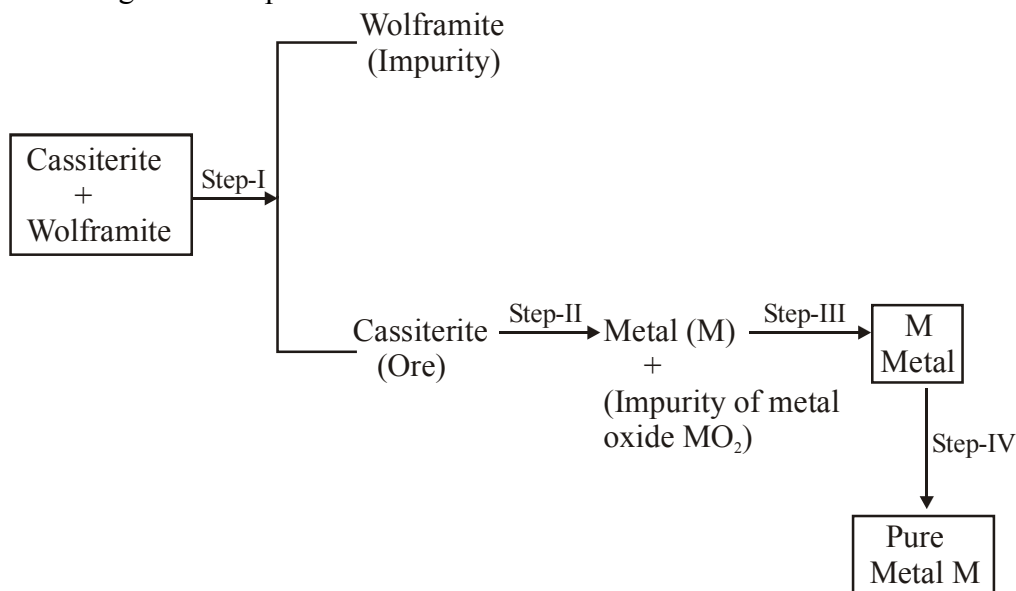
- (a) Zinc from ZnCO_3
- (b) Lead from PbS
- (c) Cu from CuFeS_2
- (d) Tin from cassiterite

Column - II

(Process involved for extraction of metal)

- (P) Calcination
- (Q) Removal of iron
- (R) Froath floatation process
- (S) Poling
- (T) Gravity separation

29. Consider following extraction process and match Column-I with Column-II :



Column - I

- (a) Magnetic separation
- (b) Liquation
- (c) Poling
- (d) Carbon reduction

Column - II

- (P) Step-I
- (Q) Step-II
- (R) Step-III
- (S) Step-IV

30 Match the column :-

Column - I

- (a) Downs cell
- (b) Dow sea water process
- (c) Hall - Heroult
- (d) Moissan process

Column - II

- (P) Fused MgCl_2
- (Q) Fused $\text{Al}_2\text{O}_3 + \text{Na}_3\text{AlF}_6$
- (R) Fused KHF_2
- (S) Fused 40 % NaCl and 60% CaCl_2

31 Match the column :-

Column - I

Ore

- (a) ZnS
- (b) FeCr_2O_4
- (c) SnO_2
- (d) $\text{Cu}_2\text{S} \cdot \text{Fe}_2\text{S}_3$

Column - II

Process for benefactions

- (P) Magnetic separation
- (Q) Froath floatation
- (R) Gravity separation
- (S) Roasting

3.20 METALLURGY

32 Match the column :-

Column - I

Conversion

- (a) $\text{Cu}_2\text{S} \rightarrow \text{Cu}_2\text{O}$
- (b) $\text{CaCO}_3 \rightarrow \text{CaO}$
- (c) $\text{ZnS} \rightarrow \text{Zn}$
- (d) $\text{PbS} \rightarrow \text{Pb}$

Column - II

Process involved

- (P) Roasting
- (Q) Calcination
- (R) Carbon reduction
- (S) Self reduction

33. Match the column :-

Column - I

Definition

- (a) Naturally occurring chemical substance in the earth's crust obtainable by mining
- (b) Substance from which metal can be extracted economically and conventionally
- (c) Undesired earthy substance contaminated with desired earthy substance ore
- (d) Scientific and technological process used for isolation of metal

Column - II

Term

- (P) Metallurgy
- (Q) Gangue
- (R) Ore
- (S) Mineral

34. Chemical changing metallurgy :-

Column - I

Phenomenon

- (a) Calcium phosphate
- (b) Hard steel is heated to bright redness and then allowed to cool very slowly
- (c) Many impurities C/Si/P/S
- (d) SO_2 evolution take place and pores are developed

Column - II

Term

- (P) Blister copper
- (Q) Pig iron
- (R) Thomas slag
- (S) Annealing

35. Match the column:-

Column - I

Different types of iron

- (a) Cast iron
- (b) Wrought iron
- (c) Steel
- (d) Pig iron

Column - II

Features

- (P) Purest form of iron with C % $\rightarrow$ 0.1 - 0.15%
- (Q) Molten iron from blast furnace having more than 4% carbon
- (R) Carbon % varies from 0.15 - 1.5%
- (S) Carbon % varies from 2 - 4%

EXERCISE # IV

❑ Integer Type :

1. If a mixture of ores consist of:

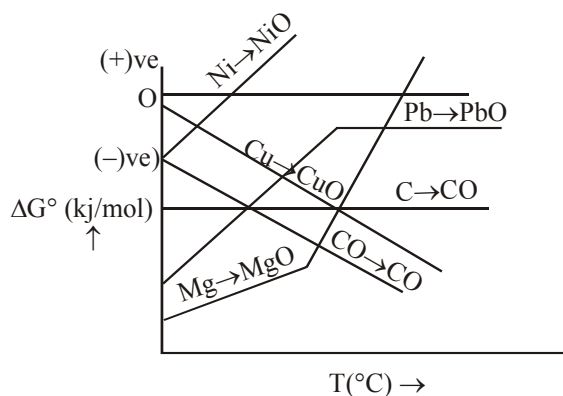
- (i) Zinc blend (ii) Argentite (iii) Haematite (iv) Chromite (v) Dolomite
- (vi) FeWO_4 (vii) Cassiterite

Then find out number of ores which are collected in container when allowed to move towards magnetic roller.

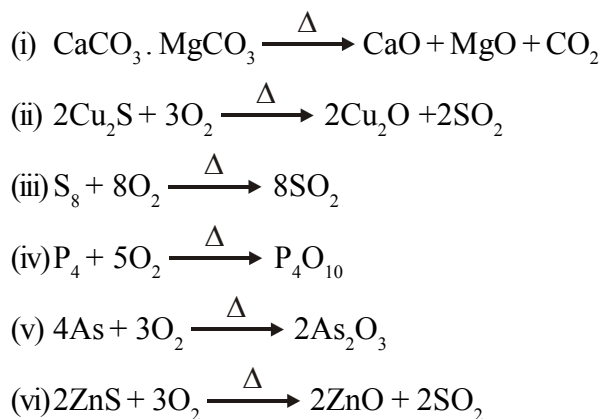
2. Calcination of how many ore produce CO_2 .
 (a) Limonite (b) Siderite
 (c) Malachite (d) Cerrusite
 (e) Azurite (f) Dolomite
3. Of the following metal that can not be obtained by electrolysis of the aqueous solution of their salts using inert electrode :
 Ag ; Cu ; Cr ; Mg ; Al ; Na ; Ba ; Ca
4. Find the number of compounds which are added to acidic impurities to form slag.
 (a) SiO_2 ; MgO ; CaO ; CaCO_3 ; MgCO_3 ; P_2O_5 ; B_2O_3
5. How many metals are commercially obtained by auto reduction.
 Fe ; Na ; Mg ; Mg ; Ca ; Pb ; Sn ; Cu ; Hg ; Au
6. How many metals are commercially obtained by carbon reduction of their oxide .
 Fe ; Al ; Mg ; Ca ; Sn ; Pb ; Zn
7. How many metals can be commercially extracted by hydrometallurgy i.e., displacement reaction takes place in aqueous solution.
 Au ; Ag ; In ; Zn ; Pb ; Al ; Cu
8. How many statements are correct?
 (i) Graphite rod is used as anode but not diamond because mobile electrons are present in graphite layer which helps in electrical conductivity.
 (ii) If the impurity in a metal has greater affinity for oxygen then the purification of metal may be carried out by cupellation.
 (iii) Fe and Pt can't form alloy with Hg.
 (iv) During roasting of sulfide ore at high temperature oxide is formed and at low temperature sulphate is formed.
 (v) During Sn/Pb/iron metallurgy CaSiO_3 is formed as slag.
 (vi) Wolframite is ferro magnetic, therefore attracted by magnet.
 (vii) Magnetic siderite, chromite, Wolframite are of Fe(II) but limonite, Haematite are of Fe(III).
 (viii) Bauxite is considered as oxide as well as hydroxide of Al.
 (ix) In cyanide extraction process of silver from argentite ore the oxidising and reducing agent used as O_2 and Zn.
9. How many of the following terms are associated for purification of iron.
 (a) Puddling process
 (b) Thomas slag
 (c) Cupellation
 (d) Bessemerisation
 (e) Amalgamation
10. Find the number of ores in which any metal is present in +1 oxidation state.
 Copper pyrite, Chalcocite, Chile Salt petre, Cryolite, Cuprite, Sylvine
11. Find the number substance which contain metal is in its elemental form mainly.
 Blister copper, Pig iron, Wrought iron, Steel, Copper matte, Alumina, Spelter
12. Find the number of ores which can be roasted or calcined followed by reduction with carbon to give respective metal.
 $\text{Fe}_2\text{O}_3 \cdot 3\text{H}_2\text{O}$; ZnS ; ZnCO_3 ; PbS (low grade)
13. Find the number of impurities which are deposited as anode mud in the electrorefining of copper.
 Antimony ; Selenium ; Tellurium ; Silver ; Gold ; Platinum
14. Find the number of ores which can be concentrated by magnetic separation.
 Galena ; Copper pyrites ; Haematite ; Siderite
15. Find the number of ores in which roasting process is used in metallurgy of corresponding metal.
 Galena ; Haematite ; Calamine ; Zinc Blende ; Cinnabar, Horn Silver ; Lime stone.
16. Find the number of steps in the following used during the extraction of spelter from zinc blend.
 Poling ; Electrorefining ; Roasting ; Froath floatation ; Smelting ; Magnetic separation
17. The number of following pairs is correctly matched:
 (i) Van Arkel method - Zirconium
 (ii) Amalgamation - Lead
 (iii) Distillation method - Zinc
 (iv) Poling process - Copper
 (v) Mond process - Titanium
 (vi) Van Arkel method - Zirconium

3.22 METALLURGY

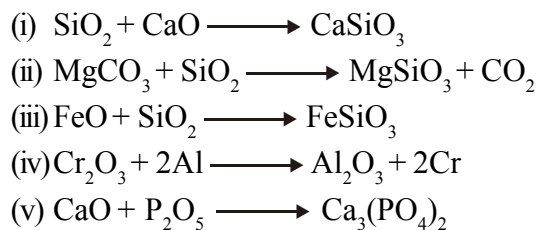
18. Find the number of curves which are wrongly presented in the Ellingham diagram.



19. Find the number of following reactions which are involved in roasting process :



20. How many reactions can show slag formaton process from the given reaction ?



21. How many elements can be purified by liquation method ?

Sn ; Pb ; Bi ; Zn ; Cd ; Fe

22. How many elements can be purified by Zone refining method ?

Ge ; Si ; Ga ; Se ; Te ; Tl ;

22. How many elements can be purified by distillation?

Zn ; Pb ; Cd ; Hg ; Mg ; Na ; Ka

23. Find out the number of layers developed in Hoop's process which is used for electrolytic refining of aluminium.

24. Find the number of unstable oxide out of the following oxide which undergo decomposition below 200°C.

CaO ; Al₂O₃ ; MgO ; Fe₂O₃

25. Find the number of oxide out the of the following oxide which undergo decomposition below 400°C.

CaO ; Al₂O₃ ; MgO ; Fe₂O₃ ; Ag₂O and HgO

26. What is the coordination number of aluminium in mineral Cryolite

27. How many water of crystallisation present in the ore Carnalite.

28. In iron metallurgy for preparation pig iron from Haematite ore the number of reducing agent involed.

29. Find out number of ores which are examples of oxy salt ore but not oxide ore ?

Epsom salt ; Glauber salt ; Gypsum ; Anglesite ; Rutile ; Cuprite ; Plaster of paris

30. Find the number of ores in which atleast one metal is in +3 oxidation state :-

Bauxite ; Diaspore ; Gibbsite ; Limonite ; Cryolite ; Haematite ; Magnetite ; Chromite

31. Find the number of ores in which atleast one metal is in +2 oxidation state :-

Baryte ; Fluorspar ; Chromite ; Magnetite ; Iron pyrite ; Copper pyrite ; Carnalite

32. % of silver in German silver -

33. Number of metals present in Alnico alloy -

34. Number of metals present in Devardas alloy -

35. For how many elements amalgamation process is applicable for purification of metals -

Ag ; Au ; Cu ; Zn

36. For the reaction $\text{Al}_2\text{O}_3 \cdot 2\text{H}_2\text{O} + x\text{NaOH} + \text{H}_2\text{O} \rightarrow x\text{Na}[\text{Al}(\text{OH})_4]$

The value of x

37. Find the total number of acidic refracories out of the following compounds.

CaO ; Quartz ; Dolomite ; Bone ash ; Magnesite.

38. Among the following number of compound that would require electrolysis process to get their respective metal is .

Al₂O₃, MgCl₂, Fe₂O₃

39. Find out the number of layers developed in Park's process which is used for purification of lead metal..
40. How many statements are correct
- Steel are Brittle
 - Cast iron is soft
 - Pewter is an Alloy of Pb and Sn

EXERCISE # V(A) (JEE-MAIN)

1. Aluminium is extracted by the electrolysis of :-

[AIEEE - 2002]

- Bauxite
- Alumina
- Alumina mixed with molten cryolite
- Molten cryolite

2. Pyrolusite is a / an :-

[AIEEE - 2002]

- Oxide ore
- Sulphide ore
- Carbide ore
- Not an ore

3. Which one of the following ores is best concentrated by froth - flotation method :-

[AIEEE - 2004]

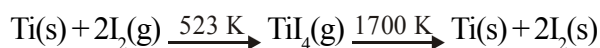
- Galena
- Cassiterite
- Magnetite
- Malachite

4. Which one of the following factors is of no significance for roasting sulphide ores to the oxides and not subjecting the sulphide ores to carbon reduction directly?

[AIEEE - 2008]

- Metal sulphides are thermodynamically more stable than CS_2
- CO_2 is thermodynamically more stable than CS_2
- Metal sulphides are less stable than the CS_2
- CO_2 is more volatile than CS_2

5. Which method of purification is represented by the following equation :-

[AIEEE - 2012]

- Van Arkel
- Zone refining
- Cupellation
- Poling

6. The substance used as froth stabilisers in froth - floatation process is :-

[Jee-Mains -2012(online)]

- Copper sulphat
- Aniline
- Sodium cyanide
- Potassium ethyl xanthate

7. Which of the oxide groups among the following cannot be reduced by carbon :-

[Jee-Mains - 2012(online)]

- Fe_2O_3 , ZnO
- PbO , Fe_2O_3
- Cu_2O , SnO_2
- CaO , K_2O

8. In Goldschmidt aluminothermic process which of the following reducing agents is used :-

[Jee-Mains - 2013(online)]

- Calcium
- Coke
- Sodium
- Al-power

9. Calcination is the process in which :-

- Ore is heated strongly below its melting point in the presence of excess of air and is used for the conversion of carbonates and hydrated oxide ores to their respective oxides.
- Ore is heated strongly below its melting point in the absence or limited supply of air and is used for conversion of sulphide ores to their respective oxides.
- Ore is heated strongly below its melting point either in the limited or absence of air and is used to convert carbonates and hydrated oxide ores to their respective oxides.
- Ore is heated strongly above its melting point in the limited supply of air to convert sulphide ores to their respective oxides.

10. The metal that cannot be obtained by electrolysis of an aqueous solution of its salts is :-

[Jee-Mains - 2014]

- Cu
- Cr
- Ag
- Ca

11. The form of iron obtained from blast furnace is :-

[Jee-Mains - 2014(On line)]

- Steel
- Wrought iron
- Cast Iron
- Pig iron

3.24 METALLURGY

12. In the context of the Hall-Heroult process for the extraction of Al, which of the following statements is false? [Jee-Mains - 2015]

(a) Al^{3+} is reduced at the cathode to form Al
 (b) Na_3AlF_6 serves as the electrolyte
 (c) CO and CO_2 are produced in the process
 (d) Al_2O_3 is mixed with CaF_2 which lowers the melting point of the mixture and brings conductivity.

EXERCISE # V(B) (ADVANCED)

- In extractive metallurgy of zinc partial fusion of ZnO with coke is called and reduction of the ore to the molten metal is called [Jee'88]
- Carnallite does not contain :-
 (a) K
 (b) Ca
 (c) Mg
 (d) Cl
- During initial treatment, preferential wetting of ore by oil and gangue by water takes place in :-
 (a) Levigation (gravity separation)
 (b) Froth floatation
 (c) Leaching
 (d) Bessemerisation
- Which of the following is true for calcination of metal ore ?
 (a) It makes the ore more porous
 (b) The ore is heated to a temperature when fusion just begins
 (c) Hydrated salts lose their water of crystallisation
 (d) Sulphur in sulphides is oxidised to SO_2
 (e) Heating with carbon leads to better calcination
- In the commercial electrochemical process for aluminium extraction, the electrolyte used as :- [Jee-1999]
 (a) $\text{Al}(\text{OH})_3$ in NaOH solution
 (b) An aqueous solution of $\text{Al}_2(\text{SO}_4)_3$
 (c) A molten mixture of Al_2O_3 and Na_3AlF_6
 (d) A molten mixture of $\text{AlO}(\text{OH})$ and $\text{Al}(\text{OH})_3$
- The chemical process in the production of steel from haematite ore involve :- [2000 Qualifying]
 (a) reduction
 (b) oxidation
 (c) reduction followed by oxidation
 (d) oxidation followed by reduction
- Electrolytic reduction of alumina to aluminium by Hall - Heroult process is carried out :- [2000 Qualifying]
 (a) in the presence of NaCl
 (b) in the presence of fluorite
 (c) in the presence of cryolite which forms a melt with lower melting temperature
 (d) in the presence of cryolite which forms a melt with higher melting temperature
- The chemical composition of "slag" formed during the smelting process in the extraction of copper is :- [2001 Qualifying]
 (a) $\text{Cu}_2\text{O} + \text{FeS}$ (b) FeSiO_3
 (c) CuFeS_2 (d) $\text{Cu}_2\text{S} + \text{FeO}$
- Which of the following process is used in extractive metallurgy of magnesium ? [2002 Qualifying]
 (a) Fused salt electrolysis
 (b) Self reduction
 (c) Aqueous solution electrolysis
 (d) Thermite reduction
- In the process of extraction of gold, [2003 Qualifying]

$$\text{Roasted gold ore} + \text{CN}^- + \text{H}_2\text{O} \xrightarrow{\text{O}_2} [\text{X}] + \text{OH}^-[\text{X}] + \text{Zn} \longrightarrow [\text{Y}] + \text{Au}$$
 Identify the complexes [X] and [Y] :-
 (a) $\text{X} = [\text{Au}(\text{CN})_2]^-$, $\text{Y} = [\text{Zn}(\text{CN})_4]^{2-}$
 (b) $\text{X} = [\text{Au}(\text{CN})_4]^{3-}$, $\text{Y} = [\text{Zn}(\text{CN})_4]^{2-}$
 (c) $\text{X} = [\text{Au}(\text{CN})_2]^-$, $\text{Y} = [\text{Zn}(\text{CN})_6]^{4-}$
 (d) $\text{X} = [\text{Au}(\text{CN})_4]^-$, $\text{Y} = [\text{Zn}(\text{CN})_4]^{2-}$
- The methods chiefly used for the extraction of lead and tin from their ores are respectively :- [JEE - 2004]
 (a) self reduction and carbon reduction
 (b) self reduction and electrolytic reduction
 (c) carbon reduction and self reduction
 (d) cyanide process and carbon reduction

12. Which ore contains both iron and copper ?

[JEE - 2004]

- (a) Cuprite (b) Chalcocite
(c) Chalcopyrite (d) Malachite

13. Extraction for zinc from zinc blende is achieved by:-

[JEE - 2007]

- (a) electrolytic reduction
(b) roasting followed by reduction with carbon
(c) roasting followed by reduction with another metal
(d) roasting followed by self-reduction

14. Native silver metal forms a water soluble complex with a dilute aqueous solution of NaCN in the presence of :-

[JEE-2008]

- (a) Nitrogen (b) Oxygen
(c) Carbon dioxide (d) Argon

Passage for Q. 15 to 16

Copper is the most noble of the first row transition metals and occurs in small deposits in several countries. Ores of copper include chalcantite ($\text{CuSO}_4 \cdot 5\text{H}_2\text{O}$), atacamite ($\text{Cu}_2\text{Cl}(\text{OH})_3$), cuprite (Cu_2O), copper glance (Cu_2S) and malachite ($\text{Cu}_2(\text{OH})_2\text{CO}_3$). However, 80% of the world copper production comes from the ore chalcopyrite (CuFeS_2). The extraction of copper from chalcopyrite involves partial roasting, removal of iron and self-reduction. [JEE-2010]

15. Partial roasting of chalcopyrite produces :-

- (a) Cu_2S and FeO
(b) Cu_2O and FeO
(c) CuS and Fe_2O_3
(d) Cu_2O and Fe_2O_3

16. Iron is removed from chalcopyrite as :-

- (a) FeO
(b) FeS
(c) Fe_2O_3
(d) FeSiO_3

17. In self-reduction, the reducing species is :-

- (a) S (b) O^{2-}
(c) S^{2-} (d) SO_2

18. Match the extraction processes listed in column I with metals listed in column II :- [JEE-2006]

Column-I

Column-II

- (a) Self reduction (P) Lead
(b) Carbon reduction (Q) Silver
(c) Complex formation and displacement by metal (R) Copper
(d) Decomposition of iodide (S) Boron

19. Match the conversions in column I with the type(s) of reaction(s) given in column II. Indicate your answer by marking the appropriate bubbles of the 4×4 matrix given in the ORS :- [JEE-2008]

Column-I

Column-II

- (a) $\text{PbS} \rightarrow \text{PbO}$ (P) Roasting
(b) $\text{CaCO}_3 \rightarrow \text{CaO}$ (Q) Calcination
(c) $\text{ZnS} \rightarrow \text{Zn}$ (R) Carbon reduction
(d) $\text{Cu}_2\text{S} \rightarrow \text{Cu}$ (S) Self reduction

20. In extractive metallurgy of zinc partial fusion of ZnO with coke is called and reduction of the ore the molten metal is called (smelting, calcining, roasting, sintering) :- [JEE-1988]

21. Extraction of metal from the ore cassiterite involves:- [JEE2011]

- (a) carbon reduction of an oxide ore
(b) self-reduction of a sulphide ore
(c) removal of copper impurity
(d) removal of iron impurity

22. Oxidation states of the metal in the minerals haematite and magnetite, respectively, are:-

[JEE-2011]

- (a) II, III in haematite and III in magnetite
(b) II, III in haematite and II in magnetite
(c) II in haematite and II, III in magnetite
(d) III in haematite and II, III in magnetite

23. In the cyanide extraction process of silver from argentite ore, the oxidizing and reducing agents used are :- [JEE-2012]

- (a) O_2 and CO respectively
(b) O_2 and Zn dust respectively
(c) HNO_3 and Zn dust respectively
(d) HNO_3 and CO respectively

3.26 METALLURGY

24. Sulphide ores are common for the metals :-

[JEE-2013]

- (a) Ag, Cu and Pb (b) Ag, Cu and Sn
(c) Ag, Mg and Pb (d) Al, Cu and Pb

25. The carbon - based reduction method is NOT used for the extraction of :-

[JEE-2013]

- (a) Tin from SnO_2
(b) Iron from Fe_2O_3
(c) Aluminium from Al_2O_3
(d) Magnesium from $\text{MgCO}_3 \cdot \text{CaCO}_3$

26. Upon heating with Cu_2S , the reagent(s) that give copper metal is / are :-

[JEE-2014]

- (a) CuFeS_2 (b) CuO
(c) Cu_2O (d) CuSO_4

27. Copper is purified by electrolytic refining of blister copper. The correct statement (s) about this process is (are) :-

[JEE-Adv.2015]

- (a) Impure Cu strip is used as cathode
(b) Acidified aqueous Cu_2SO_4 is used as electrolyte
(c) Pure Cu deposit at cathode
(d) Impurities settle as anode - mud

28. Match the anionic species given in column I that are present in the ore(s) given in column II :-

[JEE-Adv. 2015]

Column-I	Column-II
(a) Carbonate	(P) Siderite
(b) Sulphide	(Q) Malachite
(c) Hydroxide	(R) Bauxite
(d) Oxide	(S) Calamine
	(T) Argentite

■ ■

QUALITATIVE SALT ANALYSIS

PROBLEMS BASED ON GIVEN TOPICS

Tests for acid radicals

- Action of dilute acids
- Tests for CO_3^{2-} / HCO_3^- and SO_3^{2-} / HSO_3^- radicals
 - Distinction between carbonate and bicarbonate
 - Distinction of carbonate and bicarbonate when both are present together
 - Distinction between sulphide and bisulphite
 - Some other tests for SO_3^{2-} ions
- Tests for sulphide (S^{2-}) radical
- Tests for thiosulphate ($\text{S}_2\text{O}_3^{2-}$) radical
- Tests for nitrite (NO_2^-) radical
- Tests for Acetate, Formate and Oxalate radicals
 - Specific test for acetate (cacodyl oxide test)
 - Specific test for formate (mercury (II) formate test)
 - Specific test for oxalate
- Tests for Halide (Cl^- , Br^- , I^-) radicals
 - Specific test for Cl^- (chromyl chloride test)
 - Specific test for Br^- and I^- (layer test)
 - Other test for Br^-
 - Other test for I^-

- Tests for nitrate (NO_3^-) radical
- Tests for sulphate (SO_4^{2-}) radical
- Tests for borate (BO_3^{3-}) radical
- Tests for phosphate (PO_4^{3-}) radical
- Tests for chromate (CrO_4^{2-}) and dichromate (CrO_7^{2-}) radicals
- Tests for permagnate (MnO_4^-) and magnate (MnO_4^{2-}) radicals

Tests for basic radical

- Dry tests for basic radical
 - Heating effect on the dry sample
 - Flame test
 - Borax bead test
 - Sodium carbonate bead test
- Wet tests for basic radicals
 - Classification of cations (group analysis)
- Some general test for cations
 - Test for group V cations
 - Test for group VI and zero group cations
- **Specific tests for some cations**

Heating effect

Qualitative Salt Analysis

EXERCISE # I

❑ Only one correct answer :

1. Which of the following compound(s) slowly disappears with prolonged passage of CO_2 in aqueous solution ?

(a) BaCO_3 (b) CaCO_3
(c) Both (a) and (b) (d) None of these

2. (I) Salt of 'X' + CaCl_2 (excess) $\rightarrow$ 'Y' + Filtrate
(II) Filtrate + $\text{NH}_3 \rightarrow$ 'Y'
(III) Filtrate + Boiling $\rightarrow$ 'Y'

Here salt 'X' is :-

(a) mixture of HCO_3^- and CO_3^{2-}
(b) only CO_3^{2-}
(c) only HCO_3^-
(d) None of these

3. $(\text{C}) \xleftarrow{\text{Hg}_2(\text{NO}_3)_2} (\text{A}) \xrightarrow{\text{dil. H}_2\text{SO}_4} (\text{B})$
yellow ppt. (Salt) Odourless gas
 $\downarrow \text{CaCl}_2 \text{ sol.}$
(D)
White ppt.

Salt 'A' is :

(a) Na_2CO_3 (b) Na_2SO_3
(c) $\text{Na}_2\text{C}_2\text{O}_4$ (d) Na_2SO_4

4. What colour of ppt. is formed when KHCO_3 reacts with excess AgNO_3 followed by boiling ?

(a) white (b) green
(c) brown (d) none of these

5. Which of the following does not form volatile product with conc. H_2SO_4 ?

(a) CO_3^{2-} (b) SO_4^{2-}
(c) Cl^- (d) NO_3^-

6. Black (A) + $\text{H}_2\text{SO}_4 \rightarrow$
(B) (smell of rotten egg) + (C)
(C) + $\text{K}_3[\text{Fe}(\text{CN})_6] \rightarrow$ Blue (D)

A would be

(a) HgS (b) FeS
(c) Bi_2S_3 (d) CuSO_3

7. (I) Salt 'A' + dil. $\text{H}_2\text{SO}_4 \rightarrow$ 'B' gas ['B' has suffocating order.]

(II) Salt 'A' + $\text{Pb}(\text{OAc})_2 \rightarrow \text{C}(\downarrow) \xrightarrow{\Delta/\text{O}_2}$
'D'($\downarrow$) ['C' & 'D' both have white ppt.]

Have 'A' and 'D' are :-

(a) HSO_3^- , PbSO_4 (b) SO_3^{2-} , PbSO_3
(c) SO_3^{2-} , PbSO_4 (d) CO_3^{2-} , PbSO_3

8. Aqueous solution of a gas 'X' decolourises an acidified $\text{K}_2\text{Cr}_2\text{O}_7$ solution and on passing H_2S , the solutions gives white turbidity 'Y'. Here 'X' and 'Y' are :-

(a) SO_2 & S (b) CO_2 & S
(c) SO_2 & $\text{Cr}(\text{OH})_3$ (d) CO_2 & $\text{Cr}(\text{OH})_3$

9. $\text{Na}_2\text{S}_2\text{O}_3 \xrightarrow{\text{dil. H}_2\text{SO}_4} \text{Na}_2\text{SO}_4 + \text{S}(\downarrow) + \text{'B' (gas)}$

'B' + $\text{Ba}(\text{OH})_2 \rightarrow$ 'C'($\downarrow$) + H_2O
(white ppt.)

$\downarrow$ 'B'
Clear solution 'D'

Here 'B' and 'D' are :-

(a) H_2S & BaSO_3
(b) SO_2 & BaSO_3
(c) H_2S & $\text{Ba}(\text{HSO}_3)_2$
(d) SO_2 & $\text{Ba}(\text{HSO}_3)_2$

4.4 QUALITATIVE SALT ANALYSIS

10. The sulphide ion gives purple colour with sodium nitroprusside. The purple colour is due to the formation of :-

- (a) $\text{Na}_2[\text{Fe}(\text{CN})_3(\text{NOS})_3]$
 (b) $\text{Na}_4[\text{Fe}(\text{CN})_4(\text{NOS})_2]$
 (c) $\text{Na}_4[\text{Fe}(\text{CN})_5(\text{NOS})]$
 (d) $\text{Na}_4[\text{Fe}(\text{NOS})_6]$

11. Which of the following statement is correct, about "on passing H_2S gas through acidified $\text{K}_2\text{Cr}_2\text{O}_7$ solution"?

- (a) White turbidity is due to SO_3^{2-}
 (b) White turbidity is due to SO_4^{2-}
 (c) White turbidity is due to S
 (d) White turbidity is due to Cr^{3+}

12. ' X ' + $\text{CuSO}_4 \rightarrow \text{I}_2 + \text{White ppt. [Y]}$



$\text{NaI} + [\text{Z}] \text{ colourless compound.}$

Here compound $[\text{Z}]$ is -

- (a) $\text{Cu}_2\text{S}_2\text{O}_3$ (b) Cu_2I_2
 (c) $\text{Na}_2\text{S}_4\text{O}_6$ (d) $\text{Na}_4[\text{Cu}_6(\text{S}_2\text{O}_3)_5]$

13. $\text{FeCl}_3 + \text{'X' (salt solution)} \longrightarrow \text{'Y' (dark violet colouration)}$



$[\text{colour disappears}]$

Here ' X ' and ' Y ' are -

- (a) $\text{Na}_2\text{S}_2\text{O}_3$ & FeS_4O_6
 (b) $\text{Na}_2\text{S}_2\text{O}_3$ & $[\text{Fe}(\text{S}_2\text{O}_3)_2]^-$
 (c) Na_2SO_3 & FeSO_4
 (d) Na_2S & $(\text{FeS} + \text{S})$

14. Which of the following compound is insoluble in water ?

- (a) KNO_2 (b) AgNO_2
 (c) $\text{Ba}(\text{NO}_2)_2$ (d) NaNO_2

15. Why few drops conc. HNO_3 is added to IInd group filtrate before proceeding for IIIrd group radicals?

- (a) Because it oxidises the dissolved H_2S to colloidal sulphur

- (b) Because it oxidises Fe^{2+} to Fe^{3+} so that Fe^{3+} can be completely precipitated as its hydroxide
 (c) Because precipitation of cations belonging to IIIrd group by group reagent takes place only in presence of conc. HNO_3
 (d) None of these

16. Sodium nitroprusside produces violet coloration when react with Na_2S due to :-

- (a) Oxidation of Fe^{2+} to Fe^{3+}
 (b) Reduction of Fe^{2+} to Fe^{1+}
 (c) Formation of soluble complex without changing oxidation state of iron
 (d) Precipitation of iron sulphide from sodium nitroprusside

17. Which of the following reagent is preferred to give vinegar-like odour on warming with acetate ion?

- (a) dil. H_2SO_4
 (b) conc. $\text{CH}_3\text{CO}_2\text{H}$
 (c) (conc. $\text{H}_2\text{SO}_4 + \text{C}_2\text{H}_5\text{OH}$)
 (d) conc. HNO_3

18. Anion that can not decolorise acidic solution of KMnO_4 .

- (a) HCO_3^- (b) HSO_3^-
 (c) SO_3^{2-} (d) S^{2-}

19. Aq. solution of ' A ' $\xrightarrow[\text{solution}]{\text{Na}_2\text{S}_2\text{O}_3}$ white ppt.

which is not ' A ' from the following :-

- (a) $\text{Pb}(\text{OAc})_2$ (b) dil. HCl
 (c) $[\text{Ni}(\text{en})_3](\text{NO}_3)_2$ (d) CuSO_4

20. Which of the following reagent is used with KI solution to give pure HI ?

- (a) conc. H_2SO_4 (b) dil. H_2SO_4
 (c) conc. H_3PO_4 (d) dil. HCl

21. When ' A ' salt solution is added carefully to a saturated solution of $\text{Fe}(\text{II})$ sulphate acidified with dilute H_2SO_4 a brown ring is formed at the junction of two liquids. The compound A is

- (a) NaNO_3 (b) NaNO_2
 (c) NaCl (d) All of these

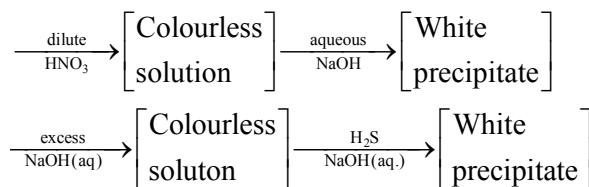
22. An inorganic salt solution 'M' gives white precipitate with $\text{Pb}(\text{OAc})_2$ solution. The ppt. dissolves on warming but becomes needle like crystal on cooling. The anionic part of salt 'M' is :-

- (a) Chloride (b) Bromide
(c) Iodide (d) Sulphite

23. The mixture of salts NaCl , NaBr , NaI on adding conc. H_2SO_4 and followed by warming gives gases respectively.

- (a) Cl_2 , Br_2 , I_2 (b) HCl , HBr , HI
(c) HCl , Br_2 , I_2 (d) CrO_2Cl_2 , Br_2 , I_2

24. A metal M and its compound can give the following observable changes in a consequence of reactions



- (a) Mg (b) Pb
(c) Zn (d) Sn

25. [Salt 'A'] + dil. $\text{H}_2\text{SO}_4 \longrightarrow$ No action

+ conc. H_2SO_4

[B']

Brown fumes gas

Salt 'A' and gas 'B' are :-

- (a) NaNO_2 and NO_2 (b) NaNO_3 and NO_2
(c) NaBr and HBr (d) NaCl and Cl_2

26. $\text{NaNO}_3(\text{s}) + \text{conc. H}_2\text{SO}_4 + \text{Cu. turning}$

on heating

Brown fumes of 'X' + Blue colour of 'Y'

Here 'Y' is :-

- (a) NO_2 (b) $\text{Cu}(\text{NO}_3)_2$
(c) Ag_2SO_4 (d) Na_2SO_4

27. Which of the following ion gives white precipitate with ferric chloride solution ?

- (a) F^-
(b) $\text{S}_2\text{O}_3^{2-}$
(c) CO_3^{2-}
(d) CH_3COO^-

28. Silver is not produced when :

- (a) AgNO_3 solution is treated with Cu rod
(b) AgNO_3 (ammoniacal solution) reacts with glucose
(c) AgNO_3 solution + excess KCN is reacted with Zn dust
(d) AgNO_3 solution is treated with $\text{Na}_2\text{S}_2\text{O}_3$ & followed by heating

29. Which of the following compound(s) give black precipitate with ammonia solution ?

- (a) Hg_2Cl_2 (b) AgCl
(c) PbCl_2 (d) None of these

30. Which of the following cation gives black precipitate with H_2S gas as well as soluble with excess yellow sodium disulphide ?

- (a) Hg_2^{2+} (b) Ag^+
(c) Cd^{2+} (d) Both (a) & (b)

31. Salt solution ['X'] $\xrightarrow{\text{dil. HCl}}$ white ppt. ['Y']

ppt. is washed by decantation and dilute ammonia is added.

white ppt. [Z]

Here white ppt. of [Y] is :-

- (a) PbCl_2 (b) AgCl
(c) $\text{Pb}(\text{OH})_2$ (d) Hg_2Cl_2

32. With ammonium sulphide $(\text{NH}_4)_2\text{S}$, Al^{3+} gives :-

- (a) White precipitate of Al_2S_3
(b) White precipitate of $\text{Al}(\text{OH})_3$
(c) White precipitate of $\text{Al}(\text{OH})_2(\text{CH}_3\text{CO}_2)$
(d) None

33. A white powder (A) on heating gave a non combustible gas and a white residue. The residue on heating turns yellow. The residue dissolve in dil HCl and the solution gives a white ppt with $\text{K}_4[\text{Fe}(\text{CN})_6]$. A would be

- (a) CaCO_3 (b) ZnCO_3
(c) CaSO_3 (d) CuCO_3

4.6 QUALITATIVE SALT ANALYSIS

34. Which of the following compound is not coloured yellow :-
 (a) $(\text{NH}_4)_3[\text{As}(\text{Mo}_3\text{O}_{10})_4]$
 (b) BaCrO_4
 (c) $\text{Zn}_2[\text{Fe}(\text{CN})_6]$
 (d) $\text{K}_3[\text{Co}(\text{NO}_2)_6]$
35. 3% H_2O_2 on heating converts :-
 (a) Black precipitate PbS into yellow precipitate PbSO_4
 (b) Black precipitate PbS into white precipitate PbSO_4
 (c) Black precipitate PbS into colorless solution PbSO_4
 (d) White precipitate PbS into colorless solution PbSO_4
36. The color of BiI_3 precipitate is :-
 (a) Blue (b) Brownish
 (c) Black (d) White
37. Sodium carbonate when reacts with mercuric chloride then
 (a) reddish brown precipitate is formed.
 (b) yellow precipitate is formed.
 (c) white precipitate is formed.
 (d) black precipitate is formed.
38. $\text{NO} \xrightarrow{\text{O}_2} \text{A} \rightleftharpoons \text{D}$
 $\downarrow \text{NaOH}$
 $\text{B} + \text{C}$
 A, B, C, D are respectively :-
 (a) NO_2 , N_2O_3 , NaNO_2 , NaNO_3
 (b) N_2O_3 , NO_2 , NaN_3 , NaNO_3
 (c) NO_2 , NaNO_2 , NaNO_3 , N_2O_4
 (d) NO_2 , N_2O_4 , NaNO_2 , NaNO_3
39. $\text{K}_2[\text{Ni}(\text{CN})_4]$ solution on treatment with sodium hydroxide and bromine water followed by heating gives a black residue (A). The compound A is
 (a) NiO
 (b) NaCN
 (c) Ni_2O_3
 (d) None of these
40. What colour is observed when phenolphthalein indicator is added to soluble carbonate
 (a) White (b) Blue
 (c) Pink (d) Green
41. ZnCl_2 reacts with excess of NH_3 solution to produce
 (a) a precipitate on $\text{Zn}(\text{OH})_2$
 (b) a complex ion $[\text{Zn}(\text{NH}_3)_4]^{2+}$ with tetrahedral geometry
 (c) a complex ion $[\text{Zn}(\text{NH}_3)_4]^{2+}$ with linear geometry
 (d) a complex ion $[\text{Zn}(\text{NH}_3)_4]^{2+}$ with square planar geometry
42. Which of the following compound(s) turns brownish - black precipitate, on boiling with Na_2CO_3 to their respectively aqueous salt solution?
 (a) $\text{Pb}(\text{OH})_2 \cdot 2\text{PbCO}_2$ (b) Hg_2CO_3
 (c) Ag_2CO_3 (d) $(\text{NH}_4)_2\text{CO}_3$
43. One of the products formed, when Bi_2O_3 is added to an alkaline solution of sodium hypochlorite, is
 (a) $\text{Bi}(\text{OH})_3$ (b) BiO_3^-
 (c) BiOCl (d) Bi_2Cl_3
44. A salt on treatment with dil HCl gives a pungent smelling gas and a yellow ppt. The salt gives green flame when burnt. The salt solution gives a yellow ppt. with potassium chromate. The salt is
 (a) BaS_2O_3 (b) PbS_2O_3
 (c) CuSO_4 (d) NiSO_4
45. A light blue coloured compound (a) on heating gives a black compound (b) which reacts with glucose to give red compound (c). (a), (b) and (c) are respectively.
 (a) $\text{CuSO}_4 \cdot 5\text{H}_2\text{O}$
 (b) $\text{Cu}(\text{OH})_2$, Cu_2O , CuO
 (c) $\text{Cu}(\text{OH})_2$, CuO , Cu_2O
 (d) $[\text{Cu}(\text{NH}_3)]\text{SO}_4$, CuO , Cu_2O
46. Which of the following oxide gives hydrogen peroxide on reaction with a dilute acid :-
 (a) PbO_2 (b) Na_2O_2
 (c) MnO_2 (d) TiO_2

47. $A + HCl \longrightarrow B + HNO_3$ precipitate B is soluble
ppt. in hot water (33.4 g/l at 100°C)
 $A + KI \longrightarrow$ Yellow ppt. Identify 'A'
- (a) $Pb(NO_3)_2$ (b) $AgNO_3$
(c) $Hg_2(NO_3)_2$ (d) None of these
48. $M + N_2 \rightarrow$ Nitride $\xrightarrow{H_2O} NH_3$ in above reaction metal M may be :-
- (a) only Li (b) only Mg
(c) only Cs (d) Li and Mg both
49. On the electrolysis of HCl, hydrogen gets liberated at :-
- (a) Anode (b) Cathode
(c) In air (d) Cannot be defined
50. $A +$ moderately concentrated (8M) $HNO_3 \longrightarrow$
 $B + NO + S + H_2O$;
 $B + NH_4 OH \longrightarrow$ Deep blue solution; Identify 'B'
- (a) $Pb(NO_3)_2$ (b) $Bi(NO_3)_3$
(c) $Cu(NO_3)_2$ (d) $Cd(NO_3)_2$
51. Which of the following thermal decomposition yields a basic as well as an acidic oxide :-
- (a) Na_2CO_3 (b) $CaCO_3$
(c) $NaNO_3$ (d) All the correct
52. $CaCO_3(s) + AcOH \xrightarrow{Na_2C_2O_4 \text{ solution}}$
Comment on the product of this reaction :-
- (a) $CaCO_3$ remains unaffected
(b) CaC_2O_4 will be precipitated as white ppt.
(c) $Ca(OAc)_2$ will be precipitated as white ppt.
(d) Clear solution
53. HCO_2^- and $CH_3CO_2^-$ ion can be distinguished by :-
- (a) Ethanol in conc. H_2SO_4
(b) $HgCl_2$
(c) $FeCl_3$
(d) $AgNO_3$
54. Addition of $SnCl_2$ to $HgCl_2$ gives ppt. of :-
- (a) Black turning to white silky
(b) White silky turning to greyish black
(c) White silky turning to red
(d) Black silky turning to green
55. A metal ion 'X' reacts with NaOH to gives a white ppt. Precipitate can be oxidised by 4-6 drops of conc. H_2O_2 , when yellowish brown colour solution is formed. The metal ion 'X' is :-
- (a) Hg^{2+} (b) Bi^{3+}
(c) Cu^{2+} (d) Pb^{2+}
56. Which of the following metal sulphide is soluble in both hot and conc. HNO_3 and KCN :-
- (a) CuS (b) CdS
(c) Ag_2S (d) PbS
57. Which of the following metal cation gives yellow ppt. with pyrogallol (10% freshly prepared) :-
- (a) Pb (b) Cu^{2+}
(c) Bi^{3+} (d) All the correct
58. Mixing of which will result in a white ppt. which turns blue on oxidation :-
- (a) $CuSO_4 + HgCl_2$
(b) $MgCl_2 + Cu$
(c) $Fe_2(SO_4)_3, K_3[Fe(CN)_6]$
(d) $FeSO_4, K_4[Fe(CN)_6]$
59. With 'X' cation $[Fe(CN)_6]^{4-}$ gives Prussian blue colouration due to the formation of ferr-ferrocyanide, $Fe_4[Fe(CN)_6]_3$, while with NH_4SCN , 'X' cation gives
- (a) Deep red, $[Fe(H_2O)_5(SCN)]^{2+}$
(b) Blue, $Fe_4[Fe(CN)_6]_3$
(c) Brown, $[Fe(H_2O)_5(NO)]^{2+}$
(d) Green, $[Cr(NH_3)_6]^{3+}$
60. Turnbull's blue is formed when Fe^{+2} ions are added to $K_3[Fe(CN)_6]$. Turnbull's blue is :-
- (a) $K_2Fe[Fe(CN)_6]$
(b) $Fe_3[Fe(CN)_6]_2$
(c) $Fe_2[Fe(CN)_6]$
(d) All of these
61. Find the ion that can be precipitated by the reagents NH_4Cl and aqueous NH_3/NH_4OH .
- (a) Ca^{2+}
(b) Al^{3+}
(c) Mg^{2+}
(d) Zn^{2+}

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62. Anion (A_1) / Acid radical Cation (B_1) / Basic radical

↓ conc. H_2SO_4

Gas I

↓ NaOH

Gas II

Gas I and Gas II reacts to give white fume.

A_1 and B_1 are respectively :-

- (a) SO_3^{2-} , Ca^{2+} (b) S^{2-} , Ba^{2+}
(c) I^- , Ag^+ (d) Cl^- , NH_4^+

63. Identify wrong statement with $CuSO_4$:-

- (a) $CuSO_4$ reacts with KI to give violet vapour I_2 which further reacts with I_2 to give brown solution of KI_3 .
(b) $CuSO_4$ on heating give CuO
(c) $CuSO_4$ reacts with KCl to give greenish yellow vapour Cl_2
(d) It's tartarate complex reacts with NaOH and glucose to give reddish brown precipitate of Cu_2O .

64. A white powder (a) when strongly heated it gives off a colourless, odourless gas (b) which turns lime water milky (c). A on strong heating give solid residue (e) which is yellow when hot but turns white on cooling A is :-

- (a) Ag_2CO_3 (b) $PbCO_3$
(c) $HgCO_3$ (d) $ZnCO_3$

65. When can produce Rinmann's green colour with cobalt nitrate $Co(NO_3)_2$ solution :-

- (a) ZnO (b) $ZnSO_4$
(c) $3Zn(OH)_2 \cdot ZnCO_3$ (d) All

66. A yellow solid (a) is soluble in water. Its aqueous solution turns blue litmus red. Aqueous solution of A in strong heating gives a dark brown residue (b) and a compound C which gives white fumes with NH_3 gas. When dry Cl_2 gas is passed over a heated mixture of (b) and carbon, compound D is formed. D absorbs water vapours to give compound A. A is :-

- (a) $FeCl_2$
(b) $AlCl_3$
(c) $FeCl_3 \cdot 6H_2O$
(d) $Fe_2O_3 \cdot H_2O$

67. $X + H_2S \xrightarrow{HCl} Y_{ppt.}; Y + (NH_4)_2 S_2 \rightarrow \text{Soluble};$
(YAS)

'X' may contain

- (i) As^{3+} (ii) Sb^{3+}
(iii) Sn^{2+} (iv) Cd^{2+}
(a) Only (iv) (b) (ii) or (iv)
(c) (i), (ii) or (iii) (d) Any of the four

68. $A + CH_3COOH \longrightarrow \underset{(soluble)}{B} + CO_2 + H_2O ;$

$B + (NH_4)_2 C_2O_4 \longrightarrow \text{White ppt.}$

A and B may contain

- (a) Ni^{2+} (b) Ba^{2+}
(c) Sr^{2+} (d) Ca^{2+}

69. Which of the following oxide of lead satisfy the observation?

- (1) It is an oxidising agent
(2) It is insoluble in acid
(3) In ice cold medium when conc. HCl is added to the oxide, yellow hexa-chloroplumbate (IV) ion $[PbCl_6]^{-2}$ is formed.
(a) PbO (b) PbO_2
(c) Pb_2O_3 (d) PbO_3

70. Arrange the following hydroxy compound in order of increasing acid strength

$CrO_2(OH)_2$ (I); $Cr(OH)_2$ (II); $Cr(OH)_3$ (III)

- (a) $I < II < III$ (b) $II < III < I$
(c) $III < II < I$ (d) $III < I < II$

71. A mixture contains manganous sulphide and cobalt sulphide. Dilute hydrochloric acid is added. Which of the following statement is correct?

- (a) Both are soluble in dil HCl
(b) Both are insoluble in dil HCl
(c) MnS is soluble in HCl while CoS is insoluble
(d) CoS is soluble in HCl while MnS is insoluble

72. To a solution containing metal cations A and B, K_2CrO_4 is added separately when A gives red coloured precipitate while B gave a yellow precipitate. A and B are

- (a) Cu^{+2} , Pb^{+2} (b) Ag^+ , Ba^{+2}
(c) Zn^{+2} , Cu^{+2} (d) Pb^{+2} , Cu^{+2}

73. KI gives a precipitate with all the cations given below. Choose the cation
 (a) Ag^+ , Hg_2^{2+} , Pb^{2+} (b) Cu^{+2} , Zn^{+2} , Ni^{+2}
 (c) Na^+ , Ca^{+2} , Mg^{+2} (d) Ag^+ , Ca^{+2} , Sr^{+2}
74. NH_4SCN can be used to test, one or more out of Fe^{+3} , Co^{+2} and Cu^{+2}
 (a) Fe^{+3} only (b) Co^{+2} , and Cu^{+2}
 (c) Fe^{+3} , Cu^{+2} (d) all
75. A light coloured crystalline solid A has 27.55% H_2O . A gives the following reactions.
 (i) $\text{A} \xrightarrow[\text{Solution}]{\text{BaCl}_2}$ white precipitate insoluble in conc. HNO_3
 (ii) $\text{A} \xrightarrow[\text{Solution}]{\text{K}_3[\text{Fe}(\text{CN})_6]}$ a dark-blue precipitate
 (iii) $\text{A} \xrightarrow[\text{NaOH}]{\text{K}_2\text{HgI}_4}$ Brown precipitate
 Identify A
 (a) $\text{Fe}_2(\text{SO}_4)_3 \cdot (\text{NH}_4)_2\text{SO}_4 \cdot 8\text{H}_2\text{O}$
 (b) $\text{FeSO}_4(\text{NH}_4)_2\text{SO}_4 \cdot 6\text{H}_2\text{O}$
 (c) $(\text{NH}_4)_2\text{FeSO}_4 \cdot 6\text{H}_2\text{O}$
 (d) $\text{FeSO}_4(\text{NH}_4)_2\text{SO}_4 \cdot 5\text{H}_2\text{O}$
76. Which of the following compounds change the salmon-coloured precipitate of zinc nitroprusside into red?
 (a) Na_2CO_3 (b) NaHCO_3
 (c) Moist- SO_2 (d) Moist- CO_2
77. The sulphide which is insoluble in both ammonium sulphide and HNO_3 is
 (a) Bi_2S_3 (b) CuS
 (c) HgS (d) FeS
78. To an acidified $\text{Cr}_2\text{O}_7^{2-}$ solution, Na_2O_2 is added, a blue colour is observed. The blue colour changes to green when H_2SO_4 is added. What are these blue and green coloured compounds?
 (a) Cr_2O_3 , $\text{Cr}_2(\text{SO}_4)_3$ (b) CrO_3 , $\text{CrO}(\text{O}_2)_2$
 (c) CrO_3 , $\text{Cr}_2(\text{SO}_4)_3$ (d) $\text{CrO}(\text{O}_2)_2$, $\text{Cr}_2(\text{SO}_4)_3$
79. A colourless solution contains a metal nitrate. A little solution of sodium chloride is added to it when a cloudy white precipitate appears. Identify the precipitate
 (a) PbCl_2 (b) AgCl
 (c) Hg_2Cl_2 (d) any of the three
80. In the previous question, further if the precipitate turns black with NH_4OH , it is
 (a) PbCl_2 (b) AgCl
 (c) Hg_2Cl_2 (d) (b) or (c)
81. Iron is precipitated as $\text{Fe}(\text{OH})_3$ in the third group and not as $\text{Fe}(\text{OH})_2$ because
 (a) $\text{Fe}(\text{OH})_2$ is light green
 (b) $\text{Fe}(\text{OH})_3$ is reddish brown
 (c) $\text{Fe}(\text{OH})_2$ is partially soluble and is not completely precipitated
 (d) $\text{Fe}(\text{OH})_2$ is oxidant
82. Cu^{2+} ions will be reduced to Cu^+ ion by addition of an aqueous solution of
 (a) KF (b) KCl
 (c) KI (d) KOH
83. Which of the following gives black precipitate on passing H_2S through it
 (a) acidified zinc nitrate solution
 (b) ammoniacal barium chloride solution
 (c) magnesium nitrate solution
 (d) copper nitrate solution
84. A mixture containing Cu^{2+} and Ni^{2+} can be separated for identification by
 (a) passing H_2S in acid medium
 (b) passing H_2S in alkaline medium
 (c) passing H_2S in neutral medium
 (d) all
85. Identify the correct order of solubility of Na_2S , CuS and ZnS in aqueous medium.
 (a) $\text{CuS} > \text{ZnS} > \text{Na}_2\text{S}$
 (b) $\text{ZnS} > \text{Na}_2\text{S} > \text{CuS}$
 (c) $\text{Na}_2\text{S} > \text{CuS} > \text{ZnS}$
 (d) $\text{Na}_2\text{S} > \text{ZnS} > \text{CuS}$
86. Which of the following reagent cause the separation of precipitate of $\text{Fe}(\text{OH})_3$ and $\text{Cr}(\text{OH})_3$ in IIIrd group analysis?
 (a) dil H_2SO_4
 (b) Aq. NH_3
 (c) $\text{NaOH} + \text{O}_2$
 (d) $\text{NaOH}/\text{H}_2\text{O}$

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87. In the qualitative analysis, NH_4Cl is added before NH_4OH to
- decrease OH^- ions conc.
 - increase OH^- ion conc.
 - form HCl
 - form complexes of group II radicals
88. Mercurous ions is represent as
- Hg^+
 - Hg^{2+}
 - Hg^{3+}
 - Hg_2^{2+}
89. Mark the correct statement
- I group basic radical precipitate as chlorides
 - IV group basic radical precipitate as sulphides
 - V group basic radical precipitate as carbonates
 - all the above statements are correct
90. Which of the following is not precipitated as sulphides by passing H_2S in presence of conc. HCl ?
- Copper
 - Arsenic
 - Chromium
 - Lead
91. The cation present in slightly acidic solution are Fe^{3+} , Ni^{2+} and Cu^{2+} . Which reagent, when added excess to this solution will identify and separate Fe^{3+} in one step
- 2M HCl
 - 6 M NH_3
 - 6 M NaOH
 - H_2S gas
92. During analysis of basic radicals of a mixture containing cations of group V, if a solution of NaHCO_3 is added instead of Na_2CO_3 along with NH_4OH this would result in formation of
- white ppt
 - yellow ppt
 - brick red ppt
 - no ppt at all
93. The metal whose salts do not give the borax bead test is -
- Cr
 - Ni
 - Pb
 - Mn
94. In qualitative analysis of group I radicals, a white precipitate is formed which is insoluble in boiling water but when treated with NH_4OH it turns black, the precipitate may be.
- PbCl_2
 - AgCl
 - HgCl_2
 - Hg_2Cl_2
95. On passing H_2S gas in II group, sometimes the solution turns milky. It indicates the presence of
- oxidizing agent
 - acidic salt
 - thiosulphate
 - reducing agent
96. Sodium sulphide react with sodium nitropruside to form a purple coloured compound. During the reaction, the oxidation state of iron
- change from +2 to +3
 - changes from +3 to +2
 - change from +2 to +4
 - remain unchanged
97. An inorganic salt when heated evolves coloured gas which bleaches moist litmus paper. The evolved gas is
- NO_2
 - Cl_2
 - N_2O
 - I_2
98. When hot and concentrated KOH reacts with Cl_2 , the product is
- KClO
 - KClO_4
 - KClO_3
 - None of these
99. In the reaction
- $$2\text{KI} + \text{H}_2\text{O} + \text{O}_3 \longrightarrow 2\text{KOH} + \text{O}_2 + \text{A}$$
- the compound A is:
- KIO_3
 - I_2O_5
 - HIO_3
 - I_2
100. An aqueous solution contains Hg^{2+} , Hg_2^{2+} , Pb^{2+} and Cd^{2+} . The addition of 2M HCl will precipitate
- Hg_2Cl_2 only.
 - PbCl_2 and CdCl_2
 - PbCl_2 and HgCl_2
 - PbCl_2 , Hg_2Cl_2 and HgCl_2
101. White precipitate (X) + $\text{O}_2 \rightarrow$ Brown precipitate (Hydroxide)
(X) is -
- $\text{Mn}(\text{OH})_2$
 - $\text{Fe}(\text{OH})_2$
 - $\text{Ni}(\text{OH})_2$
 - $\text{Cr}(\text{OH})_3$

102. Green precipitate (X) + $\text{H}_2\text{O}_2 + \text{OH}^- \rightarrow$ Yellow solution (Hydroxide)
(X) is -
(a) $\text{Mn}(\text{OH})_2$ (b) $\text{Fe}(\text{OH})_2$
(c) $\text{Ni}(\text{OH})_2$ (d) $\text{Cr}(\text{OH})_3$
103. Green precipitate $\xrightarrow{[\text{O}]}$ Black precipitate (X) (Hydroxide)
(X) is -
(a) $\text{Mn}(\text{OH})_2$ (b) $\text{Fe}(\text{OH})_2$
(c) $\text{Ni}(\text{OH})_3$ (d) $\text{Fe}(\text{OH})_3$
104. Which ion gives yellow precipitate on addition of KNO_2 solution ?
(a) Ni^{2+} (b) Zn^{2+}
(c) Mn^{2+} (d) Co^{2+}
105. In a mixture containing Al^{3+} , Ba^{2+} , Mg^{2+} , Cr^{3+} ; if NH_4Cl is added in ammoniacal medium, the ion pair get precipitated is
(a) Ba^{2+} , Cr^{3+} (b) Al^{3+} , Mg^{2+}
(c) Al^{3+} , Cr^{3+} (d) Mg^{2+} , Cr^{3+}
106. Correct K_{sp} value of alkaline earth sulphate
(a) $\text{BaSO}_4 > \text{SrSO}_4 > \text{CaSO}_4$
(b) $\text{SrSO}_4 > \text{CaSO}_4 > \text{BaSO}_4$
(c) $\text{BaSO}_4 > \text{CaSO}_4 > \text{SrSO}_4$
(d) $\text{CaSO}_4 > \text{SrSO}_4 > \text{BaSO}_4$
107. Chromate of which metal ion is most soluble -
(a) BaCrO_4 (b) CaCrO_4
(c) SrCrO_4 (d) All
108. Oxalate of which metal ion is least soluble -
(a) BaC_2O_4 (b) CaC_2O_4
(c) SrC_2O_4 (d) All
109. $\text{Ni}^{2+} + \text{NH}_3 \rightarrow$ product(soluble)
Product is -
(a) Brown solution
(b) sp^3d^2 hybridised
(c) Inner orbital octahedral complex
(d) Diamagnetic
110. $\text{Zn}^{2+} + \text{NH}_3 \rightarrow$ product(soluble)
Product is -
(a) Colourless solution (b) sp^3 hybridised
(c) Diamagnetic (d) All
111. Passing H_2S gas in a solution mixture containing Co^{2+} , Zn^{2+} , Pb^{2+} , Cu^{2+} in an acidified solution precipitates -
(a) CoS , PbS (b) ZnS , CoS
(c) PbS , CuS (d) CuS , ZnS
112. Aqueous solution of $\text{Ni}^{2+} + \text{HCl}(\text{excess}) \rightarrow \text{P}_1$
Aqueous solution of $\text{Zn}^{2+} + \text{HCl}(\text{excess}) \rightarrow \text{P}_2$
 P_1 and P_2 differs in -
(a) Shape (b) Magnetic moment
(c) Hybridisation (d) Colour
113. A metal ion $\xrightarrow{(\text{NH}_4)_2\text{S}}$ Pink precipitate
Metal ion can be -
(a) Mn^{2+} (b) Ni^{2+}
(c) Co^{2+} (d) both (B) and (C)
114. A metal ion $\xrightarrow{(\text{NH}_4)_2\text{S}}$ Black precipitate
Metal ion can be -
(a) Mn^{2+} (b) Ni^{2+}
(c) Co^{2+} (d) both (B) and (C)
115. A compound gives scarlet red precipitate with KI solution. The compound may have -
(a) Fe^{2+} (b) Bi^{3+}
(c) Hg^{2+} (d) Hg_2^{2+}
116. A compound gives green precipitate with KI solution. The compound may have -
(a) Fe^{2+} (b) Bi^{3+}
(c) Hg^{2+} (d) Hg_2^{2+}
117. Which metal ion gives green precipitate with KCN solution -
(a) Fe^{2+}
(b) Ag^+
(c) Hg^{2+}
(d) Ni^{2+}
118. $\text{FeCl}_3 +$ sodium acetate solution $\rightarrow$ deep red colouration (X_1)
 X_1 is -
(a) $\text{Fe}(\text{OH})_2\text{CH}_3\text{CO}_2$
(b) $[\text{Fe}(\text{OH})_2(\text{CH}_3\text{CO}_2)_4]^{3-}$
(c) $[\text{Fe}_3(\text{OH})_2(\text{CH}_3\text{CO}_2)_6]^{+1}$
(d) $[\text{Fe}(\text{OH})(\text{CH}_3\text{CO}_2)]^{+1}$

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119. FeCl_3 + sodium acetate solution $\rightarrow$

deep red colouration (X_1)

$X_1 \xrightarrow{\text{boiling}} X_2$ (reddish brown precipitate)

X_2 is -

- (a) $\text{Fe}(\text{OH})_2\text{CH}_3\text{CO}_2$
- (b) $[\text{Fe}(\text{OH})_2(\text{CH}_3\text{CO}_2)_4]^{3-}$
- (c) $[\text{Fe}_3(\text{OH})_2(\text{CH}_3\text{CO}_2)_6]^{+1}$
- (d) $[\text{Fe}(\text{OH})(\text{CH}_3\text{CO}_2)]^{+1}$

120. Red colouration is not observed when Fe^{2+} reacts with -

- (a) NH_4SCN (b) DMG
- (c) 2,2'-bipyridyl (d) 1,10-phenanthroline

EXERCISE # II

One or More Than One Correct Answer :

1. Which of the following compound gives 'X' gas with addition of dilute HCl. The gas 'X', is passed into lime water or baryta water, turns turbidity.

- (a) Na_2CO_3 (b) Na_2SO_3
- (c) $\text{Na}_2\text{S}_2\text{O}_3$ (d) BaCO_3

2. Which of the following anion(s) gives white ppt. with CaCl_2 in aqueous solution?

- (a) CO_3^{2-} (b) S^{2-}
- (c) SO_3^{2-} (d) CHCOO^-

3. Which of the following acid is/are used to evolve CO_2 gas from carbonate salt?

- (a) dil. HCl (b) dil. CH_3COOH
- (c) dil. HCN (d) dil. H_3BO_3

4. Which of the following reagent(s) form ppt. with Na_2CO_3 ?

- (a) BaCl_2 (b) CuSO_4
- (c) AgNO_3 (d) HgCl_2

5. The precipitate of Ag_2CO_3 dissolve in

- (a) dil. HNO_3 (b) KCN solution
- (c) NH_3 solution (d) None of these

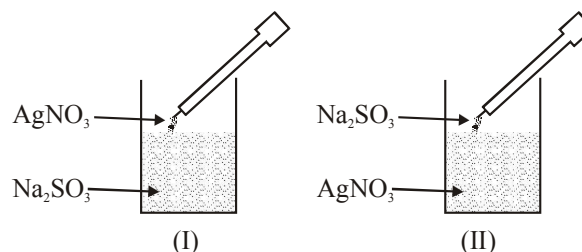
6. Which of the following interfering radical form volatile product with conc. H_2SO_4 ?

- (a) PO_4^{3-} (b) F^-
- (c) $\text{C}_2\text{O}_4^{2-}$ (d) CO_3^{2-}

7. When salt solution 'A' is treated with dilute HCl, it gives off gas 'B' which is passed through a filter paper, moistened with potassium iodate and starch solution turns deep blue colouration. The correct statement about 'A' and 'B' is /are :-

- (a) $\text{A} = \text{Na}_2\text{SO}_3$ & $\text{B} = \text{SO}_2$
- (b) $\text{A} = \text{Na}_2\text{SO}_3$ & $\text{B} = \text{CO}_2$
- (c) $\text{A} = \text{Na}_2\text{S}$ & $\text{B} = \text{H}_2\text{S}$
- (d) $\text{A} = \text{NaHSO}_3$ & $\text{B} = \text{SO}_2$

8. Observe the following experiment :-



The correct statement about set up (I) and (II) are :-

- (a) In set up (I) initially no precipitate is formed but on addition of excess of reagent and followed by boiling gives greyish back ppt.
- (b) In set up (II) initially white precipitate is formed as well as on boiling gives greyish back ppt.
- (c) In set up (II) the precipitate is soluble in dilute nitric acid, and gives gas with suffocating smell of burning sulphur which turns lime water milky.
- (d) In set up (I) the precipitate is formed with excess AgNO_3 which is soluble in ammonia solution.

9. Which of the following compounds(s) gives green solution, when treated with $\text{K}_2\text{Cr}_2\text{O}_7$ and dilute H_2SO_4 ?

- (a) KNO_3 (b) Na_2S
- (c) NaHCO_3 (d) Na_2SO_3

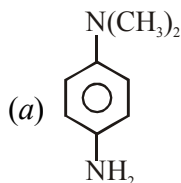
10. CO_2 & SO_2 gas can be distinguished by

- (a) $\text{KMnO}_4, \text{H}^+$ (b) $\text{Ca}(\text{OH})_2$ solution
- (c) Baryta water (d) $\text{K}_2\text{Cr}_2\text{O}_7, \text{H}^+$

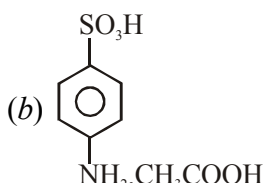
11. Which of the following compounds give precipitate with sulphide solution?

- (a) BaCl_2 (b) KI_3
- (c) Silver nitrate (d) $\text{Cd}(\text{OAc})_2$

12. Which of the following(s) is /are required to give water soluble dye stuff, methylene blue in strongly acid solution with H_2S .



(c) FeCl_3



(d) FeCl_2

13. Which of the following reagents are used to give blue ring test of solution of $\text{Na}_2\text{S}_2\text{O}_3$?

(a) $(\text{NH}_4)_2\text{MoO}_4$

(b) CuSO_4

(c) conc. H_2SO_4

(d) $[\text{Ni}(\text{en})_3](\text{NO}_3)_2$

14. Which of the following statement is / are correct?

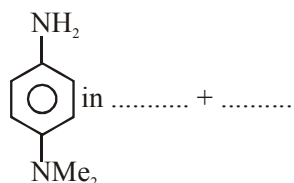
(a) In presence of sulphate, sulphite can be detected by treating dil. HCl

(b) Both sulphite and sulphate form white precipitate with BaCl_2

(c) Both sulphite and carbonate can be detected by treating acidified KMnO_4

(d) Both sulphite and carbonate can be detected by treating acidified $\text{K}_2\text{Cr}_2\text{O}_7$

15. Methylene blue is an indicator widely used in acid-base titration, it is formed on mixing :-



(a) Na_2SO_3

(b) Na_2S

(c) $\text{Fe}_2(\text{SO}_4)_3$

(d) FeSO_4

16. Select the correct statements :-

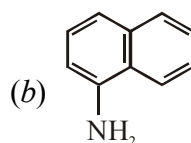
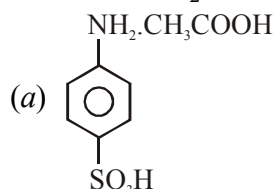
(a) The hydrogen carbonates of the alkali metal are soluble in water, but are less soluble than the corresponding normal carbonates.

(b) Solution of carbonate forms reddish brown precipitate basic mercury(II) carbonate ($3\text{HgO} \cdot \text{HgCO}_3$) when react with Mercury(II) chloride.

(c) When AgNO_3 react with sulphate ion no visible change occurs initially because of the formation of soluble sulphitoargentate ion.

(d) On standing the precipitate BaSO_3 is slowly oxidised to the sulphate.

17. Which of the following compounds give red azo dye with HNO_2



(c) $(\text{KMnO}_4 + \text{dil. H}_2\text{SO}_4)$

(d) $(\text{FeSO}_4 + \text{dil. CH}_3\text{COOH})$

18. The addition of dilute HCl acid to a solid nitrite in cold yields a transient, pale-blue liquid is obtained due to presence of following compound(s) :-

(a) free nitrous acid

(b) anhydride of nitrous acid

(c) free nitric acid

(d) condensed form of N_2O_5 & NO_2

19. Which is the following reagent(s) are used to distinguish nitrite and nitrate salt solution?

(a) Ag^+ solution

(b) $\text{FeSO}_4 + \text{dil. H}_2\text{SO}_4$

(c) KMnO_4

(d) sulphamic acid

20. Which of following reagent(s) make distinction of acetate from oxalates and formates?

(a) BaCl_2

(b) CaCl_2

(c) HgCl_2

(d) FeCl_3

21. Salt solution 'X' + $\text{FeCl}_3(\text{aq.}) \rightarrow$

'Y' (deep red colouration) + 2H^+



'Z' (Brownish red ppt.)

Correct statement(s) about the above experiment is / are :-

(a) Aqueous solution of 'X' salt gives deep red colouration, with FeCl_3 owing to formation of a complex ion 'Y' is $[\text{Fe}_3(\text{OH})_2(\text{CH}_3\text{COO})_6]^+$.

(b) Z is $\text{Fe}(\text{OH})_2(\text{CH}_3\text{CO}_2)$

(c) On boiling the red solution of 'Y' gets decomposed into brownish red precipitate of 'Z' is basic iron (III) acetate.

(d) Z is $\text{Fe}(\text{OH})_3$.

4.14 QUALITATIVE SALT ANALYSIS

22. Which of the following ions interfere chromyl chloride test :-

- (a) NO_3^- (b) Br^-
(c) I^- (d) NO_2^-

23. $\text{KI(solid)} + \text{conc. H}_2\text{SO}_4(\text{variable concentration})$

↓
On warming



Here 'X' may be -

- (a) KI_3 ($\text{I}_2 + \text{KI}$) (b) H_2S
(c) S (d) SO_2

24. $\text{Cl}^- (\text{salt solution}) + \text{conc. H}_2\text{SO}_4$

↓
On warming

only 'X' (gas) +

The correct statement about 'X' gas is :-

- (a) It is recognized by its pungent odour and production of white fumes, on blowing across the mouth of the test tube.
(b) It is recognized by formation of white clouds of ammonia chloride when a glass rod moistened with ammonia solution is held near the mouth of the test tube.
(c) It is recognised by turning litmus paper blue.
(d) 'X' gas is Cl_2 with greenish yellow fumes.

25. Which of the following compound gives white precipitate with silver nitrate solution ?

- (a) NaCl (b) Na_2CO_3
(c) Na_2SO_3 (d) $\text{Na}_2\text{S}_2\text{O}_3$

26. Which of the following salts show chromyl chloride test ?

- (a) KCl (b) CrCl_3
(c) HgCl_2 (d) FeCl_3

27. When mixture of CuCO_3 . $\text{Cu(OH)}_{2(s)}$ and $\text{Pb(NO}_3)_2(s)$ is added in dil. HCl solution -

- (a) Produces coloured precipitate and colourless solution.
(b) Produce coloured solution & white ppt.
(c) Evolve CO_2 gas
(d) Evolve NO_2 gas

28. When solid NaF is treated with conc. H_2SO_4 in a test tube and followed by warming, then dimeric form of gas 'M' is evolved, by which the test tube acquires a greasy appearance on its walls after hydrolysis which of the following compounds is/are formed.

- (a) SiF_4 (b) H_2SiF_6
(c) $\text{SiO}_2 \cdot \text{H}_2\text{O}$ (d) H_2SiO_3

29. $\text{Salt solution of 'X' + FeCl}_2(\text{aq.})$

↓

'Y' complex ion (deep red solution)

↓

on boiling

'Z' [Brownish - red ppt.]

The correct statement(s) about the flow diagram is/are :-

- (a) CH_3COONa is the salt solution 'X'
(b) 'Y' is the $[\text{Fe(H}_2\text{O)}_5(\text{SCN})^{2+}]$ ion
(c) 'Z' is the basic acetate of iron (III)
(d) Oxidation state of iron in 'Y' and 'Z' remains constant.

30. Which of the following compounds give white precipitate with dilute HCl ?

- (a) $\text{Pb(NO}_3)_2$ (b) AgNO_3
(c) $\text{Na}_2\text{S}_2\text{O}_3$ (d) CH_3COONa

31. Which of the following anionic radicals give white precipitate with magnesium nitrate reagent or magnesia mixture ?

- (a) HPO_4^{2-} (b) CO_3^{2-}
(c) CH_3COO^- (d) NO_3^-

32. Which of the following compound(s) give white precipitate with silver nitrate solution ?

- (a) Na_2CrO_4 (b) KSCN
(c) KCN (d) $\text{Na}_2\text{S}_2\text{O}_3$

33. An aqueous solution of a substance gives a white precipitate on treatment with NaCl solution. Which dissolves on heating. When H_2S gas is passed through the hot acidic solution, a black precipitate is obtained. The cations which are not associated to this test.

- (a) Mg^{2+} (b) Ag^+
(c) Hg_2^{2+} (d) Pb^{2+}

34. Depict correct flame colouration through cobalt glass for different ions.

- (a) For K^+ : Crimson colour
- (b) For Ca^{2+} : Light green colour
- (c) For Sr^{2+} : Purple colour
- (d) For Ba^{2+} : Bluish green colour

35. Select the correct statement(s) :-

- (a) Alkaline solution of potassium tetra-iodomercurate(II) gives brown ppt. with ammonium chloride solution.
- (b) On warming mixture of NH_4Cl and $NaOH$ solution, evolved gas form deep blue complex with $CuSO_4$.
- (c) On warming mixture of NH_4Cl and $NaOH$ solution, evolved gas turns $Hg(NO_3)_2$ solution black.
- (d) $NH_4^+ + NaOH \xrightarrow{\Delta}$ only gas 'X'; evolved gas 'X' turns moistened blue litmus paper red.

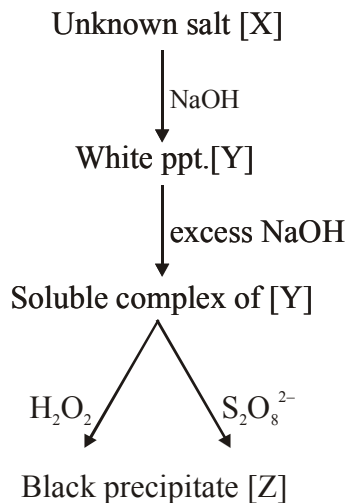
36. Which of the following reagents give white precipitate with $MgSO_4$ solution, but readily soluble in ammonium salts?

- (a) $NaOH$
- (b) $(NH_4)_2CO_3$
- (c) Na_2CO_3
- (d) Na_2HPO_4

37. Which of the following reaction is / are represented correctly?

- (a) $Pb(OAc)_2 + \text{dil. } H_2SO_4 \longrightarrow \text{'A' } \downarrow \text{ (white ppt.)} \longrightarrow \text{soluble in hot and conc. } H_2SO_4$
- (b) $Hg(NO_3)_2 + Na_2CO_{3(aq)} \longrightarrow \text{'B' } \downarrow \xrightarrow{\Delta} \text{black ppt. (yellow ppt.)}$
- (c) $2AgNO_3 + 2NH_3(aq.) \xrightarrow{+H_2O} \text{'C' } \downarrow \text{ (brown ppt.)} \longrightarrow \text{soluble in } NH_3$
- (d) $4[Ag(NH_3)_2]^+ + N_2H_4 \cdot H_2SO_4 \longrightarrow 4\text{'D' } (\downarrow) + N_2(\uparrow) + 6NH_4^+ + 2NH_3 + SO_4^{2-} \longrightarrow \text{'D' soluble in conc. } HNO_3$

38. Following observation are given :-



The correct statement is / are :-

- (a) White precipitate of Y is $Pb(OH)_2$
 - (b) The precipitate [Y] dissolves in excess $NaOH$ and forms tetrahydroxido plumbate(II) ion
 - (c) Soluble complex of [Y] gets oxidised by H_2O_2 into black ppt. (Z) of PbO_2
 - (d) Black precipitate of [Z] is $(Hg + HgO)$
39. Which of the following cations form precipitate with $K_4[Fe(CN)_6]$?
- (a) Zn^{2+}
 - (b) Cu^{+2}
 - (c) Cd^{2+}
 - (d) Al^{3+}
40. How many of the following reactions produce blue colouration or blue ppt.

- (a) $CuSO_4$ solution $\xrightarrow[\text{excess}]{NH_3 \text{ in}}$
- (b) $CuSO_4 \xrightarrow{K_4[Fe(CN)_6]}$
- (c) $NiCl_2$ solution $\xrightarrow[NH_4OH]{\text{excess}}$
- (d) $FeSO_4 \xrightarrow{K_3[Fe(CN)_6]}$

41. The correct statement about Cr^{2+} and Mn^{3+} is / are

- (a) Cr^{2+} is a reducing agent
- (b) Mn^{3+} is an oxidising agent
- (c) Both Cr^{2+} and Mn^{3+} exhibits d^4 configuration
- (d) When Cr^{2+} is used as a reducing agent, the Cr^{3+} attains d^5 electronic configuration

4.16 QUALITATIVE SALT ANALYSIS

42. Which of the following statements is/are incorrect?

- In thiourea test for nitrite, a green coloured solution is obtained.
- It is not necessary to carry out the chromyl chloride test in a dry test tube.
- In $\text{Pb}(\text{NO}_3)_2$, the brown ring test can be performed with its water extract.
- Suspension of CdCO_3 gives black precipitate with sodium sulphide solution.

43. Which of the following salt(s) produces red coloration on flame :-

- BaCl_2
- SrCl_2
- NaCl
- LiCl

44. HgS is insoluble in :-

- Water
- Hot dilute nitric acid
- Alkali hydroxides
- Ammonium sulphide (colourless)

45. Which of the following ppt. is soluble in both aqua regia and sodium disulphide (yellow) :-

- $(\text{Hg} + \text{HgS})$
- HgS
- CuS
- Ag_2S

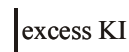
46. Using ammonia solution we can distinguish :-

- Cu^{2+} . Bi^{3+}
- Cu^{2+} . Cd^{2+}
- Bi^{3+} . Cd^{2+}
- Ba^{2+} . Ca^{2+}

47. $\text{Compound 'P'} + \text{KI}$



Precipitate of 'Q'



Soluble complex of 'R'

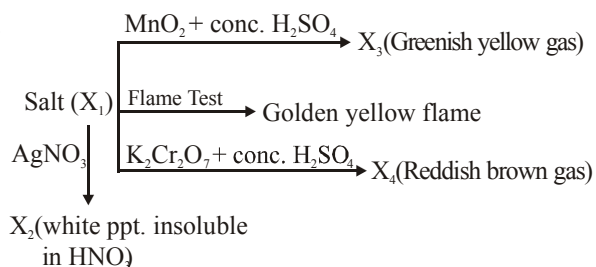
Compound 'P' may be :-

- CuSO_4
- $\text{Hg}(\text{NO}_3)_2$
- $\text{Bi}(\text{NO}_3)_3$
- CdSO_4

48. Which of the following reaction(s) give(s) blue precipitate ?

- $\text{Hg}^{2+} + \text{Co}^{2+} + 4\text{SCN}^-(\text{aq.}) \rightarrow$
- $\text{CuSO} + \text{excess NH}_3 \rightarrow$
- $\text{CuSO}_4 + \text{excess NaOH} \rightarrow$
- $\text{Cu}^{2+} + \text{KSCN} \rightarrow$

49.



Correct statements are :-

- X_2 is AgBr
- X_3 is Cl_2
- X_1 is NaCl
- X_4 is Br_2

50. Which of the following pair of species can be separated by HCl ?

- $\text{Hg}(\text{NO}_3)_2$ and $\text{Hg}_2(\text{NO}_3)_2$
- AgNO_3 and $\text{Hg}(\text{NO}_3)_2$
- $\text{Fe}(\text{OH})_3$ and $\text{Al}(\text{OH})_3$
- HgS and MnS

51. In which reaction(s) blood red coloration is observed in product :-

- $\text{FeSO}_4 \xrightarrow{\text{H}_2\text{O}_2} \dots \xrightarrow{\text{KCN}} \text{Product}$
- $\text{Na}_2\text{S}_2\text{O}_3 \xrightarrow{\text{KCN}} \dots \xrightarrow{\text{Fe}^{3+}} \text{Product}$
- $\text{NH}_4\text{SCN} \xrightarrow{\text{Ferric alum}} \text{Product}$
- $\text{NO}_2^- \xrightarrow{\text{thiourea}} \dots \xrightarrow{\text{Fe}^{2+}} \text{Product}$

52. Which is dissolved by HCl ?

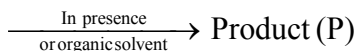
- ZnS
- MnS
- BaSO_3
- BaSO_4

53. When H_2S gas is passed through HCl containing aqueous solution of CuCl_2 , HgCl_2 , BiCl_3 , and CoCl_2 , it precipitate out :-

- CuS
- HgS
- Bi_2S_3
- CoS

54. An aqueous solution of $\text{FeSO}_4 \cdot \text{Al}_2(\text{SO}_4)_3$ and chrome alum is heated with excess of Na_2O_2 and filtered. Which can be obtained as product :-

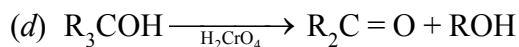
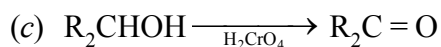
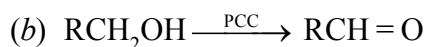
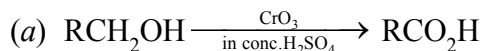
- Green filtrate
- Green Residue
- Yellow filtrate
- Brown residue

55. Acidified Chromate solution + H_2O_2 

Correct option :-

- (a) P is permanent deep blue coloured compound
 (b) Oxidation state of Cr in product P is +3
 (c) Oxidation state of Cr in product P is +6
 (d) Product has peroxy linkage

56. How many reactions, products are correctly given :-



57. For which of the ammonium salt, when they undergo heating, non redox reaction can take place :-

- (a) $(\text{NH}_4)_2\text{CO}_3$ (b) NH_4NO_3
 (c) $(\text{NH}_4)_2\text{Cr}_2\text{O}_7$ (d) $\text{CH}_3\text{CO}_2\text{NH}_4$

58. Which reactions are not applicable for obtaining greenish yellow colour Cl_2 :-

- (a) $\text{NaCl} + \text{conc. H}_2\text{SO}_4 \rightarrow$
 (b) $\text{NH}_3 + \text{HCl} \rightarrow$
 (c) $\text{NaCl} + \text{MnO}_2 + \text{H}_2\text{SO}_4 \rightarrow$
 (d) $\text{Au} + \text{H}^+ + \text{NO}_3^- + \text{Cl}^- \rightarrow$

59. The equilibrium $\text{Al}(\text{OH})_3 + \text{OH}^- \rightleftharpoons [\text{Al}(\text{OH})_4]^-$ in aqueous medium shifts towards the left in the presence of :

- (a) NH_4Cl
 (b) dilute H_2SO_4 (not in excess)
 (c) NaOH
 (d) carbon dioxide gas

60. Correct statements are :-

- (a) The compound formed in the borax bead test of Cu^{2+} ion in oxidising flame is $\text{Cu}(\text{BO}_2)_2$
 (b) NH_4NO_3 leaves no residue on heating
 (c) Mg when burnt in air and the ash on moistening smells of ammonia
 (d) CuSO_4 decolourises on addition of excess KCN, the product is $[\text{Cu}(\text{CN})_4]^{3-}$

61. n-factor of the metal centre in aqueous permanganate solution

- (a) 3 in neutral medium (b) 5 in neutral medium
 (c) 3 in alkaline medium (d) 5 in acidic medium

62. $2\text{Cr}^{3+} + 3\text{OBr}^- + 10\text{OH}^- \rightarrow 2\text{CrO}_4^{2-} + 3\text{Br}^- + 5\text{H}_2\text{O}$ Which element change its oxidation state in this reaction

- (a) Cr in Cr^{3+} (b) O in OH^-
 (c) O in OBr^- (d) Br in OBr^-

63. At 25°C , stability order

- (a) $\text{CuCl} > \text{CuCl}_2$
 (b) $\text{CuSCN} > \text{Cu}(\text{SCN})_2$
 (c) $\text{CuCN} > \text{Cu}(\text{CN})_2$
 (d) $\text{Cu}_2(\text{NO}_3)_2 > \text{Cu}(\text{NO}_3)_2$

64. Which ion gives colourless solution with excess NH_4OH solution ?

- (a) Ag^+ (b) Cu^{2+}
 (c) Ni^{2+} (d) Cd^{2+}

65. $\text{Ni}^{2+} + \text{CN}^- \rightarrow \text{Product (soluble)}$ (excess)

Product is -

- (a) Yellow colour solution
 (b) Square planar
 (c) dsp^2 hybridisation of central atom
 (d) Diamagnetic

EXERCISE # III

□ Linked Comprehension Type :

Paragraph for Q.1 to Q.3

'X' + $\text{AgNO}_3 \rightarrow$ 'Y' (white ppt.)

'Y' is soluble in nitric acid and ammonia. The ppt. becomes yellowish brown upon addition of excess reagent owing to the formation of 'Z'.

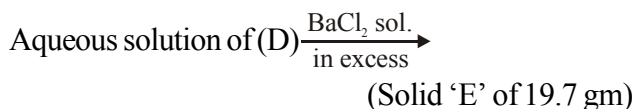
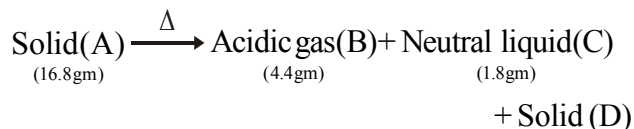
If aqueous suspension of 'Y' is boiled then it gives same product 'Z'.

1. Here 'X' and 'Y' are :-

- (a) CO_3^{2-} & Ag_2O (b) CO_3^{2-} & Ag_2CO_3
 (c) HCO_3^- & Ag_2CO_3 (d) HCO_3^- & Ag_2O

2. Here 'Z' is

- (a) Ag_2CO_3 (b) Ag
 (c) Ag_2O (d) All of these

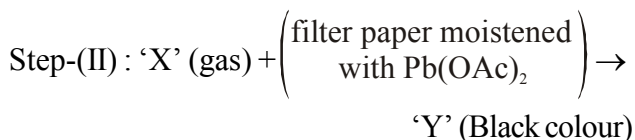
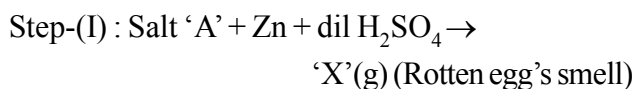
4.18 QUALITATIVE SALT ANALYSIS**Paragraph for Q.3 to Q.4**

3. Identify 'A'?

- (a) KHCO_3 (b) NaHCO_3
 (c) K_2CO_3 (d) Na_2CO_3

4. Identify 'E'?

- (a) BaCO_3 (b) Na_2CO_3
 (c) CaCO_3 (d) K_2CO_3

Paragraph for Q.5 to Q.7

5. Salt 'A' and gas 'X' may be?

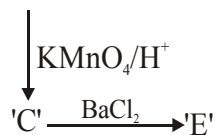
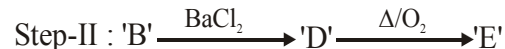
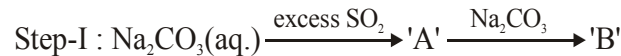
- (a) Na_2SO_3 and SO_2 (b) Na_2S and SO_2
 (c) Na_2SO_3 and H_2S (d) NaHSO_3 and SO_2

6. Here 'Y' & 'Z' are

- (a) PbS & PbSO_4
 (b) PbS & S
 (c) ZnS & PbSO_4
 (d) ZnS & ZnSO_4

7. The incorrect statement about the above experimental observation is/are :-

- (a) In step-1, salt 'A' gets reduced to 'X' gas which turns lime water milky.
 (b) 'X' gas also turns filter paper moistened with cadmium acetate to yellow coloration spot.
 (c) 'Y' is insoluble in water but with O_3 it gets completely oxidised and changed to white colour precipitate.
 (d) In step-3, white turbidity is formed due to redox reaction (comproportionation) of 'X' gas & H_2SO_3

Paragraph for Q.8 to Q.10

8. Compound 'E' and 'B' are

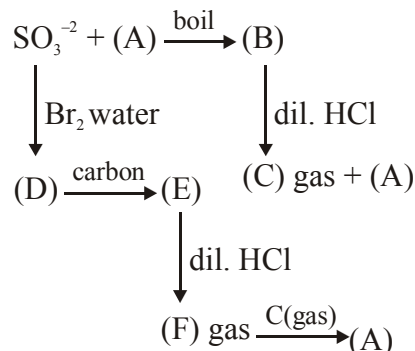
- (a) NaHCO_3 and Na_2CO_3
 (b) NaHCO_3 and Na_2SO_3
 (c) NaHSO_3 and Na_2SO_3
 (d) NaHSO_3 and Na_2CO_3

9. Here 'e' is insoluble in dilute HCl will be

- (a) BaCO_3 (b) BaSO_3
 (c) BaSO_4 (d) BaS_2O_3

10. The compound 'D' treated with dilute H_2SO_4 gives gas with pungent smell and turns lime water milky turbidity. The compound 'D' is -

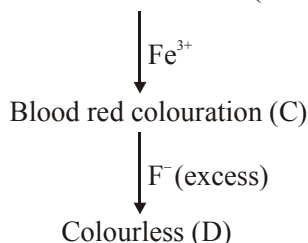
- (a) BaCO_3 (b) BaSO_3
 (c) BaSO_4 (d) Na_2SO_3

Paragraph for Q.11 to Q.1211. Identify type of reaction between $\text{F}_{(\text{gas})}$ with $\text{C}_{(\text{gas})}$:-

- (a) Ion exchange
 (b) Disproportionation
 (c) Comproportionation
 (d) None of these

12. Identify D

- (a) S^{-2}
 (b) SO_3^{-2}
 (c) $\text{S}_2\text{O}_3^{-2}$
 (d) SO_4^{-2}

Paragraph for Q.13 to Q.15Salt solution 'A' + KCN $\longrightarrow$ KSCN + 'B' (solution)**13.** Identify 'A' and 'B':-

- (a) SO_3^{2-} & $(\text{S} + \text{SO}_4^{2-})$ (b) $\text{S}_2\text{O}_3^{2-}$ & SO_3^{2-}
 (c) S^{2-} & $\text{S}_2\text{O}_3^{2-}$ (d) S^{2-} & SO_3^{2-}

14. The magnetic moment of complex 'C' is

- (a) 1.7 B.M. (b) 3.87 B.M.
 (c) 2.83 B.M. (d) 5.92 B.M.

15. The compound 'D' is

- (a) $\text{Na}_3[\text{FeF}_6]$ (b) $[\text{Fe}(\text{H}_2\text{O})_5(\text{SCN})]^{2+}$
 (c) $\text{Na}_3[\text{Fe}(\text{CN})_3\text{F}_3]$ (d) FeS_4O_6

Paragraph for Q.16 to Q.17

(I) $(\text{Na}_2\text{S} + \text{I}_2 + \text{Na}_2\text{SO}_3) \xrightarrow{\text{boil}}$
 Compound [A] + 2NaI

(II) $\text{A}(\text{aq.}) + \text{AgNO}_3 \longrightarrow \text{white ppt. [B] (unstable)}$
 $\downarrow \text{turning dark on standing}$
 Black compound (C)

(III) $\text{A}(\text{aq.}) + \text{FeCl}_3 \longrightarrow \text{dark-violet colouration [E]}$
 $\downarrow$
 [F] colour disappears on standing

16. Compound 'A' is :-

- (a) Na_2SO_4 (b) $\text{Na}_2\text{S}_2\text{O}_3$
 (c) $\text{Na}_2\text{S}_4\text{O}_6$ (d) Na_2SO_3

17. The correct statement about the above observation is / are :-

- (a) Aqueous solution of 'A' on treatment with AgNO_3 gives 'B' white ppt. of $\text{Ag}_2\text{S}_2\text{O}_3$
 (b) Aqueous solution of 'A' on treatment with AgNO_3 gives white ppt. of Ag_2SO_3
 (c) Compound 'C' is Ag_2S which is black in colour
 (d) The violet coloration of compound [E] is $[\text{Fe}(\text{S}_2\text{O}_3)_2]^{-}$, which turns into Fe^{2+} and $\text{S}_4\text{O}_6^{2-}$.

Paragraph for Q.18 to Q.19

The compound 'A' is colourless solid. It gives following tests.

Step-I : The addition of compound 'A' solution of KI, followed by acidification with dilute H_2SO_4 , result in the liberation of compound 'B', which may be identified by blue colour produced with starch paste.

Step-II : The acidified KMnO_4 solution decolorized, by a solution of compound 'A', but no gas is evolved.

18. Compound 'A' and 'B' are

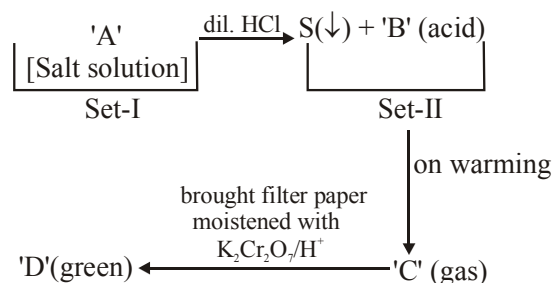
- (a) $\text{Na}_2\text{S}_2\text{O}_3$ & S (b) NaNO_2 & NO
 (c) NaNO_2 & NO_2 (d) $\text{Na}_2\text{S}_2\text{O}_3$ & I_2

19. Which of the following reaction(s) is / are incorrectly matched about step-I and step-II.

- (a) $2\text{NaNO}_3 + 3\text{KI} + 2\text{H}_2\text{SO}_4 \longrightarrow \text{KI}_3 + 2\text{NO} + 2\text{H}_2\text{O} + 2\text{SO}_4^{2-} + 2\text{K}^+ + 2\text{Na}^+$
 (b) $5\text{NaNO}_2 + 2\text{KMnO}_4 + 3\text{H}_2\text{SO}_4 \longrightarrow 5\text{NaNO}_3 + 2\text{MnSO}_4 + \text{K}_2\text{SO}_4 + 3\text{H}_2\text{O}$
 (c) Step-(I) & Step-(II) both give redox reaction
 (d) Step-(I) & Step-(II) both give precipitation reaction

Paragraph for Q.20 to Q.21

Observe the following experiment

**20.** The incorrect statement about set-II is / are :-

- (a) The acidified liquid soon becomes turbid owing to formation of sulphur and H_2SO_4 .
 (b) The sulphur first forms a colloidal solution, which is gradually coagulated by free acid present side reactions also occur giving rise to thionic acids.
 (c) On warming the solution of set up-II, the 'C' gas evolved is H_2S . Which is recognised by its action upon filter paper moistened with acidified $\text{K}_2\text{Cr}_2\text{O}_7$.
 (d) On warming the solution of set up-II, the 'C' gas evolved is SO_2 . Which is recognised by its action upon filter paper moistened with acidified $\text{K}_2\text{Cr}_2\text{O}_7$.

4.20 QUALITATIVE SALT ANALYSIS

21. Which of the following reaction is / are correctly matched about set up - I and set up -II.

- (a) $\text{Na}_2\text{S}_2\text{O}_3 + 2\text{HCl} \longrightarrow \text{S}(\downarrow) + \text{H}_2\text{SO}_3 + 2\text{NaCl}$
- (b) $\text{Na}_2\text{S}_2\text{O}_3 + 2\text{HCl} \xrightarrow{\Delta} \text{S}(\downarrow) + \text{SO}_2(\uparrow) + 2\text{NaCl} + \text{H}_2\text{O}$
- (c) $3\text{SO}_2 + \text{Cr}_2\text{O}_7^{2-} + 2\text{H}^+ \rightarrow 2\text{Cr}^{3+} + 3\text{SO}_4^{2-} + \text{H}_2\text{O}$
- (d) $\text{S} + \text{H}_2\text{SO}_3 \xrightarrow{\Delta} \text{H}_2\text{S}_2\text{O}_3$

Paragraph for Q.22 to Q.23

[Solid 'X' salt + MnO_2 (solid)]

↓ conc. H_2SO_4 added dropwise and on warming

'Y' (coloured gases)

↓ + [KI - starch paper]

[Turns KI-starch paper blue]

22. Here salt 'X' may be :-

- (a) NaCl (b) NaBr
(c) NaI (d) None of these

23. Which of the following statement(s) are correct ?

- (a) MnO_2 oxidizes halides ions into dihalogen
(b) Reducing nature of halides ions follow the order, $\text{F}^- < \text{Cl}^- < \text{Br}^- < \text{I}^-$
(c) 'Y' gas can be I_2 , which turns KI-starch paper into blue colour.
(d) 'Y' gas can be Cl_2 and Br_2 which oxidises of KI to I_2 & turns starch paper blue.

Paragraph for Q.24 to Q.25

X^- (salt solution) + $\text{AgNO}_3 \rightarrow$ 'Y' ppt. $\xrightarrow{\text{Na}_2\text{AsO}_3}$ 'Z' (yellow ppt.)

↓

Soluble in NH_4OH and $\text{Na}_2\text{S}_2\text{O}_3$

24. 'X' salt is :-

- (a) NaCl
(b) NaBr
(c) Na_2CO_3
(d) Na_2SO_3

25. The incorrect statement about above experiment is

- (a) 'Y' is AgCl and soluble in dil. HNO_3
(b) 'Z' is Ag_3AsO_3
(c) 'Y' is soluble in $\text{Na}_2\text{S}_2\text{O}_3$ and forms $\text{Na}_3[\text{Ag}(\text{S}_2\text{O}_3)_2]$
(d) AgCl forms fulminating silver (Ag_3N) with NH_3 on standing for a long time

Paragraph for Q.26 to Q.27

(I) Borax + conc. $\text{H}_2\text{SO}_4 \xrightarrow{\Delta}$ 'A' (white fumes)

↓ conc. HCl

[A] (↓) (white ppt.)

(II) [Borax + conc. $\text{H}_2\text{SO}_4 + \text{CH}_3\text{OH}$]

↓ on ignition in porcelain basin

['B' burns with green edge flames]

26. Compound 'A' is :-

- (a) HCl (b) H_3BO_3
(c) SO_2 (d) B_2O_3

27. Correct statement about 'B' is / are :-

- (a) green edge flame occurs due to formation of methyl borate
(b) 'B' is $\text{B}(\text{OCH}_3)_3$
(c) On burning, the paste of borax + CaF_2 and conc. H_2SO_4 on bunsen flame, gives same green edged flame.
(d) Borax gets hydrolysed into boric acid in acidic medium and then form ester of methyl borate with methyl alcohol.

Paragraph for Q.28 to Q.29

[KMnO_4 + 'X' salt solution]

↓ BaCl_2 (aq.) is added

('Y' pink ppt. + pink solution)

↓ H_2O_2 is added

('Z' pink ppt. + colourless solution)

28. The compound of 'X' and 'Y' are :-

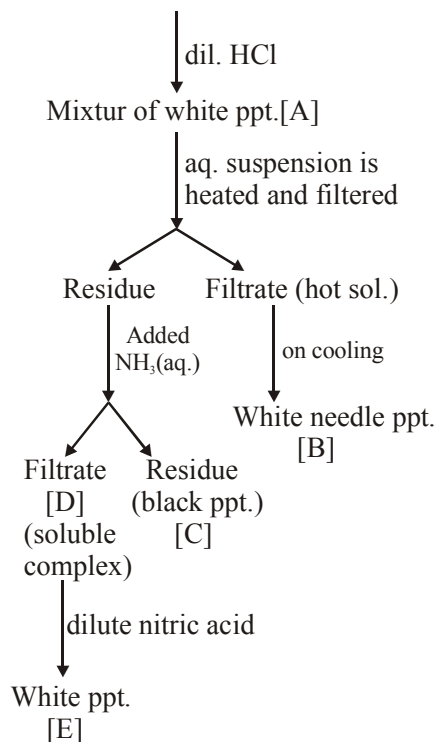
- (a) Na_2SO_3 , BaSO_4
(b) Na_2SO_4 , BaSO_4
(c) $\text{Na}_2\text{S}_2\text{O}_3$, $\text{MnO}(\text{OH})_2$
(d) Na_2SO_4 , ($\text{BaSO}_4 + \text{MnO}(\text{OH})_2$)

29. Which of the following statement is / are incorrect about of 'Y' and 'Z' ?

- The pink colour is due to adsorption of KMnO_4 on BaSO_4 surface
- H_2O_2 can not reduce adsorbed KMnO_4 on BaSO_4 surface
- The compound 'Y' also formed when BaSO_3 is exposed to air and product is added in KMnO_4 solution
- 'Y' and 'Z' are nonidentical precipitate

Paragraph for Q.30 to Q.32

Mixture of unknown salt



30. Mixture of white ppt. [A] may contain.

- | | |
|--------------------------------|--------------------|
| (I) PbCl_2 | (II) AgCl |
| (III) Hg_2Cl_2 | |
| (a) Only I | (b) Only II, III |
| (c) Only III | (d) I, II and III |

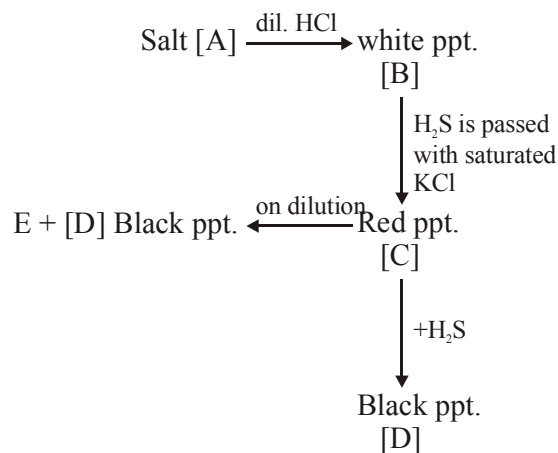
31. Compound 'B' is

- | | |
|------------------------------|-----------------------------------|
| (a) PbCl_2 | (b) AgCl |
| (c) Hg_2Cl_2 | (d) $(\text{Hg} + \text{HgCl}_2)$ |

32. Co-ordination number of central metal ion in filtrate (soluble complex) 'D' is -

- | | |
|-------|-------|
| (a) 4 | (b) 2 |
| (c) 6 | (d) 5 |

Paragraph for Q.33 to Q.34



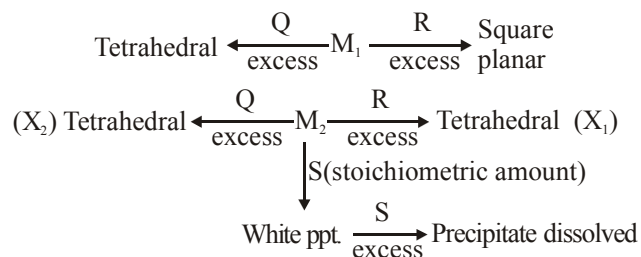
33. Compound 'D' is :-

- | | |
|--------------------------------|-------------------------------|
| (a) $(\text{Hg} + \text{HgS})$ | (b) Ag_2S |
| (c) PbS | (d) Pb_2SCl_2 |

34. The correct statement about [C] is -

- Red precipitate is formed due to formation of lead sulphochloride
- Red precipitate is formed due to formation of lead sulphochloride
- Red precipitate is formed due to formation of lead sulphochloride
- Red precipitate is formed due to formation of lead sulphochloride

Paragraph for Q.35 to Q.37



35. M_1 , Q & R respectively are :-

- | | |
|--|--|
| (a) Zn^{2+} , KCl , KCN | (b) Ni^{2+} , KCl , KCN |
| (c) Ni^{2+} , KCN , HCl | (d) Zn^{2+} , KCN , KCl |

36. M_2 , X_1 & X_2 respectively are :-

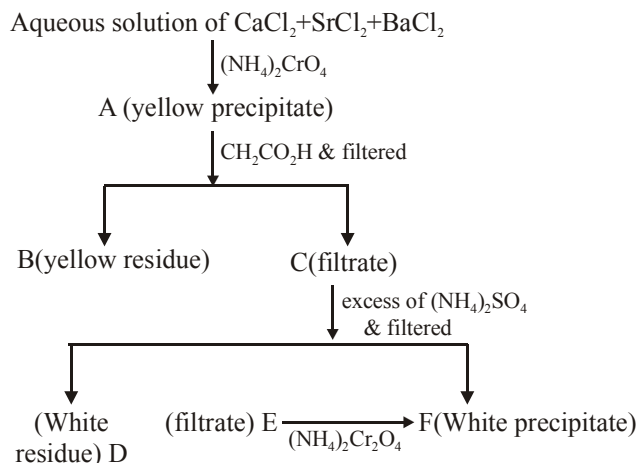
- | |
|---|
| (a) Zn^{2+} , $\text{Zn}(\text{CN})_4^{2-}$, ZnCl_4^{2-} |
| (b) Ni^{2+} , $\text{Ni}(\text{CN})_4^{2-}$, BiCl_4^{2-} |
| (c) Zn^{2+} , ZnCl_4^{2-} , $\text{Zn}(\text{CN})_4^{2-}$ |
| (d) Ni^{2+} , NiCl_4^{2-} , $\text{Zn}(\text{CN})_4^{2-}$ |

37. S is :-

- | | |
|--------------------------|-------------------|
| (a) H_2S | (b) NaOH |
| (c) HNO_3 | (d) KI |

4.22 QUALITATIVE SALT ANALYSIS

Paragraph for Q.38 to Q.40



38. Yellow precipitate A contains :-

- (a) BaCrO_4 only (b) $\text{BaCrO}_4 + \text{SrCrO}_4$
 (c) CaCrO_4 only (d) $\text{CaCrO}_4 + \text{BaCrO}_4$

39. E(aq.) is :-

- (a) $(\text{NH}_4)_2[\text{CaSO}_4]$ (b) $(\text{NH}_4)_2[\text{Ca}(\text{SO}_4)_2]$
 (c) $(\text{NH}_4)_2[\text{Ca}(\text{SO}_4)_4]$ (d) $\text{Ca}(\text{SO}_4)_2$

40. D gives :-

- (a) Apple green colour in flame test
 (b) Brick red colour in flame test
 (c) Golden yellow colour in flame test
 (d) Crimson red colour in flame test

Paragraph for Q.41 to Q.43

The elements Mg, Cr, Sr and Ba react with hydrogen to form hydrides ' MH_2 '. Beryllium hydride is difficult to prepare and less stable than the others.

41. Hydride, which act as a reducing agent :-

- (a) CaH_2 (b) SrH_2
 (c) BaH_2 (d) All

42. Which of the following hydride is polymeric and covalent in nature in solid ?

- (a) BeH_2 (b) SrH_2
 (c) BaH_2 (d) All

43. On hydrolysis with water and dilute acids, alkaline earth metal hydrides produces :-

- (a) O_2 gas
 (b) H_2 gas
 (c) O_3 gas
 (d) CO_2 gas

Paragraph for Q.44 to Q.46

The trends in properties of alkaline earth metal salts can be explained by using Fajan's rule. It is associated with incorporation of covalent character in ionic bond. It states that smaller cations form more covalent compound and ionic character explains the solubility, thermal stability, hydrolysis and other properties of these salts. Some exceptions are also there and are explained by change in lattice energy.

44. Correct order of increasing thermal stability of alkaline earth metal sulphates is :-

- (a) $\text{SrSO}_4 < \text{CaSO}_4 < \text{MgSO}_4 < \text{BeSO}_4$
 (b) $\text{BeSO}_4 < \text{MgSO}_4 < \text{CaSO}_4 < \text{SrSO}_4$
 (c) $\text{CaSO}_4 < \text{BeSO}_4 < \text{MgSO}_4 < \text{SrSO}_4$
 (d) $\text{MgSO}_4 < \text{BeSO}_4 < \text{CaSO}_4 < \text{SrSO}_4$

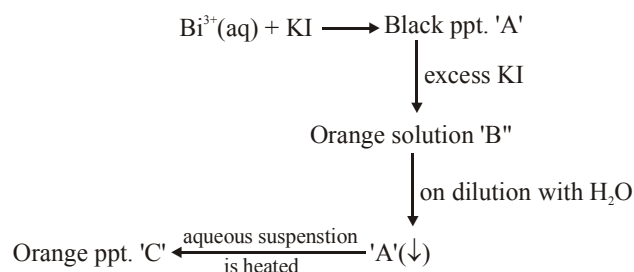
45. Which salts does not show hydrolysis at all ?

- (a) MgCl_2 (b) BeCl_2
 (c) AlCl_3 (d) BaCl_2

46. Which alkaline earth metal form superoxide ?

- (a) Be
 (b) Mg
 (c) Ca
 (d) None

Paragraph for Q.47 to Q.48



47. Orange solution 'B' is due to formation of :-

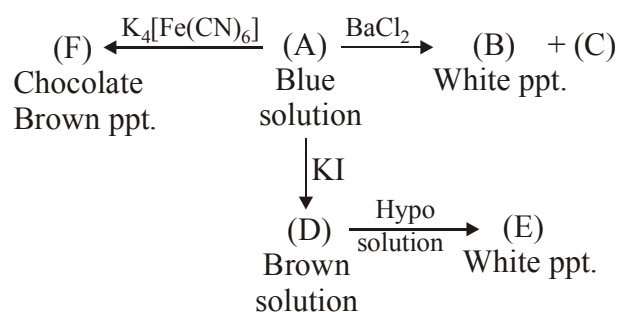
- (a) BiI_3
 (b) $[\text{BiI}_4]^-$
 (c) $\text{Bi}(\text{OH})_2\text{I}$
 (d) BiOI

48. Compound 'C' is :-

- (a) BiI_3
 (b) BiOI
 (c) $\text{Bi}(\text{OH})_2\text{I}$
 (d) $[\text{BiI}_4]^-$

Paragraph for Q.49 to Q.51

Following observations are given :



49. Correct statement :-

- (a) Compound F is $Cu_2Fe(CN)_6$
- (b) Compound E is CuI
- (c) Both (A) & (B)
- (d) None of these

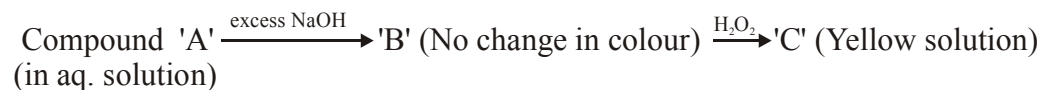
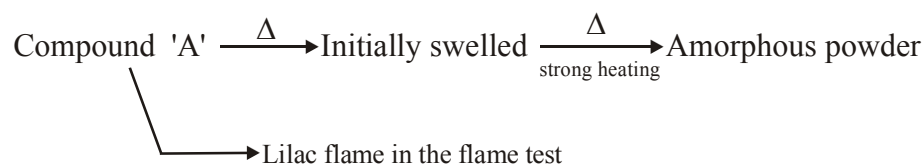
50. Brown color solution is due to presence of :-

- (a) $Cu_2[Fe(CN)_6]$
- (b) Cu_2I_2
- (c) KI_3
- (d) $CuSO_4$

51. The incorrect statement(s) about above flow diagram is / are :-

- (a) Adding an excess of hyposolution to the brown solution, 'd' is reduced to colourless solution and white precipitate becomes visible.
- (b) F is $Cu_3[Fe(CN)_6]_2$
- (c) The reduction of hyposolution yield tetrathionate ions
- (d) The co-ordination number of iron in the chocolate brown ppt.(F) is changed than the $K_4[Fe(CN)_6]$

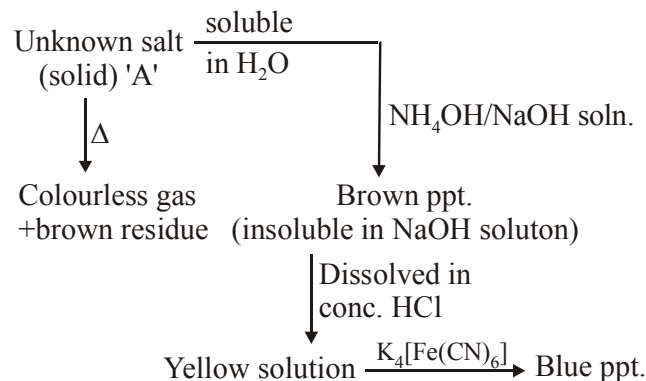
Paragraph for Q.56 to Q.58



56. Compound 'A' is having water of crystallization by the number of

- (a) 10
- (b) 20
- (c) 24
- (d) 36

Paragraph for Q.52 to Q.53



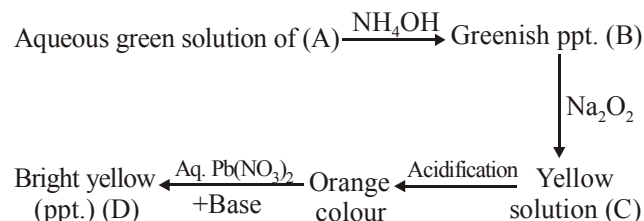
52. Identify A :-

- (a) $FeSO_4$
- (b) $Fe_2(SO_4)_3$
- (c) $FeCl_3$
- (d) $CrCl_3$

53. What is the formula of brown ppt. :-

- (a) $Cr(OH)_3$
- (b) $Fe(OH)_2$
- (c) $FeCl_3$
- (d) $Fe(OH)_3$

Paragraph for Q.54 to Q.55



54. Identify A :-

- (a) $AlCl_3$
- (b) $Cr(OH)_3$
- (c) $CrCl_3$
- (d) $FeCl_3$

55. Identify D :-

- (a) Na_2CrO_4
- (b) $PbCrO_4$
- (c) Na_2MnO_4
- (d) $Fe(OH)_3$

4.24 QUALITATIVE SALT ANALYSIS

58. The hybridization 'B' of compound 'C' is

- (a) sp^3 (b) sp^3d
(c) d^2sp^3 (d) d^3s

Paragraph for Q.59 to Q.62

Light green colour compound (A) $\xrightarrow{\Delta}$ white residue (B)

(B) $\xrightarrow[\text{temp}]{\text{high}}$ C + D + E

D and E are acidic in nature. When A is passed through $HgCl_2$, yellow precipitate is observed. D is passed through H_2S , white turbidity is obtained :-

59. D and E are :-

- (a) SO_2 , SO_3 (b) SO_2 , CO_2
(c) CO_2 , CO (d) SO_2 , CO

60. Yellow precipitate is :-

- (a) HgO
(b) Basic Hg (I) sulphate
(c) Basic Hg (II) sulphate
(d) HgI_2

61. C is soluble in :-

- (a) conc. HCl (b) dil. H_2SO_4
(c) AcOH (d) Na_2CO_3

62. Number of water of crystallization in A is :-

- (a) 0 (b) 2
(c) 7 (d) 5

Paragraph for Q.63 to Q.64

An aqueous solution of a mixture of two inorganic salts, when treated with dilute HCl, gave a precipitate (P) and a filtrate (Q). The precipitate (P) was found to dissolve in hot water. The filtrate (Q) remained unchanged, when treated with H_2S in an acidic medium. However, it gave a precipitate (R) with H_2S in an ammoniacal medium. The precipitate (R) gave a coloured solution (S), when treated with H_2O_2 in an aqueous NaOH medium.

63. The precipitate P contain :-

- (a) Pb^{2+} (b) Hg_2^{2+}
(c) Ag^+ (d) Hg^{2+}

64. The colour solution S contains :-

- (a) $Fe_2(SO_4)_3$ (b) $CuSO_4$
(c) $ZnSO_4$ (d) Na_2CrO_4

Paragraph for Q.65 to Q.68

(I) (X) $\xrightarrow{\Delta}$ glassy transparent bead (Y)

on platinum wire

(Y) + $CuSO_4 \longrightarrow$ coloured bead (Z)

(II) (X) + conc. H_2SO_4 + $CH_3CH_2OH \xrightarrow{\text{ignite}}$

green flame (W)

(III) Aqueous solution of (X) is alkaline.

65. X is :-

- (a) $NaNH_4HPO_4 \cdot 4H_2O$
(b) $Na_2B_4O_7 \cdot 10H_2O$
(c) $CuSO_4 \cdot 5H_2O$
(d) None of these

66. Y is :-

- (a) $NaPO_3$ (b) $NaBO_2$
(c) $NaBO_2 + B_2O_3$ (d) None of these

67. Z is :-

- (a) $Cu_3(PO_4)_2$ (b) $CuSO_4$
(c) $Cu(BO_2)_2$ (d) None of these

68. W is :-

- (a) $(CH_3)_3BO_3$ (b) $(C_2H_5)_3BO_3$
(c) H_3BO_3 (d) None of these

Paragraph for Q.69 to Q.71

Microcosmic salt is used to detect different metal ions. Microcosmic salt is first heated to give glassy bead of X. X when reacts with different metal oxide, different colour is observed.

69. Microcosmic salt is :-

- (a) $Na(NH_4)HPO_4 \cdot 4H_2O$
(b) $Na_2HPO_4 \cdot 4H_2O$
(c) $Na_3PO_4 \cdot 4H_2O$
(d) $Na(NH_4)HPO_4 \cdot 2H_2O$

70. The correct statement about microcosmic salt :-

- (a) It is sodium ammonium hydrogen phosphate tetrahydrate
(b) It gives canary yellow precipitate with ammonium molybdate solution
(c) Both (A) and (B)
(d) None of these

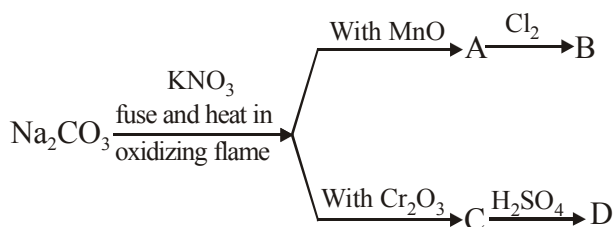
71. Microcosmic salt $\xrightarrow{\text{NaOH}/\Delta}$ Basic gas (Z) :-

Aqueous solution of Z gives :-

- (a) Blue colour precipitate with CuSO_4 solution
- (b) Brown precipitate which is called iodide of millons base with $\text{K}_2[\text{HgI}_4]$
- (c) White precipitate with ZnCl_2 which is dissolved when Z is added in excess
- (d) All of these

Paragraph for Q.72 to Q.75

Identification of cations can also be done using a dry test called sodium carbonate bead test which is similar to borax - bead test :-



72. The hybridisation and colour of A with molecular formula ?

- (a) Na_2MnO_4 , d^3s , Green
- (b) NaMnO_4 , sp^3 , Purple
- (c) Mn_2O_7 , d^3s , Red
- (d) $\text{Na}_2\text{Cr}_2\text{O}_7$, d^3s , Orange

73. The hybridisation and colour of B with molecular formula ?

- (a) Na_2MnO_4 , d^3s , Green
- (b) NaMnO_4 , sp^3 , Purple
- (c) Mn_2O_7 , d^3s , Red
- (d) $\text{Na}_2\text{Cr}_2\text{O}_7$, d^3s , Orange
- (d) None of these

74. C and its colour is :-

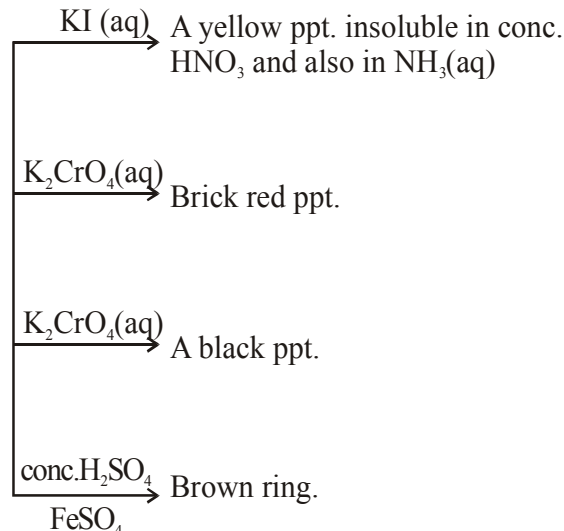
- (a) Na_2CrO_4 , orange
- (b) $\text{Na}_2\text{Cr}_2\text{O}_7$, orange
- (c) $\text{Na}_2\text{Cr}_2\text{O}_7$, yellow
- (d) Na_2CrO_4 , yellow

75. D and number of unpaired electron in D are :-

- (a) Na_2CO_3 , 1
- (b) NaMnO_4 , 2
- (c) $\text{Na}_2\text{Cr}_2\text{O}_7$, 0
- (d) Na_2CrO_4 , 0

Paragraph for Q.76 to Q.78

Crystalline salt (A) $\xrightarrow[\text{in water}]{\text{Dissolved}}$



76. Crystalline salt A is

- (a) AgNO_3
- (b) $\text{Pb(NO}_3)_2$
- (c) AgNO_2
- (d) $\text{Hg(NO}_3)_2$

77. Black ppt. and yellow ppt. are respectively

- (a) PbS , PbI_2
- (b) HgS , HgI_2
- (c) Ag_2S , AgI
- (d) $(\text{Hg} + \text{HgS})$, Hg_2I_2

78. Which of the following options are correct regarding A?

- (a) A turns our skin black
- (b) A gives white precipitate with NaOH which produces silver mirror with glucose solution
- (c) A gives white precipitate with KCN which is soluble in excess KCN
- (d) All

Paragraph for Q.79 to Q.81

A white salt (A) does not dissolve in water, but dissolve in dilute nitric acid solution. Passing H_2S gas through acidic solution of A, a black precipitate A_1 was produced which was insoluble in water, dilute HNO_3 and alkali but dissolved in concentrated solution of Na_2S as well as in aqua-regia. Adding SnCl_2 -dropwise to the solution of A, gave a white, silky precipitate (B), which turned into black precipitate (C) on adding excess reagent. Addition of KI to the acidified solution of A gave a red coloured precipitate (D) in the beginning, which is dissolved in excess reagent.

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79. A is called

- (a) Calomel (b) Corrosive sublimate
(c) Brine solution (d) Rock salt

80. The formula of B and D are respectively

- (a) Hg_2Cl_2 , HgI_2 (b) HgCl_2 , HgI_2
(c) Pb_2Cl_2 , PbI_2 (d) Hg_2Cl_2 , Hg_2I_2

81. Black ppt. A₁ and black ppt. C are respectively

- (a) Ag_2S , Ag (b) HgS , Hg
(c) Hg_2S , Hg (d) PbS , Pb

Paragraph for Q.82 to Q.84

An inorganic compound (A), transparent like glass is a strong reducing agent. Its hydrolysis in water gives white turbidity (). Aqueous solution of (A) gives white precipitate (C) with NaOH (aq) which is soluble in excess NaOH. (A) also reduce I_2 and gives chromyl chloride test.

A + corrosive sublimate $\rightarrow$ Grey mass + D.

82. Grey mass and D are respectively

- (a) Hg and SnCl_2 (b) Ag and Cu_2Cl_2
(c) Hg and SnCl_4 (d) Hg and Cu_2Cl_2

83. What is the correct option regarding A.

- (a) A reduces auric chloride to produce purple of cassius.
(b) A reduces ferric chloride to iron.
(c) A reduces cupric chloride to red deposit copper
(d) All

84. A is

- (a) CdCl_2 (b) CuCl_2
(c) SnCl_2 (d) SnCl_4

Paragraph for Q.85 to Q.87

A mixture consists of an yellow salt (A) and a white salt (B), both anhydrous. Salt mixture was dissolved in water and few drops of HCl was added to obtain a clear, yellow coloured solution. The solution was then treated with $\text{NH}_3/\text{NH}_4\text{Cl}$ solution when a reddish-brown precipitate (C), insoluble in NaOH solution, was produced. Solution was filtered off and filtrate was treated with aqueous NaOH solution when a white precipitate (D), soluble in excess reagent was formed.

Extract solution in concentrated H_2SO_4 when mixture mixed with $\text{K}_2\text{Cr}_2\text{O}_7$, gave a dense brown fumes on boiling, that turned aqueous NaOH solution yellow. Extract solution when treated with $\text{Hg}(\text{NO}_3)_2$ a yellow precipitate (E) was formed.

85. The acidic radical present in the mixture

- (a) A consists of Cl^- , B consists of SO_3^{2-}
(b) A consists of Br^- , B consists of SO_3^{2-}
(c) A consists of Cl^- , B consists of SO_4^{2-}
(d) A consists of Cl^- , B consists of CO_3^{2-}

86. Yellow precipitate is

- (a) $\text{Hg}(\text{OH})_2$, HgCO_3 (b) HgO
(c) $\text{HgSO}_4 \cdot 2\text{HgO}$ (d) HgCl_2

87. The basic radical present in the mixture

- (a) A consists of Fe^{3+} , B consists of Pb^{2+}
(b) A consists of Fe^{2+} , B consists of Pb^{2+}
(c) A consists of Fe^{3+} , B consists of Zn^{2+}
(d) A consists of Fe^{2+} , B consists of Zn^{2+}

Paragraph for Q.88 to Q.90

Substance (A) is a yellowish-white deliquescent solid which sublimes and has a vapour density of 133. (A) reacted violently with water forming solution (B). A sample of (B) gave a curdy white precipitate (C) on addition of dilute HNO_3 and AgNO_3 solution, but is readily dissolved on the addition of dilute NH_4OH , though a gelatinous white precipitate was formed in its place. (D) was filtered off and dissolved in excess NaOH, forming a clear solution (E). When CO_2 was passed into (E), compound (D) was reprecipitated.

Qualitative analysis of solution (B) gave a white gelatinous precipitate in Group III. When 0.1333 g of (A) was dissolved in water and treated with 8-hydroxyquinoline. 0.4594 g of precipitate was obtained.

88. A is

- (a) FeCl_3 (b) CrCl_3
(c) ZnCl_2 (d) AlCl_3

89. The coordination number of metal ion in E

- (a) 1 (b) 2
(c) 3 (d) 4

90. The same gelatinous white precipitate was formed when (A) reacts with

- (a) $(\text{NH}_4)_2\text{S}$ (b) Na_2S
(c) Na_2CO_3 (d) All

Paragraph for Q.91 to Q.93

A white powder turns yellow (B) on heating and evolves a gas which turns lime water milky, as well as water vapour. The yellow residue turns white on cooling but turns yellow again when heated B reacts with dilute sulphuric acid to give a colourless solution (C). If dilute NaOH is added to C, a white precipitate (D) is formed initially, which dissolves on adding excess base. With dilute ammonia solution C gives a white precipitate which dissolves in excess ammonia, giving a clear solution (E) due to complex formation.

91. A is

- (a) PbCO_3 (b) CaCO_3
(c) ZnCO_3 (d) NiCO_3

92. The coordination number of metal ion in the complex is

- (a) 2 (b) 4
(c) 6 (d) 5

93. The shape of the complex is

- (a) Square planer
(b) Tetrahedral
(c) Inner orbital Octahedral
(d) Outer orbital Octahedral

Paragraph for Q.94 to Q.95

A solid mixture consists of a reddish brown metal oxide (A) and a white hydrated salt B. The mixture was dissolved in dilute HNO_3 and a portion of this solution was treated with NH_3 solution, just to make the solution neutral and then finally treated with $\text{NH}_3/\text{NH}_4\text{Cl}$ buffer solution when a reddish brown gelatinous precipitate (C) was obtained. C was then dissolved in dilute HCl and few drops of NH_4SCN solution was added when a deep blood red colouration was observed. Filtrate obtained after removal of C was then treated with excess of $(\text{NH}_4)_2\text{S}$ solution when a pink coloured precipitate (D) was formed. D was then dissolved in dilute HNO_3 and treated with NaBiO_3 solution. A deep purple coloured solution was formed.

In a separate experiment a pinch of original salt mixture was mixed with solid $\text{K}_2\text{Cr}_2\text{O}_7$ and then dissolved in concentrated H_2SO_4 and finally boiled. A deep red fumes of (E) was obtained which made aqueous solution of NaOH yellow, when passed through it.

94. C and D are respectively :-

- (a) $\text{Fe}(\text{OH})_3$, CoS (b) $\text{Fe}(\text{OH})_3$, MnS
(c) $\text{Fe}(\text{OH})_2$, CoS (d) $\text{Fe}(\text{OH})_2$, MnS

95. The anion present in B -

- (a) NO_3^- (b) Br^-
(c) Cl^- (d) NO_2^-

Paragraph for Q.96 to Q.97

Three black powder A, B and C have to be identified. A dissolves in dilute HCl to give a blue solution which becomes deep blue with the addition of excess of NH_4OH . The solution of A in dilute HNO_3 , on evaporation yields blue crystals which, on strong heating reagents A. B dissolves completely, on boiling with concentrated HNO_3 yielding colourless brown fumes. The resulting solution after proper dilution yields no positive test for any cation. C does not dissolve in dilute HCl, but goes into solution on boiling with concentrated HCl and a crystal of KClO_3 . Evaporation of this solution yields colourless crystals which dissolve in water. The solution yields a scarlet precipitate when a little KI solution is added but redissolves in excess of KI solution.

96. A & B is -

- (a) Hg_2O , Copper (b) CuO , Coke powder
(c) Ag, Hg (d) Cu, Hg

97. C is -

- (a) PbS (b) HgS
(c) Ag_2S (d) Bi_2S_3

Paragraph for Q.98 to Q.100

A yellow deliquescent solid (a) is soluble in water, alcohol and ether. Its aqueous solution turns blue litmus red. (a) gives following reactions :

- On strong heating it gives a dark brown residue (b) water and a compound, (c) which gives white fumes with NH_3 gas. Compound (c) in aqueous solution decomposes carbonates to evolve CO_2 gas.
- When dry chlorine gas is passed over a heated mixture of (b) and carbon, one another compound (d) is formed. The colour of solid (d) is black red. It dissolves in water giving a yellow solution.
- When aqueous solution of (d) is heated, a brown precipitate (e) is formed. The hydrolysis can be checked by the addition of HCl to aqueous solution.

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- (iv) (d) absorbs water vapours to give compound (a).
 (d) also absorbs NH_3 gas to form an addition compound (F).
 (v) A solution of (d) reacts with ammonium thiocyanate to produce a red colour of compound (G).
 (vi) A solution of (d) reacts with $\text{K}_4[\text{Fe}(\text{CN})_6]$ solution to give a deep blue colour (H). A solution of (d) liberated iodine from KI solution. H_2S is oxidized to free sulphur and SO_2 to sulphuric acid.

98. A is -

- (a) FeCl_3 (b) AlCl_3
 (c) FeCl_2 (d) SbCl_3

99. Red colour compound (G) and deep blue colour (H) are respectively -

- (a) $[\text{Fe}(\text{H}_2\text{O})_5\text{SCN}]^{2+}$, $\text{Fe}_4[\text{Fe}(\text{CN})_6]_3$
 (b) $[\text{Fe}(\text{H}_2\text{O})_5\text{SCN}]^{1+}$, $\text{Fe}_3[\text{Fe}(\text{CN})_6]_2$
 (c) $[\text{Fe}(\text{SCN})_2]$, $\text{Fe}_2[\text{Fe}(\text{CN})_6]$
 (d) $[\text{Sb}(\text{SCN})_3]$, $\text{Fe}_4[\text{Fe}(\text{CN})_6]_3$

100. Brown precipitate (E) and addition compound F are -

- (a) $\text{Fe}(\text{OH})_3$, $[\text{Fe}(\text{NH}_3)_6]^{3+}$
 (b) $\text{Fe}(\text{OH})_2$, $[\text{Fe}(\text{NH}_3)_6]^{2+}$
 (c) $\text{Fe}(\text{OH})_3$, $[\text{Fe}(\text{NH}_3)_6]^{2+}$
 (d) $\text{Fe}(\text{OH})_2$, $[\text{Fe}(\text{NH}_3)_6]^{3+}$

Matrix Match Type :

101. Match the column-I with column-II :-

Column - I (Reactants)	Column - II (Products)
(a) $\text{Zn} + \text{dil. H}_2\text{SO}_4$	(P) Metallic salt
(b) $\text{Fe} + \text{dil. HNO}_3$	(Q) SO_2
(c) $\text{Zn} + \text{conc. H}_2\text{SO}_4$	(R) Oxyacid
(d) $\text{P} + \text{conc. H}_2\text{SO}_4$	(S) N_2O (T) H_2 or H_2O

102. Match the column-I with column-II :-

Column - I (Reactant)	Column - II (Product)
(a) $\text{Ag} + \text{conc. H}_2\text{SO}_4$	(P) Acidic oxide
(b) $\text{Ag} + \text{conc. HNO}_3$	(Q) Metal nitrate
(c) $\text{As} + \text{conc. H}_2\text{SO}_4$	(R) Metal sulphate
(d) $\text{Sn} + \text{conc. HNO}_3$	(S) Oxy acid (T) Water

103. Match the column-I with column-II :-

Column - I (Reagents)	Column - II (Compound)
(a) Catalytic test $[\text{NaN}_3 + \text{KI}_3]$	(P) Na_2S
(b) Sodium nitroprusside test	(Q) $\text{Na}_2\text{S}_2\text{O}_3$
(c) Blood red colour with FeCl_3	(R) NaSCN
(d) Black ppt. with AgNO_3 at room temperature	(S) Na_2CO_3 (T) Na_2SO_3

104. Match the column-I with column-II :-

Column - I (Acidic radical)	Column - II (Observations)
(a) NO_2^-	(P) Gives brown ring test
(b) NO_3^-	(Q) Gives brown ppt. with Nessler's reagent
(c) NH_4^+	(R) Gives blood red colouration with KSCN
(d) Fe^{3+}	(S) Gives red colouration with CH_3COONa (T) Radical is reduced to ammonia with $\text{Zn} + \text{conc. NaOH}$

105. Match the column-I with column-II :-

Column - I (Acidic radical)	Column - II (Observations)
(a) Cl^- salt solution	(P) Gives positive layer test (with chlorine water)
(b) Br^- salt solution	(Q) Yellow color with excess chlorine water in organic layer
(c) I^- salt solution	(R) Brown fumes with conc. H_2SO_4
(d) NO_2^- salt solution	(S) Ppt. formed with conc. AgNO_3 (T) Coloured gas with $\text{K}_2\text{Cr}_2\text{O}_7 + \text{conc. H}_2\text{SO}_4$

106. Match the column-I with column-II :-

Column - I (Reagents)	Column - II (Anion gives ppt. change in colour)
(a) $\text{FeCl}_3(\text{aq.})$	(P) $\text{SO}_4^{2-}(\text{aq.})$
(b) $\text{BaCl}_2(\text{aq.})$	(Q) $\text{CH}_3\text{COO}^-(\text{aq.})$
(c) $\text{AgNO}_3(\text{aq.})$	(R) $\text{PO}_4^{3-}(\text{aq.})$
(d) $(\text{NH}_4)_2\text{MoO}_4(\text{aq.})$	(S) $\text{S}_2\text{O}_3^{2-}(\text{aq.})$

107. Column-I lists some of group reagents, give characteristics colour / precipitate mentioned cation in column-II. Match each entry of column-I with those given in column-II :-

Column - I (Group reagents)	Column - II (Cations)
(a) H_2S in $\text{NH}_4\text{OH}(\text{aq.})$	(P) $\text{Bi}^{3+}(\text{aq.})$
(b) H_2S in $\text{HCl}(\text{aq.})$	(Q) $\text{Pb}^{2+}(\text{aq.})$
(c) $(\text{NH}_4)_2\text{CO}_3$ in $\text{NH}_4\text{Cl}(\text{aq.})$	(R) $\text{Ca}^{2+}(\text{aq.})$
(d) NH_4OH in NH_4Cl	(S) $\text{Mn}^{2+}(\text{aq.})$ (T) $\text{Al}^{3+}(\text{aq.})$

108. Match the colour of precipitate listed in Column-I with the reagent(s) listed in column-II.

Column - I (Observations)

- (a) Ag^+ gives black / brown ppt. with
 (b) Hg_2^{2+} gives black ppt. with
 (c) Pb^{2+} gives yellow ppt. with
 (d) Mg^{2+} gives pink mass

Column - II (Reagents)

- (P) $\text{Co}(\text{NO}_3)_2$ in charcoal cavity
 (Q) KI
 (R) H_2S (saturated in H_2O)
 (S) K_2CrO_4 solution
 (T) NaOH solution

109. Match the colour :-

Column - I (Basic radical)

- (a) Cu^{2+}
 (b) Fe^{3+}
 (c) Cd^{2+}
 (d) Ag^+

Column - II (Property / Test)

- (P) Produces soluble complex with excess of KCN without any change of oxidation state of metal ion
 (Q) Produces coloured soluble complex with excess of KCN
 (R) Produces no ppt. with excess of KI
 (S) Produces coloured ppt. with NaOH
 (T) Produces colourless soluble complex with excess of NH_4OH

110. Match the column-I with column-II :-

Column - I (Basic radical)	Column - II (Property / Test)
(a) Fe^{3+}	(P) Produces white ppt. with $\text{K}_4[\text{Fe}(\text{CN})_6]$
(b) Zn^{2+}	(Q) Produces soluble complex in excess of NaOH solution.
(c) Cr^{3+}	(R) Give redox reaction with $\text{Na}_2\text{S}_2\text{O}_3$ solution.
(d) Al^{3+}	(S) Produces corresponding hydroxide with aqueous Na_2S (T) Produces reddish brown ppt. with CH_3CO_2^- solution

111. Match the column-I with column-II :-

Column - I (Reagent)	Column - II (Name of product / colour)
(a) $\text{CH}_3\text{CO}_2^- + \text{As}_2\text{O}_3$	(P) Thenard Blue
(b) Na-K tartarate + CuSO_4	(Q) Tollens reagent
(c) $\text{NH}_4\text{OH} + \text{AgNO}_3$	(R) Fehling solution
(d) $\text{Al}_2\text{O}_3 + \text{Co}(\text{NO}_3)_2$	(S) Cacodyl oxide - Extremely nauseating odour

112. Match the column-I with column-II :-

Column - I (Decolourise acidic KMnO_4 solution)	Column - II (Given salt)
(a) Only acidic radical	(P) $\text{Sn}(\text{NO}_3)_2$
(b) Only basic radical	(Q) KNO_2
(c) Both radical	(R) FeC_2O_4
(d) Neither acidic nor basic radical	(S) Na_2CO_3 (T) BaSO_3

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113. Match the column-I with column-II :-

Column - I Color of the sublimate	Column - II Compound present
(a) White	(P) NH_4^+ , HgCl_2 , Hg_2Cl_2
(b) Yellow	(Q) As_2O_3
(c) Orange	(R) Sb_2O_3
(d) Black	(S) HgS

114. Match the column-I with column-II :-

Column - I (Anion)	Column - II (Property)
(a) Peroxides	(P) Diamagnetic
(b) Superoxides	(Q) Paramagnetic
(c) Oxides	(R) Bond order is 1.5
(d) Hydroxides	(S) Total number of electrons are 17
	(T) Total number of electrons are 18

Column - I Precipitate	Column - II Dissolving solution
(a) Black CuS	(P) NaOH solution
(b) White AgCl	(Q) Sodium peroxide solution
(c) Green $\text{Cr}(\text{OH})_3$	(R) 50% HNO_3
(d) White $\text{Zn}(\text{OH})_2$	(S) Excess NH_3 solution
	(T) KCN solution

116. Column - I (Compound of same metal with different oxidation state)

- (a) $\text{Cu}(\text{SCN})_2$, CuSCN
 (b) $\text{Cu}(\text{CN})_2$, CuCN
 (c) HgI_2 , Hg_2I_2
 (d) HgO , Hg_2O

Column - II (Colour of the ppt)

- (Q) Yellow, white
 (R) Red, Green
 (P) Black, white
 (S) Yellow, Black

117. Match the colour of ppt. listed in column-I with the reagent(s) in column-II :-

Column - I (Observation/Test)

- (a) Bi^{3+} gives black ppt. with
 (b) Cu^{2+} gives black ppt. with
 (c) Cd^{2+} gives white ppt. / bluish - white pp. with
 (d) Ag^+ gives black ppt. with

Column - II (Reagents)

- (P) Saturated solution of H_2S in H_2O
 (Q) $\text{KSCN}(\text{aq.})$
 (R) $\text{K}_4[\text{Fe}(\text{CN})_6](\text{aq.})$
 (S) dil. HCl
 (T) $\text{NaH}(\text{aq.})$

118. Column - I
Pair of Same colour
precipitate

- | | |
|--|------------------|
| (a) CdS , SnS_2 | (P) Brownish red |
| (b) PbS , $\text{Ni}(\text{OH})_3$ | (Q) Black |
| (c) $\text{Fe}(\text{OH})_3$, Ag_3AsO_4 | (R) Yellow |
| (d) $\text{Ni}(\text{OH})_2$, $\text{Cu}_3[\text{Fe}(\text{CN})_6]_2$ | (S) Green |

119. Match the column-I with column-II :-

Column - I (Effect on heating)	Column - II (Metal)
(a) Amphoteric metal oxide	(P) Pb
(b) Metal acetate $\xrightarrow{\Delta}$ acetone + metal carbonate	(Q) Zn
(c) Metal carbonate $\xrightarrow{\Delta}$ metal oxide + $\text{CO}_2 \uparrow$	(R) Na
(d) Metal nitrate $\xrightarrow{\Delta}$ metal oxide + $\text{NO}_2 \uparrow$ + $\text{O}_2 \uparrow$	(S) Li

120. Which of anions in the column-I shows one or more observations from the column-II :-

Column - I (Radical)	Column - II (Characteristic feature)
(a) S^{2-}	(P) White ppt. with AgNO_3
(b) NO_2^-	(Q) Evolution of pungent smell gas with (Al + conc. NaOH)
(c) SO_3^{2-}	(R) Brown fumes with conc. H_2SO_4 (hot)
(d) CH_3COO^-	(S) Decolourises acidified KMnO_4

121. Match the column-I with column-II :-

Column - I

- (a) $\mu = \sqrt{15} B.M.$
 (b) O.N. of metal = +2
 (c) Octahedral complex
 (d) d^2sp^3 hybridised metal ion

Column - II

- (P) Schweizer's salt
 (Q) Potassium ferrocyanide
 (R) Potassium ferricyanide
 (S) Brown ring complex
 (T) Sodium nitroprusside

122. Match the column-I with column-II :-

Column - I (Group reagent)

- (a) $NH_4Cl + NH_4OH$
 (b) $NH_4Cl + NH_4OH + (NH_4)_2CO_3$
 (c) $HCl(dil.) + H_2S$
 (d) $NH_4Cl + NH_4OH + H_2S$

Column - II (Group ion precipitated)

- (P) IInd group
 (Q) IIIrd group
 (R) IVth group
 (S) Vth group

123. Match the column-I with column-II :-

Column - I (Colour of precipitate)

- (a) Black ppt.
 (b) Canary yellow ppt.
 (c) Brick red ppt.
 (d) White ppt.

Column - II (Reactions)

- (P) $Bi(OH)_3 + [Sn(OH)_4]^{2-} \rightarrow$
 (Q) $Na_2S + Hg(NO_3)_2 \rightarrow$
 (R) $Na_3PO_4 + (NH_4)_2MoO_4 \rightarrow$
 (S) $AgNO_3 + K_2CrO_4 \rightarrow$
 (T) $HC \equiv CH + AgNO_3 + NH_4OH \rightarrow$

124. Match the column-I with column-II :-

Column - I**Colour of borax bead test(cold)****Oxidising flame**

- (a) Green
 (b) Violet
 (c) Yellow
 (d) Blue

Reducing flame

- Green
 Colourless
 Green
 Red

Column - II**Metal present**

- (P) Cu
 (Q) Fe
 (R) Mn
 (S) Cr

125. Match the column-I with column-II :-

Column - I**Colour of borax bead test(cold)****Oxidising flame**

- (a) Blue
 (b) Colourless
 (c) Brown
 (d) Amethyst

Reducing flame

- Blue
 Blue
 Grey
 Colourless

Column - II**Metal present**

- (P) Co
 (Q) Ti
 (R) Ni
 (S) Mn

EXERCISE # IV**Integer Type :**

- Find the number of reagent(s) which form white ppt. with CO_3^{2-} ion.
 $BaCl_2$, $CuSO_4$, Hg_2Cl_2 , $Pb(CH_3COO)_2$, $CaCl_2$
- Find the number of reagent(s) which oxidises HCO_3^- ion solution.
 MnO_4^- / H^+ , $Cr_2O_7^{2-} / H^+$, Cl_2 water, Br_2 water, I_2 water
- Find the number of carbonate having K_{sp} value greater than Li_2CO_3
 Na_2CO_3 , K_2CO_3 , Rb_2CO_3 , Cs_2CO_3
- Find out the number of anions that can not decolorise acidic solution of $KMnO_4$.
 NO_2^- ; NO_3^- ; Cl^- ; I^- ; $C_2O_4^{2-}$; HCO_2^- ; CO_3^{2-} ; S^{2-} ; SO_3^{2-}
- Find out the total number of acidic radical, to which if dilute HCl is added then volatile gases are evolved. CO_3^{2-} , SO_3^{2-} , S^{2-} , $S_2O_3^{2-}$, NO_3^- , CN^-

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6. Among the following, how many compound(s) destroy nitrite solution and N_2 gas is evolved.
 $AgNO_3$, NH_4Cl , $(NH_2CONH_2 + \text{dil. } HCl)$, Sulphamic acid, (thiourea + dil. CH_3COOH), $(FeSO_4 + \text{dil. } CH_3COOH)$; $MnO_4^- + H^+$
7. How many compounds of Mn^{2+} are **pink precipitate** :-
 MnS ; $MnO(OH)_2$; $Mn(NH_4)PO_4$; $KMnO_4$, $Mn(OH)_2$
8. How many of the following reagents can be used to distinguish between SO_2 and CO_2 ?
 $Ca(OH)_2$, (starch + KIO_3), $(K_2Cr_2O_7 + H_2SO_4)$, $(KMnO_4 + H_2SO_4)$, $Ba(OH)_2$, H_2O_2 , $FeCl_3$, $Zn[Fe(CN)_5(NO)]$
9. The total reagents are used to distinguish between sulphite and bisulphite acidic radical.
 Litmus paper, $AgNO_3$, $Pb(OAc)_2$, $CaCl_2$, $SrCl_2$, $BaCl_2$, $(K_2Cr_2O_7 + H_2SO_4)$, $(KMnO_4 + H_2SO_4)$, $(H_2O_2$ followed by litmus paper)
10. Find the total number of acidic radical, to which if conc. H_2SO_4 is added then volatile gases are evolved with specific odour. CO_3^{2-} , SO_3^{2-} , HCO_2^- , $CH_3CO_2^-$.
11. Find the total number of metals, which makes a thin protective layer of its oxide on treatment with conc. HNO_3 .
 Al , Zn , Sn , Cu , Pt , Cr , Au , Ag
12. The total number of anionic species gives coloured gas with conc. H_2SO_4 .
 CO_3^{2-} , HCO_3^- , SO_3^{2-} , NO_3^{2-} , $S_2O_3^{2-}$, NO_2^- , CH_3COO^- , SO_4^{2-}
13. Among the following find out the total number of incorrect statement(s) -
 (1) BO_3^{3-} , PO_4^{3-} , F^- & SiO_4^{4-} anions are called as interfering radicals,
 (2) $HgCl_2$ reacts with carbonate ion and gives basic mercury(II) carbonate which does not give test with bicarbonate.
 (3) SO_3^{2-} and HSO_3^- can be distinguished by addition of neutral H_2O_2 solution against litmus paper.
 (4) Zinc and sulphuric acid oxidized sulphite to sulphate.
- (5) $AgNO_3$ oxidizes suspension of sulphite solution to sulphate.
- (6) Bromine water oxidises $BaSO_3$ to $BaSO_4$.
14. Salt solution 'A' + dil. $HCl \longrightarrow$ only 'B' gas (suffocating odour of burning sulphur)
 $\downarrow BaCl_2$
 'C' [white ppt.] $\longrightarrow$ soluble in dil. HCl and gives 'B' gas
 $\downarrow$ Reagent 'M'
 white ppt. of [D] (insoluble in dil. HCl)
 Find the total number of reagents 'M':
 (conc. HNO_3), O_2 , H_2O_2 , N_2 , NH_3 , $(Br_2 + H_2O)$
15. In $Ni(DMG)_2$; how many following statements are correct ?
 (a) $Ni(DMG)_2$ is red in colour
 (b) Both 6 and 5 membered chelation takes place
 (c) H bonding is present which is intramolecular
 (d) It is diamagnetic
 (e) $Ni(+2)$ has coordination no. 4
 (f) In (DMG) , N is the donor site
16. Observe the following reaction,
 $[Ni(en)_3](NO_3)_2 + 'X'(aq) \xrightarrow[\text{alk. medium}]{\text{slightly}} 'Y'(\downarrow) \text{ violet ppt.}$
 Among the following, how many compounds decompose the nickel ethylenediamine nitrate reagent with the precipitation of NiS .
 Na_2SO_3 , Na_2SO_4 , $Na_2S_2O_3$, $Na_2S_4O_6$, H_2S , $(NH_4)_2S$, $NaSCN$
17. $Na_2S_2O_3 + \text{Reagent [A]} \longrightarrow \text{white ppt. of [B]}$
 $\swarrow \text{on warming}$ $\searrow \text{Na}_2S_2O_3 \text{ + excess}$
 [Black ppt. of metal sulphite] [Soluble complex of [B]]
 Among the following how many reagents 'A' are used to give black precipitate of metal sulphide ?
 $AgNO_3$, $BaCl_2$, $Pb(OAc)_2$, $HgCl_2$, $CaCl_2$, $CuSO_4$, $FeCl_3$

18. Among the following how many compound(s) destroy nitrite solution and does not form traces of nitrate ?

AgNO_3 , NH_4Cl , $(\text{NH}_2\text{CONH}_2 + \text{dil. HCl})$, sulphamic acid, (thiourea + dil. CH_3COOH), $(\text{FeSO}_4 + \text{dil. CH}_3\text{COOH})$

19. Salt of iron $\xrightarrow{\text{Na}_2\text{S}_2\text{O}_3}$ Violet complex (X)

$\xrightarrow{\text{On standing}}$ Green solution (Z)

Write oxidation number of Fe in (Z).

20. Find the number of cation when treated with aq. solution with aq. solution of Na_2CO_3 , ppt. of carbonate formed,

Hg^{+2} , Ba^{+2} , Cu^{+2} , Ag^+ , Pb^{+2}

21. Among the following compounds insoluble in water is/are

PbI_2 , HgI_2 , Hg_2OCl_2 , NaCl , KI , AgCl

22. The total number of following ions which interfere with the brown ring test for NO_3^- is /are

Br^- , I^- , CO_3^{2-} , NO_2^- , ClO_3^- , CrO_4^{2-}

23. $[\text{'X'} (s) (\text{salt}) + \text{K}_2\text{Cr}_2\text{O}_7(s) + \text{conc. H}_2\text{SO}_4]$

$\downarrow$ on heating
 $\text{'Z'} \xleftarrow{+\text{NaOH}}$ $\text{'Y'} (\text{gas only})$

(yellow solution)

The maximum number of difference in oxidation state between central atom of 'Y' and 'Z' is :-

24. Among the following total number of compounds, which give(s) white precipitate with calcium chloride solution.

$\text{Na}_2\text{S}_2\text{O}_3$; Na_2CO_3 ; $\text{Na}_2\text{C}_2\text{O}_4$; NaBr ; NaI

25. The total number of reagent(s) give white precipitate with solution of sodium oxalates.

$(\text{MnO}(\text{OH})_2 + \text{H}_2\text{SO}_4)$, CaCl_2 , AgNO_3 , BaCl_2 , KMnO_4 , $(\text{K}_2\text{Cr}_2\text{O}_7 + \text{H}_2\text{SO}_4)$

26. Among the following total number of anions, which decolourise the potassium permanganate solution in acid medium.

$\text{C}_2\text{O}_4^{2-}$; NO_2^- ; $(\text{BaCl}_2 + \text{SO}_4^{2-})$; SO_3^{2-} , NO_3^-

27. Find out the total number of compounds, which can be dissolved by both dil. HNO_3 and NaOH .

PbSO_3 ; $[\text{PbCO}_3 \cdot \text{Pb}(\text{OH})_2]$; PbCrO_4 ; AgCl ; Ag_2S ; Ag_2O

28. Among the following, total number of reagent(s) which gives yellow precipitate with ammonium chloride solution.

$\text{Na}_3[\text{Co}(\text{NO}_2)_6]$; $\text{H}_2[\text{PtCl}_6]$; $(\text{NaH} \cdot \text{C}_4\text{H}_4\text{O}_6)$;

NaClO_4 ; NaOH ; $\text{O}_2\text{N}-\text{C}_6\text{H}_4-\text{N}=\text{N}-\text{Cl}$

29. Colourless salt (A) $\text{NaOH}(\text{excess}) \xrightarrow{\Delta}$ Gas(B) giving white fumes with HCl + solution (C)

$(\text{C}) + \text{Zn} \xrightarrow{+\text{NaOH}}$ (B)

(A) $\xrightarrow{\Delta}$ Gas(D) + liquid (E)

D, E both are triatomic

(a) $-\text{NH}_4\text{NO}_3$;

(b) $-\text{NH}_3$;

(c) $-(\text{NaNO}_3 + \text{NaOH})$;

(d) $-\text{N}_2\text{O}$;

(e) $-\text{H}_2\text{O}$

A to E compounds are given. How many are correctly given ?

30. Read the following reaction carefully :

(A) $\text{Cu} + \text{HNO}_3(\text{dil.}) \longrightarrow \text{NO} + \text{A}$

(B) $\text{Pb}(\text{NO}_3)_2 \xrightarrow{\text{heat}} \text{PbO} + \text{B}$

(C) $\text{CuSO}_4 + \text{NH}_4\text{OH}(\text{excess}) \longrightarrow \text{C}$

(D) $\text{AgCl} + \text{NH}_4\text{OH} \longrightarrow \text{D}$

A, B, C, D are mixture of compounds which are given as follows :-

(a) $-\text{Cu}(\text{NO}_3)_2$, H_2O ;

(b) $-\text{NO}_2$, O_2 ;

(c) $-\text{[Cu}(\text{NH}_3)_4\text{]SO}_4 + \text{H}_2\text{O}$;

(d) $-\text{Ag}(\text{NH}_3)_2\text{Cl} + \text{H}_2\text{O}$

How many are correctly given ?

31. How many hydroxides are white precipitate ?

(i) $\text{Zn}(\text{OH})_2$ (ii) $\text{Cd}(\text{OH})_2$

(iii) $\text{Mn}(\text{OH})_2$ (iv) $\text{Al}(\text{OH})_3$

(v) $\text{Ni}(\text{OH})_2$ (vi) $\text{Co}(\text{OH})_2$

(vii) $\text{Pb}(\text{OH})_2$ (viii) $\text{Fe}(\text{OH})_3$

(ix) $\text{Cu}(\text{OH})_2$

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32. How many hydroxide precipitate are soluble in excess NaOH ?

- | | |
|-------------------------|--------------------------|
| (i) Zn(OH)_2 | (ii) Cd(OH)_2 |
| (iii) Mn(OH)_2 | (iv) Al(OH)_3 |
| (v) Ni(OH)_2 | (vi) Co(OH)_2 |
| (vii) Pb(OH)_2 | (viii) Fe(OH)_3 |
| (ix) Cu(OH)_2 | |

33. How many hydroxide precipitate are soluble in excess NH_4OH ?

- | | |
|-------------------------|--------------------------|
| (i) Zn(OH)_2 | (ii) Cd(OH)_2 |
| (iii) Mn(OH)_2 | (iv) Al(OH)_3 |
| (v) Ni(OH)_2 | (vi) Co(OH)_2 |
| (vii) Pb(OH)_2 | (viii) Fe(OH)_3 |
| (ix) Cu(OH)_2 | |

34. $\text{NaCl} + \text{Solid K}_2\text{Cr}_2\text{O}_7 + \text{conc. H}_2\text{SO}_4 (\text{few drops}) \longrightarrow \text{X} (\text{Reddish-brown fumes})$

How many statements are correct regarding X :-

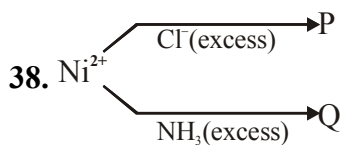
- No axial d-orbitals is involved in hybridization of X
- One s-orbital is involved in hybridization of X
- The complex does not have any unpaired electron.
- Three non axial d-orbitals are involved in hybridization of X
- Magnetic moment of X is zero

35. A solution containing several unknown cations is treated with dil. HCl and no ppt. forms. The ppt. is filtered and the filtrate at pH 1.0 is treated with H_2S , no ppt. forms. At pH 8.0 H_2S causes the formation of a ppt., the filtrate form which gives no ppt. on treatment with Na_2CO_3 . Which group of cations are present in the original solution.

36. When NaOH is added to $\text{K}_2\text{Cr}_2\text{O}_7$ solution, it becomes yellow. The change in oxidation state of Cr in this phenomenon is

37. Amongst the following, the total number of compounds soluble in concentrated NH_3 solution is:

BaSO_4 , Ni(OH)_2 , $\text{Zn}_3(\text{PO}_4)_2$, Ag_2CrO_4 , PbSO_3 , Al(OH)_3 , Mn(OH)_2 , $\text{Bi(OH)}_2\text{NO}_3$, Cu(OH)_2 , CuSO_4



Coordination number of Ni^{2+} in P and Q are respectively a and b. Find out $b - a$:-

39. How many of the following will dissolve in concentrated NaOH as well as in concentrated HCl.

Fe(OH)_2 ; Fe(OH)_3 ; Cr(OH)_3 ; Al(OH)_3 ; Zn(OH)_2 ; Sn(OH)_2

40. In how many of the following reactions, one of the products is obtained as a yellow precipitate?

$\text{Ba}^{2+} + \text{CrO}_4^{2-} \rightarrow \text{Product}$

$\text{Hg}^{2+} + \text{Co}^{2+} + 4\text{SCN}^- \rightarrow \text{Product}$

$\text{BI}^3 + \text{C}_6\text{H}_5(\text{OH})_3 \rightarrow \text{Product}$

$\text{Mn(OH)}_2 \xrightarrow{\downarrow} + \text{O}_2 \rightarrow \text{Product}$

$\text{Zn}^{2+} + 2\text{HPO}_4^{2-} \rightarrow \text{Product}$

$\text{NH}_4^+ + [\text{PtCl}_6]^{2-} \rightarrow \text{Product}$

41. $\text{Na}_2\text{S} + \text{Na}_2[\text{Fe(CN)}_5\text{NO}] \longrightarrow \text{X}$ (Purple color)

How many statements are correct regarding X :-

- Six ligands are present in compound X
- Ambidentate ligand is present in X
- Two d-orbitals are involved in hybridization of X
- Magnetic moment of X is zero
- Two axial d-orbitals are involved in hybridization of X
- The total possible linkage isomer of X is four

42. How many of the following pair of ions can be separated by H_2S in dilute HCl.

Mn^{2+} , Cd^{2+} ; Cr^{3+} , Cu^{2+} ; As^{3+} , Sn^{2+} ; Sb^{3+} , Pb^{2+} ; Bi^{3+} , Sn^{4+} ; Hg^{2+} , Fe^{3+}

43. Find the number compounds which are producing the oxides of their metal on thermal decomposition

Mg_3N_2 , Mg(OH)_2 , SrCO_3 , $\text{Ba(NO}_3)_2$, KClO_3

44. Among the following, total number of cations tend to form soluble complex with excess $\text{NH}_4\text{OH(aq.)}$ and excess NaCN(aq.)

Pb^{2+} ; Cd^{2+} ; Hg^{2+} ; Bi^{3+} ; Cu^{2+} ; Ag^+

45. Among the following, find out the total number of black metal sulphides :-
 CuS ; NiS ; CoS ; ZnS ; As_2S_3
46. Find the number of reagents in which Ag_2S is insoluble :-
 hot conc. HNO_3 ; $(\text{NH}_4)_2\text{S}$; NH_3 ; KCN ; $\text{Na}_2\text{S}_2\text{O}_3$
47. Among the following, total number of compound(s) reduce(s) the yellow solution of FeCl_3 .
 Tin(II) chloride; Potassium iodide; Hydroxylamine; Hydrazine sulphate; Ascorbic acid; Hydrogen sulphide;
 Sulphur dioxide; Potassium dichromate; Conc. nitric acid
48. Find out the total number of reagent(s) which converts chromium(III) ion to chromate ion :-
 H_2O_2 solution; $(\text{NaBO}_3 \cdot 4\text{H}_2\text{O} + \text{H}_2\text{O}_2)$; NaOBr ; FeSO_4 ; NaOH ; $\text{K}_2\text{S}_2\text{O}_8$
49. How many reagents are used to identify chromium after oxidation to chromate ion ?
 BaCl_2 ; CaCl_2 ; (Acidified H_2O_2 - Amyl alcohol); AgNO_3
50. Find total number of reagents in which oxidation of Fe^{2+} ion to Fe^{3+} takes place :
 (a) On exposure to air
 (b) On reaction with SnCl_2
 (c) On reaction with $\text{MnO}_4^- / \text{H}^+$
 (d) On reaction with $\text{Cr}_2\text{O}_7^{2-} / \text{H}^+$
 (e) On addition of concentrated HNO_3
 (f) On addition with H_2O_2
 (g) On reaction with KI
2. Which products are expected from the disproportionation of hypochlorous acid :-
 [AIEEE - 2002]
 (a) HClO_3 and Cl_2O (b) HClO_2 and HClO
 (c) HCl and Cl_2O (d) HCl and HClO_3
3. A metal M readily forms its sulphate MSO_4 which is water soluble. It forms oxide MO which becomes inert on heating. It forms insoluble hydroxide which is soluble in NaOH . The metal M is :-
 [AIEEE - 2002]
 (a) Mg (b) Ba
 (c) Ca (d) Be
4. Which statement is correct ? [AIEEE - 2003]
 (a) Fe^{3+} ions give deep green precipitate with $\text{K}_4[\text{Fe}(\text{CN})_6]$
 (b) On heating K^+ , Ca^{2+} and HCO_3^- ions, we get a precipitate of $\text{K}_2[\text{Ca}(\text{CO}_3)_2]$
 (c) Manganese salts give a violet borax bead test in reducing flame
 (d) From a mixed precipitate of AgCl and AgI ammonia solution dissolves only AgCl .
5. What would happen when a solution of potassium chromate is treated with an excess of dilute nitric acid :-
 [AIEEE - 2003]
 (a) Cr^{3+} and $\text{Cr}_2\text{O}_7^{2-}$ are formed
 (b) $\text{Cr}_2\text{O}_7^{2-}$ and H_2O are formed
 (c) $\text{Cr}_2\text{O}_7^{2-}$ is reduced to +3 state of Cr
 (d) $\text{Cr}_2\text{O}_7^{2-}$ is oxidised to +7 state of Cr
6. Ammonia forms the complex in $[\text{Cu}(\text{NH}_3)_4]^{2+}$ with copper ions in alkaline solutions but not in acidic solution. What is the reason for it :-
 [AIEEE - 2003]
 (a) In acidic solutions hydration protects copper ions
 (b) In acidic solutions protons coordinate with ammonia molecules forming NH_4^+ ion and NH_3 molecules are not available.
 (c) In alkaline solutions insoluble $\text{Cu}(\text{OH})_2$ is precipitated which is soluble in excess of any alkali
 (d) Copper hydroxide is an amphoteric substance

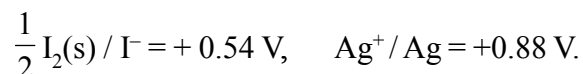
EXERCISE # V(A) (JEE-MAIN)

1. How do we differentiate between Fe^{3+} and Cr^{3+} in qualitative analysis gp. III :- [AIEEE - 2002]
 (a) By taking excess of NH_4OH
 (b) By increasing NH_4^+ ion concentration
 (c) By decreasing OH^- ion concentration
 (d) Both (b) and (c)

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7. Excess of KI reacts with CuSO_4 solution and then $\text{Na}_2\text{S}_2\text{O}_3$ solution is added to it. Which of the statement is incorrect for this reaction :-
 (a) Evolved I_2 is reduced [AIEEE - 2004]
 (b) CuI_2 is formed
 (c) $\text{Na}_2\text{S}_2\text{O}_3$ is oxidised
 (d) Cu_2I_2 is formed
8. Calomel on reaction with NH_4OH gives :-
 [AIEEE - 2004]
 (a) HgNH_2Cl (b) $\text{NH}_2 - \text{Hg} - \text{Hg} - \text{Cl}$
 (c) Hg_2O (d) HgO
9. One mole of magnesium nitride on reaction with excess of water gives :- [AIEEE - 2004]
 (a) Two mole of HNO_3 (b) Two mole of NH_3
 (c) 1 mole of NH_3 (d) 1 mole of HNO_3
10. The products obtained on heating LiNO_3 will be :- [AIEEE - 2004]
 (a) $\text{LiNO}_2 + \text{O}_2$ (b) $\text{Li}_2\text{O} + \text{NO}_2 + \text{O}_2$
 (c) $\text{Li}_3\text{N} + \text{O}_2$ (d) $\text{Li}_2\text{O} + \text{NO} + \text{O}_2$
11. What is the best description of the change that occurs when $\text{Na}_2\text{O}(\text{s})$ is dissolved in water ? [AIEEE - 2011]
 (a) Oxidation number of sodium decreases
 (b) Oxide ion accepts sharing in a pair of electrons
 (c) Oxide ion donates a pair of electron
 (d) Oxidation number of oxygen increases
12. Which of the following on thermal - decomposition yields a basic as well as an acidic oxide. [AIEEE - 2012]
 (a) NH_4NO_3
 (b) NaNO_3
 (c) KClO_3
 (d) CaCO_3
13. The correct statement for the molecule, CsI_3 , is :- [Jee(Main) - 2014]
 (a) It contains Cs^{3+} and I^- ions
 (b) It contains Cs^+ , I^- and lattice I_2 molecule
 (c) It is a covalent molecule
 (d) It contains Cs^+ and I_3^- ions
14. Fire extinguishers contain H_2SO_4 and which one of the following :-
 (a) CaCO_3 [Jee(Main)-2012, Online_P-1]
 (b) NaHCO_3 and Na_2CO_3
 (c) Na_2CO_3
 (d) NaHCO_3
15. Copper wire test for halogens is known as :- [Jee(Main)-2012, Online_P-2]
 (a) Duma's Test (b) Beilstein's Test
 (c) Lassigne's Test (d) Liebig's Test
16. The standard potentials of Ag^+ / Ag , $\text{Hg}_2^{2+} / 2\text{Hg}$, $\text{Cu}^{2+} / \text{Cu}$ and $\text{Mg}^{2+} / \text{Mg}$ electrodes are 0.80, 0.79, 0.34 and -2.37 V, respectively. An aqueous solution which contains one mole per litre of the salts of each of the four metals is electrolyzed. With increasing voltage, the correct sequence of deposition of the metals at the cathode is :- [Jee(Main)-2012, Online_P-3]
 (a) Cu, Hg, Ag only
 (b) Mg, Cu, Hg, Ag
 (c) Ag, Hg, Cu only
 (d) Ag, Hg, Cu, Mg
17. Beilstein test is used for estimation of which one of following elements :- [Jee(Main)-2012, Online_P-3]
 (a) S (b) Cl
 (c) C and H (d) N
18. In the following balanced reaction, $\text{XMnO}_4^- + \text{YC}_2\text{O}_4^{2-} + \text{ZH}^+ = \text{XMn}^{2+} + 2\text{YCO}_2 + \frac{\text{Z}}{2}\text{H}_2\text{O}$ the values of X, Y and Z respectively are :- [Jee(Main)-2012, Online_P-4]
 (a) 8, 2, 5 (b) 5, 2, 16
 (c) 2, 5, 16 (d) 5, 8, 4
19. A metal M on heating in nitrogen gas gives Y. Y on treatment with H_2O gives a colourless gas which when passed through CuSO_4 solution gives a blue colour, Y is :- [Jee(Main)-2012, Online_P-4]
 (a) NH_3 (b) MgO
 (c) Mg_3N_2 (d) $\text{Mg}(\text{NO}_3)_2$

20. Electrode potential (E°) are given below :-



Based on the above potential, strongest oxidizing agent will be :- [Jee(Main)-2013, Online_P-1]

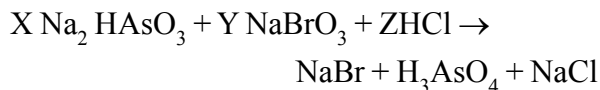
- (a) Cu^+ (b) Fe^{3+}
(c) Ag^+ (d) I_2

21. Potassium dichromate when heated with concentrated sulphuric acid and a soluble chloride, gives brown red vapours of :-

[Jee(Main)-2013, Online_P-1]

- (a) CrO_3 (b) Cr_2O_3
(c) CrCl_3 (d) CrO_2Cl_2

22. Given: [Jee(Main)-2013, Online_P-1]



The values of X, Y and Z in the above redox reaction are respectively :-

- (a) 2,1,3 (b) 3,1,6
(c) 2,1,2 (d) 3,1,4

23. Sodium carbonate cannot be used in place of $(\text{NH}_4)_2\text{CO}_3$ for the identification of Ca^{2+} , Br^{2+} and Sr^{2+} ion (in group V) during mixture analysis because :- [Jee(Main)-2013, Online_P-1]

- (a) Sodium ions will react with acid radicals
(b) Concentration of CO_3^{2-} ions is very low
(c) Mg^{2+} ions will also be precipitated
(d) Na^+ ions will interfere with the detection of Ca^{2+} , Ba^{2+} , Sr^{2+} ions :-

24. Which of the following statement is incorrect :-

[Jee(Main)-2013, Online_P-2]

- (a) Fe^{2+} ion also gives blood red colour with SCN^- ions
(b) Cupric ion reacts with excess of ammonia solution to give deep blue colour of $[\text{Cu}(\text{NH}_3)_4]^{2+}$ ion
(c) Fe^{3+} ion gives blood red colour with SCN^- ion
(d) On passing H_2S into Na_2ZnO_2 solution, a white ppt of ZnS is formed

25. Values of dissociation constant, K_a are given as follows:- [Jee(Main)-2013, Online_P-2]

Acid	K_a
HCN	6.2×10^{-10}
HF	7.2×10^{-4}
HNO_2	4.0×10^{-4}

Correct order of increasing base strength of the base CN^- , F^- and NO_2^- will be

- (a) $\text{NO}_2^- < \text{CN}^- < \text{F}^-$ (b) $\text{F}^- < \text{CN}^- < \text{NO}_2^-$
(c) $\text{NO}_2^- < \text{F}^- < \text{CN}^-$ (d) $\text{F}^- < \text{NO}_2^- < \text{CN}^-$

26. Identify incorrect statement :-

[Jee(Main)-2013, Online_P-3]

- (a) Copper (I) compounds are colourless except where colour result from charge transfer
(b) Copper (I) compounds are diamagnetic
(c) Cu_2S is black
(d) Cu_2O is colourless

27. Given :- [Jee(Main)-2013, Online_P-3]



The correct order of reducing power of the species (Cr , Cr^{3+} , Mn^{2+} and Cl^-) will be :-

- (a) $\text{Mn}^{2+} < \text{Cl}^- < \text{Cr}^{3+} < \text{Cr}$
(b) $\text{Cr}^{3+} < \text{Cl}^- < \text{Mn}^{2+} < \text{Cr}$
(c) $\text{Cr}^{3+} < \text{Cl}^- < \text{Cr} < \text{Mn}^{2+}$
(d) $\text{Mn}^{2+} < \text{Cr}^{3+} < \text{Cl}^- < \text{Cr}$

28. Which one of the following cannot function as an oxidising agent? [Jee(Main)-2013, Online_P-4]

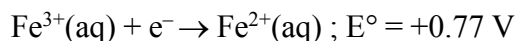
- (a) $\text{NO}_3^-(\text{aq})$
(b) I^-
(c) $\text{Cr}_2\text{O}_7^{2-}$
(d) $\text{S}(\text{s})$

29. Which of the following statement about Na_2O_2 is not correct :- [Jee(Main)-2014, Online_P-2]

- (a) Na_2O_2 oxidises Cr^{3+} to CrO_4^{2-} in acid medium
(b) It is diamagnetic in nature
(c) It is the super oxide of sodium
(d) It is a derivative of H_2O_2

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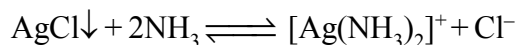
30. Given :- [Jee(Main)-2014, Online_P-2]



Considering the electrode potentials, which of the following represents the correct order of reducing power ?

- (a) $\text{Al} < \text{Fe}^{2+} < \text{Br}^{-}$ (b) $\text{Al} < \text{Br}^{-} < \text{Fe}^{2+}$
(c) $\text{Fe}^{2+} < \text{Al} < \text{Br}^{-}$ (d) $\text{Br}^{-} < \text{Fe}^{2+} < \text{Al}$

31. Consider the following equilibrium :-



White precipitate of AgCl appears on adding which of the following? [Jee(Main)-2014, Online_P-2]

- (a) NH_3 (b) Aqueous NaCl
(c) Aqueous NH_4Cl (d) Aqueous HNO_3

32. Copper becomes green when exposed to moist air for a long period. This is due to :-

[Jee(Main)-2014, Online_P-3]

- (a) the formation of a layer of cupric oxide on the surface of copper
(b) the formation of basic copper sulphate layer on the surface of the metal
(c) the formation of a layer of cupric hydroxide on the surface of copper
(d) the formation of a layer of basic carbonate of copper on the surface of copper

33. When one of the following exhibits the largest number of oxidation states ?

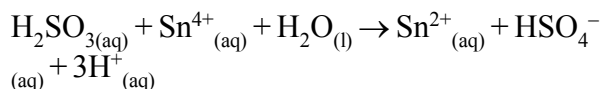
[Jee(Main)-2014, Online_P-3]

- (a) Mn(25) (b) V(23)
(c) Cr(24) (d) Ti(22)

34. Hydrogen peroxide acts both as an oxidising and as a reducing agent depending upon the nature of the reacting species. In which of the following cases H_2O_2 acts as a reducing agent in acidic medium? [Jee(Main)-2014, Online_P-3]

- (a) MnO_4^{-}
(b) SO_3^{2-}
(c) KI
(d) $\text{Cr}_2\text{O}_7^{2-}$

35. Consider the reaction :-

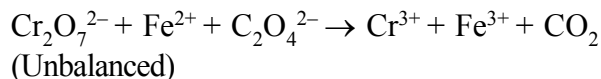


Which of the following statement is correct :-

[Jee(Main)-2014, Online_P-3]

- (a) H_2SO_3 is the reducing agent because it undergoes oxidation
(b) H_2SO_3 is the reducing agent because it undergoes reduction
(c) Sn^{4+} is the reducing agent because it undergoes oxidation
(d) Sn^{4+} is the oxidizing agent because it undergoes oxidation

36. How many electrons are involved in the following redox reaction :- [Jee(Main)-2014, Online_P-4]



- (a) 3
(b) 4
(c) 5
(d) 6

37. Amongst the following, identify the species with an atom in +6 oxidation state :-

- (a) $[\text{MnO}_4]^{-}$ [Jee(Main)-2014, Online_P-4]
(b) $[\text{Cr}(\text{CN})_6]^{3-}$
(c) Cr_2O_3
(d) CrO_2Cl_2

EXERCISE # V(B) (JEE-ADVANCED)

1. Which of the following statement(s) is (are) correct with reference to the ferrous and ferric ions :

[Jee 1998]

- (a) Fe^{3+} gives brown colour with potassium ferricyanide
(b) Fe^{2+} gives blue precipitate with potassium ferricyanide
(c) Fe^{3+} gives red colour with potassium thiocyanate
(d) Fe^{2+} gives brown colour with ammonium thiocyanate

2. Which of the following statements(s) is / are correct. When a mixture of NaCl and $\text{K}_2\text{Cr}_2\text{O}_7$ is gently warmed with conc. H_2SO_4 ? [Jee 1998]
 (a) A deep red vapours is evolved
 (b) The vapours when passed into NaOH solution gives a yellow solution Na_2CrO_4
 (c) Chlorine gas is evolved
 (d) Chromyl chloride is formed
3. An aqueous solution of a substance gives a white precipitate on treatment with dilute hydrochloric acid, which dissolves on heating. When hydrogen sulphide is passed through the hot acidic solution, a black precipitate is obtained. The substance is a : [Jee 2000]
 (a) Hg_2^+ salt (b) Cr^{2+} salt
 (c) Ag^+ salt (d) Pb^{2+} salt
4. A gas 'X' is passed through water to form a saturated solution. The aqueous solution on treatment with silver nitrate gives a white precipitate. The saturated aqueous solution also dissolves magnesium ribbon with evolution of a colourless gas 'Y'. Identify 'X' and 'Y' : [Jee 2002 (Mains)]
 (a) $\text{X} = \text{CO}_2$, $\text{Y} = \text{Cl}_2$ (b) $\text{X} = \text{Cl}_2$, $\text{Y} = \text{CO}_2$
 (c) $\text{X} = \text{Cl}_2$, $\text{Y} = \text{H}_2$ (d) $\text{X} = \text{H}_2$, $\text{Y} = \text{Cl}_2$
5. $[\text{X}] + \text{H}_2\text{SO}_4 \rightarrow [\text{Y}]$ a colourless gas with irritating smell :- [Jee-2003]
 $[\text{Y}] + \text{K}_2\text{Cr}_2\text{O}_7 + \text{H}_2\text{SO}_4 \rightarrow \text{green solution}$
 $[\text{X}]$ and $[\text{Y}]$ are :-
 (a) SO_3^{2-} , SO_2 (b) Cl^- , HCl
 (c) S^{2-} , H_2S (d) CO_3^{2-} , CO_2
6. A sodium salt of an unknown anion when treated with MgCl_2 give white precipitate only on boiling. The anion is :- [Jee 2004]
 (a) SO_4^{2-} (b) HCO_3^-
 (c) CO_3^{2-} (d) NO_3^-
7. $(\text{NH}_4)_2\text{Cr}_2\text{O}_7$ on heating gives a gas which is also given by :- [Jee 2004]
 (a) heating NH_4NO_2 (b) heating NH_4NO_3
 (c) $\text{Mg}_3\text{N}_2 + \text{H}_2\text{O}$ (d) $\text{NaNO}_2 + \text{H}_2\text{O}_2$
8. A metal nitrate reacts with KI to give a black precipitate which on addition of excess of KI convert into orange colour solution. The cation of metal nitrate is :- [Jee 2005]
 (a) Hg^{2+} (b) Bi^{3+}
 (c) Pb^{2+} (d) Cu^{2+}
9. A solution when diluted with H_2O and boiled, it gives a white precipitate. On addition of excess NH_4Cl / NH_4OH , the volume of precipitate decreases leaving behind a white gelatinous precipitate. Identify the precipitate which dissolves in NH_4OH / NH_4Cl ? [Jee 2006]
 (a) $\text{Zn}(\text{OH})_2$ (b) $\text{Al}(\text{OH})_3$
 (c) $\text{Mg}(\text{OH})_2$ (d) $\text{Ca}(\text{OH})_2$
10. CuSO_4 decolorises on addition of KCN , the product is [Jee 2006]
 (a) $[\text{Cu}(\text{CN})_4]^{2-}$
 (b) Cu^{2+} get reduced to form $[\text{Cu}(\text{CN})_4]^{3-}$
 (c) $\text{Cu}(\text{CN})_2$
 (d) CuCN
11. Consider a titration of potassium dichromate solution with acidified Mohr's salt solution using diphenylamine as indicator. The number of moles of Mohr's salt required per mole of dichromate is :- [Jee - 2007]
 (a) 3 (b) 4
 (c) 5 (d) 6
12. The species present in solution when CO_2 is dissolved in water are? [JEE - 2007]
 (a) CO_2 , H_2CO_3 , HCO_3^- , CO_3^{2-}
 (b) H_2CO_3 , CO_3^{2-}
 (c) CO_3^{2-} , HCO_3^-
 (d) CO_2 , H_2CO_3
13. Sodium fusion extract, obtained from anilone, on treatment with iron (II) sulphate and H_2SO_4 in presence of air gives a Prussian blue precipitate. The blue colour is due to the formation of :- [Jee 2007]
 (a) $\text{Fe}_4[\text{Fe}(\text{CN})_6]_3$
 (b) $\text{Fe}_3[\text{Fe}(\text{CN})_6]_2$
 (c) $\text{Fe}_4[\text{Fe}(\text{CN})_6]_2$
 (d) $\text{Fe}_3[\text{Fe}(\text{CN})_6]_3$

4.40 QUALITATIVE SALT ANALYSIS**14. Column - I****Column - II****[Jee 2007]**

- (A) $O_2^- \rightarrow O_2 + O_2^{2-}$ (P) Redox reaction
 (B) $CrO_4^{2-} + H^+ \rightarrow$ (Q) One of the products has trigonal structure
 (C) $MnO_4^- + NO_2^- + H^+ \rightarrow$ (R) Dimeric bridged tetrahedral metal ion
 (D) $NO_3^- + H_2SO_4 + Fe^{2+} \rightarrow$ (S) Disproportionation

15. A solution of a metal ion when treated with KI gives a red precipitate which dissolves in excess KI to give a colourless solution. Moreover, the solution of metal ion on treatment with a solution of cobalt (II) thiocyanate gives rise to a deep blue crystalline precipitate. The metal ion is :-

- (a) Pb^{2+} (b) Hg^{2+} **[Jee 2007]**
 (c) Cu^{2+} (d) Co^{2+}

16. A solution of colourless salt H on boiling with excess NaOH produces a non-flammable gas. The gas evolution ceases after sometime. Upon addition of Zn dust to the same solution, the gas evolution restarts. The colourless salt(s) H is (are) :- **[Jee 2007]**

- (a) NH_4NO_3 (b) NH_4NO_2
 (c) NH_4Cl (d) $(NH_4)_2SO_4$

Passage for Q. 17 to 19

p-Amino-N, N-dimethylaniline is added to a strongly acidic solution of X. The resulting solution is treated with a few drops of aqueous solution of Y to yield blue coloration due to the formation of methylene blue. Treatment of the aqueous solution of Y with the reagent potassium hexacyanoferrate(II) leads to the formation of an intense blue precipitate. The precipitate dissolves on excess addition of the reagent. Similarly, treatment of the solution of Y with the solution of potassium hexacyanoferrate(II) leads to a brown coloration due to the formation of Z. **[JEE-2009]**

18. The compound Y is :-

- (a) $MgCl_2$ (b) $FeCl_2$
 (c) $FeCl_3$ (d) $ZnCl_2$

19. The compound Z is :-

- (a) $Mg_2[Fe(CN)_6]$ (b) $Fe[Fe(CN)_6]$
 (c) $Fe_4[Fe(CN)_6]_3$ (d) $K_2Zn_3[Fe(CN)_6]_2$

20. Match each of the reactions given column I with the corresponding product(s) given in column II :- **[JEE-2009]**

Column-I**Column-II**

- (A) $Cu + \text{dil. } HNO_3$ (P) NO
 (B) $Cu + \text{conc. } HNO_3$ (Q) NO_2
 (C) $Zn + \text{dil. } HNO_3$ (R) N_2O
 (D) $Zn + \text{conc. } HNO_3$ (S) $Cu(NO_3)_2$
 (T) $Zn(NO_3)_2$

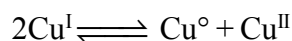
21. Passing H_2S gas into a mixture of Mn^{2+} , Ni^{2+} , Cu^{2+} and Hg^{2+} ions in an acidified aqueous solution precipitates :- **[JEE-2011]**

- (a) CuS and HgS (b) MnS and CuS
 (c) MnS and NiS (d) NiS and HgS

22. Reduction of the metal centre in aqueous permanganate ion involves **[JEE-2011]**

- (a) 3 electrons in neutral medium
 (b) 5 electrons in neutral medium
 (c) 3 electrons in alkaline medium
 (d) 5 electrons in acidic medium

23. The equilibrium :- **[JEE-2011]**



in aqueous medium at 25° shifts towards the left in the presence of :-

- (a) NO_3^- (b) Cl^-
 (c) SCN^- (d) CN^-

Passage for Q. 24 to 26

When a metal rod M is dipped into an aqueous colourless concentrated solution of compound N, the solution turns light blue. Addition of aqueous NaCl to the blue solution gives a white precipitate O. Addition of aqueous NH_3 dissolves O and gives an intense blue solution. **[JEE-2011]**

24. The metal rod M is :-

- (a) Fe (b) Cu
 (c) Ni (d) Co

25. The compound N is :-

- (a) $AgNO_3$ (b) $Zn(NO_3)_2$
 (c) $Al(NO_3)_3$ (d) $Pb(NO_3)_2$

26. The final solution contains :-

- (a) $[\text{Pb}(\text{NH}_3)_4]^{2+}$ and $[\text{CoCl}_4]^{2-}$
 (b) $[\text{Al}(\text{NH}_3)_4]^{3+}$ and $[\text{Cu}(\text{NH}_3)_4]^{2+}$
 (c) $[\text{Ag}(\text{NH}_3)_4]^{2+}$ and $[\text{Cu}(\text{NH}_3)_4]^{2+}$
 (d) $[\text{Ag}(\text{NH}_3)_2]^{2+}$ and $[\text{Ni}(\text{NH}_3)_6]^{2+}$

27. Which of the following hydrogen halides react(s) with $\text{AgNO}_3(\text{aq})$ to give a precipitate that dissolves in $\text{Na}_2\text{S}_2\text{O}_3(\text{aq})$:- [JEE-2012]

- (a) HCl (b) HF
 (c) HBr (d) HI

29. For the given aqueous reactions, which of the statement(s) is (are) true :-

[JEE-2012]

excess $\text{KI} + \text{K}_3[\text{Fe}(\text{CN})_6] \xrightarrow{\text{dilute } \text{H}_2\text{SO}_4} \text{brownish-yellow solution}$

ZnSO_4

white precipitate + brownish - yellow solution

$\text{Na}_2\text{S}_2\text{O}_3$

colourless solution

- (a) The first reaction is a redox reaction (b) White precipitate is $\text{Zn}_3[\text{Fe}(\text{CN})_6]_2$
 (c) Addition of filtrate to starch solution gives blue colour (d) White precipitate is soluble in NaOH solution

30. Upon treatment with ammonical H_2S , the metal ion that precipitate as a sulfide is

[JEE-2013]

- (a) $\text{Fe}(\text{III})$ (b) $\text{Al}(\text{III})$
 (c) $\text{Mg}(\text{II})$ (d) $\text{Zn}(\text{II})$

Passage for Q. 31 & 32

An aqueous solution of a mixture of two inorganic salts, when treated with dilute HCl , gave a precipitate (P) and a filtrate (Q). The precipitate (P) was found to dissolve in hot water. The filtrate (Q) remained unchanged, when treated with H_2S in a dilute mineral acid medium. However, it gave a precipitate (R) with H_2S in an ammonical medium. The precipitate R gave a coloured solution (S), when treated with H_2O_2 in an aqueous NaOH medium.

31. The coloured solution (S) contains :-

[JEE-2013]

- (a) $\text{Fe}_2(\text{SO}_4)_3$ (b) CuSO_4
 (c) ZnSO_4 (d) Na_2CrO_4

32. The precipitate (P) contains :-

[JEE-2013]

- (a) Pb^{2+} (b) Hg_2^{2+}
 (c) Ag^+ (d) Hg^{2+}

4.42 QUALITATIVE SALT ANALYSIS

33. Consider the following list of reagents :-

[JEE-ADV.2014]

Acidified $K_2Cr_2O_7$, alkaline $KMnO_4$, $CuSO_4$, H_2O_2 , Cl_2 , O_3 , $FeCl_3$, HNO_3 and $Na_2S_2O_3$

The total number of reagents that can oxidise aqueous iodide to iodine is :

34. Among PbS , CuS , HgS , MnS , Ag_2S , NiS , CoS , Bi_2S_3 and SnS_2 the total number of black coloured sulphide is.

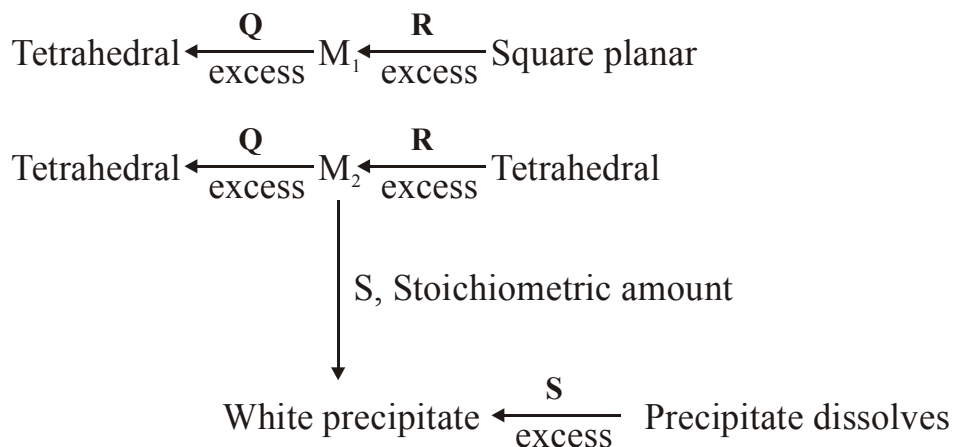
[JEE-ADV. 2014]

Passage for Q. 35 & 36

An aqueous solution of metal ion M_1 reacts separately with reagent Q and R in excess to give tetrahedral and square planar complexes, respectively. An aqueous solution of another metal ion M_2 always forms tetrahedral complex with these reagents. Aqueous solution of M_2 on reaction with reagent S gives white precipitate which dissolves in excess of S. The reaction are summarized in the scheme given below :

[JEE-ADV.2014]

SCHEME :



35. M_1 , Q and R respectively are :-

(a) Zn^{2+} , KCN and HCl

(b) Ni^{2+} , HCl and KCN

(c) Cd^{2+} , KCN and HCl

(d) Co^{2+} , HCl and KCN

36. Reagent S is :-

(a) $K_4[Fe(CN)_6]$

(b) Na_2HPO_4

(c) K_2CrO_4

(d) KOH

37. Fe^{3+} is reduced to Fe^{2+} by using :-

[JEE-Adv.-2015]

(a) H_2O_2 in presence of NaOH

(b) Na_2O_2 in water

(c) H_2O_2 in presence of H_2SO_4

(d) Na_2O_2 in presence of H_2SO_4

38. The pairs of ions where BOTH the ions are precipitated upon passing H_2S gas in presence of dilute HCl, is (are) :-

[JEE-ADV.2015]

(a) Ba^{2+} , Zn^{2+}

(b) Bi^{3+} , Fe^{3+}

(c) Cu^{2+} , Pb^{2+}

(d) Hg^{2+} , Bi^{3+}

■ ■

THE s-BLOCK ELEMENTS AND THEIR COMPOUNDS

PROBLEMS BASED ON GIVEN TOPICS

Group 1-the Alkali metals

- General properties
 - Electronic structure
 - Size of atoms and ions
 - Density
 - Ionization energy
 - Electronegativity and bond type
- Born - Haber cycle : Energy changes in the formation of ionic compounds
- Structures of the metals, Hardness and Cohesive energy
 - Melting and boiling points
- Flame colours and spectra
- Colour of compounds
- Chemical properties
 - Reaction with water
 - Reaction with air
 - Reaction with dinitrogen
- Oxides, Hydroxides, Peroxides and Superoxides
 - Reaction with air
 - Normal oxides - monoxides
 - Hydroxides
 - Peroxides and Superoxides
- Sulphides
- Oxosalts - Carbonates, Bicarbonates, Nitrates, Nitrites and Sulphates
- Halides and Polyhalides
- Hydrides
- Solubility and hydration
- Solution of metals in liquid ammonia
- Compounds with carbon
- Complexes, crowns and crypts
- Biological importance
- Difference between Lithium and the other group 1 elements

The Chlor-Alkali industry

- Sodium hydroxide
- Leblanc process
- Weldon and deacon processes
- Electrolytic processes
 - Diaphragm cell
 - Mercury cathode cell
 - Quantities
- Sodium carbonate
- The solvay (or Ammonia - Soda) process

Group 2-the Alkaline earth elements

- General properties
 - Electronic structure
 - Size of atoms and ions
 - Ionisation energy
 - Electronegativity
 - Hydration energies
 - Solubility and lattice energy
 - Solutions of the metals in liquid ammonia
- Anomalous behaviour of Beryllium
- Chemical properties
 - Reaction with water
- Hydroxides
- Hardness of water
- Reaction with acid and bases
- Oxides and peroxides
- Sulphates
- Nitrates
- Hydrides
- Halides
- Nitrides
- Carbides
- Complexes
- Biological role of Mg^{2+} and Ca^{2+}

The s-Block Elements and Their Compounds

EXERCISE # I

❑ Only one correct answer :

- Sodium pentasulphide is (Na_2S_5). Anionic part S_5^{2-} is :
 (a) Tetrahedral (b) Bent
 (c) Trigonal planar (d) Trigonal bipyramidal
- When Na reacts with NH_3 gas at 360°C , the gas evolved is :
 (a) N_2 (b) N_2O
 (c) NO (d) H_2
- Aqueous solution of Na_2CO_3 is :
 (a) Acidic (b) Basic
 (c) Neutral (d) None
- $2\text{Na} + 2\text{NH}_3 \longrightarrow 2\text{NaNH}_2 + \text{H}_2$
 In this reaction NH_3 acts as
 (a) Reducing agent (b) Dehydrating reagent
 (c) Basic gas (d) Oxidising reagent
- A soft metal (A) gives violet flame test. When burnt in excess O_2 , A give yellow powder B which on reaction with water gives alkaline solution A & B are respectively :
 (a) Na ; Na_2O_2 (b) K ; K_2O_2
 (c) K ; KO_2 (d) Li ; Li_2O
- Which of the following pair of compounds does not coexist ?
 (a) $\text{NaOH} + \text{Na}_2\text{CO}_3$
 (b) $\text{NaOH} + \text{NaHCO}_3$
 (c) $\text{NaOH} + \text{Na}$
 (d) $\text{NaOH} + \text{NaNH}_2$
- Which of the following substance is not related to the Solvay process ?
 (a) CaCO_3 (b) NaHCO_3
 (c) NH_3 (d) KCl
- Alkali metals can be kept in :
 (a) H_2O (b) CH_3OH
 (c) $\text{CH}_3\text{CO}_2\text{H}$ (d) Kerosene oil
- The ionic mobility of alkali metals in aqueous solution is maximum for
 (a) Rb^+ (b) K^+
 (c) Na^+ (d) Li^+
- The products obtained on heating LiNO_3 will be
 (a) $\text{LiNO}_2 + \text{O}_2$ (b) $\text{Li}_2\text{O} + \text{NO}_2 + \text{O}_2$
 (c) $\text{Li}_3\text{N} + \text{O}_2$ (d) $\text{Li}_2\text{O} + \text{NO} + \text{O}_2$
- When Li is combusted in air the mixture of product is :
 (a) $\text{Li}_2\text{O} + \text{Li}_2\text{O}_2$ (b) Li_3N only
 (c) $\text{Li}_2\text{O} + \text{Li}_3\text{N}$ (d) $\text{Li}_3\text{N} + \text{LiO}_2$
- Which Na is exposed to air the mixture of product is :
 (a) $\text{Na}_2\text{O} + \text{Na}_3\text{N}$ (b) $\text{Na}_2\text{O} + \text{NaO}_2$
 (c) $\text{Na}_2\text{O} + \text{Na}_2\text{O}_2$ (d) $\text{Na}_3\text{N} + \text{NaO}_2$
- What happened when to the aqueous solution of Na_2O , red litmus paper is added :
 (a) No change in colour
 (b) Red litmus paper becomes blue (permanent)
 (c) Red litmus paper becomes blue but after sometime the colour disappered
 (d) Red litmus paper becomes green
- What happened when to the aqueous solution of Na_2O_2 , red litmus paper is added
 (a) No change in colour
 (b) Red litmus paper becomes blue (permanent)
 (c) Red litmus paper becomes blue but after sometime the colour disappered
 (d) Red litmus paper becomes green

5.4 THE s-BLOCK ELEMENTS AND THEIR COMPOUNDS

15. Weakest base among the following :
 (a) Ca(OH)_2 (b) Ba(OH)_2
 (c) Be(OH)_2 (d) KOH
16. If I_2 and I_2' is the second ionisation potentials of alkali and alkaline earth metal for same period respectively then.
 (a) $I_2 = I_2'$ (b) $I_2 > I_2'$
 (c) $I_2' > I_2$ (d) $I_2' > > > I_2$
17. Bleaching powder, a mixture of $\text{CaCl}_2 + \text{Ca(OCl)}_2$ is obtained by passing Cl_2 gas with.
 (a) Ca(OH)_2 (b) CaO
 (c) CaCO_3 (d) CaCl_2
18. Hydroxide of first element of group-IA and group-IIA are respectively :
 (a) LiOH , Be(OH)_2
 (b) NaOH , Mg(OH)_2
 (c) KOH , Ca(OH)_2
 (d) RbOH , Be(OH)_2
19. The correct order of 2nd ionisation potential for Mg, Sr and K is :
 (a) $\text{K} > \text{Sr} > \text{Mg}$ (b) $\text{Mg} > \text{Sr} > \text{K}$
 (c) $\text{Sr} > \text{Mg} > \text{K}$ (d) $\text{K} > \text{Mg} > \text{Sr}$
20. The formula of soda ash is :
 (a) $\text{Na}_2\text{CO}_3 \cdot 10\text{H}_2\text{O}$ (b) $\text{Na}_2\text{CO}_3 \cdot 2\text{H}_2\text{O}$
 (c) $\text{Na}_2\text{CO}_3 \cdot \text{H}_2\text{O}$ (d) Na_2CO_3
21. The substance that gives brick red flame and breaks down on heating to give O_2 and a brown gas is
 (a) Magnesium nitrate
 (b) Calcium nitrate
 (c) Barium nitrate
 (d) Sodium nitrate
22. In the synthesis of Na_2CO_3 , the recovery of ammonia is done by heating NH_4Cl with Ca(OH)_2 . The by product obtained in this process.
 (a) CaCl_2 (b) NaCl
 (c) NaOH (d) NaHCO_3
23. Which of the following does not form hydride by direct heating with H_2 (dihydrogen) :
 (a) Be (b) Mg
 (c) Sr (d) Ba
24. To prepare BeF_2 the best route is :
 (a) Thermal decomposition of $(\text{NH}_4)_2\text{BeF}_4$
 (b) $\text{BeO} + \text{C} + \text{F}_2 \xrightleftharpoons{600^\circ\text{C}} \text{BeF}_2 + \text{CO}$
 (c) $\text{CaF}_2 + \text{BeO} \rightarrow \text{BeF}_2 + \text{CaO}$
 (d) $\text{Be} + \text{F}_2 \rightarrow \text{BeF}_2$
25. Cement clinker is :
 (a) Fused mass of clay & gypsum
 (b) Fused mass of lime & gypsum
 (c) Fused mass of clay & lime
 (d) Fused mass of Ca(OH)_2 & $\text{CaSO}_4 \cdot 1/2\text{H}_2\text{O}$
26. The milk of magnesia used as antacid is chemically
 (a) Mg(OH)_2
 (b) MgO
 (c) MgCl_2
 (d) $\text{MgO} + \text{MgCl}_2$
27. Which of the following are amphoteric in nature ?
 (a) BeO (b) Be(OH)_2
 (c) Al(OH)_3 (d) All
28. MgBr_2 , MgI_2 are soluble in acetone because of
 (a) Ionic character
 (b) Metallic nature
 (c) Covalent nature
 (d) Coordination compound
29. White heavy precipitate is formed when BaCl_2 is added to a clear solution of compound A. If precipitate is insoluble in dilute HCl then anionic part of compound A is :
 (a) Bicarbonate (b) Carbonate
 (c) Nitrite (d) Sulphate
30.
$$\text{2-Butyne} \xrightarrow[\text{Liq. NH}_3]{\text{Li in}} \text{P}_1$$

$$\text{2-Butyne} \xrightarrow[\text{BaSO}_4 + \text{Pd}]{+ \text{H}_2} \text{P}_2$$

 P_1 & P_2 are related as :
 (a) Structural isomers
 (b) Optical isomers
 (c) Geometrical isomers
 (d) Not isomers

31. Which of the following group of elements have chemical properties that is almost similar :
(a) Rb, Cs, Ca (b) Be, Al, Ba
(c) Li, Mg, K (d) Mg, Sr, Ba
32. CsI_3 , KI_3 , TlI_3 contains :
(a) Monopositive cation Cs^+ , K^+ , Tl^+ & triiodide ion I_3^-
(b) Cs^{+1} , K^{+1} , Tl^{3+} & I_3^- in CsI_3 , I_3^- in KI_3 & I^- in TlI_3
(c) Ca^{2+} , K^+ , Tl^{3+} & iodide ion I^-
(d) Cs^+ , K^{3+} , Tl^{+3} ion and I_3^- ion
33. Among the alkaline earth metal halide given below which is most covalent in nature ?
(a) SrCl_2 (b) BeCl_2
(c) MgCl_2 (d) CaCl_2
34. Which of the following reacts most vigorously with water ?
(a) Li (b) K
(c) Rb (d) Na
35. Among nitrate of alkali metal which one can be decomposed to its oxide most easily ?
(a) NaNO_3 (b) KNO_3
(c) LiNO_3 (d) RbNO_3
36. Among the carbonate of alkali metals which has highest thermal stability ?
(a) Cs_2CO_3 (b) Rb_2CO_3
(c) K_2CO_3 (d) Na_2CO_3
37. $\text{X} + \text{C} + \text{Cl}_2 \xrightarrow{\Delta} \text{Chloride salt} + \text{CO}\uparrow$
X can be :
(a) Na_2O , CaO (b) SrO , Cs_2O
(c) BeO , Al_2O_3 (d) MgO , K_2O
38. CaCO_3 can be prepared by :
(a) By passing CO_2 through slaked lime $\text{Ca(OH)}_2 + \text{CO}_2 \rightarrow \text{CaCO}_3 + \text{H}_2\text{O}$
(b) By the addition of sodium carbonate to calcium chloride $\text{CaCl}_2 + \text{Na}_2\text{CO}_3 \rightarrow \text{CaCO}_3 + 2\text{NaCl}$
(c) Both (a) & (b)
(d) None
39. The dilute solution of all alkali metal in liquid ammonia are blue in colour. The blue colour of the solution.
(a) Due to excitation of free electrons to high energy level (s $\rightarrow$ p transition). This type of transition gives absorption band near 1500°A in red region of spectrum therefore the solution looks blue in transmitted light.
(b) Due to excitation of free electrons to high energy level (s $\rightarrow$ p transition). This type of transition gives absorption band near 1500°A in yellow region of spectrum therefore the solution looks blue in transmitted light.
(c) Due to excitation of free electrons to high energy level (s $\rightarrow$ d transition). This type of transition gives absorption band near 1500°A in red region of spectrum therefore the solution looks blue in transmitted light.
(d) None of these
40. When calcium carbide (CaC_2) reacts with atmospheric N_2 at very high temperature, the product mixture formed is :
(a) $\text{Ca(CN)}_2 + \text{C}$ (b) $\text{CaCN}_2 + \text{C}$
(c) $\text{CaNC}_2 + \text{C}$ (d) $\text{Ca}_3\text{N}_2 + \text{C}$
41. The correct formula of hydrate of halide of alkaline earth metal :
(a) $\text{MgCl}_2 \cdot 2\text{H}_2\text{O}$; $\text{CaCl}_2 \cdot 4\text{H}_2\text{O}$; $\text{SrCl}_2 \cdot 6\text{H}_2\text{O}$; $\text{BaCl}_2 \cdot 8\text{H}_2\text{O}$
(b) $\text{MgCl}_2 \cdot 8\text{H}_2\text{O}$; $\text{CaCl}_2 \cdot 4\text{H}_2\text{O}$; $\text{SrCl}_2 \cdot 2\text{H}_2\text{O}$; $\text{BaCl}_2 \cdot 6\text{H}_2\text{O}$
(c) $\text{MgCl}_2 \cdot 8\text{H}_2\text{O}$; $\text{CaCl}_2 \cdot 6\text{H}_2\text{O}$; $\text{SrCl}_2 \cdot 6\text{H}_2\text{O}$; $\text{BaCl}_2 \cdot 2\text{H}_2\text{O}$
(d) None
42. $\text{X(salt)} + \text{dil. H}_2\text{SO}_4 \longrightarrow \text{Y(gas)}$
 $\text{Y(gas)} + \text{Ca(OH)}_2 \longrightarrow \text{Z(miky ppt.)}$
The salt X may be :
(a) NaNO_2
(b) Na_2SO_3
(c) Na_2CO_3
(d) Both (b) & (c)

5.6 THE s-BLOCK ELEMENTS AND THEIR COMPOUNDS

43. Percentage loss on mass on heating mixture of Na_2CO_3 & CaCO_3 containing equal mass of the two components will be :
 (a) 44% (b) 22%
 (c) 38% (d) 80%
44. If there three electron been possible to be placed in an orbital the position of potassium in periodic table would be in which block :
 (a) s (b) p
 (c) d (d) f
45. $E^\circ_{\text{Cl}_2/\text{Cl}^-} \Rightarrow +1.36 \text{ volt}$ $E^\circ_{\text{Ag}^+/\text{Ag}} \Rightarrow +0.79 \text{ volt}$
 $E^\circ_{\text{I}_2/\text{I}^-} \Rightarrow +0.53 \text{ volt}$ $E^\circ_{\text{Na}^+/\text{Na}} \Rightarrow -2.71 \text{ volt}$
 $E^\circ_{\text{Li}^+/\text{Li}} \Rightarrow -3.04 \text{ volt}$
 Correct order of reducing strength.
 (a) $\text{Li} > \text{Cl}^- > \text{Ag} > \text{I}^- > \text{Na}$
 (b) $\text{Cl}^- > \text{Ag} > \text{I}^- > \text{Na} > \text{Li}$
 (c) $\text{Li} > \text{Na} > \text{I}^- > \text{Ag} > \text{Cl}^-$
 (d) None
46. Beryllium & Magnesium are kinetically inert to oxygen & water
 (a) Because of the formation of an oxide film on their surface
 (b) Because of their large size
 (c) Because of large hydration energy of Be^{2+} & Mg^{2+}
 (d) All
47. Baking powder is used to cause the bread to swell. Baking powder is a
 (a) Mixture of sodium potassium tartarate & sodium carbonate and during reaction CO_2 gas evolved cause the bread to swell
 (b) Mixture of potassium hydrogen tartarate & sodium carbonate and during reaction CO_2 gas evolved cause the bread to swell
 (c) Mixture of sodium potassium tartarate & sodium bicarbonate and during reaction CO_2 gas evolved cause the bread to swell
 (d) Mixture of potassium hydrogen tartarate & sodium bicarbonate and during reaction CO_2 gas evolved cause the bread to swell
48. Which of the alkali metal produce hydrated perchlorate ($\text{MClO}_4 \cdot 3\text{H}_2\text{O}$)
 (a) K (b) Rb
 (c) Cs (d) Li
49. Correct order of weight percentage of calcium for same amount of gypsum & plaster of paris.
 (a) (% of Ca)_{gypsum} > (% of Ca)_{plaster of paris}
 (b) (% of Ca)_{gypsum} < (% of Ca)_{plaster of paris}
 (c) (% of Ca)_{gypsum} = (% of Ca)_{plaster of paris}
 (d) None of these
50. Aqueous solution of NaCl & fused NaH when undergo electrolysis, the same gas is evolved at cathode for NaCl (aq.) & at anode for NaH. What is that gas ?
 (a) O_2 (b) Cl_2
 (c) H_2 (d) H_2O vapours
51. Fused NaCl when undergo electrolysis using proper electrode.
 (a) Na is deposited at anode and Cl_2 is evolved at cathode
 (b) Na is deposited at cathode and Cl_2 is evolved at anode
 (c) Na is deposited at cathode and O_2 is evolved at anode
 (d) Na is deposited at anode and H_2 is evolved at cathode
52. Calcium is obtained from its ore by :
 (a) Carbon reduction
 (b) Reduction by Aluminium
 (c) Electrolytic reduction
 (d) Reduction by hydrogen gas
53. Which of the following oxalate is most insoluble ?
 (a) BeC_2O_4
 (b) $\text{K}_2\text{C}_2\text{O}_4$
 (c) CaC_2O_4
 (d) $\text{Na}_2\text{C}_2\text{O}_4$
54. Which alkaline earth metal has highest melting point ?
 (a) Be (b) Mg
 (c) Ca (d) Sr

55. Which alkaline earth metal has lowest boiling point ?
(a) Be (b) Mg
(c) Ca (d) Sr
56. Fire extinguisher contains.
(a) CaCO_3 solution
(b) H_2SO_4 and NaHCO_3 solution
(c) NaHCO_3 solution
(d) CaCO_3 and H_2SO_4 solution
57. Crown ethers & cryptands form
(a) Complex with alkali metals
(b) Hydroxides of alkali metals
(c) Salts of alkali metals used for inorganic quantitative analysis
(d) Organic salts of alkali metals
58. The hydroxides of alkaline earth metal which has lowest value of solubility product at normal temperature (25°C)
(a) $\text{Ca}(\text{OH})_2$ (b) $\text{Mg}(\text{OH})_2$
(c) $\text{Sr}(\text{OH})_2$ (d) $\text{Be}(\text{OH})_2$
59. $\text{Na} + \text{Al}_2\text{O}_3 \xrightarrow{\text{High temperature}} \text{X} \xrightarrow[\text{water}]{\text{CO}_2 \text{ in}} \text{Y}$, the compound Y is
(a) NaAlO_2 (b) NaHCO_3
(c) Na_2CO_3 (d) Na_2O_2
60. Which of the following group of element have chemical properties that are most similar
(a) Na, K, Ca
(b) Mg, Sr, Ba
(c) Be, Al, Ba
(d) Be, Cs, Ra
61. The commercial method of preparation of potassium by reduction of molten KCl with metallic sodium at 50°C is based on the fact that
(a) potassium is solid and sodium is liquid at 850°C
(b) sodium has less affinity to chloride ions than potassium
(c) potassium being more volatile so it distills off thus shifting the reaction forward
(d) reactivity of sodium is more than potassium
62. Lattice energy order
(a) $\text{LiF} > \text{NaF} > \text{KF} > \text{RbF} > \text{CsF}$
(b) $\text{LiF} < \text{NaF} < \text{KF} < \text{RbF} < \text{CsF}$
(c) $\text{NaF} > \text{KF} > \text{LiF} > \text{RbF} > \text{CsF}$
(d) $\text{RbF} > \text{CsF} > \text{KF} > \text{NaF} > \text{LiF}$
63. If NaOH is added to an aqueous solution of Al^{3+} ions, a white precipitate appears and on adding excess of NaOH, the precipitate dissolves. In the solution, aluminium exists in the
(a) anionic part
(b) cationic part
(c) both in anionic and cationic part
(d) colloidal form
64. When NaNO_3 is treated with Na:
(a) NaNO_2 and O_2 are formed
(b) Na_2O and N_2 are formed
(c) Na_2O_2 and N_2 are formed
(d) Na_2O and Na_3N are formed
65. What happens when a mixture of sodium sulphate, lime stone and coke is heated in a furnace?
(a) Black ash containing sodium carbonate and calcium sulphide is formed
(b) Carbon dioxide gas is liberated
(c) Sodium carbonate and calcium sulphate is formed
(d) (a) and (b) both
66. Alkali metals possess metallic lustre when freshly cut because :
(a) they have a hard surface and light is reflected back
(b) their crystal structure contains ordered arrangement of constituent atoms
(c) they contain loosely bound electrons which absorb the photons and then re-emit.
(d) they are obtained from the minerals on which light has been falling for years.
67. Low solubility of CsI in water is due to
(a) smaller hydration enthalpy of Cs^+
(b) smaller hydration enthalpy of I^-
(c) lower lattice enthalpy of its two ions
(d) (a) and (b) both

68. When ionic hydrides react with water, the products are :
- acidic solutions and hydrogen gas
 - acidic solutions and oxygen gas
 - basic solutions and hydrogen gas
 - basic solutions and oxygen gas
69. The stability of K_2O , K_2O_2 and KO_2 is in order $K_2O < K_2O_2 < KO_2$. This increasing stability as the size of anion increases is due to stabilisation of:
- larger cations by smaller anions through lattice energy effects.
 - larger cations by larger anions through lattice energy effects.
 - smaller cations by smaller anions through melting point.
 - smaller cations by larger anions through hydration energy effect.
70. Sodium dichromate on heating with coke yields:
- sodium carbonate
 - sodium chromite
 - sodium chromate
 - chromic anhydride
71. Barium carbide + N_2 gas $\rightarrow$ Product (P). P is
- $Ba(CN)_2$
 - $BaCN_2$
 - BaN_3
 - Ba_3N_2
72. During heating of sodium nitrate at low temperature & at very high temperature what is the common product.
- Na_2O
 - N_2O
 - NO_2
 - O_2
73. Chromate of which alkali metals is not dissolved in acetic acid
- Ca
 - Sr
 - Ba
 - all
74. Number of Cl atoms directly attached to each beryllium atom in solid $BeCl_2$ is
- 2
 - 3
 - 4
 - 5
75. Of the following the commonly used in the laboratory desiccator is .
- K_2CO_3
 - $CaCl_2$
 - NaCl
 - $CaCO_3$

EXERCISE # II**□ One or More Than One Correct Answer :**

- When KO_2 reacts with water then the products are.
 - KOH
 - H_2O_2
 - H_2O
 - O_2
- Which of the following are correct ?
 - Na_3PO_4 : Boneash
 - CaF_2 : Fluorosparg
 - $NaNH_2$: Sodamide
 - KOH : Caustic potash
- The hydration energy of Be^{2+} is :
 - More than that of Mg^{2+}
 - More than that of Li^+
 - More than that of K^+
 - More than that of Al^{3+}
- Which of the following reaction giving $MgSO_4$ as one of the product ?
 - Mg reacts with concentrated solution of H_2SO_4
 - Mg reacts with dilute H_2SO_4
 - MgO react with dilute H_2SO_4
 - Dolomite react with dilute H_2SO_4
- Which of the following reaction give hydrogen gas as a product ?
 - $Mg + \text{dilute } HCl \rightarrow$
 - $Mg + \text{dilute } H_2SO_4 \rightarrow$
 - $Mg + \text{Conc. } H_2SO_4 \rightarrow$
 - $Mg + \text{very dilute } HNO_3 \rightarrow$
- Which of the following reactions are spontaneous ?
 - $MgCl_2 + Ca(OH)_2 \longrightarrow Mg(OH)_2 + CaCl_2$
 - $BeSO_4 + BaCl_2 \longrightarrow BeCl_2 + BaSO_4$
 - $Ca(HCO_3)_2 + Ca(OH)_2 \longrightarrow CaCO_3 + 2H_2O$
 - $CaCO_3 + CO_2 + H_2O \longrightarrow Ca(HCO_3)_2$
- Which of the compounds are coloured and give violet colour to oxidising flame?
 - KO_2
 - KCl
 - KO_3
 - KNO_3

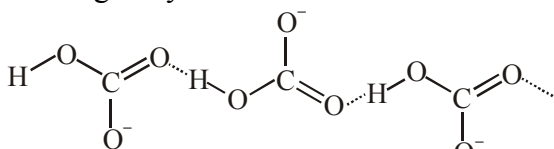
8. Which of the compounds do not have peroxy linkage and give golden yellow colour to oxidising flame.
 (a) Na_2O_2 (b) NaNO_2
 (c) BaO_2 (d) Na_2SO_3
9. Which of the halide of beryllium can't be polymerised?
 (a) BeF_2 (b) BeCl_2
 (c) BeBr_2 (d) BeI_2
10. Which of the following are example of organometallic compound?
 (a) LiCH_3 (b) NaOCH_3
 (c) BeH_2 (d) CH_3MgBr
11. Mg^{2+} is smaller than
 (a) Ba^{2+} (b) Be^{2+}
 (c) Li^+ (d) Na^+
12. Rb^+ is larger than :
 (a) Sr^{2+} (b) Ca^{2+}
 (c) K^+ (d) Cs^+
13. CaCO_3 can't react with :
 (a) Ca(OH)_2 (b) NaOH
 (c) HCl (d) H_2SO_4
14. $\text{Ca(HCO}_3)_2$ can react with :
 (a) Ca(OH)_2 (b) NaOH
 (c) HCl (d) HNO_3
15. BeH_2 can be prepared by
 (a) Reaction of BeCl_2 with HCl
 (b) Reaction of BeCl_2 with LiAlH_4
 (c) Direct combination of Be and H_2
 (d) Reaction of BeCl_2 with LiH
16. Which of the following sulphates are sufficiently soluble?
 (a) BeSO_4 (b) MgSO_4
 (c) BaSO_4 (d) SrSO_4
17. Potassium formate when allowed to heat in air, the products formed are :
 (a) K_2CO_3 (b) CO_2
 (c) H_2O (d) $\text{K}_2(\text{CO}_3)_2$
18. Correct order of basic strength :
 (a) $\text{Cs}_2\text{O} > \text{Rb}_2\text{O} > \text{K}_2\text{O} > \text{Na}_2\text{O} > \text{Li}_2\text{O}$
 (b) $\text{BaO} > \text{SrO} > \text{CaO} > \text{MgO} > \text{BeO}$
 (c) $\text{Cs}_2\text{O} > \text{K}_2\text{O} > \text{CaO} > \text{MgO} > \text{NiO}$
 (d) $\text{NaNH}_2 > \text{NaOH} > \text{NaI}$
19. Which of the following reactions give correct products?
 (a) $\text{Na} + \text{N}_2 \rightarrow \text{Na}_3\text{N}$
 (b) $\text{K} + \text{N}_2 \rightarrow \text{K}_3\text{N}$
 (c) $\text{Li} + \text{N}_2 \rightarrow \text{Li}_3\text{N}$
 (d) $\text{Mg} + \text{N}_2 \rightarrow \text{Mg}_3\text{N}_2$
20. Which of the compound has polyhalide ion?
 (a) KI_3 (b) AlI_3
 (c) ClF_3 (d) RbICl_2
21. Correct statements :
 (a) Lithium is stored in paraffin wax
 (b) Na is stored in kerosen oil
 (c) Na_2HPO_4 is stored in NaOH solution
 (d) All
22. Which salt react with NaOH to give coloured precipitate?
 (a) $\text{Ni(NO}_3)_2$ (b) FeCl_2
 (c) CuSO_4 (d) SnCl_2
23. Which compounds (oxide) react with NaOH to give products by disproportionation product?
 (a) NO_2 (b) Cl_2O_7
 (c) SO_2 (d) Cl_2O_6
24. Which of the following compound when react with NaOH to give salt by nonredox reaction?
 (a) Acetic acid ($\text{CH}_3\text{-C(OH)=O}$)
 (b) Methyl sulphonic acid ($\text{CH}_3\text{-S(O)}_2\text{(OH)}$)
 (c) Carbon dioxide
 (d) Sulphur trioxide

5.10 THE s-BLOCK ELEMENTS AND THEIR COMPOUNDS

25. Which of the following statements are not correct?
- K_2CO_3 can't be prepared by Solvay process because $KHCO_3$ is insoluble in H_2O
 - (Li^+) ionic mobility in aq. phase $>$ (Na^+) ionic mobility in aq. phase
 - Lithium is the strongest reducing alkali metals in aqueous solution
 - KOH is used as absorber of CO_2 gas

26. Correct option regarding $NaHCO_3$:

- It reacts with ethanol to evolve CO_2
- It reacts with acetic acid to evolve CO_2
- The bicarbonate ion are held together by H-bonding in crystal structure



- On heating it gives mixture of $Na_2O + H_2O + CO_2$

27. In group-IA (alkali metals), correct options are :

- Lithium fluoride is most stable fluoride
- Lithium iodides has maximum covalent character out of all halides of alkali metals
- Lithium carbonate is least thermally stable carbonate
- Lithium oxide is most basic oxide

28. Which of the following are paramagnetic?

- KO_2
- KO_3
- Na_2O_2
- RbO_2

29. Which of the following options are correct ?

- Lowest melting point among alkaline earth metal is Mg
- Highest boiling point among alkaline earth metal is Be
- Lowest melting point among alkali metal is Cs
- Highest boiling point among alkali metal is Li

30. Which of the following reactions are correctly matched ?

- $CaC_2 + N_2 \longrightarrow Ca(CN)_2$
- $BaC_2 + N_2 \longrightarrow Ba(CN)_2$
- $CaC_2 + N_2 \longrightarrow CaCN_2 + C$
- $BaC_2 + N_2 \longrightarrow BaCN_2 + C$

31. Which of the following reactions are correctly matched ?

- $Be_2C + D_2O \longrightarrow CD_4 + Be(OD)_2$
- $NaH + H_2O \longrightarrow NaOH + H_2$
- $2NaHCO_3 \xrightarrow{\Delta} Na_2CO_3 + H_2O + CO_2$
- $Na_2CO_3 \xrightarrow{\Delta} Na_2O + CO_2$

32. Select the correct statement for s-block elements and their compounds.

- All the element form basic oxide
- Potassium ion are the most abundant cations within cell fluids
- Most covalent halide is BeI_2 among all alkaline earth metals.
- Mg react with very dilute HNO_3 (2%) to give N_2O gas

33. Which of the following statement is incorrect for Na_2O_2 ?

- It absorbs CO_2
- At room temperature it produces O_2 with water
- It produces NO_2 with NH_3
- It converts green solution of Cr^{3+} to yellow solution

34. Which of the following order is /are correct ?

- $Na^+ < K^+$ (Ionic radius)
- $Na^+ < K^+$ (Hydration energy)
- $Na^+ < K^+$ (Polarising power)
- $Na^+ < K^+$ (Electrical conductance in aqueous solution)

35. Which of the following element reacts with NaOH solution to give H_2 gas?

- Zn
- Al
- Sn
- Pb

36. which salt reacts with NaOH to give white precipitate which is soluble in excess NaOH ?

- $ZnCl_2$
- $AlCl_3$
- $CuSO_4$
- $FeSO_4$

37. Which of the following options are correct ?

- LiI is less soluble than KI in ethanol
- Alkali metals can be prepared by electrolysis of their fused chloride

- (c) Sodium is found to be more useful than potassium
- (d) Lithium salts are commonly hydrated and those to the other alkali ions are usually anhydrous
38. Which of the following compounds of potassium are correctly matched ?
- (a) Carnalite : $\text{KCl} \cdot \text{MgCl}_2 \cdot 6\text{H}_2\text{O}$
- (b) Langbeinite : $\text{K}_2\text{SO}_4 \cdot \text{MgSO}_4$
- (c) Sylvine : KCl
- (d) Caustic Potash : KOH
39. Correct decreasing order of different properties :
- (a) K_{sp} : $\text{Ba}(\text{OH})_2 > \text{Sr}(\text{OH})_2 > \text{Ca}(\text{OH})_2 > \text{Mg}(\text{OH})_2 > \text{Be}(\text{OH})_2$
- (b) Electronegativity : $\text{Be} > \text{Mg} > \text{Ca} > \text{Ba}$
- (c) Heat of hydration : $\text{Ba}^{2+} > \text{Sr}^{2+} > \text{Ca}^{2+} > \text{Be}^{2+}$
- (d) Basic strength : $\text{CsOH} > \text{KOH} > \text{LiOH}$
40. Select correct statements.
- (a) NaNO_3 , KNO_3 both are soluble
- (b) NaClO_4 is water soluble but KClO_4 is insoluble
- (c) $\text{Na}_3[\text{Co}(\text{NO}_2)_6]$ is water soluble but $\text{K}_3[\text{Co}(\text{NO}_2)_6]$ is insoluble
- (d) NaCl & KCl both are insoluble
41. $2\text{M}(\text{NO}_3)_2 \rightarrow 2\text{MO} + 4\text{NO}_2 + \text{O}_2$
This decomposition pattern is applicable for which of alkaline earth metal.
- (a) Be (b) Mg
- (c) Ca (d) Sr
42. $\text{M} + (\text{x} + \text{y}) \text{NH}_3 \rightarrow [\text{M}(\text{NH}_3)_x]^{2+} + 2[\text{e}(\text{NH}_3)_y]^{-1}$... (1)
 $\text{M} + (\text{x} + \text{y}) \text{NH}_3 \rightarrow [\text{M}(\text{NH}_3)_x]^{+1} + [\text{e}(\text{NH}_3)_y]^{-1}$... (2)
Which options are correct.]
- (a) Equation (2) is applicable for alkali metal
- (b) Equation (1) is applicable for alkaline earth metal
- (c) Both the solution in product side is blue in colour
- (d) Both the solution in product side are paramagnetic
43. Which of the following compounds on hydrolysis give NH_3 gas ?
- (a) Ca_3N_2 (b) CaCN_2
- (c) Mg_3N_2 (d) Li_3N
44. The various silicates & aluminates which constitute more than 90% of the cement are :
- (a) Tricalcium silicate $3\text{CaO} \cdot \text{SiO}_2$
- (b) Dicalcium silicate $2\text{CaO} \cdot \text{SiO}_2$
- (c) Tricalcium aluminate $3\text{CaO} \cdot \text{Al}_2\text{O}_3$
- (d) Tetracalcium aluminoferrite $4\text{CaO} \cdot \text{Al}_2\text{O}_3 \cdot \text{Fe}_2\text{O}_3$
45. Be is amphoteric in nature. That's why Be reacts with several acids like H_2SO_4 ; HNO_3 . It also reacts with NaOH . How many reactions of Be with acids/base are correct ?
- (a) $\text{Be} + \text{H}_2\text{SO}_4 \rightarrow \text{BeSO}_4 + \text{H}_2$ (dilute)
- (b) $\text{Be} + 2\text{HNO}_3 \rightarrow \text{Be}(\text{NO}_3)_2 + \text{H}_2$ (hot)
- (c) $\text{Be} + 2\text{NaOH} \rightarrow \text{Na}_2\text{BeO}_2 + \text{H}_2$
- (d) $\text{Be} + 2\text{H}_2\text{SO}_4 \rightarrow \text{BeSO}_4 + \text{SO}_2 + 2\text{H}_2\text{O}$ (conc.)
46. Which of the following are polymeric in nature ?
- (a) BeCl_2 (vapour phase)
- (b) BeCl_2 (solid)
- (c) BeH_2 (solid)
- (d) BeH_2 (vapour phase)
47. Presence of which ion makes water hard ?
- (a) Na^+ (b) K^+
- (c) Mg^{2+} (d) Ca^{2+}
48. The pair of reagents that yield paramagnetic species is/are :
- (a) Na & excess of NH_3
- (b) K & excess of O_2
- (c) Cu & dilute HNO_3
- (d) O_2 & ethyl anthraquinol
49. Which of the following are ionic carbide ?
- (a) Be_2C (b) Al_4C_3
- (c) Mg_2C_3 (d) CaC_2

5.12 THE s-BLOCK ELEMENTS AND THEIR COMPOUNDS

50. Which of the following statements are correct ?
 (a) The carbonate of lithium i.e., Li_2CO_3 decomposes on heating to form lithium oxide Li_2O
 (b) Beryllium carbonate is kept in the atmosphere of carbon dioxide
 (c) Be^{2+} , Li^+ both have high polarising power
 (d) Be^{2+} , Li^+ both have high degree of hydration
51. Which of the following statements are true ?
 (a) BeCl_2 is linear molecule in the vapour phase but it is polymeric in the solid state
 (b) Oxide of Be and Al are amphoteric
 (c) Above 393 K gypsum converted into dead burnt plaster
 (d) CaO is basic oxide, so it reacts with acidic oxide like SO_3 & P_4O_{10}
52. Which of the following carbide on reaction with water to give CH_4 gas ?
 (a) Be_2C (b) Mg_2C_3
 (c) Al_4C_3 (d) CaC_2
53. Which of the following options are correct
 (a) BeCl_2 fumes in moist air.
 (b) Anhydrous calcium chloride can be used to dry NH_3
 (c) Beryllium halides dissolve in organic solvent
 (d) Mg can not form complex
54. Which of the following options are incorrect
 (a) Lithium salts are hygroscopic
 (b) BaSO_4 is reduced by carbon to give BaS & CO
 (c) Dry KOH reacts with ozone to yield black solid
 (d) LiI is more soluble than RbI in water.
55. Alkaline earth metals which does not react with cold water is
 (a) Mg (b) Ba
 (c) Be (d) Ca
56. Which of the following statements are correct?
 (a) Chloride of Be & Al are soluble in organic solvents and are strong Lewis acid.
 (b) Be & Al have strong tendency to form complexes with F^- .
 (c) Microcosmic salt is $\text{NaNH}_4\text{HPO}_4 \cdot 4\text{H}_2\text{O}$
 (d) BeCl_2 and AlCl_3 both are dimeric in liquid state
57. Which of the following options are incorrect
 (a) CaF_2 undergoes disproportionation to give CaF_3 and CaF
 (b) The product mixture formed is called lithophone when BaS reacts with ZnSO_4 solution.
 (c) BeCl_2 can exist in dimeric and polymeric state but AlCl_3 always exists as dimeric state.
 (d) $2\text{BeCl}_2 + \text{LiAlH}_4 \rightarrow 2\text{BeH}_2 + \text{LiCl} + \text{AlCl}_3$, in this reaction LiAlH_4 act as oxidising reagent.
58. Which of the following disproportionate (s) on heating with sodium hydroxide?
 (a) P_4 (b) S_8
 (c) Cl_2 (d) B
59. Hydrated sodium aluminium silicate, the zeolite, is treated with hard water, the sodium ions are exchanged with :
 (a) Ca^{2+} ions (b) Mg^{2+} ions
 (c) H^+ ions (d) SO_4^{2-} ions
60. Which of the following statements are correct?
 (a) potassium and caesium are useful in photoelectric cells
 (b) the dehydration of hydrated chloride, bromide, iodide of Ca, Sr, Ba can be achieved on heating.
 (c) KOH can be prepared by Le-Blanc process
 (d) Both LiCl & MgCl_2 are deliquescent.
61. Which of the following alkaline metals can not exist in elemental form
 (a) Li (b) Na
 (c) K (d) Si
62. Which of the following are example of water soluble compound
 (a) NaClO_4 (b) KClO_4
 (c) $\text{Na}_3[\text{Co}(\text{NO}_2)_6]$ (d) $\text{K}_3[\text{Co}(\text{NO}_2)_6]$
63. Which of the following statement(s) is/are correct with respect to alkali metal bicarbonate?
 (a) In crystalline sodium bicarbonate, the HCO_3^- anions form an infinite chain through H-bonding
 (b) Bicarbonates do not produce any colour with phenolphthalein indicator.

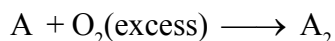
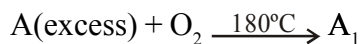
- (c) All bicarbonate except that of lithium are found in solid state
- (d) KHCO_3 is formed when CO_2 (in excess) reacts with KO_2 (medium water)
64. The more commonly used baking powder contains about 30% NaHCO_3 , 20% $\text{NaAl}(\text{SO}_4)_2$, 10% $\text{Ca}(\text{H}_2\text{PO}_4)_2$ and 40% starch. Which of the following statement(s) is/are correct?
- (a) $\text{Ca}(\text{H}_2\text{PO}_4)_2$ is acidic when moistened it reacts with NaHCO_3 evolving CO_2 gas
- (b) $\text{NaAl}(\text{SO}_4)_2$ slows down the decomposition reaction of NaHCO_3 so that CO_2 is evolved more slowly.
- (c) Starch functions as filler.
- (d) $\text{NaAl}(\text{SO}_4)_2$ and starch both acts as filler.
65. Which of the following statements are correct?
- (a) Dissolution of alkali metal hydroxide in water is exothermic.
- (b) Sea shells, dolomite, a marble statue and calcined gypsum, all contains CaCO_3 .
- (c) Lime water is a clear solution of $\text{Ca}(\text{OH})_2$ in water
- (d) Baryta water is a clear solution of $\text{Ca}(\text{OH})_2$ in water

EXERCISE # III

□ Linked Comprehension Type :

Paragraph for Q.No. 01 to 03

An element gives violet flame colour to the flame test



A_1 is colourless but A_2 is chrome yellow colour powder.

- A_1 & A_2 are respectively :
 (a) KO_2 ; KO_3 (b) K_2O ; KO_2
 (c) K_2O ; KO_3 (d) KO_3 ; K_2O
- Correct statement is :
 (a) A_1 & A_2 both are diamagnetic
 (b) A_1 & A_2 both are paramagnetic
 (c) A_1 is diamagnetic but A_2 is paramagnetic
 (d) A_2 is paramagnetic but A_1 is diamagnetic

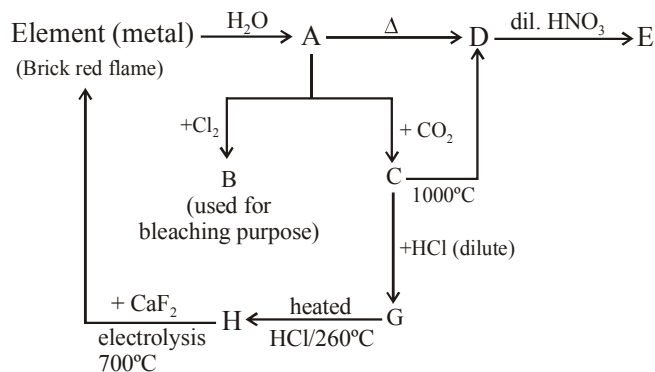
- $\text{A}_1 + \text{H}_2\text{O} \longrightarrow$ Alkaline solution
 $\text{A}_2 + \text{H}_2\text{O} \longrightarrow$ Alkaline solution + $\text{A}_3 + \text{A}_4$;
 $\text{A}_4 \xrightarrow{\Delta} \text{A}_3$ formed as one of the product
 A_3 & A_4 are respectively :
 (a) H_2 & O_2 (b) H_2O_2 ; O_2
 (c) H_2O_2 ; H_2 (d) O_2 ; H_2O_2

Paragraph for Q.No. 04 to 05

Baryta water has chemical formula $\text{Ba}(\text{OH})_2$. It is white solid. Its melting point is 78°C ; Its boiling point is 780°C .

- Which of the following option is correct regarding $\text{Ba}(\text{OH})_2$?
 (a) $\text{Ba}(\text{OH})_2$ form barium sulphate and barium phosphate with sulphuric and phosphoric acid, which are insoluble
 (b) $\text{Ba}(\text{OH})_2$ reacts with H_2S to give barium sulphide
 (c) $\text{Ba}(\text{OH})_2$ decomposes at 800°C to give BaO
 (d) All
- The reaction of barium hydroxide octahydrate with ammonium chloride is often used as a classroom chemistry demonstration because :
 (a) It produces very high temperature enough to boil water
 (b) It produces very low temperature enough to freeze water
 (c) The above reaction is highly exothermic
 (d) The above reaction is highly endothermic

Paragraph for Q.No. 06 to 08



- A is :
 (a) $\text{Mg}(\text{OH})_2$ (b) $\text{Ca}(\text{OH})_2$
 (c) CaO (d) CaH_2

5.14 THE s-BLOCK ELEMENTS AND THEIR COMPOUNDS

7. B & G are respectively :

- (a) MgCl_2 ; $\text{MgCl}_2 \cdot 6\text{H}_2\text{O}$
 (b) CaCl_2 ; $\text{CaCl}_2 \cdot 6\text{H}_2\text{O}$
 (c) $\text{Ca}(\text{OCl})\text{Cl}$; $\text{CaCl}_2 \cdot 6\text{H}_2\text{O}$
 (d) CaCl_2 ; $\text{Ca}(\text{OCl})\text{Cl}$

8. C & D are respectively :

- (a) $\text{Ca}(\text{HCO}_3)_2$; CaCO_3
 (b) CaCO_3 ; CaO
 (c) $\text{Ca}(\text{HCO}_3)_2$; CaO
 (d) $\text{Ca}(\text{NO}_3)_2$; CaSO_4

Paragraph for Q.No. 09 to 11

Sodium chloride is abundantly found in nature as rock salt. Sea water consists of 2.5% NaCl. It is used as preservative for food particles.

9. $\text{NaCl} + \text{Conc. H}_2\text{SO}_4 \longrightarrow \text{X} + \text{HCl}$

$\text{X} + \text{Coke} \longrightarrow \text{X}'$

X & X' are respectively :

- (a) Na_2SO_4 ; Na_2SO_3
 (b) NaHSO_4 ; NaHSO_3
 (c) Na_2SO_4 ; Na_2S
 (d) Na_2SO_4 ; SO_2

10. NaCl when reacts with mixture of MnO_2 & concentrated H_2SO_4 the gas evolved is

- (a) HCl (b) Cl_2
 (c) SO_2 (d) SO_3

11. Electrolysis of aqueous solution of NaCl results.

- (a) H_2 evolution at cathode
 (b) Cl_2 evolution at anode
 (c) Basic Solution
 (d) All

Paragraph for Q.No. 12 to 13

$\text{P} + \text{CO}_2 \longrightarrow \text{Q}$ (Milky cloud)

$\text{P} + \text{Na}_2\text{CO}_3 \longrightarrow \text{Q} + \text{R}$

$\text{Q} + \text{CO}_2$ (excess) $\longrightarrow \text{S}$ (clear solution)

12. P & R are respectively.

- (a) $\text{Ca}(\text{OH})_2$, NaOH (b) CaCl_2 , NaCl
 (c) NaOH, CaCl_2 (d) $\text{Ca}(\text{OH})_2$, NaCl

13. Q & S are respectively :

- (a) CaCO_3 , $\text{Ca}(\text{HCO}_3)_2$
 (b) CaCO_3 , $\text{Ca}(\text{OH})_2$
 (c) $\text{Ca}(\text{HCO}_3)_2$, CaCl_2
 (d) $\text{Ca}(\text{OH})_2$, $\text{Ca}(\text{HCO}_3)_2$

Paragraph for Q.No. 14 to 15

A colourless crystalline substance (A) gives crimson colour to the flame. On heating (A) gives a brown gas as leaving a white residue (C). C is dissolved in HCl heated with $(\text{NH}_4)_2\text{SO}_4$ gives white precipitate (D).

14. A is

- (a) $\text{Ba}(\text{NO}_3)_2$ (b) $\text{Ca}(\text{NO}_3)_2$
 (c) $\text{Mg}(\text{NO}_3)_2$ (d) $\text{Sr}(\text{NO}_3)_2$

15. D is

- (a) BaSO_4 (b) CaSO_4
 (c) MgSO_4 (d) SrSO_4

Paragraph for Q.No. 16 to 17

A white insoluble solid (A) is attacked by acid liberating a colourless, odourless gas (B) leaving a clear solution C. The solution C gives a white precipitate (D) on addition of dilute H_2SO_4 . C gives yellow precipitate E on addition of K_2CrO_4 . Gas B is bubbled through a suspension of CaCO_3 causes solution of that compound.

16. A and B is

- (a) BaCO_3 , CO_2 (b) SrCO_3 , CO_2
 (c) $\text{Ba}(\text{NO}_3)_2$, NO_2 (d) MgCO_3 , CO_2

17. C and D are respectively

- (a) SrSO_4 , SrCrO_4 (b) MgSO_4 , MgCrO_4
 (c) BaSO_4 , BaCrO_4 (d) BeSO_4 , BeCrO_4

Paragraph for Q.No. 18 to 20

A metal (A) burns in O_2 to give an orange powder B which reacts with water liberates O_2 and gives a solution C. B is a strong oxidising agent & is used in respirators of the rebreather type. When 1.42 gm B is heated with 0.32 gm of sulphur in an evacuated sealed tube, 1.74 gm of a white crystalline solid (D) is formed. D forms a hydrated double salt E with $\text{Al}_2(\text{SO}_4)_3$.

18. A & B can be

- (a) Na, Na_2O (b) K, K_2O
 (c) Na, Na_2O_2 (d) K, KO_2

19. Solution C is

- (a) Acidic (b) Basic
(c) Neutral (d) Amphoteric

20. E is

- (a) $K_2SO_4 \cdot Al_2(SO_4)_3 \cdot 24 H_2O$
(b) $Na_2SO_4 \cdot Al_2(SO_4)_3 \cdot 24 H_2O$
(c) $K_2SO_4 \cdot Al_2(SO_4)_3 \cdot 6 H_2O$
(d) $Na_2SO_4 \cdot Al_2(SO_4)_3 \cdot 6 H_2O$

Paragraph for Q.No. 21 to 23

0.347 gm of metal (A) was dissolved in HNO_3 . The solution give a red colouration to a nonluminous flame and on evaporation gave 0.747 gm metal oxide B. A also reacts with N_2 forming a compound C and with hydrogen forming D. On reacting 0.159 gm D with water, a gas E was evolved & a sparingly soluble compound F was formed which gave a strongly basic solution and requires 200 ml of 0.1(M) HCl to neutralise it.

21. A is

- (a) Li (b) Be
(c) Na (d) K

22. Sparingly soluble compound F is

- (a) $Be(OH)_2$ (b) LiOH
(c) NaOH (d) KOH

23. B, C and D are respectively

- (a) BeO ; Be_3N_2 ; BeH_2
(b) K_2O ; KN_3 ; KH
(c) Na_2O ; NaN_3 ; NaH
(d) Li_2O ; LiN_3 ; LiH

Paragraph for Q.No. 24 to 25

The elements of group 1 describe, more clearly than any other group of elements. These metals are highly electropositive, strong reducing agent, have stable oxo salts.

24. Sodium hydroxide is manufactured by the electrolysis of brine solution. The reaction by-products are :

- (a) Cl_2 and H_2
(b) Cl_2 and Na-Hg
(c) Cl_2 and NaCl
(d) Cl_2 and O_2

25. Which of the following has the highest solubility in water?

- (a) LiOH (b) KOH
(c) CsOH (d) RbOH

Paragraph for Q.No. 26 to 28

On reaction with cold water, an element (A) reacts quietly to give colourless, odourless gas B and a solution C. Lithium reacted with B yielding a solid product D which reacts with water to give a strong basic solution E. When CO_2 is bubbled through solution C, an initial precipitate F was formed. Precipitate F effervesced with concentrated HCl and gave red colour to Bunsen flame. When F was heated with carbon at $1000^\circ C$, a caustic white solid G was formed which when heated with carbon at $1000^\circ C$ gave a solid J of some commercial importance.

26. G and J are

- (a) $CaCO_3$, CaO (b) CaO, CaC_2
(c) CaO, $CaCO_3$ (d) CaO, Ca_2C_3

27. Element A and colourless gas B are respectively.

- (a) Mg, N_2 (b) Ca, CO_2
(c) Ca, H_2 (d) Mg, O_2

28. Precipitate G

- (a) will be dissolved on passing through more CO_2 gas
(b) will be ignited on passing through more CO_2 gas
(c) will remain precipitate on passing through more CO_2 gas
(d) No reaction

Matrix Match Type :

29. Column - I
(Compounds)

Column - II
(Use)

- | | |
|---|---|
| (A) $Na_2CO_3 \cdot 10H_2O$
(Washing soda) | (P) Laundering ; Cleaning ; Important laboratory reagent for qualitative analysis |
| (B) NaCl | (Q) It is used in preparation of mortar, a building material |
| (c) $Ca(OH)_2$ | (R) It is used as an antacid, mild abrasive in tooth paste |
| (D) $CaCO_3$ | (S) It is used as table salt |

- 30. Column - I**
(Compounds)
 (A) NaHCO_3
 (B) NaOH
 (c) CaO
 (D) $\text{CaSO}_4 \cdot 1/2\text{H}_2\text{O}$
- Column - II**
(Use)
 (P) Used as in fire extinguishers
 (Q) Used as laboratory reagent
 (R) Used for manufacturing cement
 (S) Used for immobilising the affected part of organ where there is a bone fracture
- 31. Column - I**
(Compounds/Aqueous solution)
 (A) K_2O
 (B) K_2O_2
 (C) N_2O_5
 (D) NH_3
- Column - II**
(Colour change on addition of litmus paper)
 (P) Red litmus paper turns blue
 (Q) Blue litmus paper turns red
 (R) Red litmus paper turns blue at first and then colourless
 (S) Blue litmus paper remains blue
- 32. Column-I**
(Heating effect)
 (A) $\text{CaSO}_4 \cdot 2\text{H}_2\text{O} \xrightarrow{120^\circ\text{C}}$
 (B) $\text{CaSO}_4 \cdot 2\text{H}_2\text{O} \xrightarrow{200^\circ\text{C}}$
 (c) $\text{CaSO}_4 \cdot 2\text{H}_2\text{O} \xrightarrow{1100^\circ\text{C}}$
 (D) $\text{CaCO}_3 \xrightarrow{\Delta}$
- Column-II**
(Products obtained in reaction)
 (P) Plaster of paris
 (Q) Dead burnt plaster
 (R) Quick lime
 (S) Gas evolved acidic in nature
- 33. Column-I**
(Compound)
 (A) Chile saltpetre
 (B) Lithophone
 (BaSO₄ + ZnS)
 (C) Brine solution
 (D) Oxone
- Column-II**
(Charasteristic Feature)
 (P) Ba^{2+} is present
 (Q) Used as pigment
 (R) Na^+ is present
 (S) $-\text{O}-\text{O}-$ linkage is present
 (T) $\text{N}-\text{O}$ linkage is present
- 34. Column-I**
(Compound)
 (A) $\text{Ca}(\text{HCO}_3)_2$
 (B) CaCO_3
 (C) NaHCO_3
 (D) Na_2CO_3
- Column-II**
(Charasteristic Feature)
 (P) Exists in solid form
 (Q) With dilute acid it gives CO_2 gas
 (R) Does not react with $\text{K}_2\text{Cr}_2\text{O}_7$ solution
 (S) Water insoluble
- 35. Column-I**
(Compounds)
 (A) KO_3
 (B) K_2O_2
 (C) SO_3
 (D) $\text{O}_2[\text{PtF}_6]$
- Column-II**
(Average oxidation state of oxygen)
 (P) -2
 (Q) $-\frac{1}{3}$
 (R) -1
 (S) $+\frac{1}{2}$
- 36. Column - I**
(Compounds)
 (A) Mg_3N_2
 (B) CaC_2
 (C) BeH_2
 (D) Mg_2C_3
- Column - II**
(Gas on hydrolysis)
 (P) NH_3
 (Q) C_2H_2
 (R) H_2
 (S) $\text{CH}_3-\text{C}=\text{CH}$
- 37. Column - I**
 (A) Hydrolith
 (B) Gypsum
 (c) Nitrolim
 (D) Dolomite
- Column - II**
 (P) Contains Ca^{2+}
 (Q) Used as fertiliser
 (R) Reacts with water to give H_2
 (S) On heating it is converted into plaster of paris

- 38. Column - I**
(Process/Experiment)
- (A) Solvay process
(B) Castner process
(C) Aq. solution is basic towards litmus
(D) On heating it gives CO_2 gas
- Column - II**
(Compound)
- (P) Na_2O
(Q) Na_2CO_3
(R) NaOH
(S) NaHCO_3
- 39. Column - I**
(Element)
- (A) Cs
(B) Li
(C) Ba
(D) Be
- Column - II**
(Characteristic feature)
- (P) Electronegativity is same as that of Al
(Q) Strongest reducing alkali metal
(R) $6s^2$ configuration
(S) Blue colour towards oxidising flame
- 40. Column - I**
- (A) $\text{Rb}_2\text{CO}_3 > \text{K}_2\text{CO}_3 > \text{Na}_2\text{CO}_3$
(B) $\text{SrSO}_4 > \text{CaSO}_4 > \text{MgSO}_4$
(C) $\text{Rb} > \text{K} > \text{Na}$
(D) $\text{Be} > \text{Mg} > \text{Ca}$
- Column - II**
- (P) solubility of salts in water
(Q) Thermal stability of salts
(R) Softness of metals
(S) Hydration energy of metals
(T) Ionisation energy of metals
- 41. Column - I**
- (A) Calcium hypochlorite
(B) Calcium sulphate hemihydrate
(C) Potassium superoxide
(D) Calcium oxide
- Column - II**
- (P) Used as air purifier in submarine
(Q) Used for 'softening' water
(R) A constituent of bleaching powder
(S) Plaster of Paris
(T) Used in preparation of sodium carbonate from caustic soda
- 42. Column - I**
(Hydrolysis)
- (A) $\text{Mg}_2\text{C}_3 + n\text{H}_2\text{O} \rightarrow \text{Mg}(\text{OH})_2 + \text{Hydrocarbon}$
(B) $\text{CaC}_2 + n\text{H}_2\text{O} \rightarrow \text{Ca}(\text{OH})_2 + \text{Hydrocarbon}$
(C) $\text{Al}_4\text{C}_3 + n\text{H}_2\text{O} \rightarrow \text{Al}(\text{OH})_3 + \text{Hydrocarbon}$
(D) $\text{Be}_2\text{C} + n\text{H}_2\text{O} \rightarrow \text{Be}(\text{OH})_2 + \text{Hydrocarbon}$
- Column - II**
(n/Hydrocarbon)
- (P) 4 ; $\text{CH}_3\text{C} \equiv \text{CH}$
(Q) 2 ; $\text{HC} \equiv \text{CH}$
(R) 12 ; CH_4
(S) 4 ; CH_4
- 43. Column - I**
- (A) BeCl_2
(B) AlCl_3
(c) MgCl_2
(D) BaCl_2
- Column - II**
- (P) Hydroxo complex in aqueous solution
(Q) Fumes in moist air
(R) Lewis Acid
(S) Soluble in water
(T) Hydrated salt formation
- 44. Column - I**
- (A) Be
(B) Al
(c) B
(D) Ca
- Column - II**
- (P) Reacts with NaOH to give H_2 gas
(Q) Forms polymeric halide
(R) Forms ionic hydride
(S) Present in colemanite ore
(T) Forms amphoteric oxide.
- 45. Column - I**
- (A) Pearl Ash
(B) MgCl_2
(c) CaSO_4
(D) CaCO_3
- Column - II**
- (P) Insoluble in water
(Q) Can exist only in aqueous solution
(R) Cause permanent hardness
(S) Contains K^+
(T) Its hydrated salts can not be made anhydrous by heating.

EXERCISE # IV

□ Integer Type :

- Find the number of s-block elements which can produce ammoniated cation and ammoniated electron in liquid ammonia :
 Li, Na, K, Rb, Cs, Ca, Sr,
- How many of the following metal chloride impart characteristic colour to the oxidising flame?
 BaCl_2 , LiCl , NaCl , KCl , BeCl_2 , MgCl_2 , CaCl_2 , SrCl_2
- Find out number of bicarbonate(s) which exist in solid form.
 (a) $\text{Ca}(\text{HCO}_3)_2$ (b) $\text{Be}(\text{HCO}_3)_2$
 (c) NaHCO_3 (d) KHCO_3
 (e) NH_4HCO_3
- What is the atomic number difference between alkali metal of 2nd period & alkaline earth metal of 4th period.
- Find out number of Cl-atoms which are directly bonded to each beryllium in solid state of BeCl_2 .
- Decomposition temperature of carbonate are given. How many of them are correct ?
 $\text{BeCO}_3 \rightarrow 25^\circ\text{C}$; $\text{MgCO}_3 \rightarrow 540^\circ\text{C}$; $\text{CaCO}_3 \rightarrow 900^\circ\text{C}$; $\text{SrCO}_3 \rightarrow 1290^\circ\text{C}$; $\text{BaCO}_3 \rightarrow 1360^\circ\text{C}$.
- $2\text{NaOH} + \text{Cl}_2 \xrightarrow{\text{Cold}} \text{X}_1 + \text{Y} + \text{Z}$
 $6\text{NaOH} + 3\text{Cl}_2 \xrightarrow{80^\circ\text{C}} \text{X}_2 + \text{Y} + \text{Z}$
 What is the difference of oxidation state of chlorine atom of anionic part of X_1 & X_2 :
- NaF is predominantly ionic and if the cationic part of the compound is replaced by potassium ion, then which of the properties decrease in the formed compound.
 (i) Melting point (ii) Lattice energy
 (iii) Thermal stability (iv) Solubility in water
 (v) Covalent character
- How many of the statements are correct ?
 (a) Fr is radioactive
 (b) Superoxide ion is stable in presence of large cations like Cs^+ , K^+
 (c) Alkali metals & alkaline earth metal are example of strong reducing agents
 (d) Photoelectric effect is maximum for caesium
 (e) LiF is less soluble because of high lattice energy
 (f) CsI is less soluble due to smaller hydration enthalpy of Cs^+ & I^-
- Epsom salt : $\text{MgSO}_4 \cdot n_1\text{H}_2\text{O}$
 Gypsum : $\text{CaSO}_4 \cdot n_2\text{H}_2\text{O}$
 Dead burnt plaster : $\text{CaSO}_4 \cdot n_3\text{H}_2\text{O}$
 Find out $n_1 + n_2 + n_3$
- How many of the following contain hydride ion ?
 (a) Hydrolith (CaH_2) (b) NaH
 (c) LiAlH_4 (d) NaBH_4
 (e) HCl (f) H_2 gas
- Find the total number of species which would undergo disproportionation in alkaline medium (NaOH).
 NO_2 ; $\text{Cr}_2\text{O}_7^{7-}$; Cl_2O_6 ; Br_2 ; P_4 ; S_8 ; Al_2O_3 ;
- Find number of substance which produce H_2 gas when react with cold water.
 Ca ; CaH_2 ; Mg ; MgCl_2 ; Ba ; LiAlH_4
- How many alkali metal on reaction with excess O_2 give superoxide as product ?
 (a) Li (b) Na
 (c) K (d) Rb
 (e) Cs
- How many compounds do not react with sodium metal ?
 (a) Ethanol ($\text{C}_2\text{H}_5\text{OH}$)
 (b) Ethyne ($\text{HC}\equiv\text{CH}$)
 (c) Dimethyl ether (MeOMe)
 (d) Phenol (PhOH)
 (e) 2-butyne ($\text{CH}_3-\text{C}\equiv\text{C}-\text{CH}_3$)
- | (Compounds) | (Oxidation state of central atom) |
|--------------------|-----------------------------------|
| (A) BaO_2 | (P) X |
| (B) KO_2 | (Q) Y |
| (c) CO_2 | (R) Z |
| (D) CrO_5 | (S) W |

 Find the value of $\text{X} + \text{Y} + \text{Z} - \text{W}$:

17. What is the maximum coordination number of beryllium?
18. Find out the number of Be–Cl–Be bonds present in product formed when BeCl_2 reacts with concentrated diethyl ether solution.
19. Beryllium gives a compound with the following percentage composition:
Be, 6.1%; N, 37.8%; Cl, 48%; H, 8.1%. One mole of the compound had a mass of 148 g. Mol. wt. (Be) = 9 gmol^{-1} . If the empirical formula = $\text{BeN}_x\text{Cl}_y\text{H}_{12}$ then find out $x + y$.
20. Find out the number of Be–O bonds present in product formed when BeCl_2 reacts with concentrated diethyl ether solution.
21. Number of alkali metal which impart colour to the flame.

Li, Na, K, Rb, Cs, Be, Mg

EXERCISE-V(A) (JEE-MAIN)

1. A metal M readily forms its sulphate MSO_4 which is water soluble. It forms oxide MO which becomes inert on heating. It forms insoluble hydroxide which is soluble in NaOH. The metal M is :- [AIEEE - 2002]
(a) Mg (b) Ba
(c) Ca (d) Be
2. KO_2 is used space and submarines because it :- [AIEEE - 2002]
(a) Absorbs CO_2 and increase O_2 concentration
(b) Absorbs moisture
(c) Absorbs CO_2
(d) Produce ozone
3. In curing cement plasters, water is sprinkled from time to time. This helps in :- [AIEEE - 2003]
(a) Hydrating sand and gravel mixed with cement
(b) Converting sand into silicate
(c) Developing interlocking needle like crystals of hydrated silicates
(d) Keeping it cool
4. The solubilities of carbonates decreases down the magnesium group due to decrease in :- [AIEEE - 2003]
(a) Inter-ionic attraction
(b) Entropy of solution formation
(c) Lattice energy of solids
(d) Hydration energy of cations
5. The substance not likely to contain CaCO_3 is :- [AIEEE - 2003]
(a) Sea shells (b) Dolomite
(c) A marble same (d) Calcined gypsum
6. One mole of magnesium nitride on reaction with excess of water gives :- [AIEEE - 2004]
(a) Two mole of HNO_3
(b) Two mole of NH_3
(c) 1 mole of NH_3
(d) 1 mole of HNO_3
7. Beryllium and aluminium exhibit many properties which are similar. But the two elements differ in :- [AIEEE - 2004]
(a) Exhibiting maximum covalency in compounds
(b) Forming polymeric hydrides
(c) Forming covalent halides
(d) Exhibiting amphoteric nature in their oxides
8. The ionic mobility of alkali metal ions in aqueous solution is maximum for :- [AIEEE - 2006]
(a) Rb^+ (b) Li^+
(c) Na^+ (d) K^+
9. The products obtained on heating LiNO_3 will be :- [AIEEE - 2011]
(a) $\text{LiNO}_2 + \text{O}_2$ (b) $\text{Li}_2\text{O} + \text{NO}_2 + \text{O}_2$
(c) $\text{Li}_3\text{N} + \text{O}_2$ (d) $\text{Li}_2\text{O} + \text{NO} + \text{O}_2$
10. What is the best description of the change that occurs when $\text{Na}_2\text{O}(\text{s})$ is dissolved in water? [AIEEE - 2011]
(a) Oxidation number of sodium decreases
(b) Oxide ion accepts sharing in a pair of electrons
(c) Oxide ion donates a pair of electrons
(d) Oxidation number of oxygen increases
11. Which of the following on thermal - decomposition yields a basic as well as an acidic oxide? [AIEEE - 2012]
(a) NH_4NO_3 (b) NaNO_3
(c) KClO_3 (d) CaCO_3

12. Fire extinguishers contain H_2SO_4 and which one of the following :- [Jee-Main - 2012 (Online)]
 (a) CaCO_3 (b) NaHCO_3 and Na_2CO_3
 (c) Na_2CO_3 (d) NaHCO_3
13. Based on lattice energy and other considerations, which one of the following alkali metal chloride is expected to have the highest melting point ? [Jee-Main - 2012 (Online)]
 (a) RbCl (b) LiCl
 (c) KCl (d) NaCl
14. Which one of the following will react most vigorously with water ? [Jee-Main - 2012 (Online)]
 (a) Li (b) K
 (c) Rb (d) Na
15. A metal M on heating in nitrogen gas gives Y. Y on treatment with H_2O gives a colourless gas which when passed through CuSO_4 solution gives a blue colour, Y is :- [Jee-Main - 2012 (Online)]
 (a) NH_3 (b) MgO
 (c) Mg_3N_2 (d) $\text{Mg}(\text{NO}_3)_2$
16. The correct statement for the molecule, CsI_3 is :- [Jee-Main - 2014]
 (a) It contains Cs^{3+} and I^- ions
 (b) It contains Cs^+ , I^- and lattice I_2 molecule
 (c) It is a covalent molecule
 (d) It contains Cs^{3+} and I_3^- ions
17. Which of the following statements about Na_2O_2 is not correct? [Jee-Main - 2014 (Online)]
 (a) Na_2O_2 oxidises Cr^{3+} to CrO_4^{2-} in acid medium
 (b) It is diamagnetic in nature
 (c) It is the super oxide of sodium
 (d) It is a derivative of H_2O_2
18. Amongst LiCl , RbCl , BeCl_2 and MgCl_2 the compounds with the greatest and the least ionic character, respectively are :- [Jee-Main - 2014 (Online)]
 (a) RbCl and MgCl_2 (b) LiCl and RbCl
 (c) MgCl_2 and BeCl_2 (d) RbCl and BeCl_2
19. The correct order of thermal stability of hydroxides is :- [Jee-Main - 2015 (Online)]
 (a) $\text{Ba}(\text{OH})_2 < \text{Sr}(\text{OH})_2 < \text{Ca}(\text{OH})_2 < \text{Mg}(\text{OH})_2$
 (b) $\text{Mg}(\text{OH})_2 < \text{Sr}(\text{OH})_2 < \text{Ca}(\text{OH})_2 < \text{Ba}(\text{OH})_2$
 (c) $\text{Mg}(\text{OH})_2 < \text{Ca}(\text{OH})_2 < \text{Sr}(\text{OH})_2 < \text{Ba}(\text{OH})_2$
 (d) $\text{Ba}(\text{OH})_2 < \text{Ca}(\text{OH})_2 < \text{Sr}(\text{OH})_2 < \text{Mg}(\text{OH})_2$
20. Which of the alkaline earth metal halides gives below is essentially covalent in nature :- [Jee-Mains - 2015 (Online)]
 (a) SrCl_2 (b) CaCl_2
 (c) BeCl_2 (d) MgCl_2
21. Which one of the following alkaline earth metal sulphates has its hydration enthalpy greater than its lattice enthalpy :- [Jee-Mains - 2015]
 (a) BaSO_4 (b) SrSO_4
 (c) CaSO_4 (d) BeSO_4

EXERCISE-V(B) (JEE-ADVANCED)

1. The species that do not contain peroxide linkage are :- [JEE-1992]
 (a) PbO_2 (b) H_2O_2
 (c) SrO_2 (d) BaO_2
2. Read the following statement and explanation and answer as per the options given below :-
Statement-1 : The alkali metals can form ionic hydrides which contain the hydride ion H^- .
Statement-2 : The alkali metals have low electronegativity ; their hydrides conduct electricity when fused and liberate hydrogen at the anode. [JEE-1994]
 (a) Both 1 and 2 are true and 2 is the correct explanation of 1.
 (b) Both 1 and 2 are true but 2 is not the correct explanation of 1.
 (c) 1 is true but 2 is true.
 (d) 1 is false but 2 is true.
3. The following compounds have been arranged in order of their increasing thermal stabilities. Identify the correct order. [JEE-1996]
 K_2CO_3 (I) MgCO_3 (II) CaCO_3 (III) BeCO_3 (IV)
 (a) $\text{I} < \text{II} < \text{III} < \text{IV}$ (b) $\text{IV} < \text{II} < \text{III} < \text{I}$
 (c) $\text{IV} < \text{II} < \text{I} < \text{III}$ (d) $\text{II} < \text{IV} < \text{III} < \text{I}$

4. Property of all the alkaline earth metals that increase with their atomic number? [JEE-1997]
(a) ionisation energy
(b) solubility of their hydroxides
(c) solubility of their sulphate
(d) electronegativity
5. Highly pure dilute solution of sodium in liquid ammonia - [JEE-1998]
(a) shows blue colour
(b) exhibits electrical conductivity
(c) produces sodium amide
(d) produces hydrogen gas
6. The set representing the correct order of first ionization potential is :- [JEE-2001S]
(a) $K > Na > Li$
(b) $Be > Mg > Ca$
(c) $B > C > N$
(d) $Ge > Si > C$

ASSERTION AND REASON

7. The questions contains statement-1 (assertion) and statement-2 (reason) and has 4 choices (a), (b), (c) and (d) out of which only one is correct.
Statement-1 : Alkali metals dissolves in liquid ammonia to give blue solutions. because.
Statement-2 : Alkali metals is liquid ammonia give solvated species of the type $[M(NH_3)_n]^+$ (M = alkali metals) [JEE-2007]
(a) Both 1 and 2 are true and 2 is the correct explanation of 1.
(b) Both 1 and 2 are true but 2 is not the correct explanation of 1.
(c) 1 is true but 2 is true.
(d) 1 is false but 2 is true.
8. The compound(s) formed upon combustion of sodium metal in excess air is (are) [JEE-2009]
(a) Na_2O_2 (b) Na_2O
(c) NaO_2 (d) $NaOH$

■ ■

THE p-BLOCK ELEMENTS AND THEIR COMPOUNDS

PROBLEMS BASED ON GIVEN TOPICS

The group 13 elements

- Oxidation state of types of bonds
 - The (+III) oxidation state
 - The (+I) oxidation state - the 'inert pair effect'
- General properties
 - Melting points, boiling points and structures
 - Size of atoms and ions
 - Electropositive characters
 - Ionization energy
- Reaction of boron
- Reaction of the other elements
 - Reaction with water and air
 - Reaction with acids and alkalis
 - Reaction with dioxygen
 - Reaction with the halogens and sulphate
 - Alums
 - Cement
- Compounds of boron and oxygen
 - Boron sesquioxide and the borates
 - Acidic properties of H_3BO_3 or $B(OH)_3$
 - Structures of borates
 - Borax
 - Sodium peroxoborate
 - Isopolyacids of B, Si and P
 - Qualitative analysis of boron compounds
 - Fluoboric acid
 - Borides
- The other group 13 oxides
 - Amphoteric behaviour - aluminates

- Tetrahydridoborates (Borohydrides)
- Halides
 - Trihalides
 - Dihalides
 - Monohalides
- Complexes
- Differences between boron and the other elements
- Boron hydrides
 - Compounds known
 - Preparation
- Reaction of the boranes
 - Hydroboration
 - Reaction with ammonia
 - Some other reactions of boranes
- Structures of the boranes
- Organometallic compounds
- The group 14 elements**
- Structure and allotropy of the elements
- Differences between carbon, Silicon and the remaining elements
 - Carbon dating
- Physical properties
 - Covalent radii
 - Ionization energy
 - Melting points
 - Metallic and non metallic character
 - Four - covalent compounds
- Chemical reactivity
 - Inert pair effect
 - Standard reduction potentials (volts)
- Graphite compounds

6.2 THE p-BLOCK ELEMENTS AND THEIR COMPOUNDS

- Carbides
 - Salt - like carbides
 - Interstitial carbides
 - Covalent carbides
 - Oxygen compounds of carbon
 - Carbon monoxide CO
 - Carbon dioxide CO₂
 - Carbon suboxides
 - Carbonates
 - The carbon cycle
 - Sulphides of carbon
 - Oxides of silicon
 - Oxides of germanium, Tin and Lead
 - Silicates
 - Occurrence in the earth crust
 - Soluble silicates
 - Principles of silicate structures
 - Classification of silicates
 - Orthosilicates (neso - silicates)
 - Pyrosilicates (soro-silicates , disilicates)
 - Cyclic silicates
 - Chain silicates
 - Sheet silicates (Phyllo - silicates)
 - Three - dimensional silicates
 - Glass
 - Organosilicon compounds and the silicones
 - Organosilicon compounds
 - Preparation of organosilicon compounds
 - Silicones
 - Hydrides of silicon
 - Complexes
 - Internal π bonding using d-orbitals
 - Halides
 - Tetrahalides
 - Catenated halides
 - Dihalides
 - Cluster compounds
 - Reaction mechanisms
 - Organic derivatives
- The group 15 elements**
- General properties and structures of the elements
 - Nitrogen
 - Phosphorus
 - Bond type
 - Metallic and non - metallic character
 - Reactivity
 - Hydrides
 - Ammonia NH₃
 - Phosphine PH₃
 - Arsine AsH₃, stibine SbH₃ and bismuthine BiH₃
 - Structure of the hydrides
 - Donor properties
 - Hydrazine N₂H₄
 - Hydroxylamine NH₂OH
 - Liquid ammonia as a solvent
 - Hydrogen azide and the azides
 - Nitrogen fixation
 - Cyanamide process
 - Haber-Bosch process
 - Fertilizers
 - Urea
 - Phosphate fertilizers
 - Halides
 - Trihalides
 - Pentahalides

- Oxides of nitrogen
 - Nitrous oxide N_2O
 - Nitric oxide NO
 - Nitrogen sesquioxide N_2O_3
 - Nitrogen dioxide NO_2 and dinitrogen tetroxide N_2O_4
 - Dinitrogen pentoxide N_2O_5
- Oxoacids of nitrogen
 - Nitrous acid HNO_2
 - Nitric acid HNO_3
- Oxides of phosphorus, Arsenic and Bismuth
 - Trioxides
 - Pentoxides
 - Other oxides
- Oxoacid of phosphorus
 - The phosphoric acid series
 - The phosphorous acid series
- Major uses of phosphate
- The group 16 elements - Chalcogens**
- General properties
 - Electronic structure and oxidation state
 - Acid rain and SO_2
 - Uses of sulphur
- Structures and allotropy of elements
 - Oxygen
 - Ozone
 - Sulphur
- Physical properties
- Chemical reactivity
 - Standard reduction potentials (volts)
 - Oxidation state (+II), (+IV), (+VI)
- Bond lengths and $p\pi-d\pi$ bonding
- Differences between oxygen and the other elements
- General properties of oxides
 - Normal oxides
 - Peroxides
 - suboxides
 - Basic oxides
 - Amphoteric oxides
 - Acid oxides
 - Neutral oxides
- Oxides of sulphur
 - Dioxides SO_2
 - Trioxides SO_3
 - Other oxides
 - Detergents
- Oxoacid of sulphur
 - Sulphurous acid series
 - Sulphuric acid series
 - Thionic acid series
 - Peroxoacid series
- Oxohalides
 - Thionyl compounds
 - Sulphuryl compounds
- Hydrides
 - Water
 - Other hydrides
 - Peroxides and polysulphides
 - Hydrogen peroxides
- Halides
- Organo derivatives

6.4 THE p-BLOCK ELEMENTS AND THEIR COMPOUNDS

The group 17 elements - The halogenes

- Extraction and uses of the elements
 - Fluorine
 - Chlorine
 - Bromine
 - Iodine
- General properties
 - Size of atoms and ions
 - Ionization energy
 - Types of bonds formed and oxidation states
 - Melting and boiling point
 - Bond energy in X_2 molecules
 - Oxidizing power
- Reaction with water
- Reactivity of the element
- Hydrogen halides HX
 - HF
 - HCl
 - HBr and HI
- Halides
 - Ionic halides
 - Molecular covalent halides
 - Bridging halides
 - Preparation of anhydrous halides
- Halogen oxides
 - Oxygen difluoride OF_2
 - Dioxygen difluoride O_2F_2
 - Dichlorine monoxide Cl_2O
 - Chlorine dioxide ClO_2
 - Chlorine perchlorate $Cl.ClO_4$
 - Dichlorine hexoxide Cl_2O_6
 - Dichlorine heptoxide Cl_2O_7

- Oxoacids
 - Hypohalous acids HOX
 - Halous acids HXO_2
 - Halic acid HXO_3
 - Perhalic acid HXO_4
 - Strength of the oxoacids

- Interhalogen compounds
 - AX compounds
 - AX_3 compounds
 - AX_5 compounds
 - AX_7 compounds

- Polyhalides
- Basic properties of the halogens
- Pseudohalogens and Pseudohalides

The group 18 elements - Noble gases

- Occurrence and recovery of the elements
- Use of the elements
- Physical properties
- Chemical properties of Helium
- Chemical properties of the noble gases
 - Molecular ions formed under excited conditions
 - Clathrate compounds
- Chemistry of Xenon
 - Xenon fluoride complexes
- Structures and bonding in xenon compound
 - XeF_2
 - XeF_4
 - XeF_6
- Valediction

The p-Block Elements and Their Compounds

EXERCISE # I

❑ Only one correct answer :

- The two main allotropes of tin are :-
 (a) grey tin, black tin (b) white tin, red tin
 (c) white tin, black tin (d) white tin, grey tin
- The colour of SnS and SnS_2 are respectively :-
 (a) Black and White
 (b) Brown and Yellow
 (c) Black and Colourless
 (d) Brown and Red
- Tin particles crumble to powder during severe winters which is said "Tin plague". It is due to :-
 (a) at low temperature, transformation of white tin into grey tin which is accompanied by volume expansion
 (b) at low temperature, transformation of grey tin into black tin which is accompanied by volume expansion
 (c) Consuming grey tin causes disease
 (d) None
- SnO is example of amphoteric oxide, supported by which of the following reaction :-
 (a) $\text{SnO} + 2\text{HCl} \rightarrow \text{SnCl}_2 + \text{H}_2\text{O}$
 (b) $\text{SnO} + 2\text{NaOH} \rightarrow \text{Na}_2\text{SnO}_2 + \text{H}_2\text{O}$
 (c) Both (a) & (b)
 (d) $\text{SnO} + \text{H}_2 \rightarrow \text{Sn} + \text{H}_2\text{O}$
- Concentrated nitric acid when reacts with tin, a white precipitate is formed. The precipitate is :-
 (a) H_2SnO_3
 (b) H_2SnO_2
 (c) $\text{H}_2\text{Sn}_5\text{O}_{11} \cdot 4\text{H}_2\text{O}$
 (d) $\text{SnO}_2 \cdot 6\text{H}_2\text{O}$
- The correct boiling point order of stannic halides is :-
 (a) $\text{SnF}_4 > \text{SnCl}_4 > \text{SnBr}_4 > \text{SnI}_4$
 (b) $\text{SnI}_4 > \text{SnBr}_4 > \text{SnCl}_4 > \text{SnF}_4$
 (c) $\text{SnCl}_4 > \text{SnI}_4 > \text{SnF}_4 > \text{SnBr}_4$
 (d) $\text{SnF}_4 > \text{SnI}_4 > \text{SnBr}_4 > \text{SnCl}_4$
- Solution of SnCl_2 and SnCl_4 can be differentiated by different reagents, because they behave in different way with different reagents. With HgCl_2 which of the following option is correct :-
 (a) SnCl_2 react with HgCl_2 to give white ppt. of Hg_2Cl_2 which is further reduced by SnCl_2 to give black precipitate
 (b) SnCl_4 react with HgCl_2 to give white ppt. of Hg_2Cl_2
 (c) SnCl_4 does not react with HgCl_2
 (d) both (a) and (c) are correct
- Which of the following statement is correct ?
 (a) SnCl_2 does not react with FeCl_3
 (b) SnCl_4 does react with FeCl_3 to give FeCl_6^{2-}
 (c) SnCl_2 reduces FeCl_3 to FeCl_2 & then when $\text{K}_3[\text{Fe}(\text{CN})_6]$ is added, then blue colour is observed.
 (d) None of these
- A mass of arborescent crystals (lead tree) slowly deposits when :-
 (a) Zinc rod is hung in a solution of lead carbonate
 (b) Iron rod is hung in a solution of lead nitrate
 (c) Copper rod is hung in a solution of lead acetate
 (d) Zinc rod is hung in a solution of lead acetate
- Lead is soluble in :-
 (a) dilute HCl
 (b) dilute H_2SO_4
 (c) cold concentrated HCl
 (d) warm dilute HNO_3

6.6 THE p-BLOCK ELEMENTS AND THEIR COMPOUNDS

11. Plumbo solvency - a serious menace where lead piping is used for water supply. It is caused by.
 - (a) Water soluble PbCO_3
 - (b) Water insoluble PbCO_3
 - (c) Slightly water soluble Pb(OH)_2
 - (d) Water soluble $\text{Pb(OH)}_2 \cdot \text{PbCO}_3$
12. Pb_3O_4 when reacts with HCl , the products formed are :-
 - (a) PbCl_2 , H_2O , Cl_2
 - (b) PbCl_2 , H_2O
 - (c) PbCl_2 , H_2O , Cl_2 , O_2
 - (d) PbO_2 , Cl_2 , H_2O
13. Pb_3O_4 when reacts with HNO_3 , the products formed are :-
 - (a) $\text{Pb(NO}_3)_2$, H_2O , NO_2
 - (b) $\text{Pb(NO}_3)_2$, H_2O
 - (c) $\text{Pb(NO}_3)_2$, O_2 , H_2O
 - (d) $\text{Pb(NO}_3)_2$, PbO_2 , H_2O
14. TEL (tetraethyl lead) is a colourless liquid which has boiling point 82°C . It is prepared by action of :-
 - (a) ethyl alcohol with lead chloride
 - (b) ethyl chloride vapour on sodium lead alloy
 - (c) ethyl ether vapour with lead metal
 - (d) ethyl benzene with lead chloride
15. The correct melting point order of plumbous halide:-
 - (a) $\text{PbF}_2 > \text{PbCl}_2 > \text{PbI}_2 > \text{PbBr}_2$
 - (b) $\text{PbI}_2 > \text{PbBr}_2 > \text{PbCl}_2 > \text{PbF}_2$
 - (c) $\text{PbCl}_2 > \text{PbF}_2 > \text{PbBr}_2 > \text{PbI}_2$
 - (d) $\text{PbF}_2 < \text{PbCl}_2 < \text{PbBr}_2 < \text{PbI}_2$
16. $\text{CH}_3\text{CH-OH} \xrightarrow{\text{lead tetraacetate}} \text{H}_2\text{O} + \text{Product(P)}.$

$\begin{array}{c} | \\ \text{CH}_3\text{CH-OH} \end{array}$

P is:-

 - (a) $\text{CH}_3\text{CO}_2\text{H}$
 - (b) $\text{CH}_3-\text{C}-\text{CH}_3$
 $\parallel$
 O
 - (c) $\text{CH}_3-\text{CH}=\text{O}$
 - (d) $\text{CH}_3-\text{C}=\text{O}$
 $|$
 $\text{H}_3\text{C}-\text{C}=\text{O}$
17. H_2O_2 converts black precipitate PbS into :-
 - (a) Elemental Pb
 - (b) Red colour Pb_3O_4
 - (c) Grey colour PbO
 - (d) White coloured precipitate PbSO_4
18. The colour of PbO_2 (lead dioxide) :-
 - (a) Yellow
 - (b) Scarlet red
 - (c) Chocolate brown
 - (d) Black
19. The colour of Pb_3O_4 :-
 - (a) Yellow
 - (b) Scarlet red
 - (c) Chocolate brown
 - (d) Black
20. When lead is heated in air, a yellow powder A is formed. When A is fused, cooled and powdered, a reddish yellow crystalline variety B is formed, A and B are respectively :-
 - (a) Massicot & Litharge
 - (b) Red lead & lead dioxide
 - (c) Litharge & Massicot
 - (d) Lead dioxide & Red lead
21. When litharge is calcinated for about 48 hrs in a current of air at a temperature almost equal to 340°C , then the product formed is :-
 - (a) PbO_2
 - (b) Pb_2O_3
 - (c) Pb_3O_4
 - (d) PbO
22. The correct formula of white lead which is used as pigment is :-
 - (a) $\text{PbCO}_3 \cdot \text{Pb(OH)}_2$
 - (b) $\text{PbCO}_3 \cdot 2\text{Pb(OH)}_2$
 - (c) $2\text{PbCO}_3 \cdot \text{Pb(OH)}_2$
 - (d) $3\text{PbCO}_3 \cdot \text{Pb(OH)}_2$
23. Lithium aluminium hydride when reacts with stannic chloride in ether medium at very low temperature, the product formed :-
 - (a) Sn_2H_6
 - (b) SnH_2
 - (c) SnH_4
 - (d) Sn_3H_8
24. PbO_2 (Pb in +4 oxidation state) oxidises MnSO_4 solution in acidic medium into :-
 - (a) pink solution of HMnO_4
 - (b) black MnO_2 ppt.
 - (c) reddish brown solution of MnO(OH)_2
 - (d) Mn_3O_8

25. $\text{Na}_2[\text{SnO}_2]$ (Sn in +2 oxidation state) reduces $\text{Bi}(\text{NO}_3)_3$ into :-
 (a) Black colour precipitate of Bi
 (b) Red colour precipitate of Bi
 (c) Brown colour precipitate of Bi
 (d) None of these
26. The lewis acidic strength order among bromo halides :-
 (a) $\text{BF}_3 > \text{BCl}_3 > \text{BBr}_3 > \text{BI}_3$
 (b) $\text{BI}_3 > \text{BBr}_3 > \text{BCl}_3 > \text{BF}_3$
 (c) $\text{BBr}_3 > \text{BCl}_3 > \text{BF}_3 > \text{BI}_3$
 (d) $\text{BF}_3 > \text{BI}_3 > \text{BBr}_3 > \text{BCl}_3$
27. Orthoborates are :-
 (a) planar unit (b) pyramidal unit
 (c) tetrahedral unit (d) angular unit
28. Product (P) is
 (a) (b)
 (c) (d)
29. Which of the following reaction, product are correctly predicted :-
 (a) $\text{B}_3\text{N}_3\text{H}_6 + \text{HCl} \longrightarrow (\text{XS})$
 (b) $\text{B}_3\text{N}_3\text{H}_3\text{Cl}_3 + \text{NaBH}_4 \longrightarrow \text{B}_3\text{N}_3\text{H}_6 + \text{NaCl} + \text{B}_2\text{H}_6$
 (c) $\text{B}_3\text{N}_3\text{H}_6 + 9\text{H}_2\text{O} \longrightarrow \text{H}_3\text{BO}_3 + \text{NH}_3 + \text{H}_2$
 (d) All
30. The oxidation state of B, N in $\text{B}_3\text{N}_3\text{H}_6$ are respectively :-
 (a) +3, -3 (b) +1, -1
 (c) -3, +3 (d) -1, +1
31. B_2H_6 can not be methylated beyond :-
 (a) $\text{Me}_2\text{B}_2\text{H}_4$
 (b) $\text{Me}_3\text{B}_2\text{H}_3$
 (c) $\text{Me}_4\text{B}_2\text{H}_2$
 (d) $\text{Me}_5\text{B}_2\text{H}$
32. Anhydrous aluminium chloride is obtained by :-
 (a) strongly heating to 1000°C a mixture of alumina and coke in a current of chlorine
 (b) by passing dry chlorine or hydrogen chloride over heated aluminium turnings
 (c) both (a) & (b)
 (d) None
33. Zinc and copper reacts with concentrated nitric acid to give
 (a) NO_2 (b) NO
 (c) N_2O (d) N_2
34. B_2H_6 is :-
 (a) electron rich molecule and paramagnetic
 (b) electron rich molecule and diamagnetic
 (c) electron deficient molecule and paramagnetic
 (d) electron deficient molecule and diamagnetic
35. Zinc reacts with dilute nitric acid to give
 (a) NO_2 (b) NO
 (c) N_2O (d) N_2
36. Aluminium triethyl in conjunction with titanium chloride TiCl_4 is called:-
 (a) Al-titanium alloy (b) Alum
 (c) Goldsmith mixture (d) Zeigler natta catalyst
37. Copper reacts with dilute nitric acid to give
 (a) NO_2 (b) NO
 (c) N_2O (d) N_2
38. Alums containing thallium one of the types :-
 (a) $\text{K}_2\text{SO}_4 \cdot \text{Tl}(\text{SO}_4)_3 \cdot 24\text{H}_2\text{O}$
 (b) $\text{Ti}_2\text{SO}_4 \cdot \text{Al}_2(\text{SO}_4)_3 \cdot 24\text{H}_2\text{O}$
 (c) $\text{K}_2\text{SO}_4 \cdot \text{Al}_2(\text{SO}_4)_3 \cdot 24\text{H}_2\text{O}$
 (d) $\text{Ti}(\text{SO}_4) \cdot \text{K}_2\text{SO}_4 \cdot 24\text{H}_2\text{O}$
39. A silvery white metal liquefies on a hot summer day but is solid on cold days. It is readily attacked by dilute acid liberating hydrogen. A is :-
 (a) B (b) Ag
 (c) Al (d) Mg
40. Which of the following is orange coloured solid :-
 (a) CCl_4 (b) SiBr_4
 (c) SnI_4 (d) SbCl_3

6.8 THE p-BLOCK ELEMENTS AND THEIR COMPOUNDS

41. A gas at low temperature does not react of the compounds. It is inert in nature. The combustion of this gas is exceptionally an endothermic reaction. The gas is :-
 (a) O_2 (b) N_2
 (c) CO (d) H_2
42. An explosive compound (a) when reacts with water, it give NH_4OH and HCl. A is :-
 (a) TNG (b) TNT
 (c) NCl_3 (d) HNO_3
43. Three allotropes A, B and C of phosphorus are transfered into each other in the following reaction :-

$$A \xrightarrow[1200\text{atm}]{470\text{K}} B \xrightarrow[CO_2\text{atm}]{570\text{K}} C$$

 (a) white, black, red (b) black, white, red
 (c) red, black, white (d) red, white, black
44. Which is incorrectly matched :-
 (a) $CsBr_3 \rightleftharpoons Cs^+ + Br_3^-$
 (b) $AgBrO_3 \rightleftharpoons Ag^+ + BrO_3^-$
 (c) $I_4O_9 \rightleftharpoons I^{3+} + (IO_3^-)_3$
 (d) $I_2O_4 \rightleftharpoons IO_2^- + IO_2^+$
45. Copper reacts with concentrated nitric acid to give
 (a) NO_2 (b) NO
 (c) N_2O (d) N_2
46. $C(OH)_4$ is unstable but $Si(OH)_4$ is a stable compound because :-
 (a) C = O bond energy is very low
 (b) C = O bond energy is very high
 (c) Si = O bond energy is very high
 (d) All of these
47. When a salt X reacts with ozone in aqueous medium, a compound Y is produced ozone also reacts with Y and produces compound Z. Z acts as an oxidising reagent. Then X, Y, Z can be :-
 (a) $X = HI$; $Y = I_2$; $Z = HIO_3$
 (b) $X = KI$; $Y = I_2$; $Z = HIO_3$
 (c) $X = KI$; $Y = I_2$; $Z = HOI$
 (d) $X = KI$; $Y = I_2$; $Z = HIO_4$
48. A gas exists in three allotropic forms α , β , γ is :-
 (a) SO_2 (b) SO_3
 (c) CO_2 (d) NH_3
49. $KOCN + (NH_4)_2SO_4 \xrightarrow{\Delta}$
 In the following reaction one of the product is :-
 (a) NH_4NO_2 (b) NH_4NO_3
 (c) NH_2CONH_2 (d) both N_2 & O_2
50. Which element become passive due to protective layer of oxide formation, when reacts with concentrated nitric acid
 (a) Fe (b) Cr
 (c) Al (d) all of these
51. The ratio of $2C-2e$ and $3C-2e$ bonds in diborane is :-
 (a) 0 (b) 1
 (c) 2 (d) 3
52. Sulphur dioxide reacts with hydrogen sulphide to give sulphur precipitate. Here SO_2 and H_2S acts :-
 (a) oxidising and reducing reagent respectively
 (b) reducing and oxidising reagent respectively
 (c) base and acid respectively
 (d) acid and base respectively
53. SO_2 and Cl_2 bleaches coloured substance :-
 (a) by reduction and oxidation respectively
 (b) by reduction and reduction respectively
 (c) by oxidation and reduction respectively
 (d) by oxidation and oxidation respectively
54. Zeolite which is hydrated sodium aluminium silicate is used as water softner because :-
 (a) the potassium ions of zeolite are exchanged with barium and strontium ions when hard water is passed through it
 (b) the sodium ion of zeolite are exchanged with barium and strontium ions when hard water is passed through it
 (c) the potassium ions of zeolite are exchanged with barium and strontium ions when hard water is passed through it
 (d) the sodium ions of zeolite are exchanged with calcium and magnesium ions when hard water is passed through it

55. Sulphur dioxide acts as more powerful reducing agent in :-
 (a) acidic medium (b) neutral medium
 (c) alkaline medium (d) None
56. The number of moles of H_2O_2 required to convert 1 mole of PbS into PbSO_4 .
 (a) 1 (b) 2
 (c) 3 (d) 4
57. Metal (M) $\xrightarrow{+\text{N}_2 \text{ gas}}$ Solids(S)
 $\text{S} + \text{H}_2\text{O} \longrightarrow \text{G}(\text{gas})$
 $\text{G} + \text{CuSO}_4 \text{ solution} \longrightarrow \text{Blue colour solution(X)}$
 Blue colour solution is :-
 (a) $[\text{Cu}(\text{NH}_3)_6]\text{SO}_4$ (b) $[\text{Cu}(\text{NH}_3)_4]\text{SO}_4$
 (c) $[\text{Cu}(\text{NH}_3)_2]\text{SO}_4$ (d) $[\text{Cu}(\text{NO}_3)_4]\text{SO}_4$
58. $\text{Xe} + \text{F}_2 \longrightarrow \text{P}_1\text{Xe} + \text{F}_2 \longrightarrow \text{P}_2\text{Xe}$
 $+ \text{F}_2 \longrightarrow \text{P}_3$
 (2:1 volume ratio)(1:5 volume ratio)(1:20 volume ratio)
 $\text{P}_1, \text{P}_2, \text{P}_3$ are respectively :-
 (a) $\text{XeF}_2, \text{XeF}_4, \text{XeF}_6$ (b) $\text{XeF}_2, \text{XeF}_6, \text{XeF}_4$
 (c) $\text{XeF}_6, \text{XeF}_4, \text{XeF}_2$ (d) $\text{XeF}_4, \text{XeF}_2, \text{XeF}_6$
59. XeO_3 in strong alkaline medium (above pH 10.5) forms :-
 (a) HXeO_4^- (perxenate ion)
 (b) XeO_6^{4-} (xenate ion)
 (c) HXeO_4^- (xenate ion)
 (d) XeO_6^{4-} (perxenate ion)
60. Among all noble gases, only Xe reacts with F_2 to give many Xenon fluoride because Xenon :-
 (a) has largest size among all noble gases
 (b) has the lowest ionisation enthalpy among all noble gases
 (c) has highest heat of vaporisation among all noble gases
 (d) is the most readily available among all noble gases
61. ClO_3 is the mixed anhydride of :-
 (a) HClO_2 and HClO_3 (b) HClO_2 and HClO_4
 (c) HOCl and HClO_3 (d) HClO_3 and HClO_4
62. Among F^- , Cl^- , Br^- , I^- :-
 (a) I^- is the least stable base
 (b) I^- is the weakest reducing reagent
 (c) I^- is the most basic
 (d) I^- is the strongest reducing reagent
63. Which of the following product is formed when sulphur dioxide gas is passed through sodium chlorate in strongly acidic solution :-
 (a) NaClO_4 (b) ClO_2
 (c) NaSO_3 (d) SO_3
64. The oxide of chlorine which has odd number electrons and paramagnetic in nature is :-
 (a) Cl_2O_7 (b) Cl_2O
 (c) ClO_2 (d) Cl_2O_6
65. The colour of vapour obtained when an iodide salt is heated with concentrated H_2SO_4 is :-
 (a) pale yellow (b) violet
 (c) reddish brown (d) greenish yellow
66. HI can be prepared by which of the following methods :-
 (a) $\text{PI}_3 + \text{H}_2\text{O} \longrightarrow$ (b) $\text{H}_2 + \text{I}_2 \xrightarrow{\text{Pt}}$
 (c) $\text{KI} + \text{H}_2\text{SO}_4 \longrightarrow$ (d) $\text{I}_2 + \text{H}_2\text{S} \longrightarrow$
67. What is the maximum coordination number of iodine in the mixture of IF_7 and CsF :-
 (a) 2 (b) 4
 (c) 6 (d) 8
68. For which halogen, reaction with water will be most exothermic :-
 (a) Fluorine (b) Chlorine
 (c) Bromine (d) Iodine
69. A greenish yellow gas reacts with KOH to give potassium halide which can be used in fire works and safety matches. The gas and half Fe respectively are :-
 (a) Br_2 ; KBrO_3 (b) Cl_2 ; KClO_3
 (c) I_2 ; NaIO_3 (d) None
70. How many statements are correct regarding neon :-
 (a) Neon is diatomic in nature
 (b) Neon does not form clathrate with quinol
 (c) Neon lights are visible even in fog and moist
 (d) All of these

6.10 THE p-BLOCK ELEMENTS AND THEIR COMPOUNDS

71. XeF_6 when reacts with cesium fluoride it acts as:-
 (a) Bronsted acid (b) Bronsted base
 (c) Lewis acid (d) Lewis base
72. A black solid X reacts with NH_3 to form a mild explosive which decomposes to give violet coloured gas. X also reacts with H_2 to give an acid Y. Y can also be prepared by heating its salt with H_2PO_4 . X and Y are :-
 (a) Cl_2 , HCl (b) F_2 , HF
 (c) Br_2 , HBr (d) I_2 , KI
73. Fluorine is obtained by the interaction of K_2MnF_6 with Lewis acid SbF_5 because :-
 (a) In the first step, K_2MnF_6 is converted into MnF_4
 (b) In the second step, MnF_4 is dissociated into MnF_2 and F_2
 (c) Both (a) & (b)
 (d) None of these
74. Which of the following product is formed when one mole periodate ion (IO_4^-) reacts with excess of iodide ions followed by the acidification of solution ?
 (a) IO^- (b) IO_3^-
 (c) I_2 (d) I_2O_5
75. Which of the following options is incorrect ?
 (a) Sodium hypochlorite is used as bleaching and sterilising agent.
 (b) All halogens except F_2 are more reactive than interhalogens.
 (c) Finely divided iron can not form ferric chloride with hydrochloric acid.
 (d) HNO_3 oxidises sulphur to H_2SO_4 but only oxidises selenium to H_2SeO_3 .
76. Ester of boric acid $\text{B}(\text{OC}_2\text{H}_5)_3$ imparts :-
 (a) Blue colour to the burner flame
 (b) Green colour to the burner flame
 (c) Red colour to the burner flame
 (d) Brown colour to the burner flame
77. The repeating structural units in silicone is :-
 (a) $\text{R}_2\text{SiO}-$ (b) SiO_2
 (c) ROSiOR (d) $-\text{SiR}_2\text{OOR}$
78. A given silicate is amphibole :

$$\text{Na}_2\overset{\text{II}}{\text{Fe}}_x\overset{\text{III}}{\text{Fe}}_y[\text{Si}_8\text{O}_{22}](\text{OH})_2$$

 The value of $x + y$ is
 (a) 5 (b) 7
 (c) 3 (d) 9
79. On contact process, one of the step is

$$2\text{SO}_2(\text{g}) + \text{O}_2(\text{g}) \rightarrow 2\text{SO}_3(\text{g})$$

 This reaction is carried out :-
 (a) at 500 K and 9 bar pressure with Pt/Rh as catalyst
 (b) at 1200 K and 5 bar pressure with Pt/Rh as catalyst
 (c) at 500 K and 7 bar pressure with V_2O_5 as catalyst
 (d) at 720 K and 2 bar pressure with V_2O_5 as catalyst
80. In ostwald's process , one of the step is

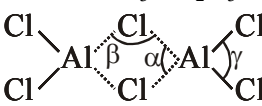
$$4\text{NH}_3(\text{g}) + 5\text{O}_2(\text{g}) \longrightarrow 4\text{NO}(\text{g}) + 6\text{H}_2\text{O}(\text{g})$$

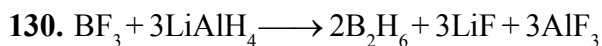
 This reaction is carried out :-
 (a) at 500 K and 9 bar pressure with Pt/Rh as catalyst
 (b) at 1200 K and 5 bar pressure with Pt/Rh as catalyst
 (c) at 500 K and 7 bar pressure with V_2O_5 as catalyst
 (d) at 720 K and 2 bar pressure with V_2O_5 as catalyst
81. $2\text{MX} + \text{H}_2\text{SO}_4 \longrightarrow 2\text{HX} + \text{M}_2\text{SO}_4$
 MX can be :-
 (a) MI (b) MCl
 (c) MBr (d) MI
82. In a mixture of HNO_3 and H_2SO_4 , HNO_3 acts as:-
 (a) base (b) acid
 (c) oxidising reagent (d) reducing agent
83. $\text{Br}_2 + 2\text{X}^-(\text{aq}) \longrightarrow \text{X}_2 + 2\text{Br}^-(\text{aq})$
 above reaction is correct when X^- is :-
 (a) F^- , Cl^- (b) I^- only
 (c) F^- , Cl^- , I^- (d) Cl^- , I^-

84. Which acid on heating produces phosphoric acid by disproportionation reaction ?
 (a) Phosphoric acid H_3PO_4
 (b) Phosphorus acid H_3PO_3
 (c) Peroxy monophosphoric acid
 (d) Metaphosphoric acid
85. In copper sulphate pentahydrate how many anion water is present.
 (a) 0 (b) 1
 (c) 2 (d) 3
86. Temporary hardness of water is caused by presence of
 (a) Soluble nitrate
 (b) Soluble nitrite
 (c) Insoluble carbonate
 (d) Soluble bicarbonate of Ca and Mg
87. The method which is not applicable for manufacturing ammonia?
 (a) $\text{NH}_2 - \text{C}(\text{O}) - \text{NH}_2 + \text{H}_2\text{O} \longrightarrow$
 (b) $\text{NH}_4\text{Cl} + \text{Ca}(\text{OH})_2 \longrightarrow$
 (c) $\text{ZnSO}_4 + \text{NH}_4\text{OH} \longrightarrow$
 (d) $(\text{NH}_4)_2\text{SO}_4 + \text{NaOH} \longrightarrow$
88. $\text{X} \xrightarrow{\text{NaOH}/\Delta} \text{Y} \xrightarrow{\text{HCl}} \text{White fume}$
 (Colourless solid)
 $\text{X} \xrightarrow{\text{CaCl}_2} \text{Z} \xrightarrow{\text{KMnO}_4} \text{Colourless solution}$
 (White ppt.)
 Salt X is :-
 (a) BaC_2O_4 (b) $(\text{NH}_4)_2\text{CO}_3$
 (c) CaC_2O_4 (d) $(\text{NH}_4)_2\text{C}_2\text{O}_4$
89. Ammonia can be dried by :-
 (a) concentrated H_2SO_4
 (b) P_4O_{10}
 (c) CaO
 (d) anhydrous CaCl_2
90. HNO_3 when reacts with P_4O_{10} it gives :-
 (a) NO_2 (b) N_2O_3
 (c) N_2O_4 (d) N_2O_5
91. H_2O_2 is not an example of :-
 (a) reducing reagent (b) oxidising reagent
 (c) dehydrating reagent (d) Bleaching reagent
92. H_2O_2 when reacts with KIO_4 and NH_2OH respectively, it acts as :-
 (a) reducing reagent & oxidising reagent
 (b) reducing reagent & reducing reagent
 (c) oxidising reagent & oxidising reagent
 (d) oxidising reagent & reducing reagent
93. When titanium oxide reacts with H_2O_2 , it gives :-
 (a) red colour due to formation of titanium oxide
 (b) yellow orange colour due to formation of titanium oxide/ titanium sulphate
 (c) black colour due to formation of titanium oxide
 (d) violet colour due to formation of titanium oxide
94. Chemical reagent added to remove temporary hardness of water is -
 (a) Na_2CO_3 (b) $\text{Ca}(\text{OH})_2$
 (c) MgSO_4 (d) CaSO_4
95. Ozonolysis is a reaction when O_3 gas is passed through :-
 (a) saturated hydrocarbon
 (b) unsaturated hydrocarbon
 (c) cyclohexane ring
 (d) cyclopentane ring
96. Ozone oxidises moist sulphur to :-
 (a) SO_2 (b) SO_3
 (c) H_2SO_4 (d) H_2SO_3
97. On heating ozone, its volume :-
 (a) decreases to half
 (b) becomes double
 (c) increases to 3/2 times
 (d) remain unchanged
98. The laboratory test "trailing of mercury" is applicable to identify :-
 (a) O_2 (b) H_2O
 (c) H_2O_2 (d) O_3
99. Which is responsible for depleting ozone layer which protects from ultraviolet light ?
 (a) Chloroform
 (b) Chlorofluoro carbon
 (c) Carbon tetrachloride
 (d) Methyl chloride

6.12 THE p-BLOCK ELEMENTS AND THEIR COMPOUNDS

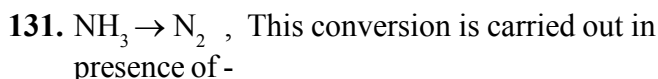
100. Aqueous solution of bleaching powder is source of :-
 (a) $O_2 + Cl^-$ (b) $OH^- + Cl_2$
 (c) $OH^- + Cl^-$ (d) $O_3 + Cl^-$
101. $CH_3COCH_3 \xrightarrow[\text{aqueous solution}]{Ca(OCl)Cl} CHCl_3 + \text{Product}$
 (P). P is :-
 (a) CH_3CH_3 (b) $CH_3CHOHCH_3$
 (c) $(CH_3CO_2)_2Ca$ (d) $(HCO_2)_2Ca$
102. The incorrect statements are :-
 (a) Bleaching powder is strong oxidising reagent due to evolution of nascent oxygen.
 (b) Bleaching powder is strong oxidising reagent due to evolution of nascent chlorine.
 (c) Bleaching powder reacts with methanol to give calcium formate
 (d) Bleaching powder oxidises NH_3 into NCl_3
103. A pale yellow solid (A) having crown shape is heated with concentrated H_2SO_4 , it gives suffocating smell of gas B, which is passed through moistened with starch iodide paper and turn it into blue. The gas B and solid A are respectively.
 (a) SO_3 & S_8 (b) SO_2 & S_8
 (c) SO_2 & S_2N_2 (d) SO_3 & SO_2
104. H_2O_2 acts as oxidising reagent when it reacts with:-
 (a) Potassium ferrocyanide in acidic medium
 (b) Potassium ferrocyanide in basic medium
 (c) Potassium ferricyanide in acidic medium
 (d) Potassium ferricyanide in basic medium
105. H_2O_2 acts as reducing reagent when it reacts with:-
 (a) Potassium ferrocyanide in acidic medium
 (b) Potassium ferrocyanide in basic medium
 (c) Potassium ferricyanide in acidic medium
 (d) Potassium ferricyanide in basic medium.
106. $Ca(OCl)Cl$ when reacts with $FeSO_4$ solution, $Ca(OCl)Cl$ is converted into :-
 (a) Cl_2 (b) $CaCl_2$
 (c) $Ca(OCl)_2$ (d) $Ca(OH)_2$
107. Calgon is used for water treatment. Calgon is:-
 (a) $Na_2[Na_4(PO_3)_6]$ (b) $Na_4[Ca(PO_3)_6]$
 (c) $Na_2[Ca_2(PO_3)_6]$ (d) $Na_4[Na_2(PO_3)_6]$
108. O-O bond length :-
 (a) $O_2 < O_3 < H_2O_2$ (b) $O_3 < O_2 < H_2O_2$
 (c) $H_2O_2 < O_3 < O_2$ (d) $O_2 < H_2O_2 < O_3$
109. The permutit process is based on
 (a) exchange of basic radical or metal ion. e.g. Ca^{2+} and Mg^{2+} by Na^+
 (b) exchange of acidic radical like HCO_3^- / CO_3^{2-} by SO_4^{2-} , NO_3^-
 (c) exchange of both acidic and basic radical
 (d) None
110. Pair of metals does not dissolve in nitric acid :-
 (a) Cr, Al (b) Ag, Au
 (c) Al, Ag (d) Pt, Hg
111. Ammonia on reaction with sodium hypochlorite gives :-
 (a) NO (b) NH_4Cl
 (c) NH_2-NH_2 (d) HNO_2
112. $NH_4Cl(s)$ is heated in test tube. Vapours are brought in contact with red litmus paper, which changes to blue and then red. It is because of :-
 (a) formation of NH_4OH and HCl
 (b) formation of NH_3 and $HOCl$
 (c) greater diffusion of NH_3 than HCl
 (d) greater diffusion of HCl from NH_3
113. Which of the following gives mixture of SO_2 and SO_3 on heating :-
 (a) $ZnSO_4$ (b) $CuSO_4$
 (c) $Fe(SO_4)_3$ (d) $FeSO_4$
114. ZSM-5-A type of zeolite is used :-
 (a) to convert starch into alcohol
 (b) to convert alcohols into gasoline
 (c) to convert petroleum products into branched hydrocarbon
 (d) to convert hydrocarbon into alcohol
115. In aluminosilicates
 (a) Aluminium atom replaces few silicon atoms in three dimensional network of silicon dioxide
 (b) Silicon atoms replaces few aluminium atom in three dimensional network of silicon dioxide

- (c) Aluminium atom replaces few silicon atoms in three dimensional network of alumina
(d) Silicon atoms replaces few aluminium atom in three dimensional network of alumina
116. Concentrated HNO_3 reacts with iodine to give :-
(a) HI (b) HOI
(c) HIO_4 (d) HIO_3
117. Blood pH is maintained between 7.26 to 7.42. It is due to -
(a) $\text{H}_2\text{SO}_3 / \text{HSO}_3^-$ buffer system
(b) $\text{H}_2\text{SO}_4 / \text{HSO}_4^-$ buffer system
(c) $\text{H}_2\text{CO}_3 / \text{HCO}_3^-$ buffer system
(d) HF / F^- buffer system
118. $2\text{E(s)} + \text{N}_2(\text{g}) \xrightarrow{\Delta} 2\text{EN(s)}$
 $2\text{E(s)} + 3\text{X}_2(\text{g}) \xrightarrow{\Delta} 2\text{EX}_3(\text{s})$
E belongs to :-
(a) Group 13 element (b) Group 14 element
(c) Group 15 element (d) Group 16 element
119. $\text{S}_2\text{O}_3^{2-} + \text{Pb}^{2+} \longrightarrow \text{PbS}_2\text{O}_3$ (Precipitate P_1)
 $\text{PbS}_2\text{O}_3 + \text{H}_2\text{O} \longrightarrow 2\text{H}^+ + \text{SO}_4^{2-} + \text{PbS}$ (Precipitate P_2)
 $\text{PbS} + \text{H}_2\text{O}_2 \longrightarrow \text{H}_2\text{O} + \text{PbSO}_4$ (Precipitate P_3)
Correct option -
(a) P_1 is black colour precipitate
(b) P_2 is white colour precipitate
(c) Precipitate P_1 is soluble in excess thiosulphate
(d) P_3 is black colour precipitate
120. Strongest acid is
(a) HI (b) HBr
(c) HCl (d) $\text{HF} + \text{SbF}_5$
121. The least atomic radius is observed in
(a) Al (b) Ga
(c) In (d) Tl
122. The highest electronegativity is observed in
(a) Al (b) Ga
(c) In (d) Tl
123. The highest and least oxidation state of element in group 16 will be -
(a) +6, +2 (b) +4, -2
(c) +5, -3 (d) +6, -2
124. A certain salt X, gives the following results
(i) Its aqueous solution is alkaline to litmus
(ii) It swells up to a glassy material Y on strong heating
(iii) When concentrated H_2SO_4 is added to a hot solution of X, white crystal of an acid Z separates out.
X is -
(a) $\text{Na}_2\text{B}_4\text{O}_7$ (b) $\text{Ca}_2\text{B}_6\text{O}_{11}$
(c) $\text{B}_2\text{O}_3 \cdot \text{TiO}_2$ (d) NaBH_4
125. Z is
(a) HBO_2 (b) H_3BO_3
(c) $\text{H}_4\text{B}_2\text{O}_7$ (d) None
126. When metal X is treated with sodium hydroxide, a white precipitate (a) is obtained, which is soluble in excess of NaOH to give soluble complex (b). Compound (a) is soluble in dilute HCl to form compound (c). The compound (a) when heated strongly gives (d), which is used to extract metal. X is :-
(a) B (b) Al
(c) Tl (d) Zn
127. B and D are respectively -
(a) $\text{Tl}(\text{OH})_2$; Tl_2O (b) $\text{Zn}(\text{OH})_2$; ZnO
(c) $\text{Al}(\text{OH})_3$; Al_2O_3 (d) $\text{B}(\text{OH})_3$; B_2O_3
128. 
(a) $\alpha > \beta > \gamma$
(b) $\beta > \gamma > \alpha$
(c) $\beta > \alpha > \gamma$
(d) $\gamma > \beta > \alpha$
129. In alum like $\text{M}^{\text{I}}\text{M}^{\text{III}}(\text{SO}_4)_2 \cdot 12\text{H}_2\text{O}$:-
(a) Six water molecule are octahedrally coordinated around M^{I}
(b) Six water molecule are octahedrally coordinated around M^{III}
(c) These water molecule are characterised as lattice water
(d) All

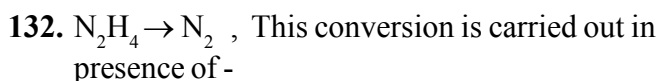
6.14 THE p-BLOCK ELEMENTS AND THEIR COMPOUNDS

Correct option for this reaction :-

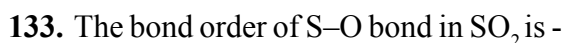
- (a) This reaction is example of redox reaction where LiAlH_4 acts as reducing agent.
- (b) This reaction is example of redox reaction where BF_3 acts as reducing agent.
- (c) This reaction is example of non redox reaction
- (d) Oxidation state of hydrogen atom in reactant side is different w.r.t. oxidation state of hydrogen atom in product side.



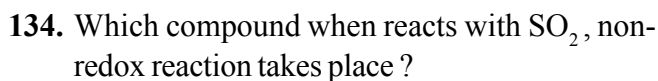
- (a) Bleaching powder (b) CuO
- (c) Both (a) and (b) (d) O_2 (air)



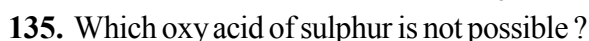
- (a) I_2 (b) KIO_3
- (c) Fe^{3+} (d) All



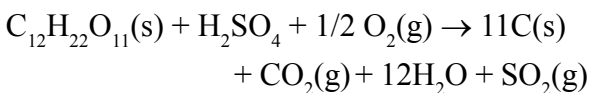
- (a) 2 (b) 1.5
- (c) 1 (d) 2.5



- (a) Cl_2 (g) (b) Fe^{3+} solution
- (c) KMnO_4 , H^+ (d) Na_2CO_3 solution

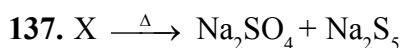


- (a) $\text{H}_2\text{S}_2\text{O}_7$
- (b) H_2SO_5
- (c) H_2SO_6
- (d) $\text{H}_2\text{S}_2\text{O}_6$



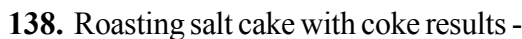
which of the following statement is correct -

- (a) Here H_2SO_4 acts as dehydrating as well as oxidising reagent
- (b) The reaction is endothermic in nature
- (c) The solution turns colourless
- (d) Colourless and odourless gas is evolved

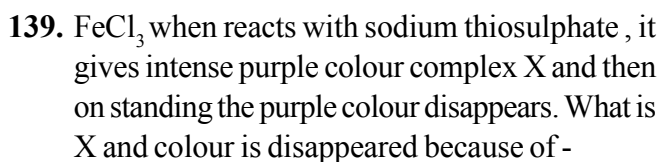


X is formed as one of the product when sulphur is boiled with caustic alkali. X is -

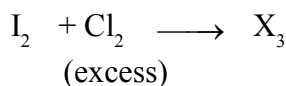
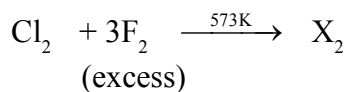
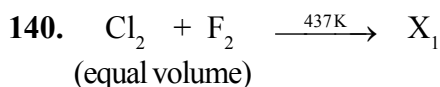
- (a) Na_2SO_3 (b) $\text{Na}_2\text{S}_4\text{O}_6$
- (c) $\text{Na}_2\text{S}_2\text{O}_3$ (d) Na_2S



- (a) formation of mixture of CO_2 & S
- (b) formation of mixture of CO_2 & Na_2S
- (c) formation of mixture of CO & Na_2S
- (d) formation of mixture of CS_2 & Na_2O

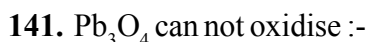


- (a) $\text{X} = [\text{Fe}(\text{S}_2\text{O}_3)_3]^{3-}$; colour is disappeared due to formation of Fe^{2+}
- (b) $\text{X} = [\text{Fe}(\text{S}_2\text{O}_3)_2]^{-1}$; colour is disappeared due to formation of Fe^{2+}
- (c) $\text{X} = [\text{Fe}(\text{S}_2\text{O}_3)_2]^{-1}$; colour is disappeared due to formation of $[\text{Fe}(\text{S}_2\text{O}_3)_2]$
- (d) $\text{X} = [\text{Fe}(\text{S}_2\text{O}_3)_3]^{3-}$; colour is disappeared due to formation of $[\text{Fe}(\text{S}_2\text{O}_3)_2]^{4-}$

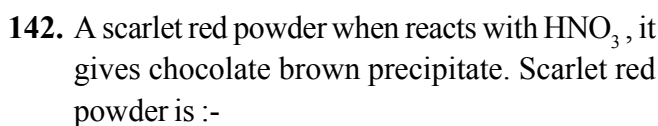


Which of the following statements are correct -

- (a) X_2 is colourless gas (b) X_3 is orange solid
- (c) X_1 is colourless gas (d) All



- (a) HCl (b) H_2SO_4
- (c) HNO_3 (d) All



- (a) PbO_2 (b) PbO
- (c) Pb_3O_4 (d) Pb_2O_3

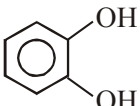
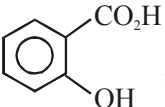
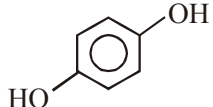
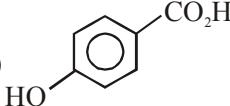
143. A scarlet red powder when reacts with HNO_3 , it gives chocolate brown precipitate. Chocolate brown precipitate is :-
 (a) PbO_2 (b) PbO
 (c) Pb_3O_4 (d) Pb_2O_3
144. The number of lone pair on Cl in HOCl , HClO_2 , HClO_3 , HClO_4 are respectively :-
 (a) 1,2,3,4 (b) 0,1,2,3
 (c) 4,3,2,1 (d) 3,2,1,0
145. Number of Cl = O bonds present in HOCl , HClO_2 , HClO_3 , HClO_4 are respectively :-
 (a) 1,2,3,4 (b) 0,1,2,3
 (c) 4,3,2,1 (d) 3,2,1,0
146. Boric acid can act as strong acid by addition of :-
 (a) Cis-1, 2-diol (b) Borax
 (c) Trans-1,2-diol (d) Na_2HPO_4
147. In aqueous solution, orthoboric acid acts as :-
 (a) Proton donor acid (b) Lewis acid
 (c) Triprotic acid (d) Strong acid
148. The transition temperature for rhombic sulphur to monoclinic sulphur is :-
 (a) 95.5°C (b) 112°C
 (c) 160.5°C (d) 190°C
149. Correct bond dissociation energy order :-
 (a) $\text{N}\equiv\text{N} > \text{P}\equiv\text{P}$ (b) $\text{N}-\text{N} > \text{P}-\text{P}$
 (c) $\text{O}-\text{O} < \text{S}-\text{S}$ (d) All
150. Chlorofluoro carbon (freons) is represented as CF_mCl_n . It can be :-
 (a) CF_2Cl_2 (b) CFCl_3
 (c) CClF_3 (d) All
2. Which of the following reactions are correctly given?
 (a) $\text{SnO}_2 + 2\text{H}_2\text{SO}_4 \rightarrow \text{Sn}(\text{SO}_4)_2 + 2\text{H}_2\text{O}$
 (b) $\text{SnO}_2 + 2\text{NaOH} \rightarrow \text{Na}_2\text{SnO}_2 + \text{H}_2\text{O}$
 (c) $\text{SnCl}_4 + \text{Hg} \rightarrow \text{Hg}_2\text{Cl}_2 + \text{SnCl}_2$
 (d) $\text{SnCl}_2 + \text{Zn} \rightarrow \text{ZnCl}_2 + \text{Sn}$
3. Which of the following names are correct :-
 (a) Crystalline SnS_2 : Mosaic Gold
 (b) $\text{SnCl}_4 \cdot 5\text{H}_2\text{O}$: Butter of tin
 (c) SnO_2 : Cassiterite
 (d) SnH_4 : Stannane
4. SnS_2 is soluble in :-
 (a) concentrated HCl (b) YAS
 (c) H_2O (d) NH_3
5. Which of the following are example of mixed oxide?
 (a) PbO (b) Pb_2O_3
 (c) Pb_3O_4 (d) Pb_2O_2
6. Which of the following reaction are correct ?
 (a) $\text{PbO}_2 + 4\text{HCl} \rightarrow \text{PbCl}_2 + 2\text{H}_2\text{O} + \text{Cl}_2$
 (b) $2\text{PbO}_2 + 4\text{H}_2\text{SO}_4 \rightarrow 2\text{PbSO}_4 + 2\text{H}_2\text{O} + \text{O}_2$
 (c) $\text{PbO}_2 + 2\text{NaOH} \rightarrow \text{Na}_2\text{PbO}_3 + \text{H}_2\text{O}$
 (d) $2\text{PbO}_2 \xrightarrow{730^\circ\text{C}} 2\text{PbO} + \text{O}_2$
7. Which can be formed as yellow precipitate ?
 (a) PbSO_4 (b) PbI_2
 (c) PbCrO_4 (d) $\text{Pb}(\text{OH})_2 \cdot \text{PbCO}_3$
8. Which of the following options are correct ?
 (a) PbS is insoluble in hot concentrated HCl
 (b) PbS is insoluble in boiling dilute HNO_3
 (c) White lead is used as pigment
 (d) $\text{Pb}(\text{OH})_2$ is dissolved in acid as well as in base
9. Which of the following reaction give NH_3 gas :-
 (a) Boron nitride is fused with potassium hydroxide
 (b) Boron nitride reacts with F_2 gas
 (c) Boron nitride undergoes hydrolysis
 (d) Boron imide undergoes decomposition

EXERCISE # II

□ One or More Than One Correct Answer :

1. Which of the following statements are correct regarding SnCl_2 ?
 (a) SnCl_2 conducts electricity
 (b) SnCl_2 is dissolved in water to form turbidity of stannous oxychloride
 (c) SnCl_2 is ionic solid
 (d) SnCl_2 is covalent angular molecule

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10. Which of the following compound forms chelating oxy derivative with $B(OH)_4^-$:-
- (a) Catechol ; 
- (b)  ; Salicylic acid
- (c) Quinol ; 
- (d)  ; p-hydroxy benzoic acid
11. The oxidation state of hydrogen atom in $B_3N_3H_6$:-
- (a) All hydrogens have same oxidation state of +1
- (b) All hydrogens have same oxidation state of -1
- (c) Hydrogens attached with boron have oxidation state of -1
- (d) Hydrogens attached with nitrogen have oxidation state of +1
12. Which of the following statements are incorrect ?
- (a) B-F bond length in BF_3 is less than B-F bond length in $NOBF_4$
- (b) N-O bond length in NO is more than N-O bond length in $NOCIO_4$
- (c) The terminal B-H bond length in B_2H_6 is more than C-H bond length in C_2H_6
- (d) The terminal B-H bond length in B_2H_6 is more than C-H bond length in C_2H_4
13. I_2 is soluble in KI solution but is insoluble in water because :-
- (a) I_2 is soft lewis acid and I^- is hard lewis base
- (b) I_2 is hard lewis acid and H_2O is hard lewis base
- (c) I_2 is soft lewis acid and I^- is soft lewis base
- (d) I_2 is soft lewis acid and H_2O is hard lewis base
14. Select correct statements :-
- (a) Borax is used as buffer
- (b) 1(M) borax solution reacts with equal volume of 2(M) HCl solution
- (c) Titration of borax can be made using methyl orange as the indicator
- (d) Coloured bead is obtained in borax bead test contains metaborate
15. The pair that yield same gaseous product on reaction with water are :-
- (a) AlN , $CaCN_2$ (b) Ca & CaH_2
- (c) F_2 & XeF_4 (d) K & KO_2
16. HNO_3 :-
- (a) oxidises S to H_2SO_4
- (b) oxidises P_4 to H_3PO_4
- (c) oxidises Se to H_2SeO_3
- (d) oxidises I_2 to HIO_3
17. Which of the following options are correct ?
- (a) HF is the only hydra acid of halogen which does not form any precipitate with $AgNO_3$.
- (b) Oxidising property of halogens $F_2 > Cl_2 > Br_2 > I_2$
- (c) The bond dissociation energy of halogen $Cl_2 > Br_2 > F_2 > I_2$
- (d) The bond dissociation energy of hydrogen halide $HF > HCl > HBr > HI$
18. Astatin is the element below iodine in the group VII A of the periodic table. Which of the following statements are not true for astatine ?
- (a) It is more electronegative than iodine
- (b) It will exhibit only -1 oxidation state
- (c) Intermolecular forces between the astatine molecules will be more than that between iodine molecules.
- (d) It is less electronegative than iodine
19. Which statements are correct about halogens ?
- (a) They are all diatomic and form univalent ions
- (b) Halogens have the smallest atomic radii in their respective periods
- (c) They are all diatomic and form divalent ions
- (d) They are all reducing agent
20. Which of the following reactions are feasible ?
- (a) $2KCl + I_2 \longrightarrow 2KI + Cl_2$
- (b) $2KI + Cl_2 \longrightarrow 2KCl + I_2$
- (c) $MnO_2 + 4HCl \longrightarrow MnCl_2 + Cl_2 + 2H_2O$
- (d) $MnO_2 + 2NaBr + 2H_2SO_4 \longrightarrow Br_2 + 2Na^+ + Mn^{2+} + 2SO_4^{2-} + 2H_2O$

21. Which of the following statements are correct ?
 (a) At ordinary temperature, the rate of disproportionation of hypohalites of chlorine, bromine and iodine follows the order $\text{IO}^- > \text{BrO}^- > \text{ClO}^-$.
 (b) Fluorine can not be prepared in aqueous medium by electrolysis, since it decomposes water with liberation of ozonised oxygen.
 (c) HI is a stronger acid than HBr because of the low dissociation energy of HI.
 (d) In aqueous solution chlorine is a stronger oxidising agent than fluorine.
22. The products formed when potassium chlorate reacts with iodine.
 (a) KIO_4 (b) KOI (c) KIO_3 (d) Cl_2
23. Correct order regarding HNO_3 and H_2SO_4 :-
 (a) acidic strength $\text{HNO}_3 > \text{H}_2\text{SO}_4$
 (b) boiling point $\text{H}_2\text{SO}_4 > \text{HNO}_3$
 (c) oxidising power $\text{HNO}_3 > \text{H}_2\text{SO}_4$
 (d) Volatility $\text{HNO}_3 > \text{H}_2\text{SO}_4$
24. Which of the following option are correct :-
 (a) Zn when reacts with very dilute HNO_3 , it gives $\text{Zn}(\text{NO}_3)_2$ and NH_4NO_3
 (b) $2\text{SO}_2(\text{g}) + \text{O}_2(\text{g}) \xrightarrow{\text{catalyst}} 2\text{SO}_3(\text{g})$, it is an example of endothermic reaction
 (c) I_2 produces HIO_3 and NO_2 with concentrated HNO_3
 (d) Ag when reacts with concentrated HNO_3 , it gives AgNO_3 and NO_2
25. Bronze when treated with hot concentrated HNO_3 :-
 (a) Cu and Zn both readily dissolves
 (b) Copper is dissolved to give blue solution
 (c) Sn is dissolved to give colourless solution $\text{Sn}(\text{NO}_2)_2$
 (d) Tin is oxidised to metastannic acid in the form of white precipitate
26. Which of the following statements are incorrect ?
 (a) To carry out brown ring test for nitrate ion, dilute H_2SO_4 should be used.
 (b) To carry out brown ring test for nitrite ion, dilute H_2SO_4 should be used.
 (c) Devarda's alloy is alloy of aluminium, copper and zinc where zinc percentage is maximum
 (d) Copper strip is added in concentrated H_2SO_4 , the solution become blue
27. Which of the following reactions give NH_3 gas ?
 (a) $\text{Zn} + \text{NaOH} + \text{NaNO}_3 \xrightarrow{\Delta}$
 (b) $(\text{NH}_4)_2\text{CO}_3 \xrightarrow{\Delta}$
 (c) $\text{NH}_4\text{ClO}_4 \xrightarrow{\Delta}$
 (d) $\text{Al} + \text{NaOH} + \text{NaNO}_3 \xrightarrow{\Delta}$
28. Which compound is not known ?
 (a) NX_5 (b) BF_6^{3-} (c) HPO_2 (d) NOBr_3
29. Which of the following statements are correct -
 (a) In ammonium nitrate, anionic part is sp^2 hybridised
 (b) In ammonium nitrate, cationic part is sp^3 hybridised
 (c) In ammonium nitrite, cationic part is sp^2 hybridised
 (d) In ammonium nitrite, anionic part is sp^2 hybridised
30. Which of the following can not coexist with solution?
 (a) $\text{H}_2\text{S} + \text{SO}_2$ (b) $\text{Fe}_2\text{S}_3 + \text{HCl}$
 (c) $\text{HNO}_2 + \text{KI}$ (d) $\text{Na}_2\text{CO}_3 + \text{NaHCO}_3$
31. Which of the reaction produces PH_3 as one of the product?
 (a) $\text{Ca}_3\text{P}_2 + \text{H}_2\text{O} \longrightarrow$
 (b) $\text{P}_4(\text{white}) + \text{NaOH} + \text{H}_2\text{O} \longrightarrow$
 (c) $\text{PH}_4\text{I} + \text{KOH} \longrightarrow$
 (d) $\text{H}_3\text{PO}_2 \xrightarrow{\Delta}$
32. $\text{A (Calcium imide)} \xrightarrow{+\text{H}_2\text{O}} \text{B} + \text{C}(\text{gas})$
 $\text{C (gas)} \xrightarrow{\text{Ca}(\text{OCl})\text{Cl}} \text{D}(\text{gas})$
 $\text{D (gas)} \xrightarrow{\text{Mg}} \text{E} \xrightarrow{+\text{H}_2\text{O}} \text{C}(\text{gas})$
 Correct option :-
 (a) C gas is N_2O (b) D gas is N_2O_3
 (c) C gas is N_2O_4 (d) D gas is N_2
33. F_2 reacts with H_2O to give :-
 (a) HOF (b) HF
 (c) O_2 (d) O_3
34. Which of the following statements are correct ?
 (a) The hydride of group 15 elements act as oxidising agents
 (b) The hydride of group 15 elements act as reducing agents
 (c) The reducing power increases is going from NH_3 to BiH_3
 (d) The basic character decreases in going from NH_3 to SbH_3

6.18 THE p-BLOCK ELEMENTS AND THEIR COMPOUNDS

35. $\text{CH}_3 - \text{CH} = \text{CH}_2 \xrightarrow{\text{HCl} + \text{H}_2\text{O}_2} \text{P}_1$
 $\text{CH}_3 - \text{CH} = \text{CH}_2 \xrightarrow{\text{HBr} + \text{H}_2\text{O}_2} \text{P}_2$
 Correct statements :-
 (a) P_1 is $\text{CH}_3\text{CH}_2\text{CH}_2\text{Cl}$
 (b) P_1 is $\text{CH}_3\text{CHClCH}_3$
 (c) P_2 is $\text{CH}_3\text{CH}_2\text{CH}_2\text{Br}$
 (d) P_2 is $\text{CH}_3\text{CHBrCH}_3$
36. Which of the followings are diamagnetic ?
 (a) H_2O_2 (b) O_3
 (c) SO_2 (d) KO_3
37. Which of the followings reagent can be used to change the colour when SO_2 gas is passed ?
 (a) Bromine water
 (b) Acidified FeCl_3 solution
 (c) Chlorine water
 (d) Acidified $\text{K}_2\text{Cr}_2\text{O}_7$ solution
38. Ozone can oxidise :-
 (a) K_2MnO_4 (b) KMnO_4
 (c) S(moist) (d) Ag
39. Aqueous solution of which ion gives black precipitate in acidic medium when H_2S gas is passed through it :-
 (a) Bi^{3+} (b) Pb^{2+}
 (c) Cu^{2+} (d) Ag^+
40. Aqueous solution of which ion gives black precipitate in basic medium when H_2S gas is passed through it :-
 (a) Fe^{2+} (b) Fe^{3+}
 (c) Co^{2+} (d) Ni^{2+}
41. H_2O_2 can be prepared by :-
 (a) Oxidation of 2-ethyl anthraquinol by air
 (b) Electrolysis of 50% sulphuric acid
 (c) CO_2 gas is passed through aqueous solution of barium peroxide
 (d) Na_2O_2 is added to dilute H_2SO_4
42. When ammonia is gas is passed through suspension of gypsum in water and then passing in carbondioxide then the products formed can be :-
 (a) H_2SO_4 (b) $(\text{NH}_4)_2\text{CO}_3$
 (c) CaCO_3 (d) $(\text{NH}_4)_2\text{SO}_4$
43. NH_3 when reacts with Cl_2 , then :-
 (a) NCl_3 is formed when Cl_2 is present in excess
 (b) N_2 is formed when Cl_2 is present in excess
 (c) N_2 is formed when NH_3 is present in excess
 (d) NCl_3 is formed when NH_3 is present in excess
44. $\text{SO}_2 + \text{Cl}_2 \rightarrow \text{Compound X}_1$
 $\text{CO} + \text{Cl}_2 \rightarrow \text{Compound X}_2$
 Correct statement regarding X_1 and X_2 ?
 (a) X_1 is used to chlorinate alkane, eg. ethane to ethyl chloride
 (b) X_2 is poisonous gas
 (c) The oxidation state of sulphur in compound X_1 is +4
 (d) The oxidation state of carbon in compound X_2 is +4
45. Dioxygen can be obtained in the laboratory by :-
 (a) Heating potassium chlorate in presence of MnO_2 at high temperature
 (b) Thermal decomposition of oxide of metal which is low in the electrochemical series, eg. Ag and Hg.
 (c) Thermal decomposition of H_2O_2
 (d) Thermal decomposition of NaNO_2
46. Which of the following statements are correct :-
 (a) SiF_4 reacts with F^- to give SiF_6^{2-}
 (b) SiCl_4 reacts with Cl^- to give SiCl_6^{2-}
 (c) O-O bond in O_2F_2 is shorter as compared to O-O bond in H_2O_2
 (d) O-F bond in O_2F_2 is longer as compared to O-F bond in OF_2
47. $\text{X} + \text{Coke powder} \xrightarrow{>2000^\circ\text{C}} \text{Y} + 2\text{CO}\uparrow$
 (3D-silicate)
 $\text{Y} + \text{Coke powder} \xrightarrow{>2000^\circ\text{C}} \text{Z}$
 What are the correct statements regarding Z ?
 (a) All atoms of Z are sp^3 hybridised
 (b) Z exhibits tetrahedral geometry
 (c) Z is yellow solid at room temperature and colourless solid when impurity is present.
 (d) Z is extremely hard substance used as abrasive

48. CO gas is absorbed by aqueous suspension of cuprous chloride forming the complex like $[\text{CuCl}(\text{CO})(\text{H}_2\text{O})_2]$.
Correct statement regarding this complex :-
(a) complex is tetrahedral
(b) complex is diamagnetic
(c) geometrical isomerism is not possible for this complex
(d) Chlorine atoms is separated by an angle 90° with respect to both water molecule
49. Which of the following ammonium salt give bronsted acid on heating ?
(a) NH_4NO_3 (b) $(\text{NH}_4)_2\text{SO}_4$
(c) $(\text{NH}_4)_2\text{MoO}_4$ (d) NH_4Cl
50. An element (X) has electronic configuration $1s^2 2s^2 2p^3$. Hydride of X is :-
(a) Water soluble
(b) Colourless gas
(c) Odourless gas
(d) Freezing point is 198.4K and 239.7K respectively
51. An element (X) has electronic configuration $1s^2 2s^2 2p^2$. Hydride of X is :-
(a) Colourless gas (b) Pungent odour gas
(c) Water insoluble (d) Water soluble
52. An element (a) has outermost electronic configuration $4s^2 4p^5$. Hydride of A is :-
(a) Strong base (b) Strong acid
(c) Brown colour gas (d) Colourless gas
53. Gun powder consists of :-
(a) NaNO_3 (b) KNO_3
(c) Sulphur (d) Charcoal
54. Which of the following statements are correct ?
(a) When dilute H_2SO_4 is added with potassium ferrocyanide, then CO gas is evolved
(b) When concentrated H_2SO_4 is added with potassium ferrocyanide, then HCN gas is evolved
(c) Urea, an nitrogenous organic matter when undergoes reaction with water, ammonia gas is evolved.
(d) For H_2SO_4 , first dissociation constant ($K_{a_1} > 10$) is very high
55. F_2 reacts with alkali to give -
(a) HOF (b) O_3
(c) F_2O (d) O_2
56. $\text{Ca}_2\text{B}_6\text{O}_{11} + \text{Na}_2\text{CO}_3 \xrightarrow{\Delta} [\text{X}] + \text{CaCO}_3 + \text{NaBO}_2$ (Unbalanced equation)
Correct statement for [X] :-
(a) Structure of anion of crystalline (X) has one boron atom sp^3 hybridised and other three boron atoms sp^2 hybridised
(b) (X) with $\text{NaOH}(\text{aq.})$ gives a compound which on reaction with H_2O_2 in alkaline medium yields a compound used as brightener in soaps
(c) Hydrolysis of (X) with HCl or H_2SO_4 yields a compound which on reaction with HF gives fluoroboric acid
(d) [X] on heating with cobalt salt in oxidising flame gives blue coloured bead
57. Which of the following reaction is correctly matched with product ?
(a) $\text{C}_2\text{H}_5\text{OH} + \text{PCl}_5 \rightarrow \text{C}_2\text{H}_5\text{Cl} + \text{POCl}_3 + \text{HCl}$
(b) $\text{CH}_3\text{CO}_2\text{H} + \text{PCl}_5 \rightarrow \text{CH}_3\text{COCl} + \text{POCl}_3 + \text{HCl}$
(c) $3\text{CH}_3\text{CO}_2\text{H} + \text{PCl}_3 \rightarrow 3\text{CH}_3\text{COCl} + \text{H}_3\text{PO}_3$
(d) $\text{Sn} + \text{PCl}_5 \rightarrow \text{SnCl}_2 + \text{PCl}_3$
58. $\text{O}_3 + 2\text{H}^+ + 2\text{e}^- \rightarrow \text{H}_2\text{O} + \text{O}_2$
 $E^\circ = 1.65 \text{ volt}$
 $\text{O}_2 + 4\text{H}^+ + 4\text{e}^- \rightarrow 2\text{H}_2\text{O}$ $E^\circ = 0.814 \text{ volt}$
This value indicates :-
(a) O_2 & O_3 are reducing reagent
(b) O_2 & O_3 are oxidising reagent
(c) O_3 is stronger oxidising reagent than O_2
(d) O_2 is stronger oxidising reagent than O_3
59. Correct statements regarding H_2O_2 :-
(a) O—O bond length is $1.49 \text{ \AA}$
(b) Boiling point is 152°C
(c) $\angle \text{HOO} = 97^\circ$
(d) Dipole moment is 2.01 Debye
60. In which of the following reactions, PCl_3 is formed as one of the product ?
(a) Reaction between P_4 and SOCl_2
(b) Reaction between P_4 and SO_2Cl_2
(c) Reaction between Ag and PCl_5
(d) Reaction between PH_3 and $\text{Ca}(\text{OCl})\text{Cl}$

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61. Correct statements regarding O_3 :-
 (a) It is deep blue in liquid form
 (b) It is violet black in solid form
 (c) It has fishy smell
 (d) Decomposition of $O_3 \rightarrow O_2$, entropy increases
62. Which of the following reaction give amorphous boron as product ?
 (a) $B_2O_3 + 3Mg \rightarrow 3MgO + B$
 (b) $3K + KBF_4 \rightarrow 4KBF + B$
 (Potassium borofluoride)
 (c) $B_2O_3 + 3H_2 \rightarrow 2B + 3H_2O$
 (d) $2BCl_3 + 3H_2 \rightarrow 2B + 6HCl$
63. AlF_3 is different from other aluminium halide. This is illustrated by :-
 (a) AlF_3 is only ionic halide but rest other halides are covalent in nature
 (b) AlF_3 hydrolyses most readily but rest other halides do not undergo hydrolysis
 (c) AlF_3 is unaffected water but other halides are easily hydrolysed
 (d) AlF_3 has least boiling point among all aluminium halides
64. Br_2 when disproportionates in hot and concentrated NaOH solution, the products formed are :-
 (a) NaBr (b) NaOBr
 (c) NaBrO₃ (d) NaClO₄
65. Which of the following are example of amphoteric oxide ?
 (a) CO (b) SnO
 (c) GeO (d) PbO
66. Which of the following are example of acidic oxide?
 (a) SiO₂ (b) PbO₂
 (c) SnO₂ (d) GeO₂
67. Which of the following compound when dissolve in water, it gives alkaline solution ?
 (a) Borax (b) Sodium sulphide
 (c) K₂CO₃ (d) HCl
68. B_2H_6 is prepared by
 (a) direct combination between elemental boron and gaseous H_2
 (b) reaction between boron trifluoride and $LiAlH_4$ in ether medium
 (c) reaction between boron trifluoride and NaH at 450 K
 (d) Oxidation of sodium borohydride with I_2
69. Which of the following options are correct ?
 (a) Concentrated HNO_3 can be transported in aluminium container
 (b) Graphite is used as lubricant
 (c) Diamond is used as abrassive
 (d) Aluminium alloys are used to make aircraft body
70. Which structure exist with central atom sp^3d^2 hybridised ?
 (a) SiF_6^{2-} (b) $SiCl_6^{2-}$
 (c) $GeCl_6^{2-}$ (d) $SnOH_6^{2-}$
71. Which of the following options are correct ?
 (a) $N_2(g) + 3H_2(g) \rightarrow 2NH_3(g)$, it is an example of endothermic reaction -
 (b) In CO_2 molecule, resonance is possible
 (c) H_2O is less polar than H_2O_2
 (d) Number of chlorine atom present in mustard gas is two
72. Which of the following gases are supporter of combustion ?
 (a) N₂O (b) CO₂
 (c) CO (d) O₂
73. $BaSO_3 + H_2O_2 \rightarrow BaSO_4 + H_2O$
 Precipitate P₁ Precipitate P₂
 Correct option :-
 (a) P₁ is white colour precipitate
 (b) P₂ is white colour precipitate
 (c) P₁ is soluble in dilute HCl
 (d) P₂ is insoluble in dilute HCl and dilute HNO_3
74. Which of the following options are correct regarding N_2O_5 ?
 (a) The total number of lone pair of electron in N_2O_5 is 10
 (b) The total number of lone pair of electron in N_2O_5 is 12
 (c) The total number of lone pairs of electron on nitrogen atom in N_2O_5 is 2
 (d) The total number of lone pairs of electron on nitrogen atom in N_2O_5 is 0

75. Which of the following options are correct regarding member of group 14 elements
- Among all elements, carbon forms the most acidic oxide
 - Among all elements, lead is most commonly found in +2 oxidation state
 - Among all elements, silicon and germanium both are semiconductor
 - Among all elements, tin has maximum melting point
76. Which property is not characterised by silicone ?
- High thermal stability
 - Chemically resistant towards oxidation
 - Water attracting in nature due to presence of polar group
 - Water repelling due to presence of non polar alkyl group
77. An aqueous solution of gas (x) shows the following reaction.
- it turns red litmus paper blue
 - when added in excess of copper sulphate solution or deep blue colour is obtained
 - on addition of the FeCl_3 solution, a brown precipitate, soluble in dilute HNO_3 is obtained
- Correct options ?
- x is PH_3
 - x is NH_3
 - Brown precipitate is $\text{Fe}(\text{OH})_3$
 - Blue solution is $[\text{Cu}(\text{NH}_3)_6]^{2+}$
78. Correct statements regarding carbide ?
- WC is example of interstitial carbide
 - Mg_2C_3 is used in propyne preparation
 - B_4C is used as abrasive
 - Cu_2C_2 & Ag_2C_2 are both white colour precipitate
79. $\text{A} + \text{O}_2 \xrightarrow{\Delta} \text{B}(\text{colourless gas})$
 $\text{B} \xrightarrow{\text{air}} \text{C}(\text{Brown colour gas})$
 $\text{B} + \text{C} \longrightarrow \text{D}$
 $\text{D} + \text{H}_2\text{O} \longrightarrow \text{E}$
 $\text{E} + \text{KI solution} \longrightarrow \text{Gas B formed.}$

A, B, C can be :-

- A, B, C can be NH_3 , N_2O , N_2O_4
 - E can be HNO_3
 - A, B, C can be NH_3 , NO , NO_2
 - E can be HNO_2
80. An inorganic salt,
- $+ \text{KOH} \longrightarrow \text{B gas} + \text{C(solution)}$
 $\text{B gas} + \text{CuSO}_4 \text{ solution} \longrightarrow \text{Black precipitate}$
 $\text{C (solution)} + \text{CuSO}_4 \longrightarrow \text{White precipitate in brown solution.}$
- Correct statements are :-
- A is NH_4Br
 - White precipitate is CuBr
 - A is PH_4I
 - White precipitate is CuI

EXERCISE # III

□ Linked Comprehension Type :

Passage for Q.1 to Q.3 :

Lead was known in ancient Egypt and Babylonia. It was largely used by Romans for water pipes. Native lead is occasionally found in traces but the chief are in galena PbS .

1. The sulphate mineral of lead is called :-

- Cerussite
- Crocoisite
- Lamarcite
- Anglesite

2. Lead is extracted from galena. Which of the following process is not involved for lead extraction.

- Froth floatation
- Calcination
- Roasting
- Smelting

Passage for Q.3 to Q.5 :

Orthoboric acid $\xrightarrow{100^\circ\text{C}}$ P_1

Orthophosphoric acid $\xrightarrow{220^\circ\text{C}}$ P_4

Orthoboric acid $\xrightarrow{140^\circ\text{C}}$ P_2

Orthophosphoric acid $\xrightarrow{320^\circ}$ P_5

Orthoboric acid $\xrightarrow[\text{heating}]{\text{very high temp.}}$ P_3

Orthophosphoric acid $\xrightarrow[\text{temperature heating}]{\text{high}}$ P_6

6.22 THE p-BLOCK ELEMENTS AND THEIR COMPOUNDS3. P_1 & P_4 are respectively :-

- (a) $H_2B_4O_7$, HPO_3 (b) HBO_2 , HPO_3
 (c) $H_2B_4O_7$, $H_4P_2O_7$ (d) HBO_2 , $H_4P_2O_7$

4. P_2 & P_5 are respectively :-

- (a) $H_2B_4O_7$, HPO_3 (b) HBO_2 , HPO_3
 (c) $H_2B_4O_7$, $H_4P_2O_7$ (d) HBO_2 , $H_4P_2O_7$

5. P_3 & P_6 are respectively :-

- (a) B_2O_5 , P_2O_5 (b) B_2O_3 , P_4O_8
 (c) B_2O_3 , P_2O_5 (d) B_2O_3 , P_2O_3

Passage for Q.6 to Q.7 :

A colourless solid A is formed on adding an ethereal solution of aluminium chloride to lithium hydride suspended in ether. A when reacts with BCl_3 & $SnCl_4$, products B and C are formed respectively. B is electron deficient but C is not. A is used as very strong reducing agent and can be used widely in organic chemistry.

6. A when reacts with $CH_3CH=O$ & CH_3CO_2H , the products are formed respectively :-

- (a) CH_3CH_2OH , $CH_3CH=O$
 (b) CH_3CH_2OH , CH_3CH_2OH
 (c) $CH_3CH=O$, CH_3CHO
 (d) CH_3CH_2OH , $CH_3CH_2CH_2OH$

7. A reacts vigorously with water to get :-

- (a) $LiOH$, $Al(OH)_3$, H_2O
 (b) $LiOH$, $Al(OH)_3$, H_2
 (c) $LiOH$, AlH_3
 (d) LiH , $Al(OH)_3$

8. $CH_3CH_2CN + A \longrightarrow P_1$ $CH_3CH_2NC + A \longrightarrow P_2$ P_1 & P_2 are related as :-

- (a) functional isomers
 (b) geometrical isomers
 (c) chain isomers
 (d) positional isomers

Passage for Q.9 to Q.11 :

A is colourless flammable gas which reduces CuO and when passed through a warm nickel tube forms a volatile liquid B. At high temperature and pressure, (a) reacts with aqueous $NaOH$ to form a salt C. A reacts in stage with Cl_2 to give finally a highly explosive liquid D.

9. A is :-

- (a) NO (b) CO
 (c) CO_2 (d) SO_2

10. B is :-

- (a) $Ni(CO)_4$ (b) $Fe(CO)_5$
 (c) $Ni_2(CO)_{10}$ (d) $Ni(NO)_4$

11. C is :-

- (a) $(CO_2Na)_2$ (b) CH_3CO_2Na
 (c) HCO_2Na (d) Na_2CO_3

Passage for Q.12 to Q.13 :

A colourless solid (a) dissolves in water. The aqueous solution gives a white precipitate (b) when $NaOH$ or NH_4OH is added. B dissolves in excess $NaOH$ but not in excess NH_4OH . $BaCl_2$ solution is added to A, a white precipitate is formed which is insoluble in dilute HCl .

Solid A, when heated with Na_2CO_3 in a charcoal cavity, followed by addition of a few drops of cobalt nitrate solution in charcoal cavity and further heating (by blow pipe flame) leaves a blue residue C.

12. A is :-

- (a) Al_2O_3 (b) $AlCl_3$
 (c) $Al(NO_3)_3$ (d) $Al_2(SO_4)_3$

13. C is :-

- (a) $CoAl_2O_3$ (b) $CoAl_2O_4$
 (c) $CoAl_2O_6$ (d) $CO_2(SO_4)_3$

Passage for Q.14 to Q.16 :

Finely powdered colemanite is treated with boiling sodium carbonate solution when insoluble X_1 is precipitated and a mixture of X_2 & X_3 remains in the solution. The insoluble X_1 is separated by filtration. To have further yield, CO_2 gas is passed through X_3 , where X_3 is converted into X_2 .

14. X_1 is :-

- (a) $CaCO_3$ (b) $NaBO_2$
 (c) $Na_2B_4O_7$ (d) Na_2CO_3

15. X_2 is :-

- (a) $CaCO_3$ (b) $NaBO_2$
 (c) $Na_2B_4O_7$ (d) Na_2CO_3

16. X_3 is :-

- (a) $CaCO_3$ (b) $NaBO_2$
 (c) $Na_2B_4O_7$ (d) Na_2CO_3

Passage for Q.17 to Q.19 :

A (organic compound) + O₂ → B + C + D.

A does not react with AgNO₃ solution. A mixture of 70% A and 30% ether is used as an anaesthetic. B is neutral oxide which turns anhydrous CuSO₄ blue. Compound C turns lime water milky and produces an acidic solution with water. D is pungent smelling gas :-

17. A and D are respectively :-

- (a) CH₂Cl₂, Cl₂ (b) CHCl₃, COCl₂
(c) CHCl₃, Cl₂ (d) CH₂Cl₂, COCl₂

18. B and C are respectively :-

- (a) N₂O and CO (b) N₂O and CO₂
(c) H₂O and CO (d) H₂O and CO₂

19. E $\xrightarrow[2.H^+]{1.Cl_2 + NaOH}$ HCO₂H + A. E can be :-

- (a) CH₃OH
(b) CH₃CH₂OH
(c) CH₃CHO
(d) both (b) and (c)

Passage for Q.20 to Q.22 :

A red liquid (a) when heated with concentrated hot NaOH gives a mixture of two salts (b) & (c) in the solution. The mixture when acidification with H₂SO₄ and distillation produces the red liquid (a) again. C has central atom with oxidation state +5.

20. Which is correct regarding liquid A ?

- (a) it is sparingly soluble in water but readily soluble in ether ; alcohol
(b) it acts as an oxidising reagent
(c) it is used to identify unsaturation present in a compound
(d) all

21. Which of the following statement is correct for salt B ?

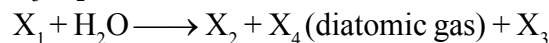
- (a) it solution in water gives yellow precipitate with silver nitrate solution
(b) it solution in water gives yellow precipitate with lead nitrate solution
(c) both (a) and (b)
(d) it is basic compound

22. Which of the following statements are correct regarding C :-

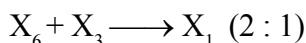
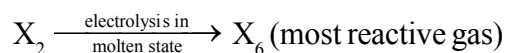
- (a) Central atom of anion of compound C has sp³ hybridisation
(b) Anion of compound C is tetrahedral in shape
(c) Both (a) and (b)
(d) None of these

Passage for Q.23 to Q.25 :

White crystalline solid X₁ react with H₂ to form a highly associated liquid X₂ & monoatomic, colourless gas X₃. X₂ is used for etching glass.



The ionisation energies of X₃ & X₄ are almost equal.



23. Correct statement regarding X₄.

- (a) X₄ oxidises acidified FeSO₄ to Fe₂(SO₄)₃
(b) X₄ oxidises acidified SnCl₂ to SnCl₄
(c) X₄ oxidises sulphurous acid to sulphuric acid
(d) All

24. The compound X₁ when reacts with sulphur, it forms a compound where hybridisation state of sulphur atom is :-

- (a) sp³ (b) sp³d
(c) sp³d² (d) sp³d³

25. Highly associated liquid X₂ when reacts with boric acid it gives :-

- (a) BF₃ (b) BCl₃
(c) BBr₃ (d) HBF₄

Passage for Q.26 to Q.28 :

A yellow solid (a) is unaffected by acids and base at room temperature. It is not soluble in water. It dissolves slowly in hot concentrated HNO₃ to give a brown gas (b). The solid (a) dissolves slowly in boiling solution of Na₂SO₃ giving a clear solution (c). Acidification of solution (c) causes a colourless gas (d). Upon concentration and cooling, solution (c) deposits as colourless crystals. Pentahydrate solution (c) is a reducing agents and reacts quantitatively with iodine solution (I₂ in KI) and acts as antichlor during bleaching:-

26. Solid A is :-

- (a) P₄ (b) S₈
(c) Pb₃O₄ (d) PbO

6.24 THE p-BLOCK ELEMENTS AND THEIR COMPOUNDS

27. Brown gas (b) & colourless gas (d) are respectively:-

- (a) NH_3 , PH_3 (b) NO_2 , PO_3
(c) NO , PbO_2 (d) NO_2 , SO_2

28. Solution C in the form of pentahydrate is :-

- (a) $\text{Na}_2\text{PbO}_2 \cdot 5\text{H}_2\text{O}$ (b) $\text{Na}_3\text{PO}_4 \cdot 5\text{H}_2\text{O}$
(c) $\text{Na}_2\text{S}_2\text{O}_3 \cdot 5\text{H}_2\text{O}$ (d) $\text{NaNH}_4\text{HPO}_4 \cdot 5\text{H}_2\text{O}$

Passage for Q.29 to Q.31 :

A compound of S, O, Cl (vapour density = 67.5) X reacts with KOH to give substance Y and Z. Y gives a white precipitate with AgNO_3 solution which is soluble in NH_4OH but insoluble in HNO_3 . Z give a white precipitate with BaCl_2 which is insoluble in HCl or HNO_3 . X is amorphous with a compound which is prepared by heating a mixture of NaCl, $\text{K}_2\text{Cr}_2\text{O}_7$ and conc. H_2SO_4

29. X is :-

- (a) SOCl_2 (b) SO_2Cl_2
(c) S_2Cl_2 (d) SO_3Cl

30. The product obtained by heating a mixture of NaCl, $\text{K}_2\text{Cr}_2\text{O}_7$ and concentrated H_2SO_4 is

- (a) CrCl_3 (b) CrO_2Cl_2
(c) CrOCl_2 (d) CrO_3Cl

31. Y & Z are respectively :-

- (a) KCl , K_2SO_3 (b) KCl , K_2SO_4
(c) ClO , KO_2 (d) KCl , K_2S

Passage for Q.32 to Q.34 :

Gas (X) is colourless diamagnetic with fishy smell. When X is cooled at -112°C , it condenses to a deep blue liquid which is dangerously explosive. X is one of the most powerful oxidising reagent. It is second to F_2 in oxidising power.

32. X is :-

- (a) H_2O_2 (b) PH_3
(c) O_3 (d) Cl_2

33. When mercury is shaken with X, the metal adheres to the glass as a mirror. It is because of formation :-

- (a) HgO (b) Hg_2O
(c) Hg_3P_2 (d) HgCl_2

34. Which of the following options are correct regarding X ?

- (a) X oxidises black PbS into white colour of PbSO_4
(b) X oxidises stannous chloride into stannic chloride
(c) X is absorbed in turpentine oil
(d) All

Passage for Q.35 to Q.37 :

Compound (Z) on reduction with LiAlH_4 gives a hydride (X) containing 21.72% hydrogen. The one mole of hydride (X) reacts with 2 moles of ammonia at high temperature gives a compound (Y). (Z) hydrolyses incompletely and forms a compound (W) and H_3BO_3 .

35. Correct statement regarding X :-

- (a) the central atom of X has trigonal planar geometry
(b) all H-atoms in X are in the same plane
(c) a three centre two-electron bond is observed in X
(d) all M-H bond lengths are identical in X

36. The hybridisation in central atom of compound W is

- (a) sp^2 (b) sp
(c) sp^3 (d) sp^3d

37. Correct statement regarding Z :-

- (a) it has trigonal planar geometry
(b) the bond lengths between the central atom and the substituent atoms is shorter than the sum of the covalent radii.
(c) the coordination geometry around central atom in compound Z and in 1 : 1 complex of Z and NH_3 is same
(d) all of these

Passage for Q.38 to Q.40 :

A is a gas with vapour density 8.5. On catalytic oxidation in presence of Pt/Rh at high pressure and temperature it gives a colourless gas (b) which rapidly turns brown in air forming gas C. B and C both are paramagnetic in nature :-

38. A, B, C are respectively :-

- (a) NH_3 , NO , NO_2 (b) NH_3 , N_2O_4 , N_2O_3
 (c) N_2H_4 , NO , NO_2 (d) NH_3 , NO_2 , N_2O

39. C is dissolved in water to give X which is concentrated by distillation upto 68% by mass. X is :-

- (a) HNO_2 and the above process to form X is called ostwald's process
 (b) NH_3 and the above process to form X is called Haber's process
 (c) HNO_2 and the above process to form X is called Birkeland - Eyde process
 (d) HNO_3 and the above process to form X is called Birkeland - Eyde process

40. Gas C on cooling ($< 0^\circ$) give :-

- (a) N_2O_3 (liquid) (b) N_2O_4 (liquid)
 (c) N_2O_4 (vap.) (d) N_2O_4 (solid)

Passage for Q.41 to Q.43 :

An unstable pale blue solution (P) which rapidly decomposes even in cold. The solution oxidises KI in acidic medium to liberate I_2 . The solution also oxidises SnCl_2 in dilute HCl . P is oxidised by bromine water to form monobasic acid Q.

41. P is :-

- (a) H_2CO_3 (b) HOCl
 (c) HNO_2 (d) HNO_3

42. P when reacts with hydrazoic acid, it gives :-

- (a) a mixture of N_2 and NO
 (b) a mixture of N_2 and N_2O
 (c) a mixture of N_2O and NO_2
 (d) a mixture of NO and NO_2

43. P reacts with which of the following compound to release N_2 gas :-

- (a) Urea (b) $\text{CH}_3\text{CH}_2\text{NH}_2$
 (c) Sulphamic acid (d) All

Passage for Q.44 to Q.46 :

A colourless solid A when heated gives off a colourless gas B which is supporter of combustion. If heated in container the whole of the solid disappears. When A is heated with NaOH , an alkaline gas C is evolved. When gas B is heated with sodamide, a colourless solid D is formed which further treated with dilute H_2SO_4 , a colourless liquid E is formed and silver salt of E are explosive.

44. A & B are :-

- (a) NH_4NO_2 , N_2 (b) NH_4NO_3 , N_2O
 (c) NH_4NO_2 , NO_2 (d) NaNO_3 , NO_2

45. C and E are respectively :-

- (a) NH_3 , HNO_2 (b) NH_3 , HNO_3
 (c) NH_3 , HN_3 (d) NH_3 , $\text{H}_2\text{N}_2\text{O}_2$

46. The anionic part of D is :-

- (a) Linear (b) trigonal planar
 (c) bent (d) tetrahedral

Passage for Q.47 to Q.49 :

A colourless solid X_1 melts at 44°C . When heated with aqueous KOH/NaOH , it gives off gaseous product X_2 , which is spontaneously inflammable in air at about 150°C . It is a powerful reducing agent.

47. X_2 when reacts with CuSO_4 then:-

- (a) Red deposit of Cu is formed
 (b) Black precipitate of Cu_3P_2 is formed
 (c) Blue precipitate of Cu_3P_2 is formed
 (d) S gets precipitated

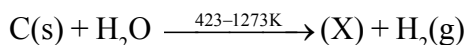
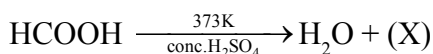
48. X_2 when reacts with AgNO_3 then :-

- (a) Brown precipitate of Ag_2O is formed
 (b) Yellow solution of Ag_3P is formed
 (c) Black precipitate of metallic Ag is formed
 (d) Black precipitate of Ag_2S is formed

49. X_1 and X_2 are respectively :-

- (a) S_8 and H_2S (b) White P_4 and P_4O_{10}
 (c) White P_4 and PH_3 (d) S_8 and SO_2

Passage for Q.50 to Q.53 :



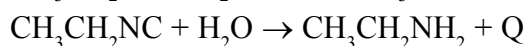
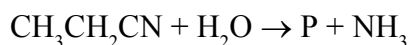
50. Select the correct statement about (X) :-

- (a) (X) is a colourless, odourless and almost water insoluble gas
 (b) (X) is highly poisonous and burns with blue flame
 (c) When (X) gas is passed through PdCl_2 solution giving rise to black ppt.
 (d) All of these

6.26 THE p-BLOCK ELEMENTS AND THEIR COMPOUNDS

51. Mixture of (X) gas + H_2 is called :-
 (a) Water gas or synthesis gas
 (b) Producer gas
 (c) Methane gas
 (d) None of these
52. In second reaction when air is used instead of steam a mixed of (X) gas and N_2 is produced which called :-
 (a) Water gas
 (b) Synthesis gas
 (c) Producer gas
 (d) Carbond dioxide gas
53. Select the correct statement about (X) :-
 (a) (X) gas is estimatated by I_2O_5
 (b) Cu_2Cl_2 is absorber of (X) gas
 (c) (X) gas is the purifying agent for Ni
 (d) All of these

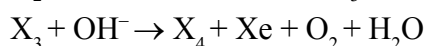
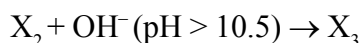
Passage for Q.54 to Q.55 :



54. P is :-
 (a) $CH_3CH_2CO_2H$
 (b) $CH_3CH_2CH_2OH$
 (c) $CH_3CH_2CH_3$
 (d) $CH_3CH_2CH_2NH_2$
55. Q is :-
 (a) CO_2
 (b) NH_3
 (c) CH_4
 (d) HCO_2H

Passage for Q.56 to Q.58 :

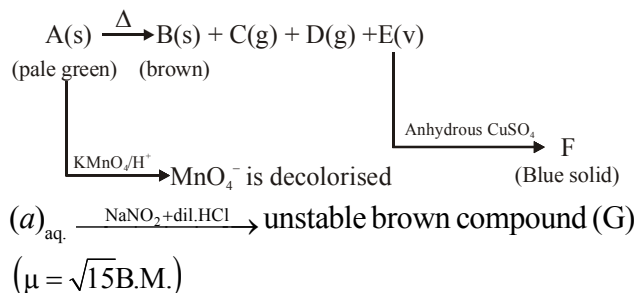
A Xenon fluoride X_1 is formed by heating Xenon with excess of fluorine at very high temperature and pressure. X_1 has a melting point 47.4° . X_1 can acts as fluoride ion donor as well as fluoride ion acceptor. X_1 undergoes hydrolysis vigorously forming an explosive compound X_2 .



56. X_1 is :-
 (a) XeF_2
 (b) XeF_4
 (c) XeF_6
 (d) None
57. X_2 is :-
 (a) XeO_3
 (b) XeO_4
 (c) XeO_6
 (d) XeO_2
58. X_3 and X_4 are respectively :-
 (a) $HXeO_4^-$ (perxenate ion), XeO_6^{4-} (xenate ion)
 (b) $HXeO_4^-$ (xenate ion), XeO_6^{4-} (perxenate ion)
 (c) XeO_6^{4-} (perxenate ion), $HXeO_4^-$ (xenate ion)
 (d) XeO_6^{4-} (xenate ion), $HXeO_4^-$ (perxenate ion)

Passage for Q.59 to Q.60 :

Consider the following reaction(s)



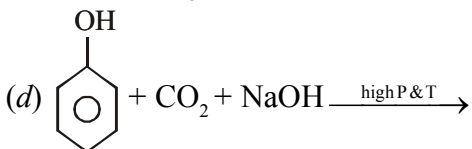
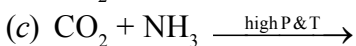
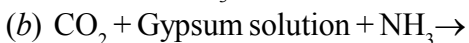
59. Sum of oxidation state of the central atom in C(g) and D(g) is :-
 (a) 6
 (b) 10
 (c) 8
 (d) 4
60. Select incorrect statement about compound F :-
 (a) It adsorb orange-red light for d-d transition
 (b) 4 water molecule are directly bonded with each Cu^{+2}
 (c) It is paramagnetic
 (d) It's $\mu = \sqrt{8} \text{ B.M.}$

Match the Column Type :

61. Match the column :-

Column-I

Reactions



Column-II

Products / Use of products

(P) Urea

(Q) Monocarboxylic acid

(R) Ammonium sulphate

(S) Product used as fertiliser

62. Match the column :

Column-I

(a) I.E. order

(b) Boiling point order

(c) Oxidising property order

(d) Acid dissociation constant

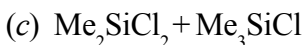
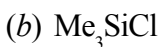
Column-II

(P) $\text{He} > \text{Ne} > \text{Ar} > \text{Kr} > \text{Xe}$ (Q) $\text{He} < \text{Ne} < \text{Ar} < \text{Kr} < \text{Xe}$ (R) $\text{F}_2 > \text{Cl}_2 > \text{Br}_2 > \text{I}_2$ (S) $\text{O} > \text{S} > \text{Se} > \text{Te}$ (T) $\text{H}_2\text{Te} > \text{H}_2\text{Se} > \text{H}_2\text{S} > \text{H}_2\text{O}$

63. Match the column :

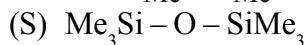
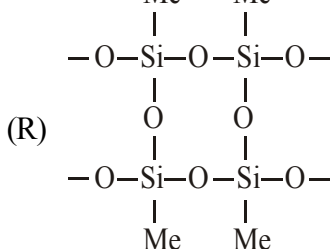
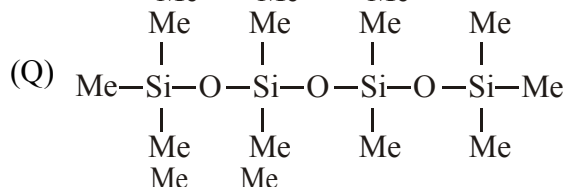
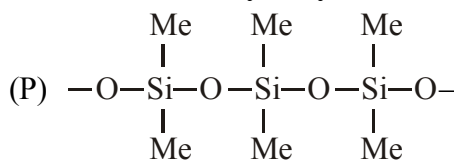
Column-I

Different types of alkyl chlorosilane



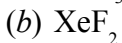
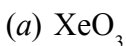
Column-II

Products on hydrolysis



64. Match the column :

Column-I



Column-II

(P) How lone pair of electron

(Q) Acts as oxidising reagent

(R) Undergoes hydrolysis with H_2O

(S) Undergoes addition reaction

(T) Give disproportionation reaction with H_2O or OH^-

6.28 THE p-BLOCK ELEMENTS AND THEIR COMPOUNDS

65. Match the column :

Column-I

Compound

- (a) XeF_2
- (b) $(\text{NH}_4)_2\text{SO}_4$
- (c) P_4
- (d) NO_2

Column-II

Features of product on treated with NaOH

- (P) O_2 is formed as product
- (Q) Disproportionation reaction
- (R) Product formed is inflammable and used in Holme's signal
- (S) Product solution reacts with Zn dust and NH_3 gas is liberated
- (T) Atleast one product is pyramidal in shape

66. Match the column :

Column-I

Acids

- (a) Hypophosphorus acid
- (b) Orthoboric acid
- (c) Hypophosphoric acid ($\text{H}_4\text{P}_2\text{O}_6$)
- (d) Sulphurous acid (H_2SO_3)

Column-II

Features

- (P) All hydrogens are ionisable in water
- (Q) Monobasic acid
- (R) sp^3 hybridised atoms present
- (S) Central atom(s) contain are lone pair of electrons

67. Match the column :

Column-I

Compounds

- (a) IF_5
- (b) ClF_3
- (c) ICl
- (d) I_2

Column-II

Features

- (P) show self ionisation
- (Q) used as potential ionising solvent
- (R) slightly conducting
- (S) hybridisation of halogen atom is sp^3d in ionised state
- (T) hybridisation of halogen atom is sp^3d^2 in ionised state

68. Match the column :

Column-I

Reaction

- (a) $\text{SiCl}_4 + \text{H}_2\text{O} \rightarrow$
- (b) $\text{Cl}_2 + \text{H}_2\text{O} \xrightarrow{h\nu}$
- (c) $\text{XeF}_4 + \text{H}_2\text{O} \rightarrow$
- (d) $\text{H}_2\text{O}_2 + \text{ClO}_2 + \text{OH}^- \rightarrow$

Column-II

Nature of reaction/Product

- (P) hydrogen halide formation
- (Q) O_2 gas formation
- (R) redox reaction
- (S) nonredox reaction
- (T) One of the product has tetrahedral shape

69. Match the column :-

Column-I

Reaction

- (a) $\text{Cl}_2\text{O}_6 + \text{H}_2\text{O} \rightarrow$
- (b) $\text{NaClO}_4 + \text{HCl}(\text{conc.}) \rightarrow$
- (c) $\text{KClO}_3 + (\text{CO}_2\text{H})_2 \rightarrow$
- (d) $\text{H}_2\text{O} + \text{Cl}_2 \xrightarrow{573\text{K}}$

Column-II

Product

- (P) ClO_2
- (Q) HClO_3
- (R) Cl_2O
- (S) HClO_4

70. Match the column :-

Column-I Compound	Column-II Shape
(a) ClF_3	(P) square pyramidal
(b) ICl_3	(Q) orange solid
(c) IF_5	(R) T shape
(d) IF_7	(S) Colourless gas
	(T) Colourless liquid

71. Match the column :-

Column-I Reaction	Column-II Feature
(a) $\text{XeF}_2 + \text{H}_2\text{O} \rightarrow$	(P) redox reaction
(b) $\text{XeF}_4 + \text{H}_2\text{O} \rightarrow$	(Q) disproportionation reaction
(c) $\text{XeF}_6 + \text{H}_2\text{O} \rightarrow$	(R) product formed is used for etching glass
(d) $\text{XeF}_3 + \text{NaOH} \rightarrow$	(S) Xe formation

72. Match the column :-

Column-I	Column-II
(a) $\text{XeF}_6 + \text{SiO}_2 \rightarrow$	(P) one of the product has square pyramidal shape
(b) $\text{P}_4(\text{white}) + \text{SO}_2\text{Cl}_2 \rightarrow$	(Q) one of the product has sp^3 hybridisation
(c) $\text{H}_2\text{SO}_4 + \text{P}_4\text{O}_{10} \rightarrow$	(R) one of the product has $\text{p}\pi\text{-d}\pi$ overlapping
(d) $\text{XeF}_6 + \text{H}_2\text{O} \rightarrow$	(S) one of the product has zero dipole moment

73. Match the column :-

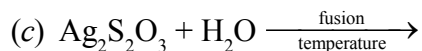
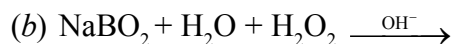
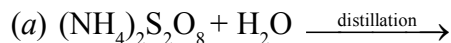
Column-I Reactions	Column-II Products formed
(a) $\text{Sn} + \text{conc. HNO}_3 \rightarrow$	(P) NO
(b) $\text{Sn} + \text{dil. HNO}_3 \rightarrow$	(Q) NO_2
(c) $\text{Ag} + \text{conc. HNO}_3 \rightarrow$	(R) NH_4NO_3
(d) $\text{Ag} + \text{dil. HNO}_3 \rightarrow$	(S) $\text{Sn}(\text{NO}_3)_2$
	(T) AgNO_3

74. Match the column :-

Column-I Compounds	Column-II Features
(a) XeF_2	(P) Can act as Lewis acid
(b) XeF_4	(Q) Can act as Lewis base
(c) XeF_6	(R) On reaction with $\text{H}_2\text{O}/\text{OH}^-$, disproportionation can take place
(d) XeO_3	(S) Has at least one lone pair of electron
	(T) At least one lone pair of electron on central atom

6.30 THE p-BLOCK ELEMENTS AND THEIR COMPOUNDS

75. Match the column :-

Column-I**Reactions****Column-II****Nature of reaction/product**

(P) Hydrolysis

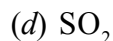
(Q) One of the products has peroxide linkage

(R) Disproportionation

(S) In one of the products the central atom has sp^3 hybridisation

(T) One of the products is used as brightner in washing powder

76. Match the column :-

Column-I**Compounds****Column-II****Features**

(P) Non-planar structure

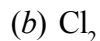
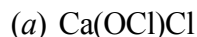
(Q) Planar structure

(R) Only reducing power

(S) Only oxidising power

(T) Has oxidising as well as reducing power

77. Match the column :-

Column-I**Compounds****Column-II****Features**

(P) Cl has negative oxidation state

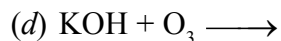
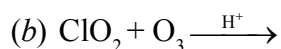
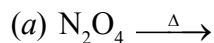
(Q) Cl has positive oxidation state

(R) oxidising reagent

(S) dehydrating reagent

(T) one of the product when Cl_2 gas is passed through basic solution

78. Match the column :-

Column-I**Reaction****Column-II****Features**

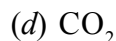
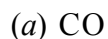
(P) One of the product is a mixed anhydride

(Q) One of the product is an acidic oxide

(R) The oxidation state of the central atom of one of the product is +6

(S) One of the product is colourless paramagnetic gas

79. Match the column :-

Column-I**Gas****Column-II****Absorber**

(P) Turpentine oil

(Q) Alkaline pyrogallol

(R) $\text{Cu}_2\text{Cl}_2 + \text{NaOH}$ (S) KOH

80. Match the column :-

Column-I**Allotropes**

- (a) Rhombic sulphur
 (b) Monoclinic sulphur
 (c) Black phosphorous
 (d) White phosphorous

Column-II**Features**

- (P) Consisting S_8 unit (8 member ring)
 (Q) Most stable allotrope of phosphorous
 (R) Unstable and ordinary temperature
 (S) Most stable allotrope of sulphur
 (T) Most reactive allotrope of phosphorous

81. Match the column :-

Column-I**Compounds**

- (a) $LiNO_3$
 (b) HNO_2
 (c) $NaNH_4HPO_4 \cdot 6H_2O$
 (d) H_3PO_3

Column-II**Products on heating**

- (P) NH_3 gas is evolved
 (Q) Oxyacid is formed as one of the product
 (R) Disproportionation reaction takes place
 (S) O_2 gas is evolved
 (T) Oxides of nitrogen is evolved

82. Match the column :-

Column-I**Ammonium salt**

- (a) NH_4IO_3
 (b) NH_4ClO_4
 (c) NH_4NO_2
 (d) NH_4NO_3

Column-II**Products on heating**

- (P) N_2 gas
 (Q) O_2 gas
 (R) N_2O gas
 (S) Cl_2 gas
 (T) I_2 gas

83. Match the column :-

Column-I**Compound**

- (a) NH_2CONH_2
 (b) NH_2NH_2
 (c) $SOCl_2$
 (d) H_2SO_4

Column-II**Feature/Use**

- (P) Chlorinating reagent
 (Q) Dehydrating reagent
 (R) Rocket fuel
 (S) Nitrogenous fertiliser

84. Match the column :-

Column-I**Reaction with ammonia**

- (a) $Cu^{2+} + NH_3(\text{excess}) \rightarrow$
 (b) $AgCl + NH_3(\text{excess}) \rightarrow$
 (c) $ZnSO_4 + NH_3(\text{less amount}) \rightarrow$
 (d) $FeCl_3 + NH_3(\text{less amount}) \rightarrow$

Column-II**Observation**

- (P) Deep blue colour solution
 (Q) Brown precipitate
 (R) Colourless solution
 (S) White precipitate
 (T) Complex formation

6.32 THE p-BLOCK ELEMENTS AND THEIR COMPOUNDS

85. Match the column :-

Column-I	Column-II
Group Number	Possible oxidation state
(a) Group 13	(P) +3
(b) Group 14	(Q) -3
(c) Group 15	(R) +1
(d) Group 17	(S) -1
	(T) +4

86. Match the column :-

$S_2O_3^{2-}$ + Different metal ion $\rightarrow$ Complex

Column-I	Column-II
Formula of the complex	Metal ion
(a) $[M(S_2O_3)_3]^{3-}$	(P) Cu^{2+}
(b) $[M(S_2O_3)_2]^{2-}$	(Q) Bi^{3+}
(c) $[M_6(S_2O_3)_5]^{4-}$	(R) Pb^{2+}
(d) $[M(S_2O_3)_2]^{3-}$	(S) Hg^{2+}
	(T) Ag^{+1}

87. Match the column :-

Column-I	Column-II
Compound	Features
(a) N_2O_3	(P) Colourless gas
(b) N_2O_4	(Q) Colourless solid / liquid
(c) N_2O_5	(R) Blue solid
(d) N_2O	(S) Acidic oxide
	(T) Neutral oxide

88. Match the column :-

Column-I	Column-II
Compound	Features
(a) P_4O_6	(P) $p\pi-d\pi$ back bonding
(b) P_4O_{10}	(Q) $p\pi-p\pi$ back bonding
(c) N_2O_4	(R) M-O bond has partial double bond character
(d) N_2O	(S) Even number of lone pair of electron
	(T) Laughing gas

89. Match the column :-

Column-I	Column-II
Order	Property
(a) $SbH_3 > NH_3 > AsH_3 > PH_3$	(P) Thermal stability
(b) $NH_3 > PH_3 > AsH_3 > SbH_3$	(Q) HEH angle
(c) $NH_3 > SbH_3 > AsH_3 > PH_3$	(R) Boiling point
(d) $SbH_3 > AsH_3 > PH_3 > NH_3$	(S) Melting point
	(T) Reducing power

90. Match the column :-

Column-I Compound	Column-II Bond parameters
(a) N_2O_5	(P) N–N length is shortest
(b) N_2O_4	(Q) N–N length is highest
(c) N_2O_3	(R) N–O–N bond length is present and N–O bond length is highest
(d) N_2O	(S) All N–O bond lengths are identical
	(T) All N–O bond lengths are not identical

EXERCISE # IV

□ Integer Type :

1. How many statements are correct regarding boron nitride :-

- (a) It has a graphite like layer structure with hexagonal ring.
 (b) B–N bond distance is $1.45\text{\AA}$ and comparable to C–C bond distance in graphite which $1.42\text{\AA}$.
 (c) Both B and N in boron nitride are sp^2 hybridised.
 (d) It is soft and lubricantes
 (e) It is a white infusible refractory solid.
 (f) It is formed by heating boron amide $B(NH_2)_3$. During this at first boron amide $B(NH_2)_3$ decomposes into boron imide $B_2(NH)_3$ and then converted into boron nitride.

2. In $B_3O_6^{3-}$ find out number of B–O–B linkage.

3. A comparative study of following properties between benzene and borazine are taken. For how many properties numerical data of benzene is more than borazene .

Melting point ; Boiling point ; Dipole moment ;
 Bond polarity ; Resonance energy ; Molecular weight

4. How many molecules result symmetrical cleavage of B_2H_6 molecule ?

CO ; $N(CH_3)_3$; $5(CH_3)_2$; $O(CH_3)_2$;  ;
 CH_3NH_2 ; $(CH_3)_2NH$

5. How many statement are correct ?

- (a) H_3PO_3 is diprotic acid
 (b) NF_3 is stable but NCl_3 and NI_3 are explosive
 (c) PCl_5 is covalent in solid stable
 (d) P–O bond length in $POCl_3$ is less than sum of the sigle bond covalent radii of P and O atom.
 (e) Metaphosphorus acid (HPO_2) is not known.
 (f) White phosphorous is more reactive than red phosphorous.

6. What is the value of n in alkyl substituted chlorisilane R_nSiCl_{4-n} , which is used for the preparation of cross linked silicone :-7. Number of $p\pi-d\pi$ bonding present in XeO_4 .

8. Find out the number of hydrogen atoms attached to phosphorus atom in hypophosphorus acid is :-

9. $XeF_4 + H_2O \rightarrow P(\text{reduced product}) + Q(\text{oxidised product}) + HF + O_2$

What in the ratio of oxidation states of Xe in reduced and the oxidised products.

10. Find out the number of following oxy acid of phosphorus, where P–O–P bond is present :-

H_3PO_4 , H_3PO_5 , H_2PO_3 , $H_3P_3O_9$, $H_4P_4O_{12}$

11. No. of S–S linkage in $S_3O_9 = m_1$ No. of P–P linkage in $(HPO_3)_3 = m_2$ No. of S–S linkage in $S_2O_8^{2-} = m_3$ No. of –O–O– linkage in $S_2O_6^{2-} = m_4$ Find out $m_1 + m_2 + m_3 + m_4$:-12. How many reagents can be used to dry NH_3 gas:-

Concentrated H_2SO_4 ; P_4O_{10} ; CaO ; Anhydrous $CaCl_2$

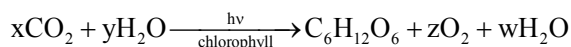
6.34 THE p-BLOCK ELEMENTS AND THEIR COMPOUNDS

13. How many statements are correct :-
- Argon is used in arc welding of metals or alloys to provide an inert atmosphere
 - XeF_2 , XeF_4 , XeF_6 are colourless crystalline solids and sublime readily at 298K.
 - XeF_2 , XeF_4 , XeF_6 all can undergo hydrolysis.
 - Se_2Cl_2 undergoes disproportionation to give SeCl_4 and Se.
 - Dioxide like MnO_2 , PbO_2 do not form H_2O_2 with dilute acids but they evolve O_2 with concentrated H_2SO_4
 - Xenon fluoride reacts with fluoride ion acceptor to form cationic species of Xe and fluoride ion donors to form fluoro anion of Xe.
14. How many statement are incorrect :-
- Barium azide on heating gives pure nitrogen gas
 - $\text{NH}_3 + \text{NaOCl}$ (Sodium hypochlorite) $\rightarrow \text{N}_2$ (gas)
 - Oxides of boron and silicon which are B_2O_3 and SiO_2 respectively, are basic in nature
 - Oxides of aluminium (Al_2O_3) and gallium (Ga_2O_3) are amphoteric in nature
 - Oxides of indium (In_2O_3) and thallium (Tl_2O_3) are acidic in nature
15. Find out the oxy acids of phosphorus in which phosphorus has lower oxidation state less than +5 contain either P–P or P–H bonds but not both in addition to P=O and P–OH bonds :-
 H_3PO_2 ; H_3PO_3 ; $\text{H}_4\text{I}_2\text{O}_5$; $\text{H}_4\text{P}_2\text{O}_6$; $\text{H}_4\text{P}_2\text{O}_7$; $(\text{HPO}_3)_3$
16. How many total number of p-block elements produce H_2 gas with sodium hydroxide :-
 Al, Sn, Pb, P, Zn
17. Among the following the total number of elements which undergo disproportionation reaction in NaOH :-
 S_8 , P_4 , Cl_2 , Br_2 , N_2
18. How many pair of compounds gives Cl_2 gas:-
- $\text{KCl} + \text{MnO}_2 \longrightarrow$
 - $\text{NaCl} + \text{conc. H}_2\text{SO}_4 \longrightarrow$
 - $\text{K}_2\text{Cr}_2\text{O}_7 + \text{HCl} \longrightarrow$
 - $\text{KMnO}_4 + \text{concentrated HCl} \longrightarrow$
 - $\text{Pb}_3\text{O}_4 + \text{HCl} \longrightarrow$
 - $\text{NaCl} + \text{conc. H}_3\text{PO}_4 \longrightarrow$
19. How many statements are correct ?
- XeF_5 is non existent
 - For drying H_2S , concentrated H_2SO_4 can be used up.
 - The inert bobble gas which is not found in atmosphere in reaction.
 - Metaphosphoric acid exists in polymeric cyclic structure.
 - SO_2 bleaches the colour of article by reduction.
 - Cl_2 bleaches the colour of article by oxidation.
 - H_3PO_3 and H_3PO_4 both are reducing in nature.
 - The first real compound of noble gas was made by Neil Bartlett.
 - When deep red PtF_6 vapour was mixed with an equal volume of Xe, the gases combined immediately at room temperature to produce a yellow solid.
20. How many products can be common product when XeF_2 reacts with H_2O and XeF_4 reacts with H_2O :-
21. Number of $\text{p}\pi\text{-d}\pi$ bonding present in SO_3 .
22. Total number of lone pairs of electrons present in HClO_3 .
23. Number of $\text{p}\pi\text{-p}\pi$ bonding present in SO_2 .
24. How many statements are correct regarding helium:-
- It is chemically inert & has very high thermal conductivity
 - It has extremely low boiling point & very low viscosity
 - It shows abnormal behaviour on liquification
 - It is less soluble in blood than nitrogen under high pressure
 - It can not form clathrate compound with quinol
 - It is a inflammable heavy gas therefore it is used in filling balloons
 - It is used to produce and sustain powerful super conducting magnets

- (h) It is used as a cryogenic agent for carrying out experiments at low temperature
- (i) Mixture of helium and oxygen are used as an artificial air for divers and other under pressure.
- (j) Because of its light weight and noninflammable character it is used in filling balloons.
25. Amongst the following, the total number of dimeric halides of group 16th elements which undergoes disproportionation is :-
 S_2F_2 , S_2Cl_2 , S_2Br_2 , Se_2Cl_2 , Se_2Br_2
- 26.
27. The number of ionisable hydrogen present in NaH_2PO_2 :-
28. In which of the following all bond lengths are identical :-
 CO_3^{2-} , PCl_5 , SO_5^{2-} , NO_2 , NO_3^- , BF_3 , O_3 , SO_2 , HCO_3^-
29. Atomicity of sulphur in rhombic sulphur is X and atomicity of phosphorus in white phosphorus is Y then find out $X - Y$:-
30. The oxidation state of S in S_8 , S_2F_2 and H_2S are x, y, z respectively. Find out $x + 2y + z$.
31. Alkyl substituted chlorosilane, $R_nSiCl_{(4-n)}$, on hydrolysis and then condensation polymerisation yields silicones. What is the value of n in the alkyl substituted chlorosilane used for the preparation of dimer silicone.
32. How many reactions give SO_2 gas as one of the product ?
 (a) $S + \text{concentrated } H_2SO_4 \rightarrow$
 (b) $Zn + \text{dilute } H_2SO_4 \rightarrow$
 (c) $Cu + \text{concentrated } H_2SO_4 \rightarrow$
 (d) $ZnS + O_2 \rightarrow$
 (e) Cerussite undergo heating
 (f) Carbon + conc. $H_2SO_4 \rightarrow$
33. 1 volume of concentrated HNO_3 and x volume of concentrated HCl mixture is called aqua regia. The value of x :-
34. How many reactions give nitric oxide gas one of the gaseous product ?
 (i) $KI + HNO_3 \longrightarrow$
 (ii) $H_2S + HNO_3 \longrightarrow$
 (iii) $NO_3^- + Al + NaOH \longrightarrow$
 (iv) $FeSO_4 + H_2SO_4(\text{conc.}) + HNO_3 \longrightarrow$
 (v) $Au + HCl + HNO_3 \longrightarrow$
35. How many reactions give N_2 gas on heating :-
 (a) $(NH_4)_2Cr_2O_7 \xrightarrow{\Delta}$
 (b) $Ba(N_3)_2 \xrightarrow{\Delta}$
 (c) $NaNO_2 + NH_4Cl \xrightarrow{\Delta}$
 (d) $NH_4NO_3 \xrightarrow{\Delta}$
 (e) HNO_3 vapours + Red hot copper $\longrightarrow$
36. No. of halide ions which change their oxidation number on heating with $MnO_2 + \text{conc. } H_2SO_4$?
 F^- ; Cl^- ; Br^- ; I^-
37. 1 mole each of H_3PO_2 , H_3PO_3 and H_3PO_4 will neutralise X mole of KOH , Y mole of $Mg(OH)_2$ and Z mole of $Cr(OH)_3$ (assuming all as strong electrolytes) respectively. Find out the value of $x+y+z$?
38. How many species does not contain peroxide ion?
 PbO_2 ; BaO_2 ; SrO_2 ; Na_2O_2 ; KO_2 ; K_2O_2
39. How many reactions involving ozone give O_2 in product side ?
 (i) $PbS(s) + O_3 \rightarrow$
 (ii) $I^-(aq.) + O_3 \rightarrow$
 (iii) $NO + O_3 \rightarrow$
 (iv) $SnCl_2 + HCl + O_3 \rightarrow$
40. How many reagents reacts with H_2O_2 but not with ozone ?
 $K_4[Fe(CN)_6]$; $K_3[Fe(CN)_6] + KOH$; PbS ; $SnCl_2 + HCl$; $KI + H^+$; $KMnO_4 + H^+$
41. How many sulphides are yellow in colour ?
 CdS ; Sb_2S_3 ; As_2S_3 ; SnS ; SnS_2
42. In crystal of polymeric boric acid, ring is formed through hydrogen bonding. What is the member of the ring.
43. In three dimensional structure of SiO_2 , rings are formed with alternate silicon and oxygen atoms. What is the member of the ring ?
44. Buckminster Fullerene (C_{60}) has x_1 vertices, x_2 six membered ring and x_3 five membered ring. Find out $\frac{x_1}{x_3} - \frac{x_1}{x_2}$.

6.36 THE p-BLOCK ELEMENTS AND THEIR COMPOUNDS

45. In photosynthesis, green plants convert atmospheric CO_2 into carbohydrate, such as glucose. The overall chemical change can be expressed as



Find out $x+z+w-y$.

46. Xe can exhibit x_1, x_2, x_3, x_4 oxidation state in its compounds.

$$x_1 < x_2 < x_3 < x_4$$

Find out $x_4 - x_3$ if $x_4 - x_3 = x_3 - x_2 = x_2 - x_1$

47. A gas G occurs in Earth's stratosphere. X is diamagnetic. It has bond angle 116.5° and bond length 0.128 nm. Dry KOH reacts with G to yield reddish brown precipitate. What is the molecular weight of G.

48. Total numbers of lone pair is present in $\text{H}_2\text{P}_2\text{O}_7$.
 49. Total numbers of P-OH bond present in $\text{H}_4\text{P}_2\text{O}_6$.
 50. Total numbers of P-OH bond present in $(\text{HPO}_3)_3$ if x, total numbers of P=O bond present in $(\text{HPO}_3)_3$ if y, total numbers of P-O-P bond present in $(\text{HPO}_3)_3$ if z then find out $x+y+z$?

EXERCISE # V(A) JEE-MAIN

- In case of nitrogen NCl_3 is possible but not NCl_5 while in case of phosphorus, PCl_3 as well as PCl_5 are possible. This is due to - [AIEEE - 2002]
 (a) Availability of vacant d-orbitals in P but not in N
 (b) Lower electronegativity of P than N
 (c) Lower tendency of H-bond formation in P than N
 (d) Occurrence of P in solid while N in gaseous state at room temperature
- Which products are expected from the disproportionation of hypochlorous acid
 -[AIEEE - 2002]
 (a) HClO_3 and Cl_2O (b) HClO_2 and HClO
 (c) HCl and Cl_2O (d) HCl and HClO_3
- Identify the incorrect statement among the following - [AIEEE - 2002]
 (a) ozone reacts with SO_2 to give SO_3
 (b) silicon reacts with NaOH(aq.) in the presence of air to give Na_2SiO_3 and H_2O

(c) Cl_2 reacts with excess of NH_3 to give N_2 and HCl

(d) Br_2 reacts with hot and strong NaOH solution to give NaBr , NaBrO_4 and H_2O

4. In XeF_2 , XeF_4 , XeF_6 the number of lone pairs on Xe is respectively -[AIEEE - 2002]

(a) 2,3,1 (b) 1,2,3

(c) 4,1,2 (d) 3,2,1

5. Aluminium is industrially prepared by - [AIEEE - 2002]

(a) Fused cryolite (b) Bauxite ore

(c) Alunite (d) Borax

6. For making good quality mirrors, plates of float glass are used. These are obtained by floating molten glass over a liquid metal which does not solidify before glass. The metal used can be -

[AIEEE - 2002]

(a) Sodium (b) Magnesium

(c) Mercury (d) Tin

7. What may be expected when phosphine gas is mixed with chlorine gas - [AIEEE - 2003]

(a) PCl_5 and HCl are formed and mixture cools down

(b) $\text{PH}_3 \cdot \text{Cl}_2$ is formed with warming up

(c) The mixture only cools down

(d) PCl_5 and HCl are formed and mixture warms up

8. Graphite is a soft solid lubricant extremely difficult to melt. The reason for this anomalous behaviour is that graphite - [AIEEE - 2003]

(a) has molecules of variable molecular masses like polymers

(b) has carbon atoms arranged in large plates of rings of strongly bonded carbon atoms with weak interplate bonds

(c) is a non crystalline substance

(d) is an allotropic form of diamond

9. Concentrated hydrochloric acid when kept in open air sometimes produces a cloud of white fumes. This is due to - [AIEEE - 2003]

- (a) strong affinity of HCl gas for moisture in air results in forming of droplets of liquid solution which appears like a cloudy smoke
(b) due to strong affinity for water, conc. HCl pulls moisture of air towards self. The moisture forms droplets of water and hence the cloud
(c) conc. HCl emits strongly smelling HCl gas all the time
(d) oxygen in air reacts with emitted HCl gas to form a cloud of Cl_2 gas
10. Aluminium chloride exists as dimer, Al_2Cl_6 in solid state as well as in solution of non-polar solvents such as benzene. When dissolved in water, it gives -[AIEEE - 2004]
- (a) $\text{Al}^{3+} + 3\text{Cl}^-$
(b) $[\text{Al}(\text{H}_2\text{O})_6]^{3+} + 3\text{Cl}^-$
(c) $[\text{Al}(\text{OH})_6]^{3-} + 3\text{HCl}$
(d) $\text{Al}_2\text{O}_3 + 6\text{HCl}$
11. The soldiers of Napoleon army while at Alps during freezing winter suffered a serious problem as regards to the tin buttons of their uniforms. White metallic tin buttons get converted to grey powder. This transformation is related to -[AIEEE - 2002]
- (a) An interaction with water vapour contained in humid air
(b) A change in crystalline structure of tin
(c) A change in the partial pressure of O_2 in air
(d) An interaction with N_2 of air at low temperature
12. Which one of the following statements regarding helium is incorrect -[AIEEE - 2004]
- (a) It is used to produce and sustain powerful superconducting magnets
(b) It is used as a cryogenic agent for carrying out experiments at low temperatures
(c) It is used to fill gas balloons instead of hydrogen because it is lighter than hydrogen and non-inflammable
(d) It is used in gas-cooled nuclear reactors
13. The number of hydrogen atoms attached to phosphorus atom in hypophosphorous acid is - [AIEEE - 2004]
- (a) Zero (b) Two
(c) One (d) Three
14. The correct order of the thermal stability of hydrogen halide (H-X) is - [AIEEE - 2005]
- (a) $\text{HI} > \text{HBr} > \text{HCl} > \text{HF}$
(b) $\text{HF} > \text{HCl} > \text{HBr} > \text{HI}$
(c) $\text{HCl} > \text{HF} > \text{HBr} > \text{HI}$
(d) $\text{HI} > \text{HCl} > \text{HF} > \text{HBr}$
15. The structure of diborane contains -[AIEEE - 2005]
- (a) Four 2c-2e bonds and four 3c-2e bonds
(b) Two 2c-2e bonds and two 3c-2e bonds
(c) Two 2c-2e bonds and four 3c-2e bonds
(d) Four 2c-2e bonds and two 3c-2e bonds
16. Heating an aqueous solution of aluminium chloride to dryness will give - [AIEEE - 2005]
- (a) AlCl_3 (b) Al_2Cl_6
(c) Al_2O_3 (d) $\text{Al}(\text{OH})\text{Cl}_2$
17. Which one of the following is the correct statement -[AIEEE - 2005]
- (a) Boric acid is a protonic acid
(b) Beryllium exhibits coordination number of six
(c) Chlorides of both beryllium and aluminium have bridged chloride structures in solid phase
(d) $\text{B}_2\text{H}_6, 2\text{NH}_3$ is known as "inorganic benzene"
18. In silicon dioxide - [AIEEE - 2005]
- (a) each silicon atom is surrounded by four oxygen atoms and each oxygen atom is bonded to two silicon atoms
(b) each silicon atom is surrounded by two oxygen atoms and each oxygen atom is bonded to two silicon atoms
(c) silicon atom is bonded to two oxygen atoms
(d) there are double bonds between silicon and oxygen atoms
19. Regular use of which of the following fertilizer increases the acidity of soil -[AIEEE - 2007]
- (a) Potassium nitrate
(b) Urea
(c) Superphosphate of lime
(d) Ammonium sulphate

6.38 THE p-BLOCK ELEMENTS AND THEIR COMPOUNDS

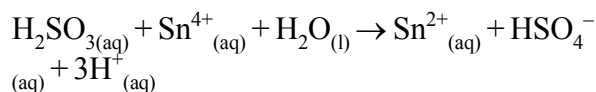
20. The stability of dihalides of Si, Ge, Sn and Pb increases steadily in the sequence
-[AIEEE - 2007]
(a) $\text{GeX}_2 \ll \text{SiX}_2 \ll \text{SnX}_2 \ll \text{PbX}_2$
(b) $\text{SiX}_2 \ll \text{GeX}_2 \ll \text{PbX}_2 \ll \text{SnX}_2$
(c) $\text{SiX}_2 \ll \text{GeX}_2 \ll \text{SnX}_2 \ll \text{PbX}_2$
(d) $\text{PbX}_2 \ll \text{SnX}_2 \ll \text{SiX}_2 \ll \text{GeX}_2$
21. Among the following substituted silanes the one which will give rise to cross linked silicone polymer on hydrolysis is - [AIEEE - 2008]
(a) R_4Si (b) RSiCl_3
(c) R_2SiCl_2 (d) R_2SiCl
22. Which one of the following reactions of Xenon compounds is not feasible - [AIEEE - 2009]
(a) $2\text{XeF}_2 + 2\text{H}_2\text{O} \rightarrow 2\text{Xe} + 4\text{HF} + \text{O}_2$
(b) $\text{XeF}_6 + \text{RbF} \rightarrow \text{Rb}[\text{XeF}_7]$
(c) $\text{XeO}_3 + 6\text{HF} \rightarrow \text{XeF}_6 + 3\text{H}_2\text{O}$
(d) $3\text{XeF}_4 + 6\text{H}_2\text{O} \rightarrow 2\text{Xe} + \text{XeO}_3 + 12\text{HF} + 1.5\text{O}_2$
23. Which of the following statements is wrong ? [AIEEE - 2011]
(a) Single N-N bond is weaker than the single P-P bond
(b) N_2O_4 has two resonance structures
(c) The stability of hydrides increases from NH_3 to BiH_3 in group 15 of the periodic table
(d) Nitrogen cannot form $d\pi-p\pi$ bond
24. Which of the following statements regarding sulphur is incorrect - [AIEEE - 2011]
(a) At 600°C the gas mainly consists of S_2 molecules
(b) The oxidation state of sulphur is never less than +4 in its compounds
(c) S_2 molecule is paramagnetic
(d) The vapour at 200°C consists mostly of S_8 rings
25. Boron cannot form which one of the following anions ? [AIEEE - 2011]
(a) $\text{B}(\text{OH})_4^-$ (b) BO_2^-
(c) BF_6^{3-} (d) BH_4^-
26. In view of the signs of $\Delta_r G^\circ$ for the following reactions -
 $\text{PbO}_2 + \text{Pb} \rightarrow 2\text{PbO}$, $\Delta_r G^\circ < 0$
 $\text{SnO}_2 + \text{Sn} \rightarrow 2\text{SnO}$, $\Delta_r G^\circ > 0$
Which oxidation states are more characteristic for lead and tin ? [AIEEE - 2011]
(a) For lead + 4, for tin + 2
(b) For lead + 2, for tin + 2
(c) For lead + 4, for tin + 4
(d) For lead + 2, for tin + 4
27. The number of S-S bonds in SO_3 , $\text{S}_2\text{O}_3^{2-}$, $\text{S}_2\text{O}_6^{2-}$ and $\text{S}_2\text{O}_8^{2-}$ respectively are - [Jee Main-Online- 2012]
(a) 1, 0, 1, 0 (b) 0, 1, 1, 0
(c) 1, 0, 0, 1 (d) 0, 1, 0, 1
28. Which one of the following depletes ozone layer - [Jee Main-Online- 2012]
(a) NO and freons (b) SO_2
(c) CO (d) CO_2
29. In which of the following arrangements, the sequence is not strictly according to the property written against it - [Jee Main-Online- 2012]
(a) $\text{CO}_2 < \text{SiO}_2 > \text{SnO}_2 > \text{PbO}_2$: increasing oxidising power
(b) $\text{B} < \text{C} > \text{O} > \text{N}$: increasing first ionisation enthalpy
(c) $\text{NH}_3 < \text{PH}_3 > \text{AsH}_3 > \text{SbH}_3$: increasing basic strength
(d) $\text{HF} < \text{HCl} > \text{HBr} > \text{HI}$: increasing acid strength
30. The compound of Xenon with zero dipole moment is - [Jee Main-Online- 2012]
(a) XeO_3 (b) XeO_2
(c) XeF_4 (d) XeOF_4
31. The formation of molecular complex BF_3-NH_3 results in a change in hybridisation of boron - [Jee Main-Online- 2012]
(a) from sp^3 to sp^3d (b) from sp^2 to dsp^2
(c) from sp^3 to sp^2 (d) from sp^2 to sp^3

32. Trigonal bipyramidal geometry is shown by
 -[Jee Main-Online- 2013]
 (a) XeO_3F_2 (b) XeOF_2
 (c) $[\text{XeF}_8]^{2-}$ (d) FXeOSO_2F
33. Oxidation state of sulphur in anions SO_3^{2-} , $\text{S}_2\text{O}_4^{2-}$ and $\text{S}_2\text{O}_6^{2-}$ increases in the orders
 -[Jee Main-Online- 2013]
 (a) $\text{S}_2\text{O}_6^{2-} < \text{S}_2\text{O}_4^{2-} < \text{SO}_3^{2-}$
 (b) $\text{SO}_3^{2-} < \text{S}_2\text{O}_4^{2-} < \text{S}_2\text{O}_6^{2-}$
 (c) $\text{S}_2\text{O}_4^{2-} < \text{SO}_3^{2-} < \text{S}_2\text{O}_6^{2-}$
 (d) $\text{S}_2\text{O}_4^{2-} < \text{S}_2\text{O}_6^{2-} < \text{SO}_3^{2-}$
34. XeO_4 molecule is tetrahedral having
 -[Jee Main-Online- 2013]
 (a) Two $\text{p}\pi\text{-d}\pi$ bonds
 (b) Four $\text{p}\pi\text{-d}\pi$ bonds
 (c) One $\text{p}\pi\text{-d}\pi$ bond
 (d) Three $\text{p}\pi\text{-d}\pi$ bonds
35. The shape of IF_6^- is
 -[Jee Main-Online- 2013]
 (a) Trigonally distorted octahedron
 (b) Pyramidal
 (c) Octahedral
 (d) Square antiprism
36. The catenation tendency of C, Si and Ge is in the order $\text{Ge} < \text{Si} < \text{C}$. The bond energies (in kJ mol^{-1}) of C–C, Si–Si and Ge–Ge bonds are respectively
 -[Jee Main-Online- 2013]
 (a) 348, 260, 297 (b) 348, 297, 260
 (c) 297, 348, 260 (d) 260, 297, 348
37. The gas evolved on heating CaF_2 and SiO_2 with concentrated H_2SO_4 , on hydrolysis gives a white gelatinous precipitate. The precipitate is
 -[Jee Main-Online- 2014]
 (a) Silica gel (b) Silicic acid
 (c) hydrofluosilicic acid (d) calciumfluorosilicate
38. Shape of certain interhalogen compounds are stated below. Which one of them is not correctly stated -
 [Jee Main-Online- 2014]
 (a) IF_7 : Pentagonal bipyramid
 (b) BrF_5 : Trigonal bipyramid
 (c) ICl_3 : Planar dimeric
 (d) BrF_3 : Planar T-shaped
39. Which of the following series correctly represents relations between the elements from X to Y?
 X $\rightarrow$ Y [Jee Main-Online- 2014]
 (a) $_{18}\text{Ar} \rightarrow _{54}\text{Xe}$ Noble character increases
 (b) $_3\text{Li} \rightarrow _{19}\text{K}$ Ionisation enthalpy increases
 (c) $_6\text{C} \rightarrow _{32}\text{Ge}$ Atomic radii increases
 (d) $_9\text{F} \rightarrow _{35}\text{Br}$ Electron gain enthalpy with negative sign increases
40. Which of the following statements about the depletion of ozone layer is correct
 -[Jee Main-Online- 2014]
 (a) the problem of ozone depletion is more serious at poles because ice crystals in the clouds over poles act as catalyst for photochemical reactions involving the decomposition of ozone by $\text{Cl}\cdot$ and $\text{ClO}\cdot$ radicals
 (b) the problem of ozone depletion is less serious at poles because NO_2 solidifies and is not available for consuming $\text{ClO}\cdot$ radicals
 (c) Oxides of nitrogen also do not react with ozone in stratosphere
 (d) Freons, chlorofluorocarbons, are inert chemically, they do not react with ozone in stratosphere
41. Which of the following xenon-OXO compounds may not be obtained by hydrolysis of xenon fluorides
 -[Jee Main-Online- 2014]
 (a) XeO_2F_2 (b) XeO_3
 (c) XeO_4 (d) XeOF_4
42. Hydrogen peroxide acts both as an oxidising and as a reducing agent depending upon the nature of the reacting species. In which of the following cases H_2O_2 acts as a reducing agent in acid medium
 -[Jee Main-Online- 2014]
 (a) MnO_4^- (b) SO_3^{2-}
 (c) KI (d) $\text{Cr}_2\text{O}_7^{2-}$

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43. Consider the reaction

-[Jee Main-Online- 2014]



Which of the following statement is correct ?

- (a) H_2SO_3 is the reducing agent because it undergoes oxidation
- (b) H_2SO_3 is the reducing agent because it undergoes reduction
- (c) Sn^{4+} is the reducing agent because it undergoes oxidation
- (d) Sn^{4+} is the oxidizing agent because it undergoes oxidation

44. The number and type of bonds in C_2^{2-} ion in CaC_2 are

-[Jee Main-Online- 2014]

- (a) two σ bonds and one π bond
- (b) two σ bonds and two π bonds
- (c) one σ bonds and two π bonds
- (d) one σ bond and one π bond

45. In the following sets of reactants which two sets best exhibit the amphoteric character of $\text{Al}_2\text{O}_3 \cdot x\text{H}_2\text{O}$?

[Jee Main-Online- 2014]

Set-1 : $\text{Al}_2\text{O}_3 \cdot x\text{H}_2\text{O}(\text{s})$ and $\text{OH}^{-}(\text{aq})$

Set-2 : $\text{Al}_2\text{O}_3 \cdot x\text{H}_2\text{O}(\text{s})$ and $\text{H}_2\text{O}(\ell)$

Set-3 : $\text{Al}_2\text{O}_3 \cdot x\text{H}_2\text{O}(\text{s})$ and $\text{H}^{+}(\text{aq})$

Set-4 : $\text{Al}_2\text{O}_3 \cdot x\text{H}_2\text{O}(\text{s})$ and $\text{NH}_3(\text{aq})$

- (a) 1 and 2
- (b) 2 and 4
- (c) 1 and 3
- (d) 3 and 4

46. Example of a three-dimensional silicate is

-[Jee Main-Online- 2014]

- (a) Beryls
- (b) Zeolites
- (c) Feldspars
- (d) Ultramarines

47. Which of the following compounds has a P-P bond ?

[Jee Main-Online- 2015]

- (a) $\text{H}_4\text{P}_2\text{O}_5$
- (b) $(\text{HPO}_3)_3$
- (c) $\text{H}_4\text{P}_2\text{O}_7$
- (d) $\text{H}_4\text{P}_2\text{O}_6$

48. Chlorine water on standing loses its colour and forms

-[Jee Main-Online- 2015]

- (a) HCl and HClO_2
- (b) HCl only
- (c) HOCl and HCl_2
- (d) HCl and HOCl

49. Which among the following is the most reactive ?

[Jee Main - 2015]

- (a) I_2
- (b) ICl
- (c) Cl_2
- (d) Br_2

50. Which one has the highest boiling point ?

[Jee Main - 2015]

- (a) Kr
- (b) Xe
- (c) He
- (d) Ne

51. From the following statements regarding H_2O_2 , choose the incorrect statement :

-[Jee Main-2015]

- (a) it has to be stored in plastic or wax lined glass bottles in dark
- (b) it has to be kept away from dust
- (c) it can act only as an oxidizing agent
- (d) it decomposes on exposure to light

EXERCISE # V(B) JEE-ADVANCED

Fill in the Blanks

- The hydrolysis of alkyl substituted chlorosilanes given [IIT-1991]
- The hydrolysis of trialkylchlorosilane R_3SiCl , yields [IIT-1994]
- Two types of bonds present in B_2H_6 are covalent and [IIT-1994]
- One recently discovered allotrope of carbon (e.g., C_{60}) is commonly known as [IIT-1994]

True/False

- Carbon tetrachloride burns in air when lighted to give phosgene. [IIT-1983]
- Graphite is a better lubricant on the moon than on the earth. [IIT-1987]
- All the Al – Cl bonds in Al_2Cl_6 are equivalent. [IIT-1989]
- Diamond is harder than graphite. [IIT-1993]
- The basic nature of the hydroxides of group 13 (Gr. IIIB) decreases progressively down the group. [IIT-1993]
- The tendency for catenation is much higher for C than for Si. [IIT-1993]

11. Complete and balance the following chemical equations - [IIT-1998, 2M]
 (i) $P_4O_{10} + PCl_5 \rightarrow$
 (ii) $SnCl_4 + C_2H_5Cl + Na \rightarrow$
12. Work out the following using chemical equations [IIT-1998, 2M]
 "Chlorination of calcium hydroxide produces bleaching powder"
13. Hydrogen peroxide acts both as an oxidizing and a reducing agent in alkaline solution towards certain first row transition metal ion. Illustrate both these properties of H_2O_2 using chemical equations -[IIT-1998, 4M]
14. In the contact process for industrial manufacture of sulphuric acid, some amount of sulphuric acid is used as a starting material. Explain briefly. What is the catalyst used in the oxidation of SO_2 ? [IIT-1998, 4M]
15. Give reasons in one or two sentences for each of the following : [IIT-1985]
 (i) Graphite is used as a solid lubricant
 (ii) Fluorine cannot be prepared from fluorides by chemical oxidation.
16. Write balanced equations for : [IIT-1990]
 (i) The preparation of crystalline silicon from $SiCl_4$
 (ii) The preparation of phosphine from CaO and white phosphorus
 (iii) The preparation of ammonium sulphate from gypsum, ammonia and carbon dioxide
17. Anhydrous $AlCl_3$ is covalent. From the data given below, predict whether it would remain covalent or become ionic in aqueous solution. [IIT-1997]
 Ionisation energy for $Al = 5137 \text{ kJ mol}^{-1}$
 $\Delta H_{\text{hydration}}$ for $Al^{3+} = 5137 \text{ kJ mol}^{-1}$
 $\Delta H_{\text{hydration}}$ for $Cl^- = -381 \text{ kJ mol}^{-1}$
18. Aluminium sulphide gives a foul odour when it becomes damp. Write a balanced chemical equation for the reaction :- [IIT-1997]
19. Draw the structure of a cyclic silicate, $(Si_3O_9)^{6-}$ with proper labelling- [IIT-1998]
20. Give reason for the following in one or two sentences only. [IIT-1999]
 "BeCl₂ can be easily."
21. Draw the molecular structures of XeF_2 , XeF_4 and XeO_2F_2 , indicating the location of lone pair(s) of electrons. [IIT-2000]
22. Give reason : [IIT-2000]
 Why elemental nitrogen exists as a diatomic molecule whereas elemental phosphorus is a tetra atomic molecule.
23. Give an example of oxidation of one halide by another halogen. Explain the feasibility of the reaction. [IIT-2000]
24. Compound X on reduction with $LiAlH_4$ gives a hydride Y containing 21.72 % hydrogen along with other products. The compound Y reacts with air explosively resulting in boron trioxide. Identify X and Y. Give balanced reactions involved in the formation of Y and its reaction with air draw the structure of Y. [IIT-2001]
25. Starting from $SiCl_4$, prepare the following in steps not exceeding the number given in parenthesis (reactions only) [IIT-2001]
 (i) Silicon (1)
 (ii) Linear silicon containing methyl group only (4)
 (iii) Na_2SiO_3 (3)
26. Write the balanced chemical equation for developing photographic films. [IIT-2001]
27. Identify (X) in the following synthetic scheme and write their structures. [IIT-2001]
 $BaCO_3 + H_2SO_4 \xrightarrow{*} X(\text{gas})$ (C denotes C^{14})
28. Write the balanced equations for the reactions of the following compounds with water [IIT-2002]
 (i) Al_4C_3 (ii) $CaNCN$
 (iii) BF_3 (iv) NCl_3
 (v) XeF_3
29. Write the balanced equations for the reactions of the following compounds with water [IIT-2002]
 (i) Al_4C_3 (ii) $CaNCN$
 (iii) BF_3 (iv) NCl_3
 (v) XeF_4

6.42 THE p-BLOCK ELEMENTS AND THEIR COMPOUNDS

30. Identify the following [IIT-2003]

$$\text{NaCO}_3 \xrightarrow[\text{aq.}]{\text{SO}_2} \text{A} \xrightarrow{\text{Na}_2\text{CO}_3} \text{B} \xrightarrow[\Delta]{\text{elemental S}} \text{C} \xrightarrow{\text{I}_2} \text{D}$$
 Also mention the oxidation state of S in all the compounds.
31. Arrange the following oxide in the increasing order of Bronsted basicity. [IIT-2004]
 Cl_2O_7 , BaO_7 , SO_3 , CO_2 , B_2O_3
32. When zeolite, which is hydrated sodium aluminium, is treated with hard water, the sodium ions are exchanged with: [IIT-1990]
 (a) H^+ ions (b) Ca^{2+} ions
 (c) SO_4^{2-} ions (d) Mg^{2+} ions
33. Which of the following halides is least stable and has doubtful existence: [IIT-1996]
 (a) CCl_4 (b) GeI_4
 (c) SnI_4 (d) PbI_4
34. In compounds of type ECl_3 , where E = B, P, As or Bi, the angles Cl–E–Cl for different E are in the order: [IIT-1999]
 (a) $\text{B} > \text{P} = \text{As} = \text{Bi}$ (b) $\text{B} > \text{P} > \text{As} > \text{Bi}$
 (c) $\text{B} < \text{P} = \text{As} = \text{Bi}$ (d) $\text{B} < \text{P} < \text{As} < \text{Bi}$
35. The number of P—O—P bonds in cyclic tetrametaphosphoric acid is: [IIT-2000]
 (a) zero (b) two
 (c) three (d) four
36. The correct order of acidic strength is: [IIT-2000]
 (a) $\text{Cl}_2\text{O}_7 > \text{SO}_2 > \text{P}_4\text{O}_{10}$
 (b) $\text{CO}_2 > \text{N}_2\text{O}_5 > \text{SO}_3$
 (c) $\text{Na}_2\text{O} > \text{MgO} > \text{Al}_2\text{O}_3$
 (d) $\text{K}_2\text{O} > \text{CaO} > \text{MgO}$
37. Amongst H_2O , H_2S , H_2Se and H_2Te , the one with the highest boiling point is: [IIT-2000]
 (a) H_2O because of hydrogen bonding
 (b) H_2Te because of higher molecular weight
 (c) H_2S because of hydrogen bonding
 (d) H_2Te because of lower molecular weight
38. Ammonia can be dried by: [IIT-2000]
 (a) conc. H_2SO_4 (b) P_4O_{10}
 (c) CaO (d) Anhydrous CaCl_2
39. Which of the following are hydrolysed: [REE-2000]
 (a) NCl_3 (b) BCl_3
 (c) CCl_4 (d) SiCl_4
40. The set with correct order of acidity is: [IIT-2001]
 (a) $\text{HClO} < \text{HClO}_2 < \text{HClO}_3 < \text{HClO}_4$
 (b) $\text{HClO}_4 < \text{HClO}_3 < \text{HClO}_2 < \text{HClO}$
 (c) $\text{HClO} < \text{HClO}_4 < \text{HClO}_3 < \text{HClO}_2$
 (d) $\text{HClO}_4 < \text{HClO}_2 < \text{HClO}_3 < \text{HClO}$
41. The reaction,

$$3\text{ClO}^-(\text{aq}) \rightarrow \text{ClO}_3^-(\text{aq}) + 2\text{Cl}^-(\text{aq})$$
 is an example of: [IIT-2001]
 (a) oxidation reaction
 (b) reduction reaction
 (c) disproportionation reaction
 (d) decomposition reaction
42. The number of S–S bonds in sulphur trioxide trimer, (S_3O_9) is: [IIT-2001]
 (a) three (b) two
 (c) one (d) zero
43. Specify the coordination geometry around and hybridisation of N and B atoms in a 1:1 complex of BF_3 and NH_3 : [IIT-2001]
 (a) N: tetrahedral sp^3 ; B: tetrahedral sp^3
 (b) N: pyramidal sp^3 ; B: pyramidal sp^3
 (c) N: pyramidal sp^3 ; B: planar sp^3
 (d) N: pyramidal sp^3 ; B: tetrahedral sp^3
44. **Statement–1**: Between SiCl_4 and CCl_4 , only SiCl_4 reacts with water. [IIT-2001]
Statement–2: SiCl_4 is ionic and CCl_4 is covalent.
 (a) Statement–1 is True, Statement–2 is True; Statement–2 is a correct explanation for Statement–1
 (b) Statement–1 is True, Statement–2 is True; Statement–2 is NOT a correct explanation for Statement–1
 (c) Statement–1 is True, Statement–2 is False
 (d) Statement–1 is False, Statement–2 is True

45. Polyphosphate are used as water softening agents because they : [IIT-2002]
 (a) form soluble complexes with anionic species
 (b) precipitate anionic species
 (c) form soluble complexes with cationic species
 (d) precipitate cationic species
46. Identify the correct order of solubility of Na_2S , CuS and ZnS in aqueous medium : [IIT-2002]
 (a) $\text{CuS} > \text{ZnS} > \text{Na}_2\text{S}$
 (b) $\text{ZnS} > \text{Na}_2\text{S} > \text{CuS}$
 (c) $\text{Na}_2\text{S} > \text{CuS} > \text{ZnS}$
 (d) $\text{Na}_2\text{S} > \text{ZnS} > \text{CuS}$
47. Identify, the correct order of acidic strength of CO_2 , CuO , CaO , H_2O : [IIT-2002]
 (a) $\text{CaO} < \text{CuO} < \text{H}_2\text{O} < \text{CO}_2$
 (b) $\text{H}_2\text{O} < \text{CuO} < \text{CaO} < \text{CO}_2$
 (c) $\text{CaO} < \text{H}_2\text{O} < \text{CuO} < \text{CO}_2$
 (d) $\text{H}_2\text{O} < \text{CO}_2 < \text{CaO} < \text{CuO}$
48. H_3BO_3 is : [IIT-2002,3]
 (a) monobasic acid and weak Lewis acid
 (b) monobasic acid and weak Bronsted acid
 (c) monobasic acid and strong Lewis acid
 (d) monobasic acid and strong Bronsted acid
49. When I^- is oxidised by MnO_4^- in alkaline medium, I^- convertes into : [IIT-2003]
 (a) IO_3^- (b) I_2
 (c) IO_4^- (d) IO^-
50. Match the column : [IIT-2003]
- | Column-I | Column-II |
|--|-----------------------|
| (a) $\text{Bi}^{3+} \rightarrow (\text{BiO})^+$ | (P) Heat |
| (b) $[\text{AlO}_2]^- \rightarrow \text{Al}(\text{OH})_3$ | (Q) Hydrolysis |
| (c) $\text{SiO}_4^{4-} \rightarrow \text{Si}_2\text{O}_7^{6-}$ | (R) Acidification |
| (d) $(\text{B}_4\text{O}_7^{2-}) \rightarrow [\text{Bi}(\text{OH})_3]$ | (S) Dilution by water |
51. $(\text{Me})_2\text{SiCl}_2$ on hydrolysis will produce : [IIT-2003]
 (a) $(\text{Me})_2\text{Si}(\text{OH})_2$
 (b) $(\text{Me})_2\text{Si} = \text{O}$
 (c) $[\text{—O—}(\text{Me})_2\text{Si—O—}]_n$
 (d) $\text{Me}_2\text{SiCl}(\text{OH})$
52. Which is the most thermodynamically stable allotropic form of phosphorus : [IIT-2004]
 (a) red (b) white
 (c) black (d) yellow
53. When PbO_2 reacts with conc. HNO_3 the gas evolved is : [IIT-2005]
 (a) NO_2 (b) O_2
 (c) N_2 (d) N_2O
54. Which of the following is not oxidised by O_3 : [IIT-2005]
 (a) KI (b) FeSO_4
 (c) N_2O_4 (d) N_2O_5
55. Which blue liquid is obtained on reacting equimolar amounts of two gases at -30°C ? [IIT-2005]
 (a) N_2O (b) N_2O_3
 (c) N_2O_4 (d) N_2O_5
56. Name of the structure of silicates in which three oxygen atoms of $[\text{SiO}_4]^{4-}$ are shared is : [IIT-2005]
 (a) Pyrosilicate
 (b) Sheet silicate
 (c) linear chain silicate
 (d) three dimensional silicate
57. $\text{B}(\text{OH})_3 + \text{NaOH} \rightleftharpoons \text{NaBO}_2 + \text{Na}[\text{B}(\text{OH})_4] + \text{H}_2\text{O}$ how can this reaction is made to proceed in forward direction ? [IIT-2006]
 (a) addition of cis 1, 2 diol
 (b) addition of borax
 (c) addition of trans 1, 2 diol
 (d) addition of Na_2HPO_4
58. The percentage of p-character in the orbitals forming P-P bonds in P_4 is : [IIT-2007]
 (a) 25 (b) 33
 (c) 50 (d) 75
59. Among the following, the paramagnetic compound is : [IIT-2007]
 (a) Na_2O_2 (b) O_3
 (c) N_2O (d) KO_2
60. **Statement-1** : Boron always forms covalent bond. [IIT-2007]
Statement-2 : The small size of B^{3+} favours formation of covalent bond.

6.44 THE p-BLOCK ELEMENTS AND THEIR COMPOUNDS

- (a) Statement-1 is True, Statement-2 is True ;
Statement-2 is a correct explanation for
Statement-1
- (b) Statement-1 is True, Statement-2 is True ;
Statement-2 is NOT a correct explanation for
Statement-1
- (c) Statement-1 is True, Statement-2 is False
- (d) Statement-1 is False, Statement-2 is True

61. Statement-1 : In water, orthoboric acid behaves as a weak monobasic acid. [IIT-2007]

Statement-2 : In water, orthoboric acid acts as a proton donor.

- (a) Statement-1 is True, Statement-2 is True ;
Statement-2 is a correct explanation for
Statement-1
- (b) Statement-1 is True, Statement-2 is True ;
Statement-2 is NOT a correct explanation for
Statement-1
- (c) Statement-1 is True, Statement-2 is False
- (d) Statement-1 is False, Statement-2 is True

Passage for Q.62 to Q.64 :

The noble gases have closed - shell electronic configuration and are monoatomic gases under normal conditions. The low boiling point of the lighter noble gases are due to weak dispersion forces between the atoms and the absence of other interatomic interaction. The direct reaction of xenon with fluorine leads to a series of compounds with oxidation number +2, +4 and +6.

XeF_4 reacts violently with water to give XeO_3 . The compounds of xenon exhibit rich stereochemistry and their geometries can be deduced considering the total number of electron pairs in the valance shell.

[IIT-2007]

62. Argon is used in are welding because of its :

- (a) low reactivity with metal
- (b) ability to lower the melting point of metal
- (c) flammability
- (d) high calorific value

63. The structure of XeO_3 is :

- (a) linear (b) planar
- (c) pyramidal (d) T-shaped

64. XeF_4 and XeF_6 are expected to be :

- (a) oxidising (b) reducing
- (c) unreactive (d) strongly basic

Passage for Q.65 to Q.67 :

There are some deposits of nitrates and phosphates in earth's crust. Nitrates are more soluble in water. Nitrates are difficult to reduce under the laboratory conditions but microbes do it easily. Ammonia forms large number of complexes with transition metal ions. Hybridisation easily explains the case of sigma donation capability of NH_3 and PH_3 . Phosphine is a flammable gas and is prepared from white phosphorus.

[IIT-2008]

65. Among the following , the correct statement :

- (a) phosphates have no biological significance in humans
- (b) between nitrates and phosphates , phosphates are less abundant in earth's crust
- (c) between nitrates and phosphates , nitrates are less abundant in earth's crust
- (d) oxidation of nitrates is possible in soil

66. Among the following , the correct statement :

- (a) Between NH_3 and PH_3 , NH_3 is a better electron donor because the lone pair of electrons occupies spherical 's' orbital and is less directional.
- (b) Between NH_3 and PH_3 , NH_3 is a better electron donor because the lone pair of electrons occupies spherical 'sp³' orbital and is more directional.
- (c) Between NH_3 and PH_3 , NH_3 is a better electron donor because the lone pair of electrons occupies 'sp³' orbital and is more directional.
- (d) Between NH_3 and PH_3 , NH_3 is a better electron donor because the lone pair of electrons occupies spherical 's' orbital and is less directional.

67. The phosphorus on reaction with NaOH gives PH_3 as one of the products. This is a :

- (a) dimerization reaction
- (b) disproportionation reaction
- (c) condensation reaction
- (d) precipitation reaction

68. The reaction of P_4 with X leads selectively to P_4O_6 .
The X is : [IIT-2009]

(a) Dry O_2
(b) A mixture of O_2 , N_2
(c) Moist O_2
(d) O_2 in the presence of aqueous NaOH

69. The nitrogen oxide(s) the contain(s) N-N bond(s) is (are) :- [JEE-2009]

(a) N_2O (b) N_2O_3
(c) N_2O_4 (d) N_2O_5

70. In the reaction $2X + B_2H_6 \rightarrow [BH_2(X)_2]^+ [BH_4]^-$ the amine(s) X is (are) [JEE-2009]

(a) NH_3 (b) CH_3NH_2
(c) $(CH_3)_2NH$ (d) $(CH_3)_3N$

71. The reaction of white phosphorus with aqueous NaOH gives phosphine along with another phosphorus containing compound. The reaction type ; the oxidation states of phosphorus in phosphine and the other product are respectively : - [JEE-2012]

(a) redox reaction ; -3 and -5
(b) redox reaction ; +3 and +5
(c) disproportionation reaction ; -3 and +1
(d) disproportionation reaction ; -3 and +3

72. Bleaching powder contains a salt of an oxoacid as one of its components. The anhydride of that oxoacid is :- [JEE-2012]

(a) Graphite is harder than diamond
(b) Graphite has higher electrical conductivity than diamond
(c) Graphite has higher thermal conductivity than diamond
(d) Graphite has higher C-C bond than diamond

74. Concentrated nitric acid upon long standing, turns yellow-brown due to the formation of :-

[JEE-2013]

(a) NO (b) NO_2
(c) N_2O (d) N_2O_4

75. The correct statement(s) about O_3 is (are) :-

[JEE-2013]

(a) O-O bond lengths are equal
(b) Thermal decomposition of O_3 is endothermic
(c) O_3 is diamagnetic in nature
(d) O_3 has bent structure

Passage for Q.76 to Q.77 :

The reaction of Cl_2 gas with cold dilute and hot concentrated NaOH in water give sodium salt of two (different) oxoacids of chlorine P and Q respectively. The Cl_2 gas reacts with SO_2 gas, in presence of charcoal to give a product R. R reacts with white phosphorus to give a compounds S. On hydrolysis, S gives as oxyacid of phosphorus T. [JEE-2013]

76. R, S and T respectively are :

(a) SO_2Cl_2 , PCl_5 and H_2PO_4
(b) SO_2Cl_2 , PCl_3 and H_3PO_3
(c) SO_2Cl_2 , PCl_3 and H_3PO_2
(d) SO_2Cl_2 , PCl_5 and H_3PO_4

77. P and Q, respectively, are the sodium salts of :

(a) Hypochlorous and chloric acid
(b) Hypochlorous and chlorous acid
(c) Chloric and perchloric acid
(d) Chloric and hypochlorous acid

78. The unbalanced chemical reactions given in List-I show missing reagent or condition (?) which are provided in List-II. Match the List-I with List-II and select the correct answer using the code given below the lists : A :- [JEE-2013]

List-I

(P) $PbO_2 + H_2SO_4 \xrightarrow{?} PbSO_4 + O_2 + \text{other product}$

(Q) $Na_2S_2O_3 + H_2O \xrightarrow{?} NaHSO_4 + \text{other product}$

(R) $N_2H_4 \xrightarrow{?} \overset{0}{N}_2 + \text{other product}$

(S) $XeF_2 \xrightarrow{?} Xe + \text{other product}$

List-II

(i) NO

(ii) I_2

(iii) Warm

(iv) Cl_2

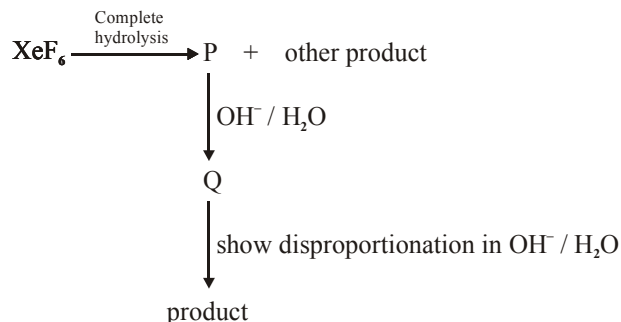
Codes :

	P	Q	R	S
(a)	1	2	3	1
(b)	3	2	1	4
(c)	1	4	2	3
(d)	3	4	2	1

6.46 THE p-BLOCK ELEMENTS AND THEIR COMPOUNDS

79. Under ambient conditions, the total number of gases released as products in the final step of the reaction scheme shown below is :-

[JEE-Adv. 2014]



- (a) 0 (b) 1
(c) 2 (d) 3
80. The product formed in the reaction of SOCl_2 with white phosphorus is :- [JEE-Adv. 2014]
(a) It behaves as a weak acid in water due to self ionization
(b) Acidity of its aqueous solution increases upon addition of ethylene glycol
(c) It has a three dimensional structure due to hydrogen bonding
(d) It is a weak electrolyte in water
82. The correct statement(s) regarding (i) HClO , (ii) HClO_2 , (iii) HClO_3 and (iv) HClO_4 is (are) :- [JEE-Adv. 2015]

- (a) The number of $\text{Cl}=\text{O}$ bonds in (ii) and (iii) together is two
(b) The number of lone pairs of electron on Cl in (ii) and (iii) together is three
(c) The hybridisation of Cl in (iv) is sp^3
(d) Amongst (i) to (iv), the strongest acid is (i)
83. When O_2 is adsorbed on a metallic surface, electron transfer occurs from the metal to O_2 . The TRUE, statements regarding this adsorption is (are) :- [JEE-Adv. 2015]
(a) O_2 is physisorbed
(b) heat is released
(c) occupancy of π_{2p}^* of O_2 is increased
(d) bond length of O_2 is increased
84. Under hydrolytic conditions, the compounds used for preparation of linear polymer and for chain termination, respectively, are :- [JEE-Adv. 2015]
(a) CH_3SiCl_3 and $\text{Si}(\text{CH}_3)_4$
(b) $(\text{CH}_3)_3\text{SiCl}$ and $(\text{CH}_3)_2\text{SiCl}_2$
(c) $(\text{CH}_3)_2\text{SiCl}_2$ and CH_3SiCl_3
(d) SiCl_4 and $(\text{CH}_3)_3\text{SiCl}$
85. Three moles B_2H_6 are completely reacted with methanol. The number of moles of boron containing product formed is :- [JEE-Adv. 2015]
86. The total number of lone pairs of electron in N_2O_3 is :- [JEE-Adv. 2015]

■ ■

THE d-BLOCK ELEMENTS AND SOME OF THEIR COMPOUNDS

PROBLEMS BASED ON GIVEN TOPICS

- Variable oxidation state
 - Stability of the various oxidation state
- Complexes
- Size of atoms and ions
- Density
- Melting and boiling points
- Reactivity and boiling points
- Ionization energies
- Colour
 - Polarization
 - Incompletely filled d or f shell
- Magnetic properties
- Catalytic properties
- Nonstoichiometry
- Abundance
- Chromate and dichromate
 - Preparation
 - Properties
- Manganate and permanganate
 - Preparation
 - Properties
- Silver and its compounds
 - Silver nitrate (AgNO_3)
- Zinc compounds
 - Zinc oxide (ZnO)
 - Zinc chloride (ZnCl_2)
 - Zinc sulphate (ZnSO_4)
- Copper compounds
 - Copper oxide (CuO)
 - Copper chloride (CuCl_2)
 - Copper sulphate (CuSO_4)
- Iron compounds
 - Iron sulphate ($\text{FeSO}_4 \cdot 7\text{H}_2\text{O}$)
 - Iron oxide (FeO)
 - Iron chloride (FeCl_2)

The d-Block Elements and Some of Their Compounds

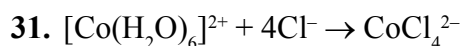
EXERCISE # I

❑ Only one correct answer :

- Which of the following chromate is brick red colour precipitate ?
 (a) BaCrO_4 (b) SrCrO_4
 (c) Ag_2CrO_4 (d) Hg_2CrO_4
- Which of the following chromate is scarlet red colour precipitate ?
 (a) BaCrO_4 (b) SrCrO_4
 (c) Ag_2CrO_4 (d) Hg_2CrO_4
- When potassium permanganate in small portion is added to well cooled concentrated sulphuric acid, a dark green solution is formed which is used as explosive. The oxidation state of manganese in product is :-
 (a) +2 (b) +4
 (c) +6 (d) +7
- The highest oxidation state of Mn is not observed in which of the following compound of Mn :-
 (a) KMnO_4 (b) K_2MnO_4
 (c) Mn_2O_7 (d) MnO_3F
- Oxalic acid in presence of sulphuric acid produces CO_2 gas when it is allowed to react with KMnO_4 solution. The reagent which is used as autocatalyst in this reaction is :-
 (a) $(\text{CO}_2\text{H})_2$ (b) KMnO_4
 (c) $\text{dil. H}_2\text{SO}_4$ (d) MnSO_4
- Which element is not affected by $1(\text{M}) \text{H}^+$, among transitional element :-
 (a) Cu (b) Fe
 (c) Co (d) Ni
- Which reaction is most spontaneous?
 (a) $\text{Mn} + 2\text{H}^+ \rightarrow \text{Mn}^{2+} + \text{H}_2$
 (b) $\text{Fe} + 2\text{H}^+ \rightarrow \text{Fe}^{2+} + \text{H}_2$
 (c) $\text{Zn} + 2\text{H}^+ \rightarrow \text{Zn}^{2+} + \text{H}_2$
 (d) $\text{Co} + 2\text{H}^+ \rightarrow \text{Co}^{2+} + \text{H}_2$
- The actual $E^\circ_{\text{Zn}^{2+}/\text{Zn}}$ is more negative than expected from the general trend. It is because of :-
 (a) very low enthalpy of hydration of Zn^{2+}
 (b) stability of Zn^{2+} due to d^5 configuration
 (c) stability of Zn^{2+} due to completely filled d subshell
 (d) None
- Lunar caustic is :-
 (a) NaOH (b) KOH
 (c) $\text{Ba}(\text{OH})_2$ (d) AgNO_3
- Bordaeux mixture is :-
 (a) $\text{CuCl}_2 + \text{Cu}(\text{OH})_2$
 (b) $\text{CuSO}_4 + \text{Ca}(\text{OH})_2$
 (c) $\text{Cu}(\text{NO}_3)_2 + \text{KOH}$
 (d) $\text{Cu}_2\text{Cl}_2 + \text{K}_2\text{CO}_3$
- The correct order of ionic radii of Y^{3+} , La^{3+} , Eu^{3+} and Lu^{3+} is
 (a) $\text{Y}^{3+} < \text{La}^{3+} < \text{Eu}^{3+} < \text{Lu}^{3+}$
 (b) $\text{La}^{3+} < \text{Eu}^{3+} < \text{Lu}^{3+} < \text{Y}^{3+}$
 (c) $\text{Y}^{3+} < \text{Lu}^{3+} < \text{Eu}^{3+} < \text{La}^{3+}$
 (d) $\text{Lu}^{3+} < \text{La}^{3+} < \text{Eu}^{3+} < \text{Y}^{3+}$
- Which of the element in first series of transition elements has highest heat of atomisation ?
 (a) Cr (b) V
 (c) Co (d) Ni

7.4 THE d-BLOCK ELEMENTS AND SOME OF THEIR COMPOUNDS

13. E° value is highest +ve for which of the following reaction :-
 (a) $\text{MnO}_4^- + e^- \longrightarrow \text{MnO}_4^{2-}$
 (b) $\text{MnO}_4^- + 4\text{H}^+ + 3e^- \longrightarrow \text{MnO}_2 + 2\text{H}_2\text{O}$
 (c) $\text{MnO}_4^- + 8\text{H}^+ + 5e^- \longrightarrow \text{Mn}^{2+} + 4\text{H}_2\text{O}$
 (d) $\text{Cr}_2\text{O}_7^{2-} + 14\text{H}^+ + 6e^- \longrightarrow 2\text{Cr}^{3+} + 7\text{H}_2\text{O}$
15. Metal (M) + dilute HCl $\rightarrow$ H_2 gas
 Metal (M) (red hot) + steam $\rightarrow$ H_2 gas + Mixed oxide
 Metal M can be :-
 (a) Copper (b) Zinc
 (c) Iron (d) Cadmium
16. In mild acidic medium manganate ion disproportionate in :-
 (a) Mn^{2+} , MnO_4^-
 (b) MnO_2 , MnO_4^-
 (c) Mn^{2+} , Mn_2O_7
 (d) MnO_2 , $\text{Mn}(\text{OH})_2$
17. Arrange the following species in the increasing order of their magnetic properties :-
 (a) $\text{VCl}_3 > \text{VOSO}_4 > \text{Na}_3\text{VO}_4 > [\text{V}(\text{H}_2\text{O})_6]\text{SO}_4 \cdot \text{H}_2\text{O}$
 (b) $[\text{V}(\text{H}_2\text{O})_6]\text{SO}_4 > \text{VCl}_3 > \text{VOSO}_4 > \text{Na}_3\text{VO}_4$
 (c) $\text{VCl}_3 > \text{VOSO}_4 > [\text{V}(\text{H}_2\text{O})_6]\text{SO}_4 \cdot \text{H}_2\text{O} > \text{Na}_3\text{VO}_4$
 (d) All
18. Oxide of which d^{10} ion is amphoteric :-
 (a) ZnO (b) CdO
 (c) HgO (d) OH
19. The aqueous solution of CuCrO_4 is green because it consists of :-
 (a) Green Cu^{2+} ion
 (b) Blue Cu^{2+} & Green CrO_4^{2-} ion
 (c) Blue CrO_4^{2-} & Green Cu^{2+} ion
 (d) Blue Cu^{2+} & Yellow CrO_4^{2-} ion
20. The highest oxidation state is exhibited by the transition metals with configuration -
 (a) $(n-1)d^3ns^2$ (b) $(n-1)d^5ns^1$
 (c) $(n-1)d^5ns^2$ (d) $(n-1)d^8ns^2$
21. $\text{RCH}=\text{O} + \text{Fehling solution} \rightarrow \text{RCO}_2\text{H} + \text{Reddish brown precipitate}$ -
 Reddish brown precipitate is
 (a) CuO (b) Pb_3O_4
 (c) Hg_2O (d) Cu_2O
22. The stability order :-
 (a) $\text{CrO}_4^{2-} > \text{MnO}_4^{2-} > \text{FeO}_4^{2-}$
 (b) $\text{CrO}_4^{2-} > \text{FeO}_4^{2-} > \text{MnO}_4^{2-}$
 (c) $\text{MnO}_4^{2-} > \text{CrO}_4^{2-} > \text{FeO}_4^{2-}$
 (d) $\text{FeO}_4^{2-} > \text{CrO}_4^{2-} > \text{MnO}_4^{2-}$
23. +4 oxidation state of Pt is observed in :-
 (a) $\text{O}_2[\text{PtF}_6]$ (b) $\text{H}_2[\text{PtCl}_4]$
 (c) $\text{H}_2[\text{PtCl}_6]$ (d) $\text{K}_2[\text{Pt}(\text{CN})_4]$
24. The colour of mercurous iodide precipitate is :-
 (a) Green (b) Black
 (c) Yellow (d) Scarlet red
25. The colour of mercuric iodide precipitate is :-
 (a) Green (b) Black
 (c) Yellow (d) Scarlet red
26. The colour of mercurous oxide precipitate is :-
 (a) Green (b) Black
 (c) Yellow (d) Scarlet red
27. The colour of mercuric oxide precipitate is :-
 (a) Green (b) Black
 (c) Yellow (d) Scarlet red
28. The aqueous solution of CuCrO_4 is green because it contains :-
 (a) green Cu^{2+} ions
 (b) green CrO_4^{2-} ions
 (c) blue Cu^{2+} ions & green CrO_4^{2-} ions
 (d) blue Cu^{2+} ions & yellow CrO_4^{2-} ions
29. The compound of pentavalent gold is ?
 (a) AuF_5 (b) AuCl_5
 (c) AuBr_5 (d) AuI_5
30. Which cuprous halide is white in colour ?
 (a) CuF (b) CuCl
 (c) CuCl_2 (d) CuBr



Colour of the complex in reactant and product side are respectively

- (a) Blue ; Pink
(b) Pink ; Blue
(c) Pink ; Yellow
(d) Dark green ; Pink

32. When acetylene is passed through cuprous chloride:-

- (a) Red colour precipitate of Cu_2C_2 is formed
(b) White colour precipitate of Cu_2C_2 is formed
(c) Blue colour precipitate of CuC_2 is formed
(d) Blue colour precipitate of Cu_2C_2 is formed

33. Microcosmic salt when heated, a transparent bead is formed. The transparent bead is :-

- (a) NaBO_2 (b) B_2O_3
(c) P_2O_5 (d) NaPO_3

34. Which of the following black colour precipitate ?

- (a) Cr_2O_3 (b) SnS_2
(c) HgS (d) AgI

35. Which one of the following ions do not give borax bead test ?

- (a) Cr^{3+} (b) Cu^{2+}
(c) Mn^{2+} (d) Zn^{2+}

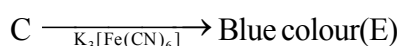
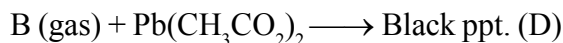
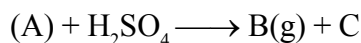
36. A white powder (A) when strongly heated, it gives off colourless odourless gas (B) which turns lime water milky (C). Also it gives solid residue D which is yellow when hot but turns white on cooling.

- (a) PbCO_3 , PbO (b) ZnCO_3 , ZnO
(c) PbCO_3 , PbO_2 (d) Pb_2CO_3 , PH_3O_4

37. Coagulation of blood is caused by :-

- (a) Alum
(b) Microcosmic salt
(c) Sodium hydrogen phosphate
(d) Borax

38. Black coloured



Here :-

- (a) A is CuS and B is H_2S
(b) A is FeS ; C is FeSO_4 ; D is PbS ; E is $\text{Fe}_3[\text{Fe}(\text{CN})_6]_2$
(c) C is $\text{Zn}_2[\text{Fe}(\text{CN})_6]$ and D is PbS
(d) D is PbS and E is $\text{Fe}_4[\text{Fe}(\text{CN})_6]_3$

39. The aqueous solution of P yields a white precipitate when treated with dilute HNO_3 and AgNO_3 . Another sample of the solution of P when treated with NaOH gives a white precipitate which is dissolved in excess of NaOH . When H_2S gas is passed through solution, a white precipitate is obtained. P is :-

- (a) CoCl_2 (b) ZnCl_2
(c) NiCl_2 (d) SnCl_2

40. The highly water soluble red violet crystalline compound $\text{K}_3[\text{Mn}(\text{C}_2\text{O}_4)_3] \cdot 3\text{H}_2\text{O}$ is octahedral complex. The magnetic moment is :-

- (a) $\sqrt{15}$ B.M. (b) $\sqrt{24}$ B.M.
(c) $\sqrt{35}$ B.M. (d) $\sqrt{8}$ B.M.

41. Which of the following form an alloy ?

- (a) $\text{Zn} + \text{Pb}$
(b) $\text{Fe} + \text{Hg}$
(c) $\text{Pt} + \text{Hg}$
(d) $\text{Fe} + \text{C}$

42. $\text{KI}_{(\text{excess})} + \text{Mercuric chloride} \rightarrow \text{Product}$.

Product along with KOH is very important laboratory reagent. The reagent and its formula -

- (a) Nessler's reagent ; $\text{K}_2[\text{HgI}_4]$
(b) Nessler's reagent ; $\text{K}_4[\text{HgI}_6]$
(c) Nessler's reagent ; $\text{K}_2[\text{Hg}_2\text{I}_4]$
(d) Nessler's reagent ; $\text{K}_2[\text{Hg}(\text{SCN})_4]$

43. H_2S gas is passed through an acidified solution of a mixture containing Cu^{2+} and Zn^{2+} ions. What will happen ?

- (a) no precipitate is formed
(b) both will get precipitated
(c) CuS gets precipitated only
(d) ZnS gets precipitated only

7.6 THE d-BLOCK ELEMENTS AND SOME OF THEIR COMPOUNDS

44. A few Cu (I) salts are coloured. The colour arises from
 (a) d-d transition
 (b) Charge transfer spectra
 (c) The large wavelengths of the rays absorbed by the solutions
 (d) The smaller wavelengths of rays is absorbed by the solution
45. Which of the following does not disproportionate?
 (a) Cu^+ (b) Au^{3+}
 (c) Ga^+ (d) Au^+
46. Acidic KMnO_4 is decolourised by
 (a) The ferric ammonium alum
 (b) Mohr's salt
 (c) A neutral ferric chloride solution
 (d) All of these
47. A colourless salt
 $\text{A} \xrightarrow{\text{MnO}_4^-/\text{H}^+}$ decolourises
 $\text{A} \xrightarrow{\text{Heat}} \text{X} + \text{Y} + \text{Z}$; X and Y are gases
 $\text{Z} + \text{water} \longrightarrow \text{B}$
 $\text{Y} + \text{B} \longrightarrow \text{milky}$
 Gas X burns with blue flame. Mark the correct choices.
- | A | X | Y | Z | B |
|------------------------------|---------------|---------------|---------------|-------------------------|
| (a) CaCO_3 | CaO | CO | CO_2 | H_2CO_3 |
| (b) CaC_2O_4 | CO | CO_2 | CaO | Ca(OH)_2 |
| (c) CaC_2O_4 | CO_2 | CO | CaO | Ca(OH)_2 |
| (d) CaSO_3 | SO_2 | O_2 | CaO | Ca(OH)_2 |
48. Statement -1: $\text{Mn}_2\text{Cl}_{10}$ will be diamagnetic.
 Statement -2: TiCl_4 is used for the test of H_2O_2 .
 Statement -3: In $\text{Cr}_3\text{O}(\text{CH}_3\text{COO})_6\text{H}_2\text{O}$, CH_3COO^- act as bridging ligand.
 (a) FTT (b) FFT
 (c) TTT (d) FFF
49. Statement -1: Actinoids show higher oxidation state than lanthanoids.
 Statement -2: All lanthanoids in M^{+3} state are coloured.
- Statement -3: Promethium is a radioactive lanthanoid.
 (a) TTF (b) TFT
 (c) FFT (d) FTF
50. Purple of cassius is :-
 (a) Colloidal sol of graphite in water
 (b) Colloidal sol of silver
 (c) Colloidal sol of gold
 (d) Colloidal sol of cellulose nitrate in ethanol
51. Which of the following has three unpaired electrons:-
 (a) Zn^{2+} (b) Cr^{3+}
 (c) Co^{2+} (d) Cu^{2+}
52. Which of the following is diamagnetic :-
 (a) La^{3+} (b) Ce^{4+}
 (c) Yb^{2+} (d) All
53. Among the following which has highest oxidising power :-
 (a) VO_2^+ (b) $\text{Cr}_2\text{O}_7^{2-}$
 (c) MnO_4^- (d) MnO_2
54. The common oxidation state of lanthanoid :-
 (a) +2 (b) +3
 (c) +4 (d) +5
55. Which of the following is negatively charged sol ?
 (a) $\text{CrO}_3 \cdot x\text{H}_2\text{O}$ (hydrated metallic oxide)
 (b) Haemoglobin (blood)
 (c) Oxides TiO_2 sol
 (d) CdS sol
56. Correct order for the wavelength of absorption in the visible region for the following :-
 (a) $[\text{Ni}(\text{H}_2\text{O})_6]^{2+} < [\text{Ni}(\text{NH}_3)_6]^{2+} < [\text{Ni}(\text{NO}_2)_6]^{4+}$
 (b) $[\text{Ni}(\text{NH}_3)_6]^{2+} < [\text{Ni}(\text{H}_2\text{O})_6]^{2+} < [\text{Ni}(\text{NO}_2)_6]^{4+}$
 (c) $[\text{Ni}(\text{H}_2\text{O})_6]^{2+} < [\text{Ni}(\text{NO}_2)_6]^{4+} < [\text{Ni}(\text{NH}_3)_6]^{2+}$
 (d) $[\text{Ni}(\text{NO}_2)_6]^{4+} < [\text{Ni}(\text{NH}_3)_6]^{2+} < [\text{Ni}(\text{H}_2\text{O})_6]^{2+}$
57. CuS precipitate is dissolved in :-
 (a) NH_4OH solution
 (b) NaOH solution
 (c) Potassium cyanide solution
 (d) Water

58. $\text{Cr}(\text{OH})_3$ precipitate is dissolved in :-
(a) Sodium hydroxide
(b) Sodium peroxide solution
(c) 50% HNO_3
(d) All
59. $\text{Zn}(\text{OH})_2$ precipitate is dissolved in :-
(a) NH_3
(b) Sodium hydroxide solution
(c) 50% HNO_3
(d) All
60. Mischmetal is an alloy which consists of :-
(a) 50% of lanthanoid metal and 50% iron & traces of S, C, Ca and Al
(b) 80% of lanthanoid metal and 20% iron & traces of S, C, Ca and Al
(c) 95% of lanthanoid metal and 5% iron & traces of S, C, Ca and Al
(d) None
61. The most stable ion in aqueous solution :-
(a) V^{3+} (b) Tl^{3+}
(c) Cr^{3+} (d) Mn^{3+}
62. $\text{Cr}_2\text{O}_3 + 2\text{Al} \rightarrow \text{Al}_2\text{O}_3 + 2\text{Cr}$, What is the ΔH for the reaction.
Given $(\Delta H_f)_{\text{Al}_2\text{O}_3} = -827 \text{ kJ}\cdot\text{mol}^{-1}$,
 $(\Delta H_f)_{\text{Cr}_2\text{O}_3} = -540 \text{ kJ}\cdot\text{mol}^{-1}$
(a) $+287 \text{ kJ}\cdot\text{mol}^{-1}$
(b) $-287 \text{ kJ}\cdot\text{mol}^{-1}$
(c) $1367 \text{ kJ}\cdot\text{mol}^{-1}$
(d) $-1367 \text{ kJ}\cdot\text{mol}^{-1}$
63. Ag^+ does not give white precipitate with :-
(a) potassium thiocyanate solution
(b) potassium chloride solution
(c) potassium ferrocyanide solution
(d) potassium iodide solution
64. Which of the following statement is correct :-
(a) the +1 oxidation state of Ni is not stable
(b) NiO is an amphoteric oxide
(c) the most important ore is melachite
(d) silver is poor conductor of heat and electricity
65. Lanthanoid contraction is caused due to :-
(a) the appreciable shielding on outer electrons by 4f electrons from the nuclear charge
(b) the appreciable shielding on outer electrons by 5d electrons from the nuclear charge
(c) the same effective nuclear charge from Ce to Lu
(d) the imperfect shielding on outer electron by 4f electrons from the nuclear charge
66. Identify the incorrect statement among the following:-
(a) d-block elements show irregular and erratic chemical properties among themselves
(b) La and Lu have partially filled d orbitals and no other partially filled orbitals
(c) The chemistry of various lanthanoids is very similar
(d) 4f and 5f orbitals are equally shielded
67. The actinoids exhibit more number of oxidation states in general than the lanthanoids. This is because :-
(a) the 5f orbitals are more buried than the 4f orbitals
(b) there is similarity between 4f and 5f orbitals in their angular part of the wave function
(c) the actinoids are more reactive than the lanthanoids
(d) the 5f orbitals extend further from the nucleus than the 4f orbitals
68. The lanthanoids contraction is responsible for the fact that :-
(a) Zr and Zn have the same oxidation state
(b) Zr and Hf have about the same radius
(c) Zr and Nb have similar oxidation state
(d) Zr and Y have about the same radius
69. More positive the value of $E_{\text{M}^{n+}/\text{M}}^\circ$:-
(a) greater is the stability of M^{n+} ion in aqueous medium
(b) less is the stability of M^{n+} ion in aqueous medium
(c) greater is the stability of M ion in aqueous medium
(d) None of these

70. Three separate test tubes contains aq. solution of dichloride of a metal :

(I) test tube -1 $\xrightarrow{\text{KOH}(\text{excess})}$ No change

(II) test tube -2 $\xrightarrow{\text{dil. H}_2\text{SO}_4}$ No change

(III) test tube -3 $\xrightarrow{\text{NH}_4\text{OH}(\text{excess})}$ No change

dichloride of metal is -

(a) SnCl_2 (b) FeCl_2

(c) CuCl_2 (d) ZnCl_2

EXERCISE # II

□ One or More Than One Correct Answer :

- When chromite ore is fused with excess of Na_2O_2 , then:-
 - brown precipitate is formed
 - Orange solution is observed
 - Green colour precipitate is observed
 - Yellow solution is observed
- CrO_5 is stabilised in ether solution as the coordination complex by :-
 - Pyridine
 - Trimethyl amine
 - Aniline
 - Amyl alcohol
- Ammonium-dichromate decompose violently on heating giving -
 - N_2
 - N_2O_3
 - H_2O
 - Cr_2O_3
- Which carboxylic acid can be oxidised by KMnO_4 solution ?
 - HCO_2H
 - $\text{CH}_3\text{CO}_2\text{H}$
 - $(\text{CO}_2\text{H})_2$
 - H_2CO_3
- Manganous sulphate is oxidised to permanganic acid by treatment with :-
 - PbO_2 is HNO_3
 - Sodium bismuthate (NaBiO_3) in HNO_3
 - KIO_4 is HNO_3
 - Ammonium perdisulphate
- What are the correct statement regarding potassium manganate solution ?
 - Potassium manganate solution is quite stable in acidic medium
 - Potassium manganate solution is quite stable in basic medium
 - Potassium manganate disproportionates in basic medium depositing MnO_2 and giving a purple solution of permanganate.
 - Potassium manganate disproportionates in acidic medium
- At least one gaseous product is formed for which compound when reacts with MnO_2 :-
 - Concentrated HCl
 - Concentrated H_2SO_4
 - Solid NH_4Cl
 - Oxalic acid in presence of H_2SO_4
- In 3d series, which element has positive electron gain enthalpy ?
 - Cr
 - Fe
 - Mn
 - Zn
- Among all d block element :-
 - Hg has lowest melting point
 - W has highest melting point
 - Fe has highest melting point
 - Zn has lowest melting point
- Which of the following statements are correct regarding use of d block element ?
 - Mo is used in X-ray tube
 - Co is used for radiotherapy of cancerous tumour
 - Pt is used as catalyst for hydrogenation of alkene
 - Ni is used in nickel plating
- Cr^{2+} is reducing and Mn^{3+} is oxidising but both are d^4 configuration. It is because :-
 - Cr^{2+} is itself oxidised to Cr^{3+} which has half filled t_{2g} orbital
 - Mn^{3+} is itself reduced to Mn^{2+} which has half filled d^5 configuration
 - Cr^{2+} is itself oxidised to Cr^{3+} which has half filled eg orbital
 - Mn^{3+} is itself reduced to Mn^{4+} which has half filled eg orbital
- Example of mixed oxide of transition elements are :-
 - Pb_3O_4
 - Mn_3O_4
 - Fe_3O_4
 - Co_3O_4

13. Among the following pair of ions, the lower oxidation state in aqueous solution is more stable than the other in :-
 (a) $\text{Cu}^+(\text{aq.})$; $\text{Cu}^{2+}(\text{aq.})$
 (b) Mn^{2+} , Mn^{3+}
 (c) Cr^{2+} ; Cr^{3+}
 (d) Tl^{+1} ; Tl^{3+}
14. The aqueous solution of the salt will be coloured. The salt can be :-
 (a) NaNO_3
 (b) ZnCl_2
 (c) $\text{Cr}_2(\text{SO}_4)_3$
 (d) $\text{Cu}(\text{NO}_3)_2$
15. Which of the following options are correct regarding ionisation potential of Ni and Pt :-
 (a) $(\text{IP}_1 + \text{IP}_2)_{\text{Ni}} < (\text{IP}_1 + \text{IP}_2)_{\text{Pt}}$
 (b) $(\text{IP}_1 + \text{IP}_2)_{\text{Ni}} > (\text{IP}_1 + \text{IP}_2)_{\text{Pt}}$
 (c) $(\text{IP}_3 + \text{IP}_4)_{\text{Ni}} < (\text{IP}_3 + \text{IP}_4)_{\text{Pt}}$
 (d) $(\text{IP}_3 + \text{IP}_4)_{\text{Ni}} > (\text{IP}_3 + \text{IP}_4)_{\text{Pt}}$
16. Ammoniacal Cu_2Cl_2 absorbs :-
 (a) CO_2 (b) CO
 (c) C_2H_4 (d) C_2H_2
17. Mn-an element which forms oxide in +2 to +7 oxidation state. Correct options is -
 (a) Oxide in +2 oxidation state is ionic
 (b) Oxide in +7 oxidation state is covalent
 (c) Oxide in +2 oxidation state is covalent
 (d) Oxide in +7 oxidation state is ionic
18. Addition of nonmetals like B and C to the interstitial sites of a transition metal results the metal :-
 (a) of more ductability
 (b) of less ductability
 (c) less malleable
 (d) more hardness
19. Which of the following options are correct ?
 (a) Hydrated Co^{2+} salt is pink.
 (b) Anhydrous Co^{2+} salt is blue.
 (c) Hybridisation of $\text{CoCl}_2 \cdot 6\text{H}_2\text{O}$ is sp^3d^2
 (d) Co^{2+} has coordination number + 8
20. Which of the following options are correct ?
 (a) $\text{CuSO}_4 \cdot 5\text{H}_2\text{O}$ is blue
 (b) Anhydrous CuSO_4 is white.
 (c) $[\text{Cu}(\text{NH}_3)_4]^{2+}$ is blue
 (d) CuO is black
21. Which reactions are spontaneous ?
 (a) $2\text{AgNO}_3 + \text{Cu} \rightarrow 2\text{Ag} + \text{Cu}(\text{NO}_3)_2$
 (b) $\text{CuSO}_4 + \text{Zn} \rightarrow \text{Cu} + \text{ZnSO}_4$
 (c) $\text{Fe}_2(\text{SO}_4)_3 + \text{Cu} \rightarrow \text{Fe} + \text{CuSO}_4$
 (d) $\text{Fe}_2(\text{SO}_4)_3 + \text{Cu} \rightarrow 2\text{FeSO}_4 + \text{CuSO}_4$
22. Which of the following options are correct ?
 (a) CuO is black colour compound
 (b) CuO is formed by heating copper nitrate $\text{Cu}(\text{NO}_3)_2$
 (c) CuO is insoluble in NH_3 solution
 (d) CuO is soluble in concentrated NH_3 solution forming deep blue Schwegler's reagent.
23. What can be possible complex formed by combination between iron and carbon monoxide?
 (a) $\text{Fe}(\text{CO})_5$ (b) $\text{Fe}_2(\text{CO})_{10}$
 (c) $\text{Fe}_3(\text{CO})_{12}$ (d) $\text{Fe}_2(\text{CO})_9$
24. KI solution can be used to identify ?
 (a) Hg^{2+} (b) Pb^{2+}
 (c) Ag^+ (d) Cu^{2+}
25. Which of the following reactions give correct product ?
 (a) $\text{Zn} + \text{KOH} \rightarrow \text{K}_2\text{ZnO}_2 + \text{H}_2$
 (b)  + $\text{FeSO}_4 + \text{H}_2\text{O}_2 \rightarrow$ 
 (c) $\text{Cu}^{2+} + \text{KI} \rightarrow \text{CuI}_2 + \text{K}^+$
 (d) $\text{Cr}(\text{OH})_3 + \text{NaOH} + \text{H}_2\text{O}_2 \rightarrow \text{Na}_2\text{Cr}_2\text{O}_7 + \text{H}_2\text{O}$
26. Which of the following pair has/have similar chemical properties?
 (a) Zirconium and hafnium
 (b) Niobium and tantalum
 (c) Cu and Zn
 (d) Fe and Cd

27. The complex of La where it shows a coordination number of more than 8 :-
 (a) $[\text{La}.\text{EDTA}.\text{(H}_2\text{O)}_4]\text{3H}_2\text{O}$
 (b) $\text{La}_2(\text{SO}_4)_3.9\text{H}_2\text{O}$
 (c) LaX_2
 (d) $\text{Cu}_2[\text{LaF}_8]$
27. Which of the following atomic number are the atomic numbers of the inner transition elements : 29, 59, 74, 95, 102, 104?
 (a) 59 (b) 95
 (c) 102 (d) 104
28. The true statements about Re–Re bonding in $[\text{Re}_2\text{X}_8]^{2-}$ are,
 (a) Re–Re bond length is abnormally short
 (b) Re–Re bond is comprised of one σ , two π and one δ -bond.
 (c) If Re–Re bond points along z-axis, then square planar ReX_4 unit will use s, p_x , p_y and $d_{x^2-y^2}$ orbitals for the formation of four Re–X σ bonds.
 (d) d_{xy} orbitals will overlap to form δ -bond.
29. Which of the following is/are correct?
 (a) Of the d^4 series, Cr^{+2} is strongly reducing while Mn^{3+} is strongly oxidising.
 (b) Cobalt (II) is stable in aqueous solution but in the presence of complexing reagents it easily oxidised.
 (c) The d^1 configuration is very unstable in ions.
 (d) $\text{Na}_2\text{Cr}_2\text{O}_7$ is preferred over $\text{K}_2\text{Cr}_2\text{O}_7$ in volumetric analysis.
30. Choose the correct statements :-
 (a) First ionisation energy of Cu(29) is higher than that of K (19)
 (b) Second ionisation of Cu is lower than that of K (19)
 (c) The third ionisation energy of Cu is higher than that of K
 (d) Third ionisation energy of Cu is lower than that of K
31. The correct statement about oxide of transition metal
 (a) Oxide of a metal may be acidic and basic
 (b) Ionic radius of M^{+2} in MO of 3d series decrease from SC to V and then increase from V to Mn
 (c) FeO is a non stoichiometric compound
 (d) Mn_2O_7 is a coloured oxide due to d-d transition
32. Cu^{2+} gives black precipitate with :-
 (a) saturated solution of H_2S
 (b) potassium iodide solution
 (c) potassium ferrocyanide solution
 (d) potassium thiocyanate solution
33. Green precipitate is formed when :-
 (a) FeSO_4 reacts with H_2S
 (b) $\text{Ni}(\text{NO}_3)_2$ reacts with H_2S
 (c) FeSO_4 reacts with NaOH
 (d) $\text{Ni}(\text{NO}_3)_2$ reacts with NaOH
34. Black precipitate is formed when :-
 (a) FeSO_4 reacts with H_2S
 (b) $\text{Ni}(\text{NO}_3)_2$ reacts with H_2S
 (c) FeSO_4 reacts with NaOH
 (d) $\text{Ni}(\text{NO}_3)_2$ reacts with NaOH
35. The oxo metal anions of the first series of the transitional metal exhibits the oxidation state equal to its group number :-
 (a) VO_3^{3-} (b) VO_4^{3-}
 (c) CrO_4^{2-} (d) MnO_4^-
36. Which of the following statements are correct :-
 (a) Thomas slag is used as fertiliser
 (b) CuO is amphoteric in nature
 (c) Cu_2O is amphoteric in nature
 (d) Thomas slag is used as fuel
37. Most common oxidation state of Ce (Cerium) are :-
 (a) +3 (b) +4
 (c) +2 (d) +5

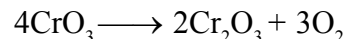
38. Which of the following statements are correct :-
- (a) the compounds of lanthanoids are less basic than those of actinoids
 - (b) the highest oxidation state by lanthanoids is +7
 - (c) lanthanoids dissolve in warm water to liberate hydrogen
 - (d) the outer electronic configuration of tungsten is of the type $(n-1)d^4ns^2$
39. Correct statements are :-
- (a) the transition metals form a number of interstitial compounds with hydrogen, carbon, nitrogen, boron etc.
 - (b) the fluorides of transition metals are ionic nature
 - (c) ferric chloride exists as a dimer (Fe_2Cl_6) at 1023 K and as a monomer at 717 K
 - (d) the chemical composition of rust is $Fe_3O_4 \cdot 6H_2O$
40. Dental amalgam used for filling teeth consists of :-
- (a) Ag
 - (b) Sn
 - (c) Au
 - (d) Hg
41. Which of the following reagents are used to convert MnO_2 to green melt of manganate salt :-
- (a) $KOH + KNO_3$
 - (b) $KOH + KClO_3$
 - (c) $KOH + O_2$
 - (d) $KOH + H_2$
42. Which ion can undergo disproportionation :-
- (a) Hg_2^{2+}
 - (b) Zn^{2+}
 - (c) Cu^{+1}
 - (d) Mn^{2+}
43. Which can produce Riemann's green with cobalt nitrate solution :-
- (a) ZnO
 - (b) $ZnSO_4$
 - (c) $3Zn(OH)_2 \cdot ZnCO_3$
 - (d) $Pb(OH)_2 \cdot PbCO_3$
44. Identify correct statement with copper sulphate :-
- (a) $CuSO_4 + KI \longrightarrow I_2$
 - (b) $CuSO_4 + KCl \longrightarrow Cl_2$
 - (c) $CuSO_4 \xrightarrow{\Delta} CuO$
 - (d) Its tartarate complex reacts with NaOH and glucose to give Cu_2O
45. The aqueous solution of the salt will be coloured in the case of :-
- (a) $Zn(NO_3)_2$
 - (b) $LiNO_3$
 - (c) $Co(NO_3)_2$
 - (d) $CrCl_3$

EXERCISE # III

□ Linked Comprehension Type :

Passage for Q.1 to Q.3

CrO_3 melts at $197^\circ C$ to a dark red liquid and begins to decompose at $200^\circ C$ giving off oxygen :



The decomposition is complete at 420° . A solution of CrO_3 in glacial acetic acid is often used as an oxidising reagent.

- CrO_3 reacts with F_2 at normal pressure at $150^\circ C$ and $220^\circ C$ respectively gives :-
 - (a) $CrOF_4$, CrO_2F_2
 - (b) CrO_2F_2 , $CrOF_4$
 - (c) $CrOF_4$, CrF_6
 - (d) CrO_2F_2 , CrF_6
- CrO_3/H_2SO_4 usually taken in acetone is called Jones reagent. Jones reagents oxidises :-
 - (a) 1° alcohol to aldehyde $RCH=O$
 - (b) 2° alcohol (R_2CHOH) to ketone $R_2C=O$
 - (c) 3° alcohol (R_3C-OH) to alkene
 - (d) can not oxidise alcohol
- CrO_3 is similar as SO_3 like :-
 - (a) CrO_3 is acidic oxide like SO_3
 - (b) CrO_3 when reacts with KOH , it gives K_2CrO_4 like SO_3 give K_2SO_4
 - (c) Crystal structure of CrO_3 consists of infinite chains of linked CrO_4 tetrahedra, $\beta-SO_3$ has also similar infinite chains of linked SO_4 tetrahedral.
 - (d) All

Passage for Q.4 to Q.6

When potassium permanganate is heated in test tube, gas (G_1) is evolved and a black residue remains behind. Upon extracting with a little water and filtering, a green solution is obtained.

- The gas evolved is :-
 - (a) O_2
 - (b) O_3
 - (c) H_2O
 - (d) CO_2
- Black residue is :-
 - (a) Mn_2O_3
 - (b) MnO
 - (c) MnO_2
 - (d) Mn_2O_7

6. Green solution is :-

- (a) $K_2Mn_2O_3$ (b) K_2MnO_4
 (c) MnO (d) $MnSO_4$

Passage for Q.7 to Q.9

$KMnO_4$ when reacts with KI in acidic medium give different products w.r.t. products formed when it reacts with KI in basic medium.

7. In acidic medium $KMnO_4$ oxidises KI to give P_1 and it is itself converted into P_2 . P_1 and P_2 are respectively :-

- (a) I_2 , Mn^{2+} (b) IO_3^- , Mn^{2+}
 (c) I_2 , MnO_2 (d) IO_3^- , MnO_2

8. In basic medium $KMnO_4$ oxidises KI to give P'_1 and it is itself converted into P'_2 . P'_1 and P'_2 are respectively :-

- (a) I_2 , Mn^{2+} (b) IO_3^- , Mn^{2+}
 (c) I_2 , MnO_2 (d) IO_3^- , MnO_2

9. When $KMnO_4$ reacts with sodium sulphite or sodium thiosulphate :-

- (a) MnO_2 is precipitated
 (b) Mn^{2+} is formed
 (c) Green MnO_4^{2-} ion formed
 (d) None

Passage for Q.10 to Q.12

One of the notable features of a transition element is the great variety of oxidation state it may show in its compounds. The element which give the greatest number of oxidation state occur in or near the middle of the series. The maximum oxidation state of reasonable stability correspond in value to the sum of the s and d electrons upto manganese.

10. Lowest oxidation state +1 is exhibited by which of the transition elements :-

- (a) Mn (b) Fe
 (c) Co (d) Cu

11. +6 oxidation states are exhibited by which of the transition elements :-

- (a) Cr (b) Mn
 (c) Fe (d) All

12. Which element can show +2, +3, +4 oxidation state :-

- (a) Fe (b) Co
 (c) Ni (d) All

Passage for Q.13 to Q.15

The d-block occupies the large middle section flanked by s & p-block in the periodic table. The very name transition given to the elements of d-block is only because of their position between s & p-block elements. The d orbitals of the penultimate energy level in their atoms receive electrons giving rise to the three rooms of the transitional metals i.e., 3d, 4d and 5d.

13. Which of the following elements are occupied in d-block but not considered as transition elements.

- (a) Zn (b) Cd
 (c) Hg (d) All

14. Re ; Os ; Ir ; Pt belongs to -

- (a) 3d series (b) 4d series
 (c) 5d series (d) 6d series

15. Mo ; Ru ; Rh ; Pd belongs to -

- (a) 3d series (b) 4d series
 (c) 5d series (d) 6d series

Passage for Q.16 to Q.18

Nearly all the transition elements display typical metallic properties such as high tensile strength, ductility, malleability, high thermal and electrical conductivity and metallic lustre.

The transition elements are very much hard and have low volatility. Their melting and boiling point are high.

16. Correct melting point order of 1st series of transition elements -

- (a) $Mn > Cr > V > Ti > Cu$
 (b) $Cr > Mn > V > Ti > Cu$
 (c) $Cr > V > Ti > Mn > Cu$
 (d) $Mn > Fe > Co > Pd > Cu$

17. The lowest melting point of 2nd series of transition elements is observed for -

- (a) Ru (b) Rh
 (c) Nb (d) Ag

18. The structure of Hg :-
 (a) Body centered cubic structure
 (b) Hexagonal close packed structure
 (c) Cubic close packed structure
 (d) A typical metal structure

Passage for Q.19 to Q.20

X (Green colour carbonate salt) $\xrightarrow{\Delta}$ Y (Black residue) + CO₂ + H₂O :

Y (Black residue) $\xrightarrow{\Delta}$ Z (Reddish brown residue) + W gas

19. X can be :-
 (a) CuCO₃.Cu(OH)₂ (b) CuCO₃.2Cu(OH)₂
 (c) both (a) and (b) (d) FeCO₃
20. Y can be :-
 (a) Cu₂O (b) FeO
 (c) CuO (d) Fe₂O₃
21. Z can be :-
 (a) Cu₂O (b) FeO
 (c) CuO (d) Fe₂O₃

Passage for Q.22 to Q.24

An orange solid X is a very important chemical used in leather industry and an oxidant for preparation of many azo compound. When X is heated, it gives yellow coloured B and green coloured C. Also during heating a colourless, odourless gas D is evolved :-

22. Compound A is
 (a) KMnO₄ (b) K₂Cr₂O₇
 (c) K₂CrO₄ (d) Cr₂O₃
23. Compound C is obtained on heating of
 (a) (NH₄)₂Cr₂O₇ (b) NH₄ClO₄
 (c) NH₄NO₃ (d) None of these
24. Gas D is :-
 (a) O₂ (b) NO₂
 (c) N₂ (d) H₂

Passage for Q.25 to Q.27

A certain metal A is boiled in dilute HNO₃ to give salt(B) and an oxide of nitrogen(C). Aqueous solution B with brine solution gives a precipitate D which is soluble in NH₄OH.

25. A is :-
 (a) Pb (b) Hg
 (c) Ag (d) Zn
26. When B reacts with hypo solution, what will be observation :-
 (a) white precipitate is formed which on boiling turns reddish brown
 (b) white precipitate is formed which on boiling turns yellow
 (c) white precipitate is formed which on boiling turns black
 (d) none
27. Cu²⁺ ion can be reduced to Cu⁺¹ by addition of an aqueous solution of :-
 (a) KF (b) KCl
 (c) KI (d) KOH

Passage for Q.28 to Q.29

A brown black solid (A) on fusion with KNO₃ is KOH gives a green compound B and a colourless compound C. Green compound B in aqueous solution on electrolytic oxidation gives a violet compound D. KOH and H₂ gas. An aqueous solution (D) in cold reacts with ethylene to give glycol and its pink solution is discharged due to precipitation of A:-

28. A and B are :-
 (a) MnO₂, K₂MnO₄ (b) Cr₂O₃, K₂CrO₄
 (c) MnO₂, KMnO₄ (d) Cr₂O₃, K₂Cr₂O₇
29. C and D are :-
 (a) NO, K₂MnO₄ (b) KNO₂, K₂MnO₄
 (c) NO, KMnO₄ (d) NO, K₂Cr₂O₇

Passage for Q.30 to Q.32

An aqueous solution of a salt upon analysis gives the following results :

- (i) It gives a white precipitate with BaCl₂ solution insoluble in water.
 (ii) Addition of excess KI gives a brown precipitate which turns starch-iodide complex blue black.
 (iii) It gives a chocolate brown coloured precipitate with potassium ferrocyanide solution.

30. What inference can be drawn from (i) above ?

- (a) Presence of SO_4^{2-} ion
 (b) Presence of Pb^{2+} ion
 (c) Presence of Zn^{2+} ion
 (d) Presence of SO_3^{2-}

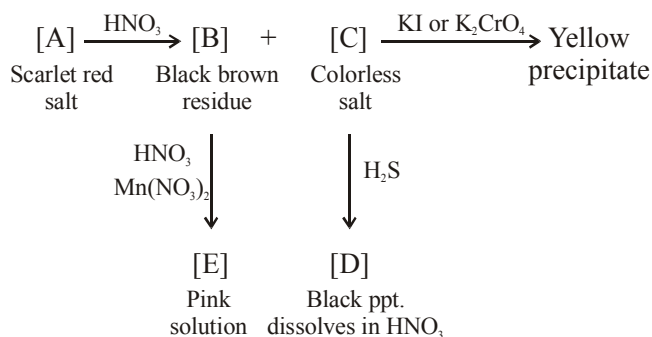
31. What is the formula of chocolate brown precipitate:-

- (a) $\text{Cu}[\text{Fe}(\text{CN})_4]$ (b) $\text{Cu}_2[\text{Fe}(\text{CN})_6]$
 (c) $\text{Cu}_3[\text{Fe}(\text{CN})_6]_2$ (d) Cu_2I_2

32. From the above inference, the formula of salt is :-

- (a) CuSO_4 (b) CuSO_3
 (c) CuCl_2 (d) $\text{Cu}(\text{OH})_2$

Passage for Q.33 to Q.35



33. The compound [A] is

- (a) An oxide (b) A chromate
 (c) A manganate (d) Chloride

34. The correct formula of compound (E) is

- (a) KMnO_4 (b) H_2MnO_4
 (c) HMnO_4 (d) K_2MnO_4

35. The yellow ppt. formed by reaction of [C] with K_2CrO_4 is also given by (other than that in C)

- (a) Pb^{2+} (b) Ca^{2+}
 (c) Ba^{2+} (d) Na^+

Passage for Q.36 to Q.38

The halides of silver are sensitive to light. This concept is utilised in photography.

36. A photographic plate consist of -

- (a) Sensitive emulsion of fine particles of AgF
 (b) Sensitive emulsion of fine particles of AgCl
 (c) Sensitive emulsion of fine particles of AgBr
 (d) Sensitive emulsion of fine particles of AgI

37. Hypo solution is used in photography to dissolve

- (a) Undissociated silver halide to give complex $[\text{Ag}(\text{S}_2\text{O}_3)_2]^{3-}$
 (b) Undissociated silver halide to give complex $[\text{Ag}(\text{S}_2\text{O}_3)]^{-1}$
 (c) Metallic silver to give complex $[\text{Ag}(\text{S}_2\text{O}_3)_2]^{3-}$
 (d) Metallic silver to give complex $[\text{Ag}(\text{S}_2\text{O}_3)]^{-1}$

38. The developer used in photography is

- (a) $\text{K}_2[\text{Fe}^{\text{II}}(\text{CO}_3)_2]$ (b) $\text{K}_2[\text{Fe}^{\text{II}}(\text{C}_2\text{O}_4)_2]$
 (c) $\text{K}[\text{Fe}^{\text{III}}(\text{C}_2\text{O}_4)_2]$ (d) $\text{K}[\text{Fe}^{\text{III}}(\text{CO}_3)_2]$

Matrix Match Type :

39. Match the column :-

Column - I

Ion

- (A) CrO_4^{2-}
 (B) $\text{Cr}_2\text{O}_7^{2-}$
 (C) MnO_4^{2-}
 (D) MnO_4^-

Column - II

Colour

- (P) Purple/Pink solution
 (Q) Green solution
 (R) Yellow
 (S) Orange

40. Match the column :-

Column - I

Compound

- (A) Cr_2O_3
 (B) CrO_3
 (C) CrO_5
 (D) CrO_2

Column - II

Colour

- (P) Green colour
 (Q) Red colour
 (R) Intense blue colour
 (S) Black colour

41. Match the column :-

Column - I

Compound

- (A) Mn_2O_7
 (B) MnO_2
 (C) MnO
 (D) Mn_3O_4

Column - II

Feature

- (P) Black colour ; most important ore of Mn
 (Q) Dark green liquid as explosive
 (R) Red oxide of manganese
 (S) Olive green powder, formed by decomposition by MnCO_3
 (T) Anhydride of HMnO_4

42. Match the column :-

Column - I	Column - II
Elements	Structure type
(A) W	(P) Transition elements
(B) Co	(Q) Hexagonal close packed
(C) Zn	(R) Body centered cubic
(D) Cd	(S) Cubic close packed
	(T) Typical metal structure

43. Match the column :-

Column - I

Conversion

- (A) $\text{MnO}_4^{2-} \longrightarrow \text{MnO}_4^-$
 (B) $\text{CrO}_4^{2-} \longrightarrow \text{Cr}_2\text{O}_7^{2-}$
 (C) $\text{CrO}_7^{2-} \longrightarrow \text{CrO}_4^{2-}$
 (D) $\text{MnO}_4^- \longrightarrow \text{MnO}_4^{2-}$

Column - II

Feature

- (P) Oxidation state remain unchanged during conversion
 (Q) $\text{pH} < 7$, conversion is carried out
 (R) $\text{pH} > 7$, conversion is carried out
 (S) Oxidation state changes during conversion
 (T) Electrolytic oxidation in alkaline medium

44. Match the column :-

Column - I

Property

- (A) Least 1st ionisation enthalpy
 (B) Highest 1st ionisation enthalpy
 (C) Highest 2nd ionisation enthalpy
 (D) Least ionic radii in M^{2+} ion

Column - II

Transitional element

- (P) Sc
 (Q) Zn
 (R) Cu
 (S) Ni

45. Match the Coloumn

Column - I	Column - II
Complex	Transitional elements
(A) $[\text{M}(\text{CN})_2]^{-x}$	(P) $x = 1$; Ag, Au
(B) $[\text{M}(\text{CN})_4]^{-x}$	(Q) $x = 3$; Mn, Fe, Co
(C) $[\text{M}(\text{CN})_4]^{-x}$	(R) $x = 2$; Zn, Cd, Hg
(D) $[\text{M}(\text{CN})_6]^{-x}$	(S) $x = 3$; Cu only

46. **Column - I**

Reaction

- (A) $\text{Fe} + 2\text{HCl} \rightarrow$
 (B) $\text{Fe} + \text{Conc. H}_2\text{SO}_4 \rightarrow$
 (C) $\text{Fe} + \text{HNO}_3$ (warm & concentrated) $\rightarrow$
 (D) $\text{Fe} + \text{HNO}_3$ (cold & dilute) $\rightarrow$

Column - II

Product nature

- (P) ferrous salt
 (Q) ferric salt
 (R) colourless, odourless gas
 (S) colourless, paramagnetic gas
 (T) Pungent odour gas

47. Match the column :-

Column - I

Reaction

- (A) $\text{N}_2\text{H}_4 + \text{CuO} \rightarrow$
 (B) $\text{Quinol} + \text{AgBr(s)} \rightarrow$
 (C) $\text{FeSO}_4(\text{s}) \xrightarrow{\Delta}$
 (D) $\text{CaCl}_2(\text{s}) + \text{K}_2\text{Cr}_2\text{O}_7(\text{s}) + \text{H}_2\text{SO}_4(\text{conc.}) \rightarrow$

Column - II

Feature

- (P) Reddish brown vapour is formed
 (Q) One of the product is +6 oxidation state
 (R) Redox reaction
 (S) One of the product is acidic oxide
 (T) Photograpy

48. Match the column :-

Column - I

Reaction

- (A) Turnbull's Blue pigment
- (B) Prussian Blue pigment
- (C) Brown ring
- (D) $\text{Na}_4[\text{Fe}(\text{CN})_5\text{NOS}]$

Column - II

Feature

- (P) Oxidation state of iron in coordination sphere $\text{Fe}_3[\text{Fe}(\text{CN})_6]_2$ is +1
- (Q) Oxidation state of iron in coordination sphere $\text{Fe}_4[\text{Fe}(\text{CN})_6]_3$ is +2
- (R) Oxidation state of iron in coordination sphere $[\text{Fe}(\text{H}_2\text{O})_5\text{NO}]\text{SO}_4$ is +3
- (S) Iron in anionic part of complex
- (T) Iron in cationic part of complex

49. Match the column :-

Column - I

Column - II

- | | |
|-----------------------------|--|
| (A) $\text{Cr} < \text{W}$ | (P) stability of the complex in +2 oxidation state |
| (B) $\text{Mn} < \text{Fe}$ | (Q) melting point |
| (C) $\text{Zn} < \text{Cu}$ | (R) magnetic moment |
| (D) $\text{Cd} < \text{Hg}$ | (S) tendency to form metal-metal bond |
| | (T) ionisation energy |

50. Match the column :-

Column - I

Column - II

- | | |
|---------------------------------------|---|
| (A) $\text{Fe}^{2+} > \text{Fe}^{3+}$ | (P) Crystal field splitting energy with strong ligand |
| (B) $\text{Pt}^{2+} > \text{Ni}^{2+}$ | (Q) Complexing tendency with dipyrldyl |
| (C) $\text{Mn}^{2+} > \text{Cr}^{2+}$ | (R) Tendency to form complex with CO |
| (D) $\text{Cu}^{2+} > \text{Cu}^+$ | (S) Magnetic moment |
| | (T) ionic radius |

51. Match the column :-

Column - I

Column - II

- | | |
|-------------------------|---|
| (A) FeCl_3 | (P) Oxidising agent and as mordant in dyeing |
| (B) FeF_3 | (Q) Exists as Fe^{3+} |
| (C) FeBr_3 | (R) d^5 ion |
| (D) FeO_4^{2-} | (S) Stable in strongly alkaline aqueous medium |
| | (T) Gives prusian blue colour with $\text{K}_4[\text{Fe}(\text{CN})_6]$ |

52. Match the column :-

Column - I

Column - II

- | | |
|----------------------|---|
| (A) Fe^{3+} | (P) Precipitate with NH_4OH |
| (B) Fe^{2+} | (Q) Precipitate with NaOH |
| (C) Cu^{2+} | (R) Complex with KCN (excess) |
| (D) Ag^+ | (S) Complex with hypo solution |
| | (T) Hydrometallurgical extraction |

53. Match the column :-

Column - I

- (A) Pb^{2+} gives yellow precipitate with
 - (B) Bi^{3+} gives black precipitate with
 - (C) Hg_2Cl_2 gives black precipitate with
 - (D) CuSO_4 gives deep blue colour with
- Column - II**
- (P) KI
 - (Q) K_2CrO_4
 - (R) NH_4OH
 - (S) Na_2SnO_2 in alkaline medium

54. Match the column :-

Column - I

Column - II

- | | |
|----------------------|--|
| (A) Cu^{2+} | (P) Form amphoteric oxide |
| (B) Zn^{2+} | (Q) Diamagnetic and colourless compounds |
| (C) Cr^{3+} | (R) Form complex with NH_3 |
| (D) Sc^{3+} | (S) Form colourless tetrahedral complex with KCN |
| | (T) Form inner orbital octahedral complex with NH_3 |

55. Match the column

Column - I

- (A) $\text{Hg}_2\text{Cl}_2 \xrightarrow{\text{NH}_3\text{OH}}$ Product/is
 (B) $\text{Hg}_2\text{Cl}_2 + \text{SnCl}_2 \longrightarrow$ Product/ is
 (C) $\text{Hg}^{+2} + \text{SnCl}_2 \longrightarrow$ Product/ is
 (D) $\text{HgCl}_2 + \text{Cu} \longrightarrow$ Product/ is

Column - II

- (P) Hg
 (Q) $\text{Hg}(\text{NH}_2)\text{Cl}$
 (R) Oxidation
 (S) Metal-metal bond in product
 (T) Reduction

EXERCISE # IV**Integer Type :**

- At what pH chromate-dichromate changes :-
- Find out the n factor of $\text{Cr}_2\text{O}_7^{2-}$ in acidic medium?
- MnO_4^- when oxidises other compounds in acidic medium, it is itself converted into Mn^{2+} . The molecular weight of MnO_4^- is x times of its equivalent weight. The value of x :-
- Find out number of electrons present in 4f orbital in Gadolinium. (At. no. 64) ?
- Find out number of electrons present in 5d orbital in Ytterbium (At. no. 70) ?
- Find out number of electrons present in 6s orbital in leutetium (At. no. 71) ?
- How many statements are correct ?
 - Permanganate titration in presence of hydrochloric acid is unsatisfactory
 - Mn^{2+} compounds are more stable than Fe^{2+} toward oxidation to their +3 state
 - Ammoniacal AgNO_3 converts glucose to gluconic acid and metallic silver is precipitated
 - Titanium and copper both in the first series of transition metals exhibits +1 oxidation stae most frequently.
 - Anhydrous ferric chloride is formed by heating metallic iron in a stream of dry chlorine gas.
- Find out the number of ions in solution which are colourless :-
 Ti^{4+} ; Ti^{3+} ; Cu^{+} ; Zn^{2+}
- The oxidation state of Mn in the product of oxidative fusion of pyrolusite ore in basic medium :-
- The oxidation state of Cr in the product of oxidative fusion of chromite ore in basic medium :-
- TiCl_4 is example of colourless liquid at room temperature.If it reacts with Zn followed by addition of water molecule give purple colour compound X. What is the number of unpaired electron present in X.
- A metal A which is strongly attracted by a magnet is attacked slowly by HCl liberating a gas and producing a blue solution. The addition of water of this solution causes it to burn pink. What is the atomic number of A. A belong to 3d series element:-
- Some transitional metals are used as traditional coinage metals in ancient world. Find out number of coinage metals.
- When 0.25 mole of hydrated ferric chloride ($\text{FeCl}_3 \cdot 6\text{H}_2\text{O}$) is made anhydrous by reacting with 1.5 mole of 2,2-dimethoxy propane :-
- What is the number of moles of SnCl_2 required for reduction of 1 mole of $\text{K}_2\text{Cr}_2\text{O}_7$ into Cr^{3+} :-
- Peacock ore is Cu_xFeS_4 . The value of x ?
- The highest oxidation state exhibited among all transitional element :-
- How many statements are correct :-
 - Metre scales are made up of invar alloy
 - Rusting iron is due to formation of $\text{Fe}_2\text{O}_3 + \text{Fe}(\text{OH})_3$
 - Transitional elements in positive oxidation state can act as Lewis acid.
 - Bordaux mixture is used as fungicide. It is a mixture of $\text{CuSO}_4 + \text{Ca}(\text{OH})_2$
 - Ce^{4+} (ceric ion) is used as an oxidising reagent in volumetric analysis

19. How many statements are incorrect :-

- Acidified KMnO_4 can be decolorised by $\text{Fe}_2(\text{SO}_4)_3$
- KMnO_4 does not act as oxidising reagents in alkaline medium
- Among ZnCl_2 , CdCl_2 , HgCl_2 maximum covalent character is shown by ZnCl_2
- Promethium is naturally found element

20. Number of moles of CrO_3 required to oxidise 9 moles of 2 butanol ?

21. An ornamental gold has 75% gold. It is x carat. The value of x :-

22. What is the basicity of Perhenic acid :-

23. A metal ion M^{2+} gives cherry blood red colour with DMG and complex formed is diamagnetic in nature. Which group of qualitative analysis metal will belong -

24. $\text{CuSO}_4 + \text{KCN} \xrightarrow{\text{(excess)}}$ Complex of Cu. The spin only magnetic moment of the complex.

25. How many reagents can be used to distinguish ferrous salt and ferric salt. :-



26. A metal ion M^{3+} gives brown red colour with $\text{K}_3[\text{Fe}(\text{CN})_6]$. With which group it may belong?

27. $\text{S}_2\text{O}_3^{2-} + \text{CN}^- \rightarrow \text{X} + \text{SO}_3^{2-} \xrightarrow{\text{Fe}^{3+}}$ Blood red colour $\xrightarrow[\text{excess}]{\text{NaF}}$ colourless complex

The coordination number of Fe^{3+} in colourless complex -

28. The general electronic configuration of transition element is $(n-1)d^{x-y}ns^{z-w}$

x = minimum number of electron present in d-orbital

y = maximum number of electron present in d-orbital

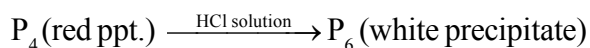
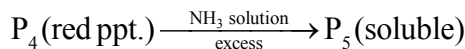
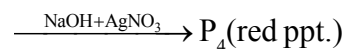
z = minimum number of electron present in s-orbital

w = maximum number of electron present in s-orbital

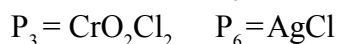
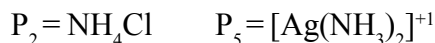
Find out the value of $y - z$:-

29. Rust consist of hydrated iron oxide. What is the oxidation state of iron in rust.

30. P_1 (imparts violet colour in flame test) $\xrightarrow[\text{conc. H}_2\text{SO}_4]{\text{comp. P}_2}$ P_3 (reddish brown gas)



How many products are correctly mentioned -



EXERCISE # V(A) JEE-MAIN

1. Which of the following arrangements does not represent the correct order of the property stated against it? [AIEEE 13]

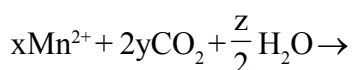
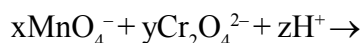
(1) $\text{V}^{2+} < \text{Cr}^{2+} < \text{Mn}^{2+} < \text{Fe}^{2+}$: paramagnetic behaviour

(2) $\text{Ni}^{2+} < \text{Co}^{2+} < \text{Fe}^{2+} < \text{Mn}^{2+}$: ionic size

(3) $\text{Co}^{3+} < \text{Fe}^{3+} < \text{Cr}^{3+} < \text{Sc}^{3+}$: stability in aqueous solution

(4) $\text{Sc} < \text{Ti} < \text{Cr} < \text{Mn}$: number of oxidation states

2. Consider the following reaction : [AIEEE 13]



The values of x, f and z in the reaction are respectively :-

(a) 5, 2 and 16

(b) 2, 5 and 8

(c) 2, 5 and 16

(d) 5, 2 and 8

3. Arrange Ce^{3+} , La^{3+} , Pm^{3+} and Yb^{3+} in increasing order of their ionic radius :- [AIEEE 02]

(a) $\text{Yb}^{3+} < \text{Pm}^{3+} < \text{Ce}^{3+} < \text{La}^{3+}$

(b) $\text{Ce}^{3+} < \text{Yb}^{3+} < \text{Pm}^{3+} < \text{La}^{3+}$

(c) $\text{Yb}^{3+} < \text{Pm}^{3+} < \text{La}^{3+} < \text{Ce}^{3+}$

(d) $\text{Pm}^{3+} < \text{La}^{3+} < \text{Ce}^{3+} < \text{Yb}^{3+}$

4. The radius of La^{3+} is $1.06 \text{ \AA}$, which of the following given values will be closest to the radius of Lu^{3+} (At. no. of Lu = 71, La = 57) [AIEEE 03]
 (a) $1.6 \text{ \AA}$ (b) $1.4 \text{ \AA}$
 (c) $1.06 \text{ \AA}$ (d) $0.85 \text{ \AA}$
5. The atomic number of vanadium (V), chromium (Cr), manganese (Mn) and iron (Fe) are 23, 24, 25 and 26 respectively. Which one of these may be expected to have the highest second ionisation enthalpy - [AIEEE 03]
 (a) V (b) Cr
 (c) Mn (d) Fe
6. What would happen a solution of potassium chromate is treated with an excess of dilute nitric acid :- [AIEEE 03]
 (a) Cr^{3+} and $\text{Cr}_2\text{O}_7^{2-}$ are formed
 (b) $\text{Cr}_2\text{O}_7^{2-}$ and H_2O are formed
 (c) $\text{Cr}_2\text{O}_7^{2-}$ is reduced to +3 state of Cr
 (d) $\text{Cr}_2\text{O}_7^{2-}$ is oxidised to +7 state of Cr
7. Among the following series of transition metal ions are ore where all metal ions have $3d^2$ electronic configuration is :- [AIEEE - 04]
 (a) Ti^{+3} , V^{2+} , Cr^{3+} , Mn^{4+}
 (b) Ti^{+} , V^{4+} , Cr^{6+} , Mn^{7+}
 (c) Ti^{4+} , V^{4+} , Cr^{2+} , Mn^{3+}
 (d) Ti^{2+} , V^{3+} , Cr^{4+} , Mn^{5+}
8. Cerium ($Z = 58$) is an important member of the lanthanoids. Which of the following statements about cerium is incorrect :- [AIEEE 04]
 (a) Cerium (IV) acts as an oxidising agent
 (b) The +3 oxidation state of cerium is more stable than the +4 oxidation state
 (c) The +4 oxidation state of cerium is not known in solutions
 (d) The common oxidation states of cerium are +3 and +4
9. The aqueous solution containing which one of the following ions will be colourless? [AIEEE 05] (Atomic number : Sc = 21, Fe = 26, Ti = 22, Mn = 25)
 (a) Sc^{3+} (b) Fe^{2+}
 (c) Ti^{3+} (d) Mn^{2+}
10. The lanthanoid contraction is responsible for the fact that :- [AIEEE 05]
 (a) Zr and Y have about the same radius
 (b) Zr and Nb have similar oxidation state
 (c) Zr and Hf have about the same radius
 (d) Zr and Zn have similar oxidation state
11. Lanthanoid contraction is caused due to :- [AIEEE 06]
 (a) the same effective nuclear charge from Ce to Lu
 (b) the imperfect shielding on outer electrons by 4f electrons from the nuclear charge
 (c) the appreciable shielding on outer electrons by 4f electrons from the nuclear charge
 (d) the appreciable shielding on outer electrons by 5d electrons from the nuclear charge
12. The actinoids exhibit more number of oxidation states in general than the lanthanoids. This because :- [AIEEE 07]
 (a) the 5f-orbitals are more buried than the 4f-orbitals
 (b) there is a similarity between 4f and 5f in their angular part of the wave function
 (c) the actinoids are more reactive than the lanthanoids
 (d) the 5f-orbitals extend further from the nucleus than the 4f-orbitals
13. Identify the incorrect statement among the following :- [AIEEE 07]
 (a) d-block elements show irregular and erratic chemical properties among themselves
 (b) La and Lu have partially filled d-orbitals and no other partially filled orbitals
 (c) the chemistry of various lanthanoids is very similar
 (d) 4f and 5f-orbitals are equally shielded
14. In context with the transition elements, which of the following statements is incorrect :- [AIEEE 09]
 (a) In the highest oxidation states of the first five transition elements (Sc to Mn), all the 4s and 3d electrons are used for bonding
 (b) Once the d^5 configuration is exceeded, the tendency to involve all the 3d electrons in bonding decreases.

- (c) In addition to the normal oxidation states, the zero oxidation state is also shown by these elements in complexes.
- (d) In the highest oxidation state, the transition metal show basic character and form cationic complexes.
15. Knowing that the chemistry of lanthanoids (Ln) is dominated by its +3 oxidation state, which of the following statements is incorrect? [AIEEE 09]
- (a) Ln(III) compounds are generally colourless
- (b) Ln(III) hydroxides are mainly basic in character
- (c) Because of the large size of the Ln(III) ions the bonding in its compounds is predominantly ionic in character
- (d) The ionic sizes of Ln(III) decrease in general with increasing atomic number
16. The correct order of $E_{M^{2+}/M}^\circ$ values with negative sign for the four successive elements Cr, Mn, Fe and Co is :- [AIEEE 2010]
- (a) Cr > Mn > Fe > Co
- (b) Mn > Cr > Fe > Co
- (c) Cr > Fe > Mn > Co
- (d) Fe > Mn > Cr > Co
17. In content of the lanthanoids, which of the following statements is not correct :- [AIEEE 2011]
- (a) because of similar properties the separation of lanthanoids is not easy
- (b) availability of 4f electrons results in the formation of compounds in +4 state for all the members of the series
- (c) there is a gradual decrease in the radii of the members with increasing atomic number in the series
- (d) all the members exhibit +3 oxidation state
18. Which series of reactions correctly represents chemical relations related to iron and its compound? [Jee-Mains 2014]
- (a) $\text{Fe} \xrightarrow{\text{Cl}_2, \text{heat}} \text{FeCl}_3 \xrightarrow{\text{heat, air}} \text{FeO}$
- (b) $\text{Fe} \xrightarrow{\text{O}_2, \text{heat}} \text{Fe}_3\text{O}_4 \xrightarrow{\text{CO}, 600^\circ\text{C}} \text{FeO} \xrightarrow{\text{CO}, 700^\circ\text{C}} \text{Fe}$
- (c) $\text{Fe} \xrightarrow{\text{dil. H}_2\text{SO}_4} \text{FeSO}_4 \xrightarrow{\text{H}_2\text{SO}_4, \text{O}_2} \text{Fe}_2(\text{SO}_4)_3 \xrightarrow{\text{Heat}} \text{Fe}$
- (d) $\text{Fe} \xrightarrow{\text{O}_2, \text{heat}} \text{FeO} \xrightarrow{\text{dil. H}_2\text{SO}_4} \text{FeSO}_4 \xrightarrow{\text{Heat}} \text{Fe}$
19. The equation which is balanced and represents the correct product(s) is :- [Jee-Mains 2014]
- (a) $[\text{Mg}(\text{H}_2\text{O})_6]^{2+} + (\text{EDTA})^{4-} \xrightarrow{\text{excess NaOH}} [\text{Mg}(\text{EDTA})]^{2-} + 6\text{H}_2\text{O}$
- (b) $\text{CuSO}_4 + 4\text{KCN} \longrightarrow \text{K}_2[\text{Cu}(\text{CN})_4] + \text{K}_2\text{SO}_4$
- (c) $\text{Li}_2\text{O} + 2\text{KCl} \longrightarrow 2\text{LiCl} + \text{K}_2\text{O}$
- (d) $[\text{CoCl}(\text{NH}_3)_5]^+ + 5\text{H}^+ \longrightarrow \text{Co}^{2+} + 5\text{NH}_4^+ + \text{Cl}^-$

EXERCISE # V(B) JEE-ADVANCED

1. Anhydrous ferric chloride is prepared by : [JEE-2002]
- (a) heating hydrated ferric chloride at a high temperature in a stream of air
- (b) heating metallic iron in a stream of dry chlorine gas
- (c) reaction of ferric oxide with HCl
- (d) reaction of metallic iron with HCl
2. When MnO_2 is fused with KOH, a coloured compound is formed, the product and its colour is : [JEE-2003]
- (a) K_2MnO_4 , green
- (b) KMnO_4 , purple
- (c) Mn_2O_3 , brown
- (d) Mn_3O_4 , black

■■

HYDROGEN AND THE HYDRIDE

PROBLEMS BASED ON GIVEN TOPICS

- Electronic structure
- Position in the periodic table
- Abundance of hydrogen
- Preparation of hydrogen
- Properties of molecular hydrogen
- Isotopes of hydrogen
- Ortho and para hydroge
- Hydrides
 - Ionic or self-like hydrides
 - Covalent hydrides
- Metallic (or interstitial) hydrides
- Intermediate hydrides
- The hydrogen ion
- Acids and Bases
 - Arrhenius theory
 - Acid and bases in proton solvents
 - Bronsted - Lowry theory
 - Lewis theory
 - The lux-fluor definition

Hydrogen and The Hydride

EXERCISE # I

❑ Only one correct answer :

- An unreactive diatomic gas A (1 mole) was treated with 3 moles of H_2 in presence of catalyst to give another gas (b) which is basic in nature. Gas (b) on further oxidation in moist condition gives a compound (c) which is part of acid rain. Gas (a) and gas (b) are respectively.
 - Cl_2 and HCl
 - Br_2 and HBr
 - P_2 and PH_3
 - N_2 and NH_3
- The radioactive isotopes of hydrogen is :-
 - Tritium
 - Deuterium
 - Protium
 - Hydronium
- Electron precise hydrides are formed by combination of hydrogen with all :-
 - Elements of group 13
 - Elements of group 14
 - Elements of group 15
 - Elements of group 16
- Elements of group 15 form :-
 - Elements deficient hydrides
 - Elements rich hydrides
 - Elements precise hydrides
 - None
- Ionisation energy of hydrogen in $kJ\ mol^{-1}$ is :-
 - 520
 - 495
 - 1681
 - 1312
- Hydrolith is :-
 - CaH_2
 - BaH_2
 - SrH_2
 - BeH_2
- Heavy water (D_2O) is used :-
 - For preparation of polymer
 - For preparation of hydro carbon
 - In atomic reactor to carry out nuclear fission
 - None
- An ionic hydride of an alkali metal has significant covalent character and is almost unreactive towards oxygen and chlorine. It is used to prepare the synthesis of other useful hydrides which acts as reducing agent. The hydride is -
 - LiH
 - BeH_2
 - CaH_2
 - KH
- $$H_2O + NH_2^- \rightleftharpoons NH_3 + OH^- \quad \dots(i)$$

$$H_2O + CH_3CO_2H \rightleftharpoons CH_3CO_2^- + H_3O^+ \quad \dots(ii)$$

In reaction (i) and (ii) water acts as :-

 - acid & acid respectively
 - acid & base respectively
 - base & acid respectively
 - base & base respectively
- Electron precise hydrides are :-
 - Pyramidal
 - Trigonal planar
 - Tetrahedral
 - Bent
- H^+ in aqueous solution exists as :-
 - OH^-
 - $H-O-O^-$
 - H_3O^+
 - H_3O^{2+}
- When Na_2O_2 is treated with dilute H_2SO_4 , the products obtained are :-
 - Na_2O_2 & H_2O_2
 - Na_2SO_4 & O_2
 - Na_2SO_4 & H_2
 - Na_2SO_4 & H_2O_2

8.4 HYDROGEN AND THE HYDRIDE

13. Hydrogen combines with other elements by :-
 - (a) Loosing an electron only
 - (b) Gaining an electron only
 - (c) Sharing an electron only
 - (d) Loosing, gaining and sharing electron
14. Hydrogen is :-
 - (a) Colourless and odourless gas
 - (b) Colourless and pungent odour gas
 - (c) Coloured and odourless gas
 - (d) Coloured and pungent odour gas
15. Heavy water is :-
 - (a) H_2O^{18}
 - (b) D_2O
 - (c) Water with specific gravity 1
 - (d) Water at 0°C
16. Heavy water freezes as :-
 - (a) 0°C
 - (b) -3.8°C
 - (c) 3.8°C
 - (d) -5°C
17. Electrolysis of molten ionic hydride gives :-
 - (a) Hydrogen at cathode
 - (b) Hydrogen at anode
 - (c) Metal at anode
 - (d) No reaction
18. Which of the following represent a pair of covalent hydrides :-
 - (a) $\text{TiH}_{1.5-1.8}$; $\text{VH}_{0.56}$
 - (b) H_2S ; HF
 - (c) KH ; NaH
 - (d) SiH_4 ; RbH
19. Which of the following represent a pair of interstitial hydrides :-
 - (a) $\text{TiH}_{1.5-1.8}$; $\text{VH}_{0.56}$
 - (b) H_2S ; HF
 - (c) KH ; NaH
 - (d) SiH_4 ; RbH
20. Which has least boiling point :-
 - (a) Liquid CO_2
 - (b) Liquid O_2
 - (c) Liquid H_2
 - (d) Liquid N_2
21. Hydrogen can not reduced :-
 - (a) Heated CuO
 - (b) Heated Fe_2O_3
 - (c) Heated SnO_2
 - (d) Heated Al_2O_3
22. Which of the following can not be oxidised by H_2O_2 :-
 - (a) PbS
 - (b) $\text{KI} + \text{HCl}$
 - (c) Na_2SO_3
 - (d) O_3
23. Deionised water is obtained by passing hard water through :-
 - (a) Cation exchanger
 - (b) anion exchanger
 - (c) Both cation and anion exchanger
 - (d) Permutit process
24. Elements of which of the following groups in periodic table do not form compound with hydrogen :-
 - (a) Group 7,8,9
 - (b) Group 13
 - (c) Group 15,16,17
 - (d) Group 14
25. The correct order of ionic character among all alkaline earth hydride :-
 - (a) $\text{BeH}_2 < \text{CaH}_2 < \text{MgH}_2 < \text{BaH}_2 < \text{SrH}_2$
 - (b) $\text{BeH}_2 < \text{CaH}_2 < \text{MgH}_2 < \text{SrH}_2 < \text{BaH}_2$
 - (c) $\text{BeH}_2 < \text{CaH}_2 < \text{BaH}_2 < \text{MgH}_2 < \text{SrH}_2$
 - (d) $\text{SrH}_2 < \text{CaH}_2 < \text{MgH}_2 < \text{BaH}_2 < \text{BeH}_2$
26. Boiling point of H_2O_2 is more than H_2O . It is due to :-
 - (a) more intermolecular hydrogen bonding in H_2O_2 w.r.t. H_2O
 - (b) more intramolecular hydrogen bonding in H_2O_2 w.r.t. H_2O
 - (c) less polarity in H_2O_2 w.r.t. H_2O
 - (d) higher specific heat in H_2O_2 w.r.t. H_2O
27. Which can not be used to prepare H_2O_2 , when BaO_2 reacts with:-
 - (a) conc. H_2SO_4
 - (b) dilute H_2SO_4
 - (c) H_3PO_4
 - (d) CO_2 in H_2O
28. Marck perhydrol is the trade name of which chemical species ?
 - (a) H_2
 - (b) H_2O
 - (c) H_2O_2
 - (d) O_2
29. Which of the following can be soluble soap ?
 - (a) $\text{C}_{17}\text{H}_{35}\text{CO}_2\text{Na}$
 - (b) $(\text{C}_{17}\text{H}_{35}\text{CO}_2)_2\text{Mg}$
 - (c) $(\text{C}_{17}\text{H}_{35}\text{CO}_2)_2\text{Ca}$
 - (d) $\text{C}_{17}\text{H}_{35}\text{CO}_2\text{Li}$

30. Temporary hardness of water is caused by the presence of :-
 (a) insoluble bicarbonates of calcium, magnesium and iron in water
 (b) soluble bicarbonates of calcium, magnesium and iron in water
 (c) insoluble carbonates of calcium, magnesium and iron in water
 (d) soluble carbonates of calcium, magnesium and iron in water
31. A clear transparent liquid is taken in a glass. Which of the following will provide clue whether it is water or not
 (a) Adding a litmus paper
 (b) Adding few drops of the liquid over anhydrous copper sulphate
 (c) Adding few drops of the liquid over anhydrous Sodium chloride
 (d) By testing its smell
32. 
 A (parallel spin of protons) B (opposed spin of protons)
 (a) A is called ortho hydrogen and B is called para hydrogen
 (b) A is called para hydrogen and B is called ortho hydrogen
 (c) A is called ortho hydrogen and B is called meta hydrogen
 (d) A is called para hydrogen and B is called meta hydrogen
33. When zeolite, which is hydrated sodium aluminium silicate is treated with hard water, the sodium ion are exchanged with :-
 (a) H^+ ions (b) Ca^{2+} ions
 (c) SO_4^{2-} ions (d) OH^-
34. Polyphosphates are used as water softening agents because they :-
 (a) form soluble complexes with anionic species
 (b) precipitate anionic species
 (c) form soluble complexes with cationic species
 (d) precipitate cationic species
35. Which one of the following processes will produce permanent hard water :-
 (a) addition of Na_2SO_4 to water
 (b) saturation of water with $CaCO_3$
 (c) saturation of water with $MgCO_3$
 (d) saturation of water with $CaSO_4$
36. The reagent commonly used to determine hardness of water titrimetrically is :-
 (a) oxalic acid
 (b) disodium salt of EDTA
 (c) sodium citrate
 (d) sodium thiosulphate
37. In context with the industrial preparation of hydrogen from water gas ($CO + H_2$), which of the following is the correct statement :-
 (a) CO is oxidised to CO_2 with steam in the presence of a catalyst followed by absorption of CO_2 in alkali
 (b) CO and H_2 are fractionally separated using differences in their densities
 (c) CO is removed by absorption in aqueous Cu_2Cl_2 solution
 (d) H_2 is removed through occlusion with Pd
38. H_2 (Ni) can not reduce :-
 (a) RCN (b) $RCHO$
 (c) RCO_2H (d) $R-CH=CH-R$
39. Organic highly branched alkyl groups and highly polar inorganic groups are :-
 (a) hydrophobic and hydrophilic respectively
 (b) hydrophilic and hydrophobic respectively
 (c) hydrophilic only
 (d) hydrophobic only
40. Roshan heard that instructions were given to laboratory attendant to store a particular chemical in dark room. It must be kept away from the dust. This chemical is important for use in the pollution control treatment of domestic and industrial effluents. The chemical is
 (a) O_3 (b) H_2O_2
 (c) H_2O (d) H_2

8.6 HYDROGEN AND THE HYDRIDE

41. Which of the following statement is incorrect :-
- Hydrogen peroxide is stored in wax lined bottles
 - Hard water does not form lather with soap
 - Tritium is a strategic material for thermonuclear weapons (Hydrogen bomb)
 - None is incorrect
42. Which of the following reaction is an examples of used of water gas in the synthesis of other compounds?
- $\text{CH}_4(\text{g}) + \text{H}_2\text{O}(\text{g}) \xrightarrow[\text{(Ni)}]{1270\text{K}} \text{CO}(\text{g}) + \text{H}_2(\text{g})$
 - $\text{CO}(\text{g}) + \text{H}_2\text{O}(\text{g}) \xrightarrow[\text{(Catalyst)}]{673\text{K}} \text{CO}_2(\text{g}) + \text{H}_2(\text{g})$
 - $\text{C}_n\text{H}_{2n+2} + n\text{H}_2\text{O}(\text{g}) \xrightarrow[\text{(Ni)}]{1270\text{K}} n\text{CO} + (2n+1)\text{H}_2$
 - $\text{CO}(\text{g}) + 2\text{H}_2(\text{g}) \xrightarrow[\text{(Catalyst)}]{\text{Cobalt}} \text{CH}_3\text{OH}(\text{l})$
43. A mixture of two gases AsH_3 and SbH_3 is passed through a tube heated at the middle portion. It is called Marsh's Test. What is the observation and why?
- Black mirror of antimany is deposited far away from heated spot because of more thermal stability of SbH_3 than AsH_3
 - Black mirror of antimany is deposited nearer from heated spot because of less thermal stability of SbH_3 than AsH_3
 - No black deposit is formed
 - Brown colour gas SbH_3 evolved but no evolution of AsH_3 takes place.
44. Which has highest dipole moment ?
- H_2
 - CH_4
 - H_2O
 - H_2O_2
45. 2000 gm aqueous solution of CaCO_3 contains 20 gm of CaCO_3 . Concentration of the solution is
- 10 ppm
 - 100 ppm
 - 1000 ppm
 - 10000 ppm
46. Excess of KI and dilute H_2SO_4 were mixed in 50 ml H_2O_2 . The I_2 liberated requires 20 ml of 0.1 (N) $\text{Na}_2\text{S}_2\text{O}_3$. The strength of H_2O_2 in gm/lit will be
- 0.12
 - 0.24
 - 0.39
 - 0.52

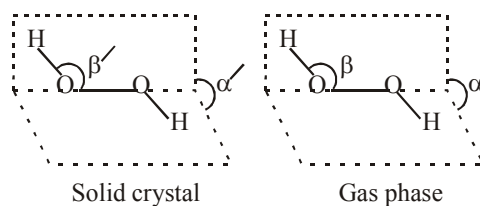
EXERCISE # II**□ One or More Than One Correct Answer :**

- Elements of group 13 form :-
 - Hydrides which are electron deficient
 - Hydrides which are electron rich
 - Hydrides which have incomplete octet
 - Hydrides which have complete octet
- What are the characteristic features of ionic hydrides :-
 - ionic hydrides do not conduct electricity in solid state
 - ionic hydrides are very good conductors of electricity in solid state
 - ionic hydrides conduct electricity in aqueous solution
 - metallic hydrides are example of ionic hydrides
- Hydrogen is released by the action of cold dilute HNO_3 on :-
 - Fe
 - Mn
 - Cu
 - Mg
- Which of the followings are example of saline hydrides :-
 - NH_3
 - NaH
 - SiH_4
 - KH
- Which elements when combined with hydrogen give products where hydrogen has negative oxidation state :-
 - Li
 - S_8
 - N_2
 - Be
- In which of the following compound hydrogen has positive oxidation state :-
 - HBr
 - NH_3
 - BaH_2
 - HI
- Which oxidation state are exhibited by hydrogen in its compound :-
 - 0
 - +1
 - +2
 - 1

8. In what aspect, hydrogen molecule resemble alkali metals?
- Hydrogen molecules & alkali metals both are diatomic with almost same ionisation energy
 - Both elements can loose electron to form mono positive ion
 - When HX and MX are electrolysed, hydrogen (H_2) and M (metal) are generated at cathode
 - Both are diatomic
9. Which property of hydrogen molecule similar to halogen :-
- Hydrogen accepts an electron to form noble gas configuration like halogen
 - Hydrogen, halogen both are diatomic molecule
 - Hydrogen, halogen both are polar molecule
 - Hydrogen, halogen both are nonpolar molecule
10. Sodium reacts with which compound to give hydrogen gas :-
- Cyclopentadine
 - Ethyl alcohol
 - Diethyl ether
 - Ethane
11. Which of the following reaction give correct product :-
- $CaC_2 + 2D_2O \rightarrow C_2D_2 + Ca(OD)_2$
 - $CH_3MgX + D_2O \rightarrow CH_3D + (OD)MgX$
 - $CD_4 + Cl_2 \rightarrow CD_3Cl + DCl$
 - $H_3C - C \equiv C - CH_3 \xrightarrow{Li \text{ in } ND_3}$
-
12. Correct increasing order :-
- $BeH_2 < TiH_2 < CaH_2$ (electrical conductance)
 - $LiH < NaH < CsH$ (ionic character)
 - $F-F < D-D < H-H$ (bond dissociation energy)
 - $H_2O < NH_3 < PH_3$ (reducing property)
13. Which of the following statements are not true for hydrogen :-
- it forms a large number of ionic compound by loosing an electron
 - it has one electron in outermost shell
 - it can loose an electron to form a cation which can freely exists
 - it can loose an electron to form a cation which can not freely exists
14. Which reaction indicate reducing property of H_2O_2 :-
- $KIO_4 + H_2O_2 \rightarrow KIO_3 + H_2O + O_2$
 - $2Fe^{3+} + 2H^+ + H_2O_2 \rightarrow 2Fe^{2+} + 2H_2O + O_2$
 - $HOCl + H_2O_2 \rightarrow H_3O^+ + Cl^- + O_2$
 - $2MnO_4^- + 6H^+ + 5H_2O_2 \rightarrow 2Mn^{2+} + 8H_2O + 5O_2$
15. Which reaction indicate oxidising property of H_2O_2 :-
- $2[Fe(CN)_6]^{4-} + 2H^+ + H_2O_2 \longrightarrow 2[Fe(CN)_6]^{3-} + 2H_2O$
 - $2I^- + 2H^+ + H_2O_2 \longrightarrow I_2 + 2H_2O$
 - $Mn^{2+} + H_2O_2 \rightarrow Mn^{4+} + 2OH^-$
 - $PbS + 4H_2O_2 \rightarrow PbSO_4 + 4H_2O$
16. Correct option regarding H_2O .
- H_2O is a weak field ligand
 - H_2O can act as lewis base
 - H_2O is used as protic solvent
 - H_2O is a very important analytical reagent to identify acidic radical
17. F_2 when reacts with H_2O , the products obtained can be
- HF
 - O_2
 - O_3
 - F_2O
18. Which of the following process, correct name is given
- $RCH = CHR + H_2 \rightarrow RCH_2 - CH_2R$ hydrogenolysis
 - $RCH = CHR + O_3 \rightarrow RCHO + RCHO$ ozonolysis
 - $RCO_2R + H_2O \rightarrow RCO_2H + ROH$ hydrolysis
 - dehydrogenation
19. H^- and He both have $1s^2$ electronic configuration
- reactivity order $H^- = He$
 - Stability order $H^- = He$
 - reactivity order $H^- > He$
 - Stability order $H^- < He$

8.8 HYDROGEN AND THE HYDRIDE

20. The structure of H_2O_2 in solid crystal and gas phase is given as follows :-



Correction option

- (a) $\alpha' > \beta'$ (b) $\beta' > \alpha'$
 (c) $\alpha > \beta$ (d) $\beta > \alpha$

EXERCISE # III**❑ Linked Comprehension Type :****Passage for Q.1 to Q.3**

Binary compounds of the elements with hydrogen are called hydrides. The type of hydride which an element forms largely depend on its electronegativity. They are broadly of three types namely salt like, molecular and interstitial hydride.

- Correct option regarding salt like hydride.
 - They are formed by elements of very low electronegativity, e.g. the alkali and alkaline earth metals of group IA and IIA.
 - They are colourless solid formed by heating the metal in H_2 at temperature 150°C to 700°C .
 - They are ionic hydrides, contain hydride ion
 - All
- Correct option regarding molecular hydride.
 - They are formed by elements of higher electronegativity, e.g. non-metal of group IIIB to VIIB
 - They are covalent, usually volatile liquid
 - They are low melting and boiling point and are nonelectrolyte in the liquid state and in solution in nonpolar solvent.
 - All
- Correct option regarding interstitial hydride.
 - Their formula are non-stoichiometric for e.g., $\text{TiH}_{1.73}$; $\text{VH}_{0.6}$
 - A number of transition metals when heated to moderate temperature absorbs large amount of hydrogen form interstitial hydride.
 - The interstitial hydrides are less than the parent metals
 - All

Passage for Q.4 to Q.6

Nascent hydrogen is hydrogen at the moment of its generation. Nascent hydrogen is more reactive than ordinary hydrogen. It is more powerful reducing agent than ordinary hydrogen.

- Correct option regarding Nascent hydrogen.
 - Nascent hydrogen is in ionic state of hydrogen
 - Nascent hydrogen is in molecular state of hydrogen
 - Nascent hydrogen is in atomic state of hydrogen
 - Nascent hydrogen is in deactivated state of hydrogen
- Which reaction can generate Nascent hydrogen?
 - Devardas alloy + $\text{NaOH} \rightarrow$
 - Sodium amalgum + $\text{H}_2\text{O} \rightarrow$
 - Zinc + dilute $\text{H}_2\text{SO}_4 \rightarrow$
 - All
- Nascent hydrogen can reduce.
 - Yellow FeCl_3 solution
 - Pink solution of KMnO_4 , H^+
 - Potassium chlorate
 - All

Passage for Q.7 to Q.9

Cold water is rapidly decomposed by alkali metals Li, Na, K etc. The vigour of the reaction may be moderate by alloying the alkali metal with lead or using an amalgum of the metal.

- The reactivity of alkali metals towards water-
 - $\text{Li} > \text{Na} > \text{K} > \text{Rb} > \text{Cs}$
 - $\text{Cs} > \text{Rb} > \text{K} > \text{Na} > \text{Li}$
 - $\text{Na} > \text{K} > \text{Li} > \text{Rb} > \text{Cs}$
 - $\text{Cs} > \text{Rb} > \text{Li} > \text{K} > \text{Na}$
- Which alkaline earth metal does not react with cold water to give hydrogen gas ?
 - Be
 - Mg
 - Ca
 - Ba
- Red hot metal when reacts with steam give mixed oxide. Which metal can not give mixed oxide?
 - Zn
 - Mg
 - Both Zn and Mg
 - Fe

Passage for Q.10 to Q.12

Pure hydrogen peroxide is a clear pale blue, viscous liquid. It is soluble in water in all proportions, also soluble in ether but is insoluble in benzene. It slowly decomposes at ordinary temperature and readily when heated into water and oxygen evolving much heat.

10. Which of the following compound / object retards the decomposition of H_2O_2 -
 (a) Metals like gold (b) Dust particles
 (c) Phosphoric acid (d) Fe_3^{+}
11. Which of the following statement is not correct regarding H_2O_2 ?
 (a) H_2O_2 has dipole moment 1.02 Debye
 (b) H_2O_2 has boiling point 152°C Debye
 (c) O-O bond distance is $1.49\text{\AA}$
 (d) $\angle \text{HOO}$ bond angle is 97° in liquid H_2O_2
12. Which reagents can be used to distinguish H_2O_2 and O_3 ? (H_2O_2 participate in the reaction to change the colour but for O_3 , no reaction)
 (a) K_2CrO_4 solution + dil. H_2SO_4
 (b) KMnO_4 solution + dil. H_2SO_4
 (c) TiO_2 solution + dil. H_2SO_4
 (d) All

Passage for Q.13 to Q.15

Water is a hydride of oxygen. Water is liquid at ordinary temperature. The properties of liquid water are highly anomalous. Its specific heat, latent heat of fusion and vaporisation are all abnormally high. It boils at 100°C .

13. Individual water molecules are only existent in -
 (a) Solid state
 (b) Liquid state
 (c) Vapour state above 100°C
 (d) All
14. X ray study of crystalline structure of ice indicates each water molecule is tetrahedrally surrounded by the other four water molecule. What type of bonding exists ?
 (a) Two by H-bonding & two by normal covalent bonding
 (b) Three by H-bonding & one by normal covalent bonding

(c) One by H-bonding & three by normal covalent bonding

(d) All four by H-bonding

15. Soap (soluble sodium salt of $\xrightarrow[\text{(hard water)}]{+\text{CaCl}_2} 2\text{NaCl}$

palmitic acid)
 + P↓ (insoluble calcium palmitate)

P is -

- (a) $(\text{C}_{17}\text{H}_{35}\text{CO}_2)_2\text{Ca}$ (b) $(\text{C}_{15}\text{H}_{31}\text{CO}_2)_2\text{Ca}$
 (c) $(\text{C}_{17}\text{H}_{32}\text{CO}_2)_2\text{Ca}$ (d) None

Matrix Match Type :

1. Match the column :-

Column - I (Reaction)

- (a) Steam + Red hot coke $\xrightarrow{1000^\circ\text{C}}$
 (b) Steam + Red hot iron $\xrightarrow{600-850^\circ\text{C}}$
 (c) 20% Castic soda $\xrightarrow{\text{electrolysis}}$
 (d) CH_4 + steam $\xrightarrow{\text{high temperature}}$

Column - II (Process/Product)

- (P) Water gas process
 (Q) H_2 preparation (industrial)
 (R) Steam iron process
 (S) Oxide formed as biproduct
 (T) Steam hydrocarbon process

2. Match the column :-

Column - I**Product formed**

- (a) D_2O
 (b) H_2O_2
 (c) (H_2+CO_2)
 (d) O_3

Column - II**Type of reaction**

- (P) electrolysis of 50% H_2SO_4
 (Q) prolonged electrolysis of water (H_2O)
 (R) water gas shift reaction
 (S) auto oxidation of 2 ethyl anthraquinol
 (T) silent discharge of oxygen gas

3. Match the column :-

Column - I (Behave as)

- (a) H_2O_2 acts as reducing agent
 (b) H_2O_2 acts as oxidising agent
 (c) H_2 acts as reducing agent
 (d) H_2 acts as oxidising agent

8.10 HYDROGEN AND THE HYDRIDE**Column - II (With elements compounds/in medium)**

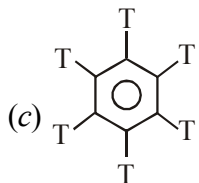
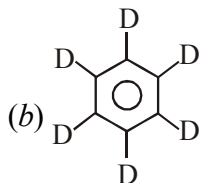
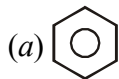
(P) with metals like Na, K, Ca

(Q) in acidic medium

(R) in basic medium

(S) with nonmetals like Cl_2

4. Match the column :-

Column - I (Molecules)(d) D_2O **Column - II (Features)**

(P) Polar molecule

(Q) dipole moment is zero

(R) most reactive towards nitration

(S) most reactive towards sulphonation

(T) dielectric constant less than water

5. Match the column :-

Column - I

(a) Use of hexametaphosphate

(b) Use of hydrogen peroxide

(c) Use of sodium borohydride

(d) Use of hydrogen (nickel)

Column - II

(P) Removal of permanent hardness

(Q) Reducing agent

(R) Hydrogenation

(S) Propellant

6. Match the column :-

Column - I**Molecules**(a) H_2O (b) H_2O_2 (c) D_2O (d) $\text{C}_2\text{H}_5\text{OH}$ **Column - II****Features**

(P) Polar solvent

(Q) Nonlinear molecule

(R) Less acidic than methanol

(S) Used in atomic reactor

7. Match the column :-

Column - I(a) $\text{H}_2 > \text{D}_2$ (b) $\text{D}_2 > \text{H}_2$ (c) $\text{H}_2\text{O} > \text{D}_2\text{O}$ (d) $\text{D}_2\text{O} > \text{H}_2\text{O}$ **Column - II**

(P) Boiling point

(Q) Freezing point

(R) Vapour pressure

(S) Latent heat of vaporisation at boiling point

(T) Latent heat of fusion at melting point

8. Match the column :-

Column - I

(a) Protium

(b) Deuterium

(c) Tritium

(d) Helium

Column - II

(P) one proton

(Q) two neutrons

(R) one neutron

(S) zero neutron

(T) two proton

9. Match the column :-

Column - I

(a) Hydrolith

(b) NaH (c) NH_3 (d) HI **Column - II**

(P) Hydrogen has oxidation state -1

(Q) Hydrogen has oxidation state +1

(R) Act as lewis base

(S) Act as bronsted acid

(T) Covalent molecule

10. Match the column :-

Column - I(a) $\text{H}_2\text{Te} > \text{H}_2\text{Se} > \text{H}_2\text{S}$ (b) $\text{HF} > \text{HI} > \text{HBr} > \text{HCl}$ (c) $\text{HF} > \text{HCl} > \text{HBr} > \text{HI}$ (d) $\text{HF} > \text{H}_2\text{O} > \text{NH}_3 > \text{CH}_4$

Column - II

- (P) Bond polarity order
 (Q) Acidic strength order
 (R) Boiling point order
 (S) Thermal stability order

11. Match the column :-

Column - I

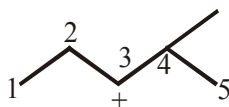
- (a) LiH
 (b) $(\text{BeH}_2)_n$
 (c) CH_4
 (d) $\text{ZrH}_{1.92}$

Column - II

- (P) Non stoichiometric
 (Q) Bridging bond present
 (R) Salt like hydride
 (S) Metallic hydride
 (T) Vander waals force of attraction exists between gaseous molecule

EXERCISE # IV**Integer Type :**

- Number of isotopes present in hydrogen -
- Number of reactions liberate hydrogen gas -
 - $\text{Zn} + \text{NaOH} \rightarrow$
 - $\text{Zn} + \text{dil H}_2\text{SO}_4 \rightarrow$
 - $\text{Zn} + \text{dil HCl} \rightarrow$
 - $\text{Ag} + \text{HNO}_3 \rightarrow$
 - $\text{Pb} + \text{NaOH} \rightarrow$
 - $\text{Sn} + \text{NaOH} \rightarrow$
 - $\text{Fe} + \text{Steam} \rightarrow$
- During carbocation rearrangement some times shift of Hydride ion takes place. From which carbon shift of hydride ion takes place.

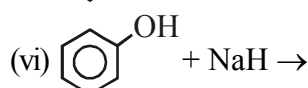
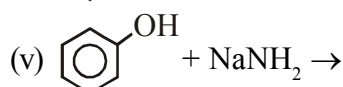
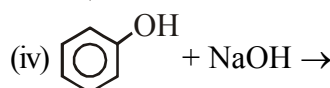
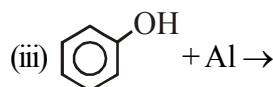
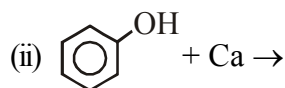
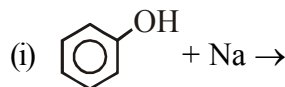


- Number of element of group 6 forms hydride -
- Number of compounds give H_2O_2 on reaction with dilute H_2SO_4
 PbO_2 ; MnO_2 ; TiO_2 ; $\text{BaO}_2 \cdot 8\text{H}_2\text{O}$; XeO_2
- Number of statements which are not true -
 - the electrolysis of molten hydrolith liberates H_2 gas at cathode.
 - Na_2SO_3 and KNO_2 when reacts with H_2O_2 , H_2O_2 acts as reducing agent.
 - In the reactions of H_2O_2 and Na_2CO_3 , H_2O_2 acts as acid.

(iv) H_2O and H_2O_2 both are ionic hydride of oxygen.

(v) H_2O_2 can be used as fuel as it produces pollution free atmosphere because its combustion product is water.

- Commercial 22.4 volume H_2O_2 solution has a molarity.
- 1.5% solution of H_2O_2 (1.5 gm H_2O_2 in 100 ml solution) is equivalent to x volume H_2O_2 . What is the value of x.
- $\text{CH}_3\text{COCH}_3 \xrightarrow{\text{OD}^-/\text{D}_2\text{O}}$ Number of hydrogen exchange by deuterium in the product.
- Find out number of reactions where phenol can generate hydrogen



- Find out number of hydrides which has more boiling point than H_2O .
 HF ; HCl ; HBr ; CH_4 ; NH_3 ; B_2H_6
- Find the number of atoms which are more electronegative than hydrogen atom.
 C; Si; N; I; Br; Be; Al; B
- Find out pH of water at 25°C .
- $\text{NiCl}_2 + 2\text{C}_6\text{H}_5\text{MgBr} \rightarrow \text{MgBr}_2 + \text{MgCl}_2 + \text{P}$
 $\text{P} + 2\text{H}_2 \rightarrow \text{Q} + \text{NiH}_2$
 Find out the number of π electrons.
- Number of elements which can adsorb large volume of hydrogen
 Ni; Pd; Pt; Zn; Hg; Cd



PERIODIC TABLE

PROBLEMS BASED ON GIVEN TOPICS

- The dual nature of electrons - particles of waves
 - The Heisenberg Uncertainty principle
 - The Schrodinger wave equation
 - Radial and Angular functions
 - Pauli exclusion principle
 - Build-up of the elements, Hund's rule
 - Arrangement of the elements in groups in the periodic table
 - Size of atoms and ions
 - Size of atoms
 - Size of ions
- Problems with ionic radii
 - Trends in ionic radii
 - Ionization energies
 - Electron affinity
 - Electronegativity
 - Pauling
 - Mulliken
 - Allred and Rochow
 - Metallic character
 - Variable valency and oxidation states
 - Horizontal, Vertical and Diagonal relationships in the periodic table.

Periodic Table

EXERCISE # I

❑ Only one correct answer :

- The correct order of first ionisation energies of the elements :-
 (a) $\text{Al} > \text{Ga}$ (b) $\text{P} > \text{S}$
 (c) $\text{Br} > \text{Cl}$ (d) $\text{Mg} > \text{Be}$
- The correct order of first ionisation energy of the elements :-
 (a) $\text{C} < \text{O} < \text{N}$ (b) $\text{C} < \text{N} < \text{O}$
 (c) $\text{O} < \text{N} < \text{C}$ (d) $\text{N} < \text{O} < \text{C}$
- The electronegativity order of carbon in different groups :-
 (a) $(\text{C})_{\text{CH}_3} < (\text{C})_{\text{CCl}_3} < (\text{C})_{\text{CF}_3}$
 (b) $(\text{C})_{\text{CF}_3} < (\text{C})_{\text{CCl}_3} < (\text{C})_{\text{CH}_3}$
 (c) $(\text{C})_{\text{CH}_3} < (\text{C})_{\text{CF}_3} < (\text{C})_{\text{CCl}_3}$
 (d) None
- The most basic compound out of following compound
 Methyl cyanide (a) ; Pyridine (B) ; Methyl amine (C) ; Aniline (D)
 (a) Methyl cyanide (A) (b) Pyridine (B)
 (c) Methyl amine (d) Aniline (D)
- The most electronegative element and least electronegative element are respectively :-
 (a) Cl, Na (b) Br, Li
 (c) I, H (d) F, Cs
- The 5th period of periodic table contains x no. of elements. The value of x:-
 (a) 2 (b) 8
 (c) 18 (d) 32
- The 6th period of periodic table contains x no. of elements. The value of x:-
 (a) 2 (b) 8
 (c) 18 (d) 32
- Which of the following order regarding electronegativity is correct ?
 (a) $\text{Mo(II)} > \text{Mo(III)} > \text{Mo(IV)} > \text{Mo(V)} > \text{Mo(VI)}$
 (b) $\text{Cl(VII)} > \text{Cl(V)} > \text{Cl(I)} > \text{Cl(III)}$
 (c) $\text{sp carbon} > \text{sp}^2 \text{ carbon} > \text{sp}^3 \text{ carbon}$
 (d) $\text{F} > \text{N} > \text{Cl} > \text{O}$
- Which of the following statement is correct ?
 (a) Be has first ionisation energy lower than that of boron
 (b) Mg has first ionisation energy lower than that of aluminium
 (c) From Zn to Hg, there is increase in ionisation energy
 (d) Among all noble gases, (group 18 elements) Xe has least ionisation energy
- Which noble gas has highest ionisation energy ?
 (a) He (b) Ne
 (c) Ar (d) Kr
- Correct order of metallic character is :-
 (a) $\text{Rb} > \text{K} > \text{Na} > \text{Li}$
 (b) $\text{K} > \text{Mg} > \text{Al} > \text{B}$
 (c) $\text{F} > \text{O} > \text{N} > \text{B}$
 (d) $\text{Ba} > \text{Be} > \text{Mg} > \text{B}$
- The most acidic oxide :-
 (a) SO_2 (b) SO_3
 (c) P_4O_{10} (d) BaO
- The most basic oxide :-
 (a) MgO (b) CaO
 (c) SrO (d) BaO

9.4 PERIODIC TABLE

14. The electronic configuration of an element is written as follows $1s^2 2s^2 2p^6 3s^2 3p^6 3d^7 4s^2$, on removal of one electron, the electronic configuration will become :-
 (a) $1s^2 2s^2 2p^6 3s^2 3p^5 3d^7 4s^2$
 (b) $1s^2 2s^2 2p^6 3s^2 3p^6 3d^6 4s^2$
 (c) $1s^2 2s^2 2p^6 3s^2 3p^6 3d^7 4s^1$
 (d) $7s^1 2s^2 2p^6 3s^2 3p^6 3d^7 4s^2$
15. The least polar bond :-
 (a) H-F (b) H-Cl
 (c) H-Br (d) H-I
16. The most polar bond :-
 (a) C-F (b) C-O
 (c) C-N (d) C-C
17. The highest melting point for oxide of alkaline earth metal is observed for :-
 (a) MgO (b) CaO
 (c) SrO (d) BaO
18. The ionisation potential of two element X_1 and X_2 are 400 and 300 kcal mol⁻¹ respectively. The electron affinities of X_1 and X_2 are 80 and 85 kcal mol⁻¹ respectively. Electronegativity order of X_1 and X_2 is :-
 (a) $X_1 > X_2$ (b) $X_1 = X_2$
 (c) $X_1 < X_2$ (d) Can not be predicted
19. The first ionisation energy of Al(g) is 577.5 kJ mol⁻¹. If second and third ionisation energy of Al are in the ratio 2 : 3 and if ΔH for $Al(g) \rightarrow Al^{3+}(g) + 3e^-$ is 5140 kJ mol⁻¹ then IE_2 and IE_3 of Al are respectively :-
 (a) 1825 kJ mol⁻¹, 2737.5 kJ mol⁻¹
 (b) 2000 kJ mol⁻¹, 3000 kJ mol⁻¹
 (c) 1200 kJ mol⁻¹, 1800 kJ mol⁻¹
 (d) 5075 kJ mol⁻¹, 2525 kJ mol⁻¹
20. Highest size will be of :-
 (a) Cl⁻ (b) Br
 (c) Br⁻ (d) Br⁺
21. The As-Cl bond distance in AsCl₃ is 2.2 Å. What is the single bond covalent radius of arsenic. (Assuming electronegativity of both to be same and radius of chlorine atom is 0.99 Å)
 (a) 0.24 Å (b) 0.77 Å
 (c) 1.21 Å (d) 3.19 Å
22. C is isomer of $PtCl_2(NH_3)_2 = A$
 Trans isomer of $PtCl_2(NH_3)_2 = B$
 If Pt-Cl bond distance is 2.32 Å, then Cl-Cl bond distance in A & B :-
 (a) 3.28 Å, 3.28 Å (b) 4.64 Å, 3.28 Å
 (c) 4.64 Å, 4.64 Å (d) 3.28 Å, 4.64 Å
23. Ionisation energy of AlCl₃ is 5137 kJ mol⁻¹
 Heat of hydration of Al³⁺ is -4665 kJ mol⁻¹
 Heat of hydration of Cl⁻ is -381 kJ mol⁻¹
 From the data, it is concluded that :-
 (a) AlCl₃ in aqueous solution do not conduct electricity
 (b) AlCl₃ in gaseous state conduct electricity due to presence of free ions.
 (c) AlCl₃ in aqueous solution conduct electricity due to presence of hydrated ions.
 (d) None of these
24. What is the electronegativity of Cl from the following data ?
 Bond energy of F₂ = 38 Kcal mol⁻¹
 Bond energy of Cl₂ = 58 Kcal mol⁻¹
 Bond energy of Cl-F = 61 Kcal mol⁻¹
 Electronegativity of F = 4
 (a) 3 (b) 2.8
 (c) 3.2 (d) 3.5
25. Which of the following does not reflect to periodicity of element ?
 (a) Electronegativity
 (b) Metallic behaviour
 (c) Neutron / Proton ratio
 (d) Ionisation potential
26. Two elements P and Q form 3 types of bonds P-P; Q-Q; P-Q. Their bond dissociation energies are respectively 81 kcal mol⁻¹; 64 kcal mol⁻¹; 76 kcal mol⁻¹. If electronegativity of Q is 2.4 then the electronegativity of P will be approximately. (P is less electronegative than Q)
 (a) 2.81 (b) 1.8
 (c) 1.99 (d) 1.33

27. If interionic distance between Na^+ and F^- ions is 2.31 Å then radii of Na^+ and F^- are :-
 (a) 0.95 Å & 1.36 Å (b) 1.155 Å & 1.155 Å
 (c) 1.36 Å & 0.95 Å (d) None of these
28. Moving from right to left in a periodic table, the atomic size is :-
 (a) increased (b) decreased
 (c) remains constant (d) None of these
29. The increasing order of electronegativity in the following elements :-
 (a) C, N, Si, P (b) N, Si, C, P
 (c) Si, P, C, N (d) P, Si, N, C
30. One element has atomic weight 39. Its electronic configuration is $1s^2, 2s^2, 2p^6, 3s^2, 3p^6, 4s^1$. The true statement for that element is :-
 (a) highest value of IE (b) transition element
 (c) isotone with $^{38}_{18}\text{Ar}$ (d) None of these
31. The number of paired electrons in oxygen is :-
 (a) 6 (b) 16
 (c) 8 (d) 32
32. The decreasing size of K^+ , Ca^{2+} , Cl^- & S^{2-} follows the order :-
 (a) $\text{K}^+ > \text{Ca}^{2+} > \text{S}^{2-} > \text{Cl}^-$
 (b) $\text{K}^+ > \text{Ca}^{2+} > \text{Cl}^- > \text{S}^{2-}$
 (c) $\text{Ca}^{2+} > \text{K}^+ > \text{Cl}^- > \text{S}^{2-}$
 (d) $\text{S}^{2-} > \text{Cl}^- > \text{K}^+ > \text{Ca}^{2+}$
33. Which of the following oxide is neutral ?
 (a) CO (b) SnO_2
 (c) ZnO (d) SiO_2
34. Which of the following has the maximum number of unpaired electrons ?
 (a) Mg^{2+} (b) Ti^{3+}
 (c) V^{3+} (d) Fe^{2+}
35. The following acids have been arranged in the order of decreasing acid strength. Identify the correct order :-
 ClOH(I) BrOH(II) IOH(III)
 (a) $\text{I} > \text{II} > \text{III}$ (b) $\text{II} > \text{I} > \text{III}$
 (c) $\text{III} > \text{II} > \text{I}$ (d) $\text{I} > \text{III} > \text{II}$
36. The incorrect statement among the following is :-
 (a) the first ionisation potential of Al is less than the first ionisation potential of Mg
 (b) the second ionisation potential of Mg is greater than the second ionisation potential of Na
 (c) the first ionisation potential of Na is less than the first ionisation potential of Mg
 (d) the third ionisation potential of Mg is greater than the third ionisation potential of Al
37. HOCl is example of acid. It can be explained by :-
 (a) The electronegativity difference between O and Cl is higher than electronegativity difference between H and O atom. So Cl – O bond will rupture.
 (b) The electronegativity difference between O and Cl is lower than electronegativity difference between H and O atom. So Cl – O bond will rupture.
 (c) The electronegativity difference between O and Cl is higher than electronegativity difference between H and O atom. So O – H bond will rupture.
 (d) The electronegativity difference between O and Cl is lower than electronegativity difference between H and O atom. So O – H bond will rupture.
38. Which of the following species is/are paramagnetic NO_2 , NO, N_2O_4 , N_2O_5 :-
 (a) Only NO_2 (b) NO_2 , NO
 (c) NO_2 , NO, N_2O_5 (d) All are paramagnetic
39. Bond order of CO_3^{2-} and CO^+ are respectively :-
 (a) 1.33, 2.5 (b) 1.5, 3
 (c) 1.75, 3.5 (d) None of these
40. Ionic radii of :-
 (a) $\text{Ti}^{4+} < \text{Mn}^{7+}$ (b) $^{35}\text{Cl}^- < ^{37}\text{Cl}^-$
 (c) $\text{K}^+ < \text{F}^-$ (d) $\text{P}^{3+} > \text{P}^{5+}$
41. Identify the correct order of acidic strengths of CO_2 , CuO, CaO, H_2O :-
 (a) $\text{CaO} < \text{CuO} < \text{H}_2\text{O} < \text{CO}_2$
 (b) $\text{H}_2\text{O} < \text{CuO} < \text{CaO} < \text{CO}_2$
 (c) $\text{CaO} < \text{H}_2\text{O} < \text{CuO} < \text{CO}_2$
 (d) $\text{H}_2\text{O} < \text{CO}_2 < \text{CaO} < \text{CuO}$

9.6 PERIODIC TABLE

42. Identify the least stable ion amongst the following:-
 (a) Li^- (b) Be^-
 (c) B^- (d) C^-
43. The set representing correct order of IP_1 is :-
 (a) $\text{K} > \text{Na} > \text{Li}$ (b) $\text{Be} > \text{Mg} > \text{Ca}$
 (c) $\text{B} > \text{C} > \text{N}$ (d) $\text{Fe} > \text{Si} > \text{C}$
44. The correct order of radii is -
 (a) $\text{N} < \text{Be} < \text{B}$ (b) $\text{F}^- < \text{O}^{2-} < \text{N}^{3-}$
 (c) $\text{Na} < \text{Li} < \text{K}$ (d) $\text{Fe}^{3+} < \text{Fe}^{2+} < \text{Fe}^{4+}$
45. Which set is expected to show the smallest difference in 1st ionisation energy?
 (a) Fe, Co, Ni (b) N, O, F
 (c) Ca, Ca^+ , Ca^{2+} (d) He, Ar, Xe
46. The formula of ferrous chloride is FeCl_2 . The formula of sodium phosphate is Na_3PO_4 . Then the formula of ferrous rphosphate
 (a) FePO_4 (b) $\text{Fe}_3(\text{PO}_4)_2$
 (c) $\text{Fe}_2(\text{PO}_4)_3$ (d) $\text{Fe}_3(\text{PO}_4)_4$
47. What is the anhydride of lime water?
 (a) CaO_2 (b) CaO
 (c) Ca (d) $\text{Ca}(\text{OH})_2$
48. What is the oxidation state of Mo in ammonium phospho molybdate?
 (a) +4 (b) +5
 (c) +6 (d) +2
49. Which configuration of metal belongs to lowest melting point?
 (a) $(n-1)d^{10}ns^2$ (b) $(n-1)d^{10}ns^1$
 (c) $(n-1)d^8ns^2$ (d) $(n-1)d^6ns^2$
50. Which element forms only one oxoacid?
 (a) F
 (b) Cl
 (c) I
 (d) N
51. The correct order of acidic strength of oxides
 (a) $\text{Na}_2\text{O} < \text{MgO} < \text{ZnO} < \text{P}_4\text{O}_{10}$
 (b) $\text{Al}_2\text{O}_3 < \text{SiO}_2 < \text{P}_4\text{O}_{10} < \text{SO}_3$
 (c) $\text{Cl}_2\text{O} < \text{ClO}_2 < \text{Cl}_2\text{O}_7$
 (d) All
52. The correct order of basic strength of oxide/hydroxide
 (a) $\text{BaO} > \text{SrO} > \text{CaO} > \text{MgO} > \text{BeO}$
 (b) $\text{Cs}_2\text{O} > \text{Rb}_2\text{O} > \text{K}_2\text{O} > \text{Na}_2\text{O} > \text{Li}_2\text{O}$
 (c) $\text{CsOH} > \text{Mg}(\text{OH})_2 > \text{Zn}(\text{OH})_2 > \text{HOCl}$
 (d) All
53. 3d series is started and ended with element -
 (a) Scandium and zinc respectively
 (b) Titanium and copper respectively
 (c) Titanium and mercury respectively
 (d) Scandium and mercury respectively
54. 4d series is started and ended with element :-
 (a) Yttrium & cadmium respectively
 (b) Ytterbium & cadmium respectively
 (c) Yttrium and mercury respectively
 (d) Scandium and mercury respectively
55. 4f series is started and ended with element :-
 (a) Cerium and lawrencium
 (b) Thorium and lawrencium
 (c) Cerium and lutetium
 (d) Thorium and lutetium respectively
56. 5f series is started and ended with element :-
 (a) Cerium and lawrencium
 (b) Thorium and lawrencium
 (c) Cerium and lutetium
 (d) Thorium and lutetium respectively
57. Element tend to lose two electrons :-
 (a) Mg (b) Ba
 (c) Ca (d) All
58. Element with five electrons in the outer subshell :-
 (a) Al (b) Si
 (c) Se (d) As
59. For beryllium, Be^{2+} is more stable than Be^{3+} from that which of the following options are correct?
 (a) $(I_2 - I_1)$ of Be is much less than $(I_3 - I_2)$ of Be.
 (b) $(I_2 - I_1)$ of Be is much more than $(I_3 - I_2)$ of Be.
 (c) $(I_2 - I_1)$ of Be is equal than $(I_3 - I_2)$ of Be.
 (d) $I_3 \ll I_2 < I_1$

60. Which elements has δ^+ charge in compound A, B, C respectively:-
 H-F ; Si-Cl ; C-O
 A B C
 (a) F, Cl, O (b) H, Si, C
 (c) H, Cl, O (d) H, Si, O
61. Ist ionisation enthalpy and IInd ionisation enthalpy of Mg are 78 and 348 kcal mol⁻¹ respectively. The ΔH for the process $\text{M(g)} \rightarrow \text{M}^{2+}(\text{g}) + 2\text{e}^-$ is:-
 (a) +170 kcal mol⁻¹ (b) +426 kcal mol⁻¹
 (c) -426 kcal mol⁻¹ (d) -170 kcal mol⁻¹
62. The largest species and smallest species among Mg, Mg²⁺, Al, Al³⁺ are respectively -
 (a) Mg, Mg²⁺ (b) Mg, Al³⁺
 (c) Al, Al³⁺ (d) Al, Mg²⁺
63. The first ionisation enthalpy values of the 3rd period elements Na, Mg and Si are respectively 496, 737, 786 kJ mol⁻¹. The first ionisation enthalpy of Al should be :-
 (a) less than 496 kJ mol⁻¹
 (b) more than 786 kJ mol⁻¹
 (c) more than 496 kJ mol⁻¹ but less than 737 kJ mol⁻¹
 (d) more than 737 kJ mol⁻¹ but less than 786 kJ mol⁻¹
64. The oxidation state of oxygen of two oxygen containing compounds OF₂ and Na₂O are respectively :-
 (a) +2, -2 (b) -2, -2
 (c) +2, +2 (d) +1, -2
65. The correct order of metallic behaviour :-
 (a) Na > Mg > Be > Si > P
 (b) Mg > Be > Na > Si > P
 (c) Be > Na > Mg > Si > P
 (d) P > Be > Si > Na > Mg
66. If C-C bond distance is 1.54 Å, C=C bond distance is 1.34 Å, C≡C bond distance is 1.2 Å. Then C-C bond distance in benzene is approximately equal to :-
 (a) 1.5 Å (b) 1.4 Å
 (c) 1.6 Å (d) 1.3 Å
67. For an element the spin only magnetic moment is 3.83 BM. The number of unpaired electron in the valence shell of this element :-
 (a) 1 (b) 2
 (c) 3 (d) 4
68. Which of the following process is exothermic ?
 (a) O(g) → O⁻(g) (b) O⁻(g) → O²⁻(g)
 (c) O(g) → O²⁻(g) (d) He(g) → He⁻(g)
69. Which of the following energy is not associated for the conversion $\text{M}^-(\text{g}) \rightarrow \text{M}^{4+}(\text{g})$:-
 (a) Ist ionisation energy of M(g)
 (b) IInd ionisation energy of M(g)
 (c) Ist electron gain enthalpy of M(g)
 (d) IInd electron gain enthalpy of M(g)
70. In the first ninety elements, the number of s-block elements is :-
 (a) 10 (b) 12
 (c) 14 (d) 16
71. Which of the following order is correct :-
 (a) F₂ > Cl₂ > Br₂ > I₂ (Bond dissociation energy)
 (b) F⁻ > Cl⁻ > Br⁻ > I⁻ (Stability)
 (c) F > Cl > Br > I (Reactivity)
 (d) F > Cl > Br > I (Electron gain enthalpy)
72. Select correct order regarding bond dissociation energy of hydrogen halide / hydrohalic acid :-
 (a) HF > HCl > HBr > HI
 (b) HCl > HBr > HF > HI
 (c) HI > HBr > HCl > HF
 (d) HF > HCl > HI > HBr
73. The Ist, IInd, IIIrd, IVth, Vth ionisation energy of an element are 7.1, 14.3, 34.5, 46.8 & 162.2 eV respectively. The element can be :-
 (a) K (b) Al
 (c) Cl (d) Si
74. In a period, the elements having least melting point:-
 (a) Pnictogens (b) Chalcogens
 (c) Halogens (d) Noble gas
75. In a period, the elements having highest atomic volume :-
 (a) Alkali metals (b) Alkaline earth metals
 (c) Halogens (d) Noble gases

9.8 PERIODIC TABLE

76. Covalent radius and Vanderwaal's radius of fluorine atom are respectively :-
 (a) 71 pm and 71 pm (b) 147 pm and 147 pm
 (c) 71 pm and 147 pm (d) 147 pm and 71 pm
77. If electronic configuration of A^- is $[Ar] 3d^{10} 4s^2 4p^6$, then which element has the configuration identical to the configuration of A^{2+} :-
 (a) Ge (b) Ga
 (c) As (d) Se
78. Decrease in size is maximum for which process :-
 (a) $O(g) \rightarrow O^+(g)$ (b) $O^-(g) \rightarrow O(g)$
 (c) $O^-(g) \rightarrow O^+(g)$ (d) All
79. Which of the following represents the correct order of increasing first ionisation enthalpy for Ca, Ba, S, Se and Ar ?
 (a) $Ba < Ca < Se < S < Ar$
 (b) $Ca < Ba < S < Se < Ar$
 (c) $Ca < S < Ba < Se < Ar$
 (d) $S < Se < Ca < Ba < Ar$
80. The first ionisation potential of Na is 5.1 eV. The value of electron gain enthalpy of Na^+ will be :-
 (a) -10.2 eV (b) +2.55 eV
 (c) -2.55 eV (d) -5.1 eV
81. Among the following oxoacids, the correct decreasing order of acid strength is :-
 (a) $HClO_4 > HClO_3 > HClO_2 > HOCl$
 (b) $HClO_2 > HClO_4 > HClO_3 > HOCl$
 (c) $HOCl > HClO_2 > HClO_3 > HOCl_4$
 (d) $HClO_4 > HOCl > HClO_2 > HClO_3$
82. Fluorine is the most reactive among all the halogens, because of its :-
 (a) high electron affinity
 (b) low dissociation energy of F-F bond
 (c) large size
 (d) high dissociation energy of F-F bond
83. In the long form of the periodic table, the valence shell electronic configuration of $5s^2 5p^4$ corresponds to the element present in :-
 (a) group 16 and period 5
 (b) group 17 and period 6
 (c) group 17 and period 5
 (d) group 16 and period 6
84. The correct order of non metallic character :-
 (a) $B > C > Si > N > F$
 (b) $Si > C > B > N > F$
 (c) $F > N > C > B > Si$
 (d) $F > N > C > Si > B$
85. The correct order of oxidising property :-
 (a) $F > Cl > O > N$ (b) $F > O > Cl > N$
 (c) $Cl > F > O > N$ (d) $O > F > N > Cl$
86. The correct order of ionisation enthalpy :-
 (a) $B > Al > Ga > In > Tl$
 (b) $B > Tl > Ga > Al > In$
 (c) $Tl > In > Ga > Al > B$
 (d) $Tl > Ga > B > Al > In$
87. Among the second period element, for which property the following order is observed :-
 $Li < B < Be < C < O < N < F < Ne$
 (a) electronegativity (b) covalent radius
 (c) ionisation energy (d) electron gain enthalpy
88. The correct order of metallic property :-
 (a) $B > Al > Mg > K$ (b) $Al > Mg > B > K$
 (c) $Mg > Al > K > B$ (d) $K > Mg > Al > B$
89. Electronegativity of nitrogen element is highest for which of its compound :-
 (a) NO (b) NH_3
 (c) NO_2 (d) N_2O_5
90. Formula of oxides of group 14 element will be :-
 (a) M_2O_5 (b) MO_3
 (c) MO_2 (d) M_2O_3
91. What is the oxidation state and covalency of Mg in Grignard reagent (alkyl magnesium halide) :-
 (a) +1, +1
 (b) +2, +2
 (c) +2, +4
 (d) +2, +3
92. Uuu and Uub is symbol of :-
 (a) Unununnium, Ununbium
 (b) Ununinium, Ununbium
 (c) Ununium, Unbium
 (d) Unununnim, Unununbium

93. The internuclear distance of Mg^{2+} and O^{2-} is 212 pm. The inter ionic distance of Mg^{2+} and S^{2-} is 256 pm. If radius of O^{2-} is 140 pm, what is the radius of S^{2-} ?
 (a) 72 pm (b) 184 pm
 (c) 82 pm (d) 116 pm
94. $(\text{Ionic radius})_{\text{O}^{2-}} = 126 \text{ pm}$ $(\text{Ionic radius})_{\text{Na}^+} = 116 \text{ pm}$. Ionic radius of F^- should be
 (a) 114 pm (b) 128 pm
 (c) 119 pm (d) 110 pm
95. $(\text{Ionic radius})_{\text{K}^+} = 152 \text{ pm}$ $(\text{Ionic radius})_{\text{Cl}^-} = 167 \text{ pm}$. Ionic radius of S^{2-} should be
 (a) 148 pm (b) 158 pm
 (c) 164 pm (d) 170 pm
96. After the filling of np orbitals, next orbital filled will be
 (a) ns (b) $(n-1)d$
 (c) $(n+1)s$ (d) nd
97. The element having atomic number 117 has not yet been discovered. It should be placed in :-
 (a) Group 14 (b) Group 15
 (c) Group 16 (d) Group 17
98. The element having atomic number 120 has not yet been discovered. It should be placed in :-
 (a) Group 1 (b) Group 2
 (c) Group 3 (d) Group 4
99. Anything that influences the valence electrons will affect the chemistry of the element. Which one of the following factors does not affect the valence shell?
 (a) Valence principal quantum number (n)
 (b) Nuclear charge (Z)
 (c) Nuclear mass
 (d) Number of core electrons
100. The size of isoelectronic species – F^- , Ne and Na^+ is affected by
 (a) Nuclear charge (Z)
 (b) Valence principal quantum number (n)
 (c) Electron-electron interaction in the outer orbitals
 (d) None of the factors because their size is the same
101. The bromine atom possesses 35 electrons. It contains 6 electrons in 2p orbital. 6 electrons in 3p orbital and 5 electrons in 4p orbital. Which of these electron experience the lowest effective nuclear charge ?
 (a) 4s (b) 4p
 (c) 4d (d) 4f
102. Which element has odd number of unpaired electron :-
 (a) P (b) Si
 (c) Cr (d) Fe
103. Increasing order of energy :-
 (a) $5s < 5d < 4f < 6p < 6d$
 (b) $5s < 4f < 5d < 6p < 6d$
 (c) $4f < 5s < 5d < 6p < 6d$
 (d) $4f < 5s < 5d < 6d < 6p$
104. The quantum number of four electrons are given below.
 (I) $n = 4, \ell = 2, m_\ell = -2, m_s = -1/2$
 (II) $n = 3, \ell = 2, m_\ell = 1, m_s = +1/2$
 (III) $n = 3, \ell = 1, m_\ell = 1, m_s = +1/2$
 (IV) $n = 4, \ell = 1, m_\ell = 0, m_s = +1/2$
 Correct order of increasing energy is -
 (a) $\text{I} < \text{IV} < \text{III} < \text{II}$ (b) $\text{III} < \text{IV} < \text{II} < \text{I}$
 (c) $\text{III} < \text{II} < \text{IV} < \text{I}$ (d) $\text{II} < \text{IV} < \text{III} < \text{I}$
105. Which of the following has least pH in aqueous solution ?
 (a) HOCl (b) HClO_2
 (c) HClO_3 (d) HClO_4
106. Which arrangement of electrons is wrongly written :-
 (a) P : $\begin{array}{c} 1s \quad 2s \quad 3p \quad 3s \quad 3p \\ \uparrow \downarrow \quad \uparrow \downarrow \quad \uparrow \downarrow \uparrow \downarrow \quad \uparrow \downarrow \quad \uparrow \downarrow \uparrow \downarrow \end{array}$
 (b) Cr : $\begin{array}{c} 1s \quad 2s \quad 2p \quad 3s \quad 3p \quad 4s \quad 3d \\ \uparrow \downarrow \quad \uparrow \downarrow \quad \uparrow \downarrow \uparrow \downarrow \quad \uparrow \downarrow \quad \uparrow \downarrow \uparrow \downarrow \quad \uparrow \downarrow \quad \uparrow \downarrow \uparrow \downarrow \uparrow \downarrow \end{array}$
 (c) Cu : $\begin{array}{c} 1s \quad 2s \quad 2p \quad 3s \quad 3p \quad 4s \quad 3d \\ \uparrow \downarrow \quad \uparrow \downarrow \quad \uparrow \downarrow \uparrow \downarrow \quad \uparrow \downarrow \quad \uparrow \downarrow \uparrow \downarrow \quad \uparrow \downarrow \quad \uparrow \downarrow \uparrow \downarrow \uparrow \downarrow \end{array}$
 (d) N : $\begin{array}{c} 1s \quad 2s \quad 2p \\ \uparrow \downarrow \quad \uparrow \downarrow \quad \uparrow \downarrow \uparrow \downarrow \end{array}$
107. 1st ionisation energy is least with which atom :-
 (a) Lead (b) Silicon
 (c) Carbon (d) Tin

9.10 PERIODIC TABLE

108. 1st ionisation energy is least with which atom :-

- (a) Boron (b) Indium
(c) Gallium (d) Thallium

109. The highest oxidising power is shown by :-

- (a) I₂ (b) B₂
(c) Cl₂ (d) F₂

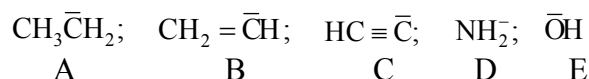
110. Hydration energy is highest for :-

- (a) Li⁺ (b) Na⁺
(c) K⁺ (d) Cs⁺

111. Hydration energy is highest for :-

- (a) Li⁺ (b) Be²⁺
(c) Mg²⁺ (d) Al³⁺

112. Correct order of stability of following anions :-



- (a) E > D > C > B > A (b) E > A > B > C > D
(c) E > C > D > B > A (d) D > E > C > B > A

113. The ionisation energy of Zn is :-

- (a) More than Cu and Ga
(b) Less than Cu and Ga
(c) More than Ga and less than Cu
(d) More than Cu and less than Ga

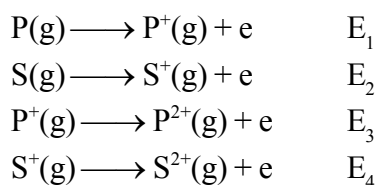
114. Which of the following property undergoes gradual steady shrinkage along the series :-

- (a) Metallic radii of lanthanide elements
(b) Covalent radii of lanthanide elements
(c) Ionic radii of lanthanide elements
(d) All

115. For a particular principal quantum number, the penetrating power of the orbitals.

- (a) nf > nd > np > ns (b) nd > nf > ns > np
(c) ns > np > nd > nf (d) np > ns > nf > nd

116. Reaction Energy involved



Correct option :-

- (a) E₁ > E₂ > E₃ > E₄ (b) E₄ > E₃ > E₁ > E₂
(c) E₄ > E₃ > E₂ > E₁ (d) E₃ > E₄ > E₁ > E₂

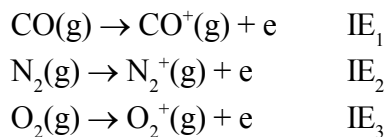
117. Which is called superhalogen ?

- (a) F (b) Cl
(c) Br (d) I

118. The preference of filling the shells runs :-

- (a) K > L > M > N (b) L > M > N > K
(c) N > M > L > K (d) N > L > M > K

119. Reaction Energy Involved



Correct option :

- (a) IE₁ > IE₂ > IE₃ (b) IE₂ > IE₃ > IE₁
(c) IE₁ > IE₃ > IE₂ (d) IE₂ > IE₁ > IE₃

120. Which of the following represent d⁹ ion :-

- (a) Cu⁺ (b) Cu²⁺
(c) Zn²⁺ (d) Ni²⁺

121. Among alkaline earth metal which one has lowest ionisation energy value :-

- (a) Mg (b) Ca
(c) Sr (d) Ba

122. The formula of magnesium nitride and sodium azide are :-

- (a) Mg₃N₂, Na₃N respectively
(b) Mg(N₃)₂, NaN₃ respectively
(c) Mg₃N₂, NaN₃ respectively
(d) Mg(N₃)₂, Na₃N respectively

123. Formula of phosphide ion and sulphate ion are respectively P³⁻ and SO₄²⁻. The formula of magnesium phosphide and aluminium sulphate are respectively :-

- (a) MgP, AlSO₄ (b) Mg₃P₂, AlSO₄
(c) Mg₃P₂, Al₂(SO₄)₃ (d) MgP, Al₂(SO₄)₃

124. Which of the following is largest cation ?

- (a) Na⁺ (b) Mg²⁺
(c) Ca²⁺ (d) Al³⁺

125. Which of the following is smallest cation ?

- (a) Li⁺
(b) Be²⁺
(c) Mg²⁺
(d) Al³⁺

126. What will be the period number, group number, block of the element having atomic number 20 ?
 (a) 4, 3, p (b) 3, 2, s
 (c) 4, 2, s (d) 3, 3, p
127. Which of the following is largest ?
 (a) Mg (b) Mg^{+1}
 (c) Mg^{2+} (d) Al^{3+}
128. The most electropositive element :-
 (a) S (b) O
 (c) Rb (d) K
129. Which of the following pair of atomic numbers represents s-block elements ?
 (a) 3, 9 (b) 11, 38
 (c) 55, 5 (d) 4, 33
130. Alkali metals and alkaline earth metals form the ion of the type :-
 (a) M^{+1} ; M^{+2} (b) M^{+2} ; M^{+1}
 (c) M^{-1} ; M^{-2} (d) M^{-2} ; M^{-1}
131. What is the electronic configuration for $\text{Co}^{3+}(\text{g})$:-
 (a) $[\text{Ar}] 3\text{d}^5$ (b) $[\text{Ar}] 3\text{d}^6$
 (c) $[\text{Ar}] 3\text{d}^7$ (d) $[\text{Ar}] 4\text{s}^1 3\text{d}^5$
132. Which of the following element is example of metalloid ?
 (a) N (b) As
 (c) Al (d) Te
133. The outermost electron resides in K, Cr, Cl, respectively :-
 (a) 4s, 4p, 3d orbital (b) 3s, 3d, 3p orbital
 (c) 3s, 4d, 4p orbital (d) 4s, 3d, 3p orbital
134. If an element X forms the highest oxide of the formula X_2O_7 , then it belongs to group :-
 (a) 14 (b) 15
 (c) 16 (d) 17
135. In Mn, the maximum number of electrons having $m_s = -\frac{1}{2}$ is :-
 (a) 15 (b) 14
 (c) 16 (d) 20
136. In Mn, the minimum number of electrons having $m_s = +\frac{1}{2}$ is :-
 (a) 10 (b) 12
 (c) 14 (d) 15
137. The ratio of unpaired electrons present in the orbital of Mn^{2+} and Cr^{3+} :-
 (a) 4 : 1 (b) 2 : 1
 (c) 5 : 2 (d) 5 : 3
138. The five successive ionisation energies of an element are 800, 2427, 3658, 25024 and 32824 kJ mol^{-1} respectively. The number of valence electrons is :-
 (a) 3 (b) 5
 (c) 1 (d) 2
139. Which of the following oxide is examples of suboxide :-
 (a) NO_2 (b) KO_2
 (c) C_3O_2 (d) N_2O
140. NaOH is example of base it can be explained by :-
 (a) The electronegativity difference between O and H is higher than electronegativity difference between Na and O atom. So Na – O bond will rupture.
 (b) The electronegativity difference between O and H is lower than electronegativity difference between Na and O atom. So Na – O bond will rupture.
 (c) The electronegativity difference between O and H is higher than electronegativity difference between Na and O atom. So O–H bond will rupture.
 (d) The electronegativity difference between O and H is lower than electronegativity difference between Na and O atom. So O–H bond will rupture.
141. The energy required to convert all atoms of Mg to Mg^{2+} ion present in 24×10^{-3} g of Mg vapour is :-
 (a) 2.67 kJ (b) 2.19 kJ
 (c) 3.56 kJ (d) 4.38 kJ
142. Which of the following pair of elements belongs to the same period :-
 (a) Ca & Cl (b) Na & Ca
 (c) Mg & Sb (d) Ca & Zn
143. Which of the following has highest second ionisation energy ?
 (a) Sc (b) Ti
 (c) V (d) Cr

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144. Which of the following has highest third ionisation energy ?

- (a) Ca (b) Tl
(c) Mn (d) V

145. Correct ionisation energy order :-

- (a) Cu > Ag > Au (b) Au > Cu > Ag
(c) Au > Ag > Cu (d) Cu > Au > Ag

146. Bond polarity order :-

- (a) F-H > O-H > Cl-H > S-H
(b) F-H > Cl-H > O-H > S-H
(c) S-H > Cl-H > O-H > F-H
(d) Cl-H > S-H > F-H > O-H

147. Which orbital is filled after filling of np orbitals :-

- (a) ns (b) nd
(c) (n-1)d (d) (n+1)s

148. If x, y, z are electronegativity on Mulliken scale, ionisation potential (in V) and electron affinity (+ in eV) respectively, then the electron affinity in the terms of electronegativity and ionisation potential will be :-

- (a) $z = \frac{x+y}{2}$ (b) $y = \frac{x+y}{2}$
(c) $x = \frac{z-y}{2}$ (d) $z = 2x - y$

149. Fluorine has the highest electronegativity among the ns² np⁵ group on the Pauling scale, but the electron affinity of fluorine is less than that of chlorine because :-

- (a) the atomic number of fluoride is less than that of chlorine
(b) fluorine being the first member of the family behaves in an unusual manner
(c) chlorine can accommodate an electron better than fluorine by utilising its vacant 3d-orbital
(d) small size, high electron density and an increased electron repulsion makes addition of an electron to fluorine less favourable than that in the case of chlorine in isolated stage.

150. First and second ionisation enthalpies of Mg are 720 kJ/mol and 1440 kJ/mol respectively. The % of Mg⁺ ions, if one gram of Mg(g) absorbs 50 kJ of energy. (Given : Atomic mass of Mg = 24 amu):-

- (a) 33.33 (b) 66.67
(c) 55.55 (d) 99.99

EXERCISE # II

□ One or More Than One Correct Answer :

1. Which of the ion pair is isoelectronic :-

- (a) F⁻, K⁺ (b) Cl⁻, Na⁺
(c) S²⁻, Ca²⁺ (d) O²⁻, Mg²⁺

2. Which process requires energy to take place ?

- (a) Ne → Ne⁻¹ (b) F → F⁻¹
(c) O⁻¹ → O²⁻ (d) Mg → Mg²⁺

3. Which of the following reactions should , are nonspontaneous in gas phase :-

- (a) Xe + He⁺ → Xe⁺ + He
(b) Si + Cl⁺ → Si⁺ + Cl
(c) F⁻ + I → I⁻ + F
(d) I⁻ + F → F⁻ + I

4. Bond dissociation energy order :-

- (a) C-C > C-H > H-H
(b) F-F > Cl-Cl > Br-Br
(c) C≡C > C=C > C-C
(d) O=O > S=S > Se=Se

5. Which of these ions are paramagnetic ?

- (a) Sr²⁺ (b) Fe³⁺
(c) Co²⁺ (d) S²⁻

6. Which of the following electronic configuration belongs to d block element ?

- (a) 1s²2s²2p⁶3s²3p⁶3d⁵4s¹
(b) 1s²2s²2p⁶3s²3p⁶3d¹⁰4s¹
(c) 1s²2s²2p⁶3s²3p⁶
(d) 1s²2s²2p⁶3s²3p⁶3d¹⁰4s²4p¹

7. Which of the following statements are correct ?

- (a) MgO is more basic than BaO
(b) Na₂(g) molecule exhibit metallic properties
(c) Ca²⁺ has smaller ionic radius than K⁺
(d) Atomic size order Cs > Na > Mg > Si > Cl

8. Which of the following statements are correct ?

- (a) Lithium is better reducing agent than Caesium
(b) Ionisation energy and sublimation energy of Li is less than Caesium
(c) 2s orbital is lower in energy than 2p orbital
(d) The first ionisation energy of Beryllium is more than that of boron

9. Which of the following statements are correct ?
 (a) The electronic configuration of Cr is $[\text{Ar}] 3d^5 4s^1$. (atomic number of Cr = 24)
 (b) The magnetic quantum number may have a negative value
 (c) In silver atom, 23 electrons have a spin of one type and 24 of the opposite type. (atomic number of Ag = 47)
 (d) The oxidation state of nitrogen in HN_3 is -3
10. In which case, 1st bond is more polar than 2nd bond?
 (a) P-Cl, P-Br (b) S-Cl, S-O
 (c) N-O, N-F (d) B-F, B-Cl
11. Diagonal relationship is observed in
 (a) F, Ne (b) C, P
 (c) B, Si (d) Li, Mg
12. Electronic configuration $ns^2(n-1)d^{0-1}(n-2)f^{1-14}$ represents
 (a) Representative elements
 (b) Lanthanides
 (c) Radioactive elements
 (d) Actinides
13. Which of the following pair is isoelectronic?
 (a) Te^{2-} , Xe (b) Mn^{2+} , Cr^{3+}
 (c) Ni, Zn^{2+} (d) Br^- , Kr
14. Which of the following properties are correctly given?
 (a) $\text{N} > \text{P} > \text{As} > \text{Sb}$ (Third ionisation energy)
 (b) $\text{Na}^+ < \text{F}^- < \text{O}^{2-} < \text{N}^{3-}$ (Ionic size)
 (c) $\text{O} > \text{F} > \text{N} > \text{C}$ (Second ionisation energy)
 (d) $\text{Mg} > \text{Al} > \text{Si} > \text{P}$ (Covalent radius)
15. Which of the following atoms has positive electron gain enthalpy?
 (a) N (b) Be
 (c) Mg (d) Mn
16. Which of the following statements are correct ?
 (a) The 5th period of periodic table contains 32 elements
 (b) The 4f and 5f inner transition series of elements are placed separately at the bottom of the periodic table
 (c) N_2 and CN^- are isoelectronic species.
 (d) Third ionisation energy of phosphorous is more than sulphur.
17. Which of the following statements are incorrect ?
 (a) Formation of Se^{2-} and Ar^- , both require absorption of energy
 (b) Metallic and covalent radii of potassium are 203 and 230 pm respectively.
 (c) If the same element is forming oxide in different oxidation state, then the oxide will be highest acidic in nature in which element will be in its highest oxidation state.
 (d) The electron gain enthalpy of oxygen is more negative than that for sulphure.
18. Elements tend to gain two electron :-
 (a) S (b) O
 (c) Te (d) Cl
19. Glenn Seaborg took part in the discovery of ten of the periodic table's chemical elements. Which element it can be :-
 (a) Nobelium (atomic number 102)
 (b) Californium (atomic number 98)
 (c) Berkelium (atomic number 97)
 (d) Curium (atomic number 96)
20. Aqueous solution of which of the following oxide are acidic in nature :-
 (a) Cl_2O_7 (b) CO_2
 (c) BaO (d) BeO
21. Which of the following options are incorrect :-
 (a) Second period element oxygen has more negative electron gain enthalpy than third period element sulphur.
 (b) Second period element lithium has more negative electron gain enthalpy than third period element sodium.
 (c) Second period element neon has more negative electron gain enthalpy than third period element argon.
 (d) Second period element fluorine has more negative electron gain enthalpy than third period element chlorine.

9.14 PERIODIC TABLE

22. In which case all M–O bond lengths are identical :-
 (a) PO_4^{3-} (b) CO_3^{2-}
 (c) MnO_4^- (d) $\text{CH}_3\text{CO}_2\text{H}$
23. Which of the following options are correct :-
 (a) Across the period effective nuclear charge decreases
 (b) Atomic gaseous of inert gases is highest in the period
 (c) Ionisation energy of Ne is greater as compared to Ne^+
 (d) Ionisation energy of Zn is higher as compared to Ga
24. Out of lithium and neon :-
 (a) IE_2 of Li is more than IE_2 of Ne
 (b) IE_2 of Li is less than IE_2 of Ne
 (c) IE_1 of Li is more than IE_1 of Ne
 (d) IE_1 of Li is less than IE_1 of Ne
25. Which of the following options are incorrect :- :-
 (a) HF bond length is smaller than sum of radius of H atom and F atom
 (b) Effective nuclear charge of inert gases is minimum
 (c) $\text{C(g)} \rightarrow \text{C}^{2+}(\text{g})$ the energy involved in this process is called 2nd ionisation energy
 (d) Electron affinity of oxygen atom is higher than that of sulphur atom
26. $2p_x$ orbital and $3p_x$ orbital of an atom have same :-
 (a) size (b) orientation
 (c) shape (d) nodal plane
27. Which of the following is associated with the removal of electron from d-subshell :-
 (a) IE_1 for Mn (b) IE_2 for Mn
 (c) IE_3 for Mn (d) IE_4 for Mn
28. The correct order of reactivity :-
 (a) $\text{Cs} > \text{Rb} > \text{K} > \text{Na} > \text{Li}$
 (b) $\text{I} > \text{Br} > \text{Cl} > \text{F}$
 (c) $\text{Li} > \text{Na} > \text{K} > \text{Rb} > \text{Cs}$
 (d) $\text{F} > \text{Cl} > \text{Br} > \text{I}$
29. Chemical elements discovered at Lawrence Berkeley National Laboratory are :-
 (a) Technetium (b) Astatine
 (c) Plutonium (d) Curium
30. Formula of oxide of group 15 elements can be :-
 (a) M_2O_3 (b) M_2O_5
 (c) M_4O_{10} (d) MO_2
31. Which formulas are correctly given :-
 (a) Compound formed by silicon and bromine : SiBr_4 (Silicon tetrabromide)
 (b) Compound formed by aluminium and sulphur : Al_2S_3 (Aluminium sulphide)
 (c) Compound formed by mercury and iodine : HgI_2 (Mercuric iodide)
 (d) Compound formed by lithium and oxygen : Li_2O (Lithium oxide)
32. Which of the following elements are called semimetals or metalloid :-
 (a) Si (b) Ge
 (c) I (d) As
34. Select correct option regarding Uuh (atomic number 116)
 (a) It belongs to 7th period element
 (b) It belongs to group 16 element
 (c) It is p-block element
 (d) It is an inert gas
35. Cl^{35} and Cl^- has
 (a) Same number of electrons
 (b) Same number of neutrons
 (c) Same number of protons
 (d) Same stability
36.
$$\begin{array}{c} \text{CH}_2 - \text{CH}_2 \\ | \quad | \\ \text{I} \quad \text{I} \end{array} \xrightarrow{-\text{I}_2} \text{CH}_2 = \text{CH}_2 ;$$
 (A) (vicinal dihalide)
- $$\begin{array}{c} \text{OH} \\ | \\ \text{CH}_2 - \text{OH} \end{array} \xrightarrow{-\text{H}_2\text{O}} \text{H}_2\text{C} = \text{O} \cdot$$
 (B) Gem diol
- Correct option
 (a) Vicinal diiodide does not exist because C–I has less bond dissociation energy
 (b) Gem diol does not exist because C=O has high bond dissociation energy
 (c) Vicinal diol does exist
 (d) Gem diol exists

37. Which of the following statements are correct :-
 (a) the size of hydride ion is greater than that of F^-
 (b) $(n+1)$ values for 5s and 4p orbitals are identical
 (c) H^- is example of stronger reducing agent as compared to hydrogen atom
 (d) The lanthanoid contraction is less as compared to actinide contraction
38. $H-C\equiv CH$ can form salt with :-
 (a) Ca metal (b) Na metal
 (c) Li metal (d) K metal
39. Which of the elements with given atomic number belong to p block element ?
 (a) Atomic number 83 (b) Atomic number 34
 (c) Atomic number 54 (d) Atomic number 64
40. Which of the elements with given atomic number belong to inner transition series ?
 (a) Atomic number 64 (b) Atomic number 42
 (c) Atomic number 79 (d) Atomic number 91
41. The order of ionic mobility :-
 (a) $K^+(aq.) > Na^+(aq.)$ (b) $Ca^{2+}(aq.) > Sr^{2+}(aq.)$
 (c) $Rb^+(aq.) > Na^+(aq.)$ (d) $Na^+(aq.) > K^+(aq.)$
42. $CH_3I + OH^- \rightarrow CH_3OH + I^-$
 Correct option regarding this reaction :-
 (a) Here OH^- attacks δ^+ charge of carbon
 (b) Here OH^- acts as nucleophile
 (c) It is example of nucleophilic substitution
 (d) $(\text{Electronegativity of carbon})_{CH_3 \text{ group}} > (\text{Electronegativity of iodine})$
43. Which of the following option are correct :-
 (a) Pauling scale electronegativity is based on bond energy value
 (b) Alred rochow scale electronegativity is based on Zeff and covalent radius
 (c) Electronegativity in Muliken's scale is based on ionisation energy and electron affinity
 (d) Electronegativity in Sanderson's scale is based on compactness of electron cloud around the nucleus
44. Ionic radii of :-
 (a) $Ti^{4+} < Mn^{7+}$ (b) $^{35}Cl^- < ^{37}Cl^-$
 (c) $K^+ < Cl^-$ (d) $P^{3+} > P^{5+}$
45. Which of the following statements is/are true for the long form of the periodic table :-
 (a) it reflects the sequence of filling the electrons in the order of sub-energy levels s,p,d and f.
 (b) it helps to predict the stable valency states of the element
 (c) it reflects trends in physical and chemical properties of the elements
 (d) it helps to predict the relative ionicity of the bond between any two elements
46. Which of the following statements are true ?
 (a) the first ionisation potential of Al is less than the first ionisation potential of Mg
 (b) the second ionisation potential of Mg is greater than the second ionisation potential of Na
 (c) the first ionisation potential of Na is less than the first ionisation potential of Mg
 (d) the third ionisation potential of Mg is greater than the third ionisation potential of Al
47. Which one of the following statements are correct in relation to ionization enthalpy?
 (a) Ionization enthalpy increases for each successive electron.
 (b) The greatest increase in ionization enthalpy is experienced on removal of electron from core noble gas configuration.
 (c) End of valence electrons is marked by a big jump in ionization enthalpy.
 (d) Removal of electron from orbitals bearing lower n value is easier than from orbital having higher n value.
48. Which of the following statements are correct for the periodic classification of elements ?
 (a) The properties of elements are the periodic functions of their atomic number
 (b) Non-metallic elements are lesser in number than metallic character
 (c) The first ionisation energy of the element along a period do not vary in a regular manner with increase in atomic number
 (d) For transition elements, the d-subshells are filled with electrons monotonically with increase in atomic number

9.16 PERIODIC TABLE

49. The correct order of radii is :-
 (a) $N < Be < B$ (b) $F^- < O^{2-} < N^{3-}$
 (c) $Na < Li < K$ (d) $Fe^{3+} < Fe^{2+} < Fe^{4+}$
50. Which of the following statements are correct ?
 (a) Among Sn, Pb, Fe, Ag ; +2 oxidation state is most stable for Pb
 (b) Among Mg^{2+} , Ti^{3+} , V^{3+} , Fe^{2+} ; maximum number of unpaired electron is present in Fe^{2+}
 (c) Among O, F, N, C ; the highest second ionisation potential is observed for oxygen
 (d) Among the following configuration $[Ne] 3s^2 3p^1$, $[Ne] 3s^2 3p^3$, $[Ne] 3s^2 3p^2$, $[Ne] 3d^{10} 4s^2 4p^3$; $[Ne] 3s^2 3p^3$ has highest ionisation energy
51. Which of the following sets contain only isoelectronic ions ?
 (a) Zn^{2+} , Ca^{2+} , Ga^{3+} , Al^{3+}
 (b) K^+ , Ca^{2+} , Sc^{3+} , Cl^-
 (c) P^{3-} , S^{2-} , Cl^- , K^+
 (d) Ti^{4+} , Ar , Cr^{3+} , V^{5+}
52. Which of the following have no unit ?
 (a) Electronegativity (b) Electron gain enthalpy
 (c) Ionisation enthalpy (d) Metallic character
53. 0 :-
 (a) 0 (b) 0
 (c) 0 (d) 0
54. Which of the following pair of atomic numbers represents p-block elements ?
 (a) 19, 33 (b) 14, 53
 (c) 6, 35 (d) 30, 31
55. Which of the following pair of atomic numbers represents d-block elements :-
 (a) 23, 49 (b) 24, 46
 (c) 56, 28 (d) 80, 21
56. Correct electron affinity order :-
 (a) $Ir > Rh > Co$ (b) $Pt > Pd > Ni$
 (c) $Au > Ag > Cu$ (d) $I > Cl > F$
57. Which of the polyatomic anion are isoelectronic :-
 (a) NO_3^- , CO_3^{2-} , BO_3^{3-}
 (b) SO_4^{2-} , PO_4^{3-} , SiO_4^{4-}
 (c) SeO_3^{2-} , SO_3^{2-} , TeO_3^{3-}
 (d) N_3^- , CN_2^{2-}
58. Which of the following elements have non metallic properties :-
 (a) Cs (b) Cl
 (c) P (d) S
59. Reducing property is shown by which element :-
 (a) Li (b) K
 (c) Ba (d) Ca
60. Which of the following are example of interhalogen compound ?
 (a) ClF_3 (b) HBr
 (c) Cl_2O_7 (d) IF_7
61. Which of the following are example of pseudohalide ?
 (a) CN^- (b) ICl_2^-
 (c) SCN^- (d) $SeCN^-$
62. Correct order of stability of following ion / radical :-
 (a) $F^- < Cl^- < Br^- < I^-$ (b) $F^+ < Cl^+ < Br^+ < I^+$
 (c) $F^\cdot < Cl^\cdot < Br^\cdot < I^\cdot$ (d) $F^- > Cl^- > Br^- > I^-$
63. Which of the following ion are example of diamagnetic ?
 (a) Co^{2+} (b) Zn^{2+}
 (c) Ti^{4+} (d) Sc^{3+}
64. Which of the following ion represent d^6 ion ?
 (a) Co^{2+} (b) Fe^{3+}
 (c) Fe^{2+} (d) Co^{3+}
65. The correct order of ionic mobility :-
 (a) $Li^+(g) > Na^+(g)$ (b) $Li^+(aq.) > Na^+(aq.)$
 (c) $Be^{2+}(g) > Mg^{2+}(g)$ (d) $Be^{2+}(aq.) > Mg^{2+}(aq.)$
66. Which of the following orbitals are represented by $n = 3, \ell = 1$?
 (a) $3p_y$ (b) $4p_x$
 (c) $3d_{xy}$ (d) $3p_x$
67. Which of the following are example of isoelectronic species ?
 (a) Cr^{6+} , Sc^{3+} (b) Te^{2-} , I^-
 (c) N_2O , CS_2 (d) NO_2^+ , O_3
68. In which pair, the first species has more size as compared to second species :-
 (a) Ar, Ne (b) I^- , Cl^-
 (c) $Na^+(aq)$, $K^+(aq)$ (d) Na, Na^+

69. Correct options are :-
(a) F^- is larger than H^-
(b) Cl is more electronegative than Br
(c) Cs is more electropositive than Na
(d) Inert gas can form cation and anion easily
70. Correct order of acidic strength :-
(a) $HNO_2 > HNO_3$
(b) $H_2SO_4 > H_2SO_3$
(c) $HClO_3 > HBrO_3 > HIO_3$
(d) $H_3PO_2 > H_3PO_4 > H_3PO_3$
71. 1 eV atom^{-1} is equivalent to :-
(a) $3.83 \times 10^{-20} \text{ Cals atom}^{-1}$
(b) $1.6 \times 10^{-19} \text{ J atom}^{-1}$
(c) $7.68 \times 10^{-19} \text{ Cals atom}^{-1}$
(d) $2.26 \times 10^{-19} \text{ Cals atom}^{-1}$
72. Which of the following are examples of mixed oxide :-
(a) Fe_3O_4 (b) Mn_3O_4
(c) N_2O_5 (d) KO_2
73. How many compound has more lattice energy than NaCl :-
(a) RbCl (b) MgO
(c) ScN (d) TiC
74. Which oxides are more basic than Li_2O :-
(a) Na_2O
(b) Al_2O_3
(c) BeO
(d) BaO
75. Which of the following ions have pseudo noble gas configuration ?
(a) Cu^{+1}
(b) Cd^{2+}
(c) Au^{+}
(d) Tl^{3+}
76. Correct option regarding As (atomic number 33)
(a) It is one of the typical element
(b) It belongs to 16th group
(c) It is a 4th period element
(d) It is a p-block element
77. The first element of a group differs in many ways from the other heavier members of the group. It is due to :-
(a) small size
(b) high electronegativity and high ionisation potential
(c) odd atomic number
(d) magic numbers of atomic weight
78. The first three ionisation energy of an element are 9.3, 18.2 and 553.8 eV. What informations are reflected by following data ?
(a) The element belongs to 15th group of Modern periodic table
(b) The element has three electrons in the valence shell
(c) The element belongs to 2nd group of Modern periodic table
(d) The element has two electrons in the valence shell
79. Correct molecular formula :-
(a) $Hg_2(N_3)_2$: Mercurous Azide
(b) HN_3 : Hydra acid which contains nitrogen
(c) Cl_2O_7 : Anhydride of $HClO_4$
(d) $Fe[Fe(CN)_6]$: Ferri-Ferri cyanide
80. Which of the following statements are correct :-
(a) The halogen that shows same oxidation state in all its compounds with other elements is chlorine.
(b) Most stable oxidation state of chromium is +3
(c) Among Na, Mg, Fe, Al ; Fe only exhibits more than one oxidation state (other than zero)
(d) The number of electrons present in its outermost shell of an element (whose most common oxidation state is -2) is six
81. What is the change in oxidation state during following conversion :-
(a) $PbS \rightarrow PbSO_4$: +2 to +6 for S
(b) $Br_2 \rightarrow BrO_3^-$: 0 to +5 for Br
(c) $H_2O_2 \rightarrow H_2O$: +1 to -2 for O
(d) $NH_3 \rightarrow NO$: -3 to +2 for N

9.18 PERIODIC TABLE

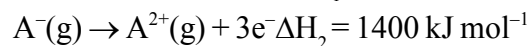
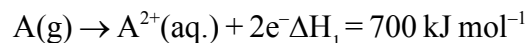
82. $3p_y$ and $4p_y$ orbital of an atom have different :-
 (a) Number of nodal plane
 (b) Quantum number
 (c) Size
 (d) Shape
83. $\begin{matrix} & 1 & 2 & 3 & 4 & 5 \\ \text{CH}_3 & - & \text{CH} & = & \text{N} & - & \text{C} \equiv \text{N} \end{matrix}$
 The correct order of electronegativity :-
 (a) $\text{N}^5 > \text{N}^3$ (b) $\text{N}^5 > \text{C}^4$
 (c) $\text{C}^4 > \text{C}^2$ (d) $\text{C}^2 > \text{C}^1$
84. Which of the following options are correct regarding $ns^2(n-1)d^{0-1}(n-2)f^{0-14}$ configuration ?
 (a) For lanthanides $n = 4$
 (b) For lanthanides $n = 5$
 (c) For actinides $n = 5$
 (d) For actinides $n = 6$
85. Which of the following sequences contain atomic numbers of only representative elements?
 (a) 3, 33, 53, 87 (b) 2, 10, 22, 36
 (c) 7, 17, 25, 37, 48 (d) 9, 35, 51, 88
86. Ionic radii vary in :-
 (a) inverse proportion to the effective nuclear charge
 (b) inverse proportion to the square of effective nuclear charge
 (c) direct proportion to the screening effect
 (d) direct proportion to the square of screening effect
87. Those elements impart colour to the flame on heating in it, the atoms of which require low energy for the ionisation (i.e., absorbs energy in the visible region of spectrum). The elements of which of the following groups in Modern periodic table will impart colour to the flame ?
 (a) 2 (b) 13
 (c) 1 (d) 17
88. Which of the following elements will gain one electron more readily in comparison to other elements of their group :-
 (a) S(g) (b) Na(g)
 (c) O(g) (d) C(g)
89. Ionisation energy of atoms A and B are 350 and 250 kcal mol^{-1} respectively. The electron affinities of these atoms are 70 and 90 kcal mol^{-1} respectively. Then:-
 (a) electron cloud is more attracted by A
 (b) electron cloud is more attracted by B
 (c) on Mulliken scale, electronegativity of A is more than B
 (d) on Mulliken scale, electronegativity of A is less than B
90. Which of the following has/have no unit ?
 (a) Electronegativity
 (b) Electron gain enthalpy
 (c) Ionisation enthalpy
 (d) Metallic character
91. A, B and C are oxides of element X, Y and Z respectively. X, Y and Z are in the same period of the Modern periodic table. A gives an aqueous solution which turns blue litmus red. B reacts with both strong acids and strong alkalis. C gives an aqueous solution which is strongly alkaline. Which of the following statements is / are true ?
 (a) All the three elements are metals
 (b) The Pauling electronegativities decrease from X to Y to Z
 (c) The atomic radius increases in the order $X < Y < Z$
 (d) X, Y and Z could be phosphorus, aluminium and sodium respectively
92. If $\frac{N_0}{2}$ atoms of X(g) are converted into $X^+(g)$ by energy E_1 and $\frac{N_0}{2}$ atoms of X(g) are converted into $X^-(g)$ by energy E_2 , then :-
 (a) Ionisation potential of X would be $\frac{2E_1}{N_0}$
 (b) Ionisation potential of X would be $2E_1$
 (c) Electron affinity of X would be $\frac{2E_2}{N_0}$
 (d) Electron affinity of X would be $2E_2$

93. Electronic configuration 1st ionisation energy
 ns^2np^1 (IE)
 ns^2np^3 IE'
 ns^2np^4 IE''
 ns^2np^5 IE'''
 then -
 (a) $IE > IE'$ (b) $IE' > IE''$
 (c) $IE'' > IE'''$ (d) $IE''' > IE$
94. Which of the following compounds exists :-
 (a) BiF_5 (b) PbO_2
 (c) PbI_4 (d) As_2O_3
95. In the long form of periodic table the valence shell electronic configuration of an element is $5s^25p^4$. The element resides in :-
 (a) Group 17 (b) Group 16
 (c) Period 5 (d) Period 6
96. Which of the following oxyacid are examples of -ic acid :-
 (a) H_2SO_4 (b) HNO_2
 (c) H_2CO_3 (d) H_2SO_3
97. Which of the following oxyacid are examples of -ous acid :-
 (a) H_2SO_4 (b) HNO_2
 (c) H_2CO_3 (d) H_2SO_3
98. Prefix pyro is attached to the names :-
 (a) $\text{S}_2\text{O}_7^{2-}$ (b) $\text{H}_4\text{As}_2\text{O}_7$
 (c) H_2SO_5 (d) $\text{H}_4\text{B}_2\text{O}_7$
99. Identify the meta acids :-
 (a) HPO_3 (b) H_2SnO_3
 (c) HMnO_4 (d) HBO_2
100. Which of the following options are correct :-
 (a) The magnitude of 1st ionisation energy of Cl(g) is same as electron gain enthalpy of $\text{Cl}^+(\text{g})$
 (b) The corresponding thioether of ethyl methyl ether is $\text{CH}_3\text{-S-C}_2\text{H}_5$
 (c) The last element of the p block in 6th period is represented by the outermost electronic configuration $4f^{14}5d^{10}6s^26p^6$
 (d) The oxidation state and covalency of sulphur in sulphur molecule (S_8) are respectively 0 and 2

EXERCISE # III

❑ Linked Comprehension Type :

Passage for Q.1 to Q.3



Electron gain enthalpy for $\text{A}^+(\text{g}) = -350 \text{ kJ mol}^{-1}$

$IE_1 + IE_2$ for $\text{A(g)} = 950 \text{ kJ mol}^{-1}$

1. What are IE_1 and IE_2 of A ?

- (a) 600, 350 kJ mol^{-1} (b) 400, 550 kJ mol^{-1}
 (c) 350, 600 kJ mol^{-1} (d) 500, 450 kJ mol^{-1}

2. Hydration energy of $\text{A}^{2+}(\text{g})$ is

- (a) - 200 kJ mol^{-1} (b) - 150 kJ mol^{-1}
 (c) - 100 kJ mol^{-1} (d) - 250 kJ mol^{-1}

3. Electron gain enthalpy of A is -

- (a) - 300 kJ mol^{-1} (b) - 350 kJ mol^{-1}
 (c) - 400 kJ mol^{-1} (d) - 450 kJ mol^{-1}

Passage for Q.4 to Q.6

In periodic table, some of the elements are given with their atomic number.

Element	Atomic number
Cl	17
F	9
Cs	55
Al	13
C	6
Xe	54

4. The element with highest electronegativity and lowest ionisation potential are respectively.

- (a) F, Xe (b) Cl, Cs
 (c) C, Al (d) Xe, C

5. The element which has smallest radius and the element whose oxide is amphoteric are respectively.

- (a) F, Al (b) F, C
 (c) Cl, C (d) Cl, Al

6. The element which has 8 electrons in outer most shell.

- (a) C (b) F
 (c) Cs (d) Xe

9.20 PERIODIC TABLE**Passage for Q.7 to Q.9**

The IE_1 and the IE_2 in kJ/mol of few elements are given below

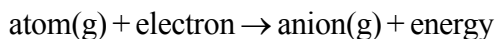
Element	IE_1	IE_2
P	2372	5270
Q	500	7300
R	900	1700
S	1680	3400

Based on the above information answer the following questions :

- Which of the above elements is likely to be a reactive metal ?
 (a) P (b) Q
 (c) R (d) S
- Which of the above elements is likely to be a reactive non metal ?
 (a) P (b) Q
 (c) R (d) S
- Which represents a noble gas ?
 (a) P (b) Q
 (c) R (d) S

Passage for Q.10 to Q.12

When an electron is added to a neutral gaseous atom energy is released. This is called electron affinity which is defined as the amount of energy released when an electron is added to an isolated gaseous atom is called electron affinity. This process is represent as



The magnitude of electron affinity measures the ability of an atom to hold an additional electron.

- Which of the following process is endothermic -
 (a) $P + e^- \rightarrow P^-$ (b) $O + e^- \rightarrow O^-$
 (c) $Li + e^- \rightarrow Li^-$ (d) $N + e^- \rightarrow N^-$
- Choose the correct pair regarding electron affinity (magnitude) -
 (a) $N > P$ (b) $Li > Na$
 (c) $Be > Li$ (d) $N > C$
- According to Millikan's, if an atom has high electronegativity than other atom then it must have -
 (a) high ionisation energy only
 (b) high electron affinity only
 (c) less s character in hybrid orbital
 (d) high (ionisation energy + electron affinity)

Passage for Q.13 to Q.15

According to Aufbau principle, the atomic orbitals fill in the order of increasing $(n + l)$ value when n is principle quantum number and l is azimuthal quantum number. Higher the $(n + l)$ value for an orbital, higher will be the energy. If the two orbitals have the same value $(n + l)$ the one which has lower value of n will be of lower energy and would be filled first.

- The correct order of energy level for different orbitals -
 (a) $6d < 8s < 5f < 7p$
 (b) $8s < 5f < 6d < 7p$
 (c) $5f < 6d < 7p < 8s$
 (d) $7p < 6d < 8s < 5f$
- For which orbitals $(n+l)$ value becomes 7
 (a) 4f (b) 5d
 (c) 6p (d) All
- The maximum number of electron if is denoted by $N_{\max}$ and if n is the principle quantum number then
 (a) $N_{\max} = 2n^2$ (b) $N_{\max} = n^2$
 (c) $N_{\max} = 3n^2$ (d) $N_{\max} = 4n^2$

Passage for Q.16 to Q.18

A solid melts when the force holding its constituent units in position is overcome by thermal energy as a consequence of increase in temperature. The melting temperature is determined by a number of different factors : nature of packing the atoms, ions or molecules the lattice energy, association through hydrogen bond etc.

- Which element of 2nd period has highest melting point?
 (a) B (b) C
 (c) N (d) O
- Which element of 3rd period has highest melting point?
 (a) Al (b) Si
 (c) Cl (d) Ar
- Correct melting point order
 (a) $P > N$ (b) $S > O$
 (c) $Cl > F$ (d) all

Passage for Q.19 to Q.21

In periodic table elements are divided with metals and non-metals. Metals comprise more than 78% of all known elements and appear on the left side on the periodic table. Non-metals which are located at the top high end of the periodic table. Metallic character increases with increasing atomic number in a group whereas decreases from left to right in a period.

19. Non-metals are usually solids or gases at room temperature with low melting and boiling point. The exception is -

- (a) F (b) B
(c) C (d) both Boron & carbon

20. Which properties are characteristic feature of metals.

- (a) Metals are good conductor of heat of electricity
(b) Metals are malleable and ductile
(c) Metals are usually solid at room temperature
(d) All

21. Which of the following is not a metalloid/semi metal ?

- (a) Bi (b) As
(c) Ge (d) Sb

Passage for Q.22 to Q.27

The first ($\Delta_i H_1$) and the second ($\Delta_i H_2$) ionization enthalpies (in kJ mol^{-1}) and the ($\Delta_{\text{eg}} H$) electron gain enthalpy (in kJ mol^{-1}) of a few elements are given below:

Element	$\Delta_i H_1$	$\Delta_i H_2$	$\Delta_{\text{eg}} H$
I	520	7300	-60
II	419	3051	-48
III	1681	3374	-328
IV	1008	1846	-295
V	2372	5251	+48
VI	738	1451	-40

22. The least reactive element.

- (a) II (b) III
(c) IV (d) V

23. The most reactive element.

- (a) II (b) III
(c) IV (d) V

24. The most reactive non metal.

- (a) VI (b) III
(c) IV (d) V

25. The least reactive non metal.

- (a) I (b) III
(c) IV (d) V

26. The metal which can form a stable binary halide of the formula MX_2 (X = halogen).

- (a) I (b) III
(c) IV (d) V

27. The metal which can form a predominantly stable covalent halide of the formula MX (X = halogen).

- (a) I (b) III
(c) IV (d) V

Passage for Q.28 to Q.30

When Schrodinger equation is solved for hydrogen atom, the solution gives the possible energy levels the electron can occupy and the corresponding wave functions (Ψ) of the electron associated with each energy level. The quantised energy states and the corresponding wave functions which are characterised by a set of quantum numbers (principal quantum number n , azimuthal quantum number l , magnetic quantum number m_l).

28. Principal quantum number n -

- (a) Can be positive or can be negative
(b) Is always positive integer
(c) Can be zero
(d) Can be fraction

29. Principal quantum number determines -

- (a) shape of the orbital
(b) the size and to large extent the energy of the orbital
(c) the spatial orientation of the orbital with respect to standard sets of coordination axis.
(d) the spin of electrons

30. The value of n is highest for

- (a) 4s orbital
(b) 6s orbital
(c) 4f orbital
(d) 3d orbital

9.22 PERIODIC TABLE**Passage for Q.31 to Q.33**

Azimuthal quantum number ' ℓ ' defines the three dimensional shape of the orbital. For a given value of n , ℓ can have n number of values ranging 0 to $n-1$. For example, when $n=1$, value of ℓ is only zero. For $n=2$, the possible value of ℓ can be 0 and 1. For $n=3$ the possible values of ℓ are 0,1,2.

31. For $\ell=4$, number of orbitals will be
 (a) 5 (b) 7
 (c) 9 (d) 11
32. Which orbitals are described by $n=5$, $\ell=3$ and $n=4$, $\ell=0$.
 (a) 5f, 4p (b) 5g, 4d
 (c) 5d, 4s (d) 5f, 4s
33. Value of $(n+\ell)=5$ is applicable for which orbital
 (a) 4p (b) 3d
 (c) 4d (d) both (a) and (b)

Passage for Q.34 to Q.36

The diagonal relationship signifies the similarities in properties of the elements of the 2nd period with the respective diagonally opposite 3rd period elements between the successive groups in the periodic table. Such diagonally opposite pairs are Li and Mg ; Be and Al ; B and Si.

34. Li and Mg resembles in which of the following way.
 (a) Both Li and Mg reacts with water to give O_2 gas
 (b) Both Li and Mg-nitride reacts with water to give NH_3 gas
 (c) Both Li and Mg-nitrate undergo heating to give N_2 gas
 (d) Both LiOH and $Mg(OH)_2$ are very much water soluble
35. Be and Al properties are similar,. It is supported by which fact
 (a) Both Be and Al reacts with caustic soda to give berylate and aluminate with molecular formula $NaBeO_2$ and $NaAlO_2$
 (b) Both Be-carbide and aluminium carbide react with water to give methane gas.
 (c) Maximum covalency of both Be and Al are 6
 (d) Maximum covalency of both Be and Al are 4

36. Similarities in the pair B and Si is illustrated by which of the following fact

- (a) Oxides of boron and silicon both are reduced in electric furnace to give elemental B and Si respectively.
 (b) Both B and Si forms oxyacid H_3BO_3 and H_4SiO_4 which are monobasic acid
 (c) Both halides of boron and silicon are reduced by $LiAlH_4$ to produce their corresponding hydrides
 (d) Maximum valency of boron and silicon are 6

Passage for Q.37 to Q.39

Phosphorous is an element with atomic number 15. It is considered as nonmetal. It forms some oxyacid like H_3PO_2 , H_3PO_3 , H_3PO_4 . It has different allotropes like white phosphorous , red phosphorous, black phosphorous.

37. Total number of orbitals which have $(n+\ell)=3$ for phosphorous atom
 (a) 3 (b) 4
 (c) 5 (d) 6
38. Total number of electrons which can have $m_l=0$ for phosphorous atom
 (a) 3 (b) 6
 (c) 9 (d) 12
39. Total number of maximum electrons which can have $s=+1/2$ for phosphorous atom
 (a) 6 (b) 9
 (c) 12 (d) 15

Passage for Q.40 to Q.42

Atomic radii increase down a group, within the s and p block, decrease from left to right across a period. The lanthanoide contraction results in a decrease in atomic radius for elements following the f -block. All monoatomic anions are lareger than their parent atoms and all monoatomic cations are smaller.

40. The correct order of increasing atomic radius of the following elements is -
 (a) $S < O < Se < C$
 (b) $O < C < Se < S$
 (c) $O < C < S < Se$
 (d) $C < O < S < Se$

41. The order of increasing ionic radius of the following is

- (a) $K^+ < Li^+ < Mg^{2+} < Al^{3+}$
 (b) $Mg^{2+} < Al^{3+} < K^+ < Li^+$
 (c) $Al^{3+} < Li^+ < Mg^{2+} < K^+$
 (d) $Al^{3+} < Mg^{2+} < Li^+ < K^+$

42. The order of increasing ionic radius of the following is

- (a) $S^{2-} < F^- < Br^- < Te^{2-}$
 (b) $F^- < Br^- < Se^{2-} < Te^{2-}$
 (c) $O^{2-} < Cl^- < I^- < Se^{2-}$
 (d) $O^{2-} < Se^{2-} < Cl^- < Te^{2-}$

Passage for Q.43 to Q.45

The polarisability of an atom is its ability to be distorted by an electric field. An atom or ion (mostly anion) is highly polarisable if its electron distribution can be distorted readily. A polarisable atom or ion is one with orbitals that lie close in energy, large, heavy atoms and tend to be highly polarisable. On the other hand, species that effectively distorted the electron distribution of an neighbouring atom or cation are described as having polarising ability.

43. The anion which is most polarisable among halide ion is

- (a) F^- (b) Cl^-
 (c) Br^- (d) I^-

44. Correct option -

- (a) Na^+ has less polarising power than Cs^+
 (b) Te^{2-} has less polarisability than O^{2-}
 (c) Ag^+ has more polarising power than K^+
 (d) Hg^{2+} has less polarising power than Ca^{2+}

45. Which of the following has highest polarisability -

- (a) F^- (b) O^{2-}
 (c) S^{2-} (d) N_3^-

Passage for Q.46 to Q.47

Allred-Rochow equation regarding electronegativity of element is given as follows

$$en = \frac{0.36Z^*}{d} + 0.744$$

d = covalent radius of atom in Å

Z = effective nuclear charge of atom

46. What is the electronegativity of Arsenic atom (having atomic number 33). Given covalent radius of Arsenic = 120 pm

- (a) 3.2 (b) 2.84
 (c) 2.61 (d) 2.32

47. According to this equation -

- (a) Down the group, electronegativity decreases
 (b) Across a period, from left to right electronegativity increases
 (c) Both (a) and (b) are correct
 (d) Electronegativity does not depend on covalent radius

Passage for Q.48 to Q.50

Hund's rule suggest that the ground state of an atom should contain maximum number of unpaired electrons (within the same subshell). Also electrons in different orbitals of the same energy will have their spins parallel.

48. Spin multiplicity for np^3 electronic configuration

- (a) 1 (b) 2
 (c) 3 (d) 4

49. Spin multiplicity for nd^5 electronic configuration

- (a) 4 (b) 5
 (c) 6 (d) 7

50. For d^7 ion number of unpaired electron with parallel spin will be

- (a) 1 (b) 2
 (c) 3 (d) 4

Passage for Q.51 to Q.53

The half filled or full filled orbitals are more stable compared to the nearest half filled or full filled orbitals respectively. This enhanced stability for half filled and full filled orbitals will be explained with the help of exchange energy. d^{10} configuration has additional exchange energy which overcomes the disfavour created due to additional pairing.

51. The actual electronic configuration of Pd (atomic number = 46) is -

- (a) $[Kr] 4d^{10} 5s^0$ (b) $[Kr] 4d^9 5s^1$
 (c) $[Kr] 4d^8 5s^2$ (d) $[Kr] 4d^{10} 5s^1$

52. The actual electronic configuration of Ag (atomic number = 47) is -

- (a) $[Kr] 4d^{10} 5s^0$ (b) $[Kr] 4d^9 5s^1$
 (c) $[Kr] 4d^8 5s^2$ (d) $[Kr] 4d^{10} 5s^1$

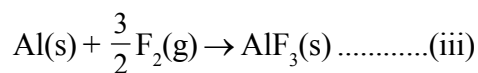
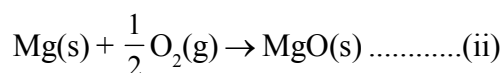
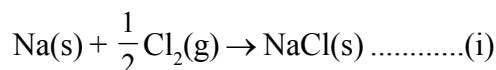
9.24 PERIODIC TABLE

53. The actual electronic configuration of Gd (atomic number = 64) is -

- (a) $[\text{Xe}] 4f^8 5d^0 6s^2$ (b) $[\text{Xe}] 4f^7 5d^1 6s^2$
 (c) $[\text{Xe}] 4f^7 5d^0 6s^2$ (d) $[\text{Xe}] 4f^8 5d^1 6s^1$

Passage for Q.54 to Q.56

Born-Haber cycle is the thermodynamic cycle which determines lattice energy of a compound. The most stable crystal structure of the compound is commonly the structure with the greatest lattice energy under prevailing condition



Given : Sublimation energy of sodium = 109 kJ mol^{-1}

Ionisation energy of sodium = 494 kJ mol^{-1}

Enthalpy of formation of NaCl(s)
 $= -414 \text{ kJ mol}^{-1}$

Electrone gain enthalpy of Cl = -347 kJ mol^{-1}

Bond dissociation energy of $\text{Cl}_2 = 242 \text{ kJ mol}^{-1}$

54. Lattice energy of NaCl(s) -

- (a) 2560 kJ mol^{-1} (b) 791 kJ mol^{-1}
 (c) 1582 kJ mol^{-1} (d) 320 kJ mol^{-1}

55. The correct expression of heat of formation of MgO with other thermodynamical data is

- (a) $\Delta H_f = S + IE_1 + IE_2 + D - EA_1 - U$
 (b) $\Delta H_f = S + IE_1 + IE_2 + D - EA_1 - EA_2 - U$
 (c) $\Delta H_f = S + 2IE_1 + D/2 - 2(EA_1) - U$
 (d) $\Delta H_f = S + IE_1 + IE_2 + D/2 - (EA_1) - (EA_2) - U$

56. To prepare AlF_3 from Al(s) and $\text{F}_2(\text{g})$, at first $\text{Al}^{3+}(\text{g})$ and $\text{F}^-(\text{g})$ has to be generated. Correct option

- (a) For $\text{Al(g)} \rightarrow \text{Al}^{3+}(\text{g})$ energy required is $3IE_1$
 (b) For $\text{Al(g)} \rightarrow \text{Al}^{3+}(\text{g})$ energy required is $IE_1 + IE_2 + IE_3$

- (c) For $\frac{3}{2} \text{F}_2(\text{g}) \rightarrow 3\text{F}^-(\text{g})$ energy required is $\frac{3D}{2} - 3EA_1$
 (d) Both (b) and (c) are correct

Passage for Q.57 to Q.58

The principal quantum number of the outermost shell of M_1, M_2 and M_3 is $n = 3$. M_1 forms the only oxide $(M_1)_2\text{O}$ with oxygen. The oxide on being dissolved in water produces strong alkali. The covalent hydride of M_2 is $(M_2)_2\text{H}_3$ and the formula of highest oxides of M_2 is $(M_2)_2\text{O}_5$. The hydride of M_3 is strong acid and on being ionised M_3 produces M_3^- ion.

57. M_2 & M_3 are respectively -

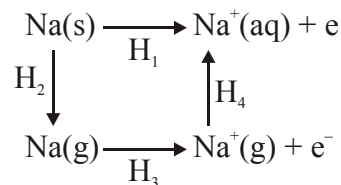
- (a) N, F
 (b) P, Cl
 (c) N, Cl
 (d) P, F

58. Incorrect option regarding M_1 -

- (a) M_1 is example of strong reducing element
 (b) M_1 is example of strong oxidising element
 (c) M_1 is highly reactive metal
 (d) Dianion of M_1 does not exist

Passage for Q.59 to Q.60

If we have Born Haber cycle with this equation, then following energies are associated :



59. H_2 and H_3, H_4 refers :-

- (a) Fusion energy of sodium, ionisation energy of sodium, hydration energy of sodium
 (b) Vaporisation energy of sodium, electron affinity of sodium, hydration energy of sodium
 (c) Sublimation energy of sodium, ionisation energy of sodium, hydration energy of sodium
 (d) Bond dissociation energy of sodium, hydration energy of sodium

60. The expression of H_1 in terms of H_2, H_3, H_4 (with sign) is -

- (a) $-H_1 = -H_2 - H_3 + H_4$
 (b) $-H_1 = H_2 + H_3 - H_4$
 (c) $-H_1 = -H_2 + H_3 - H_4$
 (d) $-H_1 = H_2 - H_3 + H_4$

Matrix Match Type :

61. Match the column :-

Column - I Property	Column - II Ordered
(a) Electronegativity order	(P) $N > B$
(b) Electron affinity order	(Q) $Se > Br$
(c) Ionisation energy order	(R) $C > B$
(d) No. of valance electron	(S) $O > S$

62. Match the column :-

Column - I Electronic configuration	Column - II Value of born exponent
(a) $1s^2$	(P) 10
(b) $2s^2 2p^6$	(Q) 9
(c) $3s^2 3p^6$ or $3s^2 3p^6 3d^{10}$	(R) 7
(d) $4s^2 4p^6$ or $4s^2 4p^6 4d^{10}$	(S) 5

63. Match the column :-

Column - I Compound	Column - II Melting point
(a) $BeCl_2$	(P) $872^\circ C$
(b) $MgCl_2$	(Q) $772^\circ C$
(c) $CaCl_2$	(R) $712^\circ C$
(d) $SrCl_2$	(S) $405^\circ C$

64. Match the column :-

Column - I (Order)
(a) $Na^+ < F^- < O^{2-} < N^{3-}$
(b) $Li^+ < Na^+ < K^+ < Rb^+ < Cs^+$
(c) $O > S > F > Cl$
(d) $Cl^- < K^+ < Ca^{2+} < Sc^{3+}$

Column - II (Property)

- (P) Electronegativity
(Q) Mobility of hydrated ions
(R) Ionisation energy
(S) Electron affinity
(T) Ionic size

65. Match the column :-

Column - I Atomic number	Column - II Feature
(a) 38	(P) s-block element
(b) 51	(Q) p-block element
(c) 29	(R) d-block element
(d) 64	(S) 5th period element
	(T) Rare earth element

66. Match the column :-

Column - I Element	Column - II Feature
(a) Hg	(P) Solid at room temperature
(b) Mn	(Q) Liquid at room temperature
(c) Zn	(R) d-block element
(d) W	(S) Transition element
	(T) $(n-1)d^{10} ns^2$ or $(n-1)d^5 ns^2$ configuration

67. Match the column :-

Column - I Element	Column - II Feature
(a) Osmium (Os)	(P) Inner transition element
(b) Promethium (Pm)	(Q) Transition element
(c) Magnesium (Mg)	(R) Highest positive oxidation state
(d) Germanium (Ge)	(S) Typical element
	(T) Used as semiconductor

68. Match the column :-

Column - I Pair of atoms	Column - II Feature
(a) S, Se	(P) Chalcogen family
(b) Cl, Br	(Q) Halogen family
(c) N, P	(R) Pnictogen family
(d) Be, Al	(S) Diagonal relationship
	(T) Non metals

69. Match the column :-

Column - I Order	Column - II Property
(a) $As > Se$	(P) Atomic number
(b) $Se < Br$	(Q) Atomic radius
(c) $Mg < Al$	(R) Electronegativity
(d) $K < Mg$	(S) Number of valance electrons
	(T) 1st ionisation energy

9.26 PERIODIC TABLE

70. Match the column :-

Column - I	Column - II
Formula of hydride	Elements
(a) MH_4	(P) Groups 16 elements
(b) MH_3	(Q) Groups 15 elements
(c) MH_2	(R) Groups 14 elements
(d) MH	(S) Groups 13 elements
	(T) Groups 17 elements

71. Notation of IUPAC nomenclature of elements

Column - I	Column - II
Name	Digit/Abbreviation
(a) nil	(P) 0/n
(b) enn	(Q) 9/e
(c) sept	(R) 7/s
(d) un	(S) 1/u

72. Match the column :-

Column - I	Column - II
C-F bond distance	Molecule
(a) 139.1 pm	(P) CH_3F
(b) 135.8 pm	(Q) CH_2F_2
(c) 133.2 pm	(R) CHF_3
(d) 132 pm	(S) CF_4

73. Match the column :-

Column - I	Column - II
Molecular formula	Atomic number
(a) AB_2	(P) Atomic number of A and B are respectively 13 and 53
(b) A_3B_2	(Q) Atomic number of A and B are respectively 12 and 07
(c) AB_3	(R) Atomic number of A and B are respectively 56 and 09
(d) AB_5	(S) Atomic number of A and B are respectively 15 and 17

74. Match the column :-

Column - I	Column - II
Metal	Metallic Radii (pm)
(a) Rb	(P) 137 pm
(b) Ca	(Q) 157 pm
(c) Mn	(R) 197 pm
(d) Li	(S) 250 pm

75. Match the column :-

Column - I	Column - II
Metal ion	Ionic radii (symmetric)
(a) Mg^{2+}	(P) 202 pm
(b) Li^+	(Q) 179 pm
(c) I^-	(R) 102 pm
(d) Br^-	(S) 92 pm

76. Match the column :-

Consider chlorine atom for given property

Column - I	Column - II
(a) Electronegativity in Pauling Scale	(P) 1.732
(b) Spin magnetic moment in BM	(Q) 0
(c) Z/e ratio	(R) 3
(d) Number of vacant orbital in outermost shell	(S) 1

77. Match the column :-

Column - I	Column - II
Element	Number of valence electrons
(a) Group 13 elements	(P) 3
(b) Group 14 elements	(Q) 4
(c) Group 18 elements	(R) 8
(d) Group 2 elements	(S) 2

78. Match the column :-

Column - I	Column - II
Order	Property
(a) $Cl^- < K^+ < Ca^{2+} < Sc^{3+}$	(P) Electronegativity
(b) $O < S < F < Cl$	(Q) Nuclear charge
(c) $Li^+ < Na^+ < K^+ < Rb^+ < Cs^+$	(R) Size
(d) $Na^+ < F^- < O^{2-} < N^{3-}$	(S) Electron affinity
	(T) Ionisation energy

79. Match the column :-

Column - I	Column - II
(a) $Fe^{3+} \rightarrow Fe^{2+}$	(P) Exothermic in nature
(b) $Br \rightarrow Br^-$	(Q) Endothermic in nature
(c) $H \rightarrow H^-$	(R) Becomes diamagnetic
(d) $N_2 \rightarrow N$	(S) μ undergoes a change
	(T) Becomes paramagnetic

80. Match the column :-

Column - I	Column - II
(a) MnO	(P) Aqueous solution is basic in nature
(b) Mn_2O_7	(Q) Aqueous solution is acidic in nature
(c) Al_2O_3	(R) Amphoteric in nature
(d) ZnO	(S) Highest oxidation state
	(T) Transition metal oxide

81. Match the column :-

Column - I	Column - II
Element	Atomic radius (pm)
(a) Be	(P) 88
(b) C	(Q) 80
(c) O	(R) 77
(d) B	(S) 66

82. Match the column :-

Column - I	Column - II
Electronic configuration	Electron gain enthalpy / kJ mol^{-1}
(a) $1s^2 2s^2 sp^6$	(P) -53
(b) $1s^2 2s^2 2p^6 3s^1$	(Q) -328
(c) $1s^2 2s^2 2p^5$	(R) -141
(d) $1s^2 2s^2 2p^4$	(S) +48

83. Match the column :-

Column - I	Column - II
Element	ΔH_1 ΔH_2 $\Delta_{\text{eg}} H$
(a) Most reactive non metal	(P) 419 3051 -48
(b) Most reactive metal	(Q) 1681 3374 -328
(c) Least reactive element	(R) 738 1451 -328
(d) Metal forming binary halide	(S) 2372 5251 +48

84. Match the column :-

Column - I	Column - II
Electrons	Z effective
(a) 4d electron in Pd	(P) 6.85
(b) 3d electron in Ni	(Q) 3.9
(c) 3p electron in Ca	(R) 8.75
(d) 2s electron in N	(S) 7.55

85. Match the column :-

Column - I	Column - II
Orbital	Number of nodal plane
(a) p_y orbital	(P) Two nodal plane present in xy and yz plane
(b) d_{xz} orbital	(Q) One nodal plane present in xz plane
(c) d_{z^2} orbital	(R) Zero nodal plane present but two nodal cones are present
(d) s orbital	(S) Zero nodal plane is present.

86. Match the column :-

Column - I	Column - II
Overlap of orbitals	Bond formation
(a) $p_x + d_{xz}$ (z axis is internuclear axis)	(P) σ bond is formed
(b) $p_x + d_{x^2-y^2}$ (x axis is internuclear axis)	(Q) π bond is formed
(c) $d_{z^2} + d_{z^2}$ (z axis is internuclear axis)	(R) No bond is formed
(d) $p_x + d_{xy}$ (z axis is internuclear axis)	(S) Head on overlap (T) Side way overlap

87. Match the column :-

Column - I	Column - II
Oxidation state of sulphur	Compound
(a) Zero	(P) Dimethyl sulphoxide
(b) +6	(Q) Sulphuryl chloride
(c) +4	(R) Sulphonyl chloride
(d) -2	(S) Sodium sulphide (T) Sulphuric acid

88. Match the column :-

Column - I	Column - II
Shell	Number of electrons in subshell or shell
(a) K	(P) 2 electrons in s subshell
(b) L	(Q) 6 electrons in p subshell
(c) M	(R) 10 electrons in d subshell
(d) N	(S) 14 electrons in f subshell (T) Even number of electron in each shell

9.28 PERIODIC TABLE

89. Match the column :-

Column - I	Column - II
Type of elements	Elements
(a) Typical elements	(P) Aluminium
(b) Eka-boron	(Q) Gallium
(c) Eka-silicon	(R) Scandium
(d) Eka-aluminium	(S) Germanium
	(T) Chlorine

90. Match the column :-

Column - I	Column - II
Period	Number and types orbitals involved / No. of elements
(a) 3 rd	(P) s,p,d,f all orbitals involved
(b) 4 th	(Q) 9, 18
(c) 6 th	(R) 4, 8
(d) 7 th	(S) 16, 32

91. Match the column :-

Column - I	Column - II
Elements	Features
(a) Chromium (At. No. 24)	(P) half filled d-orbitals
(b) Rhodium (At. No. 45)	(Q) full filled d-orbitals
(c) Platinum (At. No. 78)	(R) one electron present in ns orbitals
(d) Molybdenum (At. No. 42)	(S) eight electrons present in (n-1) d orbitals
	(T) nine electrons present in (n-1) d orbitals

92. Match the column :-

Column - I (Symbol Name)

- (a) Unniltrium
(b) Ununquadium
(c) $F > O > N$
(d) $F_2 > O_2 > N_2$

Column - II (Atomic number)

- (P) 117
(Q) 108
(R) 103
(S) 114

93. Match the column :-

Column - I	Column - II
Oxide	Nature of oxide
(a) CS_2O	(P) Acidic oxide
(b) Cl_2O_7	(Q) Basic oxide
(c) BeO	(R) Amphoteric oxide
(d) CO	(S) Neutral oxide

94. Match the column :-

Column - I	Column - II
Type of elements	Example/Features
(a) Inert gases	(P) He, Ne, Ar, Kr, Xe
(b) Representative elements	(Q) II-A to VII-A elements
(c) Transition elements	(R) Incomplete d-shell either in ground state or in most stable valency state
(d) Inner transition elements	(S) The last electron goes to the f-orbital of antepenultimate shell either in ground state or in any common oxidation state

95. Match the column :-

Column - I	Column - II
Elements	Feature
(a) Sc	(P) Not considered as transition elements
(b) Ti	(Q) Highest covalent radii
(c) Ni	(R) Least covalent radii
(d) Zn	(S) Odd number of unpaired d electrons
	(T) Even number of unpaired d electrons (excluding zero)

96. Match the column :-

Column - I	Column - II
Elements	Feature
(a) Na	(P) Transition element
(b) Ag	(Q) Member of the 4th period
(c) Co	(R) Behaves as metal
(d) Br	(S) Behaves as non metal
	(T) One unpaired electron

97. Match the column :-

Column - I (Pair of elements)

- (a) Mg, Cl
(b) Se, Te
(c) V, Cd
(d) B, Si

Column - II (Feature)

- (P) Same group
(Q) Diagonal relationship
(R) Same period
(S) Same block elements
(T) Both element in pair have atomic number less than 50

98. Match the column :-

Column - I	Column - II
Compounds	Feature

- | | |
|------------------------|--|
| (a) Magnesium oxide | (P) ns np ⁶ configuration for both cation and anion |
| (b) Potassium chloride | (Q) Isoelectronic cation and anion |
| (c) Sodium fluoride | (R) Cation and anion have +1 and -1 oxidation state respectively |
| (d) Barium sulphide | (S) Cation and anion have 2 and -2 oxidation state respectively |
| | (T) Largest internuclear distance |

99. Match the column :-

Column - I
Element

- (a) Gadolinium
(b) Potassium
(c) Chromium
(d) Gallium

Column - II

Fullfilled orbital	Half filled orbital
(P) 1s, 2s, 2p, 3s, 3p, 3d, 4s	4p
(Q) 1s, 2s, 2p, 3s, 3p,	3d, 4s
(R) 1s, 2s, 2p, 3s, 3p	4s
(S) 1s, 2s, 2p, 3s, 3p, 3d, 4s, 4p, 4d, 5s, 5p, 6s	4f

100. Match the column :-

Column - I**Reactions**

- (a) $O(g) \rightarrow O^{2-}(g)$
(b) $O(g) \rightarrow O^{2-}(g)$
(c) $O(g) \rightarrow O^{-1}(g)$
(d) $O^{+1}(g) \rightarrow O^{2+}(g)$

Column - II**Nature of reactions**

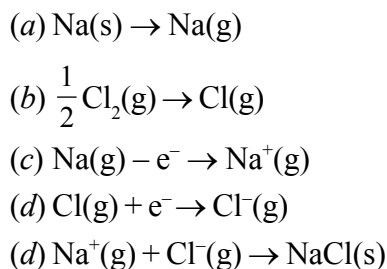
- (P) Endothermic process
(Q) Exothermic process
(R) Ist electron gain enthalpy of oxygen
(S) IInd ionisation energy of oxygen
(T) IInd electron gain enthalpy of oxygen

EXERCISE # IV**Integer Type :**

- Number of elements present in group-17.
- Number of elements present in group-18.
- An element (M) has atomic number, A M²⁺ has electronic configuration resembles with inert gas configuration. That element belongs to 2nd period. What is the atomic number of A.
- An element has electronic configuration [Kr]4d⁷5s². On the basis of this electronic configuration, find out the group number of elements according to modern periodic table :-
- Number of unpaired electron present in 4p orbital of arsenic atom.
- Ionisation potential of Be^{x+} is found to be 217.6 electron volt. What is the value of x -
- How many hydroxides are basic in nature ?
CsOH ; Se(OH)₂ ; IOH ; ClOH ; Sr(OH)₂ ; P(OH)₃
- On the Pauling electronegativity scale, what is the atomic number of the element next to fluorine atom.
- Find out electronegativity of chlorine atom on Pauling scale if ionisation energy of Cl⁻ is 4 eV and electron affinity of Cl⁺ is 13 eV.
- In the periodic table, if there are 10 periods, then what is the number of maximum elements present in 10th period.

9.30 PERIODIC TABLE

11. During formation of NaCl, from Na(s) and Cl₂(g), how many steps are endothermic



12. Total number of gaseous molecules

He ; N₂ ; O₂ ; Cl₂ ; F₂ ; H₂ ; Xe ; Kr ; Ar

13. How many columns are present in p block elements ?

14. If columns are present in d block elements is p, then find out p-1

15. Find the number of elements which has higher EA₂ than EA₁

P ; N ; C ; Se ; Br ; Na ; Be ; Mg ; O ; B

16. Count the number of properties which have higher values for nitrogen atom as compared to phosphorus atom (magnitude) :-

- (i) electronegativity
 (ii) electron affinity
 (iii) ionisation potential
 (iv) covalent radius
 (v) proton electron ratio

17. Oxidation state of nitrogen is positive integer for how many of the following compounds :

- (a) Hydrazoic acid
 (b) Magnesium nitride
 (c) Ammonium chloride
 (d) Hydrazine
 (e) Laughing gas
 (f) Nitrosyl chloride

18. Find out lattice energy of KI

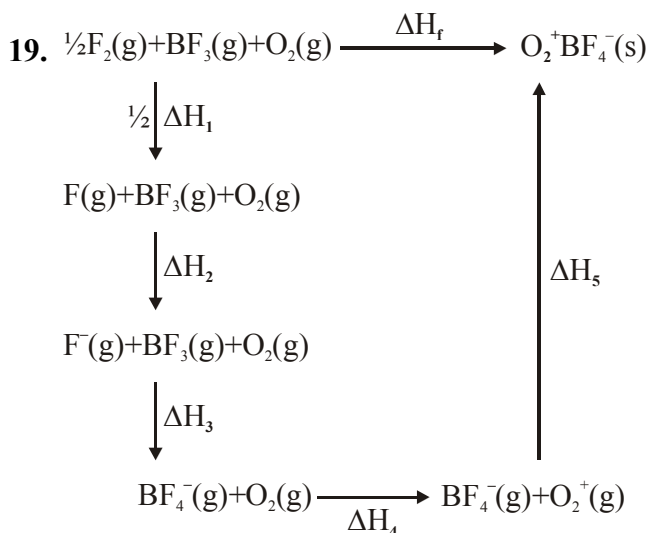
Given : Sublimation energy of I₂ = 14 kcal mol⁻¹

Sublimation energy of K = 20 kcal mol⁻¹

IE₁ of K = 92 kcal mol⁻¹

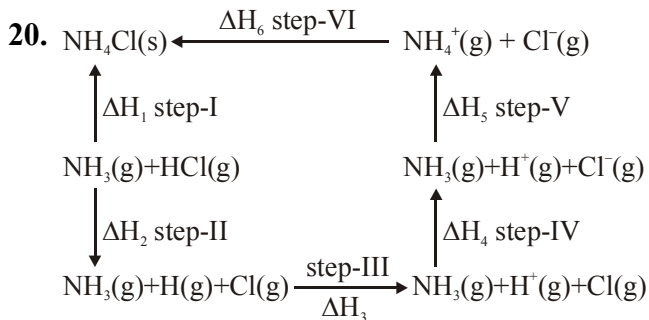
Dissociation energy of I₂ = 28 kcal mol⁻¹

Electron affinity of I₂ = -70 kcal mol⁻¹



How many statements are correct :

- (a) ΔH₁ corresponds to bond dissociation energy of F₂.
 (b) ΔH₂ corresponds to electron affinity of F.
 (c) ΔH₃ corresponds to electron affinity of BF₃.
 (d) ΔH₄ corresponds to ionisation energy of O₂.
 (e) ΔH₅ corresponds to lattice energy of O₂⁺BF₄⁻.



How many statements are correct :

- (a) ΔH₁ corresponds to heat of formation of NH₄Cl.
 (b) ΔH₂ corresponds to bond dissociation energy of HCl.
 (c) ΔH₃ corresponds to ionisation energy of H(g).
 (d) ΔH₄ corresponds to electron gain enthalpy of Cl(g).
 (e) ΔH₅ corresponds to proton affinity of NH₃.
 (f) ΔH₆ corresponds to lattice energy of NH₄Cl.
 (g) Step II and Step III are endothermic process.
 (h) Step IV and Step V are exothermic process.
 (i) Step I is exothermic process
 (j) Step VI is exothermic process.

21. Find the total number of cations for which I.P. of carbon is lower than the corresponding atom.
- (a) Na^+ (b) Mg^+
(c) Sn^{4+} (d) Fe^{2+}
(e) Cl^+ (f) S^+
22. Find out the total number of acidic compound.
- (a) $\text{ClO}_2(\text{OH})$
(b) $\text{BrO}_3(\text{OH})$
(c) $\text{Sr}(\text{OH})_2$
(d) $\text{NO}_2(\text{OH})$
(e) $\text{Mg}(\text{OH})_2$ (f) $\text{PO}(\text{OH})_3$
(g) $\text{SO}_2(\text{OH})_2$
23. How many electrons satisfy $n = 3, l = 2, s = +\frac{1}{2}$
24. Calculate Z_{eff} of valence electron of Gd (atomic number 64)
25. Ionisation enthalpies of elements of second period are given below. (in Kcal mole^{-1}) [not in sequence]
2080 ; 1086 ; 899 ; 1314 ; 801 ; 520 ; 1402 ; 1681
If the atomic number of element with ionisation energy 1314 Kcal mole^{-1} is x.
If the atomic number of element with ionisation enthalpy 1080 Kcal mole^{-1} is y.
Then find out $x - y$:
26.
$$\text{HO} - \overset{\text{O}}{\underset{\text{O}}{\parallel}}\text{S} - \text{S} - \text{S}^* - \text{S} - \overset{\text{O}}{\underset{\text{O}}{\parallel}}\text{S} - \text{OH}$$

What is the oxidation state of S^* :
27. How many statements are correct ?
- (a) Na^+ has higher value of ionisation enthalpy than Ne change both have same electronic configuration.
(b) 1st ionisation energy of nitrogen is higher than oxygen and opposite for 2nd ionisation energy.
(c) Electrons gain enthalpy of oxygen less negative than selenium.
(d) As and Sb are example of metalloid.
(e) d block elements generally exhibit variable valency.
- (f) Mercury and Bromine are liquids at room temperature.
(g) Screening effect is not observed in He^{+1} , Li^{+2} , Be^{+3} ion.
(h) Oxidation state of S in S_2Cl_2 is +1.
28. Find out number of electrons lost by Cr atom to produce an isoelectronic ion of S^{2-} ion.
29. Number of electrons in Na (atomic number = 11) having $m = 0$.
30. Number of electrons in S (atomic number = 16) having $(n+l) = 3$.
31. Number of lanthanoids in Modern periodic table.
32. The maximum number of electrons in phosphorous atom for which $n + l + m = 3$.
33. Find out maximum number of electrons in Cr atom for which $l + m = 0$.
34. Electrons of one subshell do not participate in bonding due to inert pair effect. Find out $(n+l)$ for that subshell.
35. For one element shielding effect is not possible. What is the atomic number of that element ?
36. According to Long form of modern periodic table, chalcogens are elements of group number x. Find the value of x.
37. The elements which exhibit both vertical and horizontal similarities are called transition elements. What is the atomic number of the 1st element of transition elements.
38. An element has spin magnetic moment equal to 1.73 B.M.. It has atomic number in between 22 to 30. Find out atomic number of element which is just after it along a period in the modern periodic table.
39. How many compounds having higher melting point than TiC .
 NaF , BeO , MgO , CaO , BaO , CsCl
40. How many elements have $3d^7$ electronic configuration in M^{+3} state. (Ti to Ni)
 Co , Fe , Ni , Mn , Cu , Zn
41. How many ions are diamagnetic.
 Sc^{3+} , Ti^{3+} , Ti^{2+} , V^{2+} , Cr^{3+} , Mn^{2+} , Fe^{2+} , Cu^{2+} , Co^{2+} , Zn^{2+}

9.32 PERIODIC TABLE

42. Number of 3d block elements which can have +4 oxidation state.

43.

Internuclear axis			
	x	y	z
$P_x + P_x$			
$P_z + P_z$			
$P_z + P_z$			

Number of overlap give σ bond = x.

Number of overlap giving π bond = y.

Find out x + y.

44. Total number of ions present in solution when $TiCl_3$ dissociated with ions completely.

45. How many overlaps give σ bond. (If inter nuclear axis is x axis).

- (i) s + s (v) $p_x + p_y$
(ii) s + p_x (vi) $p_x + p_x$
(iii) s + p_y (vii) $p_x + p_z$
(iv) s + p_z (viii) $p_y + p_y$

46.

Internuclear axis			
	x	y	z
$d_{xy} + d_{xy}$			
$d_{xz} + d_{xz}$			
$d_{x^2-y^2} + d_{x^2-y^2}$			
$d_z^2 + d_z^2$			

Find out total how many overlaps give π bond.

47.

Internuclear axis			
	x	y	z
$d_{xy} + d_{xy}$			
$d_{xz} + d_{xz}$			
$d_{x^2-y^2} + d_{x^2-y^2}$			
$d_z^2 + d_z^2$			

Find out number of overlaps which give δ bond.

48. If number of pentagon and number of hexagon in fullerene C_{60} is x and y then find out y - x

49. How many of the following statements related to the modern periodic table is incorrect ?

- (i) The p-block has 6 columns, because a maximum of 6 electrons can occupy all the orbitals in a p-shell.
(ii) The d-block has 8 columns, because a maximum of 8 electrons can occupy all the orbitals in a d-subshell.
(iii) Each block contains a number of columns equal to the number of electrons that can occupy that subshell.
(iv) The block indicates value of azimuthal quantum number (l) for the last subshell that received electrons in building up the electronic configuration.
(v) An element with mass number 37 possesses one unit of negative charge. If the ion contains 11.1% more neutrons than the electrons, then the symbol of ion is $^{37}_{17}Cl^{-1}$
(vi) An element with mass number 81 contains 31.7% more neutrons as compared to proton the atomic symbol of the element is $^{81}_{35}Br$.
50. If the number of electrons in the species H_2^+ , H_2 and O_2^+ are x, y, z respectively then find out $\frac{z}{x+y}$.

**EXERCISE # V
(JEE- MAIN & ADVANCED)**

1. The correct order of second ionisation potential of carbon, nitrogen, oxygen and fluorine is :

[IIT-JEE, 1981]

- (a) $C > N > O > F$ (b) $O > N > F > C$
(c) $O > F > N > C$ (d) $F > O > N > C$

2. The element with the highest first ionization potential is :

[IIT-JEE, 1982]

- (a) boron (b) carbon
(c) nitrogen (d) oxygen

3. The first ionization potentials (in electron volts) of nitrogen and oxygen atoms are, respectively, given by :

[IIT-JEE, 1987]

- (a) 14.5, 13.6 (b) 13.6, 14.6
(c) 13.6, 13.6 (d) 14.6, 14.6

4. Atomic radii of fluorine and neon (in Angstrom units) are, respectively given by : [IIT-JEE, 1987]
 (a) 0.72, 1.60 (b) 1.60, 1.60
 (c) 0.72, 0.72 (d) None of these
5. The electronegativity of the following elements increases in the order : [IIT-JEE, 1987]
 (a) C, N, Si, P (b) N, Si, C, P
 (c) Si, P, C, N (d) P, Si, N, C
6. The first ionization potentials of Na, Mg, Al and Si are in the order : [IIT-JEE, 1988]
 (a) $\text{Na} < \text{Mg} < \text{Al} < \text{Si}$ (b) $\text{Na} > \text{Mg} > \text{Al} > \text{Si}$
 (c) $\text{Na} < \text{Mg} < \text{Al} > \text{Si}$ (d) $\text{Na} > \text{Mg} > \text{Al} < \text{Si}$
7. Which of the following is smallest in size : [IIT-JEE, 1988]
 (a) N^{3-} (b) O^{2-}
 (c) F^- (d) Na^+
8. Among the following elements (whose electronic configurations are given below), the one having the highest ionization energy : [IIT-JEE, 1990]
 (a) $[\text{Ne}]3s^23p^1$ (b) $[\text{Ne}]3s^23p^3$
 (c) $[\text{Ne}]3s^23p^2$ (d) $[\text{Ne}]3d^{10}4s^24p^1$
9. Which of the following statements is not correct for the periodic classification of elements ? [IIT-JEE, 1992]
 (a) The properties of elements are the periodic functions of their atomic numbers.
 (b) Non-metallic elements are lesser in number than metallic elements
 (c) The first ionisation energies of elements along a period do not vary in a regular manner with increase in atomic number.
 (d) For transition elements, the d-subshells are filled with electrons monotonically with increase in atomic number.
10. Which of the following has the most stable +2 oxidation state ? [IIT-JEE, 1995]
 (a) Sn
 (b) Pb
 (c) Fe
 (d) Ag
11. Which of the following has the maximum number of unpaired electrons : [IIT-JEE, 1996]
 (a) Mg^{2+} (b) Ti^{3+}
 (c) V^{3+} (d) Fe^{2+}
12. Which of the following statements is wrong ? [IIT-JEE, 1997]
 (a) The first ionisation potential of Al is less than the first ionization potential of Mg.
 (b) The second ionisation potential of Mg is greater than the second ionization potential of Na.
 (c) The first ionisation potential of Na is less than the first ionization potential of Mg.
 (d) The third ionisation potential of Mg is greater than the third ionization potential of Mg.
13. The correct order of acidic strength is : [IIT-JEE, 2000]
 (a) $\text{Cl}_2\text{O}_7 > \text{SO}_2 > \text{P}_4\text{O}_{10}$
 (b) $\text{CO}_2 > \text{N}_2\text{O}_5 > \text{SO}_3$
 (c) $\text{Na}_2\text{O} > \text{MgO} > \text{Al}_2\text{O}_3$
 (d) $\text{K}_2\text{O} > \text{CaO} > \text{MgO}$
14. The correct order of radii is : [IIT-JEE, 200]
 (a) $\text{N} < \text{Be} < \text{B}$
 (b) $\text{F}^- < \text{O}^{2-} < \text{N}^{3-}$
 (c) $\text{Na} < \text{Li} < \text{K}$
 (d) $\text{Fe}^{3+} < \text{Fe}^{2+} < \text{Fe}^{4+}$
15. The incorrect statements among the following is : [JEE, 1997]
 (a) The first ionisation energy of Al is less than first ionization energy of Mg
 (b) The second ionisation energy of Mg is greater than second ionization energy of Na.
 (c) The first ionisation energy of Na is less than first ionization energy of Mg.
 (d) The third ionisation energy of Mg is greater than third ionization energy of Al.
16. Property of the alkaline earth metals that increases with their atomic number is : [JEE, 2000]
 (a) ionisation energy
 (b) solubility of their hydroxides
 (c) solubility of their sulphates
 (d) electronegativity

9.34 PERIODIC TABLE

17. Arrange the following ions in order of their increasing size : Li^+ , Mg^{2+} , K^+ , Al^{3+} . [JEE, 1997]
18. Compounds that formally contain Pb^{4+} are easily reduced to Pb^{2+} . The stability of the lower oxidation state is due to [JEE, 1997]
19. **Assertion :** F atom has a less negative electron affinity than Cl atom. [JEE, 1998]
Reason : Additional electron are repelled more effectively by 3p electrons in Cl atom than by 2p electrons in F atom.
 (a) Both Assertion and Reason are true, and Reason is the correct explanation of Assertion.
 (b) Both Assertion and Reason are true, and Reason is not correct explanation of Assertion.
 (c) Assertion is true but Reason is false.
 (d) Assertion is false but Reason is true.
20. Ionic radii of: [JEE, 1999]
 (a) $\text{Ti}^{4+} < \text{Mn}^{7+}$
 (b) $^{35}\text{Cl}^- < ^{37}\text{Cl}^-$
 (c) $\text{K}^+ > \text{Cl}^-$
 (d) $\text{P}^{3+} > \text{P}^{5+}$
21. Give reason for the following in one or two sentences only: [JEE 1999]
 CrO_3 is an acid anhydride.
22. **Assertion :** The first ionization energy of Be is greater than that of B. [JEE, 2000]
Reason : 2p orbital is lower in energy than 2s.
 (a) Both Assertion and Reason are true, and Reason is the correct explanation of Assertion.
 (b) Both Assertion and Reason are true, and Reason is not correct explanation of Assertion.
 (c) Assertion is true but Reason is false.
 (d) Assertion is false but Reason is true.
23. The set representing the correct order of first ionization potential is [JEE, 2001]
 (a) $\text{K} > \text{Na} > \text{Li}$
 (b) $\text{Be} > \text{Mg} > \text{Ca}$
 (c) $\text{B} > \text{C} > \text{N}$
 (d) $\text{Ge} > \text{Si} > \text{C}$
24. The set with correct order of acidity is : [JEE, 2001]
 (a) $\text{HClO} < \text{HClO}_2 < \text{HClO}_3 < \text{HClO}_4$
 (b) $\text{HClO}_4 < \text{HClO}_3 < \text{HClO}_2 < \text{HClO}$
 (c) $\text{HClO} < \text{HClO}_4 < \text{HClO}_3 < \text{HClO}_2$
 (d) $\text{HClO}_4 < \text{HClO}_2 < \text{HClO}_3 < \text{HClO}$
25. Identify the correct order of acidity strengths of CO_2 , CuO , CaO , H_2O : [JEE, 2002]
 (a) $\text{CaO} < \text{CuO} < \text{H}_2\text{O} < \text{CO}_2$
 (b) $\text{H}_2\text{O} < \text{CuO} < \text{CaO} < \text{CO}_2$
 (c) $\text{CaO} < \text{H}_2\text{O} < \text{CuO} < \text{CO}_2$
 (d) $\text{H}_2\text{O} < \text{CO}_2 < \text{CaO} < \text{CuO}$
26. Identify the least stable ion amongst the following : [JEE, 2002]
 (a) Li^-
 (b) Be^-
 (c) B^-
 (d) C^-
27. Arrange the following oxides in the increasing order of Bronsted basicity : [JEE, 2004]
 Cl_2O_7 , BaO , SO_3 , CO_2 , B_2O_3
28. **Assertion :** Pb^{4+} compounds are stronger oxidizing agents than Sn^{4+} compounds. [JEE, 2008]
Reason : The higher oxidation states for the group 14 elements are more stable for the heavier members of the group due to inert pair effect.
 (a) Both Assertion and Reason are true, and Reason is the correct explanation of Assertion.
 (b) Both Assertion and Reason are true, and Reason is not correct explanation of Assertion.
 (c) Assertion is true but Reason is false.
 (d) Assertion is false but Reason is true.
29. Which one of the following ions has the highest value of ionic radius - [AIEEE, 2004]
 (a) Li^+
 (b) B^{3+}
 (c) O^{2-}
 (d) F^-

30. The formation of the oxide ion $O^{2-}(g)$ requires first an exothermic and then an endothermic step as shown below : [AIEEE, 2004]
- $$O(g) + e^- = O^-(g) ; \Delta H^\circ = -142 \text{ kJmol}^{-1}$$
- $$O^-(g) + e^- = O^{2-}(g) ; \Delta H^\circ = 844 \text{ kJmol}^{-1}$$
- This is because -
- oxygen is more electronegative
 - oxygen has high electron affinity
 - O^- ion will tend to resist the addition of another electrons
 - O^- ion has comparatively larger size than oxygen atom
31. Among Al_2O_3 , SiO_2 , P_2O_3 and SO_2 the correct order of acid strength is: [AIEEE, 2004]
- $SO_2 < P_2O_3 < SiO_2 < Al_2O_3$
 - $SiO_2 < SO_2 < Al_2O_3 < P_2O_3$
 - $Al_2O_3 < SiO_2 < SO_2 < P_2O_3$
 - $P_2O_3 < SiO_2 < Al_2O_3 < SO_2$
32. Which of the following oxides is amphoteric in nature :: [AIEEE, 2005]
- CaO
 - CO_2
 - SiO_2
 - SnO_2
33. In which of the following arrangements the order is NOT according to the property indicated against it? [AIEEE, 2005]
- $Al^{3+} < Mg^{2+} < Na^+ < F^-$ - increasing ionic size
 - $B < C < N < O$ - increasing first ionisation enthalpy
 - $I < Br < F < Cl$ - increasing electron gain enthalpy (with negative size)
 - $Li < Na < K < Rb$ - increasing metallic radius
34. Which of the following factors may be regarded as the main cause of lanthanide contraction : [AIEEE, 2005]
- greater shielding of 5d-electrons by 4f-electrons
 - poorer shielding of 5d-electrons by 4f-electrons
 - effective shielding of one of 4f-electrons by another in the sub-shell
 - poor shielding of one of 4f-electrons by another in the sub-shell
35. The lanthanide contraction is responsible for the fact that : [AIEEE 2005]
- Zr and Y have about the same radius
 - Zr and Nb have similar oxidation state
 - Zr and Hf have about the same radius
 - Zr and Zn have similar oxidation state
36. The increasing order of the first ionization enthalpies of the elements B, P, S and F (lowest first) is : [AIEEE 2006]
- $F < S < P < B$
 - $P < S < B < F$
 - $B < P < S < F$
 - $B < S < P < F$
37. Which of the following statements is true ? [AIEEE 2006]
- H_3PO_4 is a stronger acid than H_2SO_3
 - In aqueous medium, HF is stronger acid than HCl
 - $HClO_4$ is a weaker acid than $HClO_3$
 - HNO_3 is a stronger acid than HNO_2
38. Lanthanoid contraction is caused due to : [AIEEE, 2007]
- the appreciable shielding on outer electrons by 4f electrons from the nuclear charge
 - the appreciable shielding on outer electrons by 5f electrons from the nuclear charge
 - the same effective nuclear charge from Ce to Lu
 - the imperfect shielding on outer electrons by 4f electrons from the nuclear charge
39. The stability of dihalides of Si, Ge, Sn and Pb increases steadily in the sequence : [AIEEE, 2007]
- $SiX_2 << GeX_2 << SnX_2 << PbX_2$
 - $PbX_2 << SnX_2 << GeX_2 << SiX_2$
 - $GeX_2 << SiX_2 << SnX_2 << PbX_2$
 - $SiX_2 << GeX_2 << PbX_2 << SnX_2$
40. The set representing the correct order of ionic radius is: [AIEEE, 2009]
- $Na^+ > Li^+ > Mg^{2+} > Be^{2+}$
 - $Li^+ > Na^+ > Mg^{2+} > Be^{2+}$
 - $Mg^{2+} > Be^{2+} > Li^+ > Na^+$
 - $Li^+ > Be^{2+} > Na^+ > Mg^{2+}$

9.36 PERIODIC TABLE

41. In which of the following arrangements, the sequence is not strictly according to the property written against it? [AIEEE, 2009]
 (a) $\text{HF} < \text{HCl} < \text{HBr} < \text{HI}$: increasing acid strength
 (b) $\text{NH}_3 < \text{PH}_3 < \text{AsH}_3 < \text{SbH}_3$: increasing basic strength
 (c) $\text{B} < \text{C} < \text{O} < \text{N}$: increasing first ionization enthalpy
 (d) $\text{CO}_2 < \text{SiO}_2 < \text{SnO}_2 < \text{PbO}_2$: increasing oxidising power
42. The correct sequence which shows decreasing order of the ionic radii of the elements is : [AIEEE, 2010]
 (a) $\text{Al}^{3+} > \text{Mg}^{2+} > \text{Na}^+ > \text{F}^- > \text{O}^{2-}$
 (b) $\text{Na}^+ > \text{Mg}^{2+} > \text{Al}^{3+} > \text{O}^{2-} > \text{F}^-$
 (c) $\text{Na}^+ > \text{F}^- > \text{Mg}^{2+} > \text{O}^{2-} > \text{Al}^{3+}$
 (d) $\text{O}^{2-} > \text{F}^- > \text{Na}^+ > \text{Mg}^{2+} > \text{Al}^{3+}$
43. The outer electron configuration of Gd (atomic number 64) is: [AIEEE, 2011]
 (a) $4f^3 5d^5 6s^2$
 (b) $4f^8 5d^{10} 6s^2$
 (c) $4f^4 5d^5 6s^2$
 (d) $4f^7 3d^1 6s^2$
44. Which of the following orders present the correct sequence of the increasing basic nature of the given oxides? [AIEEE 2011]
 (a) $\text{Al}_2\text{O}_3 < \text{MgO} < \text{Na}_2\text{O} < \text{K}_2\text{O}$
 (b) $\text{MgO} < \text{K}_2\text{O} < \text{Al}_2\text{O}_3 < \text{Na}_2\text{O}$
 (c) $\text{Na}_2\text{O} < \text{K}_2\text{O} < \text{MgO} < \text{Al}_2\text{O}_3$
 (d) $\text{K}_2\text{O} < \text{Na}_2\text{O} < \text{Al}_2\text{O}_3 < \text{MgO}$
45. The correct order of electron gain enthalpy with negative sign of F, Cl, Br and I having atomic number 9, 17, 35 and 53 respectively as : [AIEEE, 2012]
 (a) $\text{F} > \text{Cl} > \text{Br} > \text{I}$
 (b) $\text{Cl} > \text{F} > \text{Br} > \text{I}$
 (c) $\text{Br} > \text{Cl} > \text{I} > \text{F}$
 (d) $\text{I} > \text{Br} > \text{Cl} > \text{F}$
46. The increasing order of the ionic radii of the given isoelectronic species is - [AIEEE, 2012]
 (a) $\text{Cl}^-, \text{Ca}^{2+}, \text{K}^+, \text{S}^{2-}$
 (b) $\text{S}^{2-}, \text{Cl}^-, \text{Ca}^{2+}, \text{K}^+$
 (c) $\text{Ca}^{2+}, \text{K}^+, \text{Cl}^-, \text{S}^{2-}$
 (d) $\text{K}^+, \text{S}^{2-}, \text{Ca}^{2+}, \text{Cl}^-$
47. Which of the following represent the correct order of increasing first ionization enthalpy for Ca, Ba, S, Se and Ar? [AIEEE, 2013]
 (a) $\text{Ca} < \text{S} < \text{Ba} < \text{Se} < \text{Ar}$
 (b) $\text{S} < \text{Se} < \text{Ca} < \text{Ba} < \text{Ar}$
 (c) $\text{Ba} < \text{Ca} < \text{Se} < \text{S} < \text{Ar}$
 (d) $\text{Ca} < \text{Ba} < \text{S} < \text{Se} < \text{Ar}$
48. The first ionization potential of Na is 5.1 eV. The value of electron gain enthalpy of Na^+ will be: [AIEEE, 2014]
 (a) -2.55 eV
 (b) -5.1 eV
 (c) -10.2 eV
 (d) +2.55 eV
49. Among the following oxoacids, the correct decreasing order of acid strength is: [AIEEE, 2014]
 (a) $\text{HClO}_4 > \text{HClO}_3 > \text{HClO}_2 > \text{HOCl}$
 (b) $\text{HClO}_2 > \text{HClO}_4 > \text{HClO}_3 > \text{HOCl}$
 (c) $\text{HOCl} > \text{HClO}_2 > \text{HClO}_3 > \text{HClO}_4$
 (d) $\text{HClO}_4 > \text{HOCl} > \text{HClO}_2 > \text{HClO}_3$
50. In the long form of the periodic table, the valence shell electronic configuration of $5s^2 5p^4$ corresponds to the element present in : [AIEEE, 2015]
 (a) Group 16 and period 5
 (b) Group 17 and period 6
 (c) Group 17 and period 5
 (d) Group 16 and period 6
51. In an atom, the total number of electrons having quantum numbers $n = 4$, $|m_l| = 1$ and $m_s = -\frac{1}{2}$ is : [AIEEE, 2014]

■■■

ANSWERS**CH-1 CHEMICAL BONDING****EXERCISE # I**

1. (d)	2. (d)	3. (c)	4. (c)	5. (d)	6. (b)	7. (c)	8. (b)	9. (d)	10. (c)
11. (d)	12. (d)	13. (c)	14. (c)	15. (c)	16. (b)	17. (a)	18. (d)	19. (b)	20. (d)
21. (d)	22. (a)	23. (b)	24. (a)	25. (c)	26. (b)	27. (c)	28. (a)	29. (a)	30. (c)
31. (d)	32. (c)	33. (c)	34. (c)	35. (c)	36. (d)	37. (d)	38. (a)	39. (c)	40. (d)
41. (b)	42. (d)	43. (d)	44. (b)	45. (a)	46. (c)	47. (a)	48. (a)	49. (a)	50. (c)
51. (b)	52. (b)	53. (b)	54. (b)	55. (c)	56. (c)	57. (d)	58. (d)	59. (b)	60. (a)
61. (c)	62. (d)	63. (c)	64. (a)	65. (a)	66. (a)	67. (b)	68. (d)	69. (a)	70. (d)
71. (d)	72. (a)	73. (c)	74. (d)	75. (d)	76. (d)	77. (a)	78. (c)	79. (a)	80. (d)
81. (d)	82. (c)	83. (b)	84. (c)	85. (c)	86. (c)	87. (a)	88. (c)	89. (c)	90. (b)
91. (d)	92. (a)	93. (d)	94. (c)	95. (b)	96. (c)	97. (c)	98. (b)	99. (d)	100. (c)
101. (d)	102. (a)	103. (c)	104. (c)	105. (a)	106. (d)	107. (b)	108. (c)	109. (c)	110. (c)
111. (d)	112. (b)	113. (c)	114. (b)	115. (c)	116. (c)	117. (a)	118. (d)	119. (d)	120. (d)
121. (b)	122. (d)	123. (c)	124. (d)	125. (b)	126. (b)	127. (b)	128. (d)	129. (d)	130. (c)
131. (a)	132. (d)	133. (d)	134. (c)	135. (a)	136. (c)	137. (a)	138. (c)	139. (d)	140. (d)
141. (c)	142. (a)	143. (c)	144. (c)	145. (c)	146. (a)	147. (b)	148. (d)	149. (b)	150. (a)
151. (d)	152. (b)	153. (d)	154. (d)	155. (c)	156. (a)	157. (a)	158. (d)	159. (d)	160. (a)
161. (d)	162. (b)	163. (c)	164. (c)	165. (a)	166. (b)	167. (d)	168. (a)	169. (c)	170. (b)
171. (b)	172. (c)	173. (a)	174. (b)	175. (b)	176. (c)	177. (d)	178. (c)	179. (c)	180. (c)
181. (d)	182. (a)	183. (c)	184. (d)	185. (a)	186. (c)	187. (a)	188. (b)	189. (d)	190. (d)
191. (d)	192. (d)	193. (b)	194. (a)	195. (b)	196. (d)	197. (a)	198. (d)	199. (c)	200. (b)

EXERCISE-II

1. (a,b,c,d)	2. (a,c,d)	3. (a,b,d)	4. (a,c,d)	5. (b,c,d)	6. (a,b,c,d)
7. (a,b)	8. (b,c)	9. (a,d)	10. (a,b,c)	11. (b,d)	12. (a,b,d)
13. (a,d)	14. (b,c,d)	15. (a,c,d)	16. (b,c)	17. (a,b,d)	18. (a,b,d)
19. (c,d)	20. (a,d)	21. (c,d)	22. (a,b)	23. (b,d)	24. (a,c,d)
25. (b,c)	26. (a,c,d)	27. (a,c,d)	28. (a,d)	29. (a,b)	30. (b,c)
31. (a,c,d)	32. (b,c)	33. (a,b,c)	34. (a,b,c,d)	35. (a,b,c,d)	36. (b,d)
37. (a,b,c,d)	38. (a,b)	39. (c,d)	40. (a,b)	41. (a,d)	42. (b,c,d)
43. (a,c,d)	44. (a,b)	45. (b,d)	46. (b,d)	47. (a,c)	48. (a,b)
49. (b,d)	50. (b,c)	51. (a,b,c,d)	52. (a,d)	53. (c,d)	54. (a,b)
55. (a,d)	56. (c,d,f)	57. (a,c)	58. (a,b,d)	59. (a,c,d)	60. (b,c)
61. (a,b,c)	62. (b,c)	63. (a,c)	64. (c,d)	65. (a,c)	66. (a,b,c)
67. (a,b,d)	68. (a,c,d)	69. (a,b)	70. (a,b,c)	71. (b,c,d)	72. (a,d)
73. (a,b,c,d)	74. (a,b)	75. (a,b,c,d)	76. (a,b,c)	77. (b,c)	78. (c,d)
79. (a,c)	80. (a,d)	81. (a,b,d)	82. (a,d)	83. (b,c)	84. (a,b,c)
85. (a,b)	86. (a,b,c)	87. (a,b,c,d)	88. (a,b)	89. (a,c)	90. (a,c,d)

10.2 ANSWERS

91. (a,c) 92. (a,b,d) 93. (a,b,c,d) 94. (c,d) 95. (a,d) 96. (b,d)
 97. (a,b,c) 98. (a,b,d) 99. (c,d) 100. (a,b) 101. (c,d) 102. (a,d)
 103. (a,b,c) 104. (a,c) 105. (a,b,c) 106. (a,d) 107. (a,b,d) 108. (a,b,c,d)
 109. (a,b) 110. (a,d) 111. (a,b,c,d) 112. (a,b,c,d) 113. (a,c,d) 114. (a,b,c)
 115. (a,d) 116. (c,d) 117. (a,d) 118. (b,c) 119. (a,b) 120. (a,b)
 121. (a,b,c) 122. (a,d) 123. (a,b,c) 124. (c,d) 125. (a,b,d) 126. (a,b)
 127. (a,b,c,d) 128. (b,c,d) 129. (a,d) 130. (a,c,d) 131. (b,c) 132. (a,b,c)
 133. (a,b,c,d) 134. (b,c) 135. (a,d) 136. (a,b,c,d) 137. (a,c,d) 138. (b,c)
 139. (b,c) 140. (a,b,c) 141. (a,b,c) 142. (b,c,d) 143. (a,b,c,d) 144. (a,b,c)
 145. (a,c,d) 146. (a,b,c,d) 147. (a,b,d) 148. (a,b,c) 149. (c,d) 150. (c,d)

EXERCISE-III : Paragraph Type & Matrix Match Type

1. (c) 2. (c) 3. (b) 4. (a) 5. (c) 6. (d) 7. (b) 8. (c) 9. (d) 10. (d)
 11. (b,d) 12. (c) 13. (b) 14. (d) 15. (b) 16. (a) 17. (c) 18. (b) 19. (b) 20. (c)
 21. (a) 22. (c) 23. (c) 24. (d) 25. (d) 26. (d) 27. (d) 28. (a) 29. (b) 30. (c)
 31. (b) 32. (b) 33. (d) 34. (c) 35. (d) 36. (c) 37. (d) 38. (d) 39. (a) 40. (d)
 41. (d) 42. (c) 43. (a) 44. (b) 45. (a) 46. (a) 47. (d) 48. (c) 49. (b) 50. (a)
 51. (b) 52. (d) 53. (d) 54. (d) 55. (d) 56. (d) 57. (d) 58. (d) 59. (d) 60. (b)
 61. (d) 62. (a) 63. (c) 64. (b) 65. (d)
 66. (A)-P ; (B)-Q ; (C)-S ; (D)-R
 67. (A)-P ; (B)-Q ; (C)-R ; (D)-S
 68. (A)-Q ; (B)-P ; (C)-R ; (D)-S
 69. (A)-P, Q, R, S ; (B)-P, Q, R, S ; (C)-P, R, S ; (D)-P, R, S
 70. (A)-Q, R, S ; (B)-Q, R ; (C)-P, R ; (D)-Q, R, S
 71. (A)-S ; (B)-R ; (C)-Q ; (D)-P
 72. (A)-P ; (B)-Q, S ; (C)-P, R ; (D)-Q, R
 73. (A)-P ; (B)-Q, S ; (C)-P, R ; (D)-Q, S
 74. (A)-Q, R, T ; (B)-P, Q, R, S ; (C)-P, R, S ; (D)-Q, R, T
 75. (A)-P, R, S ; (B)-R ; (C)-Q, R ; (D)-Q, R
 76. (A)-P, S ; (B)-S ; (C)-Q, R ; (D)-P, Q, R
 77. (A)-R, T ; (B)-Q, S, T ; (C)-P, R, T ; (D)-P, S, T
 78. (A)-P, S ; (B)-Q, S ; (C)-R, S ; (D)-S
 79. (A)-R ; (B)-S ; (C)-Q ; (D)-P
 80. (A)-P ; (B)-P, Q ; (C)-R, S ; (D)-S, T
 81. (A)-P, S, T ; (B)-P, Q, T ; (C)-P, R, T ; (D)-P, T
 82. (A)-P, Q, R, S, T ; (B)-P, Q, R, S, T ; (C)-Q, R, T ; (D)-Q, R, T
 83. (A)-P, R ; (B)-P, Q ; (C)-P, R ; (D)-P, S
 84. (A)-S ; (B)-R ; (C)-Q ; (D)-P
 85. (A)-R ; (B)-P, Q, S, T ; (C)-P, R, S ; (D)-T
 86. (A)-P, Q ; (B)-P ; (C)-R ; (D)-S
 87. (A)-P, S, T ; (B)-P, S ; (C)-P, Q ; (D)-R
 88. (A)-P ; (B)-Q ; (C)-R, T ; (D)-S
 89. (A)-Q, S, T ; (B)-R, S ; (C)-S ; (D)-P, S
 90. (A)-P, Q ; (B)-R ; (C)-S, T ; (D)-P, S
 91. (A)-R ; (B)-Q ; (C)-P ; (D)-S
 92. (A)-R, T ; (B)-P, T ; (C)-Q, T ; (D)-P, S
 93. (A)-P ; (B)-Q ; (C)-R ; (D)-S
 94. (i) -X, Z (ii) -P, Q, S (iii) -P, R, S (iv) -T, X, Y
 (v) -T, X (vi) -T, V, X, Z (vii) -U, Y (viii) -W, X, Z
 95. (i) -Q, W (ii) -Q, W, Y (iii) -Q, U, W, Y (iv) -Q, R, U, W, Y
 (v) -P, Q, U, W, Y (vi) -Q, R, U, W, Y (vii) -Q, S, U, W, Y (viii) -Q, S, W, X, Y
 (ix) -Q, T, V, W, Y (x) -Q, T, V, W, Y

96. (A)-P,Q,R,S ; (B)-P,Q,R,S ; (C)-Q,R,S ; (D)-Q,R
 97. (A)-R ; (B)-R,S ; (C)-P,Q,R ; (D)-P,R
 98. (A)-S ; (B)-P ; (C)-R ; (D)-Q
 99. (A)-P, Q, R ; (B)-Q,R ; (C)-P,Q,R,S ; (D)-P,Q,R
 100. (A)-P ; (B)-Q ; (C)-R ; (D)-S

EXERCISE-IV : Integer Type

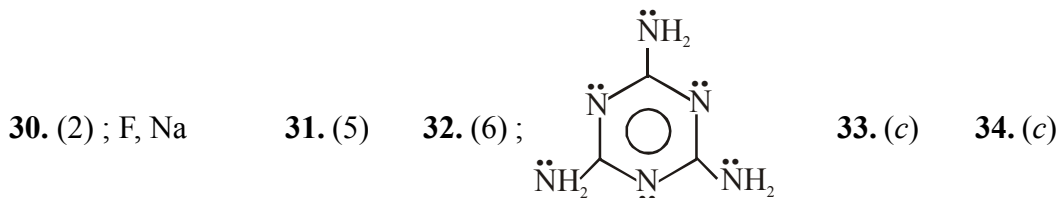
1. (5) 2. (5) 3. (6) 4. (1) 5. (3) 6. (5) 7. (4) 8. (7) 9. (7) 10. (6)
 11. (2) 12. (3) 13. (9) 14. (1) 15. (6) 16. (5) 17. (2) 18. 12 or 3 19. (1) 20. (3)
 21. (5) 22. (5) 23. (3) 24. (5) 25. (3) 26. (2) 27. (9) 28. (2) 29. (1) 30. (5)
 31. (4) 32. (3) 33. (0) 34. (0) 35. (2) 36. (3) 37. (1) 38. (3) 39. (0) 40. (7)
 41. (2) 42. (4) 43. (8) 44. (4) 45. (7) 46. (4) 47. (8) 48. (5) 49. (5) 50. (3)
 51. (6) 52. (1) 53. (2) 54. (6) 55. (3) 56. (2) 57. (5) 58. 7, 25 59. (3) 60. (9)
 61. (5) 62. (4) 63. (2) 64. (6) 65. (6) 66. (4) 67. (1) 68. (2) 69. (4) 70. (3)
 71. (9) 72. (4) 73. (5) 74. (4) 75. (4) 76. (0) 77. 24 or 6 78. (6) 79. (3) 80. (6)

EXERCISE-V(A) : (JEE-MAIN)

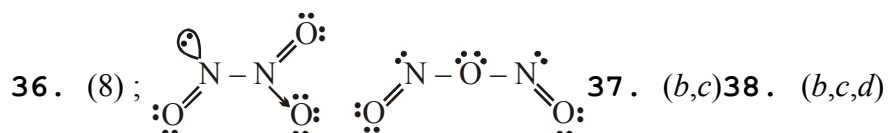
1. (3) 2. (2) 3. (2) 4. (1) 5. (1) 6. (2) 7. (3) 8. (1) 9. (4) 10. (3)
 11. (2) 12. (4) 13. (4) 14. (3) 15. (2) 16. (3) 17. (4) 18. (1) 19. (3) 20. (4)
 21. (1) 22. (1) 23. (4) 24. (1) 25. (1) 26. (3) 27. (4) 28. (2) 29. (1) 30. (4)
 31. (1) 32. (4) 33. (4) 34. (2) 35. (2) 36. (Bonus) 37. (2) 38. (4) 39. (3) 40. (3)
 41. (3) 42. (4) 43. (2) 44. (1) 45. (3) 46. (2) 47. (2) 48. (1) 49. (1) 50. (3)
 51. (3) 52. (2) 53. (1) 54. (2) 55. (4) 56. (1) 57. (3) 58. (2) 59. (3) 60. (3)
 61. (3) 62. (2) 63. (1) 64. (3) 65. (2) 66. (1) 67. (2) 68. (2) 69. (1) 70. (1)
 71. (3) 72. (1) 73. (4) 74. (2,3,4) 75. (4) 76. (2,4) 77. (2) 78. (4) 79. (2) 80. (2)
 81. (2) 82. (2) 83. (2) 84. (4)

EXERCISE-V(B) : (JEE-ADVANCED)

1. (b) 2. (b) 3. (a) 4. (a) 5. (a) 6. (a) 7. (c) 8. (a) 9. (b) 10. (d)
 11. (d) 12. (d) 13. (a) 14. (c) 15. (c) 16. (c) 17. (c)
 18. (A)-P,Q,R,T ; (B)-Q,R,S,T ; (C)-P,Q,R ; (D)-P,Q,R,S
 19. (a,b,c) 20. (b,c) 21. (d) 22. (a) 23. (b) 24. (b) 25. (d) 26. (4) 27. (0) 28. (3)
 29. (6) : H_2SO_4 ; H_3PO_4 ; H_2CO_3 ; $\text{H}_2\text{S}_2\text{O}_7$; H_2CrO_4 ; H_2SO_3



35. (D) ; BeCl_2 , N_3^- , N_2O , NO_2^+



CH-2 COORDINATION COMPOUNDS**EXERCISE # I**

- | | | | | | | | | | |
|----------|----------|----------|----------|----------|----------|----------|----------|----------|----------|
| 1. (b) | 2. (c) | 3. (c) | 4. (c) | 5. (d) | 6. (b) | 7. (d) | 8. (c) | 9. (d) | 10. (a) |
| 11. (c) | 12. (d) | 13. (c) | 14. (b) | 15. (b) | 16. (c) | 17. (d) | 18. (d) | 19. (c) | 20. (a) |
| 21. (d) | 22. (c) | 23. (c) | 24. (c) | 25. (a) | 26. (c) | 27. (b) | 28. (d) | 29. (d) | 30. (c) |
| 31. (d) | 32. (a) | 33. (c) | 34. (c) | 35. (c) | 36. (c) | 37. (d) | 38. (b) | 39. (d) | 40. (a) |
| 41. (a) | 42. (b) | 43. (a) | 44. (a) | 45. (d) | 46. (c) | 47. (d) | 48. (c) | 49. (d) | 50. (b) |
| 51. (d) | 52. (c) | 53. (a) | 54. (a) | 55. (a) | 56. (d) | 57. (c) | 58. (b) | 59. (a) | 60. (c) |
| 61. (a) | 62. (c) | 63. (c) | 64. (d) | 65. (d) | 66. (a) | 67. (b) | 68. (c) | 69. (b) | 70. (c) |
| 71. (d) | 72. (b) | 73. (a) | 74. (a) | 75. (d) | 76. (c) | 77. (d) | 78. (d) | 79. (c) | 80. (a) |
| 81. (b) | 82. (a) | 83. (a) | 84. (a) | 85. (d) | 86. (b) | 87. (c) | 88. (b) | 89. (a) | 90. (b) |
| 91. (d) | 92. (a) | 93. (d) | 94. (c) | 95. (c) | 96. (c) | 97. (a) | 98. (c) | 99. (d) | 100. (c) |
| 101. (d) | 102. (d) | 103. (d) | 104. (b) | 105. (c) | 106. (b) | 107. (b) | 108. (b) | 109. (a) | 110. (d) |
| 111. (d) | 112. (d) | 113. (a) | 114. (d) | 115. (a) | 116. (c) | 117. (a) | 118. (d) | 119. (c) | 120. (d) |
| 121. (d) | 122. (d) | 123. (c) | 124. (c) | 125. (d) | 126. (d) | 127. (d) | 128. (c) | 129. (a) | 130. (c) |
| 131. (a) | 132. (c) | 133. (b) | 134. (b) | 135. (a) | 136. (d) | 137. (b) | 138. (c) | 139. (c) | 140. (c) |
| 141. (c) | 142. (a) | 143. (d) | 144. (c) | 145. (d) | 146. (b) | 147. (b) | 148. (d) | 149. (d) | 150. (a) |

EXERCISE # II

- | | | | | | |
|---------------|---------------|---------------|---------------|---------------|---------------|
| 1. (a,b,c) | 2. (a,b,c) | 3. (b,d) | 4. (b,c,d) | 5. (a,b,d) | 6. (a,b) |
| 7. (a,d) | 8. (a,d) | 9. (b,c,d) | 10. (b,c) | 11. (a,d) | 12. (a,b) |
| 13. (a,b) | 14. (a,b,c) | 15. (a,b,c,d) | 16. (a,b,c,d) | 17. (a,b,c,d) | 18. (b,c,d) |
| 19. (a,b,c) | 20. (a,b,c,d) | 21. (b,d) | 22. (a,b,c,d) | 23. (a,b,c) | 24. (b,c) |
| 25. (a,d) | 26. (a,b,d) | 27. (a,b,c,d) | 28. (b,c) | 29. (a,b,c) | 30. (b,c) |
| 31. (c,d) | 32. (a,b,c) | 33. (b,c,d) | 34. (a,c) | 35. (a,b,c,d) | 36. (a,b,c) |
| 37. (a,b,c) | 38. (b,c,d) | 39. (b,c) | 40. (a,b,c) | 41. (a,b,c) | 42. (a,c) |
| 43. (a,d) | 44. (a,b) | 45. (a,d) | 46. (a,c) | 47. (a,c) | 48. (a,b,c,d) |
| 49. (b,c) | 50. (a,b,c) | 51. (b,d) | 52. (a,c) | 53. (b,c) | 54. (b,c,d) |
| 55. (a,b,d) | 56. (c,d) | 57. (a,b,d) | 58. (a,b) | 59. (b,c) | 60. (a,b,c) |
| 61. (b,c,d) | 62. (a,b) | 63. (b,d) | 64. (a,c,d) | 65. (b,c,d) | 66. (a,d) |
| 67. (a,b,d) | 68. (a,b,c,d) | 69. (a,b) | 70. (c,d) | 71. (c,d) | 72. (a,b,d) |
| 73. (a,c) | 74. (b,c) | 75. (c,d) | 76. (a,b,c) | 77. (b,c,d) | 78. (a,b,d) |
| 79. (b,c,d) | 80. (a,b,c,d) | 81. (a,b) | 82. (a,c,d) | 83. (a,b,c,d) | 84. (a,b,c,d) |
| 85. (a,b,c,d) | 86. (a,b) | 87. (a,b,d) | 88. (b,c,d) | 89. (b,c) | 90. (a,c,d) |
| 91. (b,d) | 92. (a,b,d) | 93. (b,c) | 94. (b,c,d) | 95. (c,d) | 96. (a,b,c,d) |
| 97. (a,b,c) | 98. (a,b) | 99. (a,b) | 100. (a,b,c) | | |

EXERCISE # III

Paragraph Type :

1. (c) 2. (c) 3. (a) 4. (d) 5. (a) 6. (b) 7. (d) 8. (a) 9. (a) 10. (c)
 11. (b) 12. (b) 13. (a) 14. (c) 15. (b) 16. (c) 17. (d) 18. (b) 19. (c) 20. (c)
 21. (b) 22. (c) 23. (d) 24. (b) 25. (d) 26. (d) 27. (d) 28. (c) 29. (a) 30. (c)
 31. (c) 32. (d) 33. (b) 34. (c) 35. (d) 36. (b) 37. (d) 38. (c) 39. (a) 40. (b)
 41. (a) 42. (c) 43. (c) 44. (a) 45. (b) 46. (d) 47. (b) 48. (a,b,c,d)
 49. (a,b,c,d) 50. (a,b,c,d) 51. (a,b,c,d) 52. (a,c) 53. (a,c) 54. (b) 55. (a)
 56. (a,b,c,d) 57. (b) 58. (c) 59. (b) 60. (d) 61. (b) 62. (b) 63. (c) 64. (c)
 65. (c) 66. (a) 67. (c) 68. (a) 69. (b) 70. (c)

Matrix Match Type :

- | | (a) | (b) | (c) | (d) | | (a) | (b) | (c) | (d) |
|------|-----------|-------------|---------|--------|------|-----------|---------|---------|--------|
| 71. | S; | -P,T ; | Q,R,T; | R,T | 72. | -P,T; | -R,S,T; | -Q,S,T; | -P |
| 73. | -P, R, S; | -P,Q,R,S,T; | -P,Q,R; | -P,Q,R | 74. | -P, R, T; | -Q,S,T; | -P,R,T; | -Q,S,T |
| 75. | -P; | -Q; | -R,T; | -S | 76. | -P; | -Q; | -R; | -S |
| 77. | -S,T; | -R,T; | -S,T; | -Q,T | 78. | -R; | -Q; | -S; | -P |
| 79. | -Q; | -P; | -S; | -R | 80. | -P,R; | -T; | -P,S; | -Q |
| 81. | -P; | -Q; | -R; | -S | 82. | P; | -Q; | -R; | -S |
| 83. | -P, Q, S; | -P,Q,S,T; | -P,Q,R; | -P,Q,R | 84. | P,R,S; | -R, S; | -R, S; | -W,R,S |
| 85. | -P,S; | -P,Q; | -R,T; | -P,Q | 86. | -Q,R,S; | -P,R; | -Q,R; | -R,S |
| 87. | -P,Q,R,S; | -Q,R; | -Q,R; | -P,R,S | 88. | Q,R; | -Q,S; | -Q,S; | -P,R,S |
| 89. | P; | -P,Q; | -S,T; | -P,Q,R | 90. | -P,Q; | -Q,R,S; | -P; | -P,Q |
| 91. | -R; | -T; | -P; | -P | 92. | -R,S; | -P,Q; | -P; | -P,Q |
| 93. | -Q,R; | -P,T; | -Q,R; | -S,T | 94. | -P; | -Q; | -R; | -S |
| 95. | -R; | -S; | -P; | -Q | 96. | -P; | -Q; | -R; | -S |
| 97. | -P; | -Q; | -R; | -S | 98. | -S; | -R; | -Q; | -P |
| 99. | -S; | -R; | -Q; | -P | 100. | -P,S; | -R; | -P,Q,S; | -P,S,T |
| 101. | -P,R,S; | -P,T; | -P,Q,T; | -P,R,S | | | | | |

EXERCISE # IV

1. (4) 2. (4) 3. (4) 4. (6) 5. (7) 6. (7) 7. (2) 8. (3) 9. (3) 10. (5)
 11. (4), a,b,f,g 12. (3) 13. (6) 14. (6) 15. (2) 16. (4) 17. (6) 18. (7) 19. (6)
 20. (4) 21. (6) 22. (5) 23. (3) 24. (8) 25. (7) 26. (0) 27. (9) 28. 32/5 29. (5)
 30. (4) 31. (5) 32. (2) 33. (2) 34. (6) 35. 5 36. 2 37. 2 38. 3 39. 8
 40. 11/2 41. 2 42. 3 43. 6 44. 0 45. 9 46. 1 47. 3 48. 4 49. 9/18
 50. 3

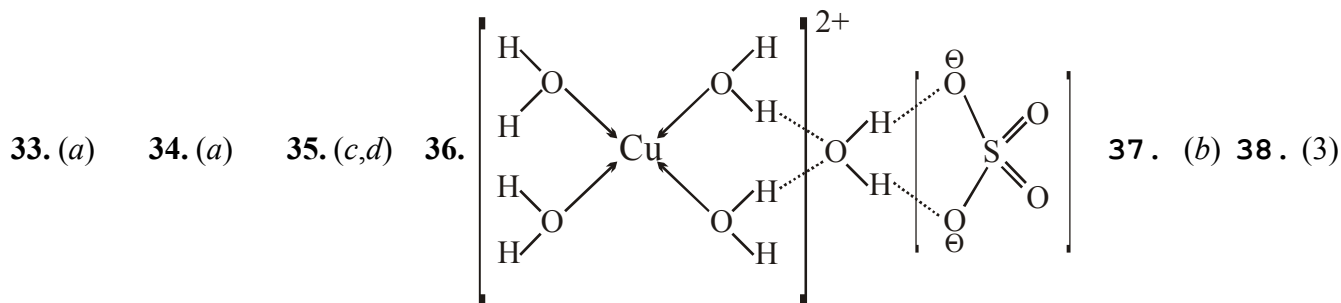
EXERCISE # V(A) JEE-MAIN

1. (b) 2. (c) 3. (d) 4. (a) 5. (c) 6. (b) 7. (a) 8. (c) 9. (d) 10. (c)
 11. (c) 12. (a) 13. (b) 14. (d) 15. (d) 16. (a) 17. (c) 18. (d) 19. (d) 20. (a)
 21. (d) 22. (b) 23. (c) 24. (b) 25. (a) 26. (a) 27. (b) 28. (c) 29. (c) 30. (c)
 31. (a) 32. (c) 33. (c) 34. (d) 35. (c) 36. (d) 37. (d) 38. (d) 39. (a) 40. (c)
 41. (c) 42. (c) 43. (b) 44. (d) 45. (d) 46. (a) 47. (c) 48. (c) 49. (d) 50. (c)
 51. (c) 52. (a) 53. (b)

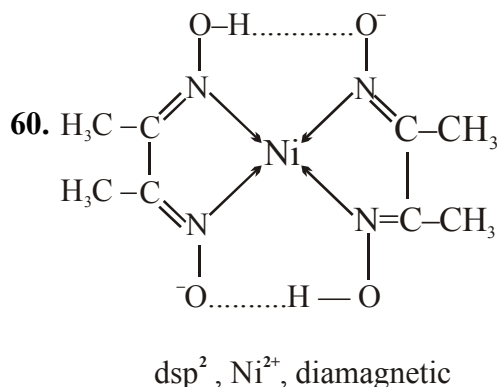
EXERCISE # V(B) JEE-ADVANCED

True/False

1. (F) 2. (b) 3. (F) 4. (a) 5. (a) 6. (b) 7. (b) 8. (b) 9. Paramagnetism
 10. Hexaamminecobalt(III)chloride 11. (c,d) 12. (d) 13. (d) 14. (c) 15. (c) 16. (a)
 17. (b) 18. (d) 19. (b) 20. (b) 21. (b) 22. (a) 23. (a) 24. (c) 25. (a) 26. (a)
 27. (d) 28. (a)-P,Q,S ; (b)-P,R,S ; (c)-Q,S ; (d)-Q,S 29. (c) 30. (c) 31. (b) 32. (b)



39. (c) 40. (b) 41. (c) 42. (6) 43. (d) 44. (a) 45. (c) 46. (b) 47. (8) 48. (b,d)
 49. (b) 50. (4) 51. Γ^- is reducing agent than Cl^- 52. (i) Pentaamminenitritocobalt(III) chloride (ii) Potassiumhexacyanochromate(III) 53. Pentaamminecaronatochromium(III)chloride
 54. (i) $[\text{Co}(\text{NH}_3)_5\text{Cl}]^{2+}$ (ii) $\text{Li}[\text{AlH}_4]$ 55. (a) $\rightarrow [\text{Cr}(\text{H}_2\text{O})_6]\text{Cl}_3$; (b) $\rightarrow [\text{Cr}(\text{H}_2\text{O})_5\text{Cl}]\text{Cl}_2 \cdot \text{H}_2\text{O}$; (c) $\rightarrow [\text{Cr}(\text{H}_2\text{O})_4\text{Cl}_2]\text{Cl} \cdot \text{H}_2\text{O}$ 56. d^2sp^3 , dsp^2 and sp^3 57. (a) $\rightarrow [\text{Cr}(\text{NH}_3)_4\text{Cl}_2]\text{Cl}$;
 (b) $\rightarrow [\text{Cr}(\text{NH}_3)_4\text{Cl}_2]\text{Br}$ In both Cr is d^2sp^3 hybridised and magnetic moment is $\sqrt{15}\text{BM}$.
 58. $[\text{NiCl}_4]^{2-} \rightarrow sp^3$, $\sqrt{8}\text{BM}$ $[\text{Ni}(\text{CN})_4]^{2-} \rightarrow dsp^2$, 0 59. Potassium amminetetracyanonitrosochromate(I) $\rightarrow d^2sp^3$, octahedral



63. (6) 64. (b) 65. (5) 66. (a)

CH-3 METALLURGY

EXERCISE # I

- | | | | | | | | | | |
|---------|---------|---------|---------|---------|---------|---------|---------|---------|----------|
| 1. (d) | 2. (d) | 3. (b) | 4. (d) | 5. (d) | 6. (a) | 7. (b) | 8. (b) | 9. (a) | 10. (c) |
| 11. (a) | 12. (d) | 13. (d) | 14. (c) | 15. (c) | 16. (c) | 17. (b) | 18. (d) | 19. (b) | 20. (a) |
| 21. (b) | 22. (c) | 23. () | 24. (a) | 25. (c) | 26. (c) | 27. (c) | 28. (d) | 29. (d) | 30. (a) |
| 31. (b) | 32. (a) | 33. (c) | 34. (c) | 35. (b) | 36. (c) | 37. (c) | 38. (c) | 39. (d) | 40. (d) |
| 41. (a) | 42. (a) | 43. (a) | 44. (c) | 45. (a) | 46. (a) | 47. (b) | 48. (c) | 49. (c) | 50. (a) |
| 51. (b) | 52. (a) | 53. (b) | 54. (c) | 55. (b) | 56. (b) | 57. (b) | 58. (a) | 59. (c) | 60. (c) |
| 61. (b) | 62. (b) | 63. (c) | 64. (a) | 65. (b) | 66. (c) | 67. (c) | 68. (d) | 69. (c) | 70. (d) |
| 71. (d) | 72. (c) | 73. (b) | 74. (a) | 75. (d) | 76. (a) | 77. (b) | 78. (a) | 79. (b) | 80. (c) |
| 81. (a) | 82. (d) | 83. (d) | 84. (c) | 85. (d) | 86. (c) | 87. (d) | 88. (c) | 89. (b) | 90. (a) |
| 91. (c) | 92. (d) | 93. (b) | 94. (d) | 95. (d) | 96. (c) | 97. (d) | 98. (c) | 99. (d) | 100. (c) |

EXERCISE # II

- | | | | | |
|-------------|---------------|---------------|---------------|---------------|
| 1. (b,c,d) | 2. (a,b) | 3. (a,v,c) | 4. (a,b,c) | 5. (a,c) |
| 6. (b,c) | 7. (a,c) | 8. (a,b,c) | 9. (a,b,c,d) | 10. (a,b,c,d) |
| 11. (a,d) | 12. (a,b,c,d) | 13. (a,b) | 14. (b,c,d) | 15. (a,b,d) |
| 16. (b,c) | 17. (a,b,d) | 18. (b,c,d) | 19. (b,c,d) | 20. (b,c) |
| 21. (a) | 22. (c,d) | 23. (a,b,c,d) | 24. (a,b,c,d) | 25. (a,b,d) |
| 26. (a,b) | 27. (a,c,d) | 28. (a,c,d) | 29. (a,b,c) | 30. (a,b,c) |
| 31. (a,b) | 32. (a,b,c) | 33. (a,b,c) | 34. (a,c,d) | 35. (a,b) |
| 36. (a,b) | 37. (a,c) | 38. (b,d) | 39. (a,b,d) | 40. (a,b,c,d) |
| 41. (b) | 42. (a,b,c,d) | 43. (a,b,c,d) | 44. (a,b,c,d) | 44. (a,c) |
| 45. (a,b,d) | 46. (a,d) | 47. (a,b,c) | 48. (b,c) | 49. (a,d) |
| 50. (c,d) | 51. (a,b,c) | 52. (a,b,c) | 53. (a,c) | 54. (c,d) |
| 55. (c,d) | 56. (a,b) | 57. (a,b,c) | 58. (b,c) | 59. (a,b,c) |
| 60. (a,c,d) | | | | |

EXERCISE # III

- | | | | | | | | | | |
|----------|---------|---------|---------|---------|---------|---------|---------|---------|---------|
| 1. (a,d) | 2. (a) | 3. (c) | 4. (c) | 5. (d) | 6. (c) | 7. (d) | 8. (b) | 9. (d) | 10. (c) |
| 11. (d) | 12. (c) | 13. (d) | 14. (d) | 15. (c) | 16. (b) | 17. (b) | 18. (c) | 19. (a) | 20. (a) |

Matrix Match Type :

- | | | | | | | | | | |
|-----|-------|-----|-------|-----|-----|---------|-----|-------|-------|
| | (a) | (b) | (c) | (d) | | (a) | (b) | (c) | (d) |
| 21. | S | P | Q | R | 22. | P | R | Q | S |
| 23. | S,T | R,T | Q,R | P,S | 24. | P, Q, S | R | R | Q,T |
| 25. | R | S | Q | P | 26. | P,R | Q | S,T | Q |
| 27. | Q,S,T | P,R | Q,R,T | R,T | 28. | P,T | R | Q,R,S | Q,S,T |
| 29. | P | S | R | Q | 30. | S | P | Q | R |
| 31. | Q,S | P | P,R | Q,S | 32. | P | Q | P,R | P,R,S |
| 33. | S | R | Q | P | 34. | R | S | Q | P |
| 35. | S | P | R | Q | | | | | |

EXERCISE # IV

1. (3) 2. (5) 3. (5) 4. (4) 5. (3) 6. (4) 7. (3) 8. (9) 9. (3) 10. (6)
 11. (5) 12. (4) 13. (6) 14. (2) 15. (4) 16. (4) 17. (4) 18. (4) 19. (6) 20. (4)
 21. (3) 22. (3) 22. (2) 23. (3) 24. (0) 25. (2) 26. (6) 27. (6) 28. (2) 29. (5)
 30. (8) 31. (7) 32. (0) 33. (3) 34. (3) 35. (2) 36. (2) 37. (1) 38. (2) 39. (2)
 40. (1)

EXERCISE # V(a) JEE-MAIN

1. (3) 2. (1) 3. (1) 4. (3) 5. (1) 6. (2) 7. (4) 8. (4) 9. (3) 10. (4)
 11. (4) 12. (2)

EXERCISE # V(b) JEE-ADVANCED**1. Smelting, Sintering**

2. (b) 3. (b) 4. (a, c) 5. (c) 6. (c) 7. (c) 8. (b) 9. (a) 10. (a)
 11. (a) 12. (c) 13. (b) 14. (b) 15. (a) 16. (d) 17. (c)
 18. (a) $\rightarrow$ (P, R) ; (b) $\rightarrow$ (P) ; (c) $\rightarrow$ (Q) ; (d) $\rightarrow$ (S) 19. (a) $\rightarrow$ (P) ; (b) $\rightarrow$ (Q) ; (c) $\rightarrow$ (P, R) ; (d) $\rightarrow$ (S)
20. Sintering, Smelting 21. (a, c, d) 22. (d) 23. (b) 24. (a) 25. (c, d)
 26. (a, c, d) 27. () 28. (a) $\rightarrow$ (P, Q, S) ; (b) $\rightarrow$ (T) ; (c) $\rightarrow$ (Q, R) ; (d) $\rightarrow$ (R)

CH-4 QUALITATIVE SALT ANALYSIS**EXERCISE # I**

1. (c) 2. (a) 3. (a) 4. (c) 5. (b) 6. (b) 7. (c) 8. (a) 9. (d) 10. (c)
 11. (c) 12. (c) 13. (b) 14. (b) 15. (b) 16. (c) 17. (a) 18. (a) 19. (c) 20. (c)
 21. (b) 22. (a) 23. (c) 24. (c) 25. (b) 26. (b) 27. (a) 28. (d) 29. (a) 30. (a)
 31. (a) 32. (b) 33. (b) 34. (c) 35. (b) 36. (c) 37. (a) 38. (c) 39. (c) 40. (c)
 41. (b) 42. (c) 43. (b) 44. (a) 45. (c) 46. (b) 47. (a) 48. (d) 49. (b) 50. (c)
 51. (b) 52. (b) 53. (d) 54. (b) 55. (b) 56. (a) 57. (c) 58. (d) 59. (a) 60. (b)
 61. (b) 62. (d) 63. (c) 64. (d) 65. (d) 66. (c) 67. (c) 68. (d) 69. (b) 70. (b)
 71. (c) 72. (b) 73. (a) 74. (c) 75. (b) 76. (c) 77. (c) 78. (d) 79. (a) 80. (c)
 81. (c) 82. (c) 83. (d) 84. (d) 85. (d) 86. (b) 87. (a) 88. (d) 89. (d) 90. (c)
 91. (c) 92. (d) 93. (c) 94. (d) 95. (a) 96. (d) 97. (b) 98. (c) 99. (c) 100. (c)
 101. (a) 102. (d) 103. (c) 104. (d) 105. (c) 106. (d) 107. (b) 108. (b) 109. (b) 110. (d)
 111. (c) 112. (b) 113. (a) 114. (d) 115. (c) 116. (d) 117. (d) 118. (c) 119. (a) 120. (a)

EXERCISE # II

1. (a, b, c, d) 2. (a, c) 3. (a, b) 4. (a, b, c, d) 5. (a, b, c)
 6. (b, c) 7. (a, c, d) 8. (a, b, c, d) 9. (b, d) 10. (a, d)
 11. (b, c, d) 12. (a, c) 13. (a, c) 14. (a, b) 15. (b, c)
 16. (a, b, c, d) 17. (a, b) 18. (a, b) 19. (a, b, c, d) 20. (a, b, c)

- | | | | | |
|---------------|---------------|---------------|---------------|---------------|
| 21. (a,b,c) | 22. (a,b,c,d) | 23. (a,b,c,d) | 24. (a,b) | 25. (a,b,c,d) |
| 26. (a,b,d) | 27. (b,c) | 28. (c,d) | 29. (a,c,d) | 30. (a,b,c) |
| 31. (a,b) | 32. (b,c,d) | 33. (a,b,c) | 34. (a,b,c,d) | 35. (a,b,c) |
| 36. (a,b,c) | 37. (a,b,c,d) | 38. (a,b,c) | 39. (a,b,c) | 40. (a,c,d) |
| 41. (a,b,c) | 42. (a,b,c,d) | 43. (b,d) | 44. (a,b,c,d) | 45. (a,b) |
| 46. (a,b,c) | 47. (b,c) | 48. (a,c) | 49. (b,c) | 50. (a,b,d) |
| 51. (a,b,c,d) | 52. (a,b,c) | 53. (a,b,c) | 54. (b,d) | 55. (a,c,d) |
| 56. (a,b,c) | 57. (a,d) | 58. (a,b,c) | 59. (a,b,d) | 60. (a,b,c,d) |
| 61. (a,c,d) | 62. (a,d) | 63. (a,b,d) | 64. (a,d) | 65. (a,b,c,d) |

EXERCISE # III

- | | | | | | | | | | |
|---------------|-------------|---------------|---------|----------|---------|---------------|---------|---------|-----------|
| 1. (b) | 2. (c) | 3. (b) | 4. (a) | 5. (b,c) | 6. (b) | 7. (a) | 8. (c) | 9. (c) | 10. (b) |
| 11. (c) | 12. (d) | 13. (b) | 14. (d) | 15. (a) | 16. (b) | 17. (a,c,d) | 18. (b) | 19. (d) | 20. (a,c) |
| 21. (a,b,c,d) | 22. (a,b,c) | 23. (a,b,c,d) | 24. (a) | 25. (a) | 26. (b) | 27. (a,b,c,d) | | | |
| 28. (b) | 29. (d) | 30. (d) | 31. (a) | 32. (b) | 33. (c) | 34. (c) | 35. (b) | 36. (a) | 37. (b) |
| 38. (b) | 39. (b) | 40. (d) | 41. (d) | 42. (a) | 43. (b) | 44. (b) | 45. (d) | 46. (d) | 47. (b) |
| 48. (b) | 49. (c) | 50. (c) | 51. (a) | 52. (b) | 53. (d) | 54. (c) | 55. (b) | 56. (c) | 57. (c) |
| 58. (d) | 59. (a) | 60. (c) | 61. (a) | 62. (c) | 63. (a) | 64. (d) | 65. (b) | 66. (c) | 67. (c) |
| 68. (b) | 69. (a) | 70. (c) | 71. (d) | 72. (a) | 73. (b) | 74. (d) | 75. (c) | 76. (a) | 77. (c) |
| 78. (d) | 79. (b) | 80. (a) | 81. (b) | 82. (c) | 83. (a) | 84. (c) | 85. (c) | 86. (c) | 87. (c) |
| 88. (d) | 89. (d) | 90. (d) | 91. (c) | 92. (b) | 93. (b) | 94. (b) | 95. (c) | 96. (b) | 97. (b) |
| 98. (a) | 99. (a) | 100. (a) | | | | | | | |

Matrix Match Type :

- | (a) | (b) | (c) | (d) | (a) | (b) | (c) | (d) |
|-------------|------------|----------|-------|------|--------|----------|----------|
| 101. P,T; | P,S,T; | P,Q,T; | Q,R,T | 102. | P,R,T; | P,Q,T; | P,S,T; |
| 103. P,Q,R; | P; | R; | P | 104. | P,T; | P,T; | Q; |
| 105. S,T; | P,Q,R,S,T; | P,S,T; | R,S | 106. | Q,R,S; | P,R,S; | P,Q,R,S; |
| 107. S; | P,Q; | R; | T | 108. | R,T; | R,T; | Q,S; |
| 109. S; | P,Q,R,S; | P,R,T; | P,S,T | 110. | R,T; | P,Q; | Q,S; |
| 111. S; | R; | Q; | P | 112. | Q,T; | P; | R; |
| 113. P; | Q; | R; | S | 114. | P,T; | Q,R,S; | P; |
| 115. Q,R,T; | P,T; | P,Q,R,S; | P,R,S | 116. | P; | Q; | R; |
| 117. P; | P,Q; | R,T; | Q,R,S | 118. | R; | Q; | P; |
| 119. P,Q; | R; | P,Q,S; | P,Q,S | 120. | S; | P,Q,R,S; | P,S; |
| 121. S; | P,Q,T; | Q,R,S,T; | Q,R,T | 122. | Q; | S; | P; |
| 123. P,Q; | R; | S; | T | 124. | S; | R; | Q; |
| 125. P; | Q; | R; | S | | | | |

EXERCISE # IV

- | | | | | | | | | | |
|---------|---------|---------|---------|---------|---------|---------|---------|---------|---------|
| 1. (3) | 2. (0) | 3. (0) | 4. (7) | 5. (3) | 6. (5) | 7. (2) | 8. (5) | 9. (7) | 10. (3) |
| 11. (2) | 12. (2) | 13. (2) | 14. (4) | 15. (6) | 16. (2) | 17. (4) | 18. (1) | 19. (2) | 20. (5) |
| 21. (4) | 22. (5) | 23. (0) | 24. (2) | 25. (3) | 26. (3) | 27. (3) | 28. (2) | 29. (5) | 30. (3) |
| 31. (5) | 32. (3) | 33. (5) | 34. (5) | 35. (4) | 36. (0) | 37. (4) | 38. (2) | 39. (4) | 40. (3) |
| 41. (6) | 42. (3) | 43. (3) | 44. (3) | 45. (4) | 46. (4) | 47. (7) | 48. (4) | 49. (3) | 50. (5) |

EXERCISE # V(A) JEE-MAIN

- | | | | | | | | | | |
|---------|---------|---------|---------|---------|---------|---------|---------|---------|---------|
| 1. (d) | 2. (d) | 3. (d) | 4. (d) | 5. (b) | 6. (b) | 7. (b) | 8. (a) | 9. (b) | 10. (b) |
| 11. (c) | 12. (d) | 13. (d) | 14. (d) | 15. (b) | 16. (c) | 17. (b) | 18. (b) | 19. (c) | 20. (c) |
| 21. (d) | 22. (b) | 23. (c) | 24. (a) | 25. (d) | 26. (d) | 27. (a) | 28. (b) | 29. (c) | 30. (d) |
| 31. (d) | 32. (d) | 33. (a) | 34. (a) | 35. (a) | 36. (d) | 37. (d) | | | |

EXERCISE # V(B) JEE-ADVANCED

- | | | | | | | | | | |
|------------|------------|---|---|---------|-----------|-------------|-------------|-------------|---------|
| 1. (a,b,c) | 2. (a,b,d) | 3. (d) | 4. (c) | 5. (a) | 6. (b) | 7. (a) | 8. (b) | 9. (a) | 10. (d) |
| 11. (d) | 12. (a) | 13. (a) | 14. $A \rightarrow P, S$; $B \rightarrow R$; $C \rightarrow P, Q$; $D \rightarrow P$ | | | | 15. (b) | 16. (a,b) | 17. (d) |
| 18. (c) | 19. (b) | 20. $A \rightarrow P, S$; $B \rightarrow Q, S$; $C \rightarrow R, T$; $D \rightarrow Q, T$ | | | | 21. (a) | 22. (a,c,d) | 23. (b,c,d) | |
| 24. (b) | 25. (a) | 26. (c) | 27. (a,c,d) | | 28. (c) | 29. (a,c,d) | | 30. (d) | 31. (d) |
| 32. (a) | 33. (7) | 34. (6) | 35. (b) | 36. (d) | 37. (a,b) | 38. (c,d) | | | |

CH-5 THE s-BLOCK ELEMENTS AND THEIR COMPOUNDS**EXERCISE # I**

- | | | | | | | | | | |
|---------|---------|---------|---------|---------|---------|---------|---------|---------|---------|
| 1. (b) | 2. (d) | 3. (b) | 4. (d) | 5. (c) | 6. (b) | 7. (d) | 8. (d) | 9. (a) | 10. (b) |
| 11. (c) | 12. (c) | 13. (b) | 14. (c) | 15. (c) | 16. (b) | 17. (a) | 18. (a) | 19. (d) | 20. (d) |
| 21. (b) | 22. (a) | 23. (a) | 24. (a) | 25. (c) | 26. (a) | 27. (d) | 28. (c) | 29. (d) | 30. (c) |
| 31. (d) | 32. (a) | 33. (b) | 34. (c) | 35. (c) | 36. (a) | 37. (c) | 38. (c) | 39. (a) | 40. (b) |
| 41. (c) | 42. (d) | 43. (b) | 44. (b) | 45. (c) | 46. (a) | 47. (d) | 48. (d) | 49. (b) | 50. (c) |
| 51. (b) | 52. (c) | 53. (c) | 54. (a) | 55. (d) | 56. (b) | 57. (a) | 58. (d) | 59. (c) | 60. (b) |
| 61. (c) | 62. (a) | 63. (a) | 64. (a) | 65. (b) | 66. (c) | 67. (d) | 68. (c) | 69. (b) | 70. (a) |
| 71. (a) | 72. (d) | 73. (c) | 74. (c) | 75. (b) | | | | | |

EXERCISE # II

- | | | | | | |
|-------------|---------------|-------------|--------------|---------------|---------------|
| 1. (a,b,d) | 2. (a,b,c,d) | 3. (a,b,c) | 4. (a,b,c,d) | 5. (a,b,d) | 6. (a,b,c,d) |
| 7. (a,c) | 8. (b,d) | 9. (a,c,d) | 10. (a,d) | 11. (a,c,d) | 12. (a,b,c) |
| 13. (a,b) | 14. (a,b,c,d) | 15. (b,d) | 16. (a,b) | 17. (a,b,c) | 18. (a,b,c,d) |
| 19. (c,d) | 20. (a,d) | 21. (a,b) | 22. (a,b,c) | 23. (a,d) | 24. (a,b,c,d) |
| 25. (a,c,d) | 26. (b,c) | 27. (b,c) | 28. (a,b,d) | 29. (b,c,d) | 30. (b,c) |
| 31. (a,b,c) | 32. (b,c) | 33. (a,d) | 34. (a,d) | 35. (a,b,c,d) | 36. (a,b) |
| 37. (b,c,d) | 38. (a,b,c,d) | 39. (a,b,d) | 40. (a,b,c) | 41. (a,b,c,d) | 42. (a,b,c,d) |

43. (a,b,c,d) 44. (a,b,c,d) 45. (a,b,c,d) 46. (b,c) 47. (c,d) 48. (a,b,c)
 49. (a,b,c,d) 50. (a,b,c,d) 51. (a,b,d) 52. (a,c) 53. (a,c) 54. (c,d)
 55. (a,c) 56. (a,b,c) 57. (a,d) 58. (a,b,c) 59. (a,b) 60. (a,b,d)
 61. (a,b,c) 62. (a,c) 63. (a,b,c,d) 64. (a,b,c) 65. (a,c,d)

EXERCISE # III : Paragraph Type & Matrix Match Type

1. (b) 2. (a) 3. (b) 4. (d) 5. (b,d) 6. (b) 7. (d) 8. (b) 9. (c) 10. (b)
 11. (d) 12. (a) 13. (a) 14. (d) 15. (d) 16. (a) 17. (c) 18. (d) 19. (b) 20. (a)
 21. (a) 22. (b) 23. (d) 24. (d) 25. (c) 26. (b) 27. (c) 28. (a)
 29. (A)→P ; (B)→S ; (C)→Q ; (D)→R 30. (A)→P ; (B)→Q ; (C)→R ; (D)→S
 31. (A)→P,S ; (B)→R ; (C)→Q ; (D)→P,S 32. (A)→P ; (B)→Q ; (C)→R ; (D)→R,S
 33. (A)→R,T ; (B)→P,Q ; (C)→R ; (D)→P,S
 34. (A)→Q,R ; (B)→P,Q,R,S ; (C)→P,Q,R ; (D)→P,Q,R
 35. (A)→Q ; (B)→R ; (C)→P ; (D)→S 36. (A)→P ; (B)→Q ; (C)→R ; (D)→S
 37. (A)→P,R ; (B)→P,S ; (C)→P,Q ; (D)→P 38. (A)→Q ; (B)→R ; (C)→P,Q,R,S ; (D)→S
 39. (A)→S ; (B)→Q ; (C)→R ; (D)→P 40. (A)→P,Q (B)→Q (C)→R (D)→S,T
 41. (A)→R (B)→S (C)→P (D)→Q,T 42. (A)→P ; (B)→Q ; (C)→R ; (D)→S
 43. (A)→P,Q,R,S,T (B)→P,Q,R,S,T (C)→P,R,S,T (D)→S,T
 44. (A)→P,Q,T (B)→P,Q,T (C)→P,S (D)→R,S 45. (A)→S (B)→Q,R,T (C)→P,R (D)→P,R

EXERCISE # IV : Integer Type

1. (7) 2. (6) 3. (3) 4. (8/17) 5. (4) 6. (5) 7. (4) 8. (4) 9. (6) 10. (9)
 11. (4) 12. (5) 13. (4) 14. (3) 15. (2) 16. (1) 17. (4) 18. (3) 19. (6) 20. (6)

EXERCISE # V(A) : (JEE-MAIN)

1. (4) 2. (1) 3. (3) 4. (4) 5. (4) 6. (2) 7. (1) 8. (1) 9. (2) 10. (3)
 11. (4) 12. (4) 13. (4) 14. (3) 15. (3) 16. (4) 17. (3) 18. (4) 19. (3) 20. (3)
 21. (4)

EXERCISE # V(B) : (JEE-ADVANCED)

1. (a) 2. (a) 3. (b) 4. (b) 5. (a,b) 6. (b) 7. (b) 8. (a,b)

CH-6 THE p-BLOCK ELEMENTS AND THEIR COMPOUNDS**EXERCISE # I**

1. (d) 2. (b) 3. (a) 4. (c) 5. (c) 6. (d) 7. (a) 8. (c) 9. (d) 10. (d)
 11. (c) 12. (a) 13. (d) 14. (b) 15. (a) 16. (c) 17. (d) 18. (c) 19. (b) 20. (a)
 21. (c) 22. (c) 23. (c) 24. (a) 25. (a) 26. (b) 27. (a) 28. (b) 29. (d) 30. (a)
 31. (c) 32. (c) 33. (a) 34. (d) 35. (c) 36. (d) 37. (b) 38. (b) 39. (c) 40. (c)
 41. (b) 42. (c) 43. (a) 44. (d) 45. (a) 46. (b) 47. (b) 48. (b) 49. (c) 50. (d)
 51. (c) 52. (a) 53. (d) 54. (d) 55. (c) 56. (d) 57. (b) 58. (a) 59. (c) 60. (b)

10.12 ANSWERS

61. (d) 62. (d) 63. (c) 64. (c) 65. (b) 66. (d) 67. (d) 68. (a) 69. (b) 70. (d)
 71. (c) 72. (d) 73. (c) 74. (c) 75. (b) 76. (b) 77. (a) 78. (a) 79. (d) 80. (a)
 81. (b) 82. (a) 83. (b) 84. (b) 85. (b) 86. (d) 87. (c) 88. (d) 89. (c) 90. (d)
 91. (c) 92. (a) 93. (b) 94. (b) 95. (b) 96. (c) 97. (c) 98. (d) 99. (b) 100. (b)
 101. (c) 102. (b) 103. (b) 104. (a) 105. (d) 106. (b) 107. (a) 108. (a) 109. (a) 110. (d)
 111. (c) 112. (c) 113. (a) 114. (b) 115. (a) 116. (c) 117. (c) 118. (a) 119. (c) 120. (d)
 121. (b) 122. (d) 123. (d) 124. (a) 125. (a) 126. (b) 127. (c) 128. (d) 129. (d) 130. (c)
 131. (c) 132. (d) 133. (b) 134. (d) 135. (c) 136. (a) 137. (c) 138. (c) 139. (b) 140. (d)
 141. (c) 142. (c) 143. (a) 144. (d) 145. (b) 146. (a) 147. (b) 148. (a) 149. (d) 150. (d)

EXERCISE # II

1. (a,b,d) 2. (a,d) 3. (a,b,c,d) 4. (a,b) 5. (b,c,d) 6. (a,b,c,d) 7. (b,c) 8. (c,d)
 9. (a,c,d) 10. (a,b) 11. (c,d) 12. (c,d) 13. (c,d) 14. (a,b,c,d) 15. (a,b,c) 16. (a,b,c,d)
 17. (a,b,c,d) 18. (a,b) 19. (a,b) 20. (b,c,d) 21. (a,b,c) 22. (c,d) 23. (b,c,d) 24. (a,c,d)
 25. (b,d) 26. (b,d) 27. (a,b,d) 28. (a,b,c,d) 29. (a,b,d) 30. (a,b,c) 31. (a,b,c,d) 32. (c,d)
 33. (b,c) 34. (b,c,d) 35. (b,c) 36. (a,b,c) 37. (a,b,d) 38. (a,b,d) 39. (a,b,c,d) 40. (a,c,d)
 41. (a,b,c,d) 42. (c,d) 43. (a,c) 44. (a,b,d) 45. (a,b,c) 46. (a,c,d) 47. (a,b,c,d) 48. (a,b,c)
 49. (b,d) 50. (a,b,d) 51. (a,c) 52. (b,d) 53. (b,c,d) 54. (c,d) 55. (c,d) 56. (b,c,d)
 57. (a,b,c) 58. (b,c) 59. (a,b,c,d) 60. (a,c,d) 61. (a,b,c,d) 62. (a,b) 63. (a,c) 64. (a,c)
 65. (b,d) 66. (a,d) 67. (a,b,c) 68. (b,c,d) 69. (a,b,c,d) 70. (a,c,d) 71. (a,b,c,d) 72. (a,d)
 73. (a,b,c,d) 74. (b,d) 75. (a,b,c) 76. (a,b,d) 77. (c,d) 78. (a,b,c) 79. (c,d) 80. (c,d)

EXERCISE # III

Paragraph Type :

1. (d) 2. (b) 3. (d) 4. (a) 5. (c) 6. (b) 7. (b) 8. (a) 9. (b) 10. (a)
 11. (c) 12. (d) 13. (a) 14. (a) 15. (d) 16. (b) 17. (c) 18. (d) 19. (d) 20. (d)
 21. (b) 22. (c) 23. (d) 24. (c) 25. (d) 26. (b) 27. (d) 28. (c) 29. (b) 30. (b)
 31. (b) 32. (c) 33. (b) 34. (d) 35. (c) 36. (c) 37. (d) 38. (a) 39. (b) 40. (d)
 41. (c) 42. (d) 43. (c) 44. (b) 45. (c) 46. (a) 47. (b) 48. (c) 49. (c) 50. (d)
 51. (a) 52. (c) 53. (d) 54. (a) 55. (d) 56. (c) 57. (a) 58. (b) 59. (b) 60. (d)

Matrix Match Type :

- | (a) | (b) | (c) | (d) | (a) | (b) | (c) | (d) |
|--------------|--------------|---------|-------|----------------|----------|----------|---------|
| 61. Q; | R,S; | P, S ; | Q | 62. P,S; | Q; | R ; | T |
| 63. P; | S; | Q; | R | 64. P,R,S; | P,Q,R,T; | P,Q,R,T; | P,Q,R,S |
| 65. P; | T; | Q,R,T; | Q,S | 66. Q,R; | Q; | P,R; | P,R,S |
| 67. P,Q,R,S; | T,P,Q,R,S,T; | P,Q,R,S | P,R,S | 68. P,S; | P,Q,R; | P,Q,R; | Q,R |
| 69. Q,S; | S; | P | R | 70. R,S; | Q,R; | P,T; | S |
| 71. P,Q,R,S; | P,Q,R,S; | R; | P,Q,S | 72. P,Q,R,S,T; | S; | P,Q,R,T; | P,Q,R |
| 73. Q; | R,S; | Q,T; | P,T | 74. P; | R,S; | Q,T; | P,T |

(a)	(b)	(c)	(d)	(a)	(b)	(c)	(d)
75. P,R,S;	P,R,S,T;	Q,R;	P,R,S	76. P,T;	Q,S;	Q,R;	Q,T
77. P,Q,R,T;	R;	P,T;	R,S	78. P,Q;	P,Q,R,S;	R;	S
79. R;	Q;	P;	S	80. P,S;	P,R;	Q;	R,T
81. R,S,T;	Q,R,T;	P;	Q,R	82. P,Q,T;	P,Q,S;	P;	R
83. S;	R;	P;	Q	84. P,T;	R,T;	S;	Q
85. P,R;	T;	P,Q,R;	P,R,S	86. Q;	R,S;	P;	T
87. R,S;	Q,S;	Q,S;	P,T	88. P,R,S;	P,R,S;	Q,R,S;	Q,R,S,T
89. R;	P,Q;	S;	T	90. R,T;	S;	Q,T;	P

EXERCISE # IV

1. (6) 2. (3) 3. (3) 4. (5) 5. (5) 6. (1) 7. (4) 8. (2) 9. (0) 10. (2)
 11. (0) 12. (1) 13. (6) 14. (2) 15. (4) 16. (3) 17. (4) 18. (4) 19. (7) 20. (2)
 21. (2) 22. (7) 23. (1) 24. (9) 25. (5) 26. (0) 27. (0) 28. (6) 29. (4) 30. (0)
 31. (3) 32. (4) 33. (3) 34. (4) 35. (4) a,b,c,d,e 36. (3) 37. (3) 38. (2) 39. (3)
 40. (2) 41. (3) 42. (8) 43. (8) 44. (2) 45. (6) 46. (2) 47. 3/48 48. 5/14 49. (4)
 50. (9)

EXERCISE # V(A) JEE-MAIN

1. (a) 2. (d) 3. (d) 4. (d) 5. (b) 6. (c) 7. (d) 8. (b) 9. (b) 10. (b)
 11. (b) 12. (c) 13. (b) 14. (b) 15. (d) 16. (c) 17. (c) 18. (a) 19. (d) 20. (c)
 21. (b) 22. (c) 23. (c) 24. (b) 25. (c) 26. (d) 27. (b) 28. (a) 29. (c) 30. (c)
 31. (d) 32. (a) 33. (c) 34. (b) 35. (a) 36. (b) 37. (b) 38. (b) 39. (c) 40. (a)
 41. (c) 42. (a) 43. (a) 44. (c) 45. (c) 46. (b,c,d) 47. (d) 48. (d) 49. (b) 50. (b)
 51. (c)

EXERCISE # V(B) JEE-ADVANCED

Fill in the blanks :

1. Silicones 2. $R_3Si(OH)$ 3. Three centre two electron bond or banana bond
 4. buckminsterfullerene

True/False :

5. (F) 6. (T) 7. (F) 8. (T) 9. (F) 10. (T)

Objective :

32. (b,d) 33. (d) 34. (b) 35. (c) 36. (a) 37. (a) 38. (c) 39. (b,d) 40. (a) 41. (c)
 42. (d) 43. (a) 44. (c) 45. (c) 46. (d) 47. (a) 48. (a) 49. (a) 51. (c) 52. (c)
 53. (a,b) 54. (c) 55. (b) 56. (b) 57. (a) 58. (d) 59. (d) 60. (a) 61. (c) 62. (a)
 63. (c) 64. (a) 65. (c) 66. (c) 67. (b) 68. (b) 69. (a,b,c) 70. (b,c) 71. (c) 72. (a)
 73. (b,d) 74. (b) 75. (a,c,d) 76. (a) 77. (a) 78. (d) 79. (c) 80. (a) 81. (b,d) 82. (b,c)
 83. (b,c,d) 84. (b) 85. (6) 86. (8)

CH-7 THE d-BLOCK ELEMENTS AND SOME OF THEIR COMPOUNDS

EXERCISE # I

- | | | | | | | | | | |
|----------------|----------------|----------------|----------------|----------------|----------------|----------------|----------------|----------------|----------------|
| 1. (c) | 2. (d) | 3. (d) | 4. (b) | 5. (d) | 6. (a) | 7. (b) | 8. (c) | 9. (d) | 10. (b) |
| 11. (b) | 12. (b) | 13. (b) | | 15. (c) | 16. (b) | 17. (b) | 18. (a) | 19. (d) | 20. (c) |
| 21. (d) | 22. (a) | 23. (c) | 24. (a) | 25. (d) | 26. (b) | 27. (c) | 28. (d) | 29. (a) | 30. (b) |
| 31. (b) | 32. (a) | 33. (d) | 34. (c) | 35. (d) | 36. (b) | 37. (a) | 38. (b) | 39. (b) | 40. (d) |
| 41. (d) | 42. (a) | 43. (c) | 44. (b) | 45. (a) | 46. (b) | 47. (b) | 48. (a) | 49. (b) | 50. (c) |
| 51. (b) | 52. (d) | 53. (c) | 54. (b) | 55. (d) | 56. (a) | 57. (c) | 58. (d) | 59. (d) | 60. (c) |
| 61. (c) | 62. (b) | 63. (d) | 64. (a) | 65. (d) | 66. (d) | 67. (d) | 68. (b) | 69. (a) | 70. (d) |

EXERCISE # II

- | | | | | | |
|-----------------------|------------------------|----------------------|-----------------------|-----------------------|------------------------|
| 1. (a,d) | 2. (a,b,c,d) | 3. (a,c,d) | 4. (a,c) | 5. (a,b,c,d) | 6. (b,d) |
| 7. (a,b,c,d) | 8. (c,d) | 9. (a,b) | 10. $a,b,c,d)$ | 11. (a,b) | 12. (b,c) |
| 13. (b,d) | 14. (c,d) | 15. (a,d) | 16. (b,d) | 17. (a,b) | 18. (b,c) |
| 19. (a,b,c) | 20. (a,b,c,d) | 21. (a,b,d) | 22. (a,b,d) | 23. (a,c,d) | 24. (a,b,c,d) |
| 25. (a,b) | 26. (a,b) | 27. (a,b) | 27. (a,b,c) | 28. (a,b,d) | 29. (a,b,c) |
| 30. (a,b,d) | 31. (a,b,c) | 32. (a,d) | 33. (c,d) | 34. (a,b) | 35. (a,c,d) |
| 36. (a,b) | 37. (a,b) | 38. (c,d) | 39. (a,b) | 40. (a,b,d) | 41. (a,b,c) |
| 42. (a,c) | 43. (a,b,c) | 44. (a,c,d) | 45. (c,d) | | |

EXERCISE # III : Paragraph Type & Matrix Match Type

- | | | | | | | | | | |
|---|----------------|----------------|----------------|----------------|---|----------------|----------------|----------------|----------------|
| 1. (b) | 2. (b) | 3. (d) | 4. (a) | 5. (c) | 6. (b) | 7. (a) | 8. (d) | 9. (a) | 10. (d) |
| 11. (d) | 12. (d) | 13. (d) | 14. (c) | 15. (b) | 16. (c) | 17. (d) | 18. (d) | 19. (c) | 20. (c) |
| 21. (a) | 22. (b) | 23. (a) | 24. (c) | 25. (c) | 26. (c) | 27. (c) | 28. (a) | 29. (a) | 30. (b) |
| 31. (b) | 32. (a) | 33. (a) | 34. (c) | 35. (a) | 36. (c) | 37. (a) | 38. (b) | | |
| 39. $A \rightarrow R ; B \rightarrow S ; C \rightarrow Q ; D \rightarrow P$ | | | | | 40. $A \rightarrow P ; B \rightarrow Q ; C \rightarrow R ; D \rightarrow S$ | | | | |
| 41. $A \rightarrow Q,T ; B \rightarrow P ; C \rightarrow S ; D \rightarrow R$ | | | | | 42. $A \rightarrow P,R ; B \rightarrow P,S ; C \rightarrow Q,T ; D \rightarrow Q,T$ | | | | |
| 43. $A \rightarrow S,T ; B \rightarrow P,Q ; C \rightarrow P,R ; D \rightarrow R,S,T$ | | | | | 44. $A \rightarrow P ; B \rightarrow Q ; C \rightarrow R ; D \rightarrow S$ | | | | |
| 45. $A \rightarrow P ; B \rightarrow R ; C \rightarrow S ; D \rightarrow Q$ | | | | | 46. $A \rightarrow P,R ; B \rightarrow Q,T ; C \rightarrow Q,R,S ; D \rightarrow P$ | | | | |
| 47. $A \rightarrow Q,R ; B \rightarrow R,T ; C \rightarrow P,Q,T ; D \rightarrow P,Q$ | | | | | 48. $A \rightarrow \quad ; B \rightarrow \quad ; C \rightarrow \quad ; D \rightarrow \quad$ | | | | |
| 49. $A \rightarrow P ; B \rightarrow Q ; C \rightarrow R ; D \rightarrow S$ | | | | | 50. $A \rightarrow P ; B \rightarrow Q ; C \rightarrow R ; D \rightarrow S$ | | | | |
| 51. $A \rightarrow P ; B \rightarrow Q ; C \rightarrow R ; D \rightarrow S$ | | | | | 52. $A \rightarrow P ; B \rightarrow Q ; C \rightarrow R ; D \rightarrow S$ | | | | |
| 53. $A \rightarrow P,Q ; B \rightarrow P,S ; C \rightarrow R ; D \rightarrow R$ | | | | | 54. $A \rightarrow P ; B \rightarrow Q ; C \rightarrow R ; D \rightarrow S$ | | | | |
| 55. $A \rightarrow P,Q,R,T ; B \rightarrow P,R,T ; C \rightarrow P,R,S,T ; D \rightarrow P,R,T$ | | | | | | | | | |

EXERCISE # IV : Integer Type

- | | | | | | | | | | |
|------------|---------|---------|---------|---------|---------|---------|---------|---------|---------|
| 1. (4) | 2. (6) | 3. (5) | 4. (7) | 5. (0) | 6. (2) | 7. (4) | 8. (3) | 9. (6) | 10. (6) |
| 11. (1) | 12. (9) | 13. (3) | 14. (3) | 15. (3) | 16. (5) | 17. (8) | 18. (5) | 19. (4) | 20. (6) |
| 21. (9/18) | 22. (1) | 23. (4) | 24. (0) | 25. (4) | 26. (3) | 27. (6) | 28. (9) | 29. (3) | 30. (7) |

EXERCISE # V(A) : (JEE-MAIN)

1. (1) 2. (1) 3. (1) 4. (4) 5. (2) 6. (2) 7. (4) 8. (3) 9. (1) 10. (3)
 11. (2) 12. (4) 13. (4) 14. (4) 15. (1) 16. (2) 17. (2) 18. (2) 19. (1)

EXERCISE # V(B) : (JEE-ADVANCED)

1. (b) 2. (a)

CH-8 HYDROGEN AND THE HYDRIDE**EXERCISE # I**

1. (c) 2. (a) 3. (b) 4. (c) 5. (d) 6. (a) 7. (c) 8. (a) 9. (b) 10. (c)
 11. (c) 12. (d) 13. (d) 14. (a) 15. (b) 16. (c) 17. (b) 18. (b) 19. (a) 20. (c)
 21. (c) 22. (d) 23. (c) 24. (a) 25. (b) 26. (a) 27. (a) 28. (c) 29. (a) 30. (b)
 31. (b) 32. (a) 33. (b) 34. (c) 35. (d) 36. (b) 37. (d) 38. (c) 39. (a) 40. (b)
 41. (d) 42. (d) 43. (b) 44. (d) 45. (c) 46. (a)

EXERCISE # II

1. (a,c) 2. (c,d) 3. (b,d) 4. (b,d) 5. (a,d) 6. (a,b,d)
 7. (b,d) 8. (b,c) 9. (a,b,d) 10. (a,b) 11. (a,b,c,d) 12. (a,b,d)
 13. (a,b,d) 14. (a,b,c,d) 15. (a,b,c,d) 16. (a,b,c) 17. (a,b,c) 18. (a,b,c,d)
 19. (c,d) 20. (b,c)

EXERCISE # III**Paragraph Type :**

1. (d) 2. (d) 3. (d) 4. (c) 5. (d) 6. (d) 7. (b) 8. (a) 9. (b) 10. (c)
 11. (a) 12. (d) 13. (c) 14. (a) 15. (b)

Matrix Match Type :

- | | (a) | (b) | (c) | (d) | | (a) | (b) | (c) | (d) |
|-----|--------|----------|--------|-------|-----|----------|--------|----------|-------|
| 1. | P,Q,S; | Q,R,S; | Q; | Q,S,T | 2. | Q; | P,S; | R; | T |
| 3. | R,S; | Q; | S; | P | 4. | Q,R,S,T; | Q,R,T; | Q,R,T; | P,T |
| 5. | P; | S; | Q; | R | 6. | P,Q,R; | P,Q; | P,Q,R,S; | P,Q,R |
| 7. | R; | P,Q,S,T; | R,S,T; | P,Q,R | 8. | P,S; | P,R; | P,Q; | T |
| 9. | P,R; | P,R; | Q,R,T; | Q,S,T | 10. | Q,R; | R; | P,Q; | P,S |
| 11. | R; | Q; | T; | P,S | | | | | |

EXERCISE # IV

1. 3 2. 6 3. 4 4. 1 5. 1 6. 4 7. 2 8. 5 9. 6 10. 6
 11. 0 12. 4 13. 7 14. 6 15. 3

CH-9 PERIODIC TABLE

EXERCISE # I

- | | | | | | | | | | |
|----------|----------|----------|----------|----------|----------|----------|----------|--------------|----------|
| 1. (b) | 2. (a) | 3. (a) | 4. (c) | 5. (d) | 6. (c) | 7. (d) | 8. (c) | 9. (c) | 10. (a) |
| 11. (a) | 12. (d) | 13. (d) | 14. (c) | 15. (d) | 16. (a) | 17. (a) | 18. (a) | 19. (a) | 20. (c) |
| 21. (c) | 22. (d) | 23. (c) | 24. (c) | 25. (c) | 26. (d) | 27. (a) | 28. (a) | 29. (c) | 30. (c) |
| 31. (a) | 32. (d) | 33. (a) | 34. (d) | 35. (a) | 36. (b) | 37. (d) | 38. (b) | 39. (d) | 40. (d) |
| 41. (a) | 42. (a) | 43. (b) | 44. (b) | 45. (a) | 46. (b) | 47. (b) | 48. (c) | 49. (a) | 50. (a) |
| 51. (d) | 52. (d) | 53. (a) | 54. (a) | 55. (c) | 56. (b) | 57. (d) | 58. (a) | 59. (b) | 60. (b) |
| 61. (b) | 62. (b) | 63. (c) | 64. (a) | 65. (a) | 66. (b) | 67. (c) | 68. (a) | 69. (d) | 70. (b) |
| 71. (c) | 72. (a) | 73. (d) | 74. (d) | 75. (a) | 76. (c) | 77. (c) | 78. (c) | 79. (a) | 80. (d) |
| 81. (a) | 82. (b) | 83. (a) | 84. (c) | 85. (b) | 86. (b) | 87. (c) | 88. (d) | 89. (d) | 90. (c) |
| 91. (b) | 92. (a) | 93. (b) | 94. (c) | 95. (d) | 96. (c) | 97. (d) | 98. (b) | 99. (a) | 100. (a) |
| 101. (b) | 102. (a) | 103. (b) | 104. (c) | 105. (d) | 106. (a) | 107. (d) | 108. (b) | 109. (d) | 110. (a) |
| 111. (d) | 112. (b) | 113. (a) | 114. (c) | 115. (c) | 116. (b) | 117. (a) | 118. (a) | 119. (d) | 120. (b) |
| 121. (d) | 122. (c) | 123. (c) | 124. (c) | 125. (b) | 126. (c) | 127. (a) | 128. (c) | 129. (b) | 130. (a) |
| 131. (b) | 132. (b) | 133. (d) | 134. (d) | 135. (a) | 136. (a) | 137. (d) | 138. (a) | 139. (a,b,c) | 140. (b) |
| 141. (a) | 142. (d) | 143. (d) | 144. (a) | 145. (c) | 146. (a) | 147. (d) | 148. (d) | 149. (d) | 150. (b) |

EXERCISE # II

- | | | | | | |
|---------------|---------------|----------------|---------------|---------------|---------------|
| 1. (c,d) | 2. (a,d) | 3. (a,b,c) | 4. (c,d) | 5. (b,c) | 6. (a,b) |
| 7. (c,d) | 8. (a,c,d) | 9. (a,b,c) | 10. (a,d) | 11. (b,c,d) | 12. (b,d) |
| 13. (a,c,d) | 14. (a,b,c,d) | 15. (a,b,c,d) | 16. (b,c,d) | 17. (b,d) | 18. (a,b,c) |
| 19. (a,b,c,d) | 20. (a,b) | 21. (a,d) | 22. (a,b,c) | 23. (b,d) | 24. (a,d) |
| 25. (b,c,d) | 26. (b,c,d) | 27. (c,d) | 28. (a,d) | 29. (a,b,c,d) | 30. (b,c,d) |
| 31. (a,b,c,d) | 32. (a,b,d) | 34. (a,b,c) | 35. (b,c) | 36. (a,b) | 37. (a,b,c,d) |
| 38. (a,b,c,d) | 39. (a,b,c) | 40. (a,d) | 41. (a,c) | 42. (a,b,c) | 43. (a,b,c,d) |
| 44. (c,d) | 45. (a,c,d) | 46. (a,c,d) | 47. (a,b,c) | 48. (a,b,c) | 49. (b,c) |
| 50. (a,b,c,d) | 51. (b,c) | 52. (a,d) | 53. (c,d) | 54. (b,c) | 55. (b,d) |
| 56. (a,b,c) | 57. (a,b,d) | 58. (b,c,d) | 59. (a,b,c,d) | 60. (a,d) | 61. (a,c,d) |
| 62. (a,b,d) | 63. (b,c,d) | 64. (c,d) | 65. (d) | 66. (a,d) | 67. (a,b) |
| 68. (a,b,c,d) | 69. (b,c) | 70. (b,c) | 71. (a,b) | 72. (a,b) | 73. (b,c,d) |
| 74. (a,d) | 75. (a,b,c,d) | 76. (c,d) | 77. (a,b) | 78. (c,d) | 79. (a,b,c,d) |
| 80. (b,c,d) | 81. (b,d) | 82. (b,d) | 83. (b,c,d) | 84. (a,c) | 85. (a,d) |
| 86. (a,c) | 87. (a,c) | 88. (a,c) | 89. (a,c) | 90. (a,d) | 91. (b,c,d) |
| 92. (a,c) | 93. (b,d) | 94. (a,b,d) | 95. (b,c) | 96. (a,c) | 97. (b,d) |
| 98. (a,b,d) | 99. (a,b,d) | 100. (a,b,c,d) | | | |

EXERCISE # III

Paragraph Type :

- | | | | | | | | | | |
|---------|---------|---------|---------|---------|---------|---------|---------|---------|---------|
| 1. (c) | 2. (d) | 3. (d) | 4. (b) | 5. (a) | 6. (d) | 7. (b) | 8. (d) | 9. (a) | 10. (d) |
| 11. (b) | 12. (d) | 13. (c) | 14. (d) | 15. (a) | 16. (b) | 17. (b) | 18. (d) | 19. (d) | 20. (d) |
| 21. (a) | 22. (d) | 23. (a) | 24. (b) | 25. (c) | 26. (d) | 27. (a) | 28. (b) | 29. (b) | 30. (b) |
| 31. (c) | 32. (d) | 33. (d) | 34. (b) | 35. (b) | 36. (c) | 37. (b) | 38. (c) | 39. (b) | 40. (c) |
| 41. (c) | 42. (b) | 43. (d) | 44. (c) | 45. (c) | 46. (d) | 47. (c) | 48. (d) | 49. (c) | 50. (c) |
| 51. (a) | 52. (d) | 53. (b) | 54. (b) | 55. (d) | 56. (d) | 57. (b) | 58. (b) | 59. (c) | 60. (b) |

Matrix Match Type :

- | | (a) | (b) | (c) | (d) | | (a) | (b) | (c) | (d) |
|-----|--------|----------|--------|---------|------|--------|----------|----------|-----------|
| 61. | P; | Q; | R; | S | 62. | S; | R; | Q; | P |
| 63. | S; | R; | Q; | P | 64. | P; | Q; | R; | S |
| 65. | P,S; | Q,S; | R; | T | 66. | Q,R,T; | P,R,S,T; | P,R,S,T; | P,R,S |
| 67. | Q,R; | P; | S; | T | 68. | P,T; | Q,T; | R,S; | S |
| 69. | Q,T; | P,R,S,T; | R,S,T; | P,R,S,T | 70. | P; | Q; | R; | S |
| 71. | P; | Q; | R; | S | 72. | P; | Q; | R; | S |
| 73. | P; | Q; | R; | S | 74. | S; | R; | P; | Q |
| 75. | R; | S; | P; | Q | 76. | R; | P; | S; | Q |
| 77. | P; | Q; | R; | S | 78. | P; | Q; | R; | S |
| 79. | Q,R,T; | P,R,S; | P,R,S; | Q,S,T | 80. | P,T; | Q,S,T; | R,S; | R,S |
| 81. | P; | R; | S; | Q | 82. | S; | P; | Q; | R |
| 83. | Q; | R; | P; | S; | 84. | P; | Q; | R; | S |
| 85. | Q; | P; | R; | S | 86. | Q,T; | P,S; | P,S; | R |
| 87. | P; | Q,T; | R; | S | 88. | P,T; | P,Q,T; | P,Q,R,T; | P,Q,R,S,T |
| 89. | P,T; | R; | S; | Q | 90. | R; | Q; | P,S; | P,S |
| 91. | P,R; | R,S; | Q,R; | P,R | 92. | P; | Q; | R; | S |
| 93. | P; | Q; | R; | S | 94. | P; | Q; | R; | S |
| 95. | Q,S; | T; | R,T; | P | 96. | R,T; | P,R,T; | P,Q,R; | Q,S,T |
| 97. | R,T; | P,S; | S,T; | Q,S,T | 98. | P,Q,S; | P,Q,R; | P,Q,R; | P,S,T |
| 99. | S; | R; | Q; | P | 100. | P; | P,T; | Q,S; | P,S |

EXERCISE # IV

- | | | | | | | | | | |
|------------|---------|---------|---------|---------|------------|------------|-------------|---------|-----------|
| 1. (5) | 2. (6) | 3. (8) | 4. (9) | 5. (3) | 6. (3) | 7. (2) | 8. (8) | 9. (3) | 10. (9)72 |
| 11. (3) | 12. (9) | 13. (6) | 14. (9) | 15. (0) | 16. (2) | 17. (2) | 18. (141/6) | 19. (8) | 20. (9) |
| 21. (6) | 22. (5) | 23. (5) | 24. (3) | 25. (2) | 26. (0) | 27. (8) | 28. (6) | 29. (7) | 30. (8) |
| 31. (14/5) | 32. (4) | 33. (4) | 34. (6) | 35. (1) | 36. (16/7) | 37. (21/3) | 38. (3/30) | 39. (0) | 40. (1) |
| 41. (2) | 42. (7) | 43. (9) | 44. (2) | 45. (3) | 46. (4) | 47. (3) | 48. (8) | 49. (1) | 50. (5) |

EXERCISE # V(JEE-MAIN & ADVANCED)

1. (c) 2. (c) 3. (a) 4. (a) 5. (c) 6. (a) 7. (d) 8. (b) 9. (d) 10. (b)
11. (d) 12. (b) 13. (a) 14. (b) 15. (b) 16. (b) 17. $\text{Al}^{3+} < \text{Mg}^{2+} < \text{Li}^+ < \text{K}^+$
18. Inert pair effect 19. (c) 20. (d)
21. CrO_3 on reaction with water produces chromic acid (H_2CrO_4). So CrO_3 is an acid anhydride.
22. (c) 23. (b) 24. (a) 25. (a) 26. (b) 27. $\text{Cl}_2\text{O}_7 < \text{SO}_3 < \text{CO}_2 < \text{B}_2\text{O}_3 < \text{BaO}$ 28. (c)
29. (c) 30. (c) 31. (d) 32. (d) 33. (b) 34. (d) 35. (c) 36. (d) 37. (d) 38. (d)
39. (a) 40. (a) 41. (b) 42. (d) 43. (d) 44. (a) 45. (b) 46. (c) 47. (c) 48. (b)
49. (a) 50. (a) 51. (6)

■ ■

Advanced Problems in Inorganic Chemistry

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About the Author

Mr. Ramashish Paul is a B.Tech graduate from Calcutta University. He has been actively involved in educating students appearing for IIT-JEE & other engineering entrance examinations for more than 17 years in various renowned IIT-JEE coaching institutes of Delhi & Kota. Mr. Paul is famous among students and well-known for his conceptual clarity in Inorganic & Organic Chemistry.



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ISBN 978-81-8355-864-8